Lecture Notes in Networks and Systems 699

Valentina Emilia Balas
Vijay Bhaskar Semwal
Anand Khandare *Editors*

Intelligent Computing and Networking

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Valentina Emilia Balas · Vijay Bhaskar Semwal ·
Anand Khandare
Editors

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Editors
Valentina Emilia Balas
Aurel Vlaicu University of Arad
Arad, Arad, Romania

Vijay Bhaskar Semwal
NIT
Bhopal, India

Anand Khandare
Thakur College of Engineering
and Technology
Mumbai, Maharashtra, India

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Preface

Thakur College of Engineering & Technology (TCET), a Graded Autonomous Linguistic Minority Institute, was established in AY 2001-02 with a clear objective of providing Quality Technical Education in tune with international standards and contemporary global requirements. The Institute is recognized by All India Council for Technical Education (AICTE) & Government of Maharashtra and is affiliated to the University of Mumbai (UOM). It has been conferred Autonomous Status by University Grant Commission (UGC) for 10 years w.e.f. A.Y. 2019–20 to A.Y. 2028–29. TCET offers 14 UG (8 in Engineering, 3 in Technology and 3 in Vocational Course), 3 PG and 3 Ph.D. (Tech.) programmes. TCET has implemented Holistic Multidisciplinary Education which is as per National requirement in building the budding Engineers for Global opportunities.

International Conference on Intelligent Computing and Networking (IC-ICN-2023) is a platform for conducting conference with an objective of strengthening the research culture by bringing together academicians, scientists, researchers in the domain of intelligent computing and Networking.

The 14th annual event IC-ICN-2023 in the series of international conferences organized by Thakur College of Engineering and Technology (TCET) under the umbrella of MULTICON 2023. IC-ICN-2023 was conducted online and offline on 24th and 26th February, 2023. It provides not only a great platform to think innovatively but also bringing in sync the theory and applications in the field of Intelligent Computing and Networking for stakeholders. This platform also helps us for the collaborations and enhancing the network with the peer universities and institutions in India and aboard. The various types of papers Research Papers, Technical Papers, Case Studies, Best and Innovative Practices, Engineering Concepts & Designs so that the applied study or research in the domain are presented in the conference. There are more than 600 papers are received this year. Need of hour is the technological development in the domain of intelligent computing and Networking and this conference facilitates this.

Year by Year IC-ICN 2023 is gaining popularity and wide publicity through our various platforms like website, social media coverage as well as the vigorous promotion by the team of faculty members to the various colleges. The IC-ICN-2023 has affiliation with Scopus Indexed journals for intelligent systems, leading publication house and conference proceeding with ISBN number.

Authors and participants appreciated efforts taken by the team of TCET for making this 4th Springer conference successful and building strong relationship and bonding by taking care of each and every participant's requirement in the event. The two days event comprises of conferences and workshops with multiple tracks. During these two days, there were 600 papers were presented by national and international authors. Papers were presented in front of delegates from reputed institutes and industries. Our sincere thanks to our management for providing us resources and infrastructure. We also thank all the members organizing and editorial committee for supporting the event and extending their cooperation to make it a grand successful.

Team-ICICN-2023

Arad, Romania — Valentina Emilia Balas
Bhopal, India — Vijay Bhaskar Semwal
Mumbai, India — Anand Khandare

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Editors and Contributors

About the Editors

Valentina Emilia Balas is currently a full professor at “Aurel Vlaicu” University of Arad, Romania. She is an author of more than 300 research papers. Her research interests are in intelligent systems, fuzzy control and soft computing. She is the editor-in-chief to International Journal of Advanced Intelligence Paradigms (IJAIP) and to IJCSE. Dr. Balas is a member of EUSFLAT, ACM and a SM IEEE, member in TC-EC and TC-FS (IEEE CIS), TC-SC (IEEE SMCS) and a joint secretary FIM.

Vijay Bhaskar Semwal is working as an assistant professor (CSE) at NIT Bhopal since February 5, 2019. Before joining NIT Bhopal, he was working at NIT Rourkela. He has also worked with IIIT Dharwad as an assistant professor(CSE) for 2 years (2016–2018), and he has also worked as an assistant professor (CSE) at NIT Jamshedpur . He has earned his doctorate degree in robotics from IIIT Allahabad (2017), M.Tech. in Information Technology from IIIT Allahabad (2010) and B.Tech. (IT) from College of Engineering Roorkee (2008). His areas of research are bipedal robotics, gait analysis and synthesis, artificial intelligence, machine learning and theoretical computer science. He has published more than 15 SCI research papers. He has received early career research award by DST-SERB under Government of India.

Dr. Anand Khandare is the Associate Professor & Deputy Head of the Department of Computer Engineering, Thakur College of Engineering and Technology, Mumbai, with 18 years of teaching experience. He completed Ph.D. in Computer Science and Engineering in the domain data clustering in machine learning from Sant Gadge Baba Amravati University. He has a total of 60+ publications in national and international conferences and journals. He has 1 copyright and 2 patents. He guided various research and funded projects. He conducted corporate training on AIML and Programming languages for the industries. He organized various conferences. He worked as a volume editor in Springer International Conference on Intelligent Computing and Networking for the Years 2020, 2021, 2022. He is also a reviewer in various journals and conferences.

Contributors

Rajanikanth Aluvalu Chaitanya Bharathi Institute of Technology, Hyderabad, India

Shashikant V. Athawale Department of Computer Engineering, AISSMS COE, SavitribaiPhule Pune University, Pune, India

Bhavna Bajpai Dr. C V Raman University, Khandwa, Madhya Pradesh, India

Shubham Bane Shivajirao S. Jondhale College of Engineering, University of Mumbai, Dombivli (E), India

Rajesh Bansode Thakur College of Engineering and Technology, Mumbai University, Mumbai, India

Sukanta Kumar Baral Indira Gandhi National University, New Delhi, India; Department of Commerce, Faculty of Commerce and Management, Indira Gandhi National Tribunal University, Amarkantak, Madhya Pradesh, India

Fabian Barreto Department of Electronics and Telecommunication, Xavier Institute of Engineering, Mumbai, India

Amrut Bhagwat Vishwakarma Institute of Technology, Pune, India

Pallavi Bharambe Shivajirao S. Jondhale College of Engineering, University of Mumbai, Dombivli (E), India

Alokam Ujwala Bharati School of Computer Science and Engineering, VIT-AP University, Andhra Pradesh, India

Diksha Bhave Department of Computer Engineering, Shivajirao S. Jondhale College of Engineering Dombivli (E), Dombivli, India

Pradnya D. Bormane Department of Computer Engineering, AISSMS College of Engineering, Pune, India

Mohit Burchunde Vishwakarma Institute of Technology, Pune, India

Piyush Charan Electronics and Communication Engineering, Manav Rachna University, Faridabad, India

Amogh Chaudhari General Practitioner, M.B.B.S., Mumbai, India

Archana Chaudhari Department of Instrumentation Engineering, Vishwakarma Institute of Technology, Pune, India

Shailesh Chaudhari Veer Narmad South Gujarat University, Surat, India

Subhasis Dash KIIT, Bhubanewar, India

Sandeep Kumar Davuluri Department of Information Technology, University of the Cumberlands, Williamsburg, KY, USA

Sheshang Degadwala Department of Computer Engineering, Sigma Institute of Engineering, Vadodara, Gujarat, India

Yash Desai Shivajirao S. Jondhale College of Engineering, University of Mumbai, Dombivli (E), India

Sarita Dhal KIIT Deemed to be University, Bhubaneswar, Odisha, India

Gaurav Dhiman Institute of Engineering and Technology, Chitkara University, Punjab, India;
Department of Computer Science and Engineering, Graphic Era Deemed to be University, Dehradun, India

Salunke Mangesh Dilip Department of Computer Engineering, GHRCEM, Pune, India

Vinitkumar Dongre Thakur College of Engineering and Technology, Mumbai, Maharashtra, India

Juan Carlos Orosco Gavilán Universidad Privada del Norte, campus virtual, Lima, Peru

Premanand Ghadekar Vishwakarma Institute of Technology, Pune, India

Hritwik Ghosh School of Computer Science and Engineering (SCOPE), VIT-AP University, Amaravati, Andhra Pradesh, India

Sanjukta Ghosh Srishti Manipal Institute of Art, Design and Technology, Manipal Academy of Higher Education, Manipal, India

Richa Goel Symbiosis Centre for Management Studies, Symbiosis International University, Noida, India

Riya Gokhale Department of Instrumentation Engineering, Vishwakarma Institute of Technology, Pune, India

G. Gokulkumari Department of E-Commerce, College of Administration and Finance, Saudi Electronic University, Riyadh, Kingdom of Saudi Arabia

Devidas Golhar Marathwada Mitramandal's College of Commerce, Pune, Maharashtra, India

Saniya Gonsalves Department of Applied Sciences and Humanities, Xavier Institute of Engineering, Mumbai, India

Reddy N. Hanuman R K University, Rajkot, India

Vaishali S. Hanwate Department of Computer Engineering, Government College of Engineering, Yavatmal, Maharashtra, India

Varun P. Herlekar Department of Computer Engineering, Government College of Engineering, Yavatmal, Maharashtra, India

Pritidhara Hota GITA Autonomous College, Bhubaneswar, Odisha, India

Tejas Indulkar Shivajirao S. Jondhale College of Engineering, University of Mumbai, Dombivli (E), India

Kunal Jadhav Vishwakarma Institute of Technology, Pune, India

Omkar Jadhav Vishwakarma Institute of Technology, Pune, India

Chinnakanu Sai Janakiram School of Computer Science and Engineering, VIT-AP University, Andhra Pradesh, India

Malik Jawarneh Faculty of Computing Sciences, Gulf College, Muscat, Oman

Malik Mustafa Mohammad Jawarneh Faculty of Computing Sciences, Gulf College, Seeb, India

M. Jayakrishna Mechanical Engineering, Sri Sivani Engineering College, Srikakulam, India

K. Jeeveth Department of School of Computer Science and Engineering (SCOPE), VIT-AP University, Amaravati, Andhra Pradesh, India

Aaron Johns Department of Information Technology, St. Xavier's College, Mumbai, India

Deepa Jose Department of Electronics and Communication, KCG College of Technology, Chennai, Tamil Nadu, India

J. Armstrong Joseph Department of Computer Science and Engineering, Sri Venkateswara College of Engineering and Technology (Autonomous), Chittoor, India

Lakshman Kumar Kanulla Department of SAP SD MM, Working for Apple Inc, Cupertino, USA

Akramjon Karimov Department of Corporate Finance and Securities, Tashkent Institute of Finance, Tashkent, Uzbekistan

Ashwini Katkar Thakur College of Engineering and Technology, Mumbai, Maharashtra, India

Amrinder Kaur Pyramid Institute of Business and Technology, Phagwara, India

Manpreet Kaur Lovely Professional University, Phagwara, India

Ismail Keshta Computer Science and Information Systems Department, College of Applied Sciences, AlMaarefa University, Riyadh, Saudi Arabia

Anand Khandare Department of Computer Engineering, Thakur College of Engineering and Technology, Kandivali, Mumbai, India

Swapnil V. Khandare Department of Computer Engineering, Government College of Engineering, Yavatmal, Maharashtra, India

Amit Khare Department of Technology, JIET Institute of Design and Technology, Jodhpur, Rajasthan, India

Syed Khasim School of Computer Science and Engineering (SCOPE), VIT-AP University, Amaravati, Andhra Pradesh, India

Aditya Kirar Vishwakarma Institute of Technology, Pune, India

D. Kirubakaran Department of EEE, St. Joseph's Institute of Technology, Chennai, Tamil Nadu, India

Samhita Kothandaraman Thakur College of Engineering and Technology, Mumbai, India

M. Vamsi Krishna Department of IT, Aditya Engineering College, Surampalem, India

Mukund Anant Kulkarni Bharati Vidyapeeth (Deemed to Be University), Institute of Management, Kolhapur, India

Ashok Kumar Department of Computer Science, Jaipur, Rajasthan, India

Athmakuri Naveen Kumar Senior Software Engineer, Glosoft Technologies Pvt Ltd, Chennai, India

Atul Kumar Dr. D. Y. Patil B-School, Pune, India

Pankaj Kumar Wipro Ltd, Patna, Bihar, India

T. Kumaresan Department of Artificial Intelligence and Data Science, Bannari Amman Institute of Technology, Sathyamangalam, India

J. Divya Lakshmi Department of Electronics and Communication Engineering, Sri Krishna Institute of Technology, Bengaluru, India

Govinda Rajulu Lanke Data Science and Engineering, Birla Institute of Technology and Science, Pilani, Rajasthan, India

Amit Lathigara R K University, Rajkot, India

Renato R. Maaliw College of Engineering, Southern Luzon State University, Lucban, Quezon, Philippines

Puspalata Mahapatra KIIT Deemed to be University, Bhubaneswar, Odisha, India

P. Y. Mallikarjun Department of Electronics and Communication Engineering, Dayananda Sagar Academy of Technology and Management, Bengaluru, India

Nishikanta Mishra KIIT Deemed to be University, Bhubaneswar, Odisha, India

Sachi Nandan Mohanty School of Computer Science and Engineering (SCOPE), VIT-AP University, Amaravati, Andhra Pradesh, India

Lalita Moharkar Department of Electronics and Telecommunication, Xavier Institute of Engineering, Mumbai, India

Rasal Mrugesh Department of Computer Engineering, Shivajirao S. Jondhale College of Engineering Dombivli (E), Dombivli, India

Mustafa Mudhafar Department of Anesthesia and Health Care Faculty of Altuff College University, Karbala, Iraq

Krishna Murthy Department of Journalism and Mass Communication, Indira Gandhi National Tribal University, Amarkantak, Madhya Pradesh, India

Tanmayi Nagale Thakur College of Engineering and Technology, Kandivali (E), India

S. R. Nalamwar School of Computing, Bharath Institute of Higher Education and Research (BIHER), Chennai, Tamilnadu, India

Bhagirathi Nayak Sri Sri University, Cuttack, Odisha, India

S. Neduncheliyan School of Computing, Bharath Institute of Higher Education and Research (BIHER), Chennai, Tamilnadu, India

Rahul P. Neve Thakur College of Engineering and Technology, Mumbai University, Mumbai, India

Sapaeva Feruza Norbaevna Urgench Branch of Tashkent University of Information, Technologies Named After Muhammad Al-Khwarizmi, Urgench, Uzbekistan

Hiral S. Padhariya Veer Narmad South Gujarat University, Surat, India

Sagar Dhanraj Pande School of Computer Science and Engineering, VIT-AP University, Amaravati, Andhra Pradesh, India

Bhavesh Pandekar KPMG India, Mumbai, India;
School of Engineering and Applied Sciences, University of Mumbai, Kalyan, India

Hetal Pandya Department of Computer Engineering, L D College of Engineering, Ahmedabad, Gujarat, India

Aniket Pardeshi Vishwakarma Institute of Technology, Pune, India

Archana B. Patankar Computer Engineering Department, Thadomal Shahani Engineering College, Bandra (West), Mumbai, India

Ashwini Patil Thakur College of Engineering and Technology, Mumbai, India

Leena Patil Department of Electronics and Telecommunication, Xavier Institute of Engineering, Mumbai, India

Megharani Patil Thakur College of Engineering and Technology, Mumbai, India

Radha Mohan Pattanayak School of Computer Science and Engineering, VIT-AP University, Andhra Pradesh, India

Nihar Ranjan Pradhan Department of School of Computer Science and Engineering (SCOPE), VIT-AP University, Amaravati, Andhra Pradesh, India

G. N. R. Prasad Department of MCA, Chaitanya Bharathi Institutute of Technology (A), Hyderabad, Telangana, India

K. L. Raghavender Reddy R K University, Rajkot, India

Irfan Sadiq Rahat School of Computer Science and Engineering (SCOPE), VIT-AP University, Amaravati, Andhra Pradesh, India

Santhosh Kumar Rajamani Department of E.N.T, MAEER MIT Pune's MIMER Medical College, Dr. BSTR Hospital, Pune, India

S. V. Ramanan Department of Electronics and Communication Engineering, PPG Institute of Technology, Coimbatore, India

Shubhangee Ramaswamy Institute of Management Education Research and Training, Pune, Maharashtra, India

Milind Rane Vishwakarma Institute of Technology, Pune, India

Khan Darakhshan Rizwan Computer Engineering Department, Thadomal Shahani Engineering College, Bandra (West), Mumbai, India

Pendurkar Rohit Department of Computer Engineering, Shivajirao S. Jondhale College of Engineering Dombivli (E), Dombivli, India

Jitendra Kumar Rout NIT, Raipur, India

Mukta Sandhu SVSU, Haryana, India

Savita Sangam Department of Computer Engineering, School of Engineering and Applied Sciences, University of Mumbai, Kalyan, India

Srushti Sankhe Department of Computer Engineering, Thakur College of Engineering and Technology, Kandivali, Mumbai, India

Snehal Sarangi KIIT, Bhubanewar, India

Vidya Sarode Department of Electronics and Telecommunication, Xavier Institute of Engineering, Mumbai, India

Mugdha Sawant Department of Computer Engineering, Thakur College of Engineering and Technology, Kandivali, Mumbai, India

Nikita Sawant Department of Instrumentation Engineering, Vishwakarma Institute of Technology, Pune, India

V. Selvakumar Department of Maths and Statistics, Bhavan's Vivekananda College of Science, Humanities and Commerce, Hyderabad, Telangana, India

Amita Shah Department of Computer Engineering, L D College of Engineering, Ahmedabad, Gujarat, India

Aakanksha Sharaff National Institute of Technology Raipur, Raipur, Chhattisgarh, India

Purva Shendge Department of Instrumentation Engineering, Vishwakarma Institute of Technology, Pune, India

Gangaprasad M. Shirale Department of Computer Engineering, Government College of Engineering, Yavatmal, Maharashtra, India

S. D. Shirbahadurkar Department of E&TC, Zeal College of Engineering and Research, Pune, India

Madhura Shirodkar Department of Electronics and Telecommunication, Xavier Institute of Engineering, Mumbai, India

Mithun Shivakoti Department of School of Computer Science and Engineering (SCOPE), VIT-AP University, Amaravati, Andhra Pradesh, India

Ankit Singh Vishwakarma Institute of Technology, Pune, India

Gurpreet Singh Faculty of Computational Science, GNA UNIVERSITY, Phagwara, India

Pavitar Parkash Singh Department of Management, Lovely Professional University, Phagwara, India

Sapan Kumar Singh Wipro Ltd, Pune, Maharashtra, India

Shweta Singh Electronics and Communication Department, IES College of Technology, Bhopal, India

Tilottama Singh Uttaranchal Institute of Management (UIM), Dehradun, India

Dhiraj D. Sirbhate Department of Computer Engineering, Government College of Engineering, Yavatmal, Maharashtra, India

Mukesh Soni Department of CSE, University Centre for Research and Development Chandigarh University, Mohali, Punjab, India

Keerthi Srinivas Altiostar Networks India, Pvt. Ltd., Bangalore, India

Singh Sumit Department of Computer Engineering, Shivajirao S. Jondhale College of Engineering Dombivli (E), Dombivli, India

Herison Surbakti Faculty of Computing and Digital Technology, HELP University, Kuala Lumpur, Malaysia

Ajay Talele Vishwakarma Institute of Technology, Pune, India

Saikumar Tara Department of ECE, BVRIT Hyderabad College of Engineering for Women, Hyderabad, India

Mahatir Ahmed Tusher School of Computer Science and Engineering (SCOPE), VIT-AP University, Amaravati, Andhra Pradesh, India

Gyananjaya Tripathy National Institute of Technology Raipur, Raipur, Chhattisgarh, India

Shailaja Udtewar Department of Electronics and Telecommunication, Xavier Institute of Engineering, Mumbai, India

Surendra Reddy Vinta School of Computer Science and Engineering, VIT-AP University, Amaravati, Andhra Pradesh, India

Uma Maheswari Viswanadhula Chaitanya Bharathi Institute of Technology, Hyderabad, India

Amar Pal Yadav CSE-AI College: Noida Institute of Engineering and Technology Greater Noida, Greater Noida, India

Mannava Yesu Babu Department of CSE, Vardhaman College of Engineering, Shamshabad, Hyderabad, Telangana, India

Analysis of Healthcare System Using Classification Algorithms

Anand Khandare, Mugdha Sawant, and Srushti Sankhe

Abstract Classification is a method of predicting similar information from the value of a categorical target or categorical variable. It is a useful technique for any kind of statistical data. These algorithms are used for different purposes like image classification, predictive modelling, data mining techniques, etc. The primary objective of supervised learning is to build a simple and clear class labelling model of the predictive features. The classifiers are then used to classify the class labels of the test cases where the values of the predictive features are noted with the value of the unknown class labels. In other words, the goal of machine learning is to build a concise model of the distribution of class labels in terms of predictor features. Assigning class labels to the generated classifier is the next step and checking the cases where the predictive feature values are known. This paper illustrates various classification techniques used in supervised machine learning that incorporate a number of parameters. We have compared seven machine learning algorithms including Decision Tree, Random Forest, K-nearest Neighbor, Support Vector Machines, Logistic Regression, Naive Bayes and LightGBM using four different datasets regarding healthcare systems. In this review article, we summarize machine learning techniques that are widely used in the region of health systems due to their data processing and analysis capabilities.

Keywords Healthcare · Classification · Machine learning

1 Introduction

A sublet of artificial intelligence known as "machine learning" is based on the belief that computers are equipped with the learning from data, spotting patterns, and making judgements without much or no human involvement. Nowadays, the field of

A. Khandare · M. Sawant (✉) · S. Sankhe
Department of Computer Engineering, Thakur College of Engineering and Technology, Kandivali, Mumbai, India
e-mail: mugdhamsawant01@gmail.com

A. Khandare
e-mail: anand.khandare@thakureducation.org

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data analysis and computing has expanded quickly, typically enabling applications to function in brilliant ways. It is frequently referred to as the most popular modern technology, providing systems with the ability to automatically learn and improve from experience without being specially programmed [1].

Algorithms are trained using analytical techniques to perform classification or prediction, revealing important findings from data mining projects [2]. This information then guides decision making across operations and companies, exquisitely impacting fundamental growth metrics. As Big Data keeps on growing and evolving, the market need for data scientists also will increase, requiring them to assist to recognize the most apt business queries and, subsequently, data to answer them [3].

In order to create intelligent methods for identifying and treating disease, scientists are focusing on machine learning. Because ML can identify disease and virus infections more precisely, patients' illnesses can be identified earlier, the risky stages of diseases may be avoided, and there will be fewer patients overall. The work of anticipating COVID-19 infection can be automated, and ML can also assist in predicting COVID-19 infection rates in the future [4]. We believe this technology will be a mighty tool for allocating medical resources and planning of medical center capacity by making accurate real-time assessments of mortality risk, taking into account the highly specific characteristics of individual patients [5].

Precision medicine, which determines which treatment programs are likely to be effective on a patient according to certain patient traits, serves as the most typical approach of standard machine learning in the healthcare industry and context for treatment, etc. For the majority of machine learning as well as precision medicine applications, the outcome variable must all be known. Supervised learning is the term used here [6]. In healthcare, the use of machine learning is increasing and helps patients and clinicians overcome industry challenges and create a more unified system to improve workflow. It requires advanced preparation before application implemented in healthcare to distinguish between types of data associations, similar types of data, where learning from data gives the proper output. The larger the data sample provided to the "machine", the more accurate the machine's output becomes [7].

In high income nations, a group at the Cambridge Center for AI in Medicine in the UK created the Cambridge Adjutorium system, which employs the highest developed ML framework to precisely estimate the rate of mortality, the need for ICU admission, and the requirement for ventilation in hospital patients with COVID-19. The system's accuracy rate ranged from 77 to 87%, which was a pretty high accuracy rate even though it was trained using extremely small datasets (CHESS data from Public Health England) [8]. The main advantages of ML/AI nowadays are diagnostic and predictive analysis, however efforts are being made to benefit additional medical problem domains. Recently, considerable advancements have been achieved in the ability of methods for machine learning, in particular deep learning algorithms, to autonomously diagnose illnesses while decreasing the cost and expanding accessibility of diagnostics [9].

2 Motivation

The research area of ML, or "machine learning," is constantly growing and offers many opportunities for investigation and application [10]. Even if machine learning has had a minimal impact on healthcare, Mr. James Collin claimed at MIT that it is the technology that would define this decade. Several new companies in the ML sector are focusing seriously on the medical field. Even Google has entered the fray and created a machine learning program for detecting cancerous tumors on mammograms. Stanford utilizes a Deep Learning algorithm to detect skin cancer. The US healthcare system generates about one trillion gigabytes of data annually. Scientists and academic specialists have identified a number of risk aspects for chronic illness as well as its distinct characteristics. Additional data stands for more learning for machines, but for higher precision, these many features need a huge quantity of samples [11].

The use of machine learning is highly done in processes like feeding the data into ML models and guiding and correcting them to make the right judgments [12]. The technology has an expanded spectrum of applications for enhancing research in clinical trials. By applying advanced predictive modelling to applicants of clinical trials, medicinal specialists can evaluate a wider range of data and reduce the cost and time of medical testing. There are various ML workings that can improve the efficiency of clinical trials in future and can help to determine the optimal sample size to increase effectiveness and minimize the likeliness of data errors. We believe machine learning technologies can help the medical and healthcare industries explore new fields and entirely rethink the emphasis on healthcare [13].

It also demonstrates the diverse smart and caring characteristics of the ML customs for its spacious healthcare services. It includes assistance from a variety of digital and intelligent tools, such as artificial intelligence and cloud data performance for healthcare services. The creation of electronic documents in the medical sector further earnestly aids the healthcare sector, that too at a sensible cost. The savvy arranged reports, advanced notes, records keeping up with, and so forth., are several additional influential areas in which ML principles investigate the quality of its services in the healthcare sector [14].

3 Related Work

Mathematical models known as machine learning classification algorithms are utilized to approach classification problems. The method of organizing machine learning algorithms involves choosing the right algorithm to tackle a specific problem and achieve efficient results based on the available input datasets and the model preparation process. A classification algorithm's initial step in operating is to confirm that the input or output variables have thereby been appropriately encoded. The next step is to disassociate the dataset in two that is the training dataset and the testing dataset after it has been processed. The next stage is to choose the model that best fits our issue after partitioning the dataset in the direction of training and testing [15].

These are the different machine learning classification algorithms that we have implemented to comprehend the datasets effectively (Fig. 1).

One of the utmost demanding techniques for expressing the categorization of classifier data is indeed the decision tree classifier. This approach is referred to as a decision tree since a tree can symbolize the collection of splitting rules used to divide the dataset [16]. It is used in text mining as a classification system to ascertain how customers feel about a product. The training data set is utilized to create a decision tree model, and the validation data lot is used to regulate the appropriate tree size needed to achieve the optimal final model. Businesses can use decision trees to determine which products will bring in more revenue when they are launched [17].

One more method is the random forest of ensemble learning that falls into the category of homogeneous basic learners with respect to the type of constructive classifier. As the name suggests, all basic learners are decision trees, so they are simpler in structure than similar methods [18]. Moreover, it works well together with massive datasets and can sustain thousands of input parameters without collapsing variables, provide estimates of important variables, and are unbiased in generalization error as the forest grows. It produces internal estimates that are free of, have an efficient method of estimating missing data, and preserve accuracy [19]. All definitive as well as numerical variables can be handled by Random Forest in prediction scenarios. One of the features is the established cross-validation capacity, which allows the independent variables to be rated from most to least productive in terms of their relationship with the result variable. As a result, the procedure of obtaining functions from multi source analysis of data is made much easier [20].

Support Vector Machine also called SVM is also known as a well-known Supervised Learning algorithm based on statistical learning theory. It has played a vital role in pattern recognition, a very popular and active research field among researchers. Research in some areas where SVM works well has spurred the progression of various applications just like using SVM for the work of bigger data sets, multiple classifications, and imbalanced data sets [21]. With as much as 90% of the compounds being correctly categorized, they have been used to classify proteins. The categorization aspect of SVMs is broadening its application to cancer genomics today as improvements in advanced technologies result in the production of massive quantities of the genomic as well as epigenetics data. This has resulted in the finding of innovative

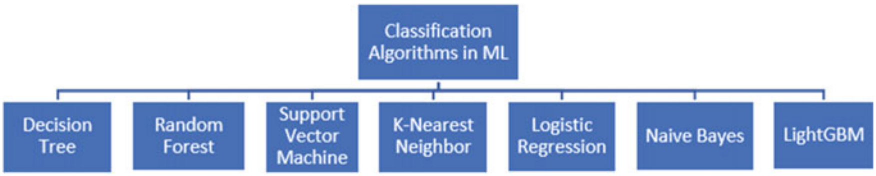

Fig. 1 Classification algorithms in ML

biomarkers, new drug project goals, and a greater sense of genes resulting in cancer [22].

A monitored machine learning method known as K-nearest neighbor (KNN) is mainly employed in classification techniques. KNN, a managed algorithm, considers the characteristics and marks of the training data to predict the categorization of unmarked data. In general, KNN algorithms can classify records adopting a good training model almost the same as the test query by considering the data points for k nearest training (neighborhoods) that are closest to the query under test. By evaluating the most probable gene expressions, it can estimate the risk of prostate, cancer and heart attacks [23]. The newly input data are grouped by kNN into consistent clutches or subsets based on their similarity to the trained data before it. The taken input is placed in the class with the largest number of its closest neighbors [24].

Another is the logistic Regression algorithm for classifying purposes. It focuses on how various independent variables affect a single result variable. Finding the association between independent factors and a dependent variable is done using regression analysis. It is a technique for predictive modelling that uses an algorithm to forecast continuous outcomes. The logistic function's curve shows the possibility of several things, such as checking obesity of the mouse depending on its weight or whether or not the cells are malignant, etc. However, this algorithm is a notable example of an algorithm that does not neatly fit into this straightforward distinction. Because it involves predicting outcomes based on quantitative relationships between variables, while it is a member of the regression family. However, in contrast to linear regression, it produces qualitative results and accepts both continuous and discrete variables as input [25].

A family of classifiers known as "Bayes classifiers" is based on the Bayes theorem application and strong parameter independent requirements. They are probabilistic in nature. The Bayes theorem can be utilized as a classifier's decision rule for disease prediction. Using the Bayes method, the likelihood of the disease is based on the number of symptoms that can be predicted [26]. The Bayes theorem draws conclusions based on the information at present and takes "conditional probabilities" into account. Prior probabilities may result in posterior probabilistic distributions that are biased. While making clinical nursing judgements and decisions, Bayesian statistics can be effective. In order to ascertain whether the information found may be employed as reliable information in the form of a prior distribution, direct comparisons of nursing diagnoses in distinct realities are possible according to the proverbial establishments of Bayesian preface [27].

Tree-based learning calculations, respected as being exceedingly viable when it comes to computation, are utilized by LightGBM, which stands for Light Angle Boosting Machine, a quick preparation method. It uses a gradient boosting framework built on decision trees to boost show execution whereas devouring less memory. It accomplishes almost the same exactness by quickening the preparing handle of the conventional GBDT (Gradient Boosting Decision Tree) by up to 20 times [28]. LightGBM algorithm develops vertically, which means it grows in the direction of the leaves whereas other algorithms trees expand horizontally that is level-wise.

There aren't many educational resources that concentrate on particular business difficulties, which is surprising given the variety of business problems that a data scientist must tackle. This essay seeks to cover the various components of one of the most common business difficulties by particularly addressing the question, "What classification is in machine learning?". In general, the nature and qualities of the data, as well as the success of the learning algorithms, determine the effectiveness and performance of a machine learning solution. Techniques such as classification analysis, regression, data clustering, feature engineering and dimensionality reduction, association rule learning, or reinforcement learning are accessible within the field of machine learning calculations to productively build data-driven frameworks [29].

A straightforward expansion of this calculation prompts strategic model trees. When there's no more structure within the information that can be modeled with a linear-logistic relapse work, the boosting handle closes. In any case, in case consideration is limited to subsets of the information, as is the case when information is part employing a standard choice tree basis like data pick up, straight models may still be able to fit the structure. As a result, the data are divided and boosted separately in each subset once it is determined that adding more straightforward linear models will not yield any additional benefits. For each subset of data, this procedure refines the previously generated logistic model separately. Each subset undergoes cross-validation once more to determine the appropriate number of iterations to carry out in that subset [30].

The performance of the classification models for a given set of experimental data is evaluated using a matrix known as the confusion matrix. Only after knowing the true values of the test data can they be determined. Confusion matrix is a performance measurement technique for machine learning classification. It is a type of spreadsheet that helps determine the performance of a classification model using a set of test data with known true values [31]. The need for confusion matrix and parameters is, it assesses how well classification models perform when they make predictions based on test data and indicates how effective our classification model is selecting fewer features can improve classification and help better understand the underlying process that generates the data, in addition to reducing the size of the data [32].

We have discussed and studied seven classification algorithms which are Decision Tree, Random Forest, Support Vector Machine and K-Nearest Neighbour, Logistic Regression, Naive Bayes and LightGBM. We computed the various model parameters using the confusion matrix on the four different datasets and got the analysis accordingly.

Some of the confusion matrix and parameters that are progressively used are mentioned below:

A. *Precision*
B. *Recall*
C. *F1-score*
D. *Support*
A. *Precision*

Precision is calculated by dividing the total number of true positives by the total number of true positives + false positives. A false positive is when the model incorrectly classifies a case as positive when it is indeed negative. The accuracy of a good classifier should preferably be 1 (high). These indicators inform users about the percentage of positive predictions that are actually positive [33] (Fig. 2).

The outcome demonstrates that the precision is the highest for logistic regression that is above 85% and the lowest for KNN algorithm which is around 70%. It is a display of very distinct values for the various algorithms selected.

B. *Recall*

Recall is a metric that measures the percentage of correct positive predictions among all possible positive predictions. Recall gives an indicator of missed positive predictions, unlike precision, which only comments on the accurate positive predictions out of all positive predictions. Recall measures the capacity to locate all relevant instances of a class in a data set. It evaluates the completeness of the classifier's results. It is the ratio of the total number of positive examples reported to the total number of truly positive examples [34] (Fig. 3).

The recall value also gives the same results like the precision values ranking logistic regression as the highest and KNN as the lowest.

C. *F1-score*

Also known as F-Score or F-Measure, this metric calculates the performance of an algorithm by considering both precision and recall. While F1 is generally beneficial over precision, especially if you have an unequal class distribution, it's intuitively not as straightforward as precision. This metric balances recall and accuracy because

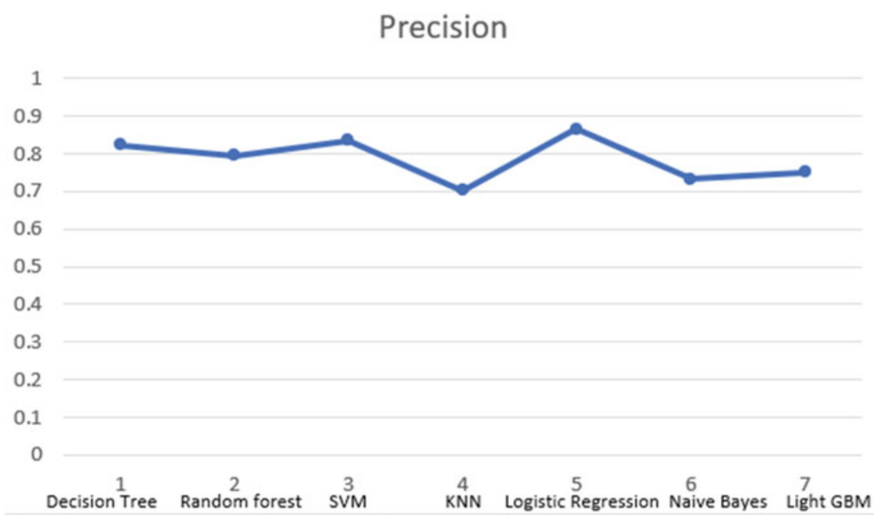

Fig. 2 Precision comparison of algorithms

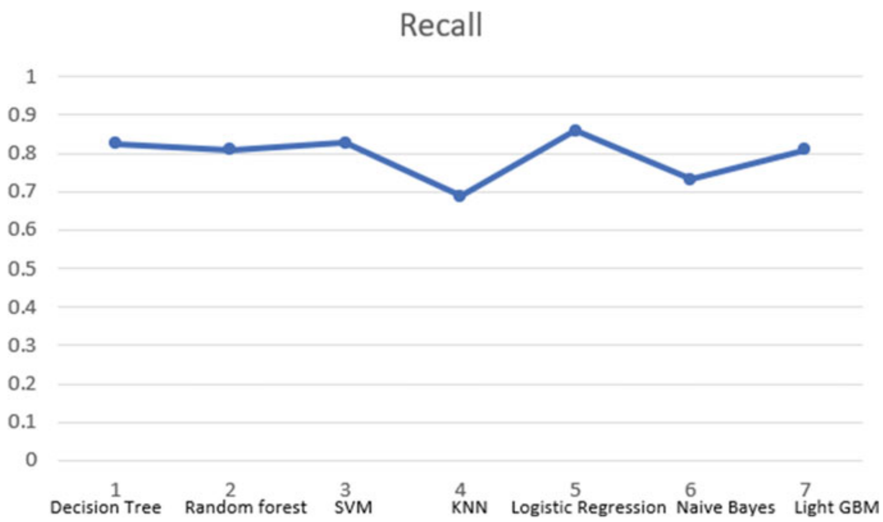

Fig. 3 Recall comparison of algorithms

it is difficult to compare two models when available, the classifier is not a balance. So here are the weighted average recall and precision values [35] (Fig. 4).

This graph is also very accurate and the rank can be determined very clearly and even these values represent the exact same results as the precision and recall values.

D. *Support*

Support seems to indicate that it is the number of occurrences of each particular class in the true answers (the test set answers). It can be calculated by summing the rows

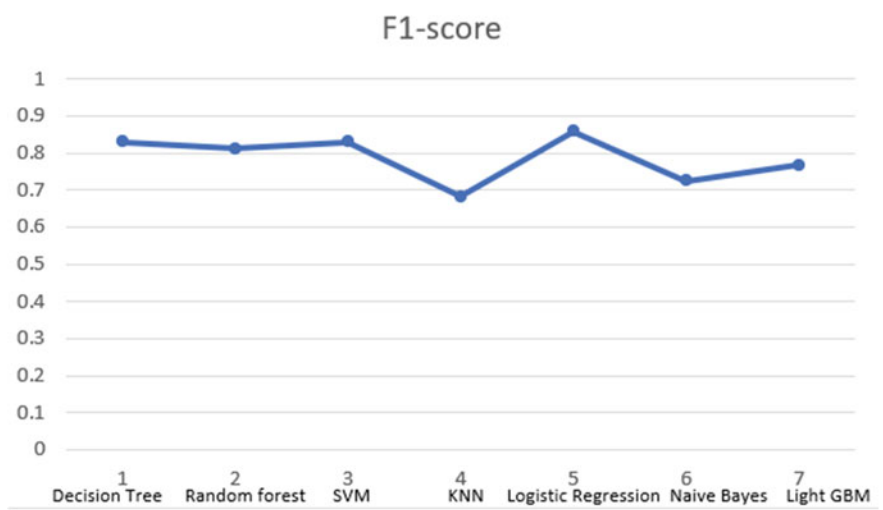

Fig. 4 F1-score comparison of algorithms

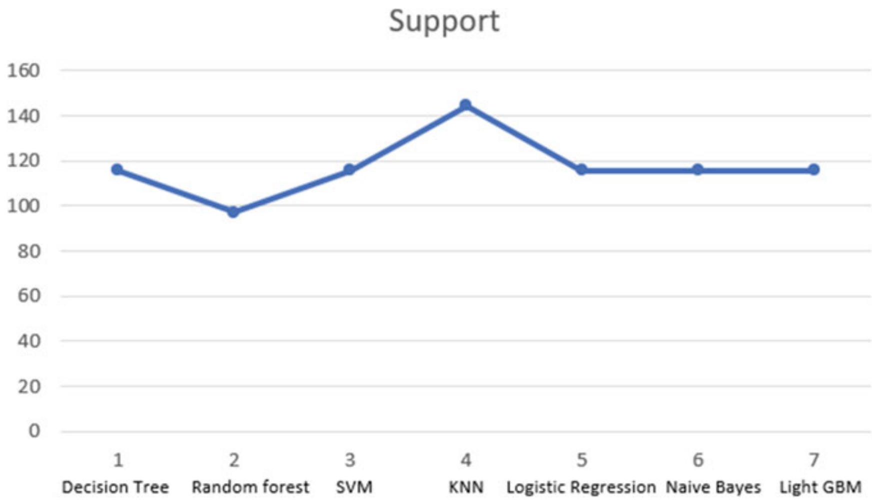

Fig. 5 Support comparison of algorithms

of the confusion matrix. Disproportionate support in the training data may indicate structural weakness in the scores reported by the classifier and may indicate a need for stratified sampling or recalibration. Support is that the actual number of response samples that fall into this class is the actual number of occurrences of the class in the dataset. There is no difference between models just to diagnose the performance evaluation process [36] (Fig. 5).

There is a difference in support where KNN presents the highest values and the results focus on the random forest algorithm with the lowest score. Therefore, a very detailed analysis was performed by looking at all values from the four datasets and using their mean to calculate a graph focused on comparison and identification.

3.1 ML for Healthcare: Challenges

The identification and prediction of populations at high risk of experiencing particularly adverse health outcomes, as well as creating public health interventions specifically for these populations, are important applications of ML in public health. The medical curriculum has to incorporate a variety of ML-related topics so that medical personnel may successfully direct and analyse research in this field. Any ML model needs high-quality data that are typical of the population to which the model's outputs should be applied if it is to produce reliable results.

(1) Safety Issues: Good performance in a controlled lab environment (a common practice in the ML community) is not proof of security. The ML/DL system's patient safety is assessed using its level of safety. The majority of a clinician's daily responsibilities are monotonous, and the patients they see have common medical issues.

They have the responsibility of diagnosing uncommon, subtle, and concealed medical illnesses that affect one in a million people. The weaknesses of these systems to adversarial ML attacks have been addressed in several latest studies. Moreover, ML/DL-based medical systems are already being targeted by similar attacks. The robustness of ML/DL models is essential in building reliability and openness in ML/DL driven healthcare applications since these problems create numerous concerns regarding the safety of ML/DL powered systems [37]. Existing systems need to be secure by allowing ML to work efficiently on hidden layers, exceptions, edges, and nuanced cases.

(2) Privacy Issues: Using user data by ML systems to make predictions raises privacy concerns, which is one of the main challenges in data-driven healthcare. Users (i.e., patients) anticipate that the providers of their healthcare services will take the appropriate precautions to protect their inalienable right to the privacy of their private information, such as their age, sex, date of birth, and health information. Two different forms of potential privacy dangers exist: disclosing private data and using data improperly (perhaps by unauthorized agents). Privacy is based on the traits and nature of the data being gathered, the setting it was developed in, and the patient population. As a result, it's crucial to use the right techniques to mitigate privacy breaches because they can have serious consequences.

(3) Ethical machine learning: Ensuring the ethical use of data is crucial in ML applications that focus on users, such as healthcare. Before gathering information to create ML models, explicit steps should be taken to comprehend the targeted user group and their social characteristics. Furthermore, it's crucial to recognise how data collecting can undermine a patient's wellbeing and dignity in this context. If ethical issues are not taken into consideration, applying ML in realistic settings will have unfavourable outcomes. Additionally, it is critical to have a comprehensive understanding of the system in uncertain and complex settings in order to ensure the fair and ethical operation of automated systems.

(4) Access to High-Quality Data: One of the biggest problems in healthcare is the lack of representative, diversified, and high-quality data. For instance, compared to the many collections of large-scale, multi-modal patient data created on a daily basis by various small and large-size healthcare institutions, the quantity of data that is available to the research community is quite modest in size and restricted in scope. On the other hand, it can be extremely difficult and expensive to collect high-quality data that closely reflects actual clinical circumstances. The ability to forecast diseases and make treatment-related decisions is made possible by the availability of high-quality data. Subjectivity, duplication, and bias are just a few problems with the data that are gathered in practice.

4 Proposed Work

Without the use of any human intelligence, different data science models can quickly and accurately produce outcomes in the health care sector. In addition to generating precise and effective recommendations, machine learning models also free up healthcare professionals' time to focus on research and perform better in emergency situations by decreasing the manual work they must do.

For finding out which algorithm is best for a certain analysis of a health issue, this study was carried out. Machine learning classification algorithms were implemented on four different healthcare datasets and a detailed evaluation was performed. The first dataset is a heart disease prediction dataset where the "target" field pertains to the existence of the patient's heart problems. 0 means there is no disease, while 1 means there is a disease. The diabetes disease is the second dataset. The goal is to determine whether a patient has diabetes based on diagnostic parameters where the "outcome" field indicates if the person tested positive or negative for diabetes. The third is the chronic kidney disease prediction dataset. Based on the specific diagnostic measures included in the dataset, the objective of the dataset is to diagnose whether a patient has chronic kidney disease. "Class" serves as the data set's target variable. The last data set deals with the prediction of breast cancer in patients. "Classification" variable acts as the indicator of the disease. Analysis was carried out by running the algorithms on these datasets.

The initial process is to import the dataset and verify the data dimension. The information is then divided into attributes and labels, and then into training and testing sets as part of the data pre-processing. The training is done on the training data and then predictions are made. Lastly, the algorithm is evaluated and the confusion matrix for the same is derived for displaying the values of the important metrics used in the classification tasks (Fig. 6).

The steps followed during the classification:

A. Importing the essential libraries

Libraries are collections of commands and operations written in a particular language. A strong collection of libraries can help programmers complete complicated jobs

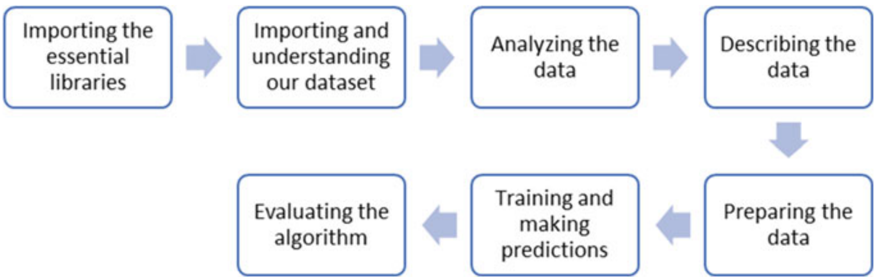

Fig. 6 Steps during classification process

faster and with fewer lines of code changes. As per our dataset we imported the basic libraries as numpy, pandas, matplotlib and seaborn.

B. Importing and understanding our dataset

Understanding data is being aware of its existence, the needs it will address, its location, and its content. After reading the csv file we will get to know what are contained in the rows and columns by using word dataset. We imported datasets of health related to heart, diabetes, kidney and breast cancer. Further verifying it as a dataframe object in pandas.

C. Analysing the data

The entire data analysis procedure is automated by machine learning to deliver richer, quicker, and more thorough insights [38]. The shape of the dataset will show the number of rows and columns present in our dataset. And then printing out a few columns to stay organized with the contents present in our dataset.

D. Describing the data

With the project definition completed prior to data preparation and the evaluation of machine learning algorithms completed later, this approach offers a context in which we may think about the data preparation necessary for the project.

E. Preparing the data

The process of preparing raw data for further processing and analysis is known as data preparation. Collecting, preparing, and labelling raw data in a format suitable for machine learning (ML) algorithms, followed by data exploration and visualization, are critical stages.

F. Training and making predictions

A very sizable dataset called training data is utilized to instruct a machine learning model. Prediction models that employ machine learning algorithms are taught how to extract attributes that are pertinent to particular business objectives using training data.

G. Evaluating the algorithm

An algorithm will be tested against the test set after being trained on the training dataset. Any project must have a machine learning algorithm evaluation. Your model may produce satisfactory results when measured against one metric, such as accuracy score, but unsatisfactory results when measured against another metric, such as loss of data. sign. After performing the following steps, this table is created, showing comparisons based on the parameters.

5 Result and Discussion

The algorithms were evaluated and the best possible and suitable solution was tried to be achieved. It was observed that Logistic Regression and SVM algorithms were proved to be the most effective for the four varied healthcare datasets (Fig. 7).

There are three different colors of graphs used representing the f1-score, recall and precision for the classification algorithms. The f1-score and the recall values for support vector machines and decision tree seem to be almost similar through the chart above. But it is the precision value that is important when comparing them because the support vector machine takes the lead according to the decision tree algorithm. The logistic regression algorithm is ranked first in all values over the others. The recall values of random forest and LightGBM are almost identical but the random forest is more precise than LightGBM and also has a greater f1-score. The least precision value is held by KNN and it is also lesser than Naive Bayes including the recall and f1-score too.

The precision for Logistic Regression is above 80% followed by SVM and decision tree algorithms, declaring them as the most effective algorithms for our datasets. After that, the accuracy for the algorithms was also considered and it was found that Logistic Regression is the most accurate with a score of 86%, shortly followed by the SVM with 83% and decision tree with 82%. The lowest accuracy was for KNN with a value of 68% (Fig. 8).

Time complexity can be considered as a scale for how quickly or slowly an algorithm will operate given an input size. Time complexity is often expressed in terms of an input size like n. Runtime complexity defines how well an algorithm performs, or how much extra time or processing power is needed to run the algorithm.

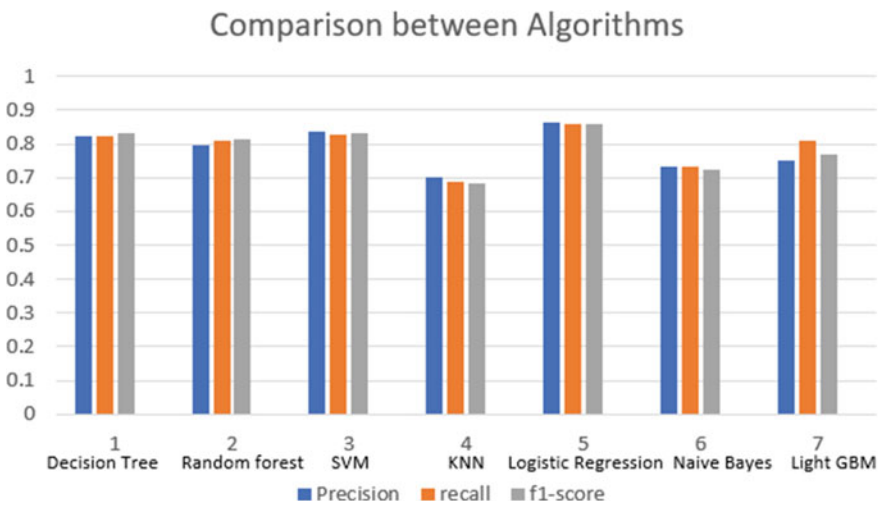

Fig. 7 Comparison between algorithms based on various parameters

Parameter	Decision tree	Random Forest	Support Vector	K-Nearest Neighbor	Logistic Regression	Naïve Bayes	LightGBM
Training time complexity	O(depth)	O(k*n*log(n)*m)	O(n^2) with n the amount of training instances.	Ω(1) which means it is constant	O(n*m)	O(nm)	O(tdxlogn), t is number of trees, d is the height of the trees, x is number of non- missing entries in the training data.
Run time complexity	O (maximum depth of the tree)	O (depth of tree* k)	O(k*d) k= number of supports	O(nd)	O(d)	O(d*c) where d is the query vector's dimension, and c is the total classes.	Relatively slow due to the model training that must follow a sequence.
Accuracy	0.82	0.77	0.83	0.68	0.86	0.72	0.70
Advantages	Easy to visualize, manages data which is both numerical and categorical.	Can deal with big datasets and missing values, will produce a variable's importance.	Optimal for highest dimensional spaces, memory efficient.	Fast computation time, there are no data assumptions made.	Easier to implement, interpret, very efficient to train.	It handles both continuous and discrete data, fast, can be used to make real-time prediction.	Lots of flexibility - can optimize on different loss functions.
Disadvantages	Occasionally fails to generalize well, unstable when input data changes.	Somewhat uncontrollable like a black box, slow real-time prediction and complicated algorithms.	Will not offer probability predictions, can estimate probabilities but takes a lot of time.	Expensive in terms of computation, needs plenty of memory, extremely sensitive to insignificant aspects.	May lead to overfitting if observations are lesser than features.	Faces the 'zero frequency problem'.	As there can be multiple outliers in a dataset, gradient boosting cannot avoid them completely.

Fig. 8 A detailed comparison between classification algorithms used in the study

The time complexities are compared in the above table for these distinct algorithms based on some parameters like k which defines the number of neighbors and d which stands for the dimension of the data. The training time complexity demonstrates that random forest is used for a large amount of data with reasonable features. Also it can be observed that SVM should not be used if n is large and KNN loops through every training observation and computes the distance d. Also, the run time complexity as observed through the above table is relatively slow because of its model training as it is necessary for it to follow a proper sequence. The pros and cons are as well discussed for estimating the algorithms based on various aspects. There is a certain ‘zero frequency problem’ discussed in the cons for Naive Bayes Algorithm. The entire information in those other probabilities is lost due to the “Zero Conditional Probability Problem.“ To solve this issue, other sample correction methods exist, including “Laplacian Correction” [39].

6 Conclusion

All of the information obtained during a medical visit with a doctor is applied by machine learning approaches to extract specific patient traits. In our study, we tried to estimate which algorithm works the best on our datasets in the healthcare field by calculating their accuracy, precision and other factors too. Thus, we were able to conclude that the Logistic Regression worked way better than the SVM, decision tree and other algorithms after evaluating it on various parameters. If numerous fields are considered, there is a much higher need for specialists than there are qualified candidates. This causes a great deal of anxiety for doctors, and delayed diagnoses of patients who need life-saving treatment that are common [15]. However, considerable advancements have been achieved in the ability of machine learning algorithms, in particular deep learning algorithms, to intelligently diagnose illnesses while decreasing the cost and expanding usability of diagnostics.

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Dollar Price Prediction Using ARIMA

Alokam Ujwala Bharati, Chinnakanu Sai Janakiram, Radha Mohan Pattanayak, Deepa Jose, and Sachi Nandan Mohanty

Abstract The proposed project analyses and forecasts the exchange rates on the Indian rupee by using time series data concepts from the year 2020 to 2022, using the most popular Box-Jenkins ARIMA model technique. Based on the research study presented, the ARIMA model's test results depict that the proposed model is very accurate in showing the results and hence works well for forecasting the USD exchange rates. Forecasting the exchange rates plays a significant role in minimizing risks and maximizing profits for the people working in the financial markets, trading as well as general public across the world. ARIMA uses the stationary time series dataset for providing accurate predictions. The real time series data used in this study has been obtained from Yahoo Finance, calculated and analyzed dollar exchange rate for the following day, subsequent 15 days, 30 days, 60 days respectively from the current date. In addition to that, we were able to achieve a small MAPE score/forecast accuracy i.e. 0.923 which indicates that the model gives better accuracy. The Daily exchange rates from 5th June 2020 to the current date were used for the prediction.

Keywords Autoregressive Integrated Moving Average (ARIMA) · Forecasting · USD · INR · Analysis

A. U. Bharati (✉) · C. S. Janakiram · R. M. Pattanayak · S. N. Mohanty
School of Computer Science and Engineering, VIT-AP University, Andhra Pradesh, India
e-mail: bharati.20bcd7180@vitap.ac.in

C. S. Janakiram
e-mail: janakiram.20bce7016@vitap.ac.in

D. Jose
Department of Electronics and Communication, KCG College of Technology, Chennai, Tamil Nadu, India

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1 Introduction

As per the recent data provided by the Census Bureau's 2018 American Community Survey (ACS), there has been an increase of nearly 50 percent in the number of Indians in US from 2010 to 2018 [1]. The IT workface from India is easily employed in countries like United States of America, United Kingdom etc. The sophisticated lifestyle and the exchange rate are the two main reasons for the migration.

The foreign exchange rate influences a variety of decisions made by participants in the foreign exchange market like investors, marketers, governments, exporting sectors, commercial banks, corporations, policy makers as well as tourists. It's one of the vital influencing factors for both developing and developed countries [1]. Exchange rate forecasting is essential for international finance enthusiasts and researchers, especially for floating exchange rates [2].

The significance of exchange rates forecasting from a practical perspective is that accurate forecasts can provide valuable information to people in various sectors such as firms, investors and central banks regarding asset allocation, risk hedging, and policy formulation. The real time importance of accurate currency exchange rate forecasting is that it has significant impact for the development of efficient market hypotheses and theoretical models in the international finance sector [3].

When it comes to buying and selling of goods, forecasting is regarded as one of the most key things when choosing an investment market. Every prediction that gives an output needs to calculate accuracy. The accuracy and prediction is influenced by many errors and one such error is human calculation error. It wastes a lot of time and effort for the financial experts. We need a way to overcome this problem that is quite difficult to be solved manually [4, 5].

Periodic data refers to the data observed over a period of time. The periodic data collected over a certain period of time is used for analyzing the patterns and forecasting of the outcome. ARIMA (Autoregressive Integrated Moving Average) models are models that completely ignore the independent variable when making predictions [6]. This is a model used to measure statistics to measure events occurring over a period of time. Models are used to make sense of past data and predict future outcomes. The used model is a mixture of two models: an autoregressive (AR) model along with a moving average (MA) model.

In the current research paper, the work has been fragmented into several sections. Section 1 describes about the need of scientific and well-defined approach for this study. The similar, related research by various authors is discussed in Sect. 2. In Sect. 3, the methodology/procedure for the same is presented. Experimental evaluation and discussion of results are mentioned in Sects. 4 and 5. Finally, conclusion and the future research scope of the work is mentioned in the Sect. 6.

2 Related Work

The models on time series prediction are developed based on the hypothesis that past data affects the forecasts of future statistical data points [7–9]. With the forecast item as exchange rate, ARIMA is demonstrated to be an excellent answer for prediction [10–12]. Nyoni [13] has undertaken an intensive survey of empirical research in Nigeria and different international locations that use ARIMA primarily based on total time collection tactics to forecast alternate rates [10]. El-Masry et al. [14] analysed the impact of organization length and overseas operations at the alternate charge publicity of UK non-monetary companies. They discovered that that a better percent of UK corporations are uncovered to contemporaneous alternate charge modifications than the ones stated in previous research. A majority of the researchers who have analysed this time series forecasting concluded that ARIMA is the best model for time series data [11].

Research that been achieved via way of means of Ahmad Amiruddin Anwary in 2011 has analysed "Prediction Rupiah against US dollar", with the help of Fuzzy Time Series" and obtained an accuracy degree of the expected outcomes calculated via way of means of the Average Forecasting Error Rate (AFER) cost. This research was performed on little amount of statistics, so the prediction will become barely much less than the optimal value [15].

In these days, many authors are interested about the forex rate problem in selecting the most accurate model for forecasting [6, 16, 17]. Among the ones, He and Jin [18] made use of the inverse Autoregressive Integrated Moving Average—Grey Model for forecasting foreign exchange rate for (US Dollar/Japanese Yen). Their hypothesis stated that their combination of models had better accuracy. Researchers Thuy and Thuy carried out the autoregressive disbursed lag (ARDL) bounds technique to analyse the relation between exchange rate volatility and exports carried out. The outcomes confirmed that the alternate charge volatility may have an opposite impact for the export magnitude over a long period of time [19].

The study proposed by Pattanayak and Panigrahi in 2020 explains that the performance of FTS forecasting model using the support vector machine is robust. It states that using FLRs (Fuzzy Logic Relationships), the statistical superiority obtained by the model is high when compared to other models obtaining a high mean rank value of 70.30 [13].

The research paper by Nonita Shrama, Monika Mangla, SN Mohanty, entitled "Employing Stacked Ensemble Approach for Time Series Forecasting" demonstrates that the accuracy of forecasting model could be improved significantly with the help of stacked generalization ensemble model. The findings of the study reveal that the model outperforms traditional methods such as Auto ARIMA, NNAR, ETS, and HW [20].

The study proposed by Febi Satya in 2015 shows the outcomes display that the Mean Absolute Percentage Error (MAPE) score was pretty high, which is 6.03% [21].

Whereas the MAPE score we achieved for our study is around 92.3%. It is a fact that rise in the volatility of a variable and or the usage of vulnerable forecasting approach within the monetary marketplace is dangerous to monetary improvement because of their damaging effect on global exchange and overseas investments [22]. Hence, variable forecasting in the monetary markets is an issue of vital importance, in particular within countries such as United States of America [20, 22, 23].

Recent research study "Forecasting of Sunspot Time Series Using a Hybridization" proposed by Panigrahi and Pattanayak shows that their proposed PHM-MAX-ARIMA-ETS-SVM gives a significant efficiency as compared to the existing hybrid and individual models [24]. The research work titled "A novel probabilistic intuitionistic fuzzy set based model for high order fuzzy time series forecasting" that was proposed by Pattanayak and Behara shows an intuitionistic fuzzy time series forecasting (PIFTSF) model with the help of a support vector machine (SVM) to solve the problem of both uncertainty and non-determinism that is related with the time series data. Such methods work effectively when compared to other traditional methods [25–29].

3 Methodology

In this section, we discuss the techniques used for collecting the data, developing the ARIMA model of the bellow-mentioned time-series data.

The study is divided into four stages: (1) The Dataset, (2) Data Preprocessing, (3) Using the ARIMA approach, (4) Accuracy Testing.

A. The Dataset

The data utilized is the Dollar (USD)—Rupee (INR) exchange rate was obtained from Yahoo Finance. Seven attributes make up the dataset: open price, close price, high, low, adjusted price, volume, and date. The time series data has maximum of 657 records which starts from June5, 2020 to December 10, 2022. (Fig. 1 shows a dataset collected from yahoo finance) (Fig. 2).

B. Data Preprocessing

The data preparation/Preprocessing stage must be completed before the prediction process can begin. This stage involves the following steps (a) load data, (b) attribute reduction, (c) data cleaning, and (d) stationarity testing.

Load data refers to the process of loading data in.csv format. Following the successful completion of the data import procedure, the following results were shown in Fig. 3.

The procedure of deleting non-essential data attributes is called Attribute reductions. The researchers opted to eliminate the following attributes open price, closing price, high, low, and volume. As a result, the experiment only employs the two remaining attributes: adjusted price and date attribute. The adjusted price column

	INR=X.open	INR=X.high	INR=X.low	INR=X.close	INR=X.adjclose	INR=X.volume
2021-06-07	72.9098	72.9160	72.7411	72.9097	72.9097	0
2021-06-08	72.8094	72.9787	72.7440	72.8094	72.8094	0
2021-06-09	72.9753	73.0135	72.8700	72.9753	72.9753	0
2021-06-10	72.9853	73.2385	72.9390	72.9853	72.9853	0
2021-06-11	73.0123	73.3184	72.8950	73.0123	73.0123	0
2021-06-14	73.2317	73.2940	73.1005	73.2316	73.2316	0
2021-06-15	73.1853	73.3763	73.1596	73.1808	73.1808	0
2021-06-16	73.3439	73.3870	73.2600	73.3447	73.3447	0
2021-06-17	73.8241	74.2385	73.2680	73.8241	73.8241	0
2021-06-18	74.2539	74.5223	73.8070	74.2539	74.2539	0
2021-06-21	74.1440	74.4408	74.0456	74.1439	74.1439	0
2021-06-22	74.1352	74.3876	74.0205	74.1352	74.1352	0
2021-06-23	74.3322	74.5448	73.5108	74.3322	74.3322	0
2021-06-24	74.2469	74.2628	74.1044	74.2469	74.2469	0
2021-06-25	74.1903	74.3910	74.1168	74.1903	74.1903	0
2021-06-28	74.2130	74.3283	74.0625	74.2130	74.2130	0
2021-06-29	74.2856	74.3378	74.1890	74.2854	74.2854	0

Fig. 1 Dollar versus rupee exchange rate dataset (*Source* yahoo finance)

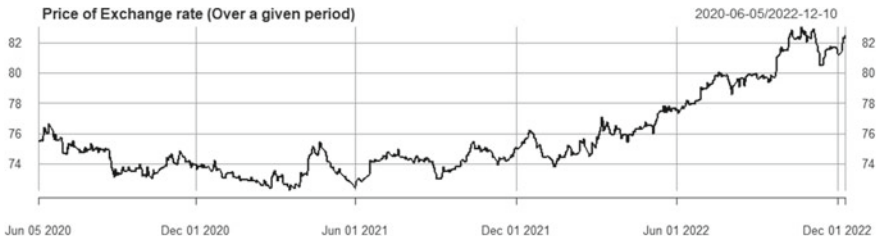


Fig. 2 Price of exchange rate (over the period 2020-06-05 to 2022-12-10)

```
> INR = pdfetch_YAHOO("INR=X",from = as.Date('2020-06-5'), to= as.Date('2022-12-09'))
> summary(INR)
     Index                INR=X.open      INR=X.high      INR=X.low       INR=X.close
 Min.   :2020-06-05   Min.   :72.38   Min.   :72.54   Min.   :72.27   Min.   :72.30
 1st Qu.:2021-01-21   1st Qu.:73.78   1st Qu.:73.94   1st Qu.:73.52   1st Qu.:73.78
 Median :2021-09-08   Median :74.74   Median :74.95   Median :74.53   Median :74.75
 Mean   :2021-09-07   Mean   :75.69   Mean   :75.93   Mean   :75.48   Mean   :75.69
 3rd Qu.:2022-04-26   3rd Qu.:76.52   3rd Qu.:76.82   3rd Qu.:76.34   3rd Qu.:76.54
 Max.   :2022-12-10   Max.   :83.00   Max.   :83.39   Max.   :82.67   Max.   :83.00
 INR=X.adjclose   INR=X.volume
 Min.   :72.30   Min.   :0
 1st Qu.:73.78   1st Qu.:0
 Median :74.75   Median :0
 Mean   :75.69   Mean   :0
 3rd Qu.:76.54   3rd Qu.:0
 Max.   :83.00   Max.   :0
```

Fig. 3 Load data

Fig. 4 The dataset after reduction

	Price
2021-06-07	72.9097
2021-06-08	72.8094
2021-06-09	72.9753
2021-06-10	72.9853
2021-06-11	73.0123
2021-06-14	73.2316
2021-06-15	73.1808
2021-06-16	73.3447

was selected as it represents the steady price of the Dollar exchange on a given day. This column has been renamed Price. Figure 4 depicts the dataset after reduction.

In the data cleaning phase, the researchers have identified all the null values in the dataset and removed them.

The time series statistical property with remains constant with respect to time is called Stationarity. The performance of model depends mainly on the Stationarity of the data in time series analysis [30–33]. Therefore, converting the non-stationary into stationary is a crucial part of the analysis. The researchers used the ADF test (Augmented Dickey–Fuller test) to assess the data stationarity (If the value of p is less than that of 0.05, the data is termed as stationary; otherwise, the data is not.) (Fig. 5).

In this study the researchers use differencing to transform non-stationary data to stationary data. Data is turned into stationary data via differencing that indicates the variation or change in value of observation. The process of subtracting unit root from the time series is knowns as the differencing process [14, 34].

The order of coefficient d on ARIMA will depend on how much differencing has been done. The data is referred to as non-stationer homogeny level d ARIMA if it has undergone a multiple differencing process to remain stationary (0, d, 0) (Fig. 6).

C. Preparing the model

ARIMA (Autoregressive Integrated Moving Average) is an acronym that combines the moving average and regression models. The moving average model predicts

```
> adf.test(INR[,1])

        Augmented Dickey-Fuller Test

data:  INR[, 1]
Dickey-Fuller = -2.1545, Lag order = 7, p-value = 0.512
alternative hypothesis: stationary
```

Fig. 5 The test result of ADF on the adjusted price

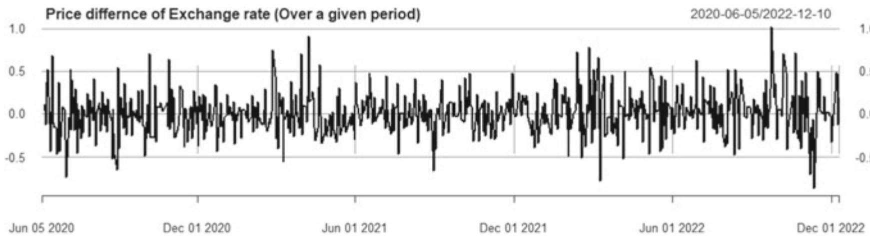


Fig. 6 The data after converting it into stationary

future values by leveraging prior forecast inaccuracy. The ARIMA equation, which combines the AR and MA equations, is abbreviated as given below.

$$X_t = \mu + \varnothing_1 X_{t-1} + \varnothing_2 X_{t-2} + .. + \varnothing_p X_{t-p} + e_t - \varnothing_1 e_{t-1} - \varnothing_2 e_{t-2} \cdots - \varnothing_q e_{t-q}$$

where:

$\varnothing_p =$ Autoregressive parameters related to p
$e_t =$ White Noise, error value at time t
$\mu =$ independent variable
$\varnothing_q =$ Parameter of Moving Average

For the training and testing we have split the original dataset in 30:70 ratio. We fit the ARIMA on the training data with the appropriate P, q, and d values to obtain the best ARIMA model. By charting the PACF (Partial autocorrelation function) and ACF (Autocorrelation function) graphs, the order of AR(p) and MR(q) may be established (Fig. 7). After determining the optimal model, we forecast the future price of the USD v INR exchange rate.

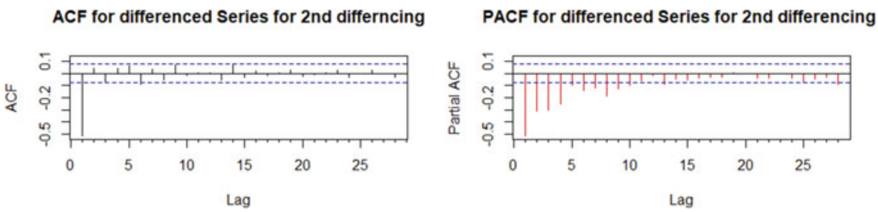


Fig. 7 PACF and ACF for the 2nd differencing

4 Experimental Evaluation

4.1 Evaluation of Performance

The following measures are helpful in assessing the model's accuracy:

MAE, MPE, ME, RMSE,MAPE.

Model that has the least MAPE is chosen for the best fit model after we assess the model's performance on the test dataset.

4.2 Experimental Setup Used

The experiment was carried out on a conventional laptop which consists of Windows 11 operating system, i7 processor, 16 GB of RAM. The result displayed in this study was the best result obtained when experiment was carried out.

5 Results and Discussion

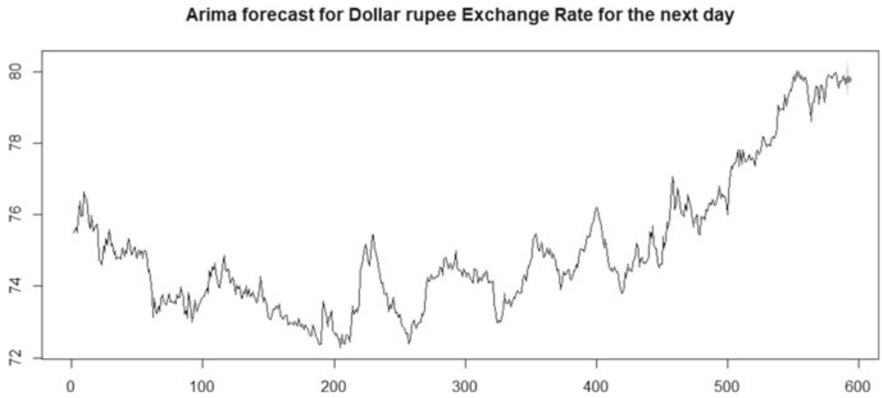
The predictions were made by specifying the number of days that we are anticipating. We have forecasted Exchange rates for the following day, 15 days, 30 days, and 60 days. Figure 8 displays the prediction's outcome.

From the below test result, we can observe that the performance of the model is with a MAPE score 0.9232047 (Fig. 9).

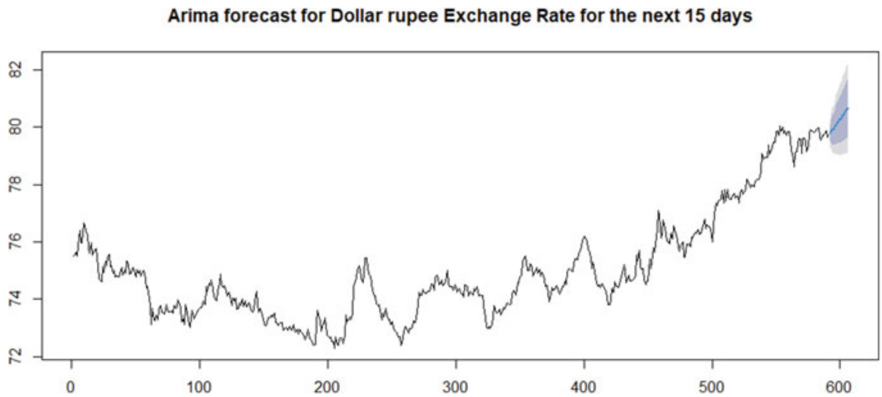
In this study, we have used ARIMA approach to predict the price of the dollar price with appropriate orders of auto regression and moving average which were observed form PACF and ACF graphs respectively. In our the study we have observed the p(Order of Auto regression) value to be 20 and q(Order of Moving average) value to be 6.

As the data provided is not stationary we convert it into stationary data in our case we have a 2nd difference to make the data stationary and this gives the order of d.

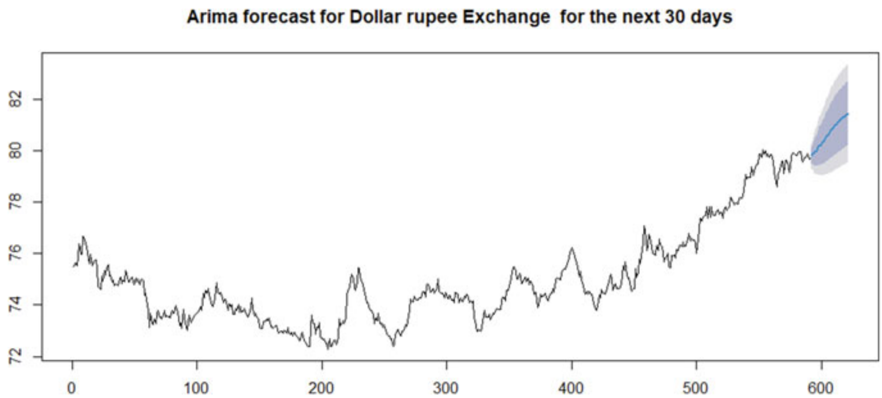
From the obtained accuracy of the model, we can conclude that the proposed model is performing very well with a low MAPE score of 0.923. According to the anticipated statistics, the dollar price will be slight rise during the following 60 days, beginning on December 11, 2022. Traders may rely on the model for preliminary dollar price predictions since it is both cost and time efficient.



(a)

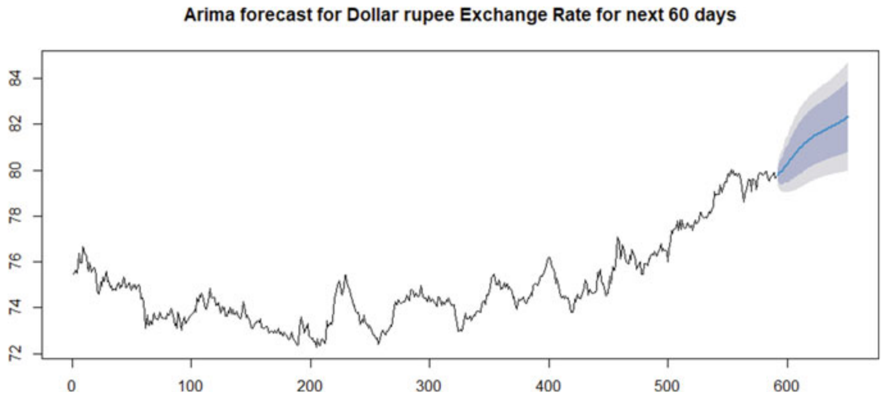


(b)



(c)

Fig. 8 The result of the next day (**a**), 15 days (**b**), 30 days (**c**) and 60 days (**d**) chart predictions



(d)

Fig. 8 (continued)

```
> accuracy(preds, INR_return_test)
                ME     RMSE       MAE       MPE      MAPE
Test set 0.1995858 0.850831 0.7549628 0.2371641 0.9232047
>
```

Fig. 9 The test results

6 Conclusion and Future Scope

Based on the findings, we arrive at a conclusion that ARIMA (20,2,6) model is the most accurate model to forecast the rupee value against the US dollar price. The MAPE test outcomes for the proposed ARIMA (20,2,6) model is 0.92322047, and we can conclude that this model is workable to forecast the value of rupee price against US dollar price value. The findings found in the present study indicated that the exchange rate of rupee versus dollar increased slightly for the current day, 15 days, 30 days, and 60 days starting on December 10, 2022.

Since we have achieved a remarkable increase in accuracy of the model, this price rate time series analysis can be extremely useful for various traders of FOREX market by making accurate predictions and reducing the risk factor in trading. The future scope of this research includes using various economic factors that influence exchange rate such as GDP (Gross Domestic Product), inflation and money supply. By using these factors various trends or patterns of movement are created that impact the exchange rate. Dua and Suri [28] analysed the relationship between four exchange rates and found a significant bidirectional causality. Predicting these exchange rates in a Vector Autoregressive Integrated Moving Average (VARIMA) model can give better results, as a result improveing the model's accuracy. Such time series forecasts provides governments and policy makers a direction to design policies in the light of forecasts.

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An Exploratory Study on the Impact of Digital Marketing and Innovations on E-commerce Mechanism

Bhagirathi Nayak, Tilottama Singh, Sukanta Kumar Baral, Richa Goel, and Pritidhara Hota

Abstract The banking sector is an important subset of our economy and plays a very crucial and significant part in its growth and development. The expansion and credibility of the banking sector are attributed to the fact that it caters to their financial needs and accordingly provides products and services to the customers. The innovation. Technology has brought about a paradigm shift in all aspects of banking like routine operations, transactions, products and processes, delivery of services in terms of various e-channels, credit administration and credit management, audit and compliance and Fin Techs partnership. The study attempts to highlight the role of technology in banking and how technology-led innovations and initiatives will determine the business of banking as a whole. The significance of these technology-driven innovations and transformations is that they put both bankers and customers in a win–win situation. The objective of this study is to investigate the digital innovations adopted by the Public Sector Banks and Private Sector Banks for e-commerce Practices in India. This study draws on existing literature in the form of scholarly articles annual reports of various Banks, Newsletters and various websites related to digital banking.

Keywords Digital innovations (DI) · Indian banking system (IBS) · Block chain (BC) · Artificial intelligence (AI) · FinTech

B. Nayak (✉)
Sri Sri University, Cuttack, Odisha, India
e-mail: bhagirathi.n@srisriuniversity.edu.in

T. Singh
Uttaranchal Institute of Management (UIM), Dehradun, India

S. K. Baral
Indira Gandhi National University, New Delhi, India
e-mail: sukanta.baral@igntu.ac.in

R. Goel
Symbiosis Centre for Management Studies, Noida, India

P. Hota
GITA Autonomous College, Bhubaneswar, Odisha, India

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1 Introduction

At present, everyone desires a healthy, robust and sustainable banking system that is critical for the growth and development of our economy. The banking system is the mainstay of our economy and Technology has become the backbone of modern-day banking. The banking industry has witnessed a tremendous shift since 1969 after the nationalization of commercial banks. In 1997 the concept of digitalization was introduced in banking way back with ICICI as the first bank. Further to offer a competitive edge in the market almost every bank started shifting to digitalization and a more customer-friendly interface. In a fast-moving scenario, the banks have been paving a multifaceted platform for their customer clientele. Now people can do banking from anywhere at any time with very minimal cost. The development of electronic home banking has significantly necessitated the reorganization of banking services and operations. Digital modes have replaced the brick–mortar model of branches.

2 Research Objectives

(a) To understand the changing digital scenario in the Banks in India
(b) To examine the Digital Banking new trends and Innovations in Indian Banks.
(c) To analyse the various Digital Banking Products and Services, Digital Banking Channels offered by Indian Banks.
(d) To analyse the role of FinTech in Indian banks towards digital transformation.

2.1 Problem Formulation

Objective 1: Use of various digital apps for providing digital innovation services to the customer in an efficient way core banking with secure and service.

Objective 2: Providing various digital modes of service for better and safe digital banking which is innovative and preventive for money transfer and distribution.

2.2 Hypothesis Development

The researchers have taken two types of hypotheses for enhancing the digital innovation banking service in Indian Banking such as;

Ho: Null hypothesis has taken that the digital Innovation mode of banking is not providing safe and secure banking service to their esteemed customer.

H1: There is a need for the use of a digital innovation mode of the banking system for a better banking service in Fin Tech areas of banking service.

3 Research Methodology

The current study is exploratory, descriptive and based on secondary data. Research studies and information have been collected from scholarly articles, yearly reports of the particular banks, 'RBI Working Group Reports, RBI Bulletin, RBI Annual Reports, Reports published by various Sites of the Government of India and various other Websites. For the present research work, various banking journals and other financial newspapers have also been referred to various studies on this area by several research agencies have also been considered.

4 Review of Literature

In the literature Review section, the researcher has to try their level best for proving the authenticity of the research work reliably as related to the aforesaid research title "Role of Digital Marketing and Innovations on E-Commerce Practices—An Exploratory Study" Henceforth, they have taken some of previous Authors' research paper as a reference and trying to further extend the research work in a better way of study concerning the paper of Rajeshwari and Shettar [1] in his artefact, he focused on the benefits of digital banking. He also stated that the use of digital banking will reduce the operating cost of the bank.

In their work "A Literature Review on the Impact of Digitalization on Indian Rural Banking System and Rural Economy," Paria and Giri [2] state that, in addition to the advantages of simplicity of use and low costs, digital banking has a significant potential to promote financial inclusion. In their article "A Study on Digital Payments in India with Perspective of Consumer's Adoption," Suma Vally and Hema Divya [3] highlighted how the use of digital technology in the payments system has improved the standard of banking services, enabling the realisation of the goal of a cashless society. Mohana Sujana claims that "digitalization in the banking sector" [4] The paper focused on the role of digital banking and the advantages and disadvantages of the online banking sector. Innovation is a crucial component for entities to generate value and offer competitive benefits [5]. Organizations adopt innovative models to reengineer their operations and increase their competitive strengths. Competitive advantage can be maintained by embracing innovation [6]. Innovation envisages the introduction of novel methods, new techniques, and reengineered processes to produce extraordinary results in the form of superior services and products [7]. So far as the word digital banking is concerned, 'Digital Banking' can be defined as the digitalization of various traditional banking operations and performance and processes that were earlier available to the customers by a visit to the bank's branch or ATM only. Electronic funds transfer means any transfer of funds which is initiated by a person by way of customers' mandate, authorization or instruction to their bank to initiate banking transactions by debit or credit to the bank account through electronic, online, internet, mobile apps and includes ATM, Point of sale, Card Payments, Direct deposits, transfers and withdrawal of funds." Digital Banking activities include online account/ fixed deposit opening, funds transfer, credit card payments, requests for cheque book, change/block pin numbers, Loan applications, bill payments and investments. Artificial intelligence is having a positive impact on the banking sector. The advanced presentations of AI and automation undoubtedly give useful skills. Language translation, chatbots, and virtual reality are just a few of the well-established claims, but many more are demonstrating a significant influence and effectiveness when used across industries [8].

The history and evolution of Digital Banking are traced back to the 1960s with the launch of cards and ATMs. Developments of the World Wide Web and Internet Banking have also seen the simultaneous evolution of Digital Banking. Online Banking emerged in the 1990s. At that time, it was an idea, taking into consideration the security and safety of customer financial data. The USA was the first country in the world to start online banking in October 1994 (Table 1).

Table 1 Digitalization in banking

1980	• The mechanisation of payment system processes, computerization • Standardisation of cheques, encoders, MICR implementation and minimal use of bank drafts and cheques
1990	• Computerization of branches, expansion of products and services, connectivity with other branches • Core banking systems, ATMs and electronic funds transfer, online banking
2010	• Internet banking, mobile banking, real time gross settlement, national electronic funds • Transfer (NEFT) and national electronic clearing services (NECS)
2011	• Emerging financial technology (fintech), e- collaborations with fin tech and adoption of • New technology, biometrics and cheque truncation systems
2012	• Rupay and national automated clearing house (NACH) introduced by NPCI. rupay-an alternative to visa and master
2013	• Aadhar enabled payment system (AEPS)
2014	• *99# USSD unstructured supplementary service data
2016	• Unified payment interface (UPI), bharat bill payment system (BBPS), National electronic toll • Collection system, bharat interface for money (BHIM)
2017	• Bharat QR developed by NPCI, master card, visa • Integrated payment system—money is transferred directly to the user's linked account
2018	• Release of RBI report of the working group on fin tech and digital banking and draft enabling framework for regulatory sand box
2019	• RBI payment system vision 2019–2021 released
2020	• Availability of NEFT on a 24 × 7x365 basis i.e., 16.12.2019

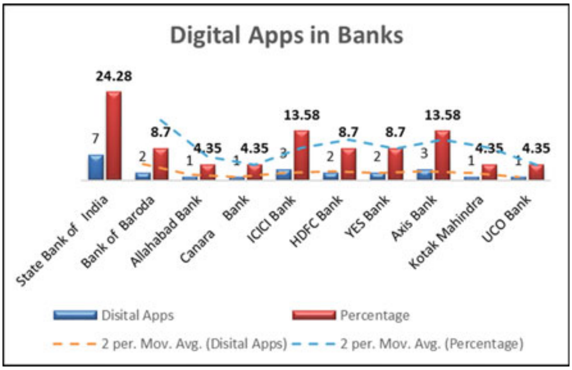

Fig. 1 Application usage in banks. *Source* Authors

The Banking System in India has witnessed a radical transformation from the conventional banking system to the electronic banking system of convenience. Payment Systems in India keep on evolving from manual to electronic to digital banking with new emerging technologies and Innovations. It started in the 1980s with the need for Computerization, Electronic Payment Systems in 2010 and move on to Digital and Fin- Tech E-collaboration. Figure 1 reflects the timelines of the evolution and achievements of Payment Systems from the 1980s till date. RBI Payment System Vision 2018 was Infinity. India started adopting digital innovations in line with innovations adopted by its global peers. Based on the four fundamental strategic pillars of strong infrastructure, responsive regulation, customer-centricity, and effective supervision, ICICI Bank was the first bank in India to launch an Internet banking revolution in 1997 under the brand name. Checks, demand drafts, and pay orders are examples of paper-based clearing instruments. These take three to four days longer to settle because they must be physically transported from the deposit location to the clearing centre and then delivered to the drawee bank before they can be paid. The Reserve Bank of India (RBI), National Payments Corporation of India (NPCI), Government of India, and individual banks all took steps to promote electronic and digital banking products, such as:

- **Real-Time Gross Settlement (RTGS)**: Real-time, gross, electronic funds transfer that does not involve any netting. This method of transfer is utilised for transfers above Rs. 2 lahks, and the recipient receives the money promptly.
- **Electronic Clearing Service (ECS)**: An ECS Debit and Credit transaction allows for the transfer of monies from one bank to multiple bank accounts and multiple bank accounts to multiple ECS Debit and Credit transactions as a single deposit into a single bank account.
- **National Electronic Funds Transfer (NEFT)**: Direct electronic funds transfer between bank accounts in India, facilitating transactions between people, businesses, and financial institutions.
- **Immediate Payment Service (IMPS)**: It's a plan that would make it so money could be transferred instantly via ATM, mobile, and the internet between any two banks in India, at any time of the day or night.

- **Mobile Banking Mobile**: Through Internet banking or a mobile data connection, banks provide consumers with access to their services around the clock, seven days a week.
- **Card Payment System**: Card purchases, Internet and mobile banking transactions, and cash withdrawals from ATMs are all on the rise among Indian consumers.
- **Bharat Interface for Money (BHIM)**: A UPI-based money transfer app that facilitates simple, fast, and secure financial transactions. Customers can use their mobile phone numbers and UPI ID to send and receive money with the BHIM app. The volume of BHIM transactions in FY2019-2020 may be seen on the NPCI Platform.
- **ATMs and Point of Sales (POS)**: The tally continues to rise. As of March 2020, there will be a total of 210,760 ATMs and 5,137,822 POS. In 2019–2020, the total amount of transactions at ATMs is 26769158 lakhs, and the total at point-of-sale terminals is 4764615 lakhs.
- **Aadhaar-Based Retail Payment Systems (ABRPS)**: include APBS Credit and Debit Cards and Aadhar-Enabled Payment Systems (Inter Bank) via Micro ATMs (e.g. Cash Deposit/Cash Withdrawal). (UIDAI number-driven Disbursement) (Table 2).

The Indian Banking and Payment System is witnessing various new emerging technologies like Blockchain, Artificial Intelligence (AI), Robotic Process Automation (RPA), Biometrics, Chatbots, Machine Learning, and the use of Big Data and Predictive Analysis offered by FinTech Companies. "AI is best understood as the overarching field that seeks to create complex machines that can exhibit all characteristics of real human intelligence," wrote Odinet (2018) to define Artificial intellect. AI facilitates increased revenue, reduced costs, better compliance and higher profits. A Report by Accenture (2018) has pointed out that AI has the potential to add USD 957 billion by 2035 (1.3%of GDP) to the Indian economy. Thus, an illustrative use of AI by banks operating in India is presented in the Table 3 (Fig. 2).

(a) **Data Analytics**: Data analytics techniques assist us to take raw data and uncover patterns to derive valuable insights from it. Several Indian banks like HDFC

Table 2 Significant innovation in fintech

Clearing, settlement and payments	Leading, capital raising and deposits	Market provisioning	Management of investment	Risk management and data analytics
Digital currencies mobile and web-based payment distributed ledger	Peer-to-peer lending crowdfunding digital currencies distributed ledger	Cloud computing smart contracts e-aggregators	Smart contracts robo advice e-trading	Artificial intelligence. robotics and big data

Source Gomber et al. (2018) stated that fin-tech or digital innovations have played a pivotal role in transforming

Table 3 Digital apps in banks

Name of banks	No. of digital apps using	Use of AI	Percentage
State Bank of India	07	• Partnered with pay jo to launch SBI intelligent assistant (SIA) an AI-powered chat assistant designed to address customer enquiries • Partnered with hitachi for payment services that will use the AI technology of hitachi for SBI payment services • Utilizing IBM watson to perform a variety of jobs • AI-based solution presently in use developed by chapdex • Thechatassistant, knownas SBI intelligent assistant, or SIA, will help customerswith everyday • Deploying artificial intelligence (AI) in a big way to improve efficiency,detect human behaviour and reduce operational costs • Artificial intelligence (AI) and robotic process automation (RPA) can be used to improve the effectiveness of internal banking procedures	24.28
Bank of Baroda	02	• Using AI-empowered robot 'Baroda Brainy' • Using AI solution by quadratyx	8.70
Allahabad bank	01	• Using AI-enabled app 'emPower' for e-commerce payments	4.35
Canara bank	01	• Using humanoid robot mitra and candi	4.35

(continued)

Bank, SBI, ICICI, Bank of Baroda, and Kotak Mahindra Bank have already initiated data analytics initiatives.

(b) **Cryptocurrency**: It's a decentralised (private) digital payment system where transactions are recorded in a public ledger.

(c) **Chatbots**: It facilitates quicker resolution of customer-related issues. It is more effective than traditional methods like email, phone etc.

Table 3 (continued)

Name of banks	No. of digital apps using	Use of AI	Percentage
ICICI bank	03	• Using iPal, an AI-based chatbot • AI features such as facial and voice recognition, bots etc.are being leveraged • Deployed robotics • Software to ease over 200 of its processes	13.58
HDFC bank	02	• Launched IRA 2.0, an interactive humanoid • 'Eva' is an AI-based chatbot, developed by Sense forth AI Research	8.70
YES bank	02	• Using YES TAG • Using AI solution by quadratyx	8.70
Axis bank	03	• Launched an AI-enabled app developed by Active AI • "Thought Factory"—an innovation lab based on innovative AI technology • Solutions for the banking sector	13.58
Kotak Mahindra bank	01	• Using AI solution by quadratyx	4.35
UCO bank	01	• Using AI solution by quadratyx	4.35
Total bank-10	23		100%

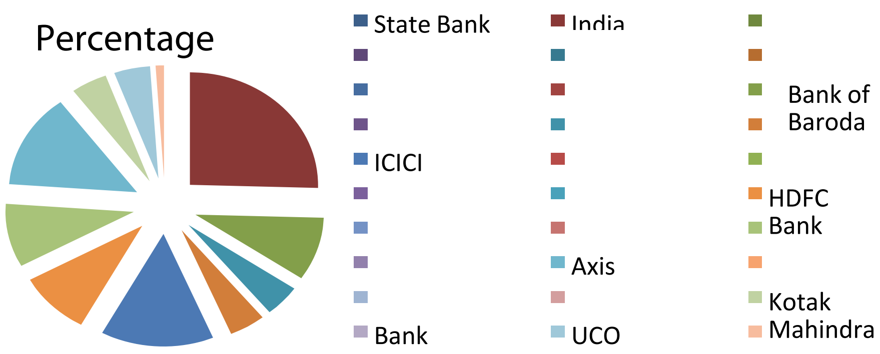

Fig. 2 Semiotic models of pie chart for using apps by various banks and their percentage

(d) **Big Data**: Big data technology aids in the analysis and processing of various transactions and data for extracting crucial information for sustaining competitive advantage. It helps in recognizing recent market movements and reorganizing operational issues and processes to manage risks.
(e) **Blockchain Technology**: It holds the potential to influence various banking transactions like Trade Finance, Cross Border Payments, Digital Identities etc. Operational efficiency, simplicity, transparency and the customer experience in banking can be improved by the use of Black chain technology.
(f) **Cyber Security**: Sometimes online transactions are susceptible to the risk of a data breach. Hence, banks need to strengthen cyber security in banking processes. Cyber security in banking transactions helps in protecting customer assets.
(g) **Robotic Process Automation (RPA)**: Robotic process automation has streamlined back-office processes which used to be performed by Bank employees. With the shift from man to machine, banks have witnessed improvement in efficiency, cost reduction and manpower-related issues.
(h) **Cloud Computing**: Rapid development of products and services is possible due to the adoption of Cloud Computing Technology. With the usage of Cloud Computing, services like Core banking, Net Banking, Mobile Banking, wallets and Card Management are handled properly. Productivity, performance and profitability can be improved by Cloud Computing. NASSCOM reports that the cloud market in India is also expanding and is expected to grow to $7.1 billion by 2022. State Bank of India has collaborated with Oracle and the Bank of Baroda has collaborated with IBM for an acceleration of Cloud Innovation in their respective Banks.
(i) **Distributed Ledger Technology (DLT)**: A distributed ledger is a database allowing the recording of the transaction of assets and their details in multiple places simultaneously. There is no central data store or administration functionality in Distributed Ledger Technology which is available in traditional databases.
(j) **Predictive Analytics** is used as a decision-making tool in several industries and disciplines, such as insurance, banking and marketing. Predictive analytics can provide a holistic view of the customer's journey with the bank and further help strengthen the relationship. Banks need to embrace the capabilities of data and analytics to improve risk modelling and fraud detection.
(k) **Machine Learning**: Machine learning models can also predict which banking tools individual members might use and recommend them so that customers can make better financial decisions. These solutions can be used by banks for risk prediction, risk prevention, fraud detection, investments, investment modelling customer services, customer modelling, etc. Artificial Intelligence (AI) Application is adopted in Machine Learning.

Paria and Giri (2004) explain the customer concerns about banking products and services have been alleviated in part thanks to digital banking and the digitalization of the banking industry. With demonetization and digitalization pushing India towards a cashless economy, the preference for online banking transactions is greatly in the interest of the country's consumers [9].

Carbo-Valverde [10] define the financial services sector is not immune to the widespread disruptions caused by the rise of digital technologies in the business world. Strategically, digitization presents a challenge for many financial institutions, despite the widely held belief that technological progress inevitably leads to simplification [10].

The opinion of Kimble and Milolidakis [11] The Big data has the potential to completely alter the way businesses are managed by providing unprecedented insight into consumer and competitor behaviour. Big data is a buzzword in the business world, but its meaning is often misunderstood [11].

As per Malyadri and Sirisha [12] the financial system is crucial to any growing economy. It is widely acknowledged that a robust banking sector is essential for long-term economic expansion. The number of branches, the number of employees, the amount of business done per employee, the amount of profit made per employee, the bank's net worth, the bank's deposits, investments, advances, interest income, other income, interest spent, operating expenses, and the cost of funds (CoF) all show that Indian banks have been struggling in recent years. Ratios of return on assets, CRAR, net nonperforming assets, and net advances to costs of funds In order to safeguard against economic setbacks and keep things steady [12].

Yoo et al. [13] explain the nature of product and service innovation is shifting dramatically as digital technologies permeate every facet of many businesses' operations, goods, and services. Programmability and data homogeneity are the cornerstones of digital technology. Convergent and generative ideas are fostered by their shared environment of open and adaptable affordances [13].

Nylén and Holmström (2014) discussed the widespread effects of digital technology have led to the fundamental restructuring of entire industries, making it increasingly vital to achieve business goals. It is hardly unexpected that managing digital innovation is of great interest to managers [14].

Estrin and Khavul (2017) Investors can benefit from the synergy between signalling and network effects in a lowcost setting, as proposed by the design of equity crowdfunding platforms. The world's largest equity crowdfunding platform's whole transaction history is used to put assumptions to the test. It demonstrates how the dynamic process of providing money is sensitive to and adaptable to new information without being explosive [15].

As per Chen et al. [16] adoption rates for various digital technology, especially mobile phones, in Asia have surpassed those in the West during the previous decade. The use of ATMs has exploded across Asia, and the "consumer decision journey" has shifted online across all age groups. Currently, most customers conduct preliminary research online before making a purchase at a branch. However, more and more transactions are being completed entirely via the Internet. The current status of regulation in many countries is a major barrier to the expansion of this trend since it requires customers to finalise transactions by signing documents in branches, in the presence of branch workers [16].

4.1 The Emergence of New Emerging Financial Technology (Fin-Tech) and E-collaborations a New Paradigm in the Indian Banking Industry

"Fin-Tech" is defined as In the coming years, revolutionary changes in the production, distribution, and use of financial products and services are anticipated as a result of technological advances that will further fuel the growth of the FinTech industry. When it comes to the banking sector, Blockchain ranks among the most cutting-edge Emerging Technologies. The Bitcoin industry is where you'll most often hear this word utilised. Blockchain technology has several practical applications for businesses, including data security, identity verification, transaction recording, contract signing, and enhanced traceability. FinTech's explosive expansion can be attributed to a confluence of factors, including new technologies, shifting social norms, and supportive government policies. In the future, fintech will permeate every sector, from retail to healthcare to agriculture to real estate, and beyond.

Challenges and Opportunities in the Implementation of Digital Innovations:

(a) **Sustainable Competitive Advantage**: The execution process of digital innovations in Indian banks confronts the challenges of sustainability and continuity. Customer education and Financial Literacy should be continued to make more people digitally inclusive.
(b) **Customer Retention**: Has become another important challenge in today's post-loyalty world. Consumer demands are changing in this tech-savvy and highly digital era as they are not tied up with traditional banks. Technology giants offering superior digital offerings like Amazon, Facebook, and Google to attract customers. Banks need to innovate or partner with them.
(c) **Digital Native, Intelligent, Social, connected (DISC) Approach:** needs to be adopted by the banks today to understand customer context and provide fresh, agile and relevant digital solutions to consolidate their digital leadership. Merchant Acceptance and Infrastructure need to be increased both in rural and urban spaces.
(d) **Skilled Resources**: Automation in the banking process through Digitalization, Artificial Intelligence (AI) and Robotic Process Automation (RPA) has brought about sociological challenges. Banks need to understand the impact on jobs and train the workforce with new IT skills,
(e) **Cyber Security Risks**: The Centre for Software and IT Management (CSITM) in their study conducted at IIM Bangalore has pointed out that one of the most important challenges in building trust among customers is cyber security risks. In their study, they have identified the potential risks in Digital wallets, specific bank apps for account holders, direct links with user's banks, and basic USSD services. Other types of security risks are phishing, vishing, hacking, credit card fraud, cloning etc.
(f) **Lack of Technological Infrastructure**: Adopting the right technology and providing the necessary support for its growth and use necessitates a country

with a suitable degree of infrastructure. But India's infrastructures still have room for improvement.

(g) **Opportunities**:

  (a) **New Innovative Product Design**: Due to various initiatives taken by RBI, the Government of India and Banks in promoting digital innovations, there has been increasing adoption and usage of digital banking by customers. The new-age customers are aware of the benefits and ready for new digital solutions. Banks need to focus more on product design, service delivery, and customer support. Bharat Bill Payment, an Interbank Web-based Platform, an E-mandate part of NACH and a Digital platform for high-value electronic transactions are the proposed new launches by NPCI.
  (b) **Better Regulatory Environment**: Data Connectivity and the Spread of Smartphones have improved the digital ecosystem by providing an efficient and effective regulatory environment.
  (c) **Leveraging the power of social media**: Social Media Technology, Digital Assistants and third-party channels like Facebook, and Twitter for leveraging internal capabilities are likely to become the primary channels by 2022 apart from online and internet banking.

4.2 Hypothesis Testing

In the section of Hypothesis Testing, the researcher has taken several response models for justifying the hypothesis, therefore, they have taken 10 Banks of different branches with taken 500 Customers from Bhubaneswar smart city, Odisha (India) (*Each having 50 respondents) with gender discrimination and different profession, Business, Cultivator, Govt, Service, Private service, Company Service etc. After obtaining the data will be presented in the appropriate data table and then classified for its accurate analysis and interpretation thereon. Here, various response models are given for the reader's kind perusal and perception.

As per the aforesaid data table, the mean value for female customers is 42.0, for the male customer is 58.0 and the Mean difference between them is 16. 0, and the number of banks is 10 d.f is ($N - 1 = 09$). It is proved that more customers are preferred use of a Digital innovation banking system. Critical values of the Student's t distribution were therefore calculated with the help of the cumulative distribution function and recorded in the table. The t distribution is symmetric so that $t1 - \alpha,\nu = -t\alpha,\nu$.

Thus, the t table can be used for both one-sided (lower and upper) and two-sided tests using the appropriate value of α. Below is a graph depicting a t distribution with 10 degrees of freedom, which illustrates the significance level. The $\alpha = 0.05$ significance level is the standard. Calculating $1 - \alpha /2$, for a two-tailed test yields 0.975 when $\alpha = 0.05$. If the test statistic's absolute value is larger than the critical value (0.975), then the null hypothesis is rejected. We only provide the positive critical values because the t distribution is symmetric. Therefore, the use of AI in digital innovation in the banking sector is justified, and the null hypothesis has been rejected by an alternative hypothesis that has been accepted due to its high significance and justification in both the levels of significance at the 0.01 and 0.05 alpha level. The Financial Stability Board (FSB) of the Bank for International Settlements is concerned about technologically enabled financial innovation that could lead to new business models, applications, processes, or products and has an associated material effect on financial markets and institutions and the provision of financial services. (BIS).

5 Suggestions and Recommendation

- Loss of data can be avoided by employing well-trend IT experts.
- For the general public and rural populations to get the most from bank services, strict regulations against cyberattacks and workshops and seminars on digital payments must be established. As a result, rural and underdeveloped communities should implement Digital Banking Literacy programs.
- Technology needs to be improved and made available to more people, especially those who live in distant places.

6 Findings

- Digital innovations help in reducing operating costs and widening the customer base.
- Reports can be generated and analysed for various purposes at different points in time.
- It is the safer way to handle financial transactions and to prevent misuse of DBT under Digital Banking.
- The Indian Banking and Payment System is witnessing various new emerging technologies like Blockchain, Distributed Ledger Technology (DLT), Artificial Intelligence (AI) Machine Learning, Robotic Process Automation (RPA), Biometrics, Chatbots and the use of Big Data and Predictive.

7 Limitations

- The banking industry in India is changing quickly, and more changes are to come with the advent of FinTech firms and Payment Banks.
- To become stronger organisations than other Non-Banking Financial Companies (NBFCs) that are also entering the banking industry, the public sector banks have been merging.

- Over the coming years, keeping a close eye on how things are developing is important.
- Since there aren't many contemporary players in banking using digital innovation, it might be wise to include more in the study.

8 Conclusion

In conclusion, we may conclude that the Digital India plan aims to make India a knowledge-based nation with an empowered digital society. The above statistics and discussion show the rising trends in the adoption of digital innovations and new Fin Tech emerging products. While there will always be challenges, opportunities also exist for banks and financial institutions ready to innovate and offer more digital financial products to the customer. The Digital Payment Ecosystem with Fin Tech collaboration and global technology giants are acting as aggregators for retail transactions. Measurement of Digital Payments should be carried out at regular intervals to study and monitor the progress. The new emerging financial technologies in the Indian Payment System will continue to be evolving, reaching global heights with regulatory compliance and risk management to achieve the vision of Digital India. This is to ensure improved customer service and achieve the objective of a cashless economy, through the digital India programme. Banks in the public and private sectors are both evolving towards consumer services that rely heavily on technology. The banking sector is expanding alongside digital innovation as a result of the benefits and challenges presented by digital technology. Due to the COVID-19 effect, digital banking has grown in popularity during the pandemic time of lockdown.

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Digital Pharmacy in Industry 4.0: A Case of Consumer Buying Behaviour Pattern Using TAM Model

Sanjukta Ghosh, Sachi Nandan Mohanty, Richa Goel, Tilottama Singh, and Sukanta Kumar Baral

Abstract The research seeks to identify the elements of consumer purchase behavior to shop through digital or E pharmacy through social media influence in India: based Technology Acceptance Model (TAM) which is popularly used in the field of behavioral science. The study is to determine the impact and relationship among the independent variables trust, perceived usefulness and perceived intention on online purchase of medicines through social media influence which is considered as a dependent variable in this study. A descriptive qualitative exploration was conducted through a structured open ended questionnaire from the 100 respondents across different age groups pertaining to Indian context especially millennial using a structured, self-administered and internet mediated survey. The study is cross sectional which studies the relationship between variables and influence of the social media on the purchase of online medicines. The findings shows that there is a positive relationship amongst the independent variables trust, perceived usefulness and perceived intention on the online purchase of medicines through social media influence but perceived intention comes out to be more important construct for buying medicines online along with the trust.

Keywords Digital pharmacy · Consumer behaviour · TAM model

S. Ghosh (✉)
Srishti Manipal Institute of Art, Design and Technology, Manipal Academy of Higher Education, Manipal, India
e-mail: sanjukta.ghosh@manipal.edu

S. N. Mohanty
School of Computer Science and Engineering (SCOPE), VIT-AP University, Amaravati, Andhra Pradesh, India

R. Goel
SCMS Noida, Symbiosis International University, Pune, India

T. Singh
Uttaranchal University, Dehradun, India

S. K. Baral
Department of Commerce, Faculty of Commerce and Management, Indira Gandhi National Tribunal University, Amarkantak, Madhya Pradesh, India
e-mail: sukanta.baral@igntu.ac.in

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1 Introduction

After the outbreak of the unpredictable and ferocious pandemic in 2019 various verticals of ecommerce have seen an outrageous demand out of them is one excellently demanding and forthcoming online pharmacy market also known as digital or E pharmacy [1]. Contact less delivery became priority during COVID 19 pandemic. New innovative ways of creating a delivery ecosystem became the major survival need for the players in ecommerce sector. Pharmacy being considered as the life saviour category became the central point of attraction among the business community [1]. The players were all evolving with various strategies to minimize movement and point of contacts in the entire supply chain process. Patient safety risk is considered as the most primary factor in this sector unlike other ecommerce category [2]. Over the past decade internet has emerged as the favorite market place for the consumers, distributors and retailers due to various reasons. Therefore it is evitable to study the consumer behavior and attitude regarding the online purchase of goods and services. It is also important to understand and develop effective communication and persuasion strategies to reach and penetrate the target groups. Researchers have majorly worked on various domains under consumer behavior. However very little emphasis has been given to the online selling of medications. The sector needs to consider various quality standards and legal protocol which restrict and regulate the transaction of medications online. Only legalized Over the Counter drugs, supplements and vitamins could be sold through internet [3]. However we saw some emerging e pharmacies like 1 mg in India which is doing potentially good. They have their greatest selling during the pandemic. So the pharmacy retailers are now going through transformation and convergence of the digital revolution with pharmacy. It uses digital tools and technologies to improve business, practice and patient care. We also use E pharmacy or online pharmacy alternatively as they are synonymous. It is a platform which provides drugs in both lawful and unlawful manner. India's digital market is booming at a rapid speed with increasing number of Internet users, with the total time invested by Indians being more than 17 h in 7 days [4].

Social media have catered a greater demand in the business market with huge number of consumers available on it [5]. Social media marketing or SMM is a public entertaining platform which also works as great marketing tool [6]. Social media marketing emphasises on the promotion and selling of the goods and services through various social media platforms, but it can be a time-consuming activity for businesses as one has to be consistent, but it is a great place to understand consumer behavior [7]. Advanced features like demographic of the user can help the advertisers to segment their population and advertisements accordingly. Social media marketing and its power to influence could be a great opportunity and an effective platform for the online pharmacy owners to sell drugs and reach out the potential customers through certain lawful regulations [8]. However, the regulations and various ethical protocols need to be worked out very carefully in to avoid various mal practices around the pharmaceutical industry. The mindful practices of the pharmacy need to be in place to harness the platform and make an effective usage of it. SMM for e pharmacy helps in generating suggestions through paid promotions, as people visits more on social media sites advertisement. This can increase the visibility of one's brand. It also helps in increasing customer retention and loyalty. The best way of using the Social Media Marketing is to run pilot testing where one can know about their consumer's preferences at low cost. Despite the advantages of using Social Media Marketing for selling drugs online there are many rules and legal regulations which control the usage of e pharmacy. As per 2019 report there were approximately 8,50,000 pharmacy retailers or offline brick and mortar stores which counts for 99% sales of drugs whereas online pharmacy contributes only 1% sale of drugs in India [9]. But after the outbreak of the pandemic, increasing digital usage and awareness there is a potential increase seen the acceptance of online pharmacy among the government, investors as well as consumers [10]. Unlike any e commerce store the concept of online pharmacy is very different due to the rigid rules and regulation. Globally various mal practices around health care has resulted to serious challenges and concern. Health care has become a highly profitable industry and the consumers are being exploited through mere capitalism. All the major stakeholders like government, multi-national corporations, health care professionals and various other big and small players in the eco system should work together with strong ethic and integrity [11]. In year 20,005 the online pharmacy startups came together to make an IPA—Indian Pharmacy association which promises an ethical trade of medicines through online platform in India, and there are other multiple laws by the government of India which regulate the online selling of drugs like Drug and Cosmetic Act 1940, Information Technology Act 2009 and the Indian Medical Act 1950. We can observe a positive shift towards online pharmacy due to the increasing chronic diseases and busy lifestyle of people. But due to the security and other reasons consumers are very slow to adopt to new online drugs but steady and slowly there will a major boost in the digital pharmacy. The preceding researcher's emphases on the variables like trust, risk, and usefulness to study the consumer buying behavior directly on the online purchase of medicines but with the ongoing pandemic and advancement of technology benefits in the context of the impact of social media on digital pharmacy as increasing social media users as well as growth of social media marketing which conditionally helps to understand the consumer buying behavior.

2 Objectives of the Study

The study envisaged on the following objectives;

- To find the relationship and impact of trust on online purchase of medicines through social media influence
- To find the relationship and impact of perceived usefulness on online purchase of medicines through social media influence
- To find the relationship and impact of perceived intention on online purchase of medicines through social media influence

- To find the relationship and impact of social media influence on online purchase of medicines.

3 Literature Review

Each individual is unique in nature therefore we cannot consider the consumer as the homogeneous group. Identify the variety in their behavioral pattern which controls their decision making is important for market research professionals. The researcher also confirms the connection between social media and consumer buying behavior to purchase particular products. Ahmad Juwaini's [12] research focused on the online transportation services through social media influence. The study aimed at find how trust is related to purchase intention towards services through social media marketing. The research showed the positive relation between purchase intention and persuasion through social media, whereas there is no significant relationship between trust and intention to purchase online services. Various researchers have shown the positive relationship between persuasion through social media and consumer decision making process [13]. The researchers showed the framework on the meditating effect of social media with relationship between perceived usefulness, reliability on online shopping behaviour [14]. Advanced usage of social media positively impacted small players in the health care sector. Various forms of text messages used by them effectively improved patient communication. WhatsApp is the most used platform for communicating with the customers and also being effectively used by many online or ecommerce players [15]. Remuneration as well as the social motivation impacts the consumer buying behavior positively and have a direct impact on their purchase decisions and indirectly mediated through trust [16]. The study shows key factors for the success of e-commerce and social networking including trust generated from the peers in the form of reflection related to perceived usefulness of the product and the social networking sites [17]. The use of social media and its tools helps the organizations in increased branding of their products and reach out to more segments of customers also increasing the brand awareness [18]. E pharmacy adds to the convenience for the elderly people by improving consumers' experience suffering from chronic conditions who aren't able to reach the offline pharmacy. Studies have highlighted three aspects of social media marketing. They are trendiness, word of mouth and customization. These aspects can directly influence consumers' loyalty and trustworthiness towards a brand [19, 20]. The researcher used technology acceptance model (TAM) and Self Determination theory (SDT) to study the intention of the consumers towards electronic healthcare. Understanding the level of acceptance towards technology is one of the popularly used model in the field of social science and user behaviour. Human motivation and personality can be captured through Self Determination Theory. The research shows the relationship between intention to use internet and e-healthcare. The researchers used the technology acceptance model to determine that perceived ease of use and perceived usefulness and trust as the factors influencing the intention to adopt e-Health in developing countries [21]. Online pharmacy leads to minor financial losses with considerable legal and health hazards which changes consumers' behavior towards purchase of medications online. Lack of personal touch, quality check and illegal traders also marks up to the dangerous and contaminated online pharmacy. Crilly et al. [22]. Pharmacists use social media platforms especially Facebook to interact with the patients by providing them links to various health resources and giving compliance with tips of using the drugs and other products which promotes meaningful health and consumer behavior changes [23]. With the advancement of technology it is inevitable to separate the online world from social media and consumer behavior; there is no escape route from social media these days for both the individuals and the companies. In the digital stores where the consumers couldn't sense the products they purchase, due the lack of physical transaction. Consumers usually develop low trust and perceive highly elevated risk [24].

It is apparent that most of the active social media users comprise of young adults whereas the people with long term health conditions typically are not so active on social media which is major drawback for the online health platforms [24]. Social Media has played an important role in areas like medicine information, safer usage, usage in chronic diseases, evidence based guidelines etc. This have started impacting an increase and improvement in the patient's outcome. The familiarity of social media to public and its economical nature along with the ability to distribute information rapidly creates an opportunity for the pharmacists who wanted to provide innovative heath care facilities both at the individual and public level.

4 Research Methodology

A well designed questionnaire was developed to capture descriptive inquiries. Open ended qualitative questions were used to collect data from the 100 respondents across different age groups pertaining to Indian context especially millennial. Self-administered online survey was conducted. The study is cross sectional which studies the relationship between variables and influence of the social media on the purchase of online medicines. The sample was selected based on convenience sampling, while sampling the frame was difficult due to time and financial constraints. Internet is used to distribute the survey seems to be sensible for the research during the pandemic. The research potentially covers the social media users who are online consumers, and reduces the uninformed occurrence of responses. There are defined hypothesis prominent to 4 variables which are defined under Table 1. The hypothesis is framed based on technology acceptance model (Fig. 1). The study aims to explore the relationship among the variables like perceived usefulness of online media, trust in an online platform and perceived intention on the online purchase behavior of buying medicines online through social media influence (dependent variables). The variables are adopted from TAM and existing theories are applied to the specific situations. TAM known as the technology acceptance model is considered to be the most influential model of technology acceptance despite of the criticism it faces, it focuses

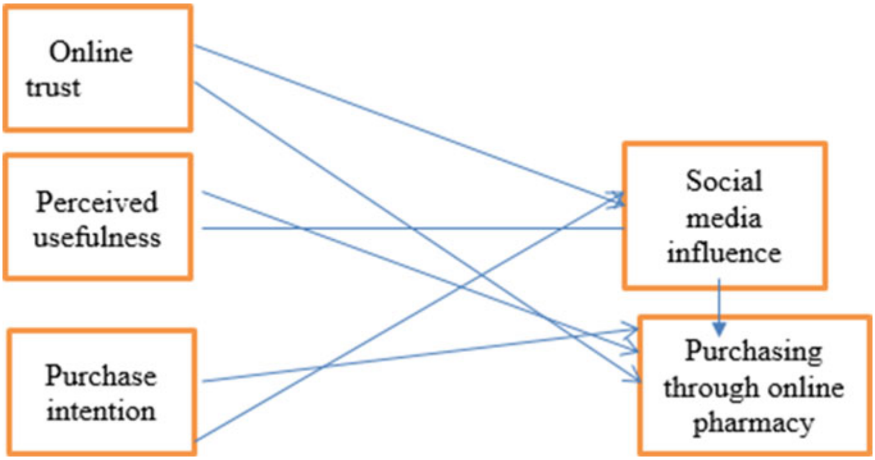

Fig. 1 Technology acceptance model

Table 1 About the key constructs

Constructs	Definations	References
1 Online trust (TRUST)	A significant factor determining successful online interaction predicting the behavior of an entity and selection of an entity	Mark Perry
2 Perceived usefulness (PU)	Consumer behavior is determined by their attitudes towardsthe	Barki
3 Purchase intention (INT)	adoption of technology i.e., perceived usefulness and perceived ease of use while interacting during the online transactions	Renko
	A person's commitment, decisions and plans to carry out	Eagly and Chaiken

on two factors which influence the user's intention towards usage of new innovative technology, perceived ease of use and perceived usefulness. TAM forecasts individual adoption of technology and its voluntary use. It focuses on the technology users and how they can adopt to the new technology.

The model is primarily focusing on the potential of the users. The two major constructs use in the model are perceived usefulness and perceived ease of use. The TAM has been continuously studied and also has been expanded and upgraded. It has also been used in the ecommerce context with inclusion of trust and perceived risk. Reliability is checked by Cronbach's Alpha using Microsoft Excel 2010 and further regression analysis is used to determine the relationship between variables.

4.1 Hypothesis Development

Rajković et al. [25] shows that social media can help in building trust among consumers and towards the business. The researcher showed the positive relationship within intention to purchase and trust. Researchers also confirm the connection between social media and consumer buying behaviour to purchase particular products.

H1 There is a positive impact of trust on social media influence.

H1A There is positive relationship between trust and online purchase of medicines.

Yahaya Nasidi et al. [26] in their study showed the framework on the meditating effect of social media on perceived usefulness, reliability on shopping behaviour on the online platform. The researchers used the technology acceptance model to determine the that perceived ease of use, trust and perceived usefulness are important factors to influence intention towards adoption of e-Health.

H2 There is positive impact of perceived usefulness on social media influence.

H2B There is positive relationship between perceived usefulness and online purchase of medicines.

Irshad et al. [15] in their study mentioned that remuneration as well as the social motivation impact the consumer buying behavior positively and have a direct impact on their purchase decisions and indirectly mediated through trust.

H3 There is a positive impact of purchase intention on social media influence.

H3C There is a positive relationship between purchase intention and online purchase of medicines.

Crilly et al. [22] in their study mentioned that pharmacists uses social media platforms especially Facebook to interact with the patients by providing them links to various health resources and giving compliance with tips for using rugs and other products which promotes the meaningful health and consumer behavior changes.

4.2 H4 There is Positive Effect of Social Media Influence on Online Purchase of Medicines

More than one items are used to define the same variable. All the variables are opinion variables measured in a scale with one as strongly disagree and five as strongly agree. The demographic variables like age and income have multiple choice options to categorize from whereas gender category follows a simple scale (dichotomous male/female). The ethical issues are taken care of at all stages of research by keeping anonymity and confidentiality with the volunteering nature. Due to social media influence more emphasis is given to the social media users and millennial with no information.

5 Result and Analysis

More than one item is defined for some variable. All the variables are opinion variables measured on in a scale with one as strongly disagree and five as strongly agree. The demographic variables information like age and income have multiple choice options to categorize from whereas gender category follows a simple scale (dichotomous male/female). The ethical issues are taken care of at all stages of research by keeping anonymity and confidentiality with the volunteering nature. Due to social media influence more emphasis is given to the social media users and millennial with no information harmful being asked while filling the questionnaire, and there was no financial reward given to the respondents. The value of Cronbach's alpha for the entire questionnaire comes out to be 0.879 i.e., 88% of which is considered to be a very good outcome and the data stands reliable because the benchmark is 0.70 i.e., 70%. Therefore, we can move forward with our data collected from 100 respondents. The results below were derived from the reliably test or Cronbach's Alpha test which was applied on the data using Microsoft Excel 2010. The value of alpha in each case was reliable as it was above 0.70 or 70% and can be further used for the study.

Male respondents are higher than the female respondents, and more respondents are between the age 18 and 24 and are active social media users. The income is distributed evenly and 23% of the respondents prefer not to disclose their income. Correlation of single determination shows 0.681293 which is 69% of variation in trust on the social media influence that can be explained by the regression; similarly 40% of variation in perceived usefulness on social media influence and 52% of variation in perceived intention on social media influence can be explained by the simple regression analysis. Whereas social media has direct and positive relationship or impact on trust ($p < 0.05$), perceived usefulness ($p < 0.05$) and perceived intention ($p < 0.05$) thus H1, H2 and H3 are supported. Due to less variation the data doesn't shows much higher correlation between the variables.

Similarly, the 85% of variation in trust on the online purchase of medicines can be explained by the regression analysis, 79% of variation in perceived usefulness, 88% of variation in perceived intention and 76% of variation social media influence on online purchase of medicines that can be explained by the regression analysis. Online purchase of medicines has direct and positive impact on trust ($p < 0.05$), perceived usefulness ($0.05 > p$), perceived intention ($p < 0.05$) and social media influence ($p < 0.05$) thus H1A, H2B, H3C and H4 stand supported.

6 Conclusion and Recommendations

Correlation analysis reflects the strength of the relationship among all the variables hence found the six-hypothesis supported. Regression analysis also supported and found out all the constructs are statistically significant predictors to know the impact of social media influence on the purchase of medicines online or the use of digital pharmacy. Purchase intentions of the consumers towards online pharmacy has a larger effect along with the trust. But it is significantly true that with social media message and its popularity among the users have great impact and opportunity for the digital pharmacy to study their consumer behavior, needs and wants. The research also contributes to the existing studies as it uses the researchers' patterns through internet tools to know the consumer behavior towards the smaller segmentation of pharmaceutical products. The research applied Technology Acceptance Model a theoretical model with general validation. The research is unique under the online shopping of medicines through social media influence that can be knowledgeable and resourceful for the existing literature contributing to study different constructs towards the digital pharmacy and consumer behavior. Our discoveries give a few important experiences to the buyers to acknowledge the e- drug store by utilizing online media impact. E-retailers and administrators completely comprehend the significance of web-based media impact during the time spent presenting e-drug store; such firms ought to promote their meds and grandstand the utilization of meds on various online media which can fulfill the buyers' requirements. Findings of the research suggest that more people are becoming aware of the social media and Millennial are the active users of different social media platforms and Trust is one of the importance factors impacting the online purchase of medicines along with the intention to purchase due to the pandemic. Most importantly it is recommended to the e- sellers to focus on the factors which authentically shape and attract online consumers like the type of advertising, benefits, loyalty programs etc. Trust can become one of the major factors which hinders the online purchase process, e- pharmacy should focus on the authenticity of the consumer data and also following all the norms, rules and regulations provided by the Government and medical authorities. Perceived usefulness and socioeconomics are the progression forward towards internet purchasing conduct through online media impact, so the perceived usefulness and socioeconomic parameters are impacted by the web-based posts across various societies which sway internet purchasing conduct of medication (e-drug store). The impact of perceived usefulness on the goal in the past investigations has been differed due to various reasons and has a significant affect on the purchase behaviour [27, 28].

7 Limitations of the Study and Future Scope

The study has been considered in Delhi NCR which represents urban cosmopolitan population and does not cover the semi urban and rural region. This signifies that the database does not have any representation from rural India which has majority of our country population. The study has been conducted in purely Indian context and not being validated in a global context. It is important to consider cultural parameters to conduct the study in the other region.

There are several suggestions for the future researchers to study the consumer behavior and pattern towards purchase of medicines online not only by the social media influence but other influences like EWOM and more. Researchers can study different variables or constructs to determine the consumer behavior towards digital pharmacy. Also, other variables like perceived ease of use, innovations, shopping attitude of the consumers along with the risk should also be studied as proposed by different theoretical models. Last but not the least population outside Delhi should be studied with the probability sampling covering the rural population also.

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Apple Stock Price Prediction Using Regression Techniques

Mithun Shivakoti, K. Jeeveth, Nihar Ranjan Pradhan, and Mannava Yesu Babu

Abstract In the course of working on this project, we conducted a regression analysis on the stock price of apple between the years 1980 and 2021. For this particular study, we relied on the techniques of Linear, Poisson and lasso regressions, from which we derived the accompanying graphs. These graphs are utilised to make projections on the value of Apple shares for the following year. The model will provide either numbers that are approximately accurate or values that are predictions. The dataset can be acquired from Kaggle, and the reference section will include a link to the site where it can be found. We will also be trying various regression approaches, and we will demonstrate that the regression obtained by linear regression is the most effective method for analysing this dataset.

Keywords Apple · Stock · Prediction · Linear regression · Poisson regression · Lasso regression

1 Introduction

Each company's stock price reflects its performance [1]. To maximise profits, businesses and investors would like to identify the future direction of stock prices so that investors can respond in advance, remain abreast of market opportunities, and gain enormous financial profits with comparatively low risk. As a result, effectively anticipating stock prices is critical, and it attracts the attention of a diverse range

M. Shivakoti (✉) · K. Jeeveth · N. R. Pradhan
Department of School of Computer Science and Engineering (SCOPE), VIT-AP University, Amaravati, Andhra Pradesh, India
e-mail: mithun.shivakoti11@gmail.com

N. R. Pradhan
e-mail: nihar.pradhan@vitap.ac.in

M. Yesu Babu
Department of CSE, Vardhaman College of Engineering, Shamshabad, Hyderabad, Telangana, India
e-mail: mannavababu@vardhamn.org

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of experienced professionals to investigate approaches for determining the possible direction of the stock market [2]. However, it is a difficult task since tonnes of trade occur every second, which is influenced by politics, unforeseen events, the global economic situation, which is volatile and uncertain, resulting in the emergence of automated, pre-programmed analytic procedures to execute orders. According to Fama's Efficient Market Theory [3] forecasting the financial market is difficult because pricing has already represented all available information. He also believed that stock prices follow a stochastic process that is random and unpredictable [4]. Nevertheless, numerous research have been conducted to anticipate stock values, ranging from simple methods to complex models, indicating that the stock market may be predicted to some extent [5].

Investors no longer view investing in traditional financial institutions, such as banks, which frequently give a low return on investment [6, 7], as a lucrative alternative. The stock market is regarded as one of the most important components of the complete financial system [8]. The practise of purchasing shares of publicly traded companies is gaining ground in today's society. Almost all of us are drawn to the idea of being able to increase one's wealth in a very short amount of time while also realising returns that are often higher than average. In return, businesses that sell stocks earn capital that can be put toward expanding their operations. On the other hand, it is frequently a lengthy haul, and the money that is invested may not be recovered for several decades at a time. In other instances, the amount of money that is invested does not result in a profit at all but rather a loss. As a result, each step needs to have serious consideration put into it, and the acquisition of stocks needs to be done with the appropriate amount of consideration. Any potential investor in the stock market hopes to acquire the ability to accurately forecast the quantity and timing of stock purchases. However, the quantity of stocks an investor can purchase is also contingent on the total value of their assets. The next crucial stage in the appraisal of the invested funds is to analyse and evaluate the past movement of stock prices and the company whose stocks we wish to buy. This is the next crucial stage in determining how much those investments are worth in the long run. This comprises its overall economic status as well as its turnover and introduction of new products to the market. There are a lot of different things that can cause a gain or a drop in value in the long run. If we are able to accurately forecast how the stock price of a certain firm will evolve, we stand a good chance of realising a substantial financial gain [9]. This sensation, along with the capacity to make accurate predictions, is what differentiates successful investors from those who are not successful. The value of stocks is actually determined by the amount of interest in them. The supply and demand for stocks are the driving forces behind the development of stock prices. If there is a high demand for a stock but the supply is limited, then the price of that stock will go up. This is true for every commodity that can be purchased on the market. As a result of fluctuations in supply and demand brought on by a variety of circumstances, the price of a stock naturally moves about over time. As a result of the inherent character of the financial world and the interplay of several recognised elements, stock values are thought to be very dynamic and prone to quick swings. This is how things are because of the structure of our financial system. One is able to create predictions and judgements based on these criteria by utilising a range of methods to approach the data as well as the procedures. It is generally agreed that the so-called time series method is one of the most convincing statistical approaches for studying change and evolution through time. It is vital to have a time series of changes in stock prices in order to discover the factors that caused changes in stock prices and defined their performance in the past, as well as for the purpose of estimating how stock prices will vary in the future.

2 Literature Review

Apple Inc. is one of the richest and well-known firm across the globe. In April of 2013, the company pledged to return $100 billion to shareholders by the end of 2015 [10] through stock buybacks and cash dividends. Khan et al. [7] suggest that investment strategies should incorporate knowledge-based decision support systems for financial management. They reasoned that investors avoid banks and other conventional alternative investments because of the poor returns available through these institutions. One of the most common places for people to put their money these days is in the stock market. The ability of day traders, investors, and data scientists to accurately forecast the price of stocks is getting increasingly challenging. These are complicated functions that are determined by a large variety of interacting elements that influence the dynamic progression of price changes [11]. The equilibrium between supply and demand [12] determines the global perception of stocks, which in turn is affected by political and social prospects, sales and other socio-economic aspects. Gupta and Chen [13] agree, stating that sentiment is a major factor in stock prices and the financial markets. Thus, studies are being conducted to determine whether public emotion displayed on social media platforms like Facebook and Twitter may be used to predict future stock market patterns. This is also confirmed by Gupta and Chen [13]. They argue that investor sentiment is a major factor in stock prices and the financial markets. By factoring in price fluctuations and understanding how individuals are feeling, it's feasible to generate more precise predictions [14]. Non-linear shifts in a firm's ROI over time make it difficult to anticipate stock return, according to the research of Chaudhari and Ghorpade [15]. This is one of the reasons why predicting stock return is difficult. Inaba [16] conducted an analysis of 37 developed and developing countries' participation in the global market for commodities and found that developed countries had larger profits on their international stock prices than underdeveloped countries did. On the other hand, due to factors that change over time, the return on investment for stock investments increased more quickly in developing countries.

Stock price predictions have been made using a wide range of methods. Both Chen et al. [17] and Long et al. [18] relied solely on past stock prices to predict future values. On the other hand, Singh et al. [19] and Patel et al. [20] have augmented the historical data with several technical indicators. Random Forest, Naive Bayes, Decision Trees, and Support Vector Machine were among the several algorithms used by Gonadliya et al. [21] to assess the categorization accuracy of stock market fluctuations (SVM). For accurate stock price forecasting, Bharadwaj et al. [22] emphasised the use of sentiment analysis.

For the first time, Moghaddam et al. [23] developed a model that incorporates the day of the week as a feature over 2 different time periods (4 and 9 days, respectively). The comparison of OSS and tansgig transfer functions was the final step of a study that compared Multilayer Perceptron and Recurrent Neural Networks. Using a shorter timestep of four days, the former fared better than the latter, which used a nine-day timestep.

Using Twitter, news articles, Kesavan et al. [24] collected necessary data. They used LSTM and lexicon-based NLP methods to determine the polarity of sentiments with reference to the Google Glove Dictionary. The results of the LSTM regression were eventually compared to the polarity of the sentiments.

To extract interesting parts of articles and tweets, Liu et al. [25] used a capsule-based hierarchical network. When compared to the original baseline HCAN and StockNet models, they were 2.45% more accurate. To evaluate news and tweets, Liu et al. [25] used basic GRU whereas Mousa et al. [26] proposed a model with bi-directional GRU after the word embedding. They have also provided comments on the final capsule's representation of candlesticks and bollinger bands. The proposed model is shown to have superior performance than LSTM, GRU, and bi-GRU. The algorithms published between 2013 and 2018 are covered in a review by Gandhmal et al. [27]. So far, only neural network-based algorithms have proven successful at stock market forecasting.

Several technical indicators were used by Nabipour et al. [28], including the n-day moving average, RSI, and CCI. The models in their work are LSTM, TREE-BASED, ANN, and RNN. They found that LSTM has the best performance using root-mean-squared error but that it needs a lot of time to train. This finding prompted the incorporation of GRU into the model proposed in this study. Mustafa et al. [29] found no relationship between the price of stocks and the price of commodities in their studies of the region encompassing South Asia. A variety of models, including linear regression, SVM, neural networks, Twitter sentiment analysis, the 2012 terrorist incident, and Brexit, have been tested by Maqsood et al. [30]. When compared to other methods, they found that neural networks were superior. When comparing CNN to a pre-trained Google model (BERT), Sousa et al. [31] found that BERT was more effective for sentiment analysis. BERT, LSTMs with attention, and SVM were also used to analyse investor sentiment in the stock market by Li et al. [32]. The results showed that BERT with attention outperformed LSTM with attention and SVM in terms of accuracy score.

3 Objectives

To design a model that would predict the stocks of Apple Inc using regression techniques.

4 Proposed System

A. *Dataset Description*

Individual data are gathered from Kaggle [33] for the purpose of conducting research on time series. This page details the evolution of Apple Inc.'s stock values over time, specifically providing daily data over the span of 1980 to 2021. Excel from Microsoft is used to perform processing on the data that was acquired. There are 10,467 rows totally in the dataset. It contains 7 attributes namely date, open, high, low, close, adjclose and volume. The below images depict the trend of all the attributes across 41 years (Figs. 1, 2, 3, 4, 5 and 6).

B. *Methodology*

Linear Regression

The method of linear regression analysis was the one that we utilised for this purpose. A quantitative outcome variable (y) can be predicted using linear regression, also known as the linear model. This prediction can be based on one or more predictor variables (x). To construct a mathematical formula that defines the variable y as a function of the variable x is the objective of this endeavour. After we have constructed a model that is statistically significant, we will be able to use it to make predictions about the future based on the new x values.

Fig. 1 Opening trends across 41 years

Fig. 2 Closing trends across 41 years

Fig. 3 Highest price trends across 41 years

Fig. 4 Lowest price trends across 41 years

Fig. 5 Adjusted closing trends across 41 years

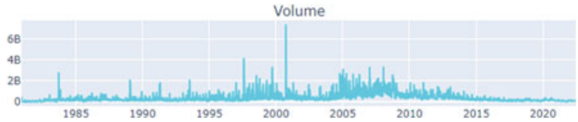


Fig. 6 Volume trends across 41 years

When you create a regression model, one of the steps that you must take is to evaluate how well the predictive model works. In other words, you need to evaluate how accurate the model is in predicting the result of a new test using data that was not used in the construction of the model. This data was not included in the initial data set.

When evaluating the accuracy of a predictive regression model, it is common practise to consider the following two key metrics.

Poisson Regression

One of the essential subjects is known as the Poisson distribution (Eq. 1). It is put to use in the process of calculating the probabilities of an event using the rate of value on average. It is a discrete probability distribution.

Poisson distribution function:

$$P(U = u) = \frac{\lambda^u e^{-\lambda}}{u!} \tag{1}$$

where,

e is Euler's number
u is a Poisson random variable
λ is mean number of occurrences.

Lasso Regression

Lasso Regression is one of the types of linear regression that incorporates a regularisation component into the loss function. This helps to minimise the size of the coefficients and prevent overfitting. The Lasso Regression may successfully perform feature selection and make the model easier to interpret by reducing some coefficients to zero. In addition to preventing overfitting, Lasso Regression can help to improve the model's stability and interpretability by lowering the number of input variables.

Mean Absolute Error

Mean Absolute Error (Eq. 2) is a metric used to evaluate a regression model's performance. It measures the average magnitude of the errors in a set of predictions without taking into account their direction. It represents the average absolute difference between the predicted and actual values. MSE is a more helpful statistic for fine-tuning the model because it does not provide information on the direction of the mistakes.

$$\mathrm{MAE} = \left|\left(\mathrm{s_i} - \mathrm{s_j}\right)\right| / \mathrm{j} \tag{2}$$

where,

- j is number of datapoints,
- $\mathrm{s_i}$ is actual value and
- $\mathrm{s_j}$ is the predicted value.

Mean Squared Error

Mean Squared Error (Eq. 3) is a widely utilized metric for assessing the performance of a regression model. It is found by taking the mean squared deviation between the predicted and observed values. MSE is a measure of prediction error that can be used with either continuous or discrete data. Lower MSE values indicate improved model performance and a closer correlation between predicted and real data.

MSE is calculated as:

$$\mathrm{MSE} = 1/\mathrm{h} * \sum_{i=1}^{h} \left(a(i) - \hat{a}(i)\right)^2 \tag{3}$$

where,

- h is the count of number of data points,
- a(i) is the actual value of the datapoint and
- aˆ(i) is the corresponding predicted value of the datapoint.

Root Mean Squared Error

Root Mean Squared Error [Eq. 4] determines how accurate the model's predictions are. It is the difference, on average, between the values of the outcome that have been observed and known and the value that has been predicted by the model. RMSE is calculated as follows:

$$\mathrm{RMSE} = \sqrt{\sum_{i=1}^{F} \left\|a(i) - \hat{a}(i)\right\|^2 \div F} \tag{4}$$

where,

F is the count of number of data points,
a(i) is the actual value of the datapoint and
aˆ(i) is the corresponding predicted value of the datapoint.

Coefficient of Determination (R^2)

R^2 (Eq. 5) represents the squared correlation between the observed values of the known outcomes and the predicted values of the model. The correlation range lies between 0.0 and 1.0. The more effective the model, the greater the R2 value should be.

$$R = \frac{m\left(\sum cd\right) - \left(\sum c\right)\left(\sum d\right)}{\sqrt{\left[m\sum c^2 - \left(\sum c\right)^2\right]\left[m\sum d^2 - \left(\sum d\right)^2\right]}} \tag{5}$$

where,

m Total number of observations
Σc Sum of values of variable c
Σd Sum of values of variable d
Σcd Sum of the Product of variables c and d
Σc^2 Sum of the Squares of the variable c
Σd^2 Sum of the Squares of the variable d

The coefficient of determination = (correlation coefficient)^2 = R^2

Firstly, we imported the data set and divided it into train and test sets with a ratio of 80:20.As the dataset appeared to be complete and free of null value no or little data preprocessing was performed. The plots for the same are illustrated in the section that follows.

Secondly, we have done predictions [34, 35] using Linear, Poisson and Lasso Regressions and the results have been graphically represented in the below section.

5 Results and Discussions

The linear regression model has given a root mean squared error of 0.59 and 0.56 for target variables Close and Low and R^2 score of 0.99 for both target variables.

Using Poisson regression, the RMSE came out to be 3.8 and 4 and R^2 score of −7.24 and −8.12 for both the target variables but Poisson distribution cannot be used as our data is clearly linear.

Finally Lasso regression's RMSE was 2.76 and 2.53 for both the target variables and R^2 score 0.99 in both the cases.

S. no	Model	Target	MAE	MSE	RMSE	R2 score
1	Linear regression	Close	0.32	0.35	0.59	0.99
		Low	0.30	0.31	0.56	0.99
2	Poisson regression	Close	6.7	1.51	3.8	−7.24
		Low	7	1.64	4	−8.12
3	LASSO regression	Close	2.10	7.65	2.76	0.99
		Low	2.02	6.43	2.53	0.99

Figures 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23 and 24 depicts performance of all 3 regression models across both the target variables on the training set.

Fig. 7 Linear regression analysis between open and close

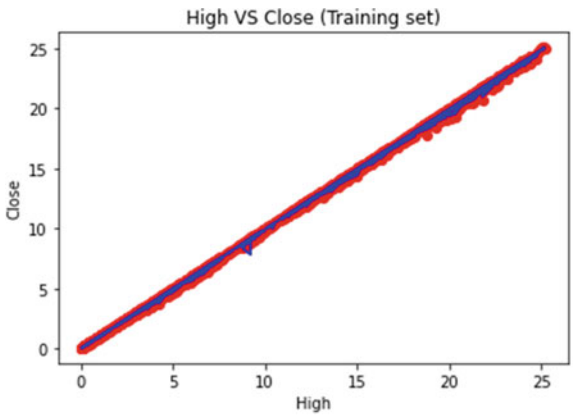


Fig. 8 Linear regression analysis between high and close

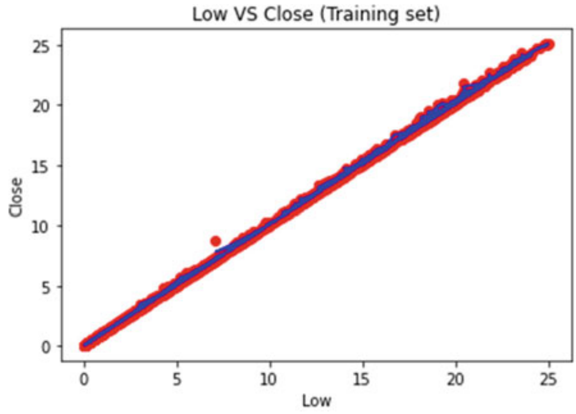


Fig. 9 Linear regression analysis between low and close

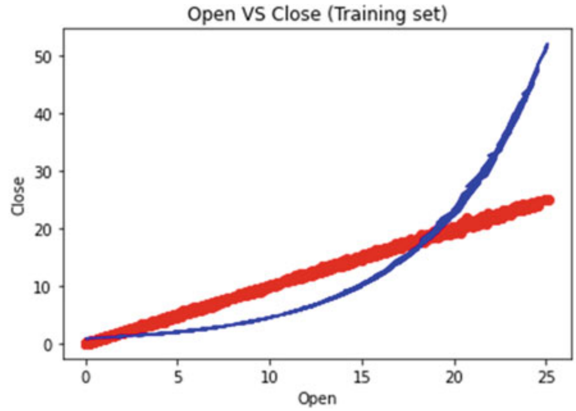


Fig. 10 Poisson regression analysis between open and close

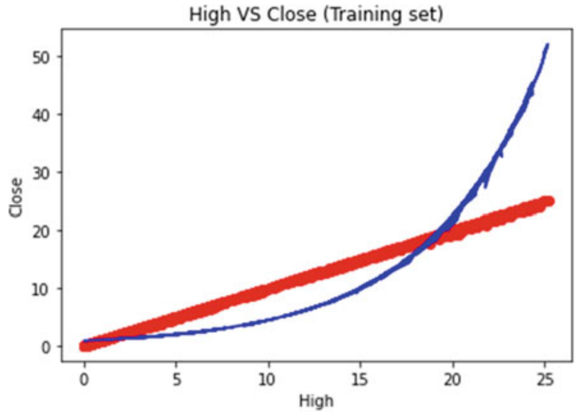


Fig. 11 Poisson regression analysis between high and close

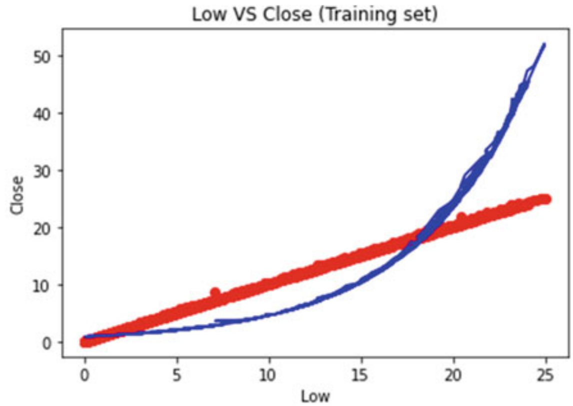


Fig. 12 Poisson regression analysis between low and close

Fig. 13 Lasso regression analysis between open and close

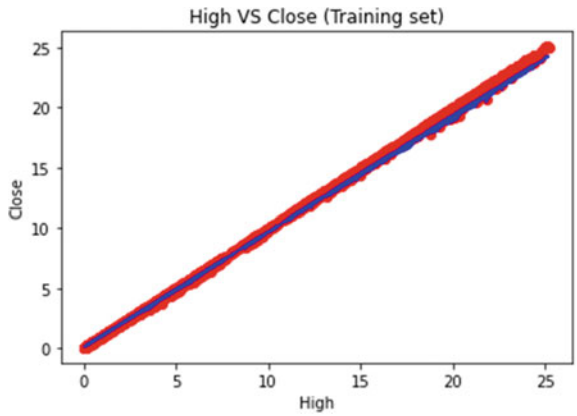


Fig. 14 Lasso regression analysis between high and close

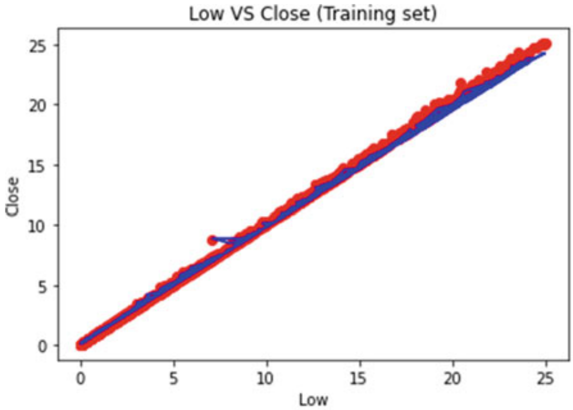


Fig. 15 Lasso regression analysis between low and close

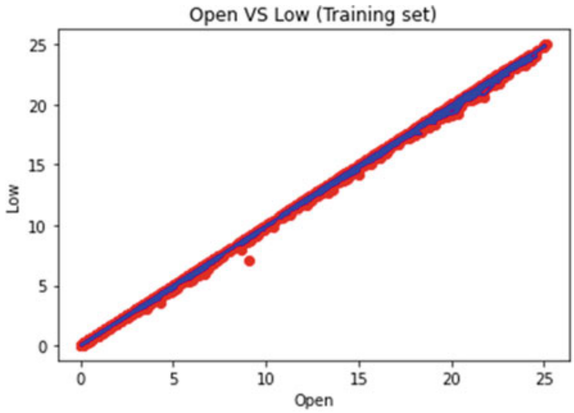


Fig. 16 Linear regression analysis between open and low

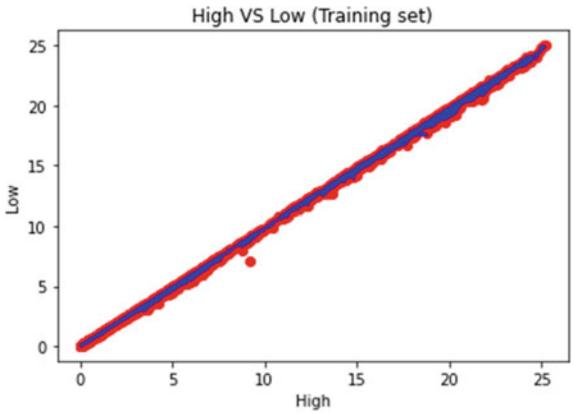


Fig. 17 Linear regression analysis between high and low

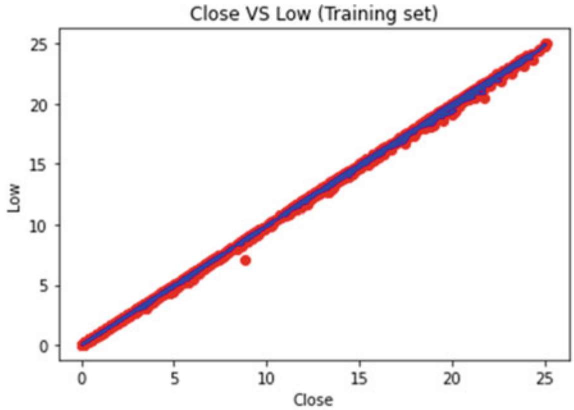


Fig. 18 Linear regression analysis between close and low

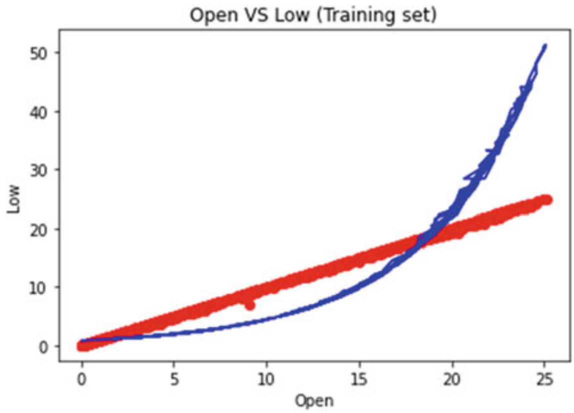


Fig. 19 Poisson regression analysis between open and low

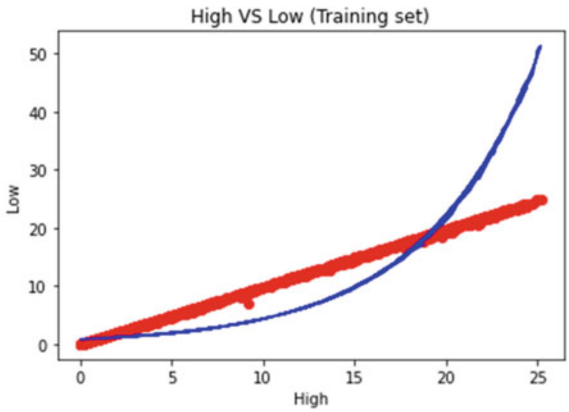


Fig. 20 Poisson regression analysis between high and low

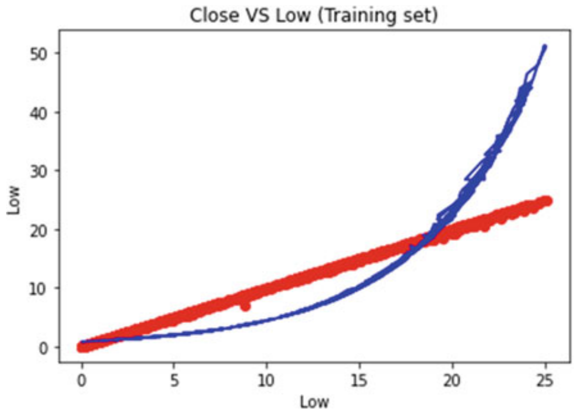


Fig. 21 Poisson regression analysis between close and low

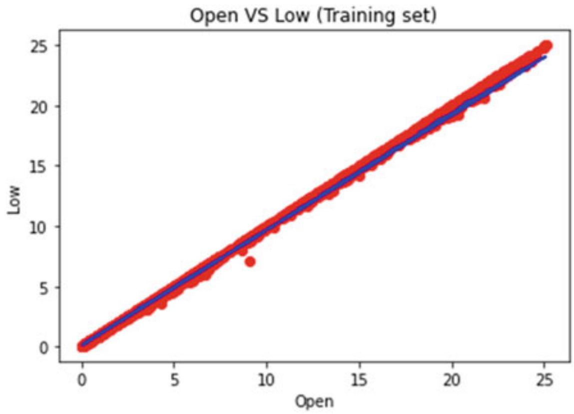


Fig. 22 Lasso regression analysis between open and low

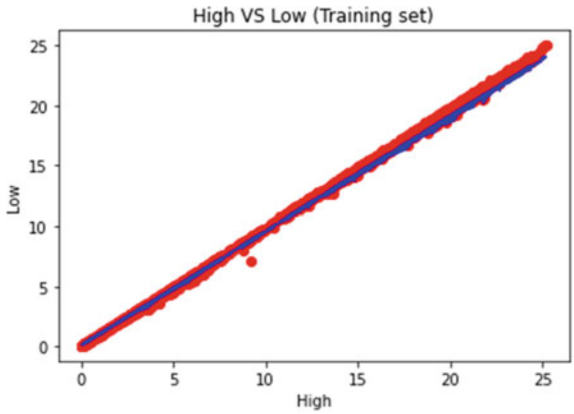
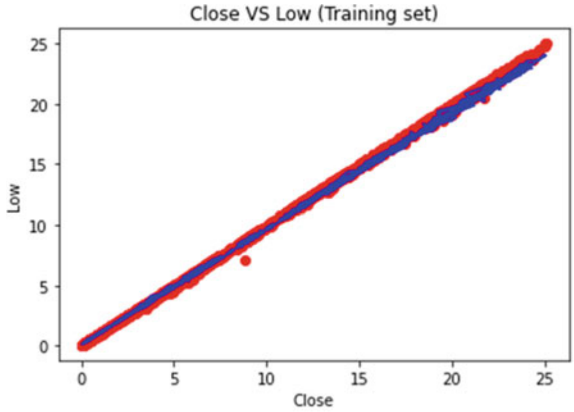


Fig. 23 Lasso regression analysis between high and low

Fig. 24 Lasso regression analysis between close and low



6 Future Scope and Limitation

The extended scope of this project will require the addition of more criteria and aspects, such as the financial ratios and various instances, amongst other things. The greater the number of characteristics that are considered, the higher the level of accuracy. The algorithms can also be used to analyse the content of public comments and, as a result, identify trends and links between customers and the employees of businesses. The application of conventional algorithms and data mining methods is another method that can assist with performance structure forecasting for the entire organisation. In the not too distant future, one of our goals is to combine neural network analysis with additional methodologies like genetic algorithm or fuzzy logic. It is possible to employ a genetic algorithm to determine the most effective network architecture and training settings. The predictions made by neural networks are accompanied with an element of uncertainty, which can be taken into consideration with fuzzy logic's help. The application of their applications in conjunction with neural networks [34] may result in an improvement to the prediction of the stock market. Because of the constraints of this research, our choice of statistical methods is limited to either linear regression or multiple linear regression. Due to the fact that the data is linear, it is not possible to utilise other regression methods such as Poisson regression or logistic regression.

7 Conclusion

On the basis of the aforementioned findings, we can conclude that the linear regression model is the most effective. Because these data points are linearly separable, it is impossible to apply the Poisson distribution to them and attempting to do so resulted in a poorly performing model. In addition, linear regression produced encouraging results, with the error being as minimal as 0.3. Therefore, we can conclude that linear regression is effective because it accurately predicts the outcomes.

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Water Quality Assessment Through Predictive Machine Learning

Hritwik Ghosh, Mahatir Ahmed Tusher, Irfan Sadiq Rahat, Syed Khasim, and Sachi Nandan Mohanty

Abstract Water quality monitoring is crucial for safeguarding the environment and human health. With the advent of artificial intelligence, water quality classification and prediction have seen significant improvement. This study aims to develop a reliable approach for forecasting water quality and distinguishing between potable and non-potable water by employing several ML models. G-Naive Bayes, B-Naive Bayes, SVM, KNN, X Gradient Boosting, Random Forest constitute the classifiers and stacking ensemble models under scrutiny. The historical dataset used in this study consists of 3277 samples collected over 9 years from various locations in Andhra Pradesh, India, and was obtained from the Andhra Pradesh Pollution Control Board (APPCB).This study's findings indicate that machine learning models can provide an effective approach to water. Water Quality Assessment Through Predictive ML. The dataset comprises pH, dissolved oxygen, BOD, and TDS. The Random Forest model acquired the greatest accurateness of 78.96% among the employed models, whereas the SVM model displayed the lowest accurateness of 68.29%. quality assessment and classification. The use of precision-recall curves enables the selection of the best model for water quality prediction, which is essential in ensuring the availability of potable water for human consumption.

Keywords Water quality monitoring · Artificial intelligence · ML models · Historical dataset · Water quality parameters · Precision-recall curves

H. Ghosh (✉) · M. A. Tusher · I. S. Rahat · S. Khasim · S. N. Mohanty
School of Computer Science and Engineering (SCOPE), VIT-AP University, Amaravati, Andhra Pradesh, India
e-mail: hritwik.21bce8973@vitapstudent.ac.in

M. A. Tusher
e-mail: ahmed.21bce8971@vitapstudent.ac.in

I. S. Rahat
e-mail: rahat.21bce9985@vitapstudent.ac.in

S. Khasim
e-mail: profkhasim@gmail.com

S. N. Mohanty
e-mail: sachinandan09@gmail.com

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1 Introduction

Having access to safe and easily obtainable water is essential for various human undertakings, such as drinking and household purposes, crop cultivation, and leisure activities. Without reliable access to clean water, human health can be severely impacted, and economic growth can be hindered. The availability of safe and clean drinking water is critical for human health and well-being. However, despite having vast reserves of water, many regions of the world still struggle with water scarcity and pollution, leading to a lack of drinkable water. In India, for example, the majority of surface water is not fit for consumption, affecting the health and livelihoods of millions. Addressing these water quality challenges is essential for poverty reduction and economic growth, as better management of water resources and improved sanitation can have a significant impact on a country's development. Waterborne diseases such as cholera, dysentery, and hepatitis A, are directly linked to poor water quality and inadequate sanitation. However, the traditional method of manually collecting water samples and performing lab analysis is often inefficient, time-consuming, and costly. To address these challenges, the use of intelligent systems for real-time water quality monitoring is increasing rapidly. Machine learning falls under the umbrella of artificial intelligence, and it imparts the ability to a system to learn and enhance itself without requiring any manual intervention through experiences. The procedures that are used in machine learning are trained to capture trends and pursuant to update themselves. In water studies, Machine learning paves the way to assess, classify and predict water quality indicators. For instance, we can successfully simulate hydro-logical processes subject to the accessibility of bountiful sets data. However, using this water potability dataset of Vijayawada, Andhra Pradesh [1], and applying machine learning to it, we are going to distinguish between Potable and Non-Potable water using some parameters such as pH value, chloramines, sulfate, conductivity, organic carbon, hardness, solids, conductivity, Trihalomethanes, turbidity, potability.

2 Literary Survey

Over the past few years, machine learning techniques have garnered significant interest in diverse domains, particularly water quality assessment. This research paper employs multiple machine learning classifiers to evaluate water potability in a specific region, including some ml algorithms. This literature review delves into some pertinent works on machine learning models applied to water quality assessment.

Ahmadi et al. (2020) were among the pioneers who employed machine learning models for water quality assessment. In their study, they utilized Support Vector Machine (SVM) and Random Forest algorithms to predict water quality in an Iranian river, with SVM demonstrating superior accuracy [2]. Gaurav et al. (2016) conducted a similar study in an Indian river, using Decision Tree, SVM, and Random Forest algorithms, where SVM yielded the highest accuracy [3].

In more recent studies, Yu et al. (2020) utilized KNN, Random Forest, and Gradient Boosting algorithms to predict algae concentration in a Chinese lake, with Gradient Boosting providing the highest accuracy [4]. Multiple other studies employed various machine learning models to assess water quality in different rivers across Iran, India, and China, each identifying a specific model with the highest predictive accuracy [1, 5–12].

Apart from water quality assessment, machine learning models have been used to predict the concentration of pollutants in Indian rivers, where SVM demonstrated the highest accuracy [12]. Other studies examined the prediction of heavy metals concentration in Iranian rivers, with Random Forest emerging as the most accurate model [13]. Some research has also been conducted on chronic diseases, with Deep Neural Network classifiers predicting Chronic Kidney Disease with high accuracy [14, 15].

Several other studies applied machine learning models to predict water quality in rivers from India, Pakistan, and China, with various models exhibiting the highest accuracy, including Naive Bayes and Random Forest [16–19]. In addition, machine learning models have been used to speculate the concentration of nitrate in groundwater and heavy metals in lakes in China [18, 19].

In summary, machine learning models have gained substantial traction in water quality assessment. Research indicates that algorithms can be employed to predict water quality and pollutant concentrations with high accurateness. These models have been successfully implemented across various countries and rivers, offering potential improvements in water quality management and public health protection [20].

3 Proposed Methodology

We have followed a systematic approach in our study to ensure the accuracy and reliability of our results. Before applying machine learning models, we prepared the input data by cleaning the dataset and dividing it into training and testing sets. The dataset was cleaned by removing inaccurate values and replacing missing data with the median of the input variables. We utilized multiple algorithms to evaluate the performance of our machine learning models. To visualize the proposed system, we have presented a framework diagram (Fig. 1) that outlines the key components and their relationships in our approach.

3.1 Data Description

Our dataset includes data for 3277 different water bodies, each with 10 water quality metrics such as pH value, chloramines, sulfate, conductivity, organic carbon, hardness, solids, conductivity, Trihalomethanes, turbidity, and potability. We began by

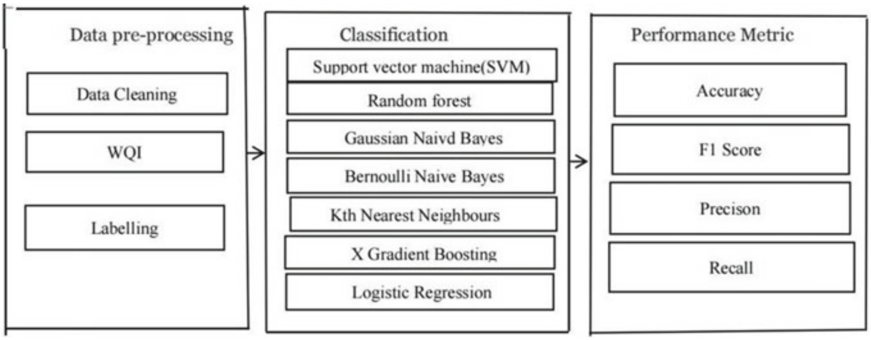

Fig. 1 Methodology of the proposed system

importing the necessary libraries for machine learning model training and data visualization. Then, we used Pandas' read_csv() function to load the dataset and performed exploratory data analysis. This involved checking the shape of the dataset, handling null values, and checking the value counts of the target feature Potability. We were able to visualize the potability using a countplot function of Seaborn (Fig. 2) and display the entire dataset using the hist method (Fig. 3). Although we explored the correlation between all the features using a heatmap function of Seaborn, we found no relation (Fig. 4), leading us to conclude that the dimensionality of the dataset cannot be reduced.

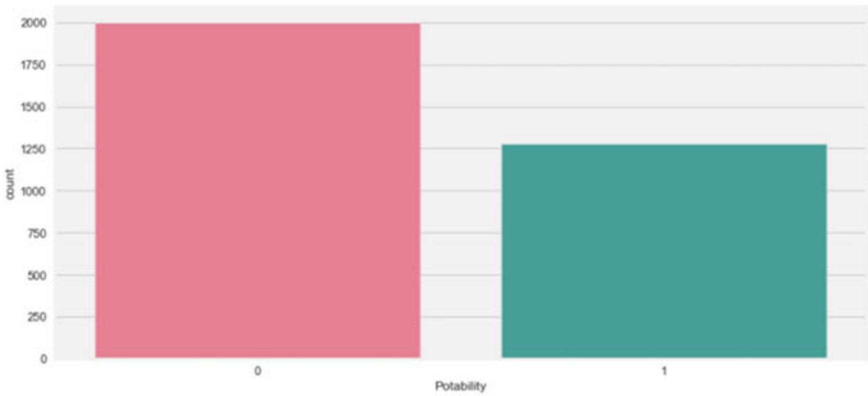

Fig. 2 Visualization of the potability

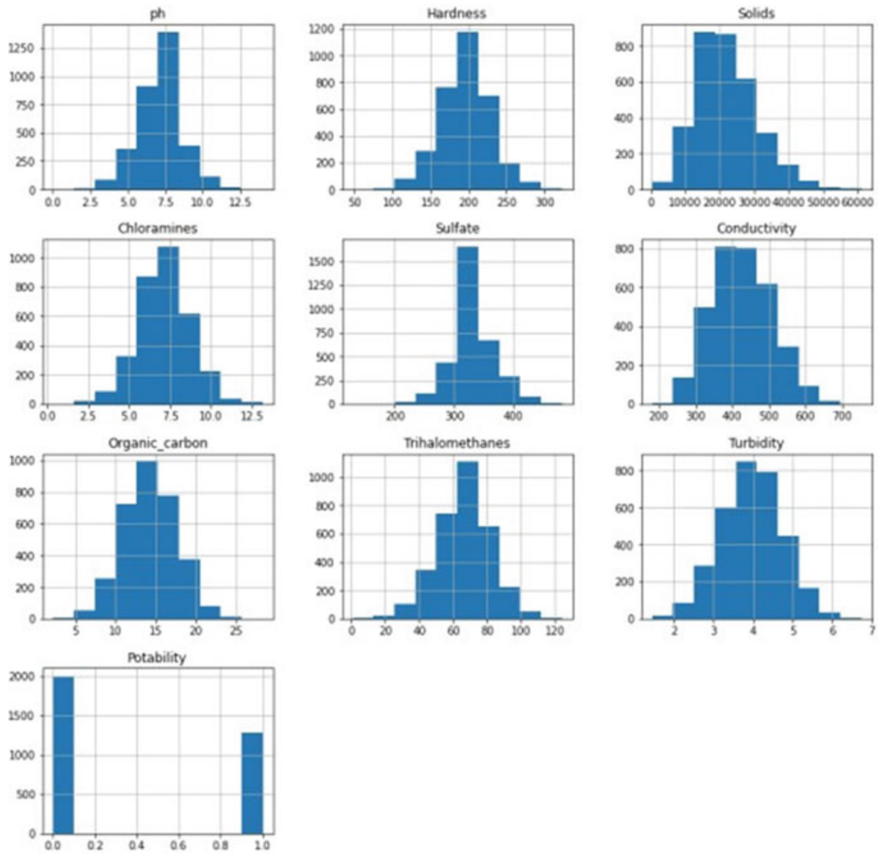

Fig. 3 Visualization of the entire dataset using hist method

3.2 *Data Preparation*

In data preparation, we have divided the data into two categories, dependent and independent features. Except potability, all our features are independent. The train test split function was then used to divide the dataset into training and testing sets. Subsequently, we utilized the training dataset (X_train, Y_train) to establish the decision tree classifier model and train it. Using the test data set (X_test), we tested the model. Finally using the accuracy score, we have evaluated the model, confusion matrix and classification report. The techniques of evaluation take two parameters; the first one is the predicted data and the second one is the actual data. Here we are demonstrating a graph of models vs accuracy (Fig. 5).

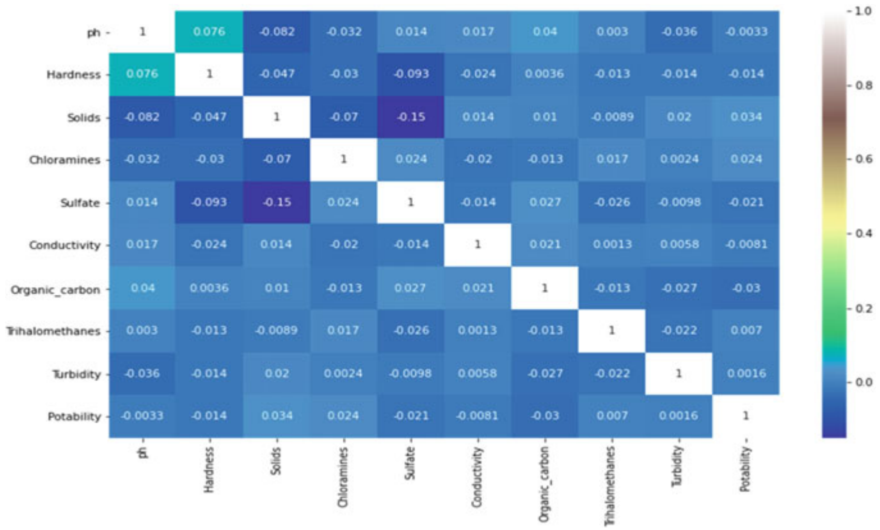

Fig. 4 Correlation heatmap

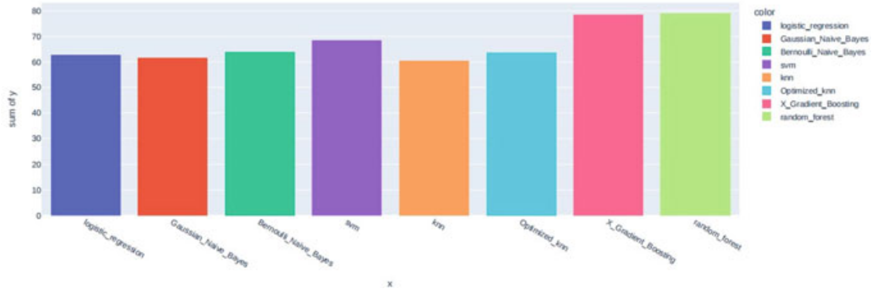

Fig. 5 Models versus accuracy

4 Experimental Analysis

Using the boxplot function, we can find outliers where these are contained by the solid feature (Fig. 6). But these outliers cannot be removed. Point to be noted, the water will be safe to drink if the outliers are removed from the solid feature. Basically these outliers in solid features make the water impure. Presence of high amount of solid particles make the water unsafe to drink. But we cannot remove this outlier to train the model.

We will evaluate the performance of our ml models including Logistic Regression, G-Naive Bayes, Support Vector Machine, KNN, XGB etc. using performance metrics like accuracy, precision, recall, and f1 score. The outcomes of our analysis will be showcased in (Tables 1, 2, 3, 4, 5, 6, 7 and 8).

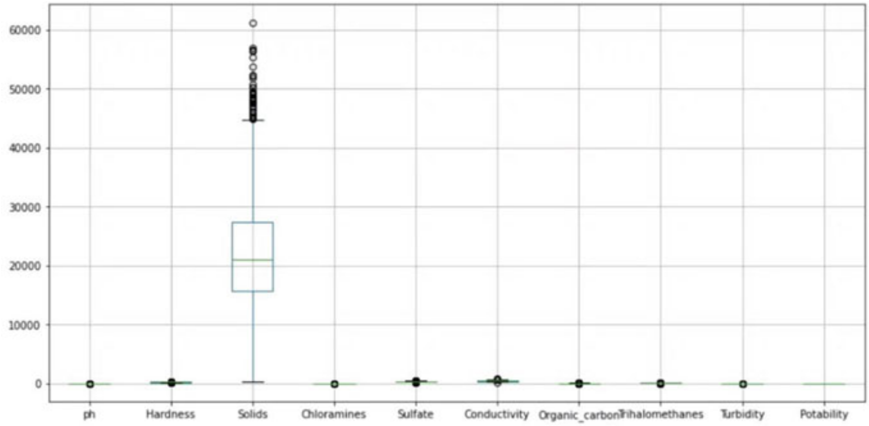

Fig. 6 Outliers in the solid feature

Table 1 Performances of our used ML models

Model name	Value of accuracy	Value of precision	Value of recall	Value of F1_Score
LR	0.63	0.67	1.00	0.80
GNB	0.61	0.65	0.85	0.74
BNB	0.64	0.68	0.81	0.73
SVM	0.68	0.68	0.93	0.89
KNN	0.63	0.68	0.69	0.73
XGB	0.78	0.81	0.86	0.83
RF	0.79	0.81	0.88	0.84

Table 2 Performance metrics of LR

Value of precision		Value of recall	Value of F-1 score	Value of support
0	0.67	1.00	0.80	412
1	0.00	0.00	0.00	246
Macro avg	0.31	0.50	0.39	656
Weighted avg	0.39	0.63	0.48	656

Table 3 Performance metrics of Gaussian Naive Bayes

Value of precision		Value of recall	Value of F1 score	Value of support
0	0.56	0.85	0.74	412
1	0.46	0.22	0.30	246
Accuracy			0.62	656
Macro avg	0.56	0.53	0.52	656
Weighted avg	0.58	0.62	0.57	656

Table 4 Performance metrics of Bernoulli Naive Bayes

Value of precision		Value of recall	Value of F1 score	Value of support
0	0.68	0.80	0.73	412
1	0.52	0.37	0.43	244
Accuracy			0.64	656
Macro avg	0.60	0.58	0.58	656
Weighted avg	0.62	0.64	0.62	656

Table 5 Performance metrics of SVM

Value of precision		Value of recall	Value of F-1 score	Value of support
0	0.68	0.93	0.79	412
1	0.69	0.27	0.38	244
Accuracy			0.68	656
Macro avg	0.69	0.60	0.59	656
Weight avg	0.69	0.68	0.64	656

Table 6 Performance metrics of KNN

Value of precision		Value of recall	Value of F-1 score	Value of support
0	0.68	0.71	0.69	412
1	0.47	0.43	0.45	244
Accuracy			0.61	656
Macro avg	0.57	0.57	0.57	656
Weighted avg	0.60	0.61	0.60	656

Table 7 Performance metrics of X Gradient Boosting

Value of precision		Value of recall	Value of F-1 score	Value of support
0	0.81	0.86	0.83	412
1	0.74	0.65	0.69	244
Macro avg	0.77	0.76	0.76	656
Weighted avg	0.78	0.79	0.78	656

Table 8 Performance metrics of random forest

Value of precision		Value of recall	Value of F-1 score	Value of support
0	0.80	0.88	0.84	412
1	0.76	0.64	0.69	244
Accuracy			0.79	656
Macro avg	0.78	0.76	0.77	656
Weighted avg	0.79	0.79	0.79	656

4.1 Logistic Regression(LR)

We use this regression model to predict or calculate the probability of a binary (yes/no) event occurring. Its function is a simple S-shaped curve that is generally used to convert data in binary expression (0 or 1).

$$h\Theta(x) = 1/1 + e - (\beta o + \beta 1X) \tag{1}$$

Here, in our study, 0 represents “non-potability”and 1 represents “potability”. We are going to demonstrate its performance metrics.

4.2 Gaussian Naive Bayes

Gradient boost machines (GBM) are a promising algorithm for water quality assessment through predictive machine learning. GBM is one of the most effective and mention-worthy algorithms of supervised machine learning, and X Gradient Boost is one of its implementations. In addition to solving regression and classification problems, GBM can be utilized to capture complex relationships between water quality parameters and identify potential sources of contamination. By processing large datasets and identifying patterns and trends, GBM can also help predict water quality issues before they become significant problems.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \tag{2}$$

4.3 Bernoulli Naive Bayes

Bernoulli Naive Bayes is classified as a supervised ml algorithm. Being connected with numerous attributes it can predict the probability of different classes. It indicates the likelihood of occurrence of an event. We also know it as conditional probability. Bernoulli Naive Bayes is an extension of naive Bayes.

4.4 SVM

Support Vector Machines (SVM) is an efficient algorithm used in supervised learning for classification and regression analysis that can be successfully utilized in water quality assessment through predictive machine learning. By building a decision boundary or hyperplane in an n-dimensional space, the support vector machine (SVM) divides the data points into several classes with the maximum margin., resulting in improved accuracy when predicting future data points and reducing the risk of overfitting.

4.5 KNN

KNN is a commonly utilized algorithm for predictive ml in the field of water quality assessment. It is recognized for its effectiveness and simplicity. The algorithm classifies new data points by their proximity to existing data points, with the value of "K" indicating the nearest neighbors employed for classification. This algorithm has been widely used in water quality studies to classify water samples based on various parameters such as pH, turbidity, and conductivity.

4.5.1 X Gradient Boosting

Gradient boost machines (GBM) are a promising algorithm for water quality assessment through predictive machine learning. GBM is one of the most effective and mention-worthy algorithms of supervised machine learning, and X Gradient Boost is one of its implementations. In addition to solving regression and classification problems, GBM can be utilized to capture complex relationships between water quality parameters and identify potential sources of contamination. By processing large datasets and identifying patterns and trends, GBM can also help predict water quality issues before they become significant problems.

4.6 Random Forest

The Random Forest algorithm is a frequently utilized technique in supervised ml remaining its capability to handle vast datasets and deliver outstanding precision. In the context of water quality assessment through predictive machine learning, Random Forest can be used to classify and predict water quality based on various input parameters. By constructing multiple decision trees and combining their outputs, Random Forest can provide accurate predictions while minimizing the risk of overfitting.

5 Conclusion

The Bureau of Indian Standards (BIS) declared the upper limit of total dissolved solids (TDS) levels in water is 500 ppm. For having values of TDS on an average of 40 times as much as the upper limit 0 the safe drinking water, the solid level seem to contain some descripancy. Equal number of basic and acidic pH level water samples are contained in our data. Very less correlation coefficients have been noticed between the features. Among our used models, Random Forest provided the highest accuracy with a percentage of 78.96, where SVM provided us the least amount of accuracy with a percentage of 68.29. But in case of training the model, XGBoost and Random forest performed the best. Both of the models gave us f1 score (Balanced with recall and precision) around 76%. Our investigation revealed that the utilization of machine learning algorithms can be an effective approach for water quality assessment. Various algorithms such as KNN, Random Forest, SVM, XGBoost, and Naive Bayes were able to accurately differentiate between potable and non-potable water samples based on different parameters. However, among these models, Random Forest had the highest accuracy in classifying the water samples. It is concerning to note that our dataset showed a significant number of water samples with TDS levels above the safe limit set by the BIS. This highlights the need for better management of water resources and improved water supply and sanitation in order to ensure safe and potable water for all.

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Comprehensive Review of Lie Detection in Subject Based Deceit Identification

Tanmayi Nagale and Anand Khandare

Abstract With the increase in crime, the issue of deception identification has become more significant. The main task at hand is to separate the innocent and culpable groups in the EEG data for lie detection. To categorise EEG data, various techniques have been created; deep belief networks are infrequently used. In order to extract the time and frequency domain characteristics of the data, this study employs a deep learning method that combines a constrained Boltzmann machine with a wavelet. Four RBMs stacked together lead to the formation of a dense belief network. Considering legal and security aspects, analysing deceit behaviour of humans is a major issue. For classification of EEG data, deep belief networks are rarely used. Our work will propose the techniques to perform analysis on EFG data. First, apply a pre-processing technique to utilize only a small fragment of the EEG image instead of the whole image. The model describes a temporal feature map of the EEG signals measured during the lie detection test. It will improve the performance by providing smaller size input data. Different classifiers are used for identifying deception. On CIT-based EEG data, an unstructured deep belief network method has been used. Four constrained Boltzmann machines are stacked and softmax regression is applied at the output layer to create a DBN. Time frequency components that are derived from EEG data are used for learning because they contain more information than unprocessed EEG data. EEG data was captured through the use of a deception detection test.

Keywords Deceit deception · Subject based identification · Lie detection · EEG image, deep learning, machine learning

T. Nagale (✉) · A. Khandare
Thakur College of Engineering and Technology, Kandivali (E), India
e-mail: tanmayinagale13@gmail.com

A. Khandare
e-mail: anand.khandare@thakureducation.org

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1 Introduction

The problem of deceit detection has grown in importance with the rise in crime. The primary job at hand is to divide the EEG data for lie detection into innocent and guilty categories. Deep belief networks are rarely used, and numerous techniques have been created previously to categorise EEG data. In order to acquire the time and frequency domain details of signals, this study employs a deep learning method that combines a restricted Boltzmann machine with a wavelet. Stacking four RBMs results in the development of a deep belief network. At the output layer, softmax regression is used to categorise EEG data into being culpable and harmless. By carrying out a "Concealed Information Test," an experiment has been carried out on EEG data that was captured. Concealed Information Test is used for a series of questions to determine whether or not a subject is telling the truth. Both pertinent and irrelevant inquiries about crime are included in these queries. The inspected person's response is then noted. In earlier polygraph tests, the subject's bodily characteristics, such as heart rate, blood pressure, perspiration, etc., were recorded. The same is true because masculine motives can govern extrinsic behaviours [1]. New techniques have therefore been created to read brain signals and assess the conduct of guilty and innocent persons. The non-invasive method of electroencephalography (EEG) is used to collect brain signals for later analysis.

Event Related Potential (ERP) is created. When a subject respond to the stimuli, such as when lying, he or she must focus on the questions, which causes ERP. The P300 wave is the ERP that is produced when a subject is exposed to a meaningful or relevant stimulus. This wave is produced 300 ms after the stimuli start. The individual is exposed to three different types of stimuli such as irrelevant, probe, and target. Target is the stimulus that is offered to both innocent and guilty subjects and is primarily concerned with keeping the subjects' attention. Test stimuli [2]. The individual is exposed to three different types of stimuli such as irrelevant, probe, and target. Target is the stimulus that is offered to both innocent and guilty subjects and is primarily concerned with keeping the subjects' attention. Only guilty subjects respond to the probe's crime-related stimuli, which causes a P300 response; the probe's irrelevant stimuli do not cause a P300 response. There is enhancement of the research method in the interface between humans and robots. Bootstrapping method is used for comparing how similarly different subjects' brains respond to the same stimuli [3].

The untruth seriously undermines the deceitful actions of numerous victims of fraud. The general populace frequently tells lies. Some limbs that automatically exhibit a distinct reaction when someone is lying can be used to uncover a lie.

2 Related Work

A. Bablani, D. Reddy, D. Tripathi, Venkatanareshbabu K have proposed Empirical Mode Decomposition such as EMD. Since polygraph tests for deception detection are regulated by humans, EEG-based lie detectors are gaining popularity these days. These devices examine any subject's covert behaviour using ERP EEG components. Empirical Mode Decomposition such as EMD, which provides information about a signal's temporal and frequency domains, has been used in this paper to extract EEG features. The EEG data has also been subjected to a variety of classifiers to determine if the patient is guilty or innocent. On subject-specific EEG data, classifiers like SVM, QDA, KNN, and decision trees are utilised. With a variety of subjects, an innovative series of experiments is run to determine if the participants are telling the truth or lying. As part of an experiment, individuals are shown particular visuals as stimuli, and their reactions are recorded and subsequently examined. SVM outperformed other methods on the recorded EEG dataset for the majority of the participants [1]. Andrew Chadwick and James Stanyer have proposed a method for enhancing the study of information, disinformation and misperceptions. Research that acknowledges the existence of erroneous and misleading information but stops short of describing their influence will benefit from an emphasis on deception. Additionally, by concentrating on how actors' deceptive tactics are crucial in their attempts to wield authority, it helps advance research on the cognitive and attitudinal biases that make people prone to error in perception. We outline the key concepts in the study of deception: media-systemic informational distortions; relational interactions that both create and activate cognitive biases; and the characteristics, tactics, and strategies of deceitful entities. The summary typology of the 10 primary factors and their 57 focal indicators serves as their conclusion [2]. Abeer A. and H. Mahmoud have proposed a model that describes an analysis of the temporal features of the EEG data obtained during the lie detector test. During the learning phase, a deep learning attention model (V-TAM) retrieves the temporal map vector. Through the use of this method, overfitting in Deep Learning architectures is lessened and computation time is decreased. We provide a deep learning convolutional neural network-based model of cascading attention (CNN). Local features and global features are extracted by the V-TAM model via distinct spatial and temporal routes. In addition, a novel Visual-Temporal Attention Model (V-TAM) is suggested to improve the EEG segmentation accuracy [3]. Annushree Bablani, Damodar Reddy Edla, Venkatanareshbabu K, Shubham dedia have proposed novel Deceit Identification Test based on the Electroencephalogram signals to identify and analyze the human behaviour. This test is done with the help of P300 signals. It helps to get results in term of good classification accuracy using P300 signals. Using "symlet" Wavelet Packet Transform (WPT) technique, features can be extracted. EFG signals are recorded and analysed using BrainVision recorder and analyser. At the end they got 95% accuracy [4]. Jennifer Tehan Stanley, Britney A, Webster have to write a review paper on deceit deception. They compared the efficacy of two training techniques to help older persons become more adept at spotting deception: valid face clues to deception and valid verbal cues to deception. 150 senior citizens were randomly assigned to a control condition, verbal training, or facial training. The participants finished their allocated training, a pre-test deceit detection task, and a post-test deceit detection task. Following training, both training groups greatly improved at identifying their respective trained cues. The control group showed increased deceit detection accuracy from pre-test to post-test while the facial cue training group showed decreased deceit detection accuracy post-test compared to pre-test. These findings are in line with the body of deception research that indicates people still operate at about chance accuracy even after training. Training can enhance older persons' ability to recognise facial and verbal cues, but these enhancements did not lead to more accurate deception detection and even hindered performance in the facial condition. Perhaps because this condition encouraged implicit rather than explicit assessments of dishonesty, older persons demonstrated the greatest advantage from simple practise at detecting deception (in the control condition) [5].

D. Barsever and E. Neftci have proposed BERT's classification to develop examples of misleading and true writing that further illustrated the distinctions between the two, highlighting the fundamental similarities of deceptive text in terms of the part-of-speech composition. Knowing how to spot lies or misleading statements in text is an important ability. This is partially due to poor knowledge surrounding the patterns underlying deceptive writing. The purpose of this research is to find patterns that describe misleading text. Training a classifier using the Bidirectional Encoder Representations from Transformers i.e. BERT network is an important step in this method. On the Ott Deceptive Opinion Spam corpus, BERT outperforms the state-of-the-art in terms of deception classification accuracy [6]. Abootalebi, Moradi, and Khalilzadeh have implemented the approach for lie detection based on EEG feature extraction in p300. It has been proposed to use P300-based Guilty Knowledge Test as an alternative to traditional polygraphy. The goal of this study was to improve a pattern recognition technique that has already been used for this application's ERP assessment. This enhancement was accomplished by both expanding the feature set further and using a mechanism for choosing the best features. Several volunteers completed the planned GKT paradigm, and their corresponding brain signals were captured as part of the method's evaluation. The P300 detection method was then put into practise using a few characteristics and a statistical classifier. Some morphological, frequency, and wavelet properties, and with the help of genetic algorithm, the best feature set was chosen and utilised to classify the data. 86% of subjects in terms of guilty and innocent were correctly identified, which was a higher rate than other previously employed techniques [7]. Arasteh, Moradi, and Janghorbani have proposed a novel method on empirical mode decomposition for P300-Based detection of deception. They are using alternatives for conventional polygraphy such P300-based guilty knowledge test. EMD technique is used to extract features from the EFG signal. Features are extracted from Pz channel for synergistic incorporation of channel information. Using the genetic algorithm features are selected and this is challenging for input space dimension increase. Using this approach, the accuracy of guilty and innocent subjects was 92.73% [8]. Annushree B., D. Reddy, V. K. have proposed restricted Boltzmann machine with the help of a deep learning approach to obtain the time and frequency domain information of signals. Stacking four RBMs results in the development of a deep belief network. At the output layer, softmax regression is used to categorise EEG data into guilty and innocent. By carrying out a "Concealed Information Test," an experiment is performed on EEG data that was captured. In the test, subjects are shown some important and irrelevant photographs on a simulated crime scene. After flashing these visuals, the EEG signals produced are captured and examined [9]. J. Gao, L. Gu, X. Min, Pan Lin have proposed a method for analysing the underlying the cognitive process and mechanism in deception. They are using the guilty knowledge test protocol for categorising the groups in term of guilty and innocent group of people. They have taken randomly 30 subjects for identifying the subjects based on functional connectivity patterns. The authors have proposed FTRPS technique to avoid false synchronization. The results show that four intuitive brain fingerprinting graphs (BFG) on delta, theta, alpha and beta bands were generated. Deception behaviour is identified with FTRPS technique in real application [10].

R. Hari N.; M. Nasrun; Casi have proposed a method for lie detection. Every person is unique in their characteristics and habits. There are many persons in our world who frequently tell lies. The falsehood must create comprehension in other people, but the understanding it creates is incorrect. Pupils can be a good indicator of whether a person is lying or not since, in accordance with psychological science, dilated pupils are a sign of depression, which includes lying. The frequency of eye blinks might also be a clue as to whether or not someone is lying. Circular Hough Transform method to observe changes in the dilated pupil and the frame difference approach to count the number of blinks, accuracy is reached. According to the study, a person who lies will have their pupils dilated by 4% to 8% of their initial diameter and will blink their eyes more frequently-up to 8 times—than they did before the question was posed. The lie detector system's accuracy in this study is 84% [11]. Sinead v. Fernandes and Muhammad s. Ullah have proposed Levenberg–Marquardt classification method and the long short-term memory classification method. Based on the nine different training and testing combinations from the three separate sessions and their derived cestrum and spectral energy features, it is necessary to assess the efficacy of deception detection. To lower the dimensionality of the retrieved features for further optimization, the principal component analysis is used. The four types of characteristics' projected main components exhibited increased precision in their ability to discern between honest and dishonest speech patterns [12]. Hannah Shaw & Minna Lyons have proposed a method for accuracy of deception detection that has been the subject of conflicting results in the literature, and the majority of the experiments have relied on small-scale lying. It's not yet apparent whether being aware of these indicators actually helps people detect lies, despite recent studies suggesting that high-stakes scenarios can provide reliable cues to dishonesty. Prior to seeing the film, participants were randomly assigned to either the cue condition (where they were shown previously recognised cues to deceit) or the no cue condition (where they were told to rely only on their gut feelings). Participants were asked if they were familiar with the case, whether they believed the appealer was telling the truth or lying, and how sure they were in their judgement. Participants recorded qualitative comments on the cues they employed during lie detection at the conclusion of the experiment. There was a positive correlation between accuracy and age, accuracy scores were not substantially predicted by cue knowledge or confidence. The ability to spot deception was dramatically improved among those who used emotion-based clues. The findings are examined in light of the existence of trustworthy cues [13]. Y. Xie, R. Liang, H. Tao, Y. Zhu, and L. Zhao have designed convolutional bidirectional long short-term memory for deception detection. To maintain the temporal information in the original voice, the method extracts frame-level acoustic features whose dimension dynamically vary with the length of the speech. To learn the context dependences in speech, bidirectional LSTM was performed to match temporal features with changing dimension. Additionally, in the conventional LSTM, which is used to extract time–frequency mixed data, the convolution operation takes the place of multiplication. The experiment's average accuracy on the Columbia-SRI-Colorado corpus is 70.3%, which is higher than it was in earlier works using non-contacting modalities [14]. Z. Labibah, M. Nasrun, and C. Setianingsih have proposed the lie detection method. The eyes are used to identify lies in this final task, specifically eye tracking and changes in pupil size using the Wavelet Transform to Gabor Image Processing technique, followed by classification to assess whether or not someone is lying using a Decision Tree. It is anticipated that having access to this lie detector will be beneficial for those who need to spot lying. The last test's outcomes are precise. The precision, recall, and accuracy values for this study are 97%, 94%, and 95% respectively [15].

Jun-Teng Yang, Guei-Ming Liu, Scott C.-H. Huang have proposed several data preparation techniques for contaminated data and useless frames in the ground truth data. Visual and aural modality information can be extracted using a multimodal deception detection system. To analyse deception detection tasks, they present a novel emotional state-based feature called the EST feature. The change in emotional state in frames and audio segments is represented spatially and temporally by the EST feature. The video's deception detection accuracy and ROC-AUC are increased to 92.78% and 0.9265, respectively [16]. Harun Bingol and Bilal Alatas have proposed a methodology for detecting deceptive contents in social networks using text mining and machine learning algorithms is proposed, and the problem of deception detection in online social networks is modelled as a classification problem. This approach performs text mining operations and transforms unstructured data sets into structured data sets because the content is text-based. The structured data sets are then subjected to customised applications of supervised machine learning algorithms. The structured data sets are then subjected to customised applications of supervised machine learning algorithms. This study compares a variety of algorithms using real public data sets, including Support Vector Machine, k-Nearest Neighbor (k-NN), Naive Bayes (NB), Random Forest, Decision Trees, Gradient Boosted Trees (GBT), and Logistic Regression. The GBT algorithm produced the highest average accuracy value in Dataset 1, at 74.4%, while the NB algorithm produced the highest average accuracy value in Dataset 2, at 71.2% [17]. Junfeng Gao, Xiangde Min, Qianruo Kang have investigated the most significant information flows (MIIFs) that occur during deception among several brain cortices. First, the guilty knowledge test protocol was used, and electroencephalogram (EEG) signals from 30 subjects were recorded on 64 electrodes (15 guilty and 15 innocent). On the 24 regions of interest, cortical current density waveforms were then estimated (ROIs). After that, the cortical waveforms were subjected to a partial directed coherence (PDC) analysis and an effective connectivity (EC) analysis to determine the brain's EC networks for the four bands of delta (1–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), and beta (13–30 Hz). The network parameters in the EC network with significant differences were extracted as features to distinguish the two groups using the graph theoretical analysis. The four bands' high classification accuracy showed that the suggested method was appropriate for lie detection. The "hub" regions of the brain were also identified based on the classification mode's best features, and the MIIFs between the guilty and innocent people differed significantly. Furthermore, among all MIIFs at the four bands, the frontoparietal network was discovered to be the most dominant. The roles of all MIIFs were further examined by combining the neurophysiology significance of the four frequency bands, which may aid in elucidating the underlying cognitive processes and mechanisms of deception [18]. According to Merylin Monaro, Stephanie Maldera research, facial micro expressions can be used to reliably spot deception. Micro expressions are recognised using machine learning (ML) techniques, which have been trained to discern between truthful and dishonest claims. More encouraging outcomes recently emerged from the artificial intelligence (AI) discipline, where machine learning (ML) techniques are utilised to recognise micro expressions and are trained to differentiate between truthful and dishonest comments. It demonstrates that artificial intelligence outperforms humans at lie detection tasks, even when people have access to more data. The best performing approach was demonstrated by the results, which showed that support vector machines (SVMs) and OpenFace were responsible (AUC = 0.72 for videos without cognitive load; AUC = 0.78 for videos with cognitive burden). All of the evaluated classifiers performed better when the interviewee was subjected to an increased cognitive burden, demonstrating that this technique makes it easier to detect deceit. Human judges achieved an accuracy of 57% in the identical task [19]. Nidhi Srivastava, Sipi Dubey have proposed strategies for questioning. It is used in lie detection, also known as deception detection, to determine truth and untruth in responses. Truth and falsehood are determined using physical characteristics and voice traits. The linear detector model is modelled using the Mel Frequency Cepstrum Coefficient, Energy, Zero Crossing Rate, Fundamental Frequency, frame function of speech signal, and physiological parameters like heart rate, blood pressure, and respiration rate. Support vector machines and artificial neural networks are used to validate the outcomes [20].

Khandare A.& Pawar R. have reviewed K-mean, Agglomerative, and DBSCAN clustering techniques. Clustering has emerged as one of the most crucial activities to categorise the data due to the growth in databases of all types. Data points are clustered or divided according to how similar they are to one another. An unsupervised data mining approach that describes the nature of datasets is clustering. Getting groupings of comparable things is the basic goal of data clustering. There are several clustering techniques, including grid-based, model-based, partition-based, hierarchical, and method-based. This essay offers a thorough examination of clustering and its operational procedures. In addition to the fundamentals, validity metrics that are necessary for algorithm evaluation are described in detail. K-Means, Agglomerative, and DBSCAN clustering techniques are reviewed in this study [21]. Khandare, A., Alvi, A.S. have reviewed Data mining technique called clustering divides the data items into n groups. In areas including e-commerce, bioinformatics, image segmentation, voice recognition, financial analysis, and fraud detection, clustering techniques can be applied. There is a lot of untapped potential in clustering applications and research, and different clustering methods are constantly being improved. Numerous clustering algorithms are studied and surveyed through experimentation on diverse data sets, followed by an analysis of the gaps and potential for algorithm improvement and scalability. Then it is suggested to use enhanced k-means to reduce these gaps. This enhanced approach creates initial centroids better than random selection and automatically determines the number of clusters. The experimentation revealed that the suggested technique minimises empty clusters, improves cluster quality, and requires less iterations [22]. Many different scientific, engineering, and technological sectors employ clustering algorithms. Examples of unsupervised clustering applications include the grouping of gene data and medical picture applications. One such application is the k-means method. For the purpose of improving the fundamental k-means clustering method, much research has been conducted. But just a few of k-means' drawbacks became the subject of inquiry. This study reviewed some of the research on enhanced k-means algorithms, outlined their drawbacks, and pinpointed areas that may still be improved to make them more scalable and effective for big data. This article examined distance, validity and stability metrics from the literature, initial centroids selection techniques, and k value computation strategies [23].

To determine whether or not someone is guilty, Syed Kamran Haider, Malik Imran Daud, Aimin Jiang, Zubair Khan have proposed the LDA method. The linear discriminant analysis (LDA) classification method is used to separate the positive and negative samples from the signals received from sensors. The use of MATLAB and the Xilinx tool is used to create the entire project. The entire system is built on an FPGA for efficiency. In both guilty and innocent individuals, there is an 85% accuracy rate in deception detection. Compared to earlier methods, it is a simpler and more practical strategy [24]. S. K. Haider, M. I. Daud, A. Jiang, and Z. Khan have proposed ERP technique. The ERP technique is used to determine whether or not an individual is lying. There are three stages in it, and signal P300 is used as an identifier. A Matlab-based programme is created to take over the processes for the sake of ease. Eleven guys between the ages of 20 and 27 were used in the study. To create several models, the collected data were then split to training sets and test sets. Then, based on precision and processing time, they were reduced using the SVM technique. The resulting model, which is used in the software, was demonstrated to be able to distinguish between all of the subjects despite having a comparatively poor accuracy. The differences in brain wave activity while a person is speaking the truth or lying will be investigated in this research. A brain wave activity-based EEG-P300 component will watch a subject's responses to queries about a fake theft situation while they first give true answers and then false ones. The trial involved eleven boys who were between 24 and 3 years old. Independent component analysis and support vector machine techniques were used for extraction and categorization. In order to create several models, the collected data were then divided into training sets and test sets. The findings indicate that there was a greater increase in the P300 component when the individual was told to hide the watch they had selected. In evaluating the reliability of using an EEG in deception detection, the results of these studies have been encouraging [25]. Rosenfeld, J. P., Hu, X., & Pederson, K have proposed P 300 based detection methods with two sets. Using two sets, P300-based detection of hidden material (1) Control subjects were shown a random sequence of rare queries (the subjects' hometowns), frequent irrelevants (other towns), and rare targets—irrelevant stimuli that required reactions on the Button 1—all of which were irrelevant. Button 2 answers were necessary for probes and extraneous information that wasn't the goal. Throughout the exercise, controls were prompted to make sure they properly conducted target/non-target discrimination. (2) Deception participants were informed of their deception (pressing a non-recognition button to probes) both before and during the run while still receiving the same stimulus sequence and reaction directions. It appears that the deception awareness modification increases test sensitivity because the deception group had significantly larger disparities between the probe and irrelevant P300s than controls and significantly more individual detections (10/10) than controls (5/10) [26]. I. Simbolon, A. Turnip, J. Hutahaean, Y. Siagian, and N. Irawati have proposed ERP technique to determine whether or not an individual is lying. There are three stages in it, and signal P300 is used as an identifier. A Matlab-based programme is created to take over the processes for simplicity. Eleven participants between the ages of 20 and 27 were used in the study. To create several models, the collected data were then split into training and test sets. Using the SVM technique, they were then whittled down based on precision and processing speed. The resulting model, which is used in the software, was demonstrated to be able to distinguish between all of the subjects despite having a comparatively poor accuracy [27].

3 Gap Identification

Following Table1 shows gap identification in different research papers.

4 Methodology

Bablani [1, 4, 9] has used the concept of Deep Belief Network using Restricted Boltzmann Machine. The following test will be performed on EFG data.

Table 1 Research gap identification

Sr. no	Paper title	Methodology used	Gaps undentified
1	Building a better lie detector with BERT: the difference between truth and lies	Bidirectional encoder representations from transformer	To identify patterns that characterize deceptive text
2	Subject based deceit identification using empirical mode decomposition	CIT (Concealed Information Test) based on unsupervised network	accuracy
3	A new approach for EFG feature extraction in p300-based lie detection	P300-based guilty knowledge test (GKT)	Rates of correct detection in guilty and innocent subjects
4	A novel method based on empirical mode decomposition for p300-based detection of deception	Empirical mode decomposition (EMD)	Less efficient feature selection, Classification accuracy
5	Deceit identification test on EEG data using deep belief network	Concealed information test	Accuracy
6	Brain fingerprinting and lie detection: a study of dynamic functional connectivity patterns of deception using EEG phase synchrony analysis	Few-trials-based relative phase synchrony (FTRPS)	Classification accuracy
7	An efficient deep learning paradigm for deceit identification test on EEG signal	Wavelet packet transform Band-pass Filter	Classification accuracy
8	Convolutional bidirectional long short-term memory for deception detection with acoustic features	CovBiLSTM model Convolutional bidirectional long short-term memory (LSTM)	Accuracy and time efficiency
9	A deep learning model for EEG-based lie detection test using spatial and temporal aspects	Visual-temporal attention model	Accuracy and time efficiency
10	Lie detector with pupil dilation and eye blinks using hough transform and frame difference method with fuzzy logic	Hough transform and frame difference method	Accuracy and time efficiency

(1) Pre-processing is performed on raw EFG data.
(2) Wavelet transformation is applied on Concealed Information Test detector with pupil dilation and eye blinks using Hough transform and frame difference method with fuzzy.

At the end result with 4 RBM will be found. [1]

Restricted Boltzmann Machine	Deep Belief Network	Wavelet Transform (WT)
• Classification, • Dimension reduction, • Feature Learning	• Record data for deceit identification	• Decomposes EEG signal into variable frequency range in term of approximation coefficient and detail coefficient.

Following Table 2 shows that different classifiers are used for feature extraction and accuracy is calculated.

Nugroho and Nasrun [11] has designed a fuzzy logic algorithm for lie detection There are some sample questions to be asked to subjects. Based on answers given by subjects, lie detection test can be decided whether the person is telling truth or not. According to survey they are asking the following question.

1. You study at Telkom University, is that your choice or your parents
2. You enter Telkom University, is your choice or as a backup? Why?
3. Do you still have love for your ex-boyfriend or girlfriend?
4. Are you currently confident when dealing with someone you like?
5. What do you with your current body size that is bigger than before?

Based on above question, response will be taken from set of respondent in terms of Eye movement, recognition of respondent. Eve movement can be right and left. Algorithm will capture eve movement and calculate enlargement of pupil and they will recognize whether the person is telling truth or not.

Labibah et al. [15] have done processing on training dataset. Eyes can detect if the person is honest or a liar. They get the following observation on datasets. They are checking for enlargement of pupils. If it is greater than 8 then the person is honest otherwise telling lie. The following table shows results based on 2 respondents (Table 3).

Figure 1 shows that analysis of pupil enlargement. Graph shows with the help of following parameters such as early pupils, average pupil, pupil enlargement, total blinks, and respondent.

Table 2 Different classifiers used for features extraction

Classifiers	Feature extraction	Data set	Channel	Accuracy
QDA (Quadratic discriminant analysis)	Various (Power, wavelet, Hjorth parameter, etc.)	Emotion dataset	64	35.9
LDA (linear discriminant analysis)	EMD	CIT datasets	3 channels (Cz, Fz, Pz)	80.0
FDBN (fractional deep belief network)	Wavelet transform	Motor imagery	3 channels (C3, Cz, C4)	75.0
Deep belief network	Wavelet transform	CIT data	16	81.03

Table 3 Observations based on datasets

Sr. no	File name	Early pupils	Average pupil	Pupil enlargement	Total blinks	Respondent
1	Respondent 1	2.65	3.25	0.6	28	Honest
2	Respondent 2	2.12	3.32	1.2	7	Lie

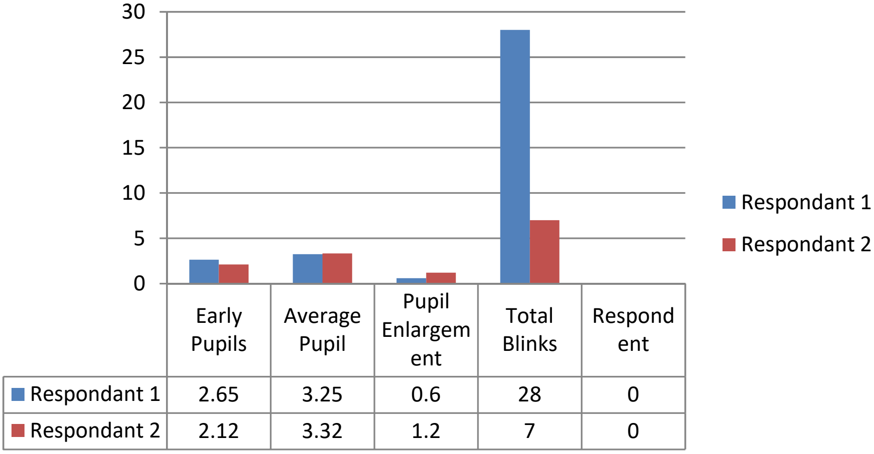

Fig. 1 Analysis of pupil enlargement

Fernandes and Ullah [12] have given comparison of the different features classification method resulting with Recognition time. They invented their own classification methods such as Levenberg–Marquardt and LSTM. This algorithm extracts the features such as Time Difference Enery, Delta Energy, Time Difference Cepstrum, Delta Cepstrum. By using this algorithm during their experiments, and they repeatedly trained and evaluated it using LSTM and Levenberg–Marquardt. The time-difference energy feature yielded identical results (i.e. 100%) each time it was run. Overfitting might have happened as a result of the amount of the data used. The recognition rate, however, can alter from 100% if we increase the size of the corpus for additional study and verification of the test results of our suggested extracted feature using the same procedure and the classifier methods. Levenberg–Marquardt and LSTM methods are used for high accuracy and speed for feed-forward neural networks.

5 Proposed Work

The following are the key objectives should be considered during my work.

(1) To design enhanced algorithm to identify the persons based on different subject.

(2) To design efficient algorithm to categories the gender for predicting deception, predicting lie for females and truth for males.
(3) To design novel approach of feature extraction algorithm for the analysis of eye blinks
(4) To design optimum algorithm for linguistic based approach for the lie detection
(5) To design enhanced approach to analysing the lying behaviour of the person under instructed and spontaneous circumstances based on EFG data.

Figure 2 shows that flowchart of deception detection. In that process input data should be taken in the form of video. It can be processed using the deep belief network. During processing Band pass filters are used. After processing, features can be selected and extracted. At the end result can be obtained in the form of extracted features. There are two key findings obtained from the above process.

(1) Pupil Detection
(2) Eye Blink Detection

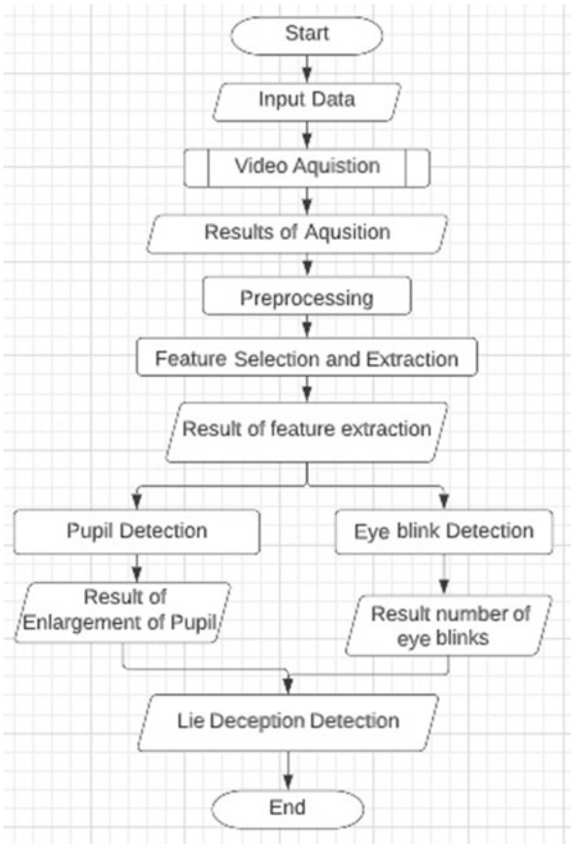

Fig. 2 Flowchart of deception detection

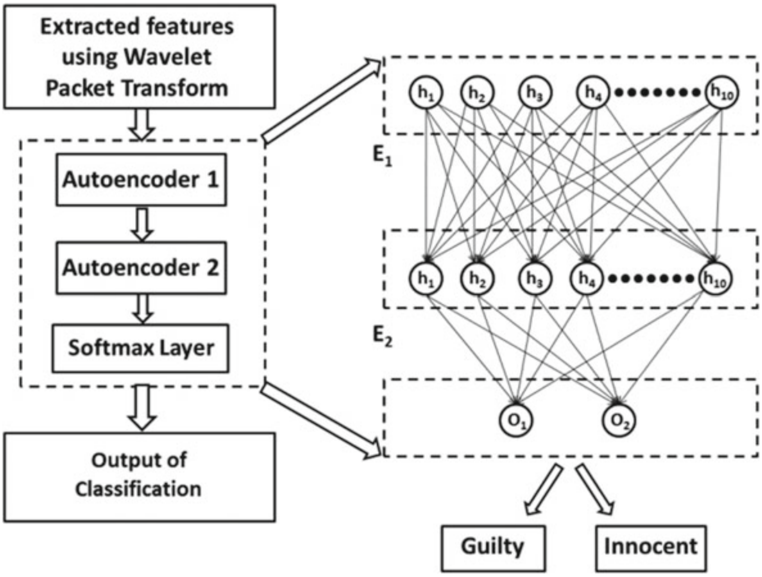

Fig. 3 Framework for deceit identification using EEG

Based on result of enlargement of pupil threshold and number of eye blink analysis, the system will identify the detection of behaviour of humans in terms of Honest and Lie (Fig. 3).

Figure 2 shows that Framework for deceit identification using EFG. For extracting the features Wavelet Packet Transform [3] is used. Auto encoders are used compressing the data. This data can be used using different classifiers like QDA, LDA, FDBN, Deep Belief Network, etc. By using different classifiers, accuracy can be obtained with the help of following technique.

TP is the number of honesty correctly identified, TN is the number of deception correctly identified, FN is the number of honesty identified as deception, and FP is the number of deception identified as honesty.

average_accuracy = (TP + TN)/TP + FP + FN + TN).

honesty_accuracy = TP/(TP + FN).

deception_accuracy = TN/(TN + FP).

Figure 4 shows that process of deception detection based on Guilty and Innocent Person. BrainVision Recorder [4] and analyser are used to gain the EFG signals. Band pass filters are used to process the signals and using this unwanted noise can be removed. Using Wavelet Packet transformation, features can be extracted and features can be classified using the deep neural network.

Fig. 4 Process of deception detection based on guilty and innocent person

6 Conclusion

The lie is very detrimental to the fraudulent acts of many people who were cheated. The lies are common in the general population. To be able to reveal a lie we can detect through some limbs that unconsciously will show a different reaction when someone is lying. Among them, through organs of our eyes can detect the person is honest or lie. The current approaches to lie detection are very imprecise and reliant on behavioural and physiological trends. The development of a computational model to handle lie detection has received less study attention. Future research can explore the development of algorithms and systems that can analyze brainwave patterns in real-time, allowing for immediate detection and response to deceptive behavior.

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Tanmayi Nagale She is completed M.E. in IT. Currently pursuing PhD in Computer Engineering from Thakur College of Engineering and Technology affiliated to University of Mumbai. She has total 4 publications in the domain of BigData and 1 publication in Deep Learning. She has presented 4 papers in conferences held at MIT Pune and Thakur College of Engineering and Technology, Kandivali.

Anand Khandare Dy. HOD, Computer Engineering, Thakur college of engineering and Technology, Mumbai with 15 years of teaching experience. He completed Ph.D. in Computer Science and Engineering in the domain Data Clustering in Machine Learning from Sant Gadge Baba Amravati University. He has total 60+publications in national, international conferences and journals. He has 1 copyright and 2 patents. He guided various research and funded projects. He is worked as volume editor in springer conference on Intelligent Computing and Networking 2020. He is also reviewer in various journal and conferences.

Medical Image Processing and Machine Learning: A Study

Hiral S. Padhariya and Shailesh Chaudhari

Abstract Machine Learning Field is a rapidly growing field in the research sector since 2 decades and a growing field in the medical imaging processing sector. So, to diagnose the diseases at the early stage machine learning provides various techniques and algorithms such as supervised learning technique, unsupervised learning technique, Reinforcement learning technique, Active learning, Semi-supervised learning, Evolutionary learning and lastly deep learning. It enables machine to automatically learn data from the past experience and predict things without any need of being explicit programmed. First we need to provide input of the past data then train the data using some sort of machine learning algorithm and then build logical models and finally providing output. The main aim of these survey paper is to give brief detail regarding how machine learning can be useful in the medical sector.

Keywords Medical imaging · Machine learning · Image processing · Supervised learning · Un-supervised learning · Reinforcement · Semi-supervised learning · K-nearest neighbor · Naïve bayes algorithm · Random forest · K-mean clustering

1 Introduction

Machine Learning is the field in computer science and it is a branch of artificial intelligence. Before 40–50 years the machine learning concept was not in the way but now it has become a necessityand part of our daily life. In 1936 Alan Turing gave a theory on how machine can execute some sort of the paper on "Computer Machinery and Intelligence". Today, when data are in the form of the images there are Machine Learning algorithms which interpret the images for further processing [1, 2].

H. S. Padhariya (✉) · S. Chaudhari
Veer Narmad South Gujarat University, Surat, India
e-mail: hspadhariya@vnsgu.ac.in

S. Chaudhari
e-mail: sachaudhari@vnsgu.ac.in

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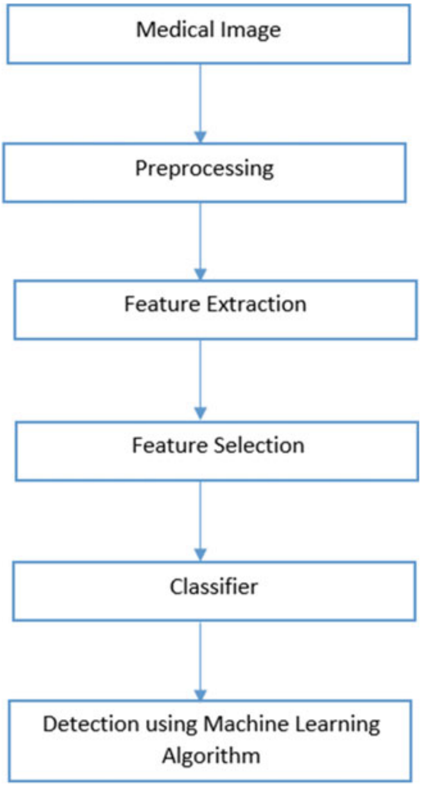

Fig. 1 Image data processing steps

Machine Learning in these days has been used in research advancements and has changed the revolution of a human. It is used across various domains such as in the deep learning applications for health sector, also COVID-19 pandemic has presented the environment with the latest advancement of machine learning techniques [3].

Image Processing is a computer technology that is applied to the images for further processing, analyzing and get meaningful information from them.

Some of the data processing steps involved are represented as diagrammatic in Fig. 1.

The Life cycle of Machine Learning involves the phases as shown Fig. 2.

2 Machine Learning in Medical Imaging

As now a days medical issues have become a common part of human life so identification of diseases at the early stage is too important so that precautions and care can be taken. For early detection of the diseases machine learning is a way which helps us

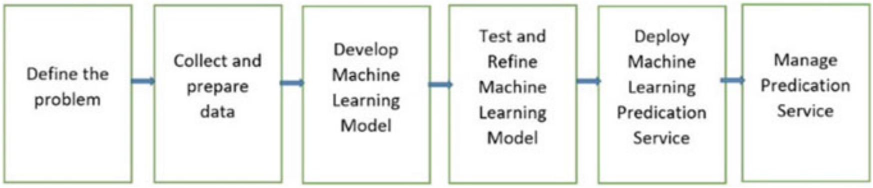

Fig. 2 Machine learning life cycle

by giving us various algorithms to study specific diseases in medical imaging, since there are many entities such as lesions in medical image which cannot be shown correctly using some mathematic solution [4, 5].

So in some recent years if we talk about skin diseases it has become common in almost the entire region of our world. One of the major skin diseases that is skin cancer is found to be the deadliest disease in humans. Skin cancer is found of 2 types that is Malignant Melanoma and Non- Melanoma. So; if we talk about the Malignant Melanoma it is found to be a dangerous skin cancer and about 4% population is affected, while 75% death cases are noted by this skin cancer [6, 7].

So to prevent such types of diseases early detection and treatment must be provided to the patient. The presence of Melanocytes which is present in the body is the cause of Melanoma Diseases So; one of the way to examine structure of skin is dermoscopy and till now, the accuracy for detection of this skin cancer is 75–85%. So some of the stages for dermoscopy image analysis system are as follows:

1. Collecting Dataset
2. Hair Removal
3. Shading Removal
4. Glare Removal

Also, to identify and classify different data based on different classes; the SVM algorithm is used which is a type of supervised learning algorithm. The performance of these algorithms is very accurate even for small amount of data as compared with other classification algorithms [7, 8].

Stage 1: Computer Vision: For identification of the skin diseases based on features extracted from the image by using various image processing techniques. The computer vision consists of two phases. In the first phase the input of the image is taken from any smart devices or camera and some important features are extracted. In the second phase the features that we extracted in phase 1 helps us to identify the skin diseases by using various different models. Some features are detected, such as color of the infected area with respect to healthy skin, and shape by using edge detection of the infected area.

Stage 2: Machine Learning: For detection and classification of the image various machine learning algorithms are used. The attributes are taken as input such as exocytosis and acanthosis after that it is trained using the dataset and finally testing is done by the algorithm. Image processing are now a days have become advances in field such as instrumentation and diagnostics and most of these field are based on image processing. Machine learning and artificial intelligence have been rapidly progress in the recent years. For supporting activities such as medical image processing, image fusion, image segmentation, retrieval of image analysis of the image. It helps doctor to assist diagnosing and prediction risk of the diseases at the early stage.

Some of the machine learning algorithm such as SVM, KNN, NN they are being limited in processing of image also some time consuming. So, if we talk about the other algorithm such as CNN, KNN in which raw data in fed, having automatic features which automatically learn rapidly [9].

Although to identify the diseases or to detect the diseases based on CNN in medical image have been given significant accuracy, now a day there are new advancement in machine learning and one of the concept is of deep learning. Basically it is used for the recognition of various objects.

Deep learning applied to medical imaging now it has become a more popular technology in era of the digital world. Applying deep learning based algorithm to medical imaging has become a growing era. For cancer detection and continuous monitoring of diseases deep learning is used.

The detection of diabetic Retinopathy now a days and to manage it manually it is too difficult and also it takes a lot of time. At the beginning of these diseases there are hardly no symptoms at the early stages which actually delay in the treatment. The automated detection of Diabetic Retinopathy provides better accuracy Deep learning also provides accurate result for cardiac imaging, some of the application CT and MRI scans are most of the recent use techniques of deep learning [10, 11].

Mhaske et al. use some of the supervised learning algorithms such as neural network, K-mean clustering, support vector machine to detect and classify skin cancer diseases. After that some sort of technique is applied on the image then with use of various classifiers the classification is done. So, if we look for the unsupervised learning technique by using K-mean algorithm about 52.63% accuracy is obtained. For neural network using back propagation about 60–70% accuracy is obtained and by using SVM 80–90% accuracy is obtained [12].

Deep learning is a part of machine learning where the different learning is used as supervised, un-supervised or semi-supervised. It is generally used when a huge amount of dataset is used for the learning process, also with increase in the large dataset the training time also gets increased [13].

With the use of Deep learning based methods for identifying melanoma diseases, first constructing a deep fully convolutions network to segment skin lesions according to the result based on segmentation, a deep model network is prepared for better accuracy and efficient skin lesion segmentation and further a fully convolutional network is constructed having advantage of FCRN by making it pixel-wise prediction, for performing skin lesion identification also each single pixel is considered as an independent training sample [14].

So, in order to work with various machine learning algorithms, machine Learning is divided as follows:

A. Supervised Machine Learning
B. Un-supervised Machine Learning
C. Semi-supervised Learning
D. Reinforcement Learning.

2.1 Supervised Machine Learning

It teaches the machine with the labelled data after that model make predictions and providew the output which is shown in Fig. 3. Basically it helps us to solve real-world problems such as spam detection [15]. Further, it is categorized into two types:

1. Regression
2. Classification

Some commonly used algorithm are:

1. Naïve Bayes
2. Linear Regression
3. Logistic Regression
4. K-Nearest Neighbor
5. Random Forest

K-Nearest Algorithm: KNN is a type of supervised learning algorithm and it is one of the simplest algorithm to implement. Based on the similarity of the new data and available data it classifies a new data point. This algorithm is also called aa lazy algorithm because it does not directly learn from the training dataset but it stores data at the time of classification and actions are performed on the dataset. Suppose we have two different categories named as category A and category B and a new data point X. so to identify that new point lies in which one of the category KNN algorithm can be used [16].

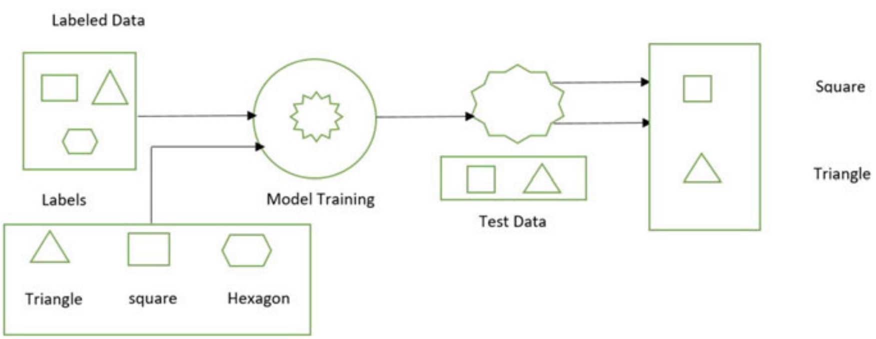

Fig. 3 Supervised learning model

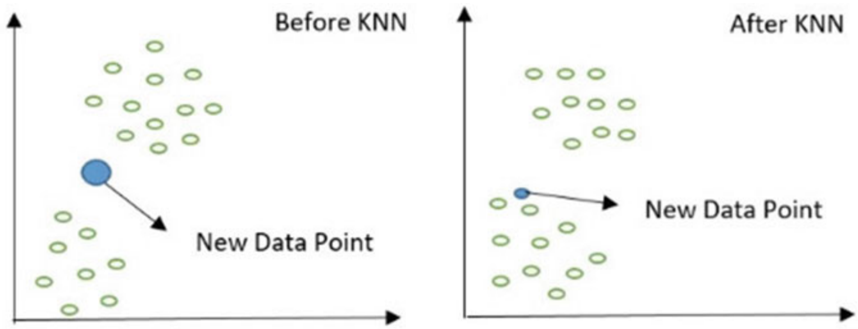

Fig. 4 KNN classifier

KNN is useful where we want to recognize the patterns to classify objects shown in Fig. 4. As value of K generally depends on type of the problem. The most preferable value of K = 5. By selecting the multiple value of K it may lead to overfitting.

A good example in the real life scenario is for heart diseases prediction as in a normal routine life heart disease is a common disease for death of humankind. According to report of US the death ratio is up to 35% so to avoid such type of issues quality and improvement should be done in the health sector. KNN is one of the frequently used algorithm in prediction of the diseases [17]. Accurate Tumor and metastasis especially in gastric cancer which is one of the biggest issue in medical image analysis, KNN classifier is used for analysis of gastric cancer which gives accuracy up to 96.33% as compared to traditional methods [18].

KNN and genetic algorithm helps us to improve the accuracy for the given disease's dataset. The algorithm make use of the two parts:

1. It deals with evaluating attributes using genetic search
2. Select the attributes with the high ranked
3. Applying KNN with genetic on subset of attributes which provides maximum accuracy [19].

Naïve Bayes Algorithm: It is a supervised learning technique which is used for solving classification problems and it is based on Bayes theorem. generally, for text classification where we have a high amount of dataset and for building fast machine learning model that make quick prediction Naive Bayes is useful. It is a probabilistic algorithm which predicts on basis of probability of object [20, 21].

Bayes theorem is called as Bayes Rule used to determine probability and formula for Bayes theorem is stated as:

$$P(A|B) = P(B|A).\,P(A)/P(B)$$

where, P(A|B) is posterior probability that is probability of A on Event B.

P(B|A) is likelihood probability which means probability of this hypothesis is true.

P(A) is prior probability where it observes evidence before hypothesis.

P(B) is Marginal probability.

Some of the advantages of using Naïve Bayes is that it is one of the fastest algorithm to predict dataset and also it can be used for multi-class classification.

Applications where Naïve Bayes have been used in such as in Face Recognition, Prediction of weather, Medical sector, Classification of different news etc.

The real scenario in the medical field where this algorithm is useful to predict Hemoglobin protein sequences. So based on the protein sequence information which we are getting from the patient for getting better accuracy result these algorithms are used to classify normal and diseased proteins. The data which we are getting from the patient are patient id, patient name and protein sequence related information. After that it will be compared with the training data and classified accordingly. So the result which we get from these is by taking dataset size as 2,0 accuracy value is 85%, and for dataset size 50, the accuracy size is 84% [22].

Random Forest Algorithm: It is popular algorithm in machine learning which belongs to the supervised learning technique. This algorithm can be used for classification as well as for regression problems. It combines multiple classification for solving a complex problem to enhance the performance of a model.

Here, in this algorithm it does not rely on one of the decision tree, it takes predictions from multiple trees and based on the majority it predicts the output. If the tree leads to a great number, then high accuracy it will show and it also prevents overfitting shown in Fig. 5 [23].

As random forest predict dataset is based on the multiple trees, some predict the correct output while some not. But all together predict the correct output. So, two assumptions for better performance are stated as:

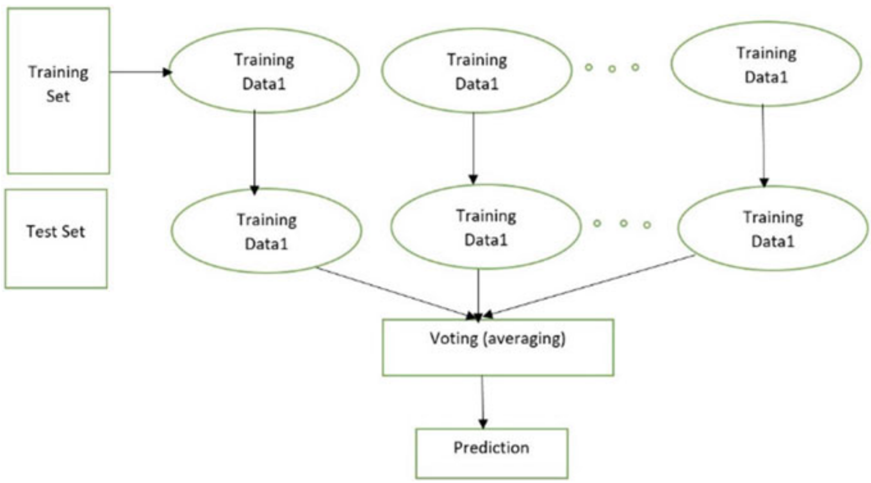

Fig. 5 Random forest model

- Some actual value must be there in the feature variable of the dataset.
- Prediction done from each tree should have very low correlations.

This algorithm takes less training time compared to all other algorithms. The accuracy is high for the output which it predicts and even for the large amount of dataset it runs efficiently. Also, it can maintain the accuracy where at some point large amount of data is mining.

How actually Random Forest Works?

As the random forest works in a two-phase. In the First phase we create a random forest by combining a multiple decision tree, and second is to make predictions for each single tree created in phase one.

Step 1: Selection of any random "K" data points from the given training set.

Step 2: Building the decision tree with the selected data points.

Step 3: Choosing number as N for the decision tree that we want to build.

Step 4: Processing same step 1 and step 2.

Step 5: For new data points, finding the prediction for each decision tree and assign-ing new data point to the category with majority votes.

Some Real life applications of the Random Forest Algorithm include in Banking, Health care, Stock Market, E-commerce etc. [24].

2.2 Un-supervised Machine Learning

It is a type of Machine Learning in which the model is trained with the unlabeled dataset. Here, these technique is used to find the hidden pattern from the dataset and group together as per some similar data point shown in Fig. 6 [25].

Unsupervised learning is categorized into two types:

1. Clustering
2. Association

K-Means Clustering Algorithm: It is a type of unsupervised machine learning technique that helps us to solve the clustering problems in machine learning. Generally, it groups the unlabeled dataset in multiple clusters. Here, K is the pre-defined

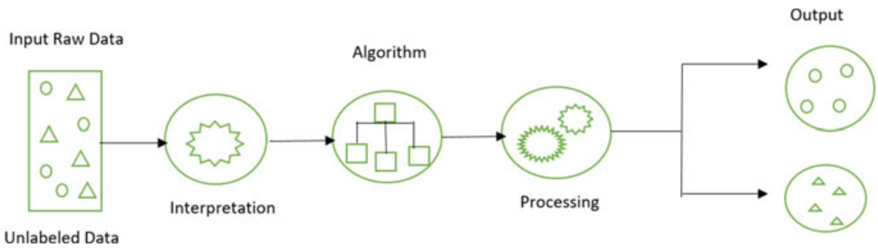

Fig. 6 Un-supervised model

cluster which is created in the process. Let's say if value of K = 2, it has two clusters, and for value of K = 3, there will be three clusters. It is called as centroid based algorithm, where every cluster is associated with the centroid. These algorithms minimize the sum of distance between data points and their corresponding clusters. As in the input it takes the unlabeled dataset and then divides into K-number of clusters and this process is repeated until it does not find the best clusters [23, 26].

The mainly two tasks performs by K-Means clustering algorithm are:

1. It determines the best value of K centroids by an iterative process.
2. Assigns each data points to the closest value of the K-center and points near to the K-center create a cluster.

2.3 Semi-supervised Machine Learning

This Machine Learning Technique is the combination of the supervised and unsupervised learning techniques. It combines the algorithm with unlabeled dataset which is used in unsupervised learning and also makes use of the labeled dataset similar to supervised learning technique. It improves the limitation of the supervised and unsupervised learning techniques. In medical image analysis most of data are unlabeled without any annotations and for labelling data it requires knowledge and training in the medical field, so semi-supervised learning improves the performance in order to develop accurate Computer-Aided Diagnosis [27, 28].

2.4 Reinforcement Machine Learning

In this method it provides rewards for the good one and punishing for undesired ones, also rewards are gained through the experiences when it interacts with the environment. The components involved in the RL system are reward signal, policy, value function and model of the environment [29].

The AI Systems is implemented in the medical sector where some of the decisions are ignored and systems makes decision based on current state of patients, so to solve such problems RL provides systems where it not only it provides an effective treatment to the patient but also the patient has long-term benefits. One of the application of RL in medical sector is it is used to develop treatment for lung cancer patients with proper strategies [30].

3 Conclusion

The need of machine learning skills are growing faster in few years. Currently machine learning is in use and they are in the learning process. The use of machine learning in the field of medical image processing plays important inferences for the medication. Along with this Deep Learning is also used in the identification of diseases. Along with this based on the different algorithms in machine learning techniques different algorithms such as KNN, Naïve Bayes have been used for various detection of the diseases. Digital Image processing helps us to identify the image and based on the input data of the image various operations are performed on the image and finally it is trained with the model by applying various different machine learning techniques. One of the deadest diseases such as skin cancer can now be detected at the early stage using machine learning. Here, there are some algorithms listed for different disease detection. From the study we can conclude that using machine learning algorithms medical practitioners can predict more accurately various image based diseases as compared to the traditional approach.

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Security and Privacy Policy of Mobile Device Application Management System

Pankaj Kumar, Sapan Kumar Singh, Shubhangee Ramaswamy, Devidas Golhar, and D. Kirubakaran

Abstract Using a variety of sensors, automatic activity recognition systems can continuously monitor a wide variety of physiological signals from sensors connected to the user or the user's surroundings. When applied to healthcare, this can greatly benefit areas like automated and intelligent monitoring of everyday activities for the elderly. This article represents a novel approach to analyzing the data using Artificial intelligence. The data was collected from the smartphone's internal sensors using the feature ranking algorithm. These sensors collect data regarding human activities. Classify the collected information using random forests, ensemble learning, and lazy learning techniques. The suggested method may lead to intelligent and autonomous real-time development, as shown by extensive tests utilising a public database 1 of human activity using smartphone inertial sensors. Human activity tracking for use in eHealth situations, including the elderly, the handicapped, and other populations with unique healthcare requirements.

Keywords Security · Random forecasting · Machine learning · Smartphones

P. Kumar (✉)
Wipro Ltd, West Boring Canal Road, Patna, Bihar 800001, India
e-mail: gempankaj08@gmail.com

S. K. Singh
Wipro Ltd, 315, Deep Complex Bharat Colony, Pune, Maharashtra 421305, India

S. Ramaswamy
Institute of Management Education Research and Training, Karvenagar, Pune, Maharashtra 411052, India

D. Golhar
Marathwada Mitramandal's College of Commerce, Pune, Maharashtra 411004, India

D. Kirubakaran
Department of EEE, St. Joseph's Institute of Technology, Chennai, Tamil Nadu 600119, India

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1 Introduction

Most app developers and smartphone owners have helped propel the mobile application (app) ecosystem to the forefront of global economic growth. According to recent statistics, the most downloaded apps in 2022 by users will be according to their requirements [1–4].

Studies show that although internet users are increasingly dependent on smart mobile devices (such as smartphones and tablets) for their day-to-day activities and demands, they may not always be aware of or able to influence the processing of personal data using these tools [5–7]. Moreover, it is frequently difficult to analyze apps' privacy and security features because of the complexity involved in understanding how they function because of their dynamic environment, reuse of software libraries, and interaction with other networks and systems. Inadequate data protection and security procedures are often the results of app developers' lack of awareness, expertise, or understanding of how to properly arrange for an engineer privacy and security needs into their products, rather than any deliberate neglect of such standards [8, 9].

The General Data Protection Regulation governs the collection, storage, and disclosure of personal information using applications (EU) [4, 10, 11].

The present Data Protection Directive 95/46/EC [2] is being replaced by a new one that is immediately applicable in all Member States [12, 13]. While The General Data Protection Regulation (GDPR) strengthens the privacy and security guarantees established by the Data Protection Directive offers supplementary safeguards that give users more say over their private information, which is extremely difficult to do on a mobile device or the internet [14, 15]. Beyond the requirements of GDPR, this also applies to The EU's Directive on Privacy and Electronic Communications also has implications for mobile applications. Transmissions EU Directive 2002/58/EC (ePrivacy) [3] are being revised at the moment and compliant with the General Data Protection Regulation [16, 17]. In January this year, the European Commission proposed a new ePrivacy Regulation [18, 19]. Commission and is being discussed at the moment in the European Parliament and the Council. Essential safeguards for users' personal information in mobile applications, including procedures for ensuring the anonymity of users' data and the prevention of data breaches exchange of messages and information, installation of apps and files (cookies) on end-user gadgets and rules governing tracking's effect on users' privacy preferences [20].

Combining these numbers with the blurring of barriers between personal and company-owned mobile devices, it's clear that mobile app analysis is becoming more vital for businesses. In reality, these applications may expose sensitive information about workers, which cybercriminals might use for social engineering, the practice of tricking people into giving up sensitive information to steal it, or even for exfiltrating data or installing malware (malware). Companies are right to be wary of the ever-expanding cyber threat environment, which now includes mobile apps, due to the sensitivity of the data they collect from their customers (personally identifiable information, or PII) [21]. The term "mobile device management" refers to a method businesses use to deploy and control mobile devices like smartphones and tablets. Rules and an application make up this framework usually, with the latter used to manage policies that limit an employee's access to install mobile apps and enforce security measures [2, 22]. These regulations are implemented to prevent unauthorized access to sensitive information, such as financial records, social security numbers, and intellectual property, and to ensure that malware is updated and cannot compromise systems (IP) [23]. An incident responder or mobile device manager may remotely erase all data on a lost or stolen employee's mobile device [24, 25].

2 Experimental Validation with Activity Recognition Process

We utilized a public activity recognition (AR) database 1 for experimental validation of our method. Thirty people between the ages of 18 and 50 have provided labeled data for this database. Every single individual performed. Wearing a smartphone belt case while doing six things, including but not limited to: walking on level surfaces, climbing stairs, swimming, playing a video game, and playing a board game sitting, standing, and lying down, as well as walking up and down stairs. In this case, we utilized a smartphone to sensor package with a 3-axis linear acceleration measuring accelerometer and angular velocity measuring gyroscope angular velocity, both at a fixed rate of 50 Hz, which is more than enough for recording human motion. The two components of the database are the raw sensor data that has not been processed and the collection of features data that is processed and extracted [26, 27].

At first, 2.56 s fixed-width sliding windows with 50% overlap was used to sample the noise filters of the raw data. There are 17 parameters, that may be extracted from the time- and frequency-domain analysis of accelerometer data [28, 29].

Another data set has 2.32 s-long vectors with 552 features apiece. Each vector stores information about the interval, including the average, maximum, lowest acceleration and angular velocity along all three axes and other, more advanced features like the Fourier, transform and autoregressive coefficients. We utilized this dataset to test our activity recognition method's efficacy. Figure 1 depicts the block diagram for this dataset's processing. The preceding section will go on to discuss the relevant context [30].

3 Motivation of the Research Work

There are several benefits of using cell phones for automated activity identification, including the convenience of the device's mobility and the absence of the need for bulky, unpleasant permanent equipment.

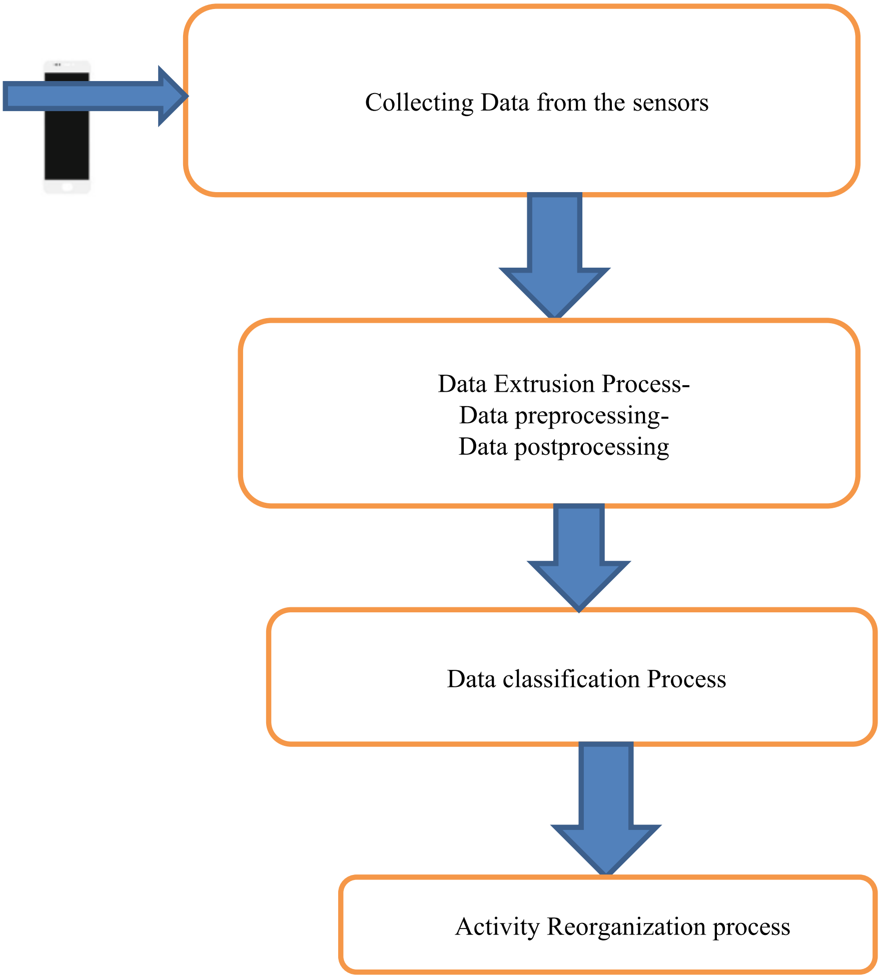

Fig. 1 Frame work—activity reorganization in step by step

Other well-established methods of activity identification include body sensor networks and dedicated hardware configurations. For example, While such sophisticated setups may improve the effectiveness of activity detection, it is unreasonable to expect individuals to wear them regularly in most residential settings due to the effort, time, and convenience involved in doing so. Smartphones have a leg up on the competition for activity detection because of their portability, simplicity of use, and the ability to utilize the phone's many sensors. Smartphone sensor outputs need the development of practical machine learning and data mining technologies for automated and intelligent activity detection. While several machine learning approaches are developed, the best algorithm for recognising mobile phone activities remains unclear. It would be a massive boon to the eHealth field if automated activity identification systems could be constructed using the intelligent processing of numerous smartphone sensor characteristics. They focus on remote activity monitoring and recognition in the elderly and disability care sectors.

This article compares the naive Bayes classifier and the unsupervised k-means clustering method. The comparison for activity recognition on several novel machine learning and data mining approaches in smartphones, Adopting the random forests and random committees on the decision tree method. A publicly available smartphone activity identification database 1 reveals that integrating machine learning and data mining significantly increases recognition performance over earlier smartphone-based activity detection methods. I'll explain the machine learning methods that allowed a mobile app-based activity recognition system.

4 Data Mining Approach to Classify the Activities

In the next stage, work will organize in the form of data mining. Initially, based on the ranking system preprocessing is implemented. For preprocessing technique, Consider real-time smartphones (561 parts). In this case, the data is used to determine the relative value of various constituents, with less important ones being disregarded. In the setting of our very high-dimensional datasets, our attribute selection technique has performed exceptionally well, allowing us to employ almost half as many characteristics as before while maintaining the same level of recognition performance. A battery of tests was conducted using a range of feature rankings derived from an information-theory-based ranking technique. These varied from the use of a Naive Bayes classifier as a starting point to those of Decision Trees, Random Forests, Ensemble Learning, and Lazy Learning. Several different classifiers were evaluated for this work, and we offer summaries of each below.

4.1 Bayesian Classifier

This classifier uses Bayes' theorem at its core to make inferences based on probabilities. Naive Bayes, the most straightforward Bayesian approach, is explained as a specific example of an algorithm that requires no modification for continuous data. Since it is supervised learning and easily trained, it provides a valuable starting point for evaluating other methods in terms of accuracy and generalization.

4.2 K-Mean Classifier Approach

In this case, the dataset may be unlabeled since clustering is an unsupervised learning technique. Instances are stored into two categories: those that are the same or connected and those that are distinct. K-Means is the most well-known and straightforward technique for determining whether instances may classify based on some criteria. Because of its ease of use and the fact that it can process unlabeled data, it might serve as a benchmark against measuring the performance of other classifiers.

4.3 Decision Tree Classifier Approach

The dependent variable, or the desired outcome for a new sample, is calculated using a decision tree classifier based on predictive machine learning algorithms. Here, the internal nodes of the decision tree represent distinct qualities, and the branches between the nodes represent the range of values that each attribute may take in the data samples being analyzed. The terminal nodes further indicate the dependent variable's ultimate values (classification). In statistics, the word "dependent variable" refers to the variable whose value is being predicted; in this context, it is the attribute whose values are being anticipated. Consequently, the independent characteristics are the independent variables in the dataset, contributing to predicting the dependent variable's value. The J48 Decision tree classifier employed in our tests has a straightforward algorithmic approach. Whenever it has to categorize anything new, it must first generate a new decision tree using the attribute values from the existing training data. Attributes that differentiate across samples are used to identify the next batch of items added to the training set. This attribute gives us the most helpful data since it allows us to easily distinguish between different occurrences of the data, which is essential for accurate categorization.

4.4 Random Forecasting Approach

When it comes to ensemble learning techniques for classification and regression, Random Forests are a decision tree ensemble. Additionally, they may be seen as a kind of closest neighbor predictor, as they build many decision trees during training and use the class mean as their output. Random Forests is an ensemble of decision trees (a term used by Leo Breiman 15). Through averaging and balancing the two extremes, Random Forests aim to mitigate bias and variation problems. In addition, there aren't many knobs and dials to fiddle with in a Random Forest; in most cases, you may get good results by just utilizing the default settings. Because of these benefits, Random Forests may frequently be used directly from the box to produce a decent, quick, and efficient model, without the requirement for extensive handcrafting or modeling in comparison to other classifiers.

4.5 Random Type Forecasting Approach

As with other types of ensemble learning methods, the premise of the random committee is that the addition of more classifiers will lead to better overall results. Each individual classifier in this sort of classifier is built from the same underlying data but with a distinct random number seed. To determine the final output class, it takes the mean of the predictions made by all of the basic classifiers.

4.6 Lazy Learning Classifier

Classifiers based on lazy learning store the training instances during the training period and use them later for classification. The IBC classifier resembles the k-nearest neighbor classifier in many respects. However, there are ways to speed up the determining closest neighbors by utilising various search methods since most of the learning occurs during the classification phase when these models are often at their slowest. While this study used a linear search strategy, the performance may have been improved using kD-trees, commonly known as cover trees. As the measure of separation, we used Euclidean distance. There was no distance-based weighting applied, and just one neighbor was utilized.

5 Result

Use the 2.56 s chunk of data from dataset 1 that has been pre-processed to include a collection of 561 characteristics to evaluate the efficacy of the recommended data mining approach for independently identifying human activities from smartphone data. Each track displays the following traits:

Body acceleration estimation and triaxial acceleration from the accelerometer.

- The gyroscope's reading of the triaxial angular velocity.
- A time and frequency-domain vector of 561 features.

That describes its function.

Individual subject identification for the researcher conducting the study.

In the relative merits of various features, as measured by their contribution to the Model's performance and the amount of time spent on its construction under various classifier learning strategies. This chart displays the results of our analysis of 2, 8, 16, 32, 64, 128, 256, and 561 features (all features), ordered by information gain.

About 10,000 samples were used in total throughout training and testing. Because there were so many samples in this dataset (10,000 total), we utilized fivefold cross validation to split it into separate training and testing sets. Each classifier's Accuracy in Classification and Model Building Time, while Tables 1, 2 and 3 provide additional metrics, including TPR, FPR, PR, RC, F-m, and ROC. Table 3 displays the confusion matrix for the best-performing IBk classifier with 128 and 256 ranked features, respectively.

The Naive Bayes Classifier takes the least amount of time (5.76 s) to finish building the model, and its 79% accuracy on the massive dataset. Among the ensemble learning methods, however, random forests provide the best combination of accuracy (96.3% on average) and speed (14.65 s on average) when developing a model. The other classifiers in the ensemble learning system also perform well in classification accuracy (96%) (random committee and random subspace). As an unsupervised method, K-Means clustering does poorly, with a classification accuracy of just 60% and a time to solution of 582 s.

In contrast, the slow-learning-based IBk classifier outperforms all other classifiers by a wide margin (90%+ accuracy for 128 and 256 features). To attain maximum performance, a smartphone-based activity identification system must trade between accuracy and the time it takes to create a model since real-time activity monitoring requires a model to be generated dynamically from the obtained data. Aside from TPR and FPR, other performance measures include Precision, Recall, and F-measure.

Since real-time activity monitoring necessitates the construction of a model dynamically from the obtained data, an activity identification system implemented on a smartphone must strike a balance between accuracy and the time required to create the model. Additional measures such as false positive rate (FPR), false negative rate (FNR), precision, recall, F-measure, and ROC area should be considered when choosing an algorithm for an automated activity identification system. Table 1 displays these additional indices of productivity.

The top-performing IBk classifier's confusion matrix on datasets with 128 and 256 sorted features is shown in Table 2. The confusion matrix (Actual Class (AC-0 to

Table 1 Individual subject identification for the researcher conducting the study

	Km	Nb	J48	RF	RC	IBK
2	40	50.45	55.32	56.62	61.1	53.12
8	80.8	48.25	62.32	62.01	61.23	60.18
16	81.82	46.57	68,304	72.23	72.10	67.82
32	72.00	51.24	71.24	75.16	74.22	71.72
64	58.00	55.23	77.28	76.21	84.71	72.51
128	56.00	54.22	92.48	95.22	95.25	92.91
256	62.00	52.76	93.81	96.62	96.22	97.52
561	64.0	78.20	85	97.83	96.92	97.89

Table 2 Supplementary efficiency indicators

Feature	TPR	FPR	PR	RC	FM	ROC
IBK-256	0.965	0.005	0.956	0.965	0.975	0.986
RC-256	0.956	0.005	0.956	0.985	0.456	0.995
RF-256	0.952	0.008	0.985	0.963	0.875	0.998
RC-128	0.995	0.002	0.987	0.956	0.965	0.996
RF-128	0.945	0.003	0.994	0.945	0.996	0.988
J48-256	0.978	0.0021	0.932	0.956	0.975	0.989
IBK-128	0.92	0.0013	0.92	0.92	0.92	0.956
J48-128	0.94	0.0012	0.94	0.94	0.95	0.986
RF-64	0.948	0.148	0.941	0.845	0.921	0.971
RC-64	0.949	0.034	0.886	0.842	0.845	0.956
IBK-64	0.886	0.49	0.886	0.883	0.844	0.975
J48-64	0.889	0.058	0.884	0.851	0.851	0.862
RC-32	0.854	0.064	0.845	0.845	0.845	0.915
RF-32	0.818	0.072	0.845	0.818	0.871	0.51
IBK-32	0.813	0.06	0.83	0.823	0.815	0.953
RF-16	0.823	0.071	0.812	0.821	0.823	0.891
RC-16	0.823	0.071	0.814	0.801	0.801	0.925
J48-32	0.812	0.072	0.801	0.73	0.782	0.951
J48-16	0.71	0.073	0.71	0.685	0.785	0.921
IBK-16	0.723	0.072	0.781	0.785	0.785	0.951
RF-8	0.73	0.072	0.72	0.73	0.74	0.90
RC-8	0.723	0.081	0.723	0.712	0.721	0.902
J48-8	0.723	0.193	0.72	0.725	0.722	0.979
IBK-8	0.713	0.195	0.717	0.713	0.714	0.812
J48-2	0.674	0.113	0.676	0.674	0.687	0.999
NB-64	0.672	0.115	0.68	0.672	0.695	0.989
RF-2	0.667	0.114	0.669	0.667	0.668	0.968
NB-128	0.664	0.11	0.755	0.664	0.634	1.003
NB-256	0.641	0.111	0.641	0.641	0.616	1.005
IBK-2	0.643	0.118	0.628	0.643	0.644	0.965
NB-32	0.634	0.213	0.628	0.634	0.564	0.985
NB-2	0.516	0.219	0.623	0.516	0.529	0.952
NB-16	0.593	0.23	0.515	0.597	0.528	0.97
NB-8	0.588	0.22	0.589	0.587	0.523	0.911

Table 3 Well developed matrix

	RC-0	RC-1	RC-2	RC-3	RC-4	RC-5
AC-0	1717	3	2	0	0	0
AC-1	12	1513	19	0	0	0
AC-2	9	25	1372	0	0	0
AC-3	0	0	0	1471	235	71
AC-4	0	0	0	231	1658	17
AC-5	0	0	0	71	28	1845

AC-5) versus Recognized Class) displays the classifier's ambiguities and misclassifications (RC-0 to RC-5). Class 1 (walking) and Class 6 (lying) exhibited the fewest discrepancies in the classifier's identification of the six activities tested. There is some ambiguity between sitting and standing, and it's downright impossible to differentiate between ascending and descending flights of steps. When compared to earlier attempts at activity identification using a single smartphone worn at the waist, this is a significant advancement.

6 Conclusion

In this article, by using machine learning, identify smartphone activities. Based on the ranking theory, machine learning automatically takes actions and records. This work is validated using well-known techniques like Lazy learning, random forests, and ensemble learning. For analysis purposes, prepare datasets according to the classification technique. Due to the classification technique, the dataset will develop accurately, concerning time, and a well matrix will create. Data mining technique is implemented in different smartphone to analyze various activities. An unsupervised algorithm is needed to construct intense real-time work on smartphones. At the end of the research IBk classifier performed well at 128 and 256 features.

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IoT Based Smart Medical Data Security System

Lakshman Kumar Kanulla, G. Gokulkumari, M. Vamsi Krishna, and Santhosh Kumar Rajamani

Abstract Health 4.0 is an approach to healthcare innovation using IoT and other sensors and devices. The result is an array of intelligent health applications that are more equipped to improve people's health and well-being in practical ways while also being more reliable, scalable, and economical. However, IoT based healthcare systems may pose problems without proper oversight, especially regarding security concerns like exposed application interfaces. Primary challenge is to learn about the architecture and security needs of IoT based multi-sensor systems and healthcare infrastructures. In addition, it has to propose lightweight, easily implementable, and efficient designs. This research introduces the Internet of Things (IoT) in healthcare and a thorough analysis of practical, novel health frameworks that use a wide range of resources and limited-power sensors and devices. Additionally, this paper also focuses on the safety of these vital Internet of Things components and their wireless connections. The result is the introduction of a lightweight-based security system that uses the Lightweight Encryption Algorithm by IoT (LEAIoT). Essential creation with the proposed hardware-based method is 97% faster than with a software-based approach, and encryption/decryption is faster by 96.2%. Finally, it is competitive with other typical hardware-based cryptography designs, achieving reduced hardware usage of up to 77% with the lowest frequency with its lightweight, flexible implementation and configuration of high-speed keys.

Keywords IoT · Health 4.0 · Multi sensors · Security · Frame work

L. K. Kanulla (✉)
Department of SAP SD MM, Working for Apple Inc, Cupertino, USA
e-mail: knvslakshmankumar@gmail.com

G. Gokulkumari
Department of E-Commerce, College of Administration and Finance, Saudi Electronic University, Riyadh 13323, Kingdom of Saudi Arabia
e-mail: g.govindasamy@seu.edu.sa

M. V. Krishna
Department of IT, Aditya Engineering College, Surampalem, India

S. K. Rajamani
Department of E.N.T, MAEER MIT Pune's MIMER Medical College, Dr. BSTR Hospital, Pune, India

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1 Introduction

Medical gadgets changed healthcare. Now we can track our health without going to the hospital. We need to examine the security of such devices not withstanding this drastic development.

These technologies compromise privacy and security. Medical device security is critical since many patients' lives rely on it. Healthcare security is crucial.

Wearable Internet of Medical Things devices diagnosis patient health. These gadgets monitor physical activity, temperature, diabetes, sleep, heart rate, and more. Smart wristbands, watches, glasses, belts, necklaces, and patches are available from head to toe.

Wearable systems include sensors, memory, solar cells, and batteries. They gather, display, and wirelessly transmit data. Gadgets may communicate patients' health data directly to doctors to reduce office visits.

Modern information technologies like the Internet of Things (IoT), big data, cloud computing, and artificial intelligence have made healthcare smarter. Care of this kind is more effective, convenient, and individualized than the alternative. Electronic healthcare systems (eHealth) based on the Internet of Things (IoT), mobile healthcare (mHealth), and ambient assisted living (AltHealth) are all part of "smart healthcare" (sHealth). Smart Healthcare Systems (SHS) are widely used presently because of their practical data storage and sharing system, fast reaction times, and reduced treatment costs. With the aid of IoT devices, patients' private medical records may be securely saved in the cloud and shared with doctors and other patients. As a result, the telecare e-medical service model may provide care recommendations based on data gathered from the patient's medical monitors. Care for patients with long-term conditions, including critical care emergency services, heart patient symptom records, and more are part of this process. These IoT based monitoring devices may collect vital signs and transmit them to a cloud server using implanted sensors. In the future, members of the savvy community may share this information [1].

It is widely believed that IoT will become standard in all technologies of the next generation [1]. In this context, "interconnection" refers to the linking together of innovative items and gadgets via which they may be detected solely. Invisible sensors connected to many things around us provide IoT with a wealth of tracking data [2]. Research indicates that health monitoring is the most promising field for future wearable electronics (HM). Smart HM [3] combines innovative computing, remote HM, and the Internet of Things.

HM expands clinical monitoring and care limits (i.e., house, for instance). An HM system consists of a smartphone with internet access and an HM app, a monitoring device for submitting health data to smart contracts [4]. Wearables and IoT are important for HM and intelligent cities [5]. Wearable devices capture patient health data for healthcare administration, diagnosis, and patient care. A Big Data scenario [6] arises as medical records are analyzed and shared. A secure data interchange between organizations is also required [7].

Security is a significant consideration for any setup. There are several security definitions since individuals have different perspectives [8, 9]. In a broad sense, security may be thought of as a concept analogous to the system's overall stability. Most modern IoT-centric HM relies on wireless connectivity, posing several potential security risks [10]. These security concerns might cause significant difficulties for wireless sensor equipment [11]. Thus, medical and health data management needs lightweight block encryption techniques for medical IoT resources [12].

Predicting abnormal health changes from the Internet of Things data [13, 14] is accomplished via the use of data mining techniques, including classification and clustering [15], neural networks [16], and other machine learning approaches. The study that utilizes clouds and IoT technologies forms the foundation for a secure patient HM system using BC-XORECC and a patient monitoring system utilizing LSK-RNN, which together permit the safe transfer of data and offer accurate patient monitoring. As a result, doctors could keep tabs on the patient from afar and catch potentially fatal conditions in their earliest stages.

They are only concerned with the healthcare features of the app and the savings on deployment. The outcome is a system readily exploited due to poor security measures and inadequate upgrades. The scalability creates another potential weakness. In particular, new devices are being added to the system without any assurance that they will maintain its security. An attacker may access the more extensive system by compromising a tiny, unsecured device. Thus, even the smallest devices interacting with the centralized services need some lightweight security strategy that keeps them safe from harm.

2 IoT Based Healthcare Security System

IoT technology can develop several smart health apps, accomplishing Health 4.0 goals [4]. Figure 1 shows the integrated technology and healthcare architecture components of Health 4.0. High-quality services for persons with diverse healthcare requirements are a crucial goal that involves optimizing tools, resources, and system performance. Automation and intelligence may improve outcomes and speed up monotonous activities. Remote access and real-time answers aid medical care and monitoring. Finally, designing databases with complete and easy-to-access medical information helps improve diagnostics and tailored therapy.

Another important goal is to improve operations while reducing expenses, resource use, and energy usage. Thus, energy-constrained IoT devices may run healthcare applications. In the end, resources will balance the critical and actual requirements of the system. The best strategy maximizes performance throughput with few resources. IoT may help monitor, diagnose, and forecast illness through health sensor data.

Cloud services quickly transmit, evaluate, and store this data, making diagnosis more straightforward and accurate. Cost-effective, user-friendly, and promptly responsive health assessment procedures will relieve healthcare staff and materials.

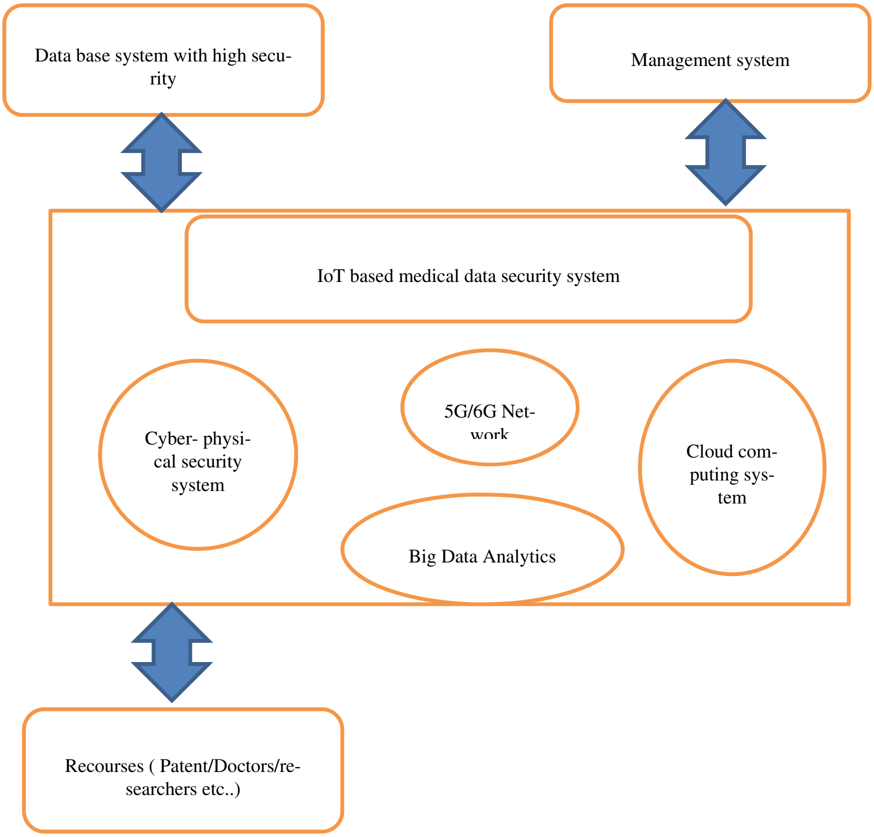

Fig. 1 Framework of the IoT based medical data security system

Finally, information exchange and cooperation between healthcare institutions and providers will be easy and timely.

2.1 Proposed Work

- First, the utilized cryptographic primitive guarantees the maintenance of the fundamental security principles while being lightweight and able to use keys ranging in size from 64 to 520 bits. The width of the key is a generalization of the measure of symmetric encryption's security; 520 bits is the minimum acceptable for most applications, as practical and crucial to the suggested setting.
- Assures efficient handling of the large amount of data sent between devices in the Internet of Things. Cryptographic primitive ran on a computer's processor

(Central Processing Unit), boosting essential creation speed by up to 99.9%, and the rate of encryption and decryption is as high as 96.2%.

- Third, four distinct key sizes are included in a single architecture, Applications may achieve four different performance rates and levels of security efficiency. Instantaneously choose based on the state of the network and the required amount of processing power for the application. As a result, the system is versatile and readily adapted to situations that vary considerably.
- In the end, implement the cryptographic technique to prove its worth as resource effectiveness without sacrificing throughput. It meets the performance criteria of intelligent health and is readily accessible when used on the different nodes of the IoT network, and it protects a national and international level organization of healthcare while keeping resources in a state of optimal performance, efficiency and safety.

3 Implementation Process

The IoT architecture comprises applications, networks, and physical/perception layers [10]. The application layer connects IoT devices [10]. E-health applications are included. Network layer protocols allow IoT components to communicate physical layer data. Popular networks include ZigBee, 5G, Wi-Fi, RFID, 6LoWPAN, and LoRaWAN. IoT-integrated WSN is another node network [14]. The physical/perception layer terminates architecture. It encompasses sensors, wearables, actuators, cellphones, antennas, and CPUs. This layer translates health signals to network data.

3.1 Infrastructure of IoT Based Medical Data Security System

Figure 2 depicts an IoT based Health infrastructure. This arrangement shows IoT-healthcare linkages. This article compares smart hospitals with near-patient/personalized systems. Individualized revolutionary health architecture comprises heterogeneous IoT devices, a wireless interface, and a cloud-based database [2, 3]. Medical equipment gets sensors first.

They use batteries, therefore maximizing efficiency is essential. To send “sensed” data, they must connect to the system’s wireless network. Multiple sensors and gadgets are used simultaneously, where their connectivity is crucial. Wearable equipment must also be lightweight and pleasant. Some implanted and wearable devices may wirelessly receive orders to modify medicine doses or gadget settings. The wireless interface must connect to the Internet to provide health data to physicians and nurses. The hospital or private clinic’s main computer or near-patient IoT system may process this data. Some wearables can analyze and wirelessly communicate

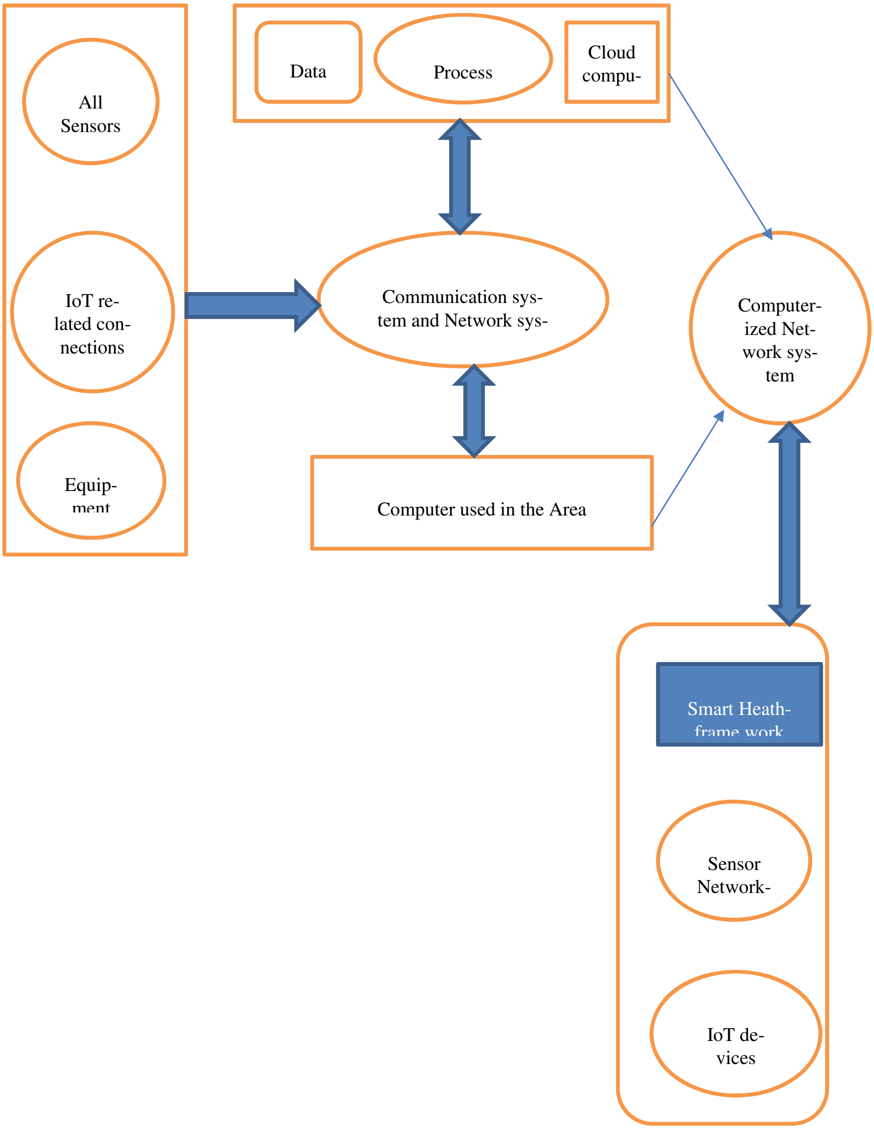

Fig. 2 IoT based health architecture

data to the Internet. Fixed or mobile devices may replace small, wearable, and implantable devices without processing capability. These intermediary devices evaluate sensor data from various IoT networks and transfer it to back-end systems and databases. They can also interface with sensors and process back-end data. These gadgets gain intelligence and real-time capability by making decisions and acting without back-end infrastructure. IoT devices lack storage. Thus, medical history is stored in databases.

IoT may improve hospital relationships and functionality.

Hospitals may use all implanted and wearable IoT devices. These sensors and gadgets must also be connected to wired and wireless networks and accept orders and sensitive data from authorized sources. Two things distinguish this hospital building:

With the help of the Internet of Things, hospital beds and other medical devices may now connect to the network and share confidential patient information for diagnostic purposes.

The hospital's medical records and healthcare database are to the IoT network. Thus, hospital staff may obtain real-time data and react to situations. All hospital devices can quickly retrieve the patient's medical history from the recorded data.

The hospital's Internet of Things (IoT) network communicates with other healthcare facilities and Internet-connected devices close to the patient.

More tailored and sophisticated health services will help healthcare professionals reduce hospital resource strain.

3.2 Security System and Scheme

High security is essential for IoT based healthcare applications. When it comes to attacks on the intelligent health infrastructure, the Internet of Things (IoT) network (Fig. 2) is the weakest link [2]. IoT network attackers may readily access devices' personal data. Eavesdropping and data transmission/traffic tracking are major data privacy breaches [10]. Data protection also affects user authentication. Unauthorized devices may access and manipulate this data. They may potentially send false health data to the IoT network. This causes misdiagnosis and inconsistent health-provider communication [17, 18].

Researchers prioritize safe communication network development. Cryptography protects data, authenticates users, and uses cyphers to encrypt and decode messages. Due to resource limits, the IoT system cannot employ cryptographic primitives [18, 19]. The cypher must not divert resources from other vital healthcare functions. Thus, IoT hardware restrictions need a lighter version. In crucial situations, low-speed algorithm implementation might delay real-time applications, which can be disastrous [20]. Therefore, must consider quickness and responsiveness. Finally, each capability must have numerous alternatives to meet the application's network and security demands. The system needs flexibility and scalability [21].

A lightweight cryptographic primitive and security method is needed to secure smart health application health data. Before IoT devices communicate data, the encryption method must encrypt it. Thus, patient data is safe from hackers. For healthcare applications, the decryption algorithm must decode this received data. The outcome is complete data content security in communication networks, particularly IoT networks and cloud-connected Internet.

4 Implementation Process of Security System and Scheme

The current system's lightweight-based security method leverages the LEAIoT cryptographic primitive to encrypt and decode data while offering variable key size and implementation speed. This approach is embedded into every Internet of Things (IoT) device in a healthcare system, securing sensitive patient information over public networks like the Internet and within private ones like smart hospitals and near-patient infrastructures. LEAIoT beats traditional encryption primitives in key generation and encryption/decryption speed. The IoT based healthcare system's complicated connection demands benefit from a lightweight design, and It enables fast end-to-end communication with little hardware. LEAIoT mixes symmetric and asymmetric encryption methods. Symmetric cryptography improves performance with fewer resources; Asymmetric primitives increase key distribution, scalability, secrecy, and authentication.

LEAIoT encrypts a made-up plaintext with an n-bit private key; the sender and recipient are well-informed. NLBC employs ciphertext with two legends: n1 and k. They are protecting encrypted content. Decryption uses the modular inverse of the three encryption keys, SSK, n1 (1), and k0. Delivered ciphertext and keys n1 (1) and k (0) are needed for asymmetric NLBC decryption. These keys are made using n1 and k's modular inverse modulo 27. Using symmetric decryption, you can acquire the plaintext if you know the modular inverse of the n SSK key. This technique continuously calculates the modular inverse modulo 27. The following material analyzes ciphering and decoding.

1. The key n multiplies the synthetic plaintext values. Modulo 27 follows;
2. The secret code k is a 3×3 matrix that keeps secret. Aside from that, the length of the key n is computed and used as the key n1;
3. In the first stage, the created text is split into sections using the critical k to identify each section. You are multiplying by k and n1 for each block bi. Here comes Module 27;
4. The secure ciphertext is the original plaintext.

Decryption sequence:

1. Use the modular inverse of the keys n1 and k modulo 27;
2. Divide the received ciphertext into blocks bi, as in step 3 of the encryption process;
3. Multiply each block with the two keys k 0 and n1 (-1). Modulo 27 follows;
4. The text is multiplied with SSK and modulo 37 again;
5. The result is displayed in plaintext.

The suggested lightweight-based security system. The Lightweight Encryption and Decryption for the Internet of Things (LEAIoT) use Symmetric and staged asymmetric encryption and decryption. Users may customize the playback duration of the key pairs with a symmetric key length of 64, 128, 256, or 520 bits.

5 Results

Simulate symmetric and asymmetric key insertion and modular inverse computation. Figure 3 demonstrate the two processes for different key sizes.

Each encryption/decryption cycle processes three 7-bit characters. Each function takes nine clock cycles. The first cycle starts with the encryption or decryption procedure, which complete in eight cycles. Table 1 represents the cycles needed in symmetric and asymmetric to generate both keys. These clock cycles provide the start signal.

Finally, the security and performance criteria will assess the design. Simulation validated the cryptographic primitive's security. It has four different vital sizes, so users may adjust it to meet their needs regarding network speed and security. A minor key size might speed up the encryption process when traffic is heavy on the

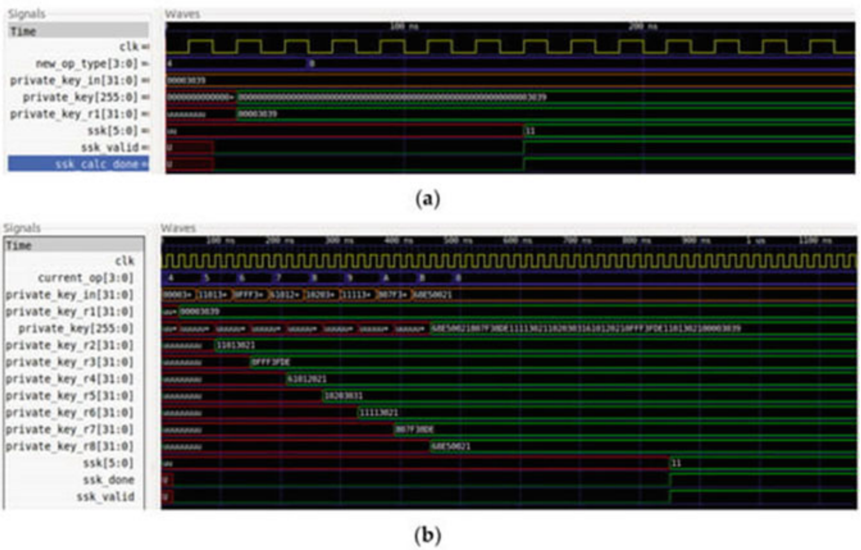

Fig. 3 The two processes for different key sizes

Table 1 Cycles needed in symmetric and asymmetric to generate both keys

Symmetric key size of the bit	Insertion of symmetric key	Inverse of the modular symmetric key	Insertion of asymmetric key	Inverse of the modular asymmetric key	Total
64	4	7	7	66	74
128	12	10	7	66	81
256	18	14	7	66	97
520	36	21	7	66	116

network. A more considerable key length may be a good option when protecting sensitive information. Transmission rates and system availability have both seen boosts thanks to faster key generation and encryption/decryption. So, it's fast enough for the Internet of Things. Finally, the synthesis results and comparisons with other hardware-based research demonstrated that the recommended design for IoT based healthcare systems is lightweight and efficient. New approaches to security are being developed to strike a good balance between availability, efficiency, and safety in an IoT based healthcare architecture.

6 Conclusion

This article presents an overview of IoT based multi-sensor architecture, the Health 4.0 design framework, and cutting-edge health infrastructure. This detailed environment overview guides IoT use in healthcare, whether for intelligent hospitals or tailored innovative health systems. The representative study helped me understand the domain's current situation.

General smart health infrastructure's top priority is data protection, and user authentication suggests a new hardware-based IoT security approach. The LEAIoT encryption/decryption algorithm gives the lightweight-based security strategy additional key selection options than existing systems. Thus, it may boost speed under network congestion. Compared to a CPU-based version, the hardware-based LEAIoT implementation is 99.9% quicker at key generation and 96.2% faster at encryption/decryption for 1000 kilobits. Compared to the lightweight cyphers AES, SNOW 3G, and ZUC, it utilizes 89.2, 64.2, and 13.4% less hardware in identical hardware devices. Even the tiniest devices can implement this architecture and be protected, making it useful in an IoT based multi-sensor ecosystem. Finally, its limited throughput and frequency reflect IoT devices' resource constraints. It is suitable for resource-efficient and fast critical generation applications. It also meets the IoT based innovative health framework's primary security and performance criteria, yielding novel outcomes and improvements.

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Investigating the Impact of Distance on the Reception in Molecular Communication

Ashwini Katkar and Vinitkumar Dongre

Abstract The development of nano-communication networks has greatly benefited from advances in nanotechnology, enabling complex tasks to be performed. These networks utilize molecular communication to enable communication be- tween transmitting and receiving nanomachines and have many medical applications, including targeted drug delivery. Nano-communication network approach minimizes the risk of adverse effects on healthy areas of the body by delivering drugs directly to the affected area. The accuracy and timeliness of drug delivery depend significantly on the distance between the transmitting and receiving nanomachines. This study explores the impact of varying this distance on the reception of molecular communication, including the maximum reception and the timing of peak reception. The results indicate that as the separation between the sender and receiver increases, the number of molecules received decreases while the latency increases.

Keywords Nanotechnology · Molecular communication · Targeted drug delivery · Nanonetworks

1 Introduction

Nanotechnology encompasses a broad range of applications that operate at the nanoscale. Molecular Communication is a field within nanotechnology that has been gaining increasing attention from researchers, as traditional communication methods such as electromagnetic and acoustic communication are inadequate at the nanoscale [1]. Molecular Communication involves using molecules as a carrier of information from the sender to the receiver. The transmitter emits molecules into its surrounding environment, which then propagate through the medium until they eventually reach

A. Katkar (✉) · V. Dongre
Thakur College of Engineering and Technology, Mumbai, Maharashtra 400101, India
e-mail: ashwini.katkar@vcet.edu.in

V. Dongre
e-mail: vinit.dongre@thakureducation.org

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the receiver [2]. Molecular Communication has numerous applications across a range of fields, including healthcare, environmental monitoring, military operations, and information and communication theory [3]. Molecular Communication is particularly effective in the field of medicine and healthcare because of its high level of biocompatibility [4]. Targeted drug delivery is a biomedical application of Molecular Communication. It involves delivering drug particles to the affected area, such as a cancerous tumor while avoiding healthy parts of the body [5]. By achieving an accurate concentration of drug molecules at the tumor site, it is possible to avoid toxicity resulting from high dosages [5–9]. If the transmission rate is too high it can result in the accumulation and loss of drug molecules. Conversely, a low transmission rate may lead to an insufficient reception of drug doses at the tumor site within the necessary timeframe, thereby hindering effective treatment. Proper placement of the transmitter and receiver is crucial to ensure the reception of an adequate dosage at the appropriate time.

This study aims to investigate how the separation between the sender and receiver impacts Molecular Communication. Specifically, the study observes the maximum reception of molecules at different time intervals depending on the distance between the transmitter and receiver. The results are then evaluated to determine the optimal placement of the transmitter and receiver to achieve maximum system throughput. The study aims to identify the exact effect of distance on the intensity of received molecules, the timing of maximum reception, and the peak reception of molecules.

2 Literature Review

The focus of much of the research in molecular communications has been on understanding the factors that influence system performance, including channel capacity, de- lay, attenuation, noise, Inter-Symbol Interference (ISI), and amplification. To achieve this, researchers have proposed and analyzed different models, such as diffusion-based, diffusion with drift, random walk, active transport models, random walk models with drift, and collision-based models [10–12]. Moore et al. [13] introduced a method where the transmitter sends molecules to the receiver, which in turn sends a feedback pattern of signal molecules. By analyzing this feedback pattern, the transmitter can measure the distance between itself and the receiver.

Al-Zubi et al. [14] have compared and evaluated different reception methods for molecular communication. The performance of passive and completely absorptive receivers was assessed by analyzing the peak amplitude and peak time. Sharma et al. [15] analyzed the 3-Dimensional molecular communication system model. The system's performance was evaluated based on receiver characteristics, erroneous alarms, capacity, and average error possibility. Lopez et al. [16] have presented a study on transmitters' location and desired signal strength. Islam et al. [17] aim to resolve the drug release synchronization issue in a local drug delivery system that employs multiple transmitters. Specifically, the objective of the study was to reduce the transmission period error resulting from propagation lag. Zhao and team [18] derived the minimum efficient intensity of drug particles from the minimum effective receiver capacity. The study of Yilmaz et al. [19] involves two main objectives: the first is to build an end-to-end simulator, which includes modeling and development, while the second objective is to evaluate the performance of proposed filtering and demodulation techniques for mitigating inter-symbol interference.

Luo et al. [20] have introduced a new method for estimating distance effectively. This method involves the release of two types of molecules by a sender nanomachine, with different diffusivities. The receiver can then use the differences in diffusivity to find the distance between the sender and the receiver. According to the authors, this method is highly accurate and has strong anti-noise abilities. To estimate the distance at which molecules are released, Huang et al. [21] have utilized a method involving the observation of the total of molecules received. This estimated distance is then used in conjunction with maximum likelihood estimation to guess the initial distance. To evaluate the efficacy of this approach, a simulation of Brownian motion using particles has been conducted.

A distance estimation protocol for a receiver nanomachine has been proposed by Lin et al. [22], which utilizes Maximum Likelihood Estimation (MLE) to achieve high accuracy. The protocol involves taking multiple samples of the concentration while considering ISI and additive noise. In order to achieve accurate distance estimation, the Newton-Raphson method has been employed. This iterative method refines the initial estimate until a maximum likelihood estimate is achieved. This ensures that the estimated distance is as close as possible to the true distance. According to the authors, this protocol provides a robust and accurate method for estimating distance in the presence of noise and interference, making it highly suitable for use in nanomachines and other applications where accurate distance measurement is crucial. Fatih et al [23] introduced a fluid dynamics approach to estimate the distance between two points. The fluid dynamics approach provides an effective way to estimate distance, which considers the behavior of liquid droplets and other factors that can affect them. Fatih et al [24] proposed that RX side can use a combination of practical methods for distance estimation. These include data analysis-based techniques and machine learning (ML) methods. According to the authors, the ML methods have demonstrated better performance compared to the data analysis-based methods at the expense of increased complexity. Su-Jin et al. [25] proposed a concentration-encoded molecular communication system using machine learning. According to the author, the ML receiver helps to minimize the bias effect and reduce (ISI). The research by Zhen et al. [26] explores the use of an amplify-and-forward method in a mobile three-dimensional (3D) environment. According to the authors, the study's results are anticipated to significantly impact the design of mobile multi-hop systems.

Maryam et al [27] offer a detailed analysis of practical techniques for generating chemical waveforms in microfluidic systems. These techniques can provide valuable insights into the development of pulse-shaping techniques for molecular communication. Ajit Kumar et al. [28] utilize an iterative maximum likelihood estimation approach to jointly estimate the locations, flow velocity, and diffusion coefficient. Oguzhan et al. [29] introduced a method for localizing multiple-point transmitters. The goal is to estimate the distances between a receiver and each transmitter point by examining the cluster sizes and the likelihood of molecules reaching the receiver within a certain timeframe and radius.

3 System Model

In the Molecular Communication study, a system model illustrated in Fig. 1 was employed. This model comprised a transmitter that acted as a point source and a receiver in a spherical shape, which were positioned at a distance of 'd' from each other in an aqueous environment. To replicate the movement of the transmitted molecules throughout the environment, Brownian motion was utilized to simulate the process of diffusion.

The diffusion coefficient, D is used to characterize Brownian motion, presented by

$$D = \frac{KbT}{6\pi \eta rc} \tag{1}$$

where,

Kb = Boltzmann Constant ($1.38*10^{-23}$J/K),

T= Temperature in kelvin,

η= Viscosity of the medium,

rc=Radius of the considered molecules

In this system, the modulation technique employed is Binary Concentration Shift Keying (BCSK). This technique works by transmitting molecules to represent the binary symbol 1, while the absence of transmission corresponds to the binary symbol 0.

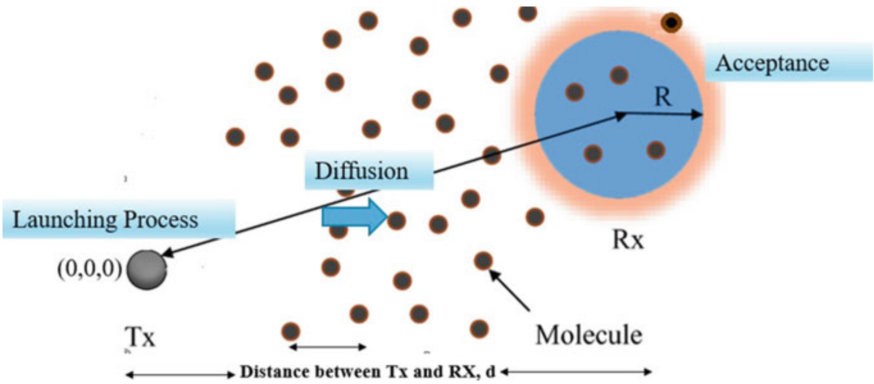

Fig. 1 System model for molecular communication

4 Methodology

The molecule diffusion and reception process shown in Fig. 2 starts with the release of molecules in the environment. Initially, they are set to zero position at the transmitter. The receiver boundary limit is set to check the number of molecules that can be absorbed by the receiver, remaining molecules outside the receiver molecules are not counted considering signal strength. Initially, the count of received molecules at the receiver is zero. Then molecule displacement is calculated by using the formula:

$$molecular\ displacement = normrnd(molecule\ position, sigma)$$

The following code will produce a random number from a normal distribution. The mean of the distribution is determined by the molecule's position parameter, while the standard deviation is determined by the sigma parameter.

$$\sigma = [\sqrt{2 * D * \delta t} \tag{2}$$

where,

D= Diffusion coefficient

The updated molecule position is obtained by adding the initial position with the displacement. Then, the position is checked to determine whether it lies within the receiver boundary. If it does, the molecule is considered successfully received, and the number of received molecules is incremented. Molecules outside the receiver boundary are not counted towards the received molecules, as they do not contribute to the signal strength. The total sum of released molecules is then checked to confirm that it is greater than zero. If this is true, a new molecular displacement is calculated, and the molecule position is updated to check whether it has reached the receiver boundary. If there are no molecules that have been released, the simulation will restart the diffusion and reception process for a new set of molecules. This approach guarantees that the system's behavior is adequately captured in the simulation, and any inaccuracies resulting from an absence of released molecules are avoided.

4.1 Implementation

Molecular communication system model considered in this study consists of a single transmitter and receiver in a 3D environment.

Table 1 describes several important simulation parameters for this study. Among these parameters, the Tx Node and Rx Node are critical components that can significantly impact the overall efficiency and reliability of the system. The transmitter (Tx) node is responsible for releasing information-bearing molecules into the environment, while the receiver (Rx) node is responsible for detecting and decoding these molecules to obtain the information they carry. The distance that a molecule

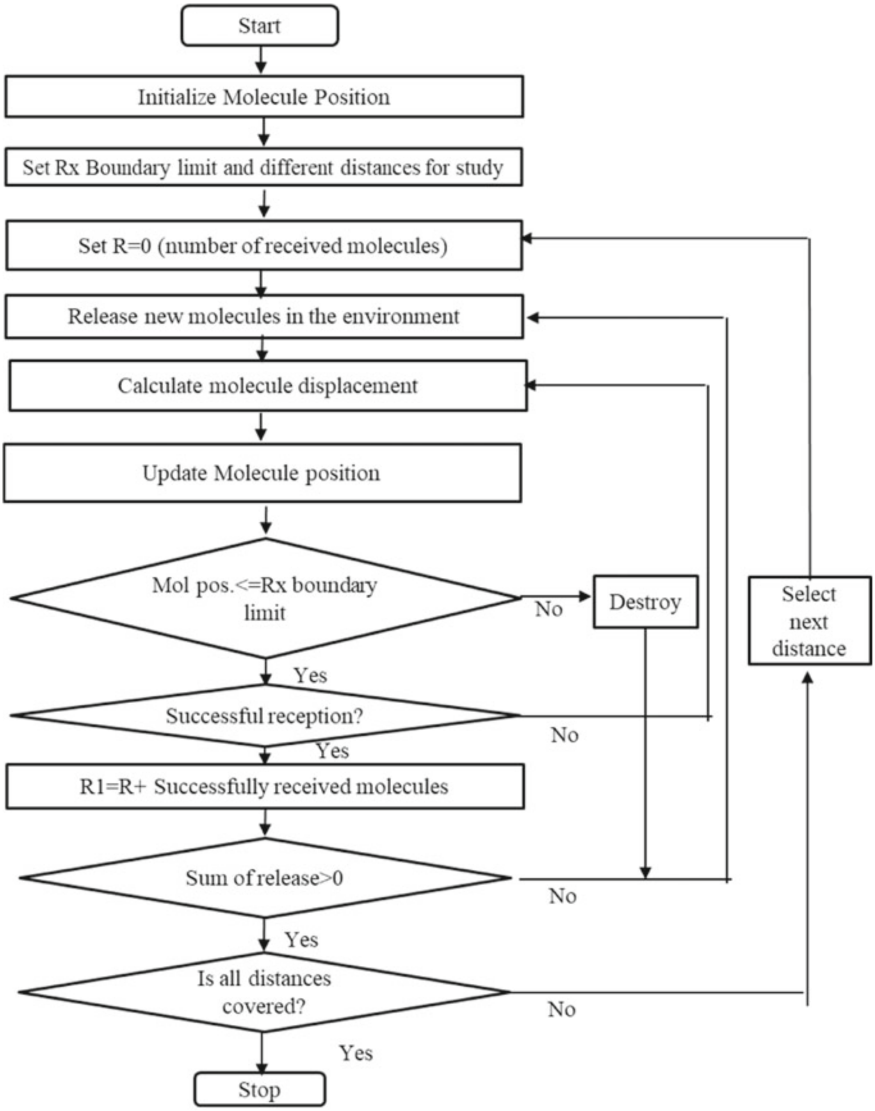

Fig. 2 Flowchart of proposed molecular communication

can travel before being destroyed or degraded is referred to as the destruction limit. In this study, a destruction limit of 60 μm was applied, meaning that any molecule that travels beyond this distance will not reach its destination as it will be degraded or destroyed before reaching the Rx Node.

The parameter Signal-to-Noise Ratio measures the quality of the received signal relative to the noise present in the communication channel. A higher SNR indicates

Table 1 Simulation parameters

S. no	Parameter name	Value
1	Tx node	1
2	Rx node	1
3	Tx node coordinates	(0,0,0)
4	Rx node coordinates	(6,0,0)
5	Destruction limit	60um
6	SNR	30 dB
7	Diffusion coefficient (D)	100 um2/s
8	Distance between Tx and Rx (d)	0.5 ~ 13 um
9	Molecule type	1
10	Symbol sequence length (nsym)	1
11	Modulation type	BCSK
12	Sampling duration (tss)	0.001 s
13	Simulation step size (delta_t)	0.001 s
14	Total simulation time (Ts)	1 s

a better-quality signal with less noise, and a value of 30dB corresponds to only one noisy molecule out of every 1000 transmitted molecules.

Another important parameter is the diffusion coefficient, which describes how quickly signaling molecules spread out in the communication medium. A moderate diffusion coefficient of 30 μm^2/sec means that signaling molecules can diffuse through the medium at a rate of 30 μm^2/sec. A higher diffusion coefficient leads to faster communication over longer distances, but it can also result in signal attenuation due to spreading over a larger area.

The symbol sequence length (nsym) refers to the number of signaling molecules transmitted to represent a single symbol in the communication system. BCSK (Binary Con- centration Shift Keying) is a modulation scheme that represents binary 0 and 1 by low and high concentrations of signaling molecules, respectively.

Sampling duration is another important parameter that determines the rate at which the receiver samples the concentration of signaling molecules to extract information from the received signal. A shorter sampling duration increases the rate of information extraction, but it also requires a higher level of reliability and throughput.

Finally, the simulation step size of 0.001sec indicates the frequency at which the simulation model updates the system variables and calculates the output values. These parameters are essential for understanding the performance of a molecular communication system and optimizing its design for specific applications.

5 Results

The simulation was carried out to understand the behavior of the system. The parametric study was carried out to observe the system's performance at various conditions.

The following figure shows signal strength received at different time intervals. Distance between Tx and Rx 2 μm, simulation step size 0.001 sec for a total simulation time of 1 sec. Figure 3 displays the signal strength received at different time intervals. The first realization and average realization show nearly equal signal strength, indicating accurate reception. The maximum signal strength observed is 88 for 100 molecules sent, which relates to the number of molecules received at 0.02 sec.

Figure 4 illustrates how the maximum reception time is impacted by the distance separating the transmitter and the receiver. Specifically, the distance between these two points has a direct influence on the maximum extent of time it takes for molecules to reach the receiver. Increasing the separation between the two components leads to a lengthier time required for molecule reception.

The impact of varying the separation between the sending and receiving entities on the maximum reception at different time intervals is shown in Figs. 5 and 6. As the gap between the two components increases, the time required for the reception also increases while the reception itself decreases.

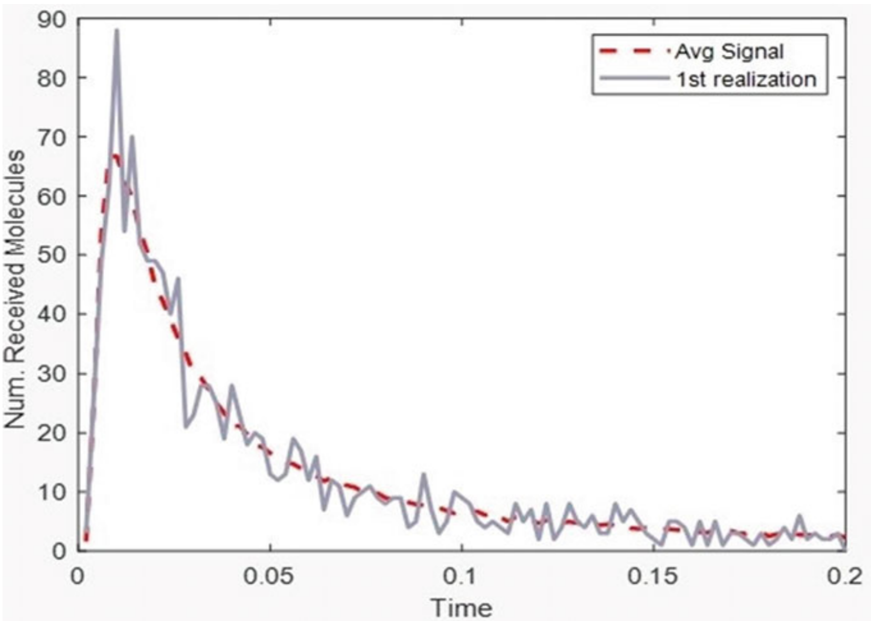

Fig. 3 Signal strength at different time intervals

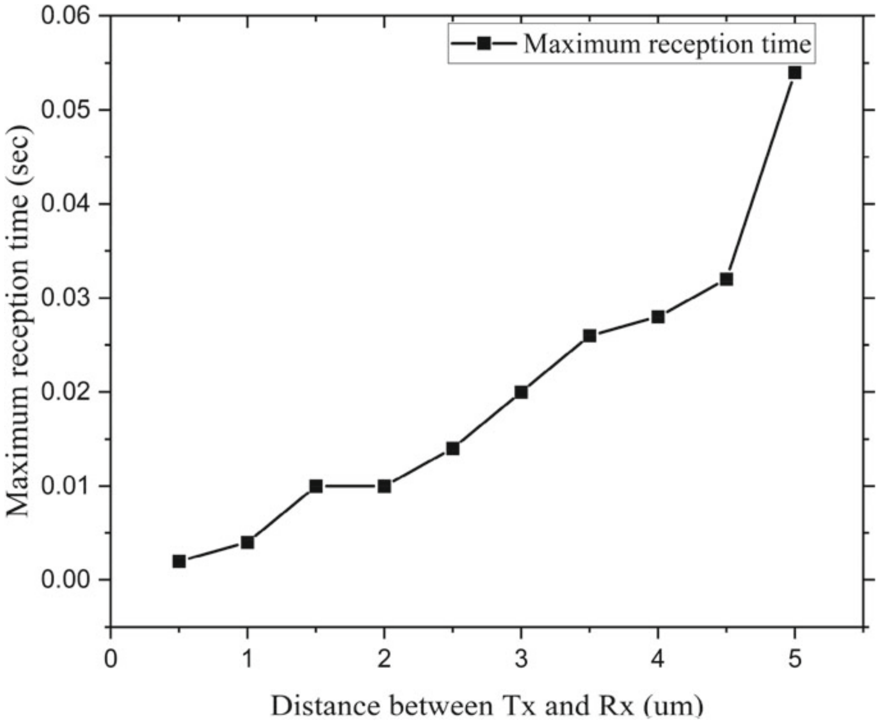

Fig. 4 Plot of Distance between Tx and Rx versus Maximum reception time

We evaluated the system's performance by varying the separation between the transmitter and receiver and analyzing the resulting peak reception of molecules. The theoretical formula for calculating Etpeak [10] is given by,

$$Etpeak = (d2/6D)Sec \tag{3}$$

Figure 7 shows the relationship between distance and Etpeak is depicted in both the analytical and simulation scenarios, with the analytical formulation validated by simulation results. An observation was made that doubling the distance between the transmitter and receiver leads to a four-fold increase in the time required to achieve the peak reception.

6 Discussions and Future Scope

The study conducted in this research paper has highlighted the influence of distance on reception in molecular communication. The outcomes of this analysis have several implications for research in this area.

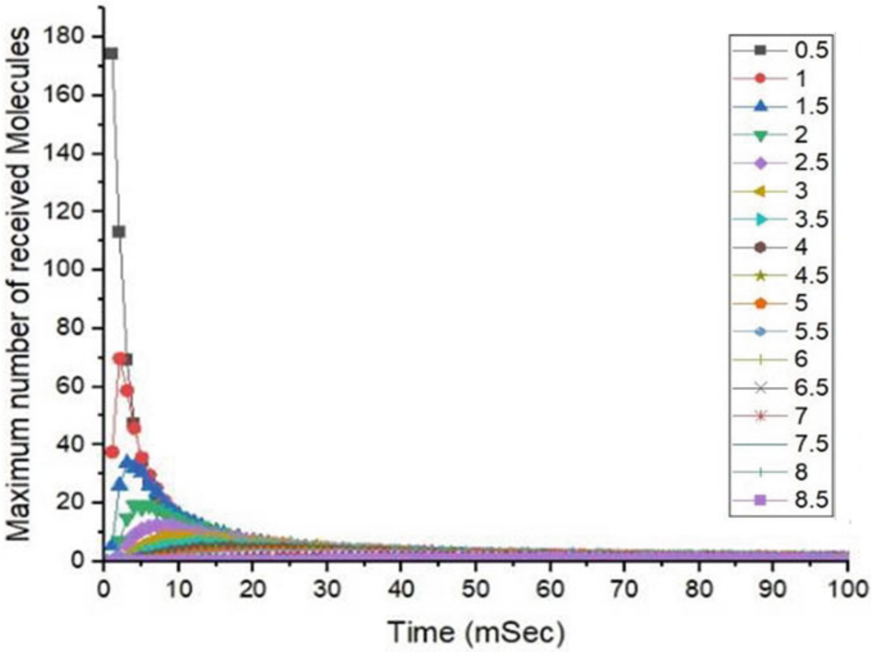

Fig. 5 Received molecules observed at different time intervals for changing distances between transmitter and receiver

One potential avenue for future research is the exploration of the effect of different environmental conditions on the reception of molecular communication. For example, researchers could investigate how temperature, humidity, and other environmental elements change the performance of the Tx Node and Rx Node. This could provide insights into the optimal operating conditions for molecular communication systems.

In addition, future research could investigate to examine the applicability of molecular communication in diverse fields, including environmental monitoring and industrial automation. Through this exploration of molecular communication potential beyond its current application in targeted drug delivery, researchers may identify novel areas where this technology could be employed.

The study conducted in this research paper focused on the reception of molecular communication. Future research could investigate the transmission of molecules and how different factors, such as the number of molecules transmitted and the duration of the transmission, affect the performance of the system. This could provide insights into the optimal design and operation of molecular communication systems. In summary, the results obtained from this research hold significant relevance for the advancement of molecular communication networks and lay the foundation for further exploration in this field.

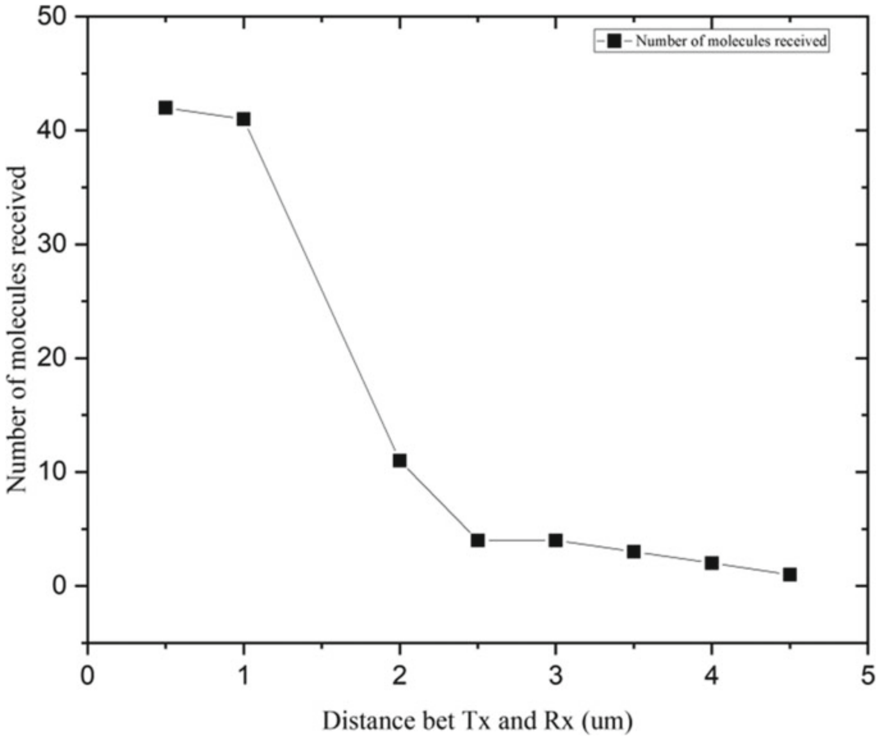

Fig. 6 Distance between Tx and Rx versus number of the received molecule

7 Conclusions

The purpose of this study is to investigate how distance impacts the maximum range at which molecules can be received in a three-dimensional molecular communication system that relies on the diffusion process. Specifically, our study aimed to investigate how the distance separating the transmitter and receiver affects the number of molecules received, peak reception time, and maximum reception. Through simulation, we found that an increase in separation between the sender and receiver results in a drop in the numberof received molecules and an increase in the time required for reception. We also observed that the time needed to observe the peak reception increases by a factor of four when the distance is doubled, which confirms the theoretical formula. Future work will investigate molecular collision during reception and transmission rate optimization.

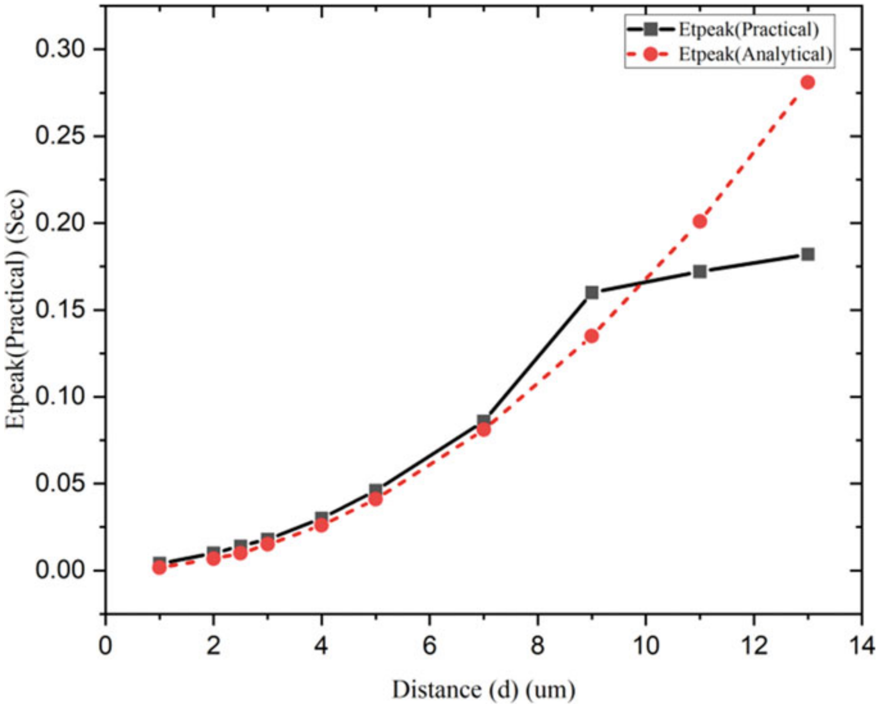

Fig. 7 Distance versus Etpeak

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Skin Cancer Multiclass Classification Using Weighted Ensemble Model

S. R. Nalamwar and S. Neduncheliyan

Abstract

Background Statistic shows that skin cancer is growing rapidly over the Globe due to the Sun ultraviolet ray radiation. Handling skin cancer is a difficult task for the dermatologist due to a lack of awareness of this disease, the Cost and time required for treatment as well as lack of clinical services and adequate expertise. Early detection of skin cancer using the Artificial based Automated system can increase the survival rate of the patient. **Methods** Our proposed system with ensemble model is developed for the detection and classification of skin cancer images in the early stage. This model used the preprocessing techniques like segmentation and black hat filtering for the removal of the hair from the image. Work on five pretrained models ResNet50, ResNet 152, Inception V2, Inception V3, VGG 16 and VGG 19. We use the HAM 10,000 open source dataset from the Kaggle. **Results** All five pretrained models work excellently on the dataset by using preprocessing techniques. ResNet 50, ResNet 152, VGG 16, VGG 19, Inception V2 and Inception V3 gives accuracy 88%,89%,82%,88%, 82% and 87% individually. Our proposed weighted average ensemble learning model improves the accuracy of the result by up to 93% by combining the different models and assigning proper weights to each model. As result shows the ensemble model performed well by giving proper weight to each model. **Conclusion** Our proposed model with ensemble weighted average is performing well as compared to the individual model performance. It is very useful to the Dermatologist for the detection of skin cancer in the early stage.

Keywords Skin cancer · Ensemble learning · Data augmentation · Transfer learning

S. R. Nalamwar (✉) · S. Neduncheliyan
School of Computing, Bharath Institute of Higher Education and Research (BIHER), Chennai, Tamilnadu, India
e-mail: sonalinalamwar77@gmail.com

S. Neduncheliyan
e-mail: dean.cse@bharathuniv.ac.in

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1 Introduction

According to the world health Organization report the global skin cancer cases are increasing drastically [1]. Cancer is developed in human beings due to the ultraviolet rays and ionizing radiation. Most people are suffered due to skin cancer due to infection of some viruses, parasites or bacteria. Genetic factors can also affect skin cancer detection. The International Agency for Research on Cancer (IARC), is continuously working on the finding of the factors for the development of skin cancer in the body of the human being. 30–50% of skin cancer can be cured if detected at an early stage and treated accurately with proper medication [2]. Main challenge in the early detection is all types of skin cancers have the same features as well as skin images shows the same patches. To improve the outcome of the automated system is the early detection of the skin lesions. Survival rate of the patient is increased due to the prediagnosis of the disease in the early stages. Melanoma cancer cases are found in men as well as women up to 4.3 million in the United States. Everyday 2 to 3 people die due to the skin cancer in the United States [3]. As the research shows that prediagnosis of the disease can help to increase the survival rate of the patient. Major problem in the diagnosis of this disease is the similarity of the images as all skin lesions images look similar. Each image requires preprocessing like skin removal and cropping for the handling of the image properly. Most of the Dermatologist diagnose the disease by seeing the skin with naked eyes which can result in the inaccuracy of the skin disease detection. Experience of the Dermatologist also plays an important role in the diagnosis of the image. To solve this problem and improve the accuracy of the diagnosis by making the automation of the system using the Deep Neural Network. Automated system helps the Doctors to detect the disease in the primary stage with accuracy. Automated system can work as a second opinion to the Doctors. Informative AI system is also introduced which can help to take the feedback from the Dermatologist and use this feedback to improve the system by changing the Hyperparameters or changing the algorithms. If it diagnoses in an early stage, then no need to do the Dermoscopy in which the image surface is evaluated and helps to diagnose the disease. Recent advancements in the technology lead to the development of automated system for the detection of diseases in the early stages. In recent days, most of the researchers are working on the development of automated image classification systems for various fields like Education, Medical, security etc. Skin cancer statistic shows that the melanoma estimated death is increasing. To handle this problem, the automated artificial intelligence model extracts the features from the Images [4–6]. Machine learning algorithms are used for the development of automated system. Machine learning algorithms are having some limitations like handling of big data. Working with images machine learning algorithm does not give accurate results as the size of the image is very large. As well as the user is going to extract the features manually so there are chances of overfitting or underfitting. All the above limitations are handled by using the Deep learning algorithm. Convolutional Neural Network (CNN) is having the Neurons as input. It is a multilayer Network consisting of Convolutional layer, Pooling Layer and Dense Layer (Fully connected Layer Classification accuracy is destructed due to the noisy images [17]. Image Enhancement is one of the most important step in the image processing by reducing the noise. Most of the images are having noise like various types of noise, packet loss and missing images can reduce the performance of the system [7]. Low light image affects the quality of the image which is becoming an obstacle in the image processing. Image acquisition and transmission are the steps in image processing that can be affected by the impulse noise [8]. Images are damaged due to the outliers in the image. These outliers affect the quality of the image due to the changing intensity of the pixels [9]. To handle a large amount of data mostly use dropout which also leads to tampered images which leads to a decrease in the quality of the images [10]. The restoration of the image from the damaged image preprocessing steps are very useful but it is very costly. To overcome this problem, we proposed an automated skin cancer detection system with the Deep Neural Network for accurate results.

2 Literature Review

In recent days, Deep Convolutional Network is a powerful tool for the classification of the images in different classes. Auto feature extraction is possible by using the deep learning approach. Handling of a large amount of data is possible using the CNN. On the kugel, the challenges are given to the researcher to solve some medical field related problems. Classification of the skin images in the early stage using the CNN can help the Dermato logist to improve their treatment. Transfer learning previously trained model help to use their experience to increase the accuracy of the model [11]. In the transfer learning models are trained previously with the different datasets we can use this for training our models. Our dataset trained with transfer learning model can make an efficient model. There are some very popular architectures like VGG-16 [12], ResNet [13], InceptionNet [14], Xception [15], DenseNet [16], MobileNet [17], ResNeXt [18], SeResNeXt [19] etc. These various architectures have different layers, size and steps to implement ImageNet [20]. It is working on more than 1000 classes so it can help to improve the efficiency of the model with the multiclass. The author developed a model using the MobileNet. The last some layers are replaced with the dropout and the last dense layer use the softmax activation function for the classification of the images [22]. Due to the activation function, number of Neurons are decreased and it helps to increase the speed of the classification. To increase the speed of the classification, we need to reduce the number of parameters. The authors investigated the effect of data augmentation and up sampling and in most of the research, researchers try to use a combination of the images of the different datasets which can lead to the accuracy of the results [23]. In this work, we are working on the Ham 10,000 dataset. But for the validation, first we are working on the preprocessing of the image using segmentation, new image reconstruction and removal of the hair. Imbalance dataset handling is very crucial work [24] which can be handled by using various methods like upscaling and downscaling [25]. The user's smoking habit also can lead to skin lesion increment [26]. Deep convolutional Network [27] with the Transfer learning[27]can definitely increase the accuracy of the result up to 92% [28]. The skin lesions detection is done using 2D and 3D skin lesion images in some diseases Chest X-rays also use to detect the disease in the early stage in place of doing the scanning [29]. Since images in the dataset are not sufficient then they handle the overfitting problem by using the methods like flipping, zooming and rotating the images [30]. As the literature shows that the first need is to do the preprocessing of the image, handling the imbalance dataset is a very important task. In this work, we are first working on the preprocessing of images and handling overloading problem by using the various data augmentation method. In the following section of the paper, all these methods are described.

The remaining paper has different sections. Section 3 Preprocessing and dataset details, Sect. 4 Implementation Details, Sect. 5 Experiment Results and Sect. 6 Conclusion and Discussion.

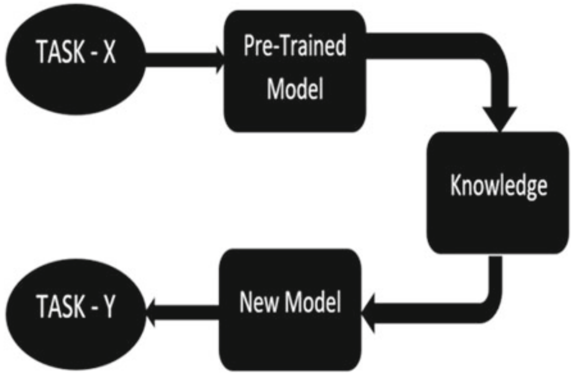
3 Methodology

This section elaborates on the methodologies used for the extraction of the features, detection and classification of skin cancer images. In our work, the first step is to do the preprocessing of the dataset by data cleaning methods, Training model, Visualization of the Model for the datasets and Standardization and Normalization of the dataset by using the scaling method. Dataset is huge so we use dropout so that it can show the results properly. First and most important task in CNN is the selection of layers' numbers. Dense layer is used in CNN for the improvement in the results. Last step is to visualize the testing results as an output with the Precision, Accuracy, Recall, F1 Score and Support. To achieve more performance in term of accuracy we used the pretrained model knowledge of the classification of images on a large amount of different datasets. In machine learning reuse of knowledge, the same old task for the execution of a new task is called Transfer Learning. It is useful to increase the prediction of new task from the knowledge learned from the old task. This method is useful for reducing the use of resources as well as training data amount required for the training of the new model. Knowledge from large amounts of data handling can be used to handle the new task without working from scratch. Figure 1 reports how to perform the Task. y knowledge pretrained model is used to develop the new task.

3.1 ResNet

In 2015 Microsoft Research team has given the concept of Residual Network [13]. In this layer hundreds of layers are present and also it keeps the effectiveness of the system. Residual Network is used mostly to train the model due to its accuracy.

Fig. 1 Transfer learning



Whenever the weight adjustment is required it uses the backpropagation and also handles the vanishing gradient problem which occurs with very less weights. It uses the skipping layers and also adds more information in the deeper layers to handle the vanishing gradient problem. It uses the 3×3 size kernel to handle the convolutional layers also the pooling layers are used after the conv layer to shrink the dataset with the selection of appropriate data from the dataset. Resnet has various versions depending on the number of layers used in the Model ResNet50 with 50 layers with the global average pooling to handle large amounts of data is followed by the Dense Layer at the last Softmax function is going to use for the classification with more accurate results. Same way ResNet 152 uses the 152 layers including Conv, Global Average pooling, Dense and Softmax functions. It is used to increase the accuracy without giving the overload on the dataset means handling computational complexity.

3.2 *Xception*

Google Researcher introduce the Xception model with the deep Neural Network. It is going to use the skip connection concept the handle the Vanishing Gradient problem which occurs due to the very small weights by Backpropagation weight adjustment. Depth wise separable Convolutional are present due to that the computation time is less and the more accuracy of the results. As the results shows that this pretrained model which is trained on the different data types can give more accuracy with the 71 layers also.

3.3 *VGG 16 and 19*

It is a Convolutional Neural Network with 16 layers deep. It's a pretrained model for image classification. This network has an image size of 224-by-224 from ImageNet

Dataset. VGG 16 with 16 layer is a pretrained model for image classification. It is more accurate and popular due to the increase in the layers to achieve more training and testing of the model. The network has an image input size of 224-by-224 from ImageNet dataset. Figure 2 shows the VGG 16 architecture with 16 layers.

Top-5 test accuracy in ImageNet is achieved up to 92.7% using VGG. It works on 100 classes including 14 million images. It is having a significant improvement over the AlexNet by using the 3*3 kernel size filters in sequence.

VGG 19 as the name indicates a total of 19 layers out of 13 layers are Convolutional layers, 5 are Pooling Layers and 3 are Dense layers to improve the efficiency of the system. As the number of layers increases the outcome of the result is also going to improve as shown in the results.

To increase the performance of the system in VGG 19 three more Convolutional layers are added. As our results also show that VGG 19 has improved accuracy over the VGG 16. VGG 19 as the name indicates 19 it has 19 layers out of which 16 are Convolutional layers, 5 are Pooling Layers and 3 are Dense layers to improve the efficiency of the system.

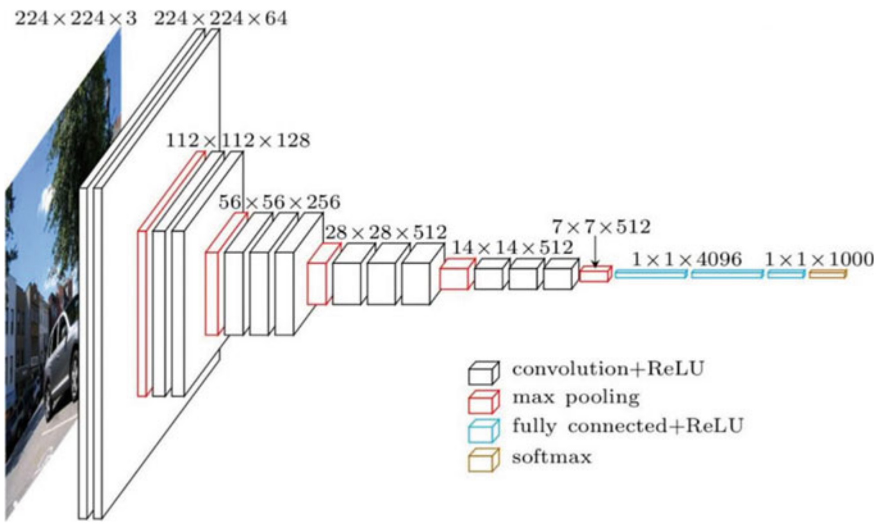


Fig. 2 VGG 16 architecture

Table 1 Dataset details

S. no	Type of cancer	No of images
1	Basal Cell Carcinoma (BCC)	541
2	Dermatofibroma (DF)	115
3	Benign Keratosis (BKL)	1099
4	Actinic Keratosis (AKIEC)	327
5	Malignant Melanoma (MEL)	1113
6	Vascular Lesion (VASC)	142
7	Melanocytic Nevus (NV)	6705

4 Implementation Details

4.1 Dataset Details

As shown in the Table 1 HAM 10,000 dataset consists of the images related to the seven different types of skin cancers. It is having a total of 10,015 images with seven different skin cancer categories. The main problem in this dataset is the imbalance of data of some classes having more images as compared to the other classes. Due to that the training is not performed properly. In this work to solve the data imbalance problem we use the upscaling and downscaling methods to generate the abstract images and try to make a balanced dataset for training. This dataset is divided into two subparts training and testing to check the performance of the system. Preprocessing on the dataset is done at the time of the prediction of the images. As Fig. 3 shows some sample images of the HAM 10,000 dataset. When these images seen with the naked eye all are mostly look similar and due to that the dermatologist confuse them with the skin lesions type detection and which lead to an increase in the time span for the proper detection of the skin cancer. The size of the image is 450*650 containing 10,015 labeled images. It includes images of the seven cancer types like Pigmented Benign Keratosis, Melanocytic nevi, Dermatofibroma, Melanoma, Vascular lesions, Vascular Lesion, Basel Cell Carcinoma and Actinic keratosis.

4.2 Data Preparation

Data preparation steps deal with the extraction of important features without any obstacle. As the images input our model has the noise as Fig. 4 shows the different type of noise present in the images. In the preprocessing step, we first remove the noise and try to make the image clear. another major obstacle is the hair present on the images. Second step is to remove the hairs from the image and try to make the image with more clear intensity. We train the model on the images those are given in the HAM 10,000 dataset having the pre-proposed data but at the time of prediction

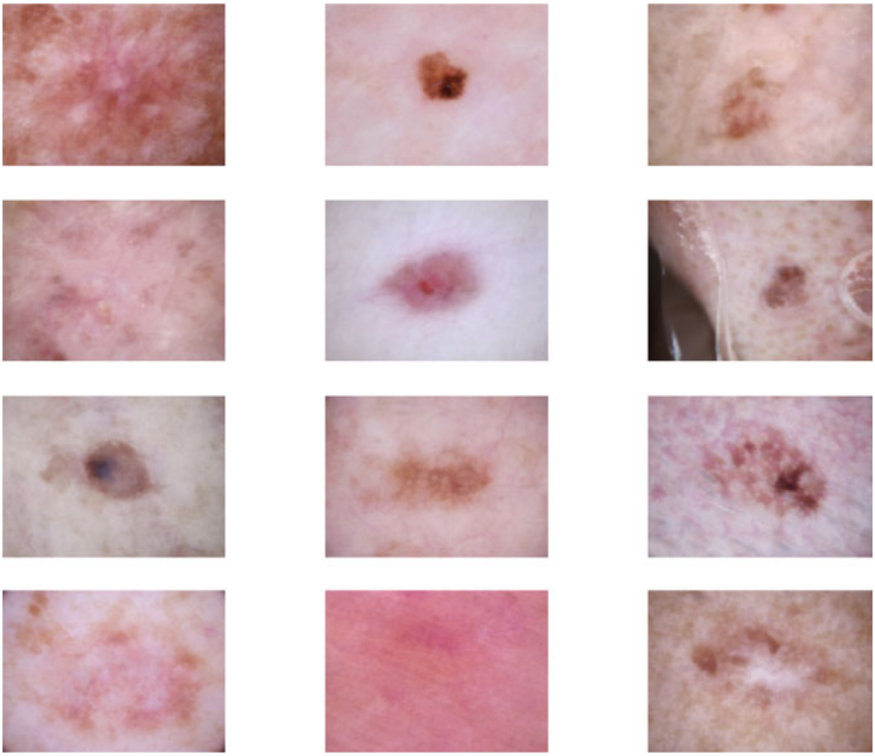


Fig. 3 Random images from HAM 10,000 dataset

we need to handle all these data preprocessing steps to make the images match with the images present in the preprocessed dataset. Data is divided into the training, testing and validation dataset. Training and testing dataset is used in the evaluation matrix to calculate the accuracy of the model individually. Validation dataset is used to check the performance of the proposed system.

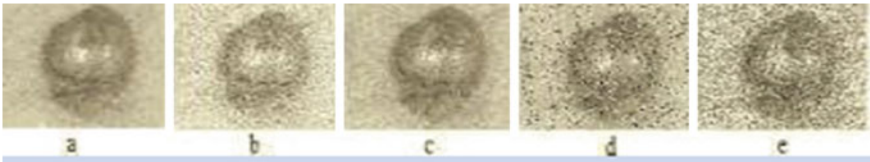


Fig. 4 **a** Image without noise **b** gaussian noise **c** poison noise **d** salt and pepper noise **e** speckle noise

4.3 Class Balancing

The first step for the preprocessing is to make the class balance. There are a number of techniques used to make class balancing like upscaling, down sampling, combine up sampling and down sampling, generating synthetic data and balanced class weight. We are working with the up sampling and the balance class weight. In up sampling the minor class samples are increased by creating the abstract or duplicate class images that are trying to match with the sampled class. Eq. 1 to calculate the class weight.

$$\text{class\_weightj} = \frac{\sum_{i=1}^{7} \text{Ni}}{Nj} -------- \quad (1)$$

where N_j is the number of samples in jth class.

4.4 Image Preprocessing

In this work, we are going to use the enhanced adaptive snake algorithm and Global threshold method for the image segmentation. We are constructing on the three dimensional (3D) skin lesion depth based feature extraction method. Most of the researchers are not constructing on the depth based feature extraction due to depth based feature extraction the 2D as well as 3D images are constructed as shown in Fig. 5. Skin cancer lesion are preprocessed and after that segmentation form the 3D reconstruction. Due to the 2D and 3D Feature selection, the performance of the classification is increased and are measured in the form of Accuracy, sensitivity, and Recall. As shown in Fig. 6 Gray scale image with the Red channel, Green channel and Blue Channel Output. Feature selection by extracting features like lesion color, shape and thickness of the lesions.

Fig. 5 Proposed CAD system for image reconstruction

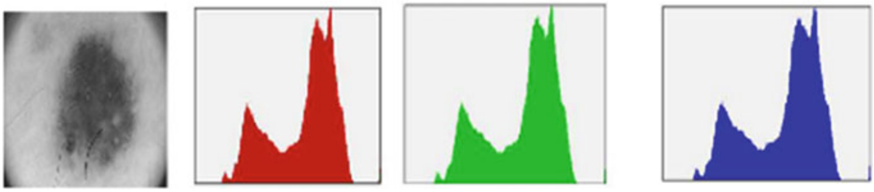


Fig. 6 **a** Gray scale **b** red channel output **c** green channel output **d** blue channel output

4.5 Data Augmentation

As in the Deep Convolutional Network work well on a large amount of the images. If the images in the dataset are not sufficient then the overfitting problem occurs means after some epochs the results and Data augmentation technique is used to increase the number of images in the dataset as well as reduce the overfitting problem. In data augmentation zooming, rotation, and flipping techniques can be used to increase the number of images. It is a very important step to handle the major problem like overfitting.

5 Result and Discussion

Deep learning requires a large amount of data and in the medical field labeled data shortage so handle the dataset with the shortage of the image by using Data augmentation. Next steps to handle the overfitting problem also initialization of the weight in place of initialization of the weight randomly. In the pretrained model, layers are going to use to select the appropriate features in each layer sequentially which can improve the performance of our model. Figure 7 shows the architecture of the model used in our system. By using Data augmentation, we are able to handle the overfitting problem. Researchers showed that the data augmentation and initialization of weights are very important for the training of the dataset with insufficient as well as imbalanced datasets. To handle the images, we use the global averaging for the handling of the pretrained models. Normalization technique like Batch normalization, to handle the data in each epoch use the Dropout so that the time required for the execution is reduced and lastly add the Dense layers with the 512 Neurons. In the last layer, 7 Neurons represent the 7 classes classification with softmax activation function. We use different batch sizes depending on our processor capacity and train the model with different batch sizes ranging from 16 to 32. Another hyperparameter we use is the learning rate ranging from 0.001 to 0.1. As the results show changing the batch size with the learning rate is improving the result of our system. Table 2 shows the results with a learning rate of 0.1 and batch size of 32. Dense layer uses the Relu activation function and the last layer uses the softmax activation function to achieve the final results. Figure 5 shows the structure of the proposed model. As

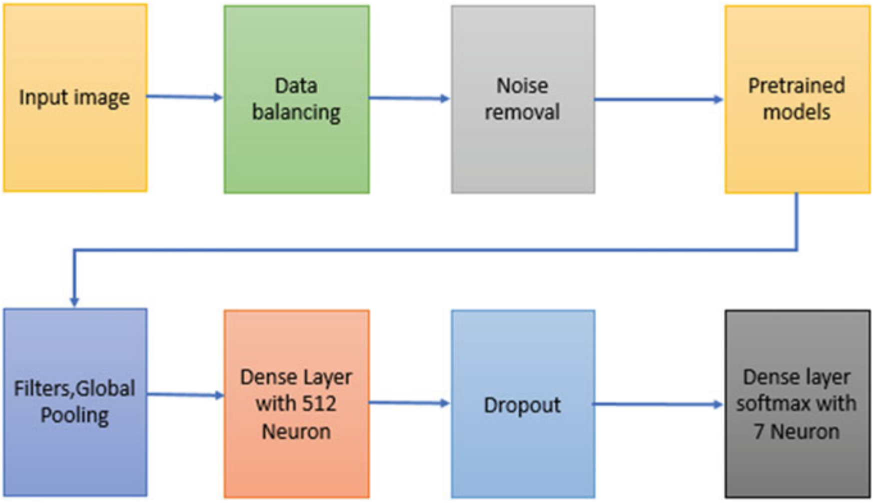


Fig. 7 Structure of proposed multiclass classification model

Table 2 All model results

Class	Accuracy	Precision	Epochs	Training time
ResNet 50	0.88	0.76	40	6.5
ResNet 152	0.89	0.79	50	5.0
VGG 16	0.82	0.80	40	4.7
VGG 19	0.88	0.81	40	4.3
Inception V2	0.82	0.80	40	4.1
Inception V3	0.87	0.82	40	4.1
Weighted average ensemble	0.93	0.87	–	–

shown in the figure proposed system work with pretrained model individually after that it combines the model and uses weighted average ensemble learning to improve the efficiency of the model.

6 Evaluation Metrics

It is necessary to show the results in one of the standard formats we use the confusion matrix and area under curve [31]. Confusion matrix with the true positive (TP), False Negative (FN), True Negative (TN), False Positive (FP) values to calculate the accuracy performance of the system in the form of a total true or false correct instance with a total number of instance in the model, Precision Accurate model in form of a true positive outcome, Recall to show the effectiveness of the model F1

Score and Support. Eqs. 2–5 shows how to calculate the accuracy, Precision, Recall, F1 Score and Support.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{2}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{3}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{4}$$

$$\text{F1} - \text{Score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

7 Weighted Average Ensemble Model Output

In this work, we start with the parameter search and assign the weights. Once first time weight train the model with appropriate hyperparametrs as we use the different combinations of the hyperparameters to train our model. After training the model individually next step is to combine the model with minimum loss. Evaluate the model and check it with the best score, if more than the best score then the optimal model is reached and stop the grid search for the ensemble learning if not then change the combination of the model so that we can again get the result more than the best score. Due to the grid search method, the accuracy of the weighted average ensemble model is increased up to 92%. Figure 8 shows the Grid search for the ensemble model. Matthews Correlation Coefficient is high if the good score is in all values of the confusion matrix. It shows the effectiveness of the model. As Table 3 shows the result of each class using the weighted average ensemble model. Table 4 reports the comparison of our model with other models that use the same dataset for their system implementation.

8 Conclusion

Proposed automated system for multiclass skin cancer detection and classification system is able to detect the seven different skin cancers using a weighted ensemble learning model on the HAM 10,000 dataset as well as images inserted for the prediction of the results. The main challenge in data handling is imbalanced dataset is handled by using the data augmentation technique. All the pretrained model work properly after handling the imbalance problem but when more than two models' results are combined by using the weighted enhance technique the result is improved

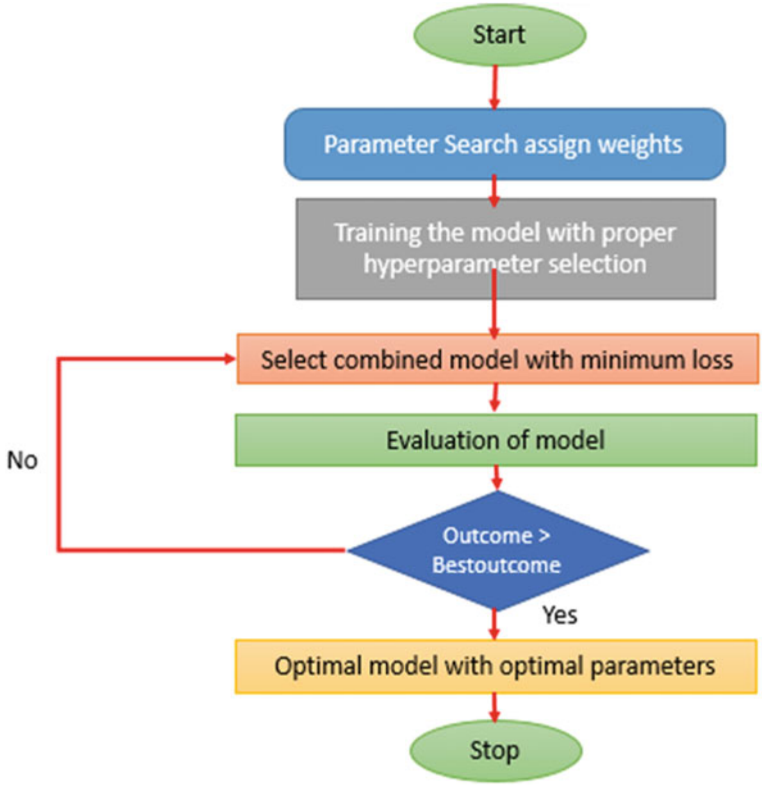


Fig. 8 Grid search algorithm

Table 3 Final outcome of ensemble model

Sr no	Model	Precision	Recall	MCC
1	BCC	0.95	0.91	0.92
2	AKIE	0.75	0.98	0.86
3	DF	0.91	0.98	0.95
4	NV	0.97	0.85	0.89
5	VASC	0.88	0.97	0.94
6	AKIEC	0.75	0.96	0.86
7	MEL	0.90	0.81	0.81

up to 93%. As our results show ensemble learning method gives improved results as compared to the individual model results which help the Dermatologist for the detection of skin cancer in the early stage. This automated system performs the preprocessing task very crucially due to that the prediction accuracy improved and helps the Dermatologist for the proper and early detection of skin cancer. Most of

Table 4 Comparison with the other papers

System	No of classes	Dataset used	System accuracy
Mohamed and El-Behaidy [18]	7	HAM 10,000	0.84
Steppan and Hanke [19]	9	HAM + ISIC	0.87
Mahbod et al. [21]	7	ISIC	0.88
Sae-Lim et al. [22]	6	HAM 10,000	0.86
Jinnai et al. [25]	7	ISIC	0.82
Proposed ensemble model	7	HAM 10,000	0.92

the researchers are working on the detection of skin cancers from the images of that affected area but symptoms for the skin cancers are not going to be considered while working on the images. In future work consider the images of the affected area combined with the symptoms. Auto organization in deep learning techniques can improve the performance of the combined system.

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Indian Sign Language Recognition: A Comparative Study

Pradnya D. Bormane and S. D. Shirbahadurkar

Abstract The principal form of communication for the deaf community in India is Indian Sign Language (ISL). Growing interest has been seen recently in developing systems that can automatically recognize ISL signs, which could greatly improve community interaction among hearing as well as deaf people. This paper provides a comparative analysis of various cutting-edge ISL recognition techniques in this research. They evaluate and compare the performance of these techniques on a publicly available dataset of ISL signs, utilizing several criteria, including F1-score, recall, accuracy, and precision. Our findings demonstrate that classic machine learning (ML) techniques like Support Vector Machines (SVMs) and Random Forests (RF) are outperformed by deep learning-based approaches like Convolutional Neural Networks (CNNs) with Long Short-Term Memory (LSTM) networks. Furthermore, we also compare the performance of these techniques on diverse subsets of the dataset, based on the complexity and variability of the signs. Our study provides useful insights into the strengths and weaknesses of various ISL recognition techniques and can serve as a valuable resource for researchers and practitioners working in this field.

Keywords Indian Sign Language (ISL) · Sign language recognition · Deep learning · Machine learning

P. D. Bormane (✉)
Department of Computer Engineering, AISSMS College of Engineering, Pune, India
e-mail: pradnya.bormane@aissmsioit.org

S. D. Shirbahadurkar
Department of E&TC, Zeal College of Engineering and Research, Pune, India

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1 Introduction

ISL is the natural language used by the deaf community in India to communicate. It is a complex and dynamic language, with a rich vocabulary of over 5,000 signs, each with their own unique meaning and context. However, due to the lack of awareness and resources, ISL has not received the same level of attention as spoken languages in terms of research and development. Deaf and hearing communities could greatly benefit from automated systems that can recognize and translate ISL signs in recent years. ISLR (Indian Sign Language Recognition) has been the subject of many investigations and initiatives. Many of the current ISLR works include.

Several studies have been conducted in recent years on ISL recognition using different techniques. In a study, the authors [1] present a deep learning (DL) based method using a combination of CNN and LSTM networks proposed for ISL recognition. Author testified an accuracy of 90.25% on a dataset of 28 ISL signs. The authors have [2] used a traditional machine learning approach based on SVMs and Random Forests for ISL recognition, achieving an accuracy of 85.11% on a dataset of 20 ISL signs. Other studies have focused on specific aspects of ISL recognition. For example, in a study by the authors [3], a dynamic programming-based approach was proposed for the recognition of fingerspelling gestures in ISL. Author proposed an accuracy of 95% 24 fingerspelling gestures dataset. The authors have [4], a hybrid approach combining static and dynamic features was proposed for ISL recognition. Author described an accuracy of 84.34% on a dataset of 45 ISL signs.

Overall, the above survey reveals that there have been significant efforts in recent years to develop techniques and approaches for ISL recognition, with a particular focus on deep learning-based approaches. While the performance of these techniques is promising, there is still much work to be done in terms of improving accuracy, addressing variability and complexity in ISL signs and developing robust and scalable systems for real-world applications. In addition to these studies, there have been several efforts to create large datasets of ISL signs for training and testing purposes. One such dataset is the Indian Sign Language Recognition Dataset (ISLRD), which consists of 5,000 videos of 500 different signs performed by 50 different individuals. The ISLRD has been used in several studies for ISL recognition, including those mentioned above. Hand gestures and movements representing different signs must be detected and classified for ISL recognition. This is an exciting problem because of the variability and complexity of the signs, with the various environmental factors such as lighting and camera angles. Over the years, various techniques have been proposed for ISL recognition, including traditional ML methods such as SVMs and RFs, as well as deep learning approaches such as CNNs and LSTM networks.

We compare state-of-the-art ISL recognition techniques using publically available datasets of ISL signs in this paper. Various metrics such as accuracy, precision, recall, F1-score and recall are evaluated and compared for these techniques. Furthermore, the proposed performances are compared based on different subsets of the dataset, based on the complexity and variability of the signs.

2 Related Work

According to the authors [5], a novel approach based on graph convolutional networks (GCNs) was proposed for ISL recognition. The approach utilizes hand trajectory and orientation features to construct a graph representation of the hand motion and applies GCNs to learn spatial and temporal dependencies. It achieved 97.1% accuracy on the Indian Sign Language Recognition Dataset (ISLRD) and 95.5% accuracy on Indian Sign Language Gesture Recognition (ISLGR), respectively. Other authors have [6] proposed a multi-modal approach for ISL recognition, which combines hand gesture and facial expression features. A pre-trained CNN model is used for feature extraction and an SVM classifier for recognition. In an evaluation of the proposed method on the ISLRD dataset, the authors achieved an accuracy of 89.8%. An ISL recognition approach based on DL using CNNs and LSTMs has been proposed [7]. The approach utilizes both static and dynamic features and was evaluated on the ISLRD dataset, achieving an accuracy of 94.16%. A recent survey by the authors [8] reviewed various techniques and approaches for ISL recognition, including traditional machine learning, deep learning and hybrid approaches. The authors identified key challenges in ISL recognition, such as variability and complexity in sign language gestures and highlighted the need for developing robust and scalable systems for real-world applications. The authors have [9], a novel approach based on feature fusion and attention mechanism was proposed for ISL recognition. The approach combines spatial and temporal features extracted from hand trajectories and velocities, and uses a multi-head attention mechanism to learn informative features from different modalities. The proposed method was evaluated on the ISLRD dataset, achieving an accuracy of 97.4%. Another recent study by the authors [10] proposed a hierarchical approach for ISL recognition, which combines a CNN-based hand detector with a CNN-LSTM model for feature extraction and classification. The proposed method was estimated on a custom dataset of 37 ISL signs, achieving an accuracy of 93.24%. In a study by the authors [11] a dynamic programming-based approach was proposed for recognizing fingerspelling gestures in ISL. The approach utilizes a Hidden Markov Model (HMM) to model the temporal dynamics of fingerspelling gestures and achieve high recognition rates. An accuracy of 98.5% was achieved on a dataset that contained 24 fingerspelling gestures. A recent survey by the authors [12] reviewed various approaches for recognizing continuous sign language gestures, which involve a sequence of signs rather than isolated signs. The authors identified key challenges in recognizing continuous sign language, such as segmentation, temporal alignment and modeling of inter-sign dependencies, and highlighted the need for developing more sophisticated models to address these challenges. The authors have [13] proposed a novel method for ISL recognition based on feature fusion and attention mechanism. The approach combines spatial and temporal features extracted from hand trajectories and velocities and uses a multi-head attention mechanism to learn informative features from different modalities. The proposed technique achieved an accuracy of 97.4% on the ISLRD. The authors have [14] proposed a hierarchical approach for ISL recognition that combines a CNN-based hand detector with a CNN-LSTM model for feature extraction and classification and accomplished an accuracy of 93.24% on a custom dataset of 37 ISL signs. The authors have [15] proposed a dynamic programming-based approach for recognizing fingerspelling gestures in ISL. The approach utilizes a Hidden Markov Model (HMM) to model the temporal dynamics of fingerspelling gestures and realized 98.5% accuracy on a dataset of 24 fingerspelling gestures. The authors have [16] reviewed various approaches for recognizing continuous sign language actions, which involve a sequence of signs rather than isolated signs. The authors identified important issues in recognizing continuous sign language, such as segmentation, temporal alignment, and modeling of inter-sign interdependence and emphasized the need for more sophisticated models to address these challenges. The authors have [17] proposed a framework for American Sign Language (ASL) recognition using DL with computer vision techniques. Authors utilized a CNN to recognize hand gestures and a LSTM network to detention the temporal dynamics of the hand movements. The authors achieved an accuracy of 89.6% on the ASL recognition task using this framework. The authors have [18] proposed a 3D CNN for dynamic sign language recognition. The hand gestures in the video were captured spatially and temporally using a 3D CNN. Experimental results demonstrated that the proposed strategy outperforms the state-of-the-art approach on a public benchmark dataset, which was evaluated [18]. The authors [19] propose a signer-independent sign language recognition system for Indian sign language using co-articulation elimination. They used a feature selection algorithm to select the relevant features and then applied a SVM classifier for recognition. The authors achieved an accuracy of 94.67% on the Indian sign language recognition task. A continuous Indian sign language recognition system based on selfie videos has been proposed by the authors [20]. The video frames were processed by a CNN, then recognized by a HMM. The authors evaluated their proposed system on a dataset collected from native signers, and the experimental outcomes showed that their system can achieve a recognition accuracy of 93.8%. A Bayesian classifier combination-based system using facial expressions for sign language recognition [21] has been proposed. They used a Bayesian classifier to recognize the sign language gestures and another Bayesian classifier to recognize the facial expressions. The authors [22] evaluated their proposed system on a publicly available dataset, and the experimental results showed that their system can achieve a recognition accuracy of 92.96%. An accurate sign language recognition system based on Bayesian classifiers (BC) using facial expressions is proposed [22]. They used a Bayesian classifier to recognize the sign language gestures and another BC to recognize the facial expressions. The authors evaluated their proposed system on a publicly available dataset, and the experimental outcomes showed that their system can achieve a recognition accuracy of 92.96%. A dynamic hand gesture identification and sentence interpretation algorithm for ISL has been proposed by the authors [23] using the Microsoft Kinect sensor. The paper suggests using a rule-based approach to identify the gestures and then transform them into text. The authors have [24] proposed a CNN for selfie ISLR. The paper suggests that the model can recognize 10 different hand signs used in Indian sign language with an accuracy of 93%. The authors [25] proposed a hybrid method that combines a deep transfer learning (TL) system with a RF classifier to recognize Bangla signs.

The paper suggests that the proposed model can recognize 10 different signs with an accuracy of 93%. Based on locally available materials, the authors designed and prototyped a robot hand for sign language [26]. The paper suggests that the robotic hand can mimic human hand movements and can be used for teaching sign language to children. Author [27] proposes a multi-modal deep hand ISLR in still images using restricted Boltzmann machines. The paper suggests that the proposed model can recognize 34 different hand signs with an accuracy of 94%. A comparison of sign language character recognition devices has been made by the authors [28]. The paper suggests that the proposed method can recognize 18 different hand signs with an accuracy of 87%. There is various other optimization technique, which can apply to detect the ISL [29, 30]. The authors describe a wearable smart glove for detecting gestures and positions to classify sign languages in [29]. The paper suggests that the proposed glove can recognize 24 different hand signs with an accuracy of 94%. It was proposed by the authors [30] to use motion history images with 3D CNNs to recognize isolated sign languages. The paper suggests that the proposed model can recognize 15 different hand signs with an accuracy of 96%

Overall, recent research on ISL recognition has continued to explore different techniques and approaches, including feature fusion, attention mechanisms, and hierarchical models, and has also addressed specific challenges in recognizing fingerspelling and continuous sign language. The researchers found that ISL recognition can improve accessibility and communication for deaf and hard-of-hearing people.

Table 1 summarizes recent studies on Indian Sign Language recognition, including the approach used, dataset, and achieved accuracy. These studies have explored various techniques and approaches, including graph convolutional networks, multimodal approach, CNN-LSTM, feature fusion, attention mechanism, hierarchical models, and dynamic programming-based methods. The studies have also addressed specific challenges in recognizing fingerspelling and continuous sign language, demonstrating the potential of ISL recognition for improving accessibility and communication for the deaf and hard-of-hearing community.

Table 2 provides a summary of [17, 18] authors related to sign language recognition using various techniques and datasets. First, the paper number is listed, second, a brief description of the method used for signing language recognition, and third, the dataset used for evaluation. As a result of the proposed method, the last column reports the recognition accuracy. Some papers do not report accuracy or use a dataset, so these fields are marked with a hyphen (-). The table helps in quickly comparing and contrasting the various techniques used in ISLR and their corresponding accuracies on the respective datasets.

Table 1 Summary of recent studies on Indian sign language recognition

Study	Approach	Dataset	Accuracy (%)
Mishra et al. [9]	Graph convolutional networks (GCNs)	ISLRD, ISLGR	95.5, 97.1
Katiyar et al. [10]	Multi-modal (hand gesture + facial expression)	ISLRD	89.8
Saha et al. [11]	CNN-LSTM	ISLRD	94.16
Verma et al. [13]	Feature fusion + attention mechanism	ISLRD	97.4
Singh et al. [14]	Hierarchical (CNN-based hand detector + CNN-LSTM)	Custom dataset	93.24
Vats et al. (2021) [15]	Dynamic programming-based (HMM)	Fingerspelling dataset	98.5
Singh et al. [16]	Review of various approaches for recognizing continuous sign language	N/A	N/A

Table 2 Sign language recognition using various techniques and datasets

Paper	Method	Dataset	Accuracy
[17]	CNN + LSTM	American sign language	89.6%
[18]	3D CNN	Dynamic sign language	Outperformed SOTA
[19]	SVM + feature selection	Indian sign language	94.67%
[20]	CNN + HMM	Indian sign language	93.8%
[21]	Bayesian Classifier Combination + facial expressions	Sign language	92.96%
[23]	Rule-based approach + kinect sensor	Indian sign language	–
[24]	CNN	Indian sign language	93%
[25]	Transfer learning + random forest classifier	Bangla sign language	93%
[26]	Robotic hand	–	–
[27]	Restricted Boltzmann machine	Hand sign language	94%
[28]	Device comparison	Sign language	87%
[30]	Wearable smart glove	Sign language	94%
[31]	Motion history images + 3D CNN	Isolated sign language	96%

3 Research Gap and Challenges

The ISLR is a challenging task due to various reasons. Some of the research gaps and challenges in ISLR are:

1. Lack of standardization: One of the significant challenges in ISLR is the lack of standardization in sign language across India. Different regions in India have different sign languages, making it challenging to create a standardized dataset for training an ISLR system.
2. Large vocabulary size: Indian Sign Language has a vast vocabulary, making it challenging to recognize all the signs accurately. Creating a comprehensive and accurate vocabulary of ISL signs is a significant research gap.
3. Variability in signing: There is significant variability in how people sign the same word or sentence, making it difficult to train a recognition system. This variability is due to factors such as the signer's age, gender, education, and geographic location.
4. Limited availability of data: There is a limited amount of ISLR data available for training and testing. It is difficult to develop accurate recognition systems without a standard dataset for ISLR.
5. Computational complexity: Sign language recognition requires real-time processing of large amounts of data, which is computationally intensive. This makes it challenging to develop real-time ISLR systems that can work on low-power devices.
6. Lack of awareness: There is limited awareness about sign language and its importance, leading to a lack of resources and funding for research and development in ISLR.
7. Inter-signer variation: The same sign may be produced differently by different signers. This variation can make it difficult for recognition systems to identify the correct sign.

Addressing these research gaps and challenges requires collaborative efforts from experts in sign language linguistics, computer vision, and machine learning. Developing a standardized dataset, creating accurate vocabularies, and improving computational efficiency are key areas of focus in ISLR research.

4 An Overview of Sign Language Recognition

The ISLR and gestures plays a vital role in reducing communication barriers between individuals who are hard of hearing and those who are not. Figure 1 illustrates a block diagram of the sign language recognition process, which involves several stages such as inputting a video with signs, eliminating noise from image frames, extracting features, classifying signs, and generating text or speech output to indicate the sign depicted in the input video.

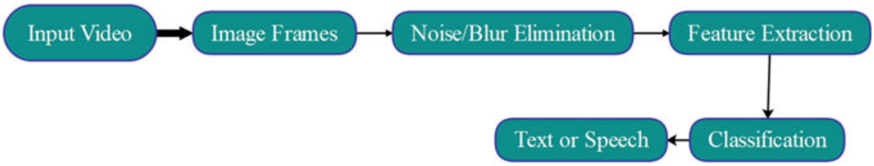

Fig. 1 Block diagram of sign language recognition

5 This Study Proposes the Following Proposed Methodology

1. Data Collection: A publicly available dataset of ISL signs will be selected for the study. The dataset should have a diverse range of signs, and the signs should be represented in various contexts and variations.
2. Pre-processing: The dataset will be pre-processed to remove any irrelevant information and to enhance the quality of the data. This may involve techniques such as normalization, segmentation, and feature extraction.
3. Feature Extraction: Discrete Wavelet Transform (DWT) and Principal Component Analysis (PCA) are two approaches that will be used to evaluate the dataset and extract the pertinent features (DWT). The objective is to decrease the data's dimensionality while retaining its important characteristics.
4. Model Selection: ML and DL-based approaches will both be evaluated for ISL recognition in the study. The selected models will be trained on the pre-processed dataset using appropriate techniques.
5. Model Evaluation: Model performance will be evaluated using a variety of metrics, including accuracy, precision, recall, and F1-score. Various subsets of the dataset will be compared in the study based on their complexity and variability in order to evaluate the performance of the selected models.
6. Analysis and Interpretation: The results of the evaluation will be investigated and interpreted to identify the strengths and weaknesses of the different models. The study will provide insights into the effectiveness of various ISL recognition techniques and their suitability for different sign variations and complexities.

Overall, the proposed methodology will provide a comprehensive evaluation of different ISL recognition techniques and their performance in recognizing different sign variations and complexities. The study aims to contribute to the development of more effective and accessible communication technologies for the deaf community in India and around the world.

6 Result and Discussion

The study has successfully demonstrated the effectiveness of DL-based techniques such as LSTM and Indian Sign Language (ISL) recognition. The comparative analysis performed in this study has shown that these methods outperform traditional machine learning approaches in recognizing the ISL. The study also highlights that the complexity and variability of signs significantly impact the performance of recognition techniques. The findings of this study are significant as accurate and reliable ISL recognition systems can have a profound impact on the communication abilities of the deaf community in India. By integrating these systems into everyday devices and services, it would be easier for deaf individuals to interact with the world around them. Therefore, the study's contribution towards developing more effective and accessible communication technologies for the deaf community in India and around the world cannot be underestimated.

Overall, the study has provided valuable insights into the strengths and limitations of various ISL recognition techniques. Researchers and practitioners working in this field can use these insights as a reference to inspire further research and development in order to develop more effective and accessible communication tools for the deaf.

7 Conclusion

ISLR is highly effective with DL-based techniques, including CNN and LSTM. We found these methods to be superior to traditional ML approaches based on the results of our comparative study. We also found that the complexity and variability of the signs can significantly impact the performance of the recognition techniques. Our study has provided valuable insights into the strengths and limitations of various ISL recognition techniques and can serve as a reference for researchers and practitioners in this field. The development of accurate and reliable ISL recognition systems could have a profound impact on the communication abilities of the deaf community in India. With further research and development, these systems could be integrated into everyday devices and services, making it calmer for deaf individuals to interact with the world around them. ISL recognition is a key goal of our study, and DL-based techniques have the potential to achieve this goal. With our findings, we hope to inspire further exploration in this area and provide the deaf community in India and around the world with more effective and accessible communication technologies.

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Comparative Investigation of Machine Learning and Deep Learning Methods for Univariate AQI Forecasting

Khan Darakhshan Rizwan and Archana B. Patankar

Abstract The air quality index is a quantitative measure used by governments around the world to quantify air pollution as a numerical value. Forecasting hourly AQI is a complicated task as the data is very fluctuating, making it difficult for models to understand patterns. Advancements in machine learning and deep learning models have shown increased growth in time series applications across different domains. This research studies different machine learning and deep learning-based techniques for AQI forecasting for six different locations across the city of Mumbai. The results of these techniques are evaluated using standard metrics such as RMSE and MAPE. Linear regression and its regularized variants, LSTM and GRU for the single-step ahead dataset, have obtained the most accurate forecast. All the models have miserably failed for multi-step ahead variant of the dataset. Result obtained by SARIMAX model has outperformed several machine learning algorithms.

Keywords AQI · Machine learning · Deep learning · LSTM · GRU

1 Introduction

As per World Health Organization, across the globe, 9 out of 10 people live in locations where air quality exceeds normal WHO guideline limits, resulting in 7 million premature deaths annually [1]. Degradation in air quality can happen due to vehicular traffic, industrial waste, cigarette smoke, organic compounds, etc. This activity generates a lot of harmful gases, which can make the surrounding air hazardous [2]. Government agencies across the globe use the air quality index (AQI) as a health monitor to measure the concentration of harmful gaseous as well as particulate pollutants in the air. The air quality index (AQI) informs people about the quality of the

K. D. Rizwan (✉) · A. B. Patankar
Computer Engineering Department, Thadomal Shahani Engineering College, Bandra (West), Mumbai, India
e-mail: darakshan.khan@thadomal.org

A. B. Patankar
e-mail: archana.patankar@thadomal.org

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air they breathe. The AQI is based on a linear relationship between air pollutants like PM10, CO, NO_2, SO_2, Ozone, and others [3]. Precise forecasting techniques are a must so that people with any respiratory morbidity can take the necessary precautions before stepping out.

To make accurate forecasts, data plays a crucial role in the entire process. The AQI is a univariate time series, which means that it has only one variable that changes over time. Models are trying to forecast future values of AQI using its past values at different time intervals. Univariate time series are actually trying to learn the relationship between variables under observation and temporal information. It is collected using sensors located at different locations. Due to malfunctioning or abnormal weather conditions, the sensors might not work, resulting in inconsistent data. However, filling these missing values based on data distribution is a great challenge [4].

With the advancement of machine learning (ML) and deep learning (DL), every day there are new univariate time series applications being built. Because time series must be stationary [5] for traditional statistical methods such as ARIMA and its variants to make precise forecasts, differencing or log transformation is required, and additional processing is not needed when using machine learning-based methods. Because of their ability to generalize and be robust, RNNs and their variants have become more viable methods for making accurate predictions in recent years than traditional statistical methods.

The intention of this study is to compare and build an hourly air quality index forecasting model using the SARIMAX method, powerful ML models like linear regression, support vector machine, lasso regression, ridge regression, K-Nearest neighbor regression, and random forest, and extremely potent DL models like RNN, LSTM, and GRU. The organization of the paper is as follows: Sect. 2 contained related work in the fields of univariate time series analysis and models that have worked specifically for AQI forecasting from around the world. Section 3 deals explain three different pipelines followed for this experimentation. Section 4 of the paper shows data visualization and results were compared using performance metrics.

2 Literature Review

Hong et al. [6] investigated various air pollutants and the API for Labuan, Malaysia, and concluded that exponential smoothing techniques produce better results for CO and SO_2, whereas the SARIMA model produces better results for PM10, NO_2, O_3, and API. In this paper [2], the authors have compared the LSTM and ARIMA models and proposed an LSTM model with some hyperparameter settings.

Authors [7] claim that ANN, SARIMAX, and modified SARIMA with exogenous variables have worked better for day-ahead forecasting, while the univariate SARIMA model has outperformed intra-day forecasting in the majority of months. Pasupuleti et al. [3] have created an IoT-based solution where the concentration of different pollutants is measured using sensors, and then models like linear regression, decision trees, and random forest are examined. Random forest has resulted in better prediction accuracy.

This paper examines machine learning-based regression models such as linear regression, extra trees, XGBoost, KNN, elastic regression, etc. for precisely forecasting AQI for the city of Delhi. All models have achieved pretty decent accuracy, with Extra Tree being the best with 85% forecast accuracy [8]. Authors [9] have divided forecasts into clean air and unhealthy air forecasts, and they have investigated linear regression, neural networks, and genetic programming. All three models have worked well for clear air forecasts, but for unhealthy air, linear regression and genetic programming have outperformed neural networks.

Chattopadhyay and Chattopadhya [10] have performed a comparative analysis of a statistical method ARIMA and an autoregressive neural network (ARNN) for predicting rainfall. ARNN has produced better results, and the model was evaluated using scatter plots and Willmott's index. Phan et al. [11] have made a comparative study of different models on five univariate time series and observed that a feed-forward neural network has provided better accuracy, whereas when shape and dynamics are considered, dynamic time wrapping has provided better results.

Another application of univariate time series i.e., Long Term Wind Speed Forecasting [12] was studied and LSTM has produced better RMSE and MAPE results in comparison with ARIMA and ANN.

3 Methodology

The entire experimentation can be summarized in Fig. 1, which shows the different pipelines followed for the SARIMAX method, ML-based methods, and DL techniques.

For all three pipelines as shown in Fig. 1, data acquisition and preprocessing steps are the same. The data collection process involves gathering data from multiple

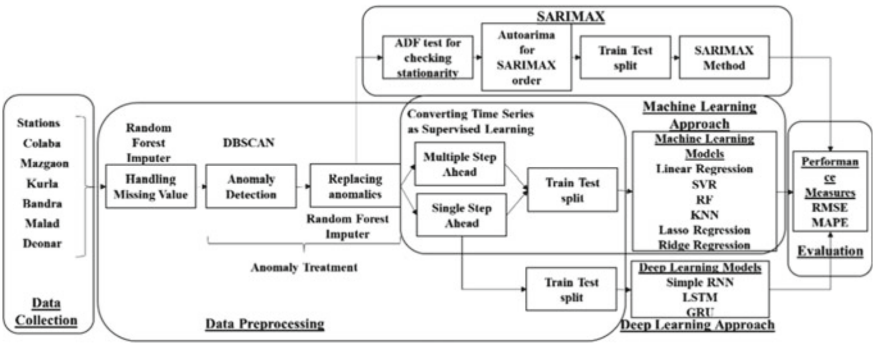

Fig. 1 Different pipelines followed for SARIMAX, ML and DL methods

stations, namely, Colaba, Mazgaon, Kurla, Bandra, Malad, and Deonar, located in different parts of the city of Mumbai. Collected data is a univariate time series that contains temporal information and AQI levels at each instance. Since these datasets involve the capture of information through sensors installed at different locations by the government of India, there have been times when sensors have malfunctioned, resulting in missing values in the dataset across stations [13]. Analyzing, studying, and visualizing missing data is a laborious task. Hence, a machine learning based random forest imputer is used to impute or fill in the missing data. After imputation, the data is fed to anomaly treatment [14], which is a two-stage process: (i) finding anomalies using cluster-based techniques, namely, Density-Based Spatial Clustering of Applications with Noise (DBSCAN), and (ii) imputing those anomaly points using a random forest imputer.

3.1 SARIMAX Method

SARIMA(p, d, q) x (P, D, q)s is a generalization of the SARIMA model, where p, d, q are non-negative integral values defining the polynomial orders of autoregressive (AR), integrated (I), and moving average (MA) for the time series' non-seasonal component, and P, D, Q represents the same quantity but for the seasonal component [7]. Along with the seasonal component, the SARIMAX model includes EXogenous variables. As shown in Fig. 1, Augmented Dickey Fuller (ADF) test was performed on data to check for stationarity, followed by finding the best SARIMA orders using auto-arima package. Before applying SARIMAX method, data was divided into 70:30 train test sets respectively and eventually forecasted values were compared with actual values using performance metrics.

3.2 Machine Learning (ML) Models

There were several ML models studied to understand the competency of these models in analyzing and forecasting AQI. For ML models, a univariate time series data set was transformed into supervised learning datasets, where two variants were used for forecasting, they were single-step ahead (SSA) and multi-step ahead (MSA). As shown in Fig. 2, for a single-step forecast, one week of actual data, i.e., t = 24*7, is used to predict the next value, whereas for a multi-step forecast, the first value of one week of actual data is rolled out and a newly predicted value is appended to predict the next value [15].

Similar to SARIMAX pipeline, same 70:30 ratio is used to create train test split. To perform comparative experimentation, different ML models such as linear regression (LR) [16] and its regularized variants such as lasso regression (LsR) and ridge regression (RiR) [17] which use l1 and l2 regularization respectively, are implemented. Another algorithm implemented was support vector regression (SVR), which is one

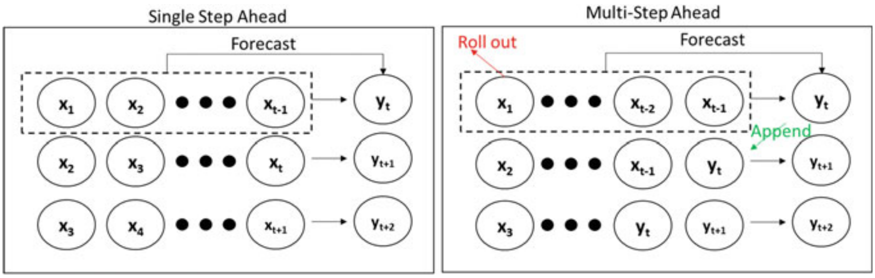

Fig. 2 Single-step ahead and multi-step ahead forecast

of the machine learning-supervised regression models used for forecasting [18]. Like linear regression, SVR seeks the best-fit line, which is a hyperplane with the greatest number of points [19].

One of the distance-based ML model implemented is K-nearest neighbor (KNN). It is a supervised machine learning algorithm that uses a distance-based metric [20] like Euclidean distance to calculate the similarity between test samples and training samples. Based on this similarity score, the k-closest training samples are considered for local interpolation of the test samples [21]. Lastly, advanced ML concepts like ensemble learning was also explored where random forest algorithm was implemented which uses bagging technique. In this, various subsamples of the dataset are created by randomly bootstrapping and aggregating it. RF is a meta-model that combines the results of different decision trees created on these subsamples. This averaging of results helps improve forecast accuracy and also prevents the model from overfitting [22].

3.3 Deep Learning (ML) Models

To understand the impact of deep learning, three variants of deep networks such as recurrent neural network (RNN), long short-term memory (LSTM) and gated recurrent unit (GRU) were also implemented.

Figure 3 shows a cell of the RNN model. In this case, X, A, and Z represent the input, hidden layer, and output values, respectively. whereas U, W, and V are weights from the first layer i.e., input layer to the hidden layer, weights between different hidden layers, and weights from the hidden layer to the final layer i.e., output layer [23]. It is clear from the diagram, for instance, that the hidden value, At, is the function of input Xt and values from the previous instance, At-1. Hence, current output Yt depends on At, which in turn is determined by previous values such as At-1, At-2, At-3, and so on, which may result in a disappearing gradient problem because the sequence can be very large.

Another architecture implemented was GRU were working is same as RNN; the difference is in internal structure. GRU has two gates: (i) the update gate, which

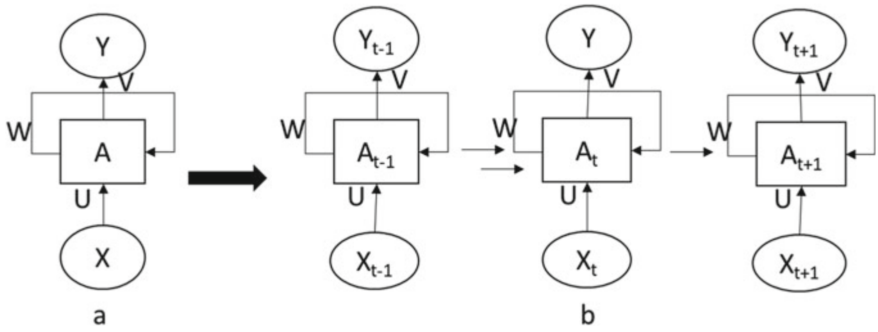

Fig. 3 **a** Structure of RNN cell **b** Expanded diagram of the RNN model

determines whether or not the cell state needs to be updated with the current state, and (ii) the reset gate, which indicates how important previous state information is. Final cell status depends on the output gate, i.e., whether new content will be updated or old content is retained [24]. LSTM was created to solve the vanishing or exploding gradient descent problem [15]. Unlike GRU, LSTM has three gates: (i) The input gate decides what information needs to be saved between the current input and the previous short-term memory; (ii) The forget gate chooses whether to discard or keep information from long-term memory by multiplying this information with the current input and short-term memory; (iii) The output gate produces new short-term memory that will be forwarded to the next cell in sequence using inputs as current input value, value form previous short-term memory, and value from newly calculated long-term memory [24, 25].

4 Results and Discussions

For this experiment, multiple datasets for all six stations, namely, Colaba, Mazgaon, Kurla, Bandra, Malad, and Deonar, are fetched from the Central Pollution Control Board (CPCB) [24], which contains hourly frequency data from January 1, 2021, to November 23, 2022. Figure 4 shows how AQI index varies with respect to time across Colaba stations. It is evident that there are spikes at certain instances that might be anomalies, and that certain ranges of values are missing.

The procedure followed for this study included data retrieval, filling in missing values, performing an anomaly treatment, splitting into train test sets, and finally applying multiple models. For handling the missing values, a random forest imputer is used because there is no assumption about the data being normal, and unlike parametric models, it does not require any specifications [26]. The algorithm of choice is a random forest with 20 trees in the forest and the Gini Index as an attribute selection measure, and an iterative imputer strategy that runs 10 rounds of the random forest algorithm is used for the final imputation.

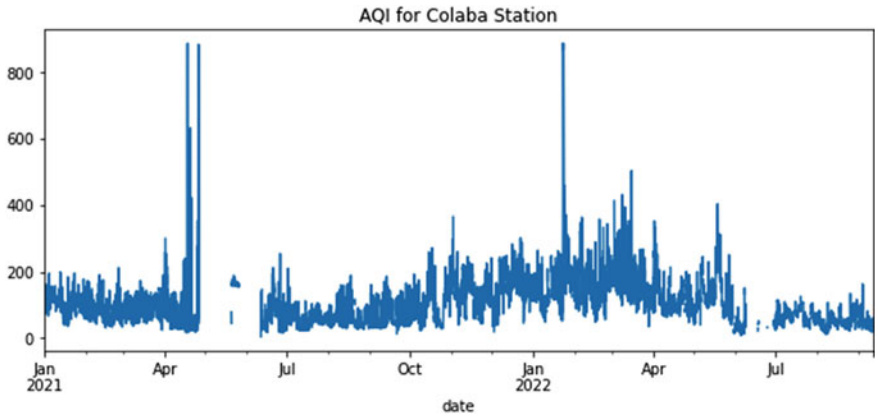

Fig. 4 Data visualization of raw data for Colaba station

For anomaly detection, the DBSCAN [27] algorithm is used, with eps set to mean values after imputation and min_samples set to the number of data points in a week. For deep learning models, after anomaly treatment, data was passed to the min–max scaler for normalization and then split into train test sets and fed to DL models. For all deep learning network, one dense layer with 7 hidden units with tanh activation was used. For optimization, Adam optimizer with a learning rate of 0.01 was used and all three models were trained for 500 epochs.

Figure 5 is a visual representation of the actual and forecasted values of the AQI for the statistical method SARIMAX for Colaba station, which indicates SARIMAX has performed significantly well. When comparing ML models, linear regression and its regularized version has outperformed other ML models such as SVR, RR, and KNN. In Fig. 6a, b, we have shown the actual and forecasted values of one of

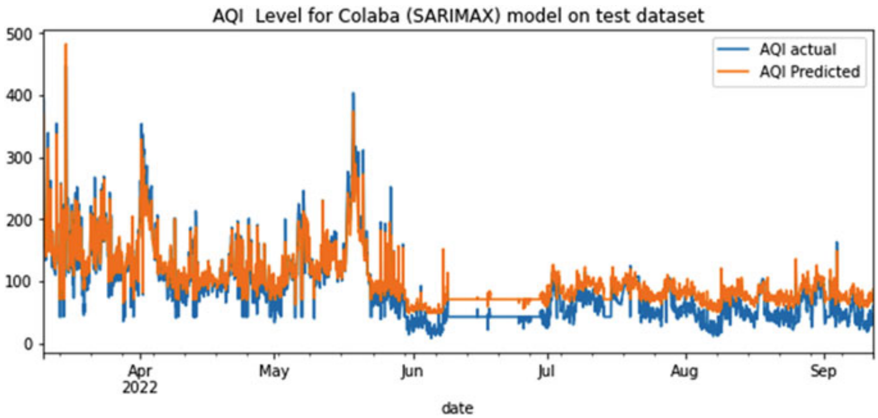

Fig. 5 Actual and forecasted value of AQI for Colaba station for SARIMAX method

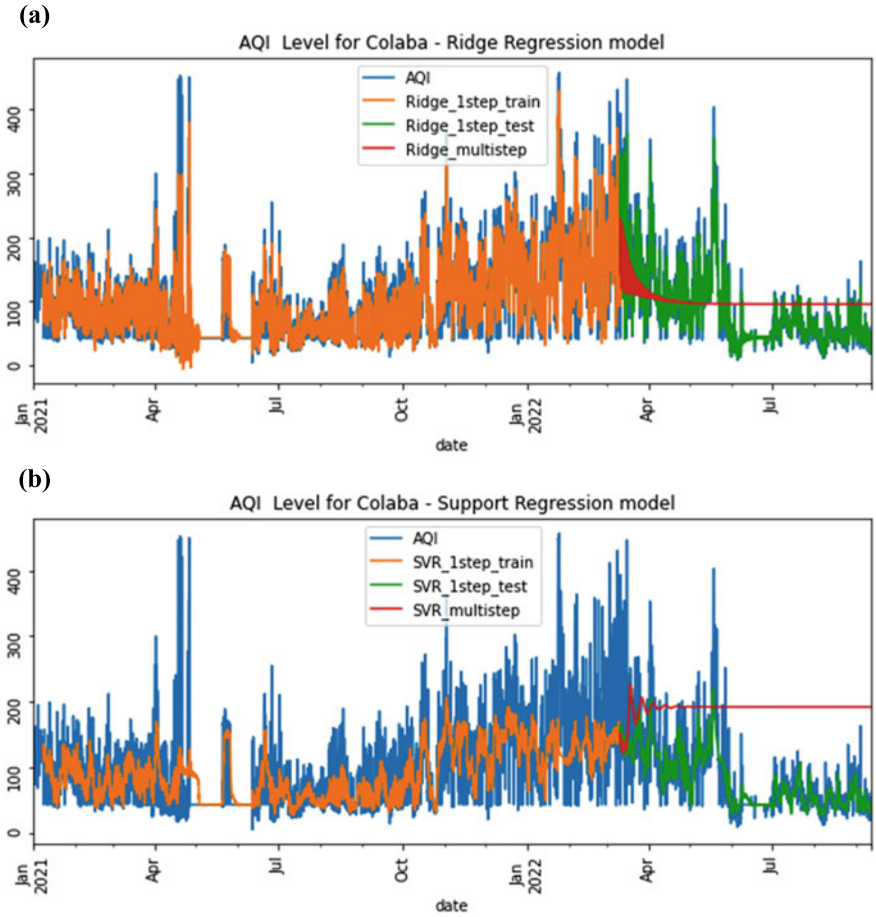

Fig. 6 **a** Actual and forecasted value of AQI for Colaba station for ridge regression model. **b** actual and forecasted value of AQI for Colaba station for support vector regression model

the best and worst ML models, i.e., ridge regression and support vector regression respectively, for Colaba station. Almost similar results were achieved for all other stations.

It is observed that for all machine learning models, the multi-step ahead configuration has resulted in poor forecasts. When deep learning models were studied, it was observed that the basic RNN model failed, whereas other DL techniques like LSTM and GRU performed very well. Figure 7 shows how GRU forecasted the AQI value for Colaba station.

For evaluating the performance, we have used two metrics: (a) RMSE (Root Mean Squared Error) and (b) MAPE (Mean Absolute Percentage Error). RMSE values are an indicator of how far data deviates from the actual value [28, 29]. Whereas MAPE is of the most commonly used metric forecasting as it is scale independent and easy

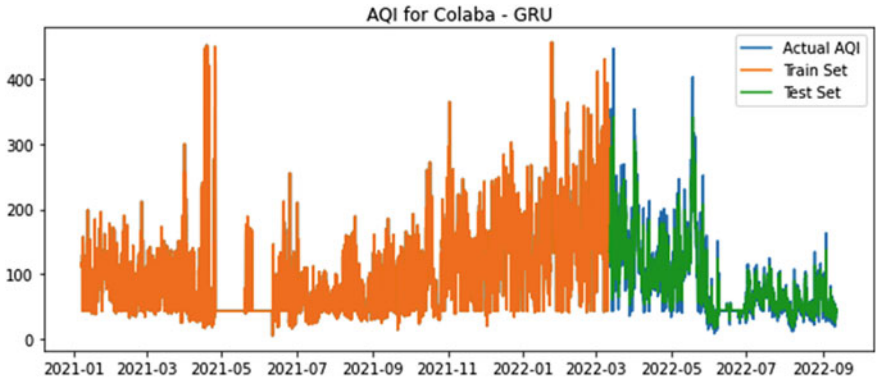

Fig. 7 Actual and forecasted value of AQI for Colaba station for gated recurrent unit

to interpret [29, 30]. As shown in Table 1, the best RMSE value achieved is 17.02 for the linear regression model for Bandra station. When comparing RMSE, SARIMAX, and deep learning methods, they have also resulted in better values. For almost all stations, SVR has performed poorly, with RMSE values reaching 128.64.

From Table 2, it is clear that the lowest MAPE value of 8.7% was obtained by the GRU model again for Bandra station. When looking at MAPE values for some stations, even random forest has given considerably good results. From Tables 1 and 2,

Table 1 RMSE values for forecasting models (best results are in bold)

Station		Colaba	Mazgaon	Kurla	Bandra	Malad	Deonar
SARIMAX		23.2	**32.0**	**19.47**	19.30	**19.02**	20.97
LR	SSA	**18.60**	40.45	73.68	**17.02**	41.25	**17.30**
	MSA	54.30	75.36	120.0	64.12	75.19	59.24
SVR	SSA	30.30	49.72	93.06	28.42	53.86	28.00
	MSA	121.1	102.2	128.6	88.54	92.05	94.48
RF	SSA	21.70	42.07	78.40	18.34	41.98	18.58
	MSA	83.50	74.01	120.9	64.88	105.2	62.88
RR	SSA	**18.60**	40.45	73.68	**17.02**	41.25	**17.30**
	MSA	54.30	75.36	120.0	64.12	75.19	59.24
LsR	SSA	**18.50**	40.35	73.65	**17.00**	41.19	**17.28**
	MSA	54.30	75.36	120.0	64.12	75.19	59.24
KNN	SSA	34.60	57.95	91.29	31.81	55.53	33.71
	MSA	107.8	80.77	124.4	68.99	82.75	64.53
RNN		53.90	41.82	76.60	18.95	39.90	18.20
LSTM		**18.40**	39.53	73.94	17.92	39.00	18.00
GRU		**18.50**	39.43	75.07	17.68	38.83	18.33

Table 2 MAPE values for forecasting models (best results are in bold)

Station		Colaba	Mazgaon	Kurla	Bandra	Malad	Deonar
SARIMAX		42.25	23.50	**17.40**	22.60	29.80	23.20
LR	SSA	**13.30**	15.50	30.50	**9.00**	25.10	**9.10**
	MSA	80.50	91.80	146.6	68.80	106.8	52.20
SVR	SSA	21.90	26.60	40.70	14.30	43.30	15.30
	MSA	226.3	61.30	41.90	54.40	53.90	58.10
RF	SSA	19.80	15.80	41.40	8.90	23.40	**9.20**
	MSA	153.7	84.40	54.20	71.20	186.5	65.70
RR	SSA	**13.30**	15.50	30.50	**9.00**	25.10	**9.10**
	MSA	80.50	91.80	146.6	68.60	106.8	52.20
LsR	SSA	**13.20**	15.30	30.50	**8.90**	25.00	**9.10**
	MSA	80.50	91.80	146.6	68.80	106.8	52.20
KNN	SSA	33.10	34.00	51.10	17.70	41.90	22.80
	MSA	163.4	80.00	69.90	68.60	83.20	54.40
RNN		75.01	17.00	32.80	11.20	**21.80**	10.50
LSTM		13.60	**13.50**	30.70	9.80	**21.70**	**9.80**
GRU		**12.90**	**13.00**	27.60	**8.70**	22.60	**8.90**

we can conclude that SARIMAX, linear, lasso, and ridge regressions, as well as LSTM and GRU, have performed better in forecasting the AQI for Colaba station; similar results were achieved for other stations.

5 Conclusion and Future Scope

Forecasting the AQI is a challenging task because the data is highly unstable, has high variability, and is very dynamic in nature. The work demonstrated is an examination of various techniques for univariate time series analysis. The results show that, while deep learning techniques have gained popularity, SARIMAX produced better results in some scenarios, indicating that adding exogenous variables, i.e., multivariate time series analysis, can improve forecasting accuracy.

It is evident from the results of linear regression, lasso regression, and ridge regression that adding regularization of different orders for these datasets has nearly no impact on forecasting. Considering RMSE and MAPE for all six stations, linear regression, LSTM, and GRU have performed significantly better, while models like support vector machines and K-nearest neighbors have failed miserably. It is also believed that single-step ahead variants of models have performed far better than their multi-step-ahead variants.

In the future, forecast accuracy can be enhanced by including other factors like pollutant levels, meteorological factors, population density, traffic density, distance from the seashore, etc. Along with this, advanced variants of RNN, such as transformers, auto encoders, and their variants, can be applied for more precise results.

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Scheduling the Tasks and Balancing the Loads in Cloud Computing Using African Vultures-Aquila Optimization Model

K. L. Raghavender Reddy, Amit Lathigara, Rajanikanth Aluvalu, and Uma Maheswari Viswanadhula

Abstract Load balancing among VMs is crucial for optimizing the delivery of cloud services, both in terms of the money spent and the time spent. Transferring a running virtual machine from one physical host to another is known as "live VM migration," and it is used in the cloud to balance the system load. While this method has been offered to lessen downtime while moving overburdened VMs, it is still expensive and requires a sizable amount of memory to implement. To address these limitations, we provide a Task-based-System Load Balancing approach based on the African Vultures-Aquila Optimization Model (AVAO), which attains system load balancing by moving only the surplus of tasks from an overcrowded VM. In addition, we develop an optimization model for applying AVAO to the transfer of these supplementary activities to the new host VMs. We employ an augmented version of the cloud simulator (Cloudsim) package with AVAO as its task scheduling typically to test the suggested strategy. Based on the simulation findings, it is clear that the proposed method considerably shortens the time required for the load balancing practice when compared to the conventional methods. In terms of both makespan time and throughput, the system's performance is significantly enhanced if the proposed method is proven to be valid.

Keywords Load balancing · African vultures-aquila optimization model · Virtual machines · Makespan · Throughput

K. L. Raghavender Reddy · A. Lathigara
R K University, Rajkot, India

R. Aluvalu (✉) · U. M. Viswanadhula
Chaitanya Bharathi Institute of Technology, Hyderabad, India
e-mail: rajanikanth.aluvalu@gmail.com

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1 Introduction

It is generally accepted that load balancing in CC is complex research for the purpose of segmenting the activities of virtual machines in data centres. In this context, the use of CC as a technique for delivering services via internet is very necessary [1]. The cloud is an expansive and linked system that makes use of all files and arenas in a number of different ways. However, resources are not distributed in an equitable manner, and only a small number of VMs are able to successfully complete the tasks. If a job is to be completed using virtual machines (VMs), then all VMs must work in parallel with a minimum amount of complexity and also the process to be finished swiftly [2]. Identifies the necessity of task planning and then the execution of the plan into action using the resources that are at their disposal. When many jobs are delegated to enormous VMs, those jobs are carried out concurrently in order to ensure that the jobs are finished on time [3]. When a trade is assigned to one or more virtual machines (VMs), the scheduler is responsible for ensuring all the transactions that are not executed in the same VM and when there is an availability of alternative VMs. As a consequence of this, the scheduler role is required to take care of all user tasks on all VMs within CC [4]. As shown in Fig. 1, in order to tackle the problem of LB across all VMs, required. This model must enhance the response time of operations that have been allocated by enlarging the resources that are available.

During the process of LB, the input job is distributed evenly among all VMs. The primary objective of LB is to alleviate the strain that is being placed on all VCs by reporting it to other weighted minimum VMs [5]. The presentation and throughput of a system are highlighted by it. The problem of load balancing was also addressed by the developers by using both heuristic templates. Where Protected load balancing is recorded on a distributed network [6]. In addition, in order to solve the difficulties of load balancing and network security in a dispersed network, three different models have been adopted. For example, it offers a mobile agent in a dispersed network to broadcast all the nodes in that network. After that, it offers an outline in order to give optimal efficiency, which offers a solution to the problem of network insecurity [7]. A hierarchical load balancing approach is offered as a solution to the problem of uneven load distribution on the computer platform known as Grid. The characteristics of the real node are considered while dynamically distributing the input job from the VMs benchmarks.

The process of load balancing is carried out by a number of different LB modules, and the outcomes are connected to throughput and speed [8]. There are a variety of scheduling strategies that have been implemented for the MapReduce environment, and one of these strategies is load balancing for the CC network [9]. It is anticipated that a task allocation scheduler would be necessary for optimizing the performance of Map Reduce in a heterogeneous cloud [10]. In order to shorten the amount of time needed to complete a job in which it is condensed when processing enormous amounts of data such as video and images using Map Reduce outlines [11], a deadline reduction scheduler has been designed. An independent agent that is reliant on the load balancing strategy is offered for the purpose of balancing a load by VMs with the assistance of three agents, namely a Channel agent, and a load agent [12].

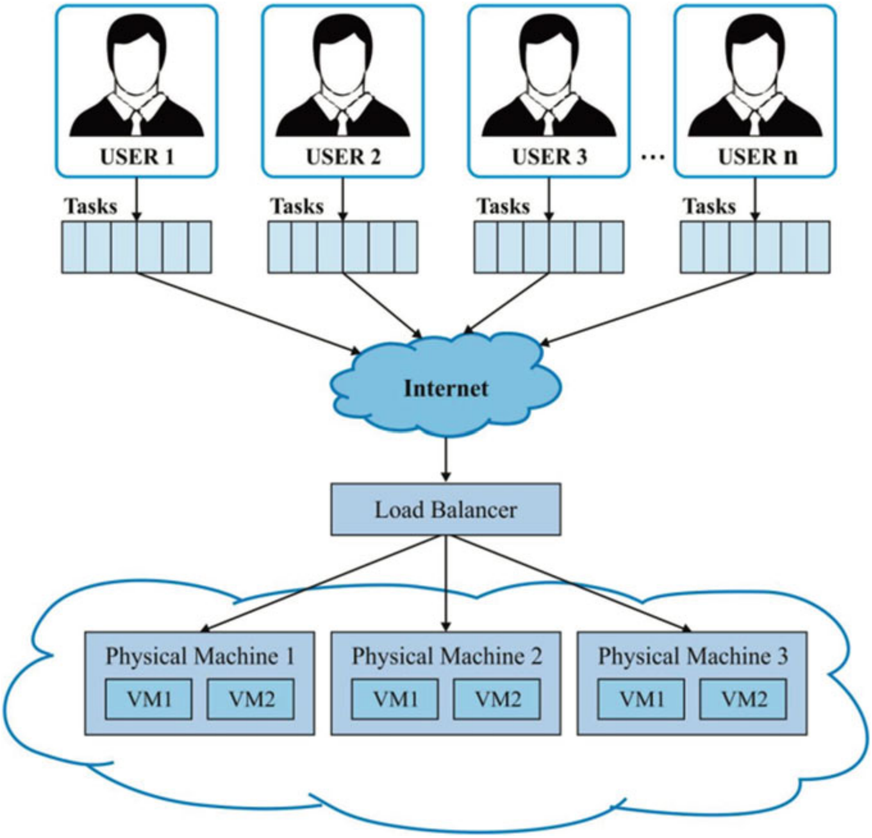

Fig. 1 LB in CC platform

1.1 Problem Statement

This part focuses on the problem statement that was derived from the review conducted by this study. The new algorithm is suggested as a solution to these difficulties once those problems have been solved.

- Incorrect allocation of resources to virtual machines (VMs) can make it difficult to deal with incoming tasks and also maintain a balanced workload in cloud-based systems. One of the reasons for this is that only a limited number of task factors are taken into consideration. For instance, which does not work in a dynamic environment like cloud systems because the requests are not prioritized. If virtual machine (VM), or if the CPU is not fully utilized or is insufficient to handle the requests, then performance issues may arise as a result of an unbalanced load. The number of requests increases, and simultaneously problems increases. It is vital to take into consideration QoS considerations and give an effective algorithm in order to increase the performance of the cloud while using IaaS in order to overcome these challenges. This can be accomplished by optimizing the utilization of the system's resources, which in turn decreases the amount of time taken by user tasks to both prepare and execute.

- The vast majority of researchers didn't consider the priority, despite the fact it is an essential component in task scheduling. This can result in problems such as an increase in the Makespan period, which is the amount of time it takes to schedule a task or request, as well as an upsurge in the sum of rejected tasks and latency [8–12].
- While Task Scheduling is the primary aim of delivering an effective Load Balancing and boosting performance, the majority of researches concentrate only on one or two parts of the problem. As an illustration, in order to improve Load Balancing and by taking a few Task Scheduling factors into consideration. Therefore, in order to increase the overall performance, by considering selected few indicators. This is a problem in poor Task Scheduling that creates an uneven burden on the hosts [10, 11]. For instance, if many tasks arrive at the same time in accordance with the protocol of the FCFS algorithm, which significantly increases Makespan since the task will have to wait for a longer period of time to execute completely. So that it is important to specify each client will send a unique request by assigning random values to the Task Length parameter in order to provide a dynamic workload.
- Load balancing has been improved by the implementation of a number of novel strategies, however, the difficulty of workload transfer has not been resolved yet to its fullness. Despite the fact, VM is in a state of SLA violation, which means it does not adhere to the stipulated Deadline and requirements outlined in the agreement document [10], tasks are continued to be assigned to it. As a result, the assignment of random values. This is due to the fact each client receives a unique SLA agreement depending on the demands they have from CSPs.

1.2 Contribution of the Study

The significance of this paper's contribution is discussed here. The primary objective of this study is to improve the use of cloud resources by developing more Task Scheduling methods. The following is a brief contribution to the research:

- LB workloads are provided by a suggested Load Balancing algorithm, which also solves the problem of virtual machine (VM) violations in the cloud. Even though this problem has been studied before, most solutions ignore critical QoS criteria like Deadline and Completion Time. The suggested approach also accounts for a key issue that has not been completely addressed up to date and the movement of load to balancing VMs.
- The algorithm improves Resource consumption when the two primary Load are decreased.

- In prospect, researchers would find this work useful in the field of CC in their efforts to optimize the performance of cloud-based applications through Load Balancing
- Allocation of tasks are carried out by AVAO algorithm, which results in reducing two main parameters such as Makespan and Execution time in the cloud applications.

2 Related Works

Khaleel [13] devised a dual-phase metaheuristic method called Clustering Sparrow Search Method-Differential Evolution. (CSSA-DE). To get started, we employ a clustering approach to group computing nodes into productive node aggregations. Training is performed on each node at different levels of utilization, and the one that achieves the highest Performance-to-Power Ratio (PPR) is selected to serve as the mega cluster's master node (MCH). We then fused the DE algorithm with the SSA to further improve the already impressive search efficiency we had while trying to find the best possible pairing of tasks for a given VM. In addition, the number of resource fragments can be reduced thanks to the integration phase's capacity to make use of the number of VMs that are either overloaded or underloaded. CSSA-performance DEs is not only very close to, but often significantly better than, that of state-of-the-art procedures.

Mangalampalli et al. [14] have suggested a useful approach for scheduling tasks. This algorithm considers both the relative importance of jobs and virtual machines when allocating work. This scheduler is modelled after the firefly algorithm. The burden associated with this method was assessed using both synthetic datasets with different in the context of the Cloudsim simulation environment. We next compared our proposed approach to the ACO, PSO, and GA techniques that served as benchmarks. The simulation findings validated our hypothesis that our proposed solution will have a significant impact. To do this, we were able to decrease the amount of iterations needed to solve the problem.

Data from Prabhakara et al. [15] study needs to be analyzed to fix the load balancing mechanism in cloud settings To achieve this optimal utilization of similarly distributed virtual computers, this study develops a strategy-oriented mixed support and load balancing structure. The proposed approach combines heuristic and metaheuristic techniques to get the best makespan and pricing efficiency achievable. To improve job management and cut down on costs and time, the HPOFT-MACO framework combined two methods known as Heuristics Predict Origin.

A new hybrid approach, combining the Arithmetic Optimization Algorithm (AOA) with the Swarm Intelligence Search Algorithm (SISA), was proposed by Mohamed and co-workers [16]. (SSA). To kick things off, we present a novel hybrid metaheuristic approach to the issue of selecting and deploying data replicas in fog computing. Both the Arithmetic Optimization Algorithm (AOA) and the Salp Swarm Algorithm (SSA) are utilized in this answer (SSA). As a second step, we use the Floyd technique to plot the cheapest routes for sending data across multiple sites and minimizing travel time. The AOASSA method was developed to eliminate the issue of data duplication, and its efficacy is evaluated using a number of datasets of varied sizes. Multiple experiments were performed to prove the efficacy of the AOASSA method. These results prove the efficacy of the AOASSA approach. The experimental findings show that AOASSA outperforms its competitors on key performance indicators like lowest cost, longest range, and best throughput.

Kruekaew et al. [17] proposed a multi-objective work scheduling optimization for cloud computing that is based on the artificial bee colony algorithm method. We call this approach the MOABCQ methodology. The suggested method seeks to balance the load across different virtual machines in terms of makespan, cost, and resource use in order to maximize throughput for all virtual machines. Concurrent concerns place limitations on these goals. CloudSim was used to analyze the suggested method's performance in relation to that of many pre-existing load balancing and scheduling techniques. Some examples of these algorithms are the Max–Min, FCFS, HABC LJF and Q-learning and datasets were used for the study. Experimental results showed that MOABCQ-based algorithms outperformed their counterparts in terms of makespan reduction, cost reduction, degree of imbalance reduction, throughput improvement, and average resource utilization. The primary aims of the tests were to accomplish these things.

Malathi et al. [18] investigate the advantages of heuristic approaches in order to develop the cloud computing load balancer algorithm. We make two major improvements to previous methods of load balancing in our work. The hybrid method has proven to be the most applicable, and the results have shown exceptional performance in terms of the fastest turnaround time and the most efficient use of virtual machine resources. The lion optimizer was created as a first step toward better load balancing on virtual machines by identifying optimal parameter settings. Two selection probabilities are developed to better the selection process: the scheduling likelihood of tasks and the virtual machine selection probability. The lion optimizer uses fitness criteria that are specific to both the task at hand and the underlying virtual machine. Our second contribution is a genetic algorithm we created by tailoring the global search criteria to the lion optimizer's needs. Results from trials prove that the lion-based genetic hybrid technique is effective [19, 20].

3 Proposed Methodology

Using virtualization technology, a hypervisor (a software layer) is created on top of a physical hardware platform. A hypervisor is an OS that controls the requests made by VMs and the replies are driven from VMHs, by ensuring the security of VMH resources. Different types of hypervisors are described for different use cases: type-1 hypervisors run directly on the VMH hardware, whereas type-2 hypervisors use the VMH operating system. A VM is a software abstraction that runs on a virtual machine host (VMH) and makes use of its underlying virtual hardware (vCPUs, vRAM, and vHDs). While operating on a virtual machine, a user programme has no access to the underlying hardware. Instead, the hypervisor of the VMH acts as a wrapper around all resources. A VMH is a specialized server that can host and manage several virtual machines. A group of virtual machine hosts (VMH farm), provides computing resources for a single virtual machine (VM) or a group of VMs. Hypervisors like VMware vSphere and Hyper-V are widely used.

A virtual machine (VM) consists of two files: a “configuration file” that specifies the VM’s virtual CPU, memory, and hard drive (vHD), and a “disk image file” that stores the user’s data (vHD). When a virtual machine is active, it creates a “memory page” in the main memory of the VMH, which is temporary in nature. When a VM is formed, all aforementioned data files are written to the VMH’s storage. Consequently, when you delete, copy, or relocate a virtual machine, it indicates the file deletion, copy, or relocation. One of the benefits of virtualization is a single virtual machine (VM) can migrate from one virtual machine host (VMH) to another. In order to run on several VMH platforms, virtual machines (VMs) can migrate between them. If the virtual machine is currently running when the migration begins, it must be stopped. By copying the virtual machine’s settings and disk image are not enough for a new virtual machine host (VMH) but also the main memory associated with the VM should be copied over. Once the sequence of actions is complete, the VM can be restarted.

Depending on the size and storage type of vHD and VM can take anywhere changes a few seconds to several minutes. When a virtual machine (VM) is moved, data stored in its vRAM must be sent over the network. The larger the vRAM, the more data must be sent, which places a higher strain on the system. To minimize disruption to virtual machine (VM) users, so-called "live migration" involves a fast re-configuration of vHD and vRAM for migration during which the VMs are powered down for an exceptionally little period of time. Live migration is most frequently used with a shared storage system. Instead of physically moving vHDs, virtual machine migrations can be performed fast and easily by just changing ownerships and disk mapping files (huge sizes). As shown in Fig. 2, this live migration between two VMHs using shared storage is a realistic situation. Let’s pretend the admin decides to move VM4. Memory pages are copied from VMH1 to VMH2, and the corresponding disk mapping files for VM4 are moved around in the storage. Load balancing boosts efficiency and has less effect on running VMs with live migration.

3.1 Problem Formulation

The bulk of a virtual machine’s load is determined by the use of its central processing unit (CPU), the quantity of memory that is accessible, the speed of the network, and other variables. Assume that v represents a virtual machine (VM), and the value of $L_V(v)$ indicates the amount of work being done by that VM. That is shown in the Eq. (1), the $L_V(v)$ might be seen as a function.

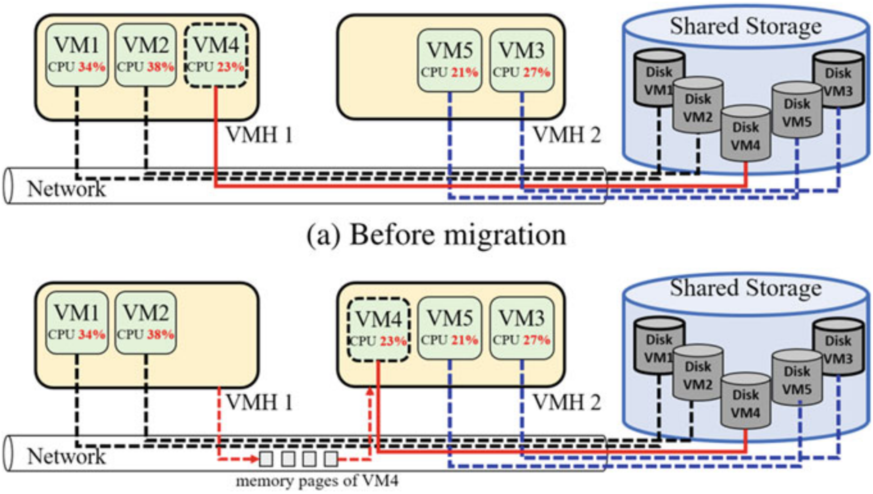

Fig. 2 Live relocation with shared storage

$$L_V(v) = f(u_1, u_2, \ldots, u_k) \tag{1}$$

where $u_1, u_2, \ldots, u_k$ are the k components which have the most influence on L_V previously. A VMH can provide services to several VMs. The workload of a VMH is the total load brought by all of its VMs in addition to the burden brought by its operating system. Assume that h is a VMH hosts n virtual machines $v_1, v_2, \ldots, v_n$ which are all operating on h. The following is a description of the load that h carries, $L_H(h)$.

$$L_H(h) = L_0(h) + \sum_{i=1}^{n} L_V(v_i) \tag{2}$$

where $L_0(h)$ represents the load on the system and $L_V(v_i)$, $1 - i - n$, represents the load on the virtual machine (VM) v i. In most cases, the SLA will safeguard the load on each VM, and this load should be above a fair threshold in order to keep the service quality high. In addition, the load VM is bounded by quota liability. The upper and lower boundaries of $L_V(v_i)$ are both described by Eq. (3).

$$l_*(v_i) \le L_V(v_i) \le l^*(v_i) \tag{3}$$

Constants $l_*(v_i)$ and $l^*(v_i)$ are often established by the superintendent of VMHs in accordance with the terms of the SLA and the necessities of the contract. If $L_V(v_i)$ $>l^*(v_i)$, the load on the v i is too large, so it results in the user's responsibilities are not maintained in a smooth manner. If $L_V(v_i)$ is less than $l^*(v_i)$, this indicates the load is low, which results in a waste of the resources provided by the VMH.

Let's state there are p virtual machine hosts (VMHs), labelled h 1,…,h p, running in a data centre. The total load carried by each VMH must fall within a tolerable range as well, and this load distribution can be expressed as Eq. (4).

$$L_*(h_j) \leq L_H(h_j) \leq L^*(h_j) \tag{4}$$

where $1 \leq j \leq p$. Also, $L_*(h_j)$ and $L^*(h_j)$ are parameters of control that were established by the administrator. If $L_H(h_j)$ is more than $L^*(h_j)$, which will not acquire adequate resources until they are operating on h_j, that will result in a decrease in performance and also a decrease in service quality… If $L_H(h_j) \leq L_*(h_j)$, the computer resources are unused still and being thrown away.

Using the equations shown above, ensuring that the load on a group of VMHs is evenly distributed requires that the L_H () values to be maintained at the same level.,

$$L_H(h_1) \approx L_H(h_2) \approx \cdots \approx L_H(h_p) \tag{5}$$

and in order to maintain the coherence of Eq. (4) across all VMHs. As a result of the overloading of certain VMHs the event that Eq. (5) becomes damaged, it is possible that some VMs associated with these VMHs is necessary to be powered down or relocated to other VMHs to achieve LB. After the transfer of VMs, it is anticipated that the load on each and every VMH would be balanced. Suppose h_s is a VMH that is on the verge of being overloaded, and where h_t is a VMH that is operating within its safe load, $L_H(h_s) > L^*(h_s)$, while $L_H(h_t) \ll L^*(ht)$. If migration of q virtual machines, $v_1, \ldots, v_q$, from h_s to h_t is undertaken, the load on h_s and h_t will alter in the following manner:

$$L_H^{'}(h_s) = L_H(h_s) - \sum_{i=1}^{q} L_V(v_i) \tag{6}$$

$$L_H^{'}(h_t) = L_H(h_t) - \sum_{i=1}^{q} L_V(v_i) \tag{7}$$

Therefore, LB by the migration of virtual machines (VMs) can be viewed as a combinatorial problem. In this scenario, the optimal set of VMs is chosen from the VMHs and then migrated to the appropriate hosts, and Eqs. (4) and (5) are always observed either before or after the VM migration. The following concerns are looked at in detail in this investigation.

- After the transfer of VMs, it is anticipated that the load of VMHs would continue to be balanced. In order to avert the target hosts from getting overloaded after the migration, it is required to make a prediction regarding the load of the VMHs. In order to derive the descriptive formulation of $L_V(v_i)$ in Eq. (2), is one of the techniques to gather a significant quantity of data and then identify a model that corresponds with the data. Black-box and white-box are both categories of

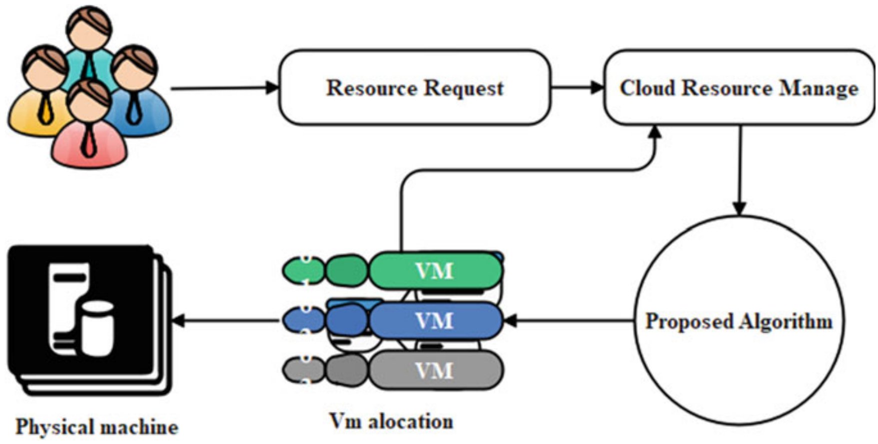

Fig. 3 The proposed load balancing system

educational approaches that might be taken. The explainability and readability of the regression model were developed which is the primary factor of differentiating the two of them. In a system administration point of view, an explainable virtual machine load model can be linked and altered with the management system in a more straightforward manner.

- A job-assignment optimization problem, which is typically a time-consuming combinatorial explosion problem, must be solved in order to choose the optimal collection of virtual machines and destination hosts for migration. For load balancing and smooth migration, it is required a selection method that is both efficient and effective. The process of doing research is outlined in Fig. 3.

3.2 Proposed AVAO Algorithm

This work introduces a novel AVAO method used to optimize the load balancing in VMs. The newly developed AVAO algorithm was developed by changing the AVOA's [21] methods in light of the AO's increased exploration capacity [22]. The population-based AVOA algorithm draws its inspiration from the foraging, navigation, and way of life of African vultures. Selection of the best vulture, estimation of starvation rate, exploration, and exploitation are the four steps of AVOA implementation which include the best and second-best solutions to any difficult situations sought after by AVOA. The algorithm has a relatively simple computational structure and is highly adaptable. In addition, between resonance and unpredictability strikes a balance in programme successfully. On the other hand, including expanded exploration, narrowed exploration, expanded exploitation, and narrowed exploitation, taking into account the predatory behaviour of Aquila are the four steps applied in AO method. Real-time applications and a quick convergence rate can be handled by the AO method successfully. Thus, excellent efficiency and quick convergence by merging both the algorithms are achieved by the AVAO algorithm. Following are the steps in the proposed AVAO algorithm.

(i) **Initialization**

Let us assume there are av number of vultures. The first step is to initialize the population of vultures in the problem space which can be represented by,

$$V = \{V_1, V_2, \ldots, V_i, \ldots, V_{av}\} \tag{8}$$

where, V_i represents the ith vulture in the population.

(ii) **Determine the best vulture**

Once the population is initialized, the best vulture is determined by considering the fitness of all the vultures. The value of fitness is calculated by using the mean square error given by the following equation.

$$\varepsilon = \frac{1}{n}\sum_{o=1}^{n}\left[U_o - U_o^*\right]^2 \tag{9}$$

Here, U_o represents the target output, U_o^* defines the output of the DMN and n designates the overall sample count.

After the fitness is computed, the best vulture of the first group is selected from the group with the best solution and the one with the second-best value of fitness is considered the second group's best vulture. The best vultures are determined by various iterations.

$$W(i) = \begin{cases} Best\ Vulture_1,\ if\ J_i = K_1 \\ Best\ Vulture_2 \quad J_i = K_2 \end{cases} \tag{10}$$

Here, K_1 and K_2 are factors that have to be calculated ahead of the search operation and has a value in the range [0,1] and the factors to be computed before the search mechanism with the measures between 0 and 1. The term J_i represents the probability of selecting the best vulture and is calculated using the roulette wheel.

(iii) **Determination of starvation rate of vultures**

Usually in search of food the vultures fly long distances when they are full and this produces high energy. But in case if they are hungry, they feel a shortage of energy of exploring long distances and they become aggressive and seek the food near the powerful vulture. Thus, the rate at which the vulture is starving determines the exploration and exploitation phases and it can be mathematically modelled by using the following equations. The satiated vulture is given by,

$$SR = (2 \times rd_1 + 1) \times w \times \left(1 - \frac{itr_1}{maxitr}\right) + C \tag{11}$$

$$C = D \times \left(Sin^{\beta}\left(\frac{\pi}{2} \times \frac{itr_i}{maxitr}\right) + Cos\left(\frac{\pi}{2} \times \frac{itr_i}{maxitr}\right) - 1 \right) \tag{12}$$

where, itr and $maxitr$ denote the present iteration count and the overall count of iterations. w, rd_1 and D are arbitrary numbers in the range [0, 1], [−1, 1] and [−2, 2] respectively. Further, β is a parameter, whose value is fixed before the searching process, and the probability of exploration enhances with the value of β. The vultures hunt for food in varied spaces and the algorithm is in exploration phase, if the value of $|SRate| > 1$, otherwise, the exploitation phase is supported.

(iv) **Exploration phase**

Vultures have good quality of eyesight and retain high capability in identifying weak animals, while hunting for food. But, searching for food is highly challenging and the vultures have to perform careful scrutiny of their surroundings for a long period over vast distances. Random areas are examined by the usage of two approaches. An arbitrary parameter I_1, , which has a value in the range [0,1] is utilized to select the approaches. Based on the subsequent equations the strategies are selected.

$$R(i+1) = W(i) - T(i) \times SR \text{ if } I_1 \geq rd_1 \tag{13}$$

$$R(i+1) = W(i) - SR + rd_2 \times ((upb - lwb) \times rd_3 + lwb) \text{ if } I_1 < rd_1 \tag{14}$$

$$T(i) = |Z \times W(i) - R(i)| \tag{15}$$

Here, $R(i+1)$ denotes the vulture position vector, Z represents the coefficient vector. rd_1, rd_2 and rd_3 are random variables in the range [0, 1]. The terms upb and lwb denotes the lower as well as the upper limits of the variable.

Relieving Eq. (13) in Eq. (15),

$$R(i+1) = W(i) - |Z \times W(i) - R(i)| \times SR \tag{16}$$

Here, $W(i) > R(i)$ and hence the above equation can be rewritten as,

$$R(i+1) = W(i) + (Z \times W(i) - R(i)) \times SR \tag{17}$$

$$R(i+1) = W(i)[1 + Z \times SR] - R(i) \times SR \tag{18}$$

In the AO algorithm, Aquila identifies the position of the prey by exploring climbing up and then determining the search area. The expanded exploration ability of the Aquila can be given by,

$$H_1(n+1) = H_{best}(n) \times \left(1 - \frac{n}{N}\right) + (H_r(n) - H_{best}(n) * rnd) \tag{19}$$

where,

$$H_r(n) = \frac{1}{T}\sum_{i=1}^{T} H_i(n) \tag{20}$$

Assume, $T = 1$

$$H_1(n+1) = H_{best}(n) \times \left(1 - \frac{n}{N} - rnd\right) + H(n) \tag{21}$$

$$\text{Consider, } H_1(n+1) = R(i+1) \tag{22}$$

$$H(n) = R(i) \tag{23}$$

$$H_{best}(n) = W(i) \tag{24}$$

$$R(i+1) = W(i) \times \left(1 - \frac{n}{N} - rnd\right) + R(i) \tag{25}$$

$$R(i) = R(i+1) - W(i) \times \left(1 - \frac{n}{N} - rnd\right) \tag{26}$$

Substituting Eq. (26) in Eq. (18),

$$R(i+1) = W(i)[1 + Z \times SR] - R(i+1) \times SR + W(i) \times \left(1 - \frac{n}{N} - rnd\right) \times SR \tag{27}$$

$$R(i+1) + R(i+1) \times SR = W(i)\Big[1 + Z \times SR + \left(1 - \frac{n}{N} - rnd\right) \times SR\Big] \tag{28}$$

$$R(i+1)[1 + SR] = W(i)\Big[1 + \left(Z + \left(1 - \frac{n}{N}\right) - rnd\right) \times SR\Big] \tag{29}$$

$$R(i+1) = \frac{W(i)\big[1 + \big(Z + \big(1 - \frac{n}{N}\big) - rnd\big) \times SR\big]}{[1 + SR]} \tag{30}$$

Here, N denotes the number of samples and rnd is an arbitrary number.

(v) **Exploitation: phase 1**

Depending on the value of SR where exploitation is performed in two phases. If the value of |SR| lies between 0.5 and 1, then phase 1 is executed. Rotating flight and siege-fight are the of two techniques comprised by the first phase. A parameter I_2 is utilized in selecting the strategies, which has to be computed ahead of searching. The parameter is compared to a random variable rd_{I_2} to select the strategies. If $I_2 < rd_{I_2}$, then rotating flight approach is implemented, else siege-fight approach is performed.

(a) **Contest for food**

The vultures are full and have high energy, if $|SR| \geq 0.5$. Brutal disputes might occur When vultures accumulate on a single food source. The highly powerful vultures wouldn't share the food with the weak vultures, whereas the weak vultures attempt to exhaust the strong vultures by assembling around them and snatching the food leading to conflicts.

$$R(i+1) = P(i) \times (SR + rnd_4) - E(t) \tag{31}$$

$$E(t) = H(i) - W(i) \tag{32}$$

Here, rnd_4 is an arbitrary number in the range [0,1].

(b) **Rotating flight of vultures**

A rotational flight is made by the vultures for modelling the spiral movement, and a spiral motion is formed among the best two vultures and the other vultures and this can be modelled as,

$$P(i+1) = W(i) - (X_1 + X_2) \tag{33}$$

$$X_1 = W(i) \times \left(\frac{rnd_5 \times R(i)}{2\pi}\right) \times Cos(R(i)) \tag{34}$$

$$X_2 = W(i) \times \left(\frac{rnd_6 \times R(i)}{2\pi}\right) \times Sin(R(i)) \tag{35}$$

where, rnd_5 and rnd_6 are arbitrary numbers in the range [0,1].

(vi) **Exploitation: phase 2**

In the second phase, by using the siege and aggressive strife strategy the food source is determined, where the other vultures aggregate over the food source following the motion of the best vultures. This phase is executed when $|SR| < 0.5$. A parameter I_3 is utilized in selecting the strategies, which has to be computed ahead of searching. The parameter is compared to a random variable rd_{I_3} to select the strategies. If $I_2 < rd_{\mathrm{I}_2}$, then the cultures are accumulated over the food source, otherwise, aggressive siege-flight strategy is performed.

(a) **Accumulation of vultures over food source**

Here, all vultures close examines the motion of the source of food is carried out. When the vultures are hungry, they compete with each other over the food source. This can be represented as,

$$O_1 = BestV_1(i) - \frac{BestV_1(i) \times R(i)}{BestV_1(i) - R(i)^2} \times SR \tag{36}$$

$$O_2 = BestV_2(i) - \frac{BestV_2(i) \times R(i)}{BestV_2(i) - R(i)^2} \times SR \tag{37}$$

Here, $BestV_1(i)$ and $BestV_2(i)$ denote the best vultures The position of the vulture in the next iteration is given by.

$$R(i+1) = \frac{O_1 + O_2}{2} \tag{38}$$

(b) **Aggressive conflicting for food**

The chief vulture becomes too weak to compete with other vultures when it is starved, when $|SR| < 0.5$, which turns aggressive and moves in multiple directions and leading to the group head in their search for food. This is modelled as,

$$R(i+1) = W(i) - |E(t)| \times SR \times Levy(E) \tag{39}$$

Here, $E(t)$ specifies the distance between a vulture and any one of the best vultures.

(vii) **Feasibility evaluation**

By finding the value of fitness the optimal solution is calculated. If the current solution found has the least fitness, then the existing solution is substituted by the current one.

(viii) **Termination**

The above steps are kept reiterated till the best solution is achieved.

4 Result Analysis

4.1 Simulation Setup

Currently, the CloudSim imitation tool is the most often utilized by academics and developers for studying cloud-related topics. It has the potential to reduce or even finish with the requirement and associated costs, computational facilities for performance assessment and modelling the research solution. Downloading and integrating this simulation tool's external framework is possible with popular IDEs like Eclipse, NetBeans, Maven, and others. NetBeans IDE 8.2 with Windows 10 runs the CloudSim toolkit to mimic the Cloud Computing environment.

To measure the quality of the suggested method in a real-world cloud setting, here it is virtualized a number of entities and computing resources to simulate a scheduling

Table 1 Hardware necessities

Component	Specification
Operating structure	Windows 64-bit OS
Processor	Intel Core™ CPU @ 1.19 GHz
RAM	16.0 GB

and load balancing situation. Two data centres, six virtual machines, and forty jobs, or cloudlets, were used in the studies, all running on a simulation platform. The number of instructions in the task is created at random up to a maximum of 1,000,000,000,000. (MI). The amount of work that can be assigned to a VM is based on its processing power, memory capacity, and network throughput. Table 1 summarises the CloudSim configuration settings.

4.2 Performance Metrics

In the context of the cloud, based on the interaction of three parameters the effectiveness of the suggested LB algorithm was evaluated. The following performance matrix is utilized in the process of measuring and analysing the presentation.

(1) **Makespan (MT)**: It refers to the amount of time that must elapse before a cloudlet can be planned. To evaluate the effectiveness of scheduling algorithms with regard to the passage of time is the primary purpose of this makespan. It has to be chopped down in order to make room for the efficient execution of other activities and to free up resources for other jobs. In the following equations, CT stands for cloudlet completion time, and n represents the total sum of VM in use. This is how it is measured.

$$MT = Max(CT) \tag{40}$$

$$MT_{avg} = \left(\frac{\sum Max(CT)}{n}\right) \tag{41}$$

(2) Execution Time (ExT): It is the precise amount of time needed to complete the tasks (cloudlets) that have been assigned to a virtual machine. It is recommended that this statistic be lowered in order to progress the algorithm's overall performance. The following equations, are measured using AcT, which stands for the total sum of cloudlets.

$$ExT = AcT \tag{42}$$

$$ExT_{avg} = \left(\frac{\sum AcT}{n}\right) \tag{43}$$

(3) Resource Utilization (RU): This is another measurable statistic, and it is dependent on the metrics discussed above. It is measured to enhance the effectiveness of utilizing the resources in a situation where hosted in the cloud. The following equations are used to compute it, where "ExT" stands for "total execution time" and "MT" stands for "total Makespan." The degree to which the suggested algorithm makes effective use of the computer's central processing unit can be evaluated that is based on the average resource usage. This measure has a range of 0 to 1, with a maximum value of 1 representing the best-case scenario, which shows all available resources are utilized, and a minimum value of 0 on behalf of the worst case scenario, which indicates that all available resources are in perfect condition.

$$RU = \left(\frac{ExT}{MT}\right) \quad (44)$$

$$RU_{avg} = \left(\frac{ExT}{MT}\right) \times 100 \quad (45)$$

The objective of performing this experiment is to provide evidence of a dynamic cloud environment can result in a shorter Makespan and execution time while simultaneously improving the amount of resource consumption. During the process of validating the algorithm, some thought has been given to the possibility of preemptively scheduling jobs. This indicates that the task may be stopped in the middle of its execution until the workload exceeds the SLA, and it can be transferred to another resource in order to finish its execution. During the process of scheduling, a number of quality of service performance criteria of cloudlets, including as:

Arrival Time: identifies the time at which cloudlets arrive or the moment at which the algorithm gets the user request. Within the CloudSim environment, this point in time is referred to as the Cloudlet start time. When using CloudSim, the default setting ensures that all cloudlets reach the broker at the same exact moment. This is known as a random Arrival Time parameter, and it has been changed in order to make adjustments in this experiment so that modifications can be made to postpone the submission of cloudlets. The logic that is implemented in this function will cause the broker to allocate the cloudlets to the VMs in a manner that is completely arbitrary. By making use of this parameter, it is able to construct an algorithm that is capable of operating in a dynamic situation in which the arrival time for each request may be varied…

Task Length: determines the size of tasks in bytes; tasks with a lower size have a greater impact on resource consumption. In CloudSim, every Cloudlet must have a length value which identifies the Cloudlet type. This number might indicate whether the cloudlet is a heavy request, a light request, or a medium request. During this experiment, the length of each Cloudlet was measured and given a value determined by chance. It is important for all cloudlets to have random values so that client requests can be distinguished from one another. This is accomplished by designating the length as a value that is chosen at random to represent the entire workload of the cloud environment. In order to calculate the load that will be placed on each Virtual Machine, the length parameter is a crucial input in the experiment. On the basis of this parameter, it is possible to identify the Time taken to Complete requests present in each VM. On the basis of this, It's able to establish whether or not there is a breach of the SLA…

Deadline: For CSPs, one of the important tasks of SLA is to establish the maximum time taken to complete the task. In the context of this experiment, each Cloudlet has a unique deadline value; thus, each customer will receive a unique SLA contract that is tailored to their specific requirements and the level of service they anticipate receiving from cloud providers. As a result, it is recommended to utilize the random deadline value rather than a static one. Here, it is determined whether a breach of the (SLA) or not by looking whether the Time is taken to Complete the requests before the Deadline.

Table 2 provides an illustration of some of the variables that comprise the workload and are determined by the two factors discussed above (Task Length and Deadline).

Here, the effectiveness of the suggested method is measured across three distinct scenarios: There are three options here: (1) two VMs housing 10–40 cloudlets, (2) four VMs housing 10–40 cloudlets, and (3) six VMs housing 10–40 cloudlets. These simulation-based variables can be increased to better simulate scheduling and workload migration between virtual machines.

As various values are evaluated for work criteria like Deadline, arrival time, and task length, the typical Makespan, Execution time, and resource consumption were recorded. All virtual machines (VMs) in the cloud used for the experiment are identical in terms of their processing power and storage space for each scenario. The allocation of resources is then modified based on the violation of a VM. Tables 3, 4 and 5 detail the changes in the proposed model's metrics as a result of varying virtual machine types.

Table 2 CloudSim simulator requirements

Type	Parameter	Value
Cloudlet (task)	Length of task (in bytes) Total number of task	Random < upper threshold (1,000,000) 1–40
Virtual machine (VM)	Count of virtual machines (VMs) CPU frequency and amount of RAM in a single VM Cloudlet scheduler The minimum sum of processor elements (PEs) that must be provided VMM	2–6 9980–15,000-MIPS 512-Mb 1000-Mb Time shared 1 Xen
Data centre	Sum of data centres Sum of host VmScheduler	2 1 Time shared

Table 3 Results were gotten for 2 VMs with 10 To 40 tasks

Number of cloudlets	Average execution time (ms)	Resource utilization (%)	Average makespan (ms)
10	195.9367204	75	261.4400938
15	289.1025095	74	388.8291983
20	366.7708264	72	512.0973245
25	418.7846801	68	616.2059895
30	527.3666282	70	753.4639850
35	618.4638792	71	870.4258042
40	607.1213388	68	892.7444428

Table 4 Consequences were gotten for 4 VMs with 10 To 40 tasks

Number of cloudlets	Average execution time (ms)	Resource utilization (%)	Average makespan (ms)
10	205.583434	76	270.8765645
15	289.853678	74	390.8272922
20	368.874635	72	514.3474892
25	427.046281	69	618.8271912
30	534.267521	71	753.5425567
35	630.6788556	72	870.2578313
40	614.9916124	69	894.7453515

Table 5 Consequences were gotten for 6 VMs with 10 To 40 tasks

Number of cloudlets	Average execution time (ms)	Resource utilization (%)	Average makespan (ms)
10	251.8378228	83	303.5410115
15	361.2242912	84	430.8096146
20	407.6729213	79	512.9966944
25	423.6572891	73	577.6690028
30	559.9376242	74	754.0924138
35	650.2356788	75	869.5038272
40	638.7329182	71	896.2638292

In the course of this investigation, Makespan Time has served as the primary comparative variable of interest. The primary purpose of the Load Balancing algorithm has been presented to improve the use and allocation of cloud resources while simultaneously reducing the amount of time needed to plan a job in order to boost the overall performance of cloud-based applications. This technique was designed in order to optimize the Makespan and allot cloudlets to Virtual Machines in the most possible effective manner. It makes effective by the use of the resources that are available in the cloud environment. This algorithm was chosen to serve as a point of comparison, that is related closely to the main aim of this research and the framework for its execution. This method was included in QoS parameters and priority as part of the future work that was planned for the Dynamic LB algorithm. The purpose of our algorithm was to highlight how the results might be modified by depending on whether these factors were employed or not. Both of these methods make use of factors like the cloudlet length and the completion time. The LB method has been developed are taken into account various arrival times and deadlines in order to follow up with the SLA document in a manner which is in line with QoS criteria. This result gives improved service for cloud applications. Makespan is used to make comparisons between the outcomes, as illustrated in Fig. 4, where the y-axis shows the value of Makespan. In conclusion, the results have been achieved for a total of forty different assignments.

The experiment took into consideration of very broad spectrum of task lengths, which is shown in the graph as a rise in Makespan for the proposed method as the number of tasks grows from 25 to 40. The LB method presented can handle requests for bigger jobs up to 1,000,000 MI in length. Because Makespan is dependent on the load that is placed on the VMs, increasing the task length will also result in an increase in Makespan. Makespan will be reduced for cases involving 25–40 jobs since the suggested approach is modified to handle a lesser size.

In addition, comparisons are made between the results based on resource usage, as shown in Fig. 5, where the y-axis represents the Resource utilization expressed as a % value and the x-axis indicates the total number of cloudlets (tasks).

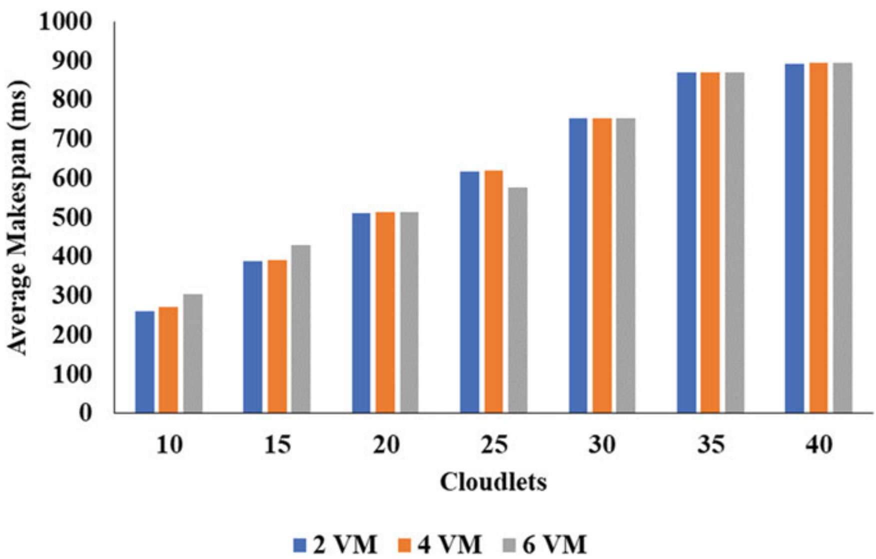

Fig. 4 Graphical representation of the proposed model in makespan

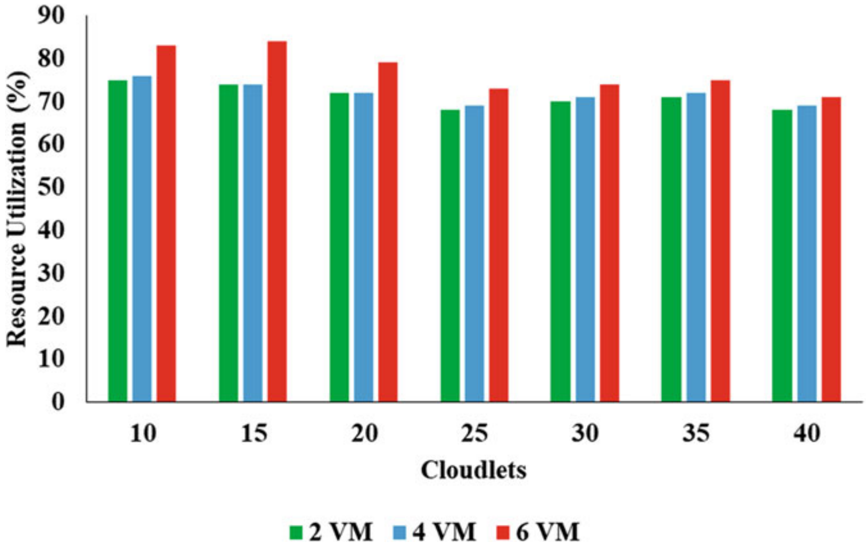

Fig. 5 Graphical representation of proposed model in resource utilization

The new method achieves a slightly higher level of resource utilization in 6 virtual machines, by reaching 78% of their total capacity. In the final analysis of the data, it was demonstrated that considering QoS criteria like the Deadline can considerably enhance the consumption of resources, hence shortening the Makespan and offering an effective allocation strategy in VMs. In addition, the Workload Balancing Algorithm for Data Centres to Optimize Cloud Computing Applications which are developed might be assistance to a variety of applications, such as location-, and so on.

5 Summary

This part serves as the paper's conclusion and summarises the findings as well as the outcomes achieved from the suggested LB algorithm. A technique for load balancing that is based on evolutionary computing is presented in this study. When building symbolic regression models of VMs using AVAO, loads on VMs as well as the resource metrics associated with those VMs are monitored and utilized as inputs. The ideal combination of virtual machine and virtual machine host (VM-VMH) assignment is determined by AVAO, which also anticipates VMH loads based on AVAO models and advises the VMs to be transferred for load balancing purposes. Hence this research gives knowledge about, task scheduling plays a significant role in ensuring that a cloud environment's workload is distributed evenly. The process of load balancing can be improved by utilizing task scheduling, which can lead to more effective use of cloud resources. A more efficient algorithm for load balancing was the focus of this particular piece of research. The findings demonstrated that the suggested approach shortens the Makespan and maximizes the consumption of resources by 78%. It also demonstrates that the suggested algorithm is capable of functioning in a dynamic cloud environment, which is characterized by a large number of variations in the duration of user requests and the arrival of requests from users in an arbitrary sequence. In comparison to the current method, the algorithm is also able to manage requests of a significant scale. By redistributing resources, the programme corrects SLA violations of VMs and the tasks are carried out effectively.

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Cyber Intrusion Detection Using a Boosting Ensemble of Neural Networks

Premanand Ghadekar, Amrut Bhagwat, Kunal Jadhav, Aditya Kirar, and Ankit Singh

Abstract With the world increasingly becoming more and more virtually connected, the threat of intrusion in systems and networks is a rapidly increasing threat as well. A lot of different ways to detect this intrusion and ultimately prevent it are being developed. The best way is the implementation of deep learning algorithms and neural networks to detect the anomalies in a given system/network. The use of LSTMs (Long Short Term Memory) has been quite popular but they have their limitations as well. The proposed system develops an ensemble with 3 different LSTM-hybrids to develop a robust and better model that detects a large number of cyber attacks and provides better accuracy compared to a single LSTM model. Out of the three LSTM-hybrids used CNN-LSTM provided high accuracy with low FAR (false alarm rate) while RNN-LSTM was able to classify 3 major attacks more accurately but it was not much robust with R2L attacks and was giving high FAR. While the MLSTM is able to classify R2L attacks in a proper manner. The proposed ensemble model of LSTM and neural networks for intrusion detection have achieved an accuracy score of 98.7% for NSL-KDD dataset and 99.3% for KDD99 dataset.

Keywords Cyber security · Neural networks · Ensemble learning · LSTM · Deep learning

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P. Ghadekar (✉) · A. Bhagwat · K. Jadhav · A. Kirar · A. Singh
Vishwakarma Institute of Technology, Pune, India
e-mail: premanand.ghadekar@vit.edu

A. Bhagwat
e-mail: amrut.bhagwat20@vit.edu

K. Jadhav
e-mail: kunal.jadhav20@vit.edu

A. Kirar
e-mail: aditya.kirar20@vit.edu

A. Singh
e-mail: ankit.singh20@vit.edu

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1 Introduction

The modern world is a completely virtual world as of now and the progress of technologies and advancements in this field are still rapidly increasing. With the revolution that computers have brought around, we humans have developed so many different things with said computers and related to said computers as well. As it stands now, humans are completely dependent on computers and the internet in day to day lives. Especially in the past few years, since the increased use of cloud computing, IoT services, and any and all sort of communication or entertainment service is most certainly over the internet today [1]. This increased reliance can also have negative repercussions if it is not thought about in a well-rounded way. Any sort of new rise brings its threats along with the opportunities as well. It would not be prudent to only focus on the endless opportunities that the past few years have brought.

One of the biggest threats with the extreme reliance on networks and the internet is obviously security [2]. Network security is a very vast and constantly evolving topic. Cyber attacks have also developed quite a bit throughout the years and hence there is a constant need for the cyber security measures to evolve as well.

With complex networks and setups, it is crucial to note the fact that any threat to the network or vulnerability will affect the entire network and can lead to huge losses and setbacks [3]. With the constant evolution of various types of cyber attacks and increasing threats to security, it is felt that just the traditional methods of security can no longer keep up. Hence the need to develop efficient network intrusion detection systems is felt quite strongly [4]. These systems should be able to provide the integrity and privacy of the data by accurately detecting the attacks and threats with a high accuracy and a low false alarm rate.

Signature based detection or misuse detection is the model where detection is based on known attacks and threats. This model has high accuracy and low false alarm rates but the major drawback with it is that it doesn't change or evolve over time [5]. Stagnant detection systems are easily overcome within a matter of a few years or months even with the rate at which all systems are advancing [6]. To prove effective, an intrusion detection system will have to be able to perform well against known and unknown attacks.

Machine learning and Deep learning are both fields that have been used for a wide variety of applications, so it should be no surprise that they have also been employed in the field of intrusion detection. Machine learning systems are based on manually extracted features and deep learning systems extract features from the available data themselves [7].

A lot of work has been done regarding the use of ML/DL techniques in intrusion detection over the last couple of years. The use of LSTMs as well as various other neural network architectures has been seen and has proved quite effective. LSTMs are especially useful in this field as they are designed for time series predictions which can be very useful here, to predict or classify the cyber attacks over time [8]. LSTM-hybrids will be the focus of this paper.

Deep learning in intrusion detection is preferred because of the feature extraction ease as well as the fact that the neural network can learn from the unstructured or unlabelled data [9]. DL algorithms are also robust and scalable with the data availability. The deep learning models are developed for complex problems like pattern recognition, natural language processing (NLP), etc. so intrusion detection systems are also efficiently developed using the deep learning algorithms.

2 Literature Survey

Intrusion over networks is a big problem now with the increase in network systems as well as the developing types of various attacks. One widely used deep learning method for this is the CNN-LSTM model. It is felt that the machine learning alternatives for intrusion detection are not sufficient and there is a need to develop deep learning models [10]. The use of datasets other than the one discussed is also a factor to be considered. Even though they are out of date, the KDD datasets are the most complete datasets for the problem of intrusion detection. It is found that CNN-LSTM models are extremely efficient and accurate for binary classification of cyber intrusion detection ie. just detecting whether a system is clean or has been attacked. But it fails for multiclass classification and cannot accurately distinguish between certain attacks. The false alarm rate for CNN-LSTM is also extremely low and hence it is good for the intrusion detection system [11].

The use of LSTMs as stated above is the focus of this paper as all the models used in the ensemble are going to be LSTM based hybrids. The use of LSTMs for temporal feature extraction in intrusion detection systems has already been carried out [12]. In a referred paper, three LSTM approaches are described. One is a plain LSTM, one is with dimensionality reduction using principal component analysis (PCA), and one is with mutual information. On the KDD99 dataset using LSTM with PCA, the model was able to get above 99% accuracy in classification.

Recurrent neural networks are also quite helpful in this field, they can be used to produce new attack types so that the model can be trained faster and in an efficient manner []. To make RNN more powerful for modeling purposes cyclic connections are made. A proposed model contributed to generating the malware mutants, the attack signature and the synthetic data gathering [13].

Another referred paper used the RNN-LSTM model on the KDD dataset. A multi-class classifier was designed using that algorithm and it gave a decent overall accuracy of around 95%. The false alarm rate of this model was also high but it gave better results when compared to similar RNN-LSTM models [14].

Intrusion over network system is a big problem to handle. Now with the increase of network systems the type of attacks are also increasing and thus the paper titled, CNN-LSTM: Hybrid Deep Neural Network for Network Intrusion Detection System elaborates that the use of ML models is not sufficient to counter the problem [15]. But with the technological advancement in deep learning and also ANN the researchers have proposed a system which uses CNN and LSTM. The ability of CNN is that it can extract important features and the LSTM has the ability to extract time based features. Thus by using CNN and LSTM, they have proposed a hybrid IDS. The dataset used was CIC-IDS-2017. Efficiency was determined by confusion matrix and performance metrics like precision, accuracy was also used [16].

In the paper A CNN-LSM Model for Intrusion Detection System from High Dimensional Data authors describe the use of deep learning techniques of CNN and LSTM for the detection of intrusion over the Internet the dataset used by them was KDD99. The proposed model did data pre-processing, then feature extraction, and after that continuous testing [17]. Accuracy achieved by the model of CNN-LSTM over the KDD99 dataset was 99.78%.

Important technology in future the software defined networking is attacked by new kin of intrusions. The combination of CNN and LSTM where one is used for the extraction of spatial features and the other is used for the extraction of temporal features is indeed increasing the accuracy to a large extent. The use of Keras was done for the training of CNN. The proposed IDS model in this paper A Hybrid CNN-LSTM based approach for Anomaly Detection Systems in SDNs was able to grab highest accuracy of 96.32%.

The proposed system describes intrusion on AMI (advanced metering infrastructure). IDS systems are able to detect abnormal activities over AMI networks. Fusion of features was done to represent the different characteristics of data. The used datasets were NSL-KDD and KDD Cup 99, the activation function used was ReLU. For NSL-KDD the accuracy was almost 99.95% while for KDD Cup 99 was 99.79%. Thus CNN-LSTM approach is efficient for IDS [18].

NSL-KDD dataset is one of the most used dataset for applying deep learning based IDS models intrusion detection. The paper describes attack categories and also features present in the dataset. Many machine learning algorithms are applied on the dataset but due to false alarm rate, the use of deep learning models becomes important as it deals with complex relationships [19]. The CNN-LSTM model used in the research was able to gain great accuracy. Also, different techniques like DenseNet and Gated Recurrent Unit with Softmax were able to achieve accuracy of 94.98% and 97.36% respectively. The proposed model was able to achieve an accuracy of 99.70% on NSL-KDD dataset.

The paper describes the use of the characteristic feature of LSTM of temporal feature extraction for IDS. The paper uses three approaches of LSTM. In the first one, LSTM is used without any dimensionality reduction then it is used with PCA and after that with mutual information. The dataset used was KDD99. The model while using LSTM with PCA was able to achieve an accuracy of 99.49%. To see the effectiveness of the model confusion matrix was used.

Recurrent Neural Networks can be helpful to produce new attack types so that the model could be trained in an efficient manner. The extension of feed-forward neural network is the RNN. To make RNN more powerful for modeling sequences they have cyclic connections. The proposed model contributes for generating malware mutants, then generating the attack signature and after that for synthetic data generation [20].

This literature survey makes a few things clear based on which the ensemble is designed and gets better results improving on the existing models. Firstly, CNN-LSTM is excellent for binary classification but fails at detecting specific types of attacks [21]. RNN-LSTM is the most well-rounded model for multiclass classification with decently high accuracy. LSTM by itself is also a very powerful tool and can be improved upon by using the Modified LSTM (MLSTM) model.

The paper uses RNN-LSTM classifier to detect intrusion on the KDD Cup dataset. The proposed system before using training dataset had normalized the instances from 0 to 1. Their input vector had 41 attributes while they were detecting 4 attacks and 1 non-attack. Thus they had an input dimension of 41 while an output dimension of five [22]. The accuracy which this proposed system achieved was 96.93%. The false acceptance rate of the proposed system was more than other models but accuracy was greater.

The proposed system uses spatiotemporal encodes which are unsupervised and with the help of these spatiotemporal encoders the spatial features are intelligently extracted from the network traffic. And after that, these extracted features are used as an input to LSTM model. Thus by using these spatial features the LSTM model detects intrusion and classifies it [23]. The proposed system again proves that using neural networks with LSTM gives far better accuracy than other traditional ways.

3 Methodology

3 different LSTM-hybrid models were created and trained individually. These hybrids were chosen so that the shortcomings or weaknesses of one can be compensated by the other. Then these models were ensemble in series with each model filtering out specific cases and passing on the ones it is not good enough to handle to the next one. As a result, all will be filtered out depending on the type of cyber attack/anomaly [24].

A. Datasets used

For the proposed model the main dataset used was the KDD Cup 1999 dataset. Even though it is a quite old dataset it is extremely well endowed and has a lot of cases for the model to train on and test. Along with that, the newer NSL-KDD dataset was also used. The data is divided into a few major types of attacks.

1. DOS: Denial of service attack. These attacks make the computing or memory resources too busy and overloaded to deny a legitimate user access to these resources of the machine.
2. U2R: User to root attack. An exploit which allows a user (legitimate or illegitimate ie. someone with access to the machine) is able to gain the root access to the system.
3. Probing: Any attempt to gather information about a network in order to bypass or circumvent its security controls and measures is called as a probing attack.

4. R2L: Remote to local attack. A remote user who doesn't have access to a machine but is able to send packets to a said machine and exploits a weakness to gain access to the local machine is called a remote to local attack.

There are a total of 22 attack types in the training set for KDD99, and 15 types in the testing set. The specifics of each attack are different but all of them are classified among these 4 broad archetypes. Along with that the data also contains systems with no anomalies ie. "Normal" states. The dataset has over 4.5 million entries, each being a vector with 41 features. Each entry is labeled as an attack or as normal.

The NSL-KDD dataset is the upgraded version of the KDD 1999 dataset. The NSL-KDD dataset aims to solve some of the major issues that are present in its predecessor [25]. One of the major things is that it removes the duplicate records in the training and testing sets so that the classifiers are not biased for the repeating records. The training and testing sets were also generated from different parts of the original KDD99 dataset to obtain authentic results while using various classifiers. The unbalanced problem of the training and testing data was also solved in this dataset to help reduce the False Alarm Rate (FAR).

All in all, these are both very well-rounded datasets but the biggest problem that is faced when using these is that they are very outdated. There is no idea about the representation of the modern norms and attacks and the respective network activities. Another issue faced is that the probability distributions of the training and testing sets for the data are not the same. This may lead to skewness in some of the classifiers that are used with this dataset [26].

B. Models used

The 3 different LSTM hybrid models implemented are as follows-

1. CNN-LSTM ()
2. RNN-LSTM ()
3. MLSTM (Modified LSTM).

CNN-LSTM:

The CNN- LSTM model is essentially a bunch of CNN (Convolutional neural network) layers on the input data followed by LSTM layers. The CNN architecture is extremely good at feature extraction and the LSTM layers support learning over time and sequence predictions. The same can be seen in Fig. 1 which is a graphical representation of the architecture. These models were first developed for visual time series applications e.g. activity recognition, image and video description etc. This model was also called a Long-term Recurrent Convolutional Network (LRCN) but are now referred to as CNN-LSTMs. These models are good with any data which has spatiotemporal features [27].

The first evaluation is done by the CNN-LSTM model. This model is a very good first evaluation tool as it can quite accurately discriminate Normal systems and DOS, U2R attacked systems with a sufficiently high accuracy. That leaves only the probing attacks and R2L attacks which are further handled by the next models in the series ensemble.

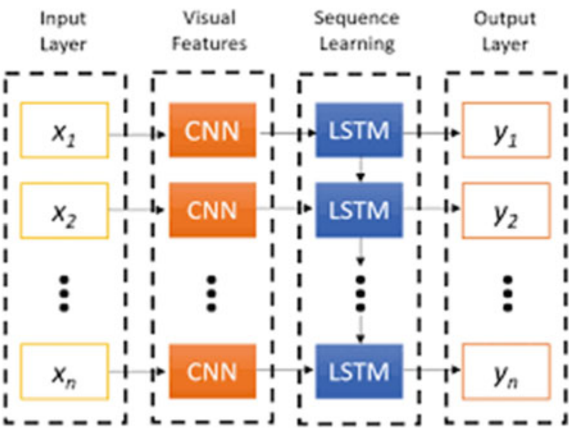


Fig. 1 CNN-LSTM architecture

RNN-LSTM:

LSTMs act like an extension for the Recurrent neural networks (RNN) in this architecture which extends the memory for the learning of the model. This model is extremely good for learning from experiences that have long time lags between them [28].

LSTM units are used as the building blocks of a recurrent neural network The LSTM cells can read, write and delete its memory. This is done by the respective gates, it is a gated cell. Input gate controls the entry of new information. Forget gate controls the deletion of information that is no longer useful. Output gate controls whether or not to let the information affect output at the current stage. The LSTM cell can be better understood with the help of Fig. 2.

Modified LSTM (MLSTM):

This model builds on the existing LSTM model to provide better flexibility in the application. It also accommodates probabilistic gate cell states in the LSTM model using sigmoid functions [29]. For e.g. Consider the output gate of the LSTM cell. The information to be passed is decided through here. Normally in a LSTM cell, it

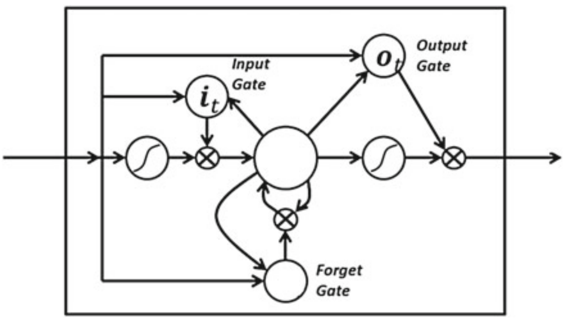
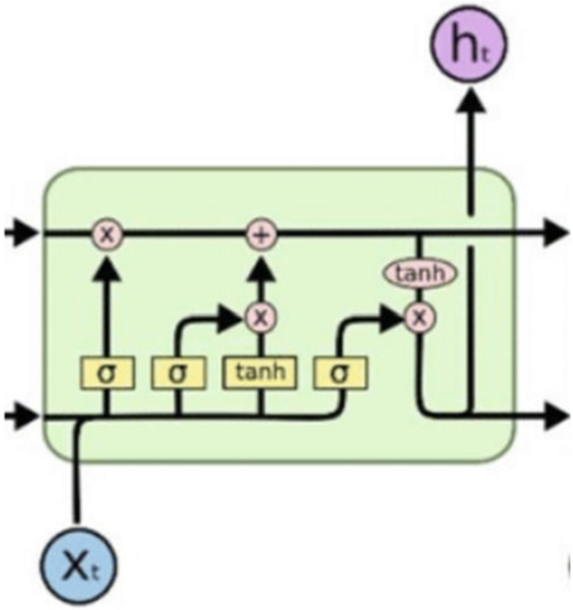


Fig. 2 LSTM gated cell

Fig. 3 LSTM architecture



is possible to only let the information pass or not pass ie. boolean logic. With the sigmoid function, a return value between 0 and 1 is obtained at each step in the cell and based on the value the amount of information to be passed is decided (0 being no information to be passed and 1 being all of the information to be passed). This is done at all the gates in the cell, as shown in the diagram below i.e. Fig. 3.

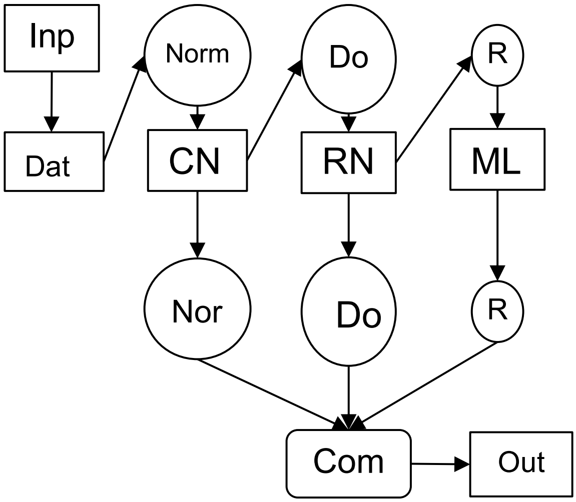
The tan(h) function is to make a vector of new values which will be added to the current state C [14].

C. Proposed model flow

As stated earlier before the explanation of each model, 3 models are used in the ensemble. First, the evaluation through CNN-LSTM is done as it is the most accurate out of all the three for a wide range of attacks as well as binary classification (Normal vs. anomaly) [30]. The accuracy of the Normal system detection through CNN-LSTM is very high along with Probing and DoS attacks. So many times CNN-LSTM can differentiate 3 out of the 5 possible outcomes by itself. Hence it is used as the first model. However, sometimes for the Probing and DoS attacks, the accuracy might not be high enough to ascertain beyond a reasonable doubt whether or not it is the attack specified. In that case, the Probing and DoS attacks are also differentiated by the next model.

The next model used in the series-ensemble is the RNN-LSTM model. This is a really good tool for the differentiation of various attacks and is used for exactly that. The Normal system vectors are already filtered out by the CNN-LSTM one step above and hence only the attacks remain to be sorted. RNN-LSTM is quite accurate

Fig. 4 Project flow diagram



with that but might fail for R2L (root to local) attacks. Hence it is given all the attack vectors as input (Probing and DoS excluded if CNN-LSTM is accurate), and it differentiates between them well enough. The drawback of the RNN-LSTM, which was the high false alarm rate, is also overcome by using this ensemble as the vectors from which it has to differentiate are almost definitely attack vectors (CNN-LSTM filters out Normal systems).

The final model in the ensemble is the MLSTM. This is the most niche model because it is trained for a very specific attack (R2L) as that is the only one left which still might not be accurately classified after the first 2 steps. The use of MLSTM gives better accuracy and efficiency in training the model as well as removes the need for dimensionality reduction to a certain degree. The flow of the project can be better understood with the diagram ie, Fig. 4.

As with any ML/DL model, one of the most important parts of the training testing process is the preparation and pre-processing of the data. Firstly the obtained data from the dataset is cleaned by removing the duplicates in it or by truncating the incomplete data entries etc. The data is then normalized and then fitted and scaled as per the requirements of the model [31].

In this system, as stated above, the KDD99 and NSL-KDD datasets have been used, which have 22 attack types in all. But based on their nature they are classified into 4 broad archetypes. The ideal case would be to run the multiclass classifier for all of the various 22 attack types included in the dataset but that would be too computationally exhaustive and hence is not feasible for the current scope of the proposed model.

D. Algorithm

Step 1: Input the data to CNN-LSTM model.

(a) The convolution layer extracts features from input data.

(b) The connected layers of CNN use data from convolution layer to create output.
(c) Considering CNN to be a mathematical model M:h($X^{(0)}$;θ). where can be input vector, and θ is weight of neuron connections.
(d) $X^{(l)}$ neuron responses on the t-th layer, which is W_l*H_l*D_l cube.
(e) $X^{(l)}$ is average over all spatial positions of $X^{(l)}$, i.e., a Dl-dimensional vector.
(f) *Equation 1*: $-x_d^{(l)} = (1/W_t \times H_t)\sum_{w=0}^{W_{t-1}}\sum_{h=0}^{H_{t-1}} X_{w,h,d}^{(l)}$
(g) Use of ReLU activation function.
(h) The output is further passed to LSTM.

Step 2: CNN-LSTM classifier working.

(a) LSTM uses a sigmoid activation function for the forget gate and input gate.
(b) The equations listed below are for forget gate, input gate and output gate respectively.
    Equation 2: $f_t = \sigma(w_f[h_{t-1}, x_t] + b_f)$
    Equation 3: $i_t = \sigma(w_i[h_{t-1}, x_t] + b_i)$
    Equation 4: $o_t = \sigma(w_o[h_{t-1}, x_t] + b_o)$
(c) The extracted features of CNN are used to predict the different types of attacks.

Step 3: Passing CNN-LSTM models output to RNN-LSTM.

(a) The RNN-LSTM is provided with CNN-LSTM output with only attacks that are normal system integer filtered.
(b) RNN-LSTM employs LSTM cells in RNN structure.

Step 4: RNN-LSTM classifier working.

(a) The RNN-LSTM works on all types of attacks and classifies them.
(b) R2L attacks are not properly handled by this classifier.

Step 5: Passing all remaining data including the major part of R2L attack.

(a) The MLSTM model classifies the R2L attack accurately.

Step 6: End.

4 Results and Discussion

The design of the ensemble can be said to be successful as it worked as intended and gave accurate results for all the classes. The main objective of this ensemble was to combine the good parts of the individual specific models and also try to cover up for the weaknesses of each of the models. Through the experimental results, it can be said that this objective has been achieved. Comparing the individual performances of the models and those of the ensemble as a whole it can be said that the ensemble has put together all the benefits of its constituent models and has also made up for the weaknesses of each individual model. The performance metrics used for the evaluation of the models are Accuracy and FAR (False alarm rate).

Table 1 CNN-LSTM performance metrics

CNN-LSTM		
Accuracy	FAR	
KDD99		
0.999	0.058	Normal
0.941	0.002	DOS
0.948	0.019	Probing
0.414	0	U2R
0.258	0.001	R2L
Overall accuracy	0.933	
NSL-KDD		
0.997	0.124	Normal
0.747	0.018	DOS
0.898	0.093	Probing
0.45	0.001	U2R
0.429	0.002	R2L
Overall accuracy	0.832	

As can be seen in Table 1, The CNN-LSTM model gives high accuracy for binary classification on whether the system is normal or attacked, but fails for U2R and R2L attacks as well as distinguishing between various attack types for the NSL-KDD dataset.

Just like seen with the CNN-LSTM model, the results for the RNN-LSTM can also be seen as described in Table 2.

MLSTM—As stated earlier, the training for MLSTM was done only for the R2L attacks and hence a separate parametric of the model was not taken.

Following are the results of the designed ensemble as a whole.

From the results of Table 3, a much higher overall as well as class wise accuracy is obtained from the ensemble than from the individual models.

Another performance metric that was later checked was the false negative rate for each attack and model. This was essential to see what percentage of the attacks were labeled as either harmless or as some other attack. This metric should be as low as possible. The results of the same for CNN-LSTM can be seen in Table 4.

Similar results for the entire ensemble can be seen in Table 5. As can be seen, the high false negative rates for U2R and R2L are cut down significantly by using the ensemble and hence it is a better way for the classification of these attacks.

This ensemble will be able to detect a wide range of attacks now without giving up any weakness towards a specific area as the other models will cover the weaknesses if one of them has any. The idea behind the making of this ensemble was hence successful.

The idea behind this ensemble was to improve upon the existing models which have a lot of weaknesses that they cannot make up for individually. When the models

Table 2 RNN-LSTM performance metrics

RNN-LSTM		
Accuracy	FAR	
KDD99		
0.963	0.091	Normal
0.987	0.072	DOS
0.969	0.044	Probing
0.978	0.063	U2R
0.688	0.061	R2L
Overall accuracy	0.969	
NSL-KDD		
0.924	0.121	Normal
0.981	0.083	DOS
0.977	0.132	Probing
0.961	0.052	U2R
0.582	0.07	R2L
Overall accuracy	0.942	

Table 3 Ensemble performance metrics

Accuracy	FAR	
KDD99		
0.999	0.058	Normal
0.987	0.002	DOS
0.969	0.019	Probing
0.978	0.063	U2R
0.991	0.022	R2L
Overall accuracy	0.993	
NSL-KDD		
0.997	0.124	Normal
0.981	0.043	DOS
0.977	0.031	Probing
0.961	0.014	U2R
0.982	0.011	R2L
Overall accuracy	0.987	

included in the ensemble are used in series they make up for each other's weak points and hence give an overall better result for the multiclass classification of the dataset.

Consider the CNN-LSTM architecture which is a widely used model and one that is used in ensembles too. It is the best deep learning model for the binary classification (normal vs attack) of the dataset. The biggest problem it faces is that the accuracy

Table 4 CNN-LSTM false negative

	FNR
KDD99	
Normal	0.001
DoS	0.059
Probing	0.052
U2R	0.586
R2L	0.742
Overall	0.067
NSL-KDD	
Normal	0.003
DoS	0.253
Probing	0.102
U2R	0.55
R2L	0.571
Overall	0.168

Table 5 Ensemble false negative

	FNR
KDD99	
Normal	0.001
DoS	0.013
Probing	0.031
U2R	0.022
R2L	0.009
Overall	0.007
NSL-KDD	
Normal	0.003
DoS	0.019
Probing	0.023
U2R	0.039
R2L	0.018
Overall	0.013

for the different attack classes is dismal. This is a big problem when multiclass classification is wanted.

Similarly, consider the RNN-LSTM architecture which is robust for multiclass classification of the attacks but the main problem is that it gives a high FAR (False Alarm Rate). When used in series after the CNN-LSTM architecture, it essentially filters out almost all of the Normal system cases hence drastically improving the

FAR scores when using RNN-LSTM. This is the way the ensemble performs better than the existing models by using their strengths for good results and also covering for their weaknesses to improve accuracy even further.

5 Conclusion

From the results, it can be concluded that using an ensemble model helps increase accuracy to a large extent. In both the datasets NSL-KDD and KDD99 models give accuracy and FAR for normal attacks and 4 types of different attacks. From the results, it can be seen that the ensemble model performs better on both the datasets comparing CNN-LSTM and RNN-LSTM individually. The ensemble performs better than the existing models yielding a higher accuracy and a low FAR. But the trade-off for this is that the proposed model is definitely more computationally exhaustive than normal single models or traditional ensembles using simpler algorithms. The use of other datasets for training and testing purposes can also be seen as a future scope. But the biggest direction in which the proposed model can be taken in the future is trying to combine simpler algorithms with the proposed model to try and reduce the computational load but also try and maximize the performance metrics as in this ensemble.

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Hyperparameter Study: An Analysis of Hyperparameters and Their Search Methodology

Gyananjaya Tripathy and Aakanksha Sharaff

Abstract Deep neural networks have significantly improved people's daily lives since they were created. Deep learning provides more rational direction than humans can in almost every aspect of daily life. Though it has been likened to alchemy, developing, and training neural networks continues to be a challenging and unpredictable process. Automated hyperparameter tuning has gained popularity in both academic and commercial circles to minimize consumer technical barriers. The most important areas of hyperparameter tuning are reviewed in this work. The primary neural network model hyperparameters are introduced in the initial section, along with their significance and methods for defining the value range. The discussion continues with the most popular optimization approaches and their efficiency. Finally, the second-order optimization and its usefulness over first-order optimization are discussed in this study.

Keywords Hyperparameter tuning · Deep neural network · Classical searching technique · Second-order optimization

1 Introduction

Neural network (NN) approaches have expanded and gained influence over the last few years in academic and industrial sectors. NNs have recently shown impressive results in the areas of image classification, objective detection, industrial control systems, and natural language understanding [1, 2]. The most popular strategy for hyperparameter (HP) tuning is usually based on experience, which implies that to develop a workable set of HPs, researchers must have previous knowledge of NN training. According to the no-free-lunch theorem, all optimization problems have a

G. Tripathy (✉) · A. Sharaff
National Institute of Technology Raipur, Raipur, Chhattisgarh, India
e-mail: gtripathy.phd2020.cse@nitrr.ac.in

A. Sharaff
e-mail: asharaff.cs@nitrr.ac.in

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fixed computing cost, and there is no shortcut to solve them [3, 4]. Specialists' prior knowledge, which helps select crucial parameters and narrows the search space, can be used in place of computational resources.

HPs are variables that are fixed throughout the training process of a NN system. Several factors, including the learning rate (LR), loss function, and optimizer, can influence how well and precisely a model is taught. The activation function and the number of hidden layers are other factors. HP tuning may be viewed as the first stage of NN training and the final stage of model development [2, 5]. To cut humans out of the machine-learning system's feedback loop, HP tuning automatically improves a model's HPs. Before the training process begins, HPs must be properly adjusted with expertise because of their impact on training accuracy and speed [6–8]. When several HPs are tuned concurrently, adjusting HPs involves a trade-off between computing resources and human effort. HP tuning has become more crucial in recent years due to two new advancements in the development of deep learning models. Upscaling NNs for greater accuracy is the first requirement. Empirical research has shown that more complex NNs with wider and deeper layers often outperform those with simple architecture [2, 9, 10]. The development of a demanding lightweight model with fewer weights and characteristics to provide satisfying accuracy is the second requirement. HP tuning is essential in both scenarios: a model with a complicated structure indicates that there are extra HPs to tune, and a well-planned structure entails that each HP must always be set to a certain range in order to repeat the accuracy. The ability to modify manually depends on experience, and researchers may always learn from previous work; thus, adjusting the HPs of a widely used model is conceivable. This is equivalent to small models. But, especially for bigger models or freshly published models, the vast array of HP possibilities demands a lot of researchers' painstaking work, as well as a lot of time and computer resources for trial and error. Figure 1 depicts the block diagram for the HP tuning process.

The major contribution of the study is as follows:

- The discussion of the crucial deep neural network (DNN) model HPs, along with their significance and default settings.
- The major HP tuning strategies are examined in depth to demonstrate their applicability to the given problem.
- The comparison of the second-order optimization technique to first-order.

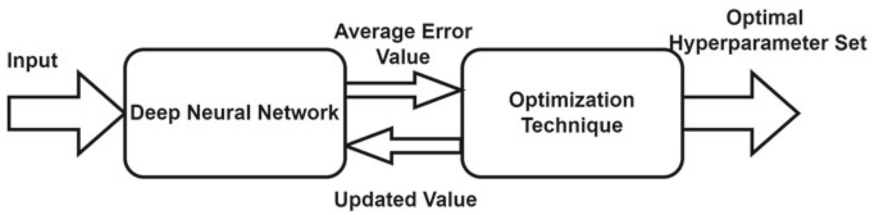

Fig. 1 Block diagram of the HP tuning process

The rest of the paper is structured as follows. The required HPs and their default values are clearly outlined in Sect. 2. The standard algorithms for HP tuning are discussed in Sect. 3. The second-order optimization applied to NN models is described in Sect. 4. Section 5 shows the experimental setup, Sect. 6 summarizes the significant ideas, while Sect. 7 concludes the study.

2 Key Hyperparameters and Its Usefulness

HPs have a significant impact on the effectiveness of NNs. HPs with a higher impact on weight during training are more important for NN training. Hence, it is given preferential attention. There are two types of HPs: those used to build models and those used to train models. NNs may learn more rapidly and perform better when the right HPs are chosen for the model training process.

Performance of the model is more influenced by several HPs such as optimizer, LR, loss function, activation function, and network depth. The previously mentioned HPs aside from batch size and LR—are supplied throughout the model-designing process. When training, the LR, and batch size are considered [9, 11]. This section further explains the HPs that are crucial for model building, training and their impact on models and suggested values.

2.1 Optimizer

Unaware of the optimizer's benefits, many people may be utilizing it to train their NNs. Optimizers, or optimization algorithms, are essential for increasing training efficiency and accuracy. The optimizer's selection, the mini-batch size, and the momentum are all HPs connected to optimizers. Making the right optimization choice is challenging. Optimizers are the algorithms used to modify the weights and LR of the NN to decrease losses and improve network performance. Gradient descent (GD), stochastic gradient descent (SGD), mini-batch gradient descent (MBGD), adaptive moment estimation (Adam), and root mean square propagation (RMS-prop) are some of the popular optimizers [12, 13].

GD is the easiest and most often used optimization technique. It is heavily utilized in classification and linear regression algorithms. Backpropagation technique in NNs also uses the GD technique. A loss function's first-order derivative is necessary for the GD method of first-order optimization. It chooses how to adjust the weights so that the function can attain a minimum. Backpropagation is used to pass the loss from one layer to the next, and the weights of the model are changed to reflect the losses in order to lessen the loss. In GD, weights are adjusted after the gradient on the full dataset has been computed. The weight is updated using Eq. 1.

$$W_{\text{new}} = W_{\text{old}} - \alpha * [\partial(\text{loss})/\partial(W_{\text{old}})] \tag{1}$$

where W_{new} and W_{old} are the updated weight and old weight respectively.

SGD is a variation of GD. It seeks to update the model's parameters more often. The model's parameters change when the loss upon every training instance is determined. Consequently, if the data contains x number of rows, SGD will update the model parameters x times during the dataset cycle as opposed to GD's single update. Since they are updated often, model parameters exhibit large variances and vary in loss functions at different intensities. To reduce the significant volatility in SGD and smooth out the convergence, momentum was developed.

The momentum term is often set to 0.9 or, as needed, 0.99 or 0.999. Oscillation is minimized by increasing the update's strength in the same direction and reducing its movement in other directions [14]. The updated value can be calculated using Eq. 2.

$$V_{\text{new}} = \eta * V_{\text{old}} - \alpha * [\partial(\text{loss})/\partial(W_{\text{old}})] \tag{2}$$

where α is a constant and η is the LR.

One of the best GD algorithm variants is called MBGD. In MBGD approach, the dataset is split up into different batches and the parameters are changed in different batches. The HP called mini-batch size has a value that is highly connected with the memory of the processing unit. Because of the GPU/CPU memory access, its value is assumed to be a power of 2. The model will run rapidly if the mini-batch size is a power of two as it accesses GPU/CPU memory with an acceptable default value of 32 [12, 15].

One of the most popular optimizers for DNN training is RMS-prop. The GD is accelerated by RMS-prop, but it performs better as the steps get smaller. The LR is modified in RMS-prop by dividing the square root of the gradient. RMS-prop speeds the horizontal movement from a considerably greater LR while slowing the vertical oscillation compared to the initial GD [16, 17]. Equation 3 is used to implement RMS-prop.

$$w = w - lr * (dw/\sqrt{S_{dw}}) \tag{3}$$

Adam deals with momentums of the first and second-order. The concept behind the Adam is that instead of rolling so rapidly just because it can jump over the minimum, it should slow down a little for a deliberate search. Additionally, Adam keeps the average of earlier squared gradients and saves the exponentially decaying average of earlier gradients [16]. The weight and bias are calculated using Eqs. 4 and 5.

$$w_t = w_{t-1} - \left[\eta \big/ \left(S_{dw_t} - \varepsilon\right) * V_{dw_t}\right] \tag{4}$$

$$b_t = b_{t-1} - \left[\left(\eta \Big/ \sqrt{S_{db_t} - \varepsilon}\right) * V_{db_t}\right] \tag{5}$$

2.2 Learning Rate

The LR is the HP in optimization algorithms that controls how much the model must change each time its weights are changed in response to the projected error. It determines how often model parameters are cross-checked and is one of the most crucial factors while building a NN. Selecting the ideal LR can be challenging because if it is excessively low, the training process may be slowed down. Nevertheless, if the LR is excessively high, the model cannot be well-tuned. Finding an ideal value for LR is difficult yet essential [18]. A small enhancement based on the constant value is setting an initial rate of 0.1 and modifying it to 0.01 when the accuracy is saturated and subsequently to 0.001 if required. Divergent behavior occurs when the LR value becomes too high. When the LR value becomes too small, it takes several updates to achieve a minimal point, the ideal LR value quickly achieves the minimal point, and severe changes occur.

Different methods are adopted to find out the value of LR. Two common methods are linear decay and exponential decay, which can be implemented using Eqs. 6 and 7, respectively.

$$lr = lr_0/(1 + kt) \tag{6}$$

$$lr = lr_0.\exp(-kt) \tag{7}$$

where lr_0 is the previous LR, k is the decay rate, and t is the training time.

2.3 Network Depth

A crucial factor in deciding the entire design of NNs, which directly affects the output, is the number of hidden units. Deep learning networks with even more layers are more likely to produce features with increased complexity and relative accuracy. It's common to practice scaling up the NN by including more layers; however, this raises the model's complexity and can occasionally result in overfitting. Overfitting happens as a result of there being more layers and neurons in each layer. Therefore, it is important to carefully calculate the model's depth [19]. Once more, regularization can be used to handle overfitting. Lasso (L1) and Ridge (L2) regularization, which may be implemented using Eqs. 8 and 9, are the two regularization norms employed to address the overfitting problem. L1 and L2 regression have their benefits and drawbacks.

$$w(\mathrm{L1}) = \sum_{i=0}^{N}\left(y_i - \sum_{j=0}^{M} x_{ij} w_j\right)^2 + \lambda * \sum_{j=0}^{M} \left|w_j\right| \tag{8}$$

$$w(\mathrm{L2}) = \sum_{i=0}^{N}\left(y_i - \sum_{j=0}^{M} x_{ij} w_j\right)^2 + \lambda * \sum_{j=0}^{M} w_j^2 \tag{9}$$

where the regularization parameter is defined as λ.

2.4 Activation Function

A neuron's activation status is determined by an activation function. According to this, it will perform some straightforward mathematical operations to decide whether the input from the neuron to the network is important or not for the prediction process. The activation function's function is to enable nonlinearity in a NN and produce output from a set of input values provided to a layer. Deep learning relies on activation functions to provide nonlinear characteristics to the output of neurons. A NN without an activation function will just be a linear regression model, unable to reflect intricate characteristics of data. The activation functions must be differentiable for backpropagation and computing weight gradients. The most well-known and often employed activation functions are sigmoid, softmax, hyperbolic tangent (tanh), and rectified linear units (ReLU) [19].

When given a number, sigmoid outputs a number between 0 and 1. It has a fixed output range, monotonicity, continuous differentiation, nonlinearity, and all the other desirable characteristics of activation functions. It is also easy to use. Typically, binary classification issues make use of this. The likelihood that a particular class will exist is provided by this sigmoid function [20]. It can be modeled mathematically using Eq. 10.

$$f(x) = 1/(1 + e^{-x}) \tag{10}$$

The activation function tanh can condense a real-valued number to a range between [−1, 1]. It varies from a sigmoid and generates zero-centered output while being nonlinear. The main advantage of the approach is that negative inputs will be actively mapped to the negative, and zero inputs will be mapped almost perfectly to zero [20]. Equation 11 is a mathematical representation of the tanh function.

$$f(x) = (e^{x} - e^{-x})/(e^{x} + e^{-x}) \tag{11}$$

ReLU is one of the activation functions that is used in applications the most. The ReLU function's gradient has the highest value of 1, which solves the vanishing gradient problem. It also solves the problem of saturating neurons because the slope of the ReLU function is never 0. The range of ReLU is 0 to infinity [19]. Equation 12 provides a mathematical illustration of the ReLU function.

$$f(x) = x,\ if\ x > 0, \text{else } 0 \tag{12}$$

Several sigmoids are combined to form the softmax function. There is a proven relative likelihood. The probability of each class or label is returned by the softmax function, just as the sigmoid activation function. The last layer of the NN in multi-class classification is most frequently used with the softmax activation function. The softmax function provides the likelihood of the present class to other classes. This indicates that it also takes other classes into account [21]. The softmax function is represented mathematically in Eq. 13.

$$f(x) = \exp(x_i) \Big/ \sum_j \exp(x_j) \tag{13}$$

2.5 Loss Function

A loss function compares the target and anticipated output values to determine how well the NN reflects the training data. During training, we make an effort to minimize this output discrepancy between the expected and the goal. The basic goal of the loss functions is to reduce the average loss by adjusting the weight and bias value. Regression loss function, which includes mean absolute error (MAE), and mean squared error (MSE), and classification loss function, which includes binary cross entropy (bc) and categorical cross entropy (cc), are the two main types of loss functions that are used [22]. The mathematical representations of the loss functions are shown in Eqs. 14–17.

$$\text{MSE} = 1/n \sum_{i=1}^{n} (y^i - \hat{y}^i)^2 \tag{14}$$

$$\text{MAE} = 1/n \sum_{i=1}^{n} |y^i - \widehat{y}^i| \tag{15}$$

$$\text{Loss}_{bc} = 1/n \sum_{i=1}^{N} -(y_i.\log(p_i)) + (i - y_i).\log(1 - p_i) \tag{16}$$

$$\text{Loss}_{cc} = -1/n \sum_{i=1}^{N} \sum_{j=1}^{M} y_{ij}.\log(p_{ij}) \tag{17}$$

3 Searching Technique

In this section, all the classical HP searching techniques are discussed.

3.1 Grid Search Technique

If enough resources are available, grid search is the simplest search algorithm that produces the most accurate predictions, and the user can always identify the ideal combination. Grid search is simple to conduct in parallel since each trial runs independently without regard to time order. Results from one experiment do not affect those from further tests. The distribution of computational resources is incredibly flexible. Grid search, however, is cursed by dimensionality since more HPs that need to be tuned consume exponentially more computational resources. When more HPs are utilized in the model, the grid search technique's major downside is the computing time [23].

3.2 Randomized Search Technique

Random search is a fundamentally improved form of grid search [24]. It denotes a randomized search over possible parameter values for HPs from certain distributions. The search procedure is carried out indefinitely or until the necessary accuracy is attained, whichever comes first. Grid search is comparable to random search; however, random search has been shown to produce superior results. When some HPs are not distributed equally, the random search may perform better.

In contrast, a longer search period cannot provide better results for a grid search. Another key benefit of random search is its parallelization simplicity and resource allocation adaptability [23]. While random search often outperforms grid search, it still requires a lot of processing.

3.3 Bayesian Search Technique

It is a sequential model-based approach that aims to identify the global optimum with the fewest trials possible. It strikes a balance between exploration and exploitation to avoid becoming trapped in the local optimum. In two ways, Bayesian search is superior to random search and grid search: first, users are not needed to have existing experience of the distribution of HPs; and second, the central concept of Bayesian search, posterior probability, beats both methods. The Bayesian search seems to be more computationally effective than random search and grid search, requiring fewer attempts to discover the ideal HP set, especially when highly expensive objective functions are observed [25]. Another outstanding benefit of Bayesian search over grid search and random search is that it may be used whether the goal function is stochastic or discrete, convex, or nonconvex.

3.4 Population-Based Search Technique

Population-based methods, such as evolutionary algorithms [26], swarm-based optimization [27], and covariance matrix adaptation evolutionary strategies, are essentially a series of random search methods based on genetic algorithms [28]. Initialization (random population), Selection (assessment of the existing population and parent selection), and Reproduction are the three key components of genetic algorithms (creation of the next generation). The population-based approach effectively uses less computing time to identify the ideal HPs.

4 Second-Order Optimization

Backpropagation is a stage in optimization approaches when an error from the calculated and intended output is monitored. The objective function is minimized during optimization in the context of NNs. The GD approach is one of the most widely utilized optimization strategies in NN training. It is a first-order derivatives approach, as seen in Eq. 18. However, the selection of LR significantly impacts how well GD performs. An excessively large LR causes significant changes with each iteration and frequently fails to find the minimum point. Better trajectories through the mountainous error surface that are frequently seen in NNs can be provided by the second-order technique using local curvature information [29].

$$w_{k+1} = w_k - \eta(\partial e/\partial w) \tag{18}$$

With the use of extra Hessian data, the second-order derivatives approach offers a superior training trajectory over the local curvature of the error surface. By adaptively changing the step size following the various learning phases, the usage of Hessian matrices simplifies the fine-tuning of HPs. The second-order derivative is represented mathematically by Eq. 19. Inverse Hessian matrices are used in the computation of the next update step.

$$w_{k+1} = w_k - \eta\left(\frac{\partial^2 e}{\partial w^2}\right)\left(\partial e/\partial w\right) \tag{19}$$

5 Experimental Setup

The experiment is carried out by taking the IMDB dataset (https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews) with a system configuration of an Intel i3 CPU, 8 GB of RAM on a windows 10 system. The performance of the model is evaluated using accuracy, precision, recall and F-score.

6 Results and Discussions

The idea of HP search has been around for a while and is a commonly utilized technique. The DNN has captured the majority of academics' interest in recent years. To adapt the model for various datasets, the HP autotuning approach is absolutely essential. Due to a large number of HPs, many classical methods are becoming impractical. In complicated scenarios, choosing HPs is highly costly.

Table 1 briefly compares the main algorithms. The discussed models are then applied to the IMDB dataset to tune the hyperparameters. Finally, the long-short term memory (LSTM) network is applied to evaluate the performance of the different searching techniques with the tuned HPs.

Table 2 shows the performance of different search algorithms using an LSTM network. The result shows the Grid search algorithm outperforms other search techniques with 85.90% accuracy, 87.3% precision, 84.62% recall, and 86.41% F-score, but the computational cost is very high. To get high-performance results with less computational cost, population-based techniques can be applied to the dataset. Figures 2 and 3 represent the graphical representation of the performance comparison of different searching techniques.

Table 1 Comparison of different hyperparameter searching algorithms

Searching technique	Advantages	Disadvantages
Grid search	– Simple and parallel execution – Almost find the optimal way	– Time consuming – Issue of overfitting
Random search	– Reduction of overfitting – Faster than grid search	– Potential for variance – Cannot guarantee for optimal
Bayesian search	– Not stock at local minima – Can handle complex problems	– Complex algorithm – Parallelism is difficult
Population-based search	– Can achieve parallelism – Not stock at local minima – Almost find the optimal way – Faster than grid search	– A bit complex to code – Extra parameters to set

Table 2 Performance comparison of different hyperparameter searching algorithms

	Accuracy	Precision	Recall	F-score	Time (mins)
Grid search	85.90	87.30	84.62	86.41	405.67
Randomized search	83.89	84.63	81.93	85.69	43.55
Bayesian search	82.65	82.30	82.36	82.47	31.78

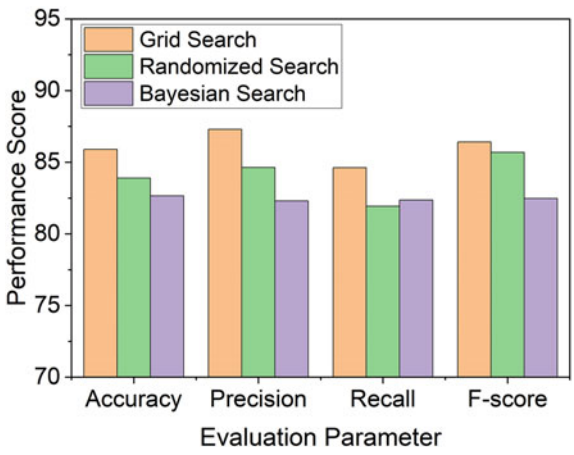

Fig. 2 Performance comparison of searching techniques

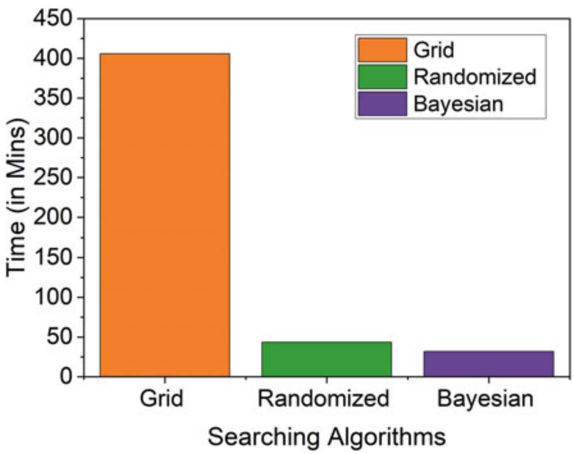

Fig. 3 Computational time comparison of searching techniques

7 Conclusion and Future Work

DNNs are being used more often in various sectors, which inspired this study. The essential HPs and the approaches for finding them are systematically reviewed in this study, with an emphasis on how to autotune HPs for NNs. Discussing various HPs and their needs served as the study's initial point. Further a brief discussion of HPs with a discussion of various autotuning searching algorithms and their application, and it concludes with a discussion of the second-order optimization methodology. This study is devoted to providing researchers and industry users with a reference summary of information on HP tuning. In future, the discussed concept can be further extended using the population-based technique with hybridization to get efficient performance.

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Virtual Machine Load Balancing Using Improved ABC for Task Scheduling in Cloud Computing

Reddy N. Hanuman, Amit Lathigara, Rajanikanth Aluvalu, and Uma Maheswari Viswanadhula

Abstract Computing resources are now more accessible, powerful, and inexpensive than ever before thanks to the widespread adoption of the Internet and rapid advancements in processing and storage technology. Thanks to this development in expertise, a new kind of computing architecture known as "cloud computing" is now a practical reality. Load balancing task scheduling is an essential issue that has a direct impact on resource utilisation in cloud settings. One of the most important goals of scheduling is to distribute work among virtual machines in such a way that no individual machine is overburdened or underutilised. Considering load balancing scheduling is crucial because of the importance it has on the backend and frontend of the cloud research sector. Any time a cloud environment is able to effectively balance its load, resource utilisation improves. The challenge with load balancing in the cloud is that it is an NP-hard optimization delinquent. To solve this issue, this research suggests a method that combines a heuristic scheduling algorithm with an artificial bee colony; this method is referred to as enhanced Artificial Bee Colony for Task Scheduling (IABC-TS). For better cloud computing scheduling of virtual machines in both homogeneous and heterogeneous contexts, this approach is used. The goal of its implementation was to shorten makespan and distribute weight more evenly. The IABC-TS cloud computing system's scheduling performance was measured against that of competing swarm intelligence techniques. In our lab tests, we used CloudSim to simulate systems with various augmentation techniques to examine their makespan and load balancing ability. Based on the results of the experiments, it is clear that the suggested IABC-TS outperformed the baseline models.

Keywords Load balancing · Virtual machines · Improved artificial bee colony · Task scheduling · CloudSim · Resource utilisation

R. N. Hanuman · A. Lathigara
R K University, Rajkot, India

R. Aluvalu (✉) · U. M. Viswanadhula
Chaitanya Bharathi Institute of Technology, Hyderabad, India
e-mail: rajanikanth.aluvalu@gmail.com

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1 Introduction

One of the most cutting-edge technologies of our day is cloud computing, in which tens of thousands of computers work together to store and process data and programmes that users access via the internet. Cloud computing combines distributed and parallel computing to enable, and data stored on remote devices. When looking at cloud services from the perspective of programmers, we may classify them into three classes: IaaS, Platform as a Service, and Software as a Service. To run services with virtualisation technology, users can take advantage of the cloud provider's infrastructure as a service [2]. High-level services can be developed using the software layer offered by the platform as a service. Cloud application creation, testing, and hosting are all simplified with the aid of this platform. (SaaS) refers to the delivery model in which consumers obtain the necessary programmes via remote access, or the cloud [3].

Software as a service is one of the most popular cloud services because it allows customers to quickly and easily deploy applications via the internet. The previous methods of using software, such as installing it on the computer that needs it, have been rendered obsolete by this new level, which removes the restrictions that previously hindered the installation of substantial software or the provision of software for the users. When users need access to shared resources, the cloud's servers in the data centre make those resources available [5, 6]. Renting virtual machines from cloud providers allows multimedia application developers to offer their customers a wider range of features and functionality. Here, subscribers can ask for and get their preferred apps sent straight from the service provider [7].

Through the use of task scheduling, users' workloads can be distributed across multiple virtual computers. Ideally, a customer-focussed scheduling algorithm would have the virtual machine do the customer's specified activities in the least amount of time possible [8]. Instead, the service provider needs a scheduler that can make the most of available resources while keeping customers happy. The onus is now on the service provider to choose an appropriate technique of job scheduling [9, 10]. Load balancing is an essential part of scheduling tasks in the cloud. In order to enhance efficiency and throughput, load balancing spreads work over multiple servers. Because of this, we may decrease waiting time and maximise virtual machine utilisation by employing an appropriate load balancing algorithm to avoid load redundancy and scarcity in certain of the serves [11].

In order to solve the task scheduling issue, many different approaches have been offered, each of which attempts to balance a different set of user and service provider requirements. Load balancing across virtual machines is a crucial part of the methods in [12], which address the scheduling problem. Different from earlier works, the proposed method not only aims to improve load balancing but also to present a dynamic multi-objective solution for job scheduling on virtual machines [13]. The goals of the proposed strategy are as follows:

First, enhancing the degree to which virtual machines load balance.
Making it last less time.
Strengthening the system's dependability, third.
Four, maximise the use of existing assets.

A scheduling mechanism for distributing work among virtual machines is developed to achieve these aims. To reduce unnecessary work [14], the suggested system assigns tasks to the most suitable virtual machines. After a task has been assigned to a virtual machine, the suggested method can make predictions about its future state. By preventing any further work from being assigned to that computer, the suggested strategy achieves load balancing among virtual machines. The proposed solution both shortens makespan and improves load distribution. However, the proposed approach monitors the current status of each virtual machine's job execution to increase the system's dependability [15, 16]. This technique chooses VMs with a history of stability and gets rid of VMs that are underperforming. In other words, the node's recent failure rate makes it less likely to accept jobs compared to nodes with better reliability. The simulation findings suggest that compared to previous research, the proposed strategy increases reliability, minimises waiting time, and improves average makespan. As a whole, the proposed method is an improvement above the prior literature because.

Improving the consistency of job scheduling using virtual machines' historical data increases virtual machine load balancing taking into explanation both the QoS for the end user and the requirements of the service provider. The remaining sections of this work are laid out as follows. In the following section, we'll examine several complementary works. The proposed algorithm and its mathematical model are presented and discussed in Sect. 3. The results of the simulations and evaluations are presented in Sect. 4, while the deductions and suggestions for further research are provided in Sect. 5.

2 Related Works

Recently, a new Credit-Based Resource Aware Balancing Scheduling method (CB-RALB-SA) was introduced by Narwal and Dhingra [17]. Using the FILL and SPILL purposes of Resource Aware and Load with the Honey bee optimisation heuristic method, we map the tasks weighted by the credit-based scheduling procedure to the resources, taking into account the load and computational capability of each resource. It has been shown through experimental evaluations and results that the suggested method improves upon the existing CBSA-LB algorithm by 16.90% in makespan time and 48.5% in processing time. Thus, it boosts overall system speed, as well as the performance of individual processes, and saves memory that was previously dedicated to RAM.

A Multi-objective trust aware scheduler was developed by Mangalampalli et al. [18], which prioritises jobs and virtual machines (VMs) and allocates them to the most suitable virtual resources, all while reducing makespan and energy consumption. We modelled our scheduler after the whale optimisation technique. Cloudsim was used for the entirety of the simulation. This simulation makes use of both synthetic and real-time workloads taken from HPC2N and NASA. We compared our method to the standard meta-heuristic methods already in use, such as ACOs, GAs, and PSOs. The simulation results showed that the makespan, energy consumption, overall) all improved significantly.

An ACO-based algorithm, ACO-RNK, has been proposed as a practical answer to the job scheduling problem by Elcock and Edward [19]. In order to lead the solutions, our method makes use of pheromones and a priority-based heuristic called the upward rank value, as well as an insertion-based policy and a pheromone ageing mechanism that tries to prevent premature convergence. We evaluated the performance of our approach to that of the HEFT algorithm and the MGACO procedure on directed acyclic networks that were constructed at random (DAGs). Our method demonstrated performance that was on par with or better than the chosen algorithms in the simulations.

Using data from the (RM), the (NM), and the Application Master, Bawankule [20] proposes a new process for scheduling tasks based on deep reinforcement learning. This method dynamically schedules tasks based on the current state of the nodes, including their load, resource availability, usage, and task information (AM). In order to handle difficult scheduling issues in a heterogeneous setting, the suggested Map Reduce Scheduling utilising the Deep-Q-Networks (MRSDQN) employs a deep reinforcement learning approach. The projected process is evaluated in terms of its efficiency in relation to Hadoop's most valuable benchmark, the HiBench benchmark suite. It validates the efficiency of the suggested method using the latest Hadoop benchmark. Finally, it contrasts the proposed strategy with Hadoop's multiple scheduling strategies, each of which places less value on the timely completion of jobs and the successful completion of tasks in a diverse computing environment. Last but not least, it expedites the completion of tasks in a diverse set of benchmarks by an average of 23 to 36%.

In order to accomplish the goal of the smallest makespan in the shortest amount of time, Yadav and Mishra [21] have created and enhanced an ordinal optimisation technique. Ordinal optimisation, which employs horse race conditions as selection rules, is applied in an improved reiterative fashion to meet the current requirement of optimal schedule, thereby achieving low overhead through the judicious allocation of work to the most talented schedule. The proposed ordinal optimisation method, in conjunction with linear regression, creates schedules that maximise productivity and minimise the time needed to produce a product. More importantly, the proposed exact equation, which was derived using linear regression, forecasts any future dynamic workload in terms of the minimum makespan period aim.

Talmale and Shrawankar's [22] suggested Cluster-Based Real-Time Catastrophe Resource Management Framework makes use of different disaster resources and emergency services. Edge computing resources are pooled together to form a cluster, and then a set of tasks is assigned to the cluster and scheduled on the edge computing cluster to increase resource utilisation and acceptance rate and decrease response time and overhead due to communiqué and migration, thereby solving the problems associated with the current partitioned scheduling.

Nematpour et al. [23] provide a new method of constructing chromosomal representation. After three iterations of ranking, clustering, and cluster scheduling, the suggested method schedules clusters using a genetic algorithm. To achieve optimal performance, the proposed genetic algorithm incorporates four heuristic philosophies: load balancing, idle time reuse, work duplication, and critical path [24, 25]. Finally, we compare the amounts of optimisation over 6 task graphs of 3 different types, and we find that in one type, the quantity of optimisation is equal to the findings of the previous best approach, but in the other 3 types, the amount of optimisation is between 4.25 and 6.88%.

3 Proposed System

3.1 System Model

The suggested method's primary purpose is to allocate tasks to VMs in accordance with the latter's available resources and current workload. The load balancing algorithms work to take work away from VMs that are too busy and distribute it to others. Each physical machine (PM) in the proposed system hosts several virtual machines (VMs) that are responsible for carrying out the user's requests. Each cloud customer has a unique workload requirement for their virtual machine. In this work, we employ a load balancing technique to distribute the jobs among the VMs. The suggested load balancing technique is constantly monitoring the workload of every virtual machine (VM) in the cloud. The VM's workload is proportional to the time required to complete each individual activity. Because each job takes a different amount of time to complete, the VM's workload is always changing.

3.2 Problem Definition with Solution Framework

Take cloud C, for example, where there are "n" data centres or physical machines (PMs) and "m" VMs in each physical host (PH) (VMs).

$$C = \{\mathrm{PM}_1, \mathrm{PM}_2, \ldots, \mathrm{PM}_n\} \tag{1}$$

In this equation, C stands for the cloud, PM 1 for the first bodily machine, and PM n for the nth. Following are several ways to designate the PM:

$$\mathrm{PM}_n = \{\mathrm{VM}_1, \mathrm{VM}_2, \ldots, \mathrm{VM}_m\} \tag{2}$$

where VM 1 refers to the very first VM and VM m to the very latest VM. It's the same in the cloud: there are I users, and each user has I tasks. The user may be identified in the following ways:

$$U_i = \left\{T_1, T_2, \ldots T_j, \right\} \tag{3}$$

The chief goal of this study is to efficiently distribute the workload across all virtual machines (VMs) in a cloud infrastructure while simultaneously reducing the task's execution time and associated costs. Specifically, we are interested in achieving three goals. First, we strive to reduce the amount of time required to complete each task (ET). Secondly, we want to reduce the time and money needed to carry out an action (EC). A third is to divide up the work among all of the cloud-based virtual machines.

Time taken to complete an operation can be determined using Eq. (4).

$$\mathrm{ET} = \frac{1}{\max(\mathrm{ET}) \times \text{number of task}} \sum_{j=1}^{\text{Number of task}} (\text{ET of corresponding VM} \times \text{Size of the task}) \tag{4}$$

The execution cost (EC) can be intended using Eq. (5).

$$\mathrm{EC} = \sum_{j=1}^{\text{Number of task}} \left[\frac{\text{Execution time} \times \text{Communication time}}{\text{Number of task}}\right] \tag{5}$$

Lode can be intended using Eq. (3).

$$\text{Load} = \frac{1}{\text{Number of task}}$$

$$\sum_{i=1}^{\text{Number of task}} \left[\frac{\text{size of VM} - ((\text{total size of VM} - \text{freespace of VM}) + \text{size of task})}{\text{size of VM}}\right] \tag{6}$$

When planning, it's important to distribute work fairly. In the absence of load balancing, the system will consume the most resources and take the longest to complete the task. In this research, we propose a method for effectively balancing loads that takes into account multiple objectives.

The planned multi-objective function (MOF) is distinct in Eq. (7).

$$\mathrm{MOF} = \min[\alpha_1(\mathrm{ET}) + a_2(\mathrm{EC}) + a_3(1 - \text{Load})] \tag{7}$$

3.3 Proposed Method

Let $VM = \{vm_1, vm_2, vm_3, \ldots, vm_m\}$, variables used in this study and Table 1 define the parameter of the study.

Configure the system to operate in a non-preemptive, hence interrupt-free mode. Heuristic algorithms were used to sort the tasks into three distinct orders before running the ABC method. By processing the tasks with the ABC method, as was previously described, the optimal setup would require the least amount of computational time. Then, the ABC algorithm used these steps to plan when tasks would access the VMs:

1. First, the population size, denoted by n, of bees was determined. Algorithm 1 depicts the random assignment of VMs to the various food sources (m) that represent them, followed by the calculation of their fitness scores.

Procedure 1:: Initialisation

1. *For i = 1 to*
2. *Send the population of bees into the system to find appropriate VMs by Random search.*
3. *Calculate the fitness of each VM by using (8)*$C_j = P_{e_j} + Mi_j + Bw_j$ (8)
where j = index of the VM that the ith bee found.
4. *End For*

Table 1 Variables used in this study and their meanings

Symbol	Definition
m	The number of virtual machines (VMs)
$VM = \{vm_1, vm_2, vm_3, .., vm_m\}$	The set of VM
K	The total sum of tasks the scheme has to achieve
$Task = \{t_1, t_2, t_3, \ldots, t_k\}$	The set of tasks
N	The total quantity of bees performing in the algorithmic procedure
C_j	The presentation of the jth virtual machine
Pe	The sum of processors in the VM
Mi	Million Instructions Per Second (MIPS)
Bw	Bandwidth of VM
F	Fitness
Tl	The distance of task in MIa
Pi	Probability that the ith food source is good which be contingent on its fitness value, $pi = \frac{\text{fitness}_i}{\sum_{i=1}^{NS} \text{fitness}_i}$ Where, NS is the size of food bases

2. Second step: the stipulations of the VM. Bees were grouped into 3 categories: Scout Bees, Employed Bees, and Onlooker Bees. A Scout Bee found the initial position of a food source. An Employed Bee went to the food source, recalculated and updated the fitness value of food source. An Onlooker Bee decided which food source was the best food source. The operation of an employed bee is presented in Algorithm 2.

Procedure 2: Employed bee phase

---

1. *For i = 1 to n*
2. *Employed bees are sent randomly to food sources (VMs in cloud computing).*
3. *Fitness of each VM is calculated based on (9)*$F_{ij} = \frac{\sum_{i=1}^{n} Tl_{ij}}{Evaluvate\, capacity\, of\, VM_j(c_j)}$
4. *Update the fitness value*
5. *End For*

---

3. *Third step: the Employed bees have searched around for food sources they brought back information about the food sources to the Onlooker bee. The Onlooker bee then recalculated the fitness values of the food sources. The operation of the onlooker bee is shown in Algorithm 3.*

Algorithm 3: Onlooker bee phase

---

1. *The onlooker bee chooses the first m food sources (VM) with the highest fitness values and perform Neighborhood Search around the food sources*
2. *An nsp number of employed bees are sent to bring back new positions of the first m food sources while a nep number of employed bees are sent to all food sources to bring back new positions of these food sources; all of them come back to relay the information to the onlooker bee*
3. *The onlooker bee calculates the new fitness values of the food sources according to (10),* $fit_{ij} = \frac{\sum_{i=1}^{n} Tl_{ij} + In_{length}}{Evaluvate\, capacity\, of\, VM_j(c_j)}$ *where* In_{length} *is the length of the task that is waiting to access a VM at that time*
4. *The onlooker bee chooses the best food source (VM) and assigns a task to the VM*

---

4. *Fourth step: the employed bee that was the owner of the best food source was then transformed into a scout bee. The operation of a scout bee is shown in Algorithm 4.*

Algorithm 4: Scout bee phase

---

1. *If food source = null then*
2. *Send the previously transformed scout bee into the system to find an appropriate VM by Random search.*
3. *Calculate the fitness value of that VM by using (8)*
4. *End If*

---

5. *Fifth step: the overall operation of HABC algorithm is described in Algorithm 5.*

Algorithm 5: Improved HABC Algorithm

Input: Dataset, Bee's Parameters
Output: Minimum makespan;
1. Initialize the individual in the population.
2. Set the default values of the food sources (VM) by using (9).
3. Initialization (Algorithm 1)
4. Repeat
5. Employed bee phase (Algorithm 2)
6. Onlooker bee phase (Algorithm 3)

Sixth, following VM task scheduling, determine the system load balance value. Standard Deviation (S.D.) in the range of 11 is used to determine this (12),

$$S.D = \sqrt{\frac{1}{n}\sum\nolimits_{j=0}^{n}\left(X_j - \overline{X}\right)^2} \tag{11}$$

$$imbalance = \frac{X_{max} - X_{min}}{X_{avg}} \tag{12}$$

The amount of time needed to process each virtual machine (Xj) is determined by (13),

$$X_j = \frac{\sum_{i=1}^{k} task\_length}{capacity_j} \tag{13}$$

To determine how long it takes the system on average to execute a virtual machine (X), we need (14), $\overline{X} = \frac{\sum_{j=1}^{n} X_j}{n}$ (14).

If the standard deviation is less than X, then the system is in equilibrium; otherwise, it is out of whack.

Stage six: scouting (Algorithm 4).

4 Results and Discussion

Cloudsim 3.0.3 is utilised for the performance evaluation proposed here. In Cloudsim, the user initiates actions by submitting requests, or tasks, that occur in a specific order within a series of clouds. Each cloudlet has its own unique set of characteristics, including the number of tasks it must do, the amount and length of any linked files, and so on. The cloudlets are then forwarded to a broker, where they are used to assign IP addresses to the desired virtual machines. This utility can be used with the broker's preexisting regulations. Hosts initiate the creation of brokers' worth of virtual machines. The TSA operates in an operational data centre broker. Code for the proposed technique is implemented in the data centre broker. Planners for data centres and virtual machines can work together in real-time or across great distances.

4.1 Performance of Proposed Scheme in Terms of Makespan

It is primarily aimed at reducing the amount of time needed, i.e. the whole duration of the work schedule, where all work has been completed or the time taken for it from the beginning to the end. The following equation demonstrates the makeup formula.

$$\text{Makespan} = \sum_{i=1}^{n} F_{t_i} \tag{15}$$

where, F is the finishing time of the task $t(n)$.

The existing techniques such as Whale Optimisation algorithm (WOA) [18], ACO [19], Butterfly Optimisation (BO) and ABC are implemented in this research work and results are averaged, which is shown in Tables 2, 3, and 4.

Analyzing the makespan of the projected and existing methods is shown in Fig. 2. Our proposed load balancing method has a shorter makespan than the state-of-the-art methods by a significant margin; for 1000 tasks, is around 395 s, while ABC's is

Table 2 Makespan comparison with a different algorithm

Number of tasks	Makespan (time in seconds)				
	WOA	ACO	BO	ABC	IABC-TS
1000	434	428	421	415	395
2000	917	921	836	816	800
3000	1302	1293	1264	1251	1229
4000	1688	1686	1635	1619	1602
5000	1979	2000	1994	1891	1875

Table 3 The comparison among the TSA using the energy consumption

Algorithms	Number of Tasks					
	1000	2000	3000	4000	5000	6000
WOA	2453	2723	3473	4142	4663	5081
ACO	2273	2657	3845	4361	4534	4921
BO	2265	2558	3518	3965	4307	4709
ABC	2159	2471	3350	3659	3878	4548
IABC-TS	**2055**	**2206**	**3032**	**3435**	**3726**	**3976**

Table 4 The comparison among the TSA using the execution overhead

Algorithms	Number of Tasks					
	1000	2000	3000	4000	5000	6000
WOA	792	875	810	826	846	884
ACO	821	854	844	861	871	892
BO	523	708	567	593	614	627
ABC	605	639	646	649	663	689
IABC-TS	**482**	**503**	**537**	**542**	**551**	**561**

around 415 s, BO's is around 421 s, ACO's is around 428 s, and WOA's is around 434 s. The system's manufacture time is inversely proportional to its throughput. The proposed model has a makespan of roughly 800 s for 2000 tasks, 1229 for 3000, 1602 for 4000, and 1875 for 5000 activities, while ABC needs 816 s, 1251 for 1619, and 1981 for 1981 tasks. The WOA model has a longer makespan need than the 917 s, 1302 s, 1688 s, and 1979s for the same amount of tasks.

Energy usage in the projected system is associated with that of existing models in Fig. 3. The system's energy usage is increasing as the number of nodes and tasks increases. Additionally, the IACO-TS method's energy efficiency improves with network size, as the percentage of total nodes visited increases in proportion to the total network size.

In order to demonstrate the efficacy of the suggested method, we benchmark our results against four additional algorithms: WOA, ACO, BO, and ABC. To be useful, a technique needs to have a shorter runtime. Our proposed technique takes less time than WOA, ACO, BO, and ABC when evaluating Fig. 4. Figure 4 provides a similar

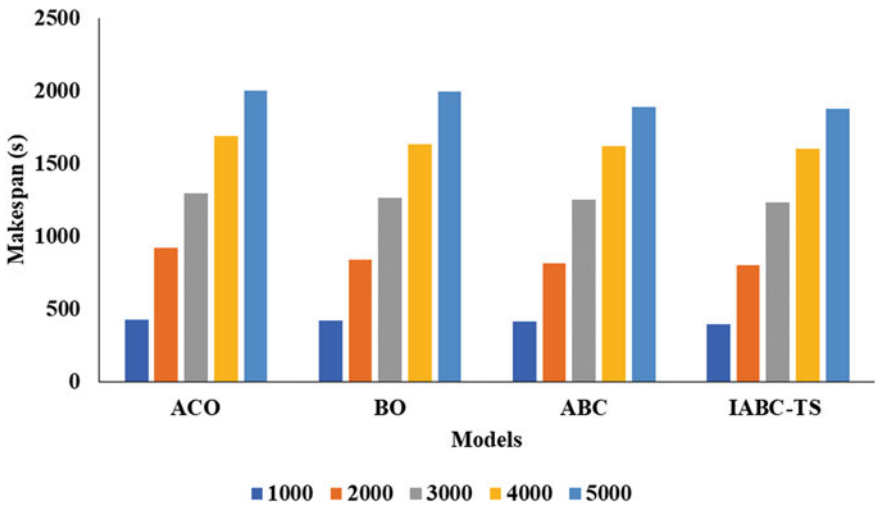

Fig. 2 Graphical representation of algorithm in terms of Makespan

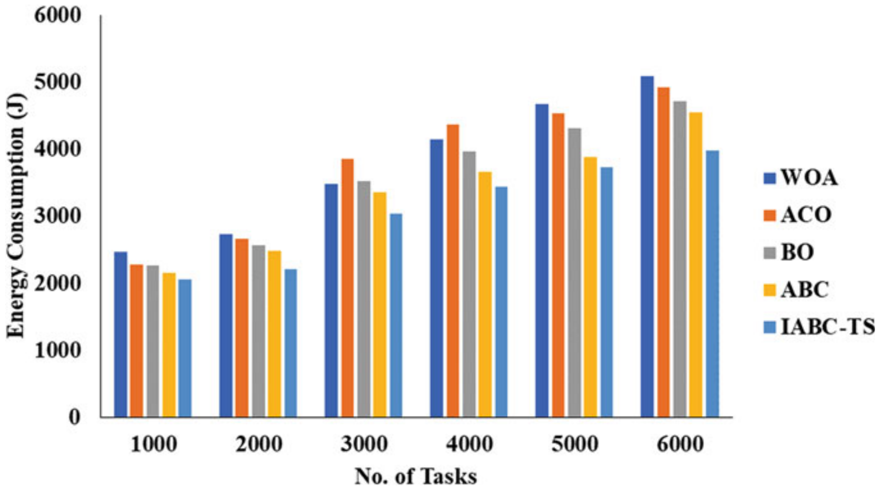

Fig. 3 Graphical representation of algorithm in terms of energy consumption

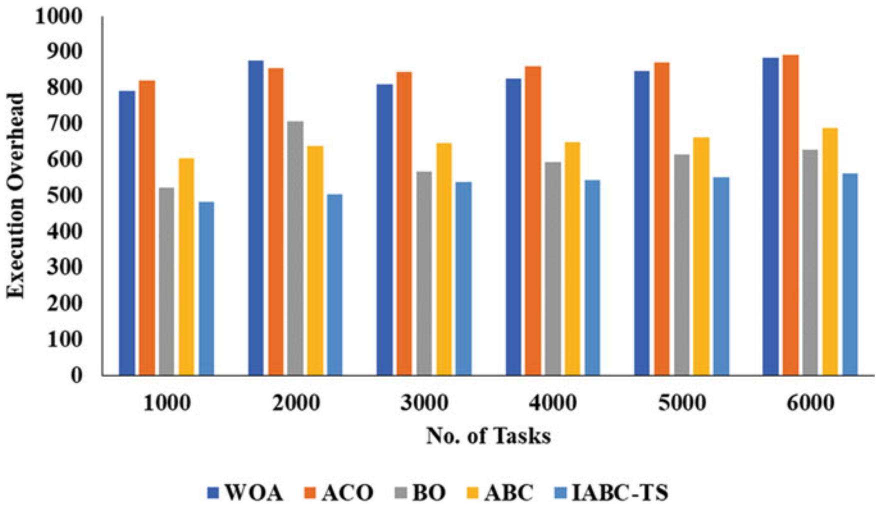

Fig. 4 Graphical representation of algorithm in terms of execution overhead

performance analysis, this time based on Execution overhead. The sum of jobs is on the *x*-axis, while the execution time is on the y-axis. When the workload size is 3000, the suggested IABC-TS method has the lowest execution cost (537), followed by BO (567), ACO (844), WOA (810), and ABC (0.649). The proposed solution is preferable than competing approaches due to its decreased execution overhead. Therefore, it is evident that the proposed IABC-TS approach has outperformed other methods.

5 Conclusion

As part of this study, we present an algorithm for cloud-based virtual machines (VMs) based on the IABC-TS for VMs in cloud computing in both contexts with the intention of enhancing job scheduling and load balancing. The implementation of the IABC-TS utilised cloud computing, which improved work scheduling and load balancing across enormous data sets. This study presents a new multi-objective Load balancing approach based on the IABC-TS standard, and puts it through its paces in a series of simulation studies. The goal of the scheduling method we offer is to provide a balanced workload among virtual machines while also minimising delivery time and cost. In addition, the benefits of using the IABC-TS to optimise resource allocation are explained by our suggested approach. Three examples of the problem are studied to gauge the efficacy of the proposed strategy. The proposed model's simulation results are associated with the standard method in terms of runtime and cost. Upcoming work will involve building a hybrid model and clustering the virtual machines (VMs) using optimisation models, both of which will improve the proposed model.

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Aspect-Based Sentiment Classification: Survey

Snehal Sarangi, Jitendra Kumar Rout, and Subhasis Dash

Abstract Through a variety of Internet applications that act as a rich source of information, Web 2.0 makes it easier for people to voice their opinions. The latent information in textual expressions can be processed and analyzed to disclose the user's or people's sentiments. Sentiment analysis, also identified as opinion mining, review mining, attitude mining, and other similar terms, entails the algorithmic extraction of opinions and perspectives from textual material. One of the three primary forms of sentiment analysis is aspect level sentiment analysis, which uses granule-level processing to determine sentiment orientation by utilizing the various characteristics of entities. Aspect-oriented sentiment analysis has been significantly impacted by the advancement of deep learning and machine learning techniques. This study provides a comprehensive overview of the present state of machine learning-based aspect-based sentiment analysis.

Keywords Sentiment analysis · Machine learning · Deep learning · Aspect-based sentiment analysis

1 Introduction

Social media is crucial for instantly distributing information about anything and everything, which encourages regular people to participate in and interact on social media. The astounding number of 500 million tweets in a year, or 6000 tweets per second, revealed by the statistics for the year 2018, confirmed Twitter's status active

S. Sarangi (✉) · S. Dash
KIIT, Bhubanewar, India
e-mail: snehal.sarangi@gmail.com

S. Dash
e-mail: sdasfcs@kiit.ac.in

J. K. Rout
NIT, Raipur, India
e-mail: jitu2rout@gmail.com

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community platform. For example, people express thoughts and feelings regarding goods or services, which eventually leads to an enormous volume of data on the Internet. We can extract much latent information from this unstructured digital data using sentiment analysis (SA). A text can genuinely change a potential customer's mindset. Processing every customer review comment provided individually is an entirely foolish idea. SA is the ideal tool for analyzing the trend underlying purchasing behavior. The technique known as SA involves using NLP to extract the attitude or sentiments from a text and categorize them according to polarity, such as positive, negative, or neutral. In contrast to most NLP study fields, SA involves various issues. Sentiment analysis is done at several levels of detail in documents, such as the sentence, aspect, and content. There are comparative reviews and surveys of SA in the literature but only a few authors, like Do et al. [1], have put together work on aspect-based sentiment analysis.

Deep learning and machine learning are common approaches for additional sentiment analysis. As a result, we seek to discuss the numerous works in this paper that have employed machine learning and deep learning methods to perform aspect-based sentiment analysis. Aspect-based sentiment analysis on key domains and a variety of datasets has been focused in this study. The key objectives of this study are:

- Analysis of aspect-based sentiment classification based on various machine learning techniques identifies which is better suited for sentiment analysis.
- Effect of deep learning to improve the accuracy further.
- To outperform a single model in terms of adaption ability performance on a hybrid model (NLP)

The remaining parts of the paper are divided up into these five sections: Sect. 2 focuses on a detailed study of sentiment analysis and annotation. Section 3 focuses on the analysis of emotions. Various domains used for sentiment analysis are discussed in Sect. 4. Detailed discussion on multiple datasets used for research is presented in Sect. 5. Different Machine Learning algorithms used in sentiment analysis have been discussed in Sect. 6. Section 7 provides the challenges that occurred during implementation. Finally, Sect. 8 concludes the paper.

2 Sentiment Analysis

Determining the polarity of the textual data is one of the key goals of the SA job. A text's inclination to favour positive or negative polarity might be observed. For instance, "That was the nastiest movie I've ever seen" has a negative polarity, while "I really loved the picture" has a positive polarity. Some sentences may not have either positive or negative polarity. Such statements fall within the heading of neutral polarity. "I neither loved nor despised that movie," for instance. Factual sentences do not fall under the category of neutral sentences (Fig. 1).

Data collection is the act of gathering raw text information from several social networking websites, like Twitter and Facebook, or e-commerce websites, like

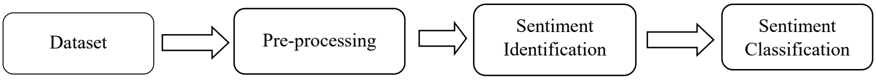

Fig. 1 Sentiment analysis procedures

Amazon and Flipkart, in the form of reviews, blogs, and discussion forums. These sources include ideas or feelings in various sizes, formats, and styles regarding various entities. The subsequent stage involves the necessary filtering procedures to retrieve pertinent data from the datasets while deleting unnecessary stuff. The most crucial phase is sentiment recognition, which simultaneously identifies sentences containing subjective expressions and implicit sentiments without ignoring them.

It might be challenging to extract opinions from unstructured textual data. An opinion or a fact may both be present in a statement; the former provides objective information about the subject at hand, while the latter provides subjective data. Identifying whether a sentence is subjective, or objective is one of the first tasks in SA. Only subjective statements containing opinions are to be continued. The problem of classifying subjective sentences into positive, negative, or neutral polarity follows the classification of subjectivity. As we work to get sentiments from textual texts, we can divide them into two categories: explicit or implicit opinions and direct or comparative opinions [2]. Straightforward language with direct opinions conveys the feelings. In contrast to direct judgments, comparative opinions compare numerous items or facets of the statement. Instead, an explicit view reveals a person's position on something completely and explicitly, parting no possibility for the reader to infer anything from the silence. However, implicit opinion lacks clarity, necessitating the reading of the underlying meaning to convey the message that was meant to be conveyed but left unstated. Contrary to the former, implicit opinions have a wide range of metaphors they can utilize, which makes the entire analytics process even more difficult because they are meant to have a lot of semantic information.

The primary objective of the SA is sentiment classification, which involves correctly categorizing subjective phrases into their appropriate polarity kinds, such as neutral, positive, or negative.

2.1 Granularity of the Text

There are many tasks to face; therefore, solving SA is a complex task. As shown in Fig. 2, sentiment analysis is done at several text granularity levels, such as documents, sentences, and aspects. The most straightforward and fundamental type of SA, document-level SA seeks to identify the whole sentiment polarity of the textual material. The first SA-related work [3] focused on document-level SA. In comparison to SA document-level, sentence level SA is a finer level of investigation. The sentiment polarity in the document's sentences is computed at the sentence level.

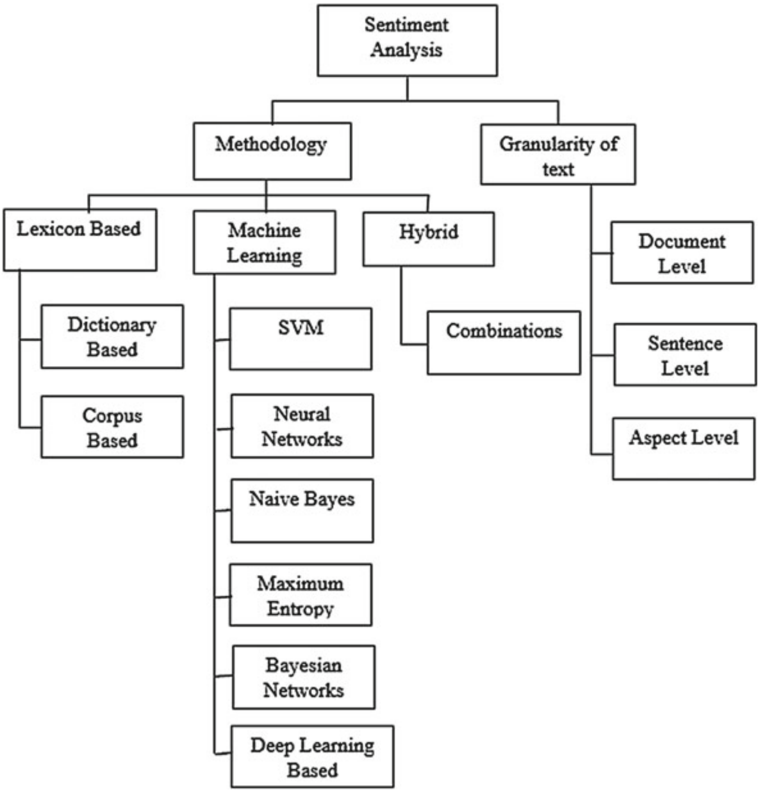

Fig. 2 Sentiment analysis classification built on granularity and methodology

The aspect level SA is a fine-grained one focusing on specific features of any entity rather than the whole sentence or document.

2.2 Methodology

The various methodologies discussed in this section provide brief insights into how features can be considered for sentiment analysis based on different datasets.

2.2.1 Lexicon-Based Approach

The lexicon-based approach was used in the initial efforts on SA [4–6] and can be used to perform the SA. Lexicon-based approaches employ annotated terms that have polarity values, which provides insight into the sentiment leaning of the text's content. This method's key benefit is that it does not require training data, making it an unsupervised learning strategy. However, the sentiment lexicons only include some expressions and terms. Here, the task of a SA is aided using lexical resources like Sent WordNet or WordNet etc. Dictionary-based and corpus-based approaches are further categories under which the lexicon-based method can be divided. The construct in the dictionary-based approach is a dictionary that includes opinion terms supported by their sentiment value, but in the corpus-based method, context knowledge replaces such a dictionary. In the corpus-based method, the likelihood that a word will be annotated or appended with positive and negative descriptive words is more important. Dictionary-based approaches and corpus-based approaches primarily differ in that the former cannot be used to identify terms with domain orientation while the latter can [7].

2.2.2 Machine Learning Approaches

By using accessible data to train models, machine learning (ML) algorithms can predict or categorize any incoming input unknown to it. These algorithms provide a better level of accuracy in their outcomes. There are primarily two types of data used in ML techniques: training and test data. An ML classifier receives the training data to begin the training process. Numerous classifiers can forecast the appropriate classifications, including neural networks, NB, K-means, SVM, etc. Selection of relevant features are done using Chi-square or Information gain. A classifier receives test data following the training phase to determine whether the machine learning model is producing the expected results.

2.2.3 Hybrid Approaches

Numerous different types of studies use hybrid methodologies that combine multiple SA approaches. A combined approach can achieve a better outcome rather than a standalone strategy. A hybrid strategy was created by many studies [8–11] by combining lexicon-based and ML-based methods.

2.3 Evaluation Metrics

The metrics Precision (P), Recall (R), F-score (F1), and Accuracy (Acc) are used to evaluate the SA as the problem primarily focuses on the categorization of words based on their sentiment polarity. Both lexicon-based and machine learning-based techniques are evaluated using the same measures as follows:

$$\text{Precision, } P = \frac{\text{True Positives}}{\text{Actual Results}} \text{or} \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

$$\text{Recall}, P = \frac{\text{True Positives}}{\text{Predicted Results}} \text{or} \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

$$F1\text{Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{Accuracy, Acc} = \frac{\text{True Positives} + \text{False Negatives}}{\text{Total}}$$

3 Analyzing Emotions Based on Perceptions

Aspect-based sentiment analysis (ABSA) [12] searches textual data for the sentiment of several features of an object. It could be anything, such as a person, place, movie, or thing. The entity may be described in the textual data using several word groups, also known as features detailing the things; these characteristics are referred to as "aspects" of the related entity. The object may be described in detail using a few different aspects and several words or phrases expressing certain feelings about those aspects. ABSA looks for its polarity or attitude to uncover pertinent details about the entity and its supporting statements. Think about the phrase, "This car looks so beautiful, but the safety features are not up to the mark." Here, "car" is the object of interest, whereas "look" and "safety features" are features under consideration. An ABSA for a statement assigns a positive feeling to the element of "appearance" and a negative sentiment to the feature of "safety ranking" as shown in Fig. 3.

ABSA employs a multistage analysis in a manner like the conventional SA. Numerous methods and approaches have already demonstrated how well they execute various jobs. The ABSA workflow is depicted in Fig. 4. Pre-processing entails putting provided data in an appropriate format so that it may be used for the task at hand. Tokenization, stop word removal, negation handling, and other processes are used in SA to clean up the data and transform it to the appropriate format. Special letters, standalone punctuation, and numerical tokens are eliminated from the text because

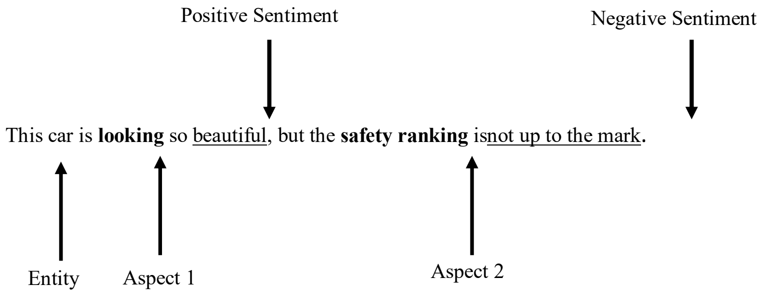

Fig. 3 Identification of aspect sentiments group text and arrows

Fig. 4 Workflow of sentiment analysis at aspect level

they do not provide any meaning. Following tokenization, stemming is used to look for provided terms' true bases in the textual material. Word embedding, a crucial step in SA employing machine learning is performed after pre-processing. By using a procedure called word embedding, we can convert a token of words into a vector representation of the words. Despite their close similarity, the words "aircraft" and "airplane" have different meanings. Word embeddings are used to change the text into a different dimension so that a machine can comprehend the differences in meaning. These vectors can also be used to extract aspects and sentiments using a machine learning model. The third stage involves finding the texts' associated features of the entity, followed by finding the contextual phrases that best express the identified aspects' feelings. At the final stage, sentiment words' exact sentiment orientation is identified.

The task of ABSA can also be split into two halves, specifically aspect category SA and feature term SA [1]. The first one is a coarser-grained level and the latter is a bit finer-grained one. The arts of music and dance are examples of aspect category SA.

The scalability of ABSA is one of its key benefits. ABSA can quickly and automatically do a fine-grained textual analysis. The manual analysis task is challenging since it is nearly impossible to process the enormous amount of text on time and at a fine-grained level. Additionally, ABSA will analyze textual elements such as reviews, comments, etc. To pinpoint the specific areas where consumers are expressing dissatisfaction or suggestions for improvement, firms and individuals can use this data. As a result, the corresponding firms or customers will see significant time and financial savings. Since ABSA falls under SA, there are multiple issues to address. ABSA produces more thorough and precise results than sentiment analysis of the document and sentence levels. Today, text analysis in ABSA is quick and simple thanks to the development of machine learning and deep learning algorithms. SVM, CNN, LSTM, and other ML techniques have been employed in several papers for the ABSA challenge [13–15].

4 Domains

The different domains that are frequently utilized in ABSA are covered in this section. When explaining the SA, the domain is crucial, particularly when the focus is on the aspect level. It is easy to extract aspects from the given text when one has domain knowledge. Otherwise, separating the characteristics from the text without a context is challenging. Think about the phrases “speaker sound is so loud that everyone in the auditorium can hear it” and “vehicle’s engine is so loud” from the speaker and car domains. According to the speaker, the loudness in the first case conveys a positive mood, whereas the loudness of the car’s engine in the second example conveys a negative opinion. It is evident that domain knowledge is essential for SA. In other

Table 1 ABSA domains and their associated datasets

Review domain	Authors and year	Associated dataset	Algorithm
TV reviews	Fu et al. [16] (2012)	Manually collected Chinese TV online reviews	K-means + Co-clustering
Product reviews	Rezaeiniaet al. [17] (2019)	CR (Amazon), SST (Stanford sentiment treebank)	LSTM
Movie reviews	Thet et al. [12] (2010)	IMDB	CNN, BiLSTM
	Anand et al. [18] (2016)	Manually collected movie reviews from IMDB & Amazon	
	Zainuddin et. al [19] (2018)	Twitter	SVM
	Rezaeiniaet al. [17] (2019)	MR (IMBD), RT (Rotten tomatoes movies reviews)	LSTM
	Zhang et al. [20] (2019)	MR, SST1, SST2, CR, AFFR	LSTM
	Piryaniet al. [21] (2017)	IMDb	Linguistic approach
Hotel reviews	Al Smadiet al. [22] (2019)	Manually collected Arabic hotel reviews	SVM
	Aktharet al. [23] (2017)	Manually collected hotel review dataset	LDA, DP, CR, and NER NLP tools
	Pham et al. [24] (2018)	Hotel reviews from tripadvisor.com	CNN
	Garcia-Pabloset al. [24] (2018)	SemEval 2016 task 5	CRF
	Al Smadiet al. [25] (2018)	Arabic hotel reviews-SemEval2016 task 5	Attention based LSTM
	Kumar et al. [26] (2019)	Manually collected reviews from booking.com	Neural network
	Garcia-Pablos et al. [24] (2018)	SemEval 2016 task 5	CRF
Restaurants	Qiuet al. [27] (2018)	Yelp	LDA
	Pham et al. [14] (2018)	Manually collected restaurant reviews	LSTM
	Garcia-Pablos et al. [24] (2018)	SemEval 2016 task 5	CRF

words, the training statistics based on many domains will impact the outcomes of SA. We can claim that SA is rather a domain determined. Based on the literature study conducted here, this conclusion was reached. There will be a distinct strategy that produces improved results on SA for each domain. Most of the SA's aspect level work focused on consumer reviews of products, including laptops, TVs, mobile phones, restaurants, hotels, and movies. Table 1 lists all the research on ABSA that has been done across several fields.

5 Datasets

Review-type data are the focus of aspect-based sentiment analysis since they are full of thoughts about the various characteristics of the products being reviewed. Lack of benchmark datasets is one of the early issues in implementing ABSA. More studies in this area have been conducted recently, and the good publicly accessible datasets that are the outcome are shown in Table 2. SemEval 2014 task 4 [28] was generated in 2014 specifically for ABSA.Pontiki et al. [29] included reviews of hotels, laptops, and restaurants—all in the English language—in the SemEval 2015 dataset. 6,940 tweets make up the target-dependent Twitter sentiment classification dataset, which was created in 2014 by Dong et al. [30]. ICWSM 2010 JDPA Sentiment corpus is a dataset that was created by Kessler et al. [31]. Documents pertaining to digital and automotive devices are included in this dataset. Darmstadt Service Review Corpus dataset, developed by Toprak et al. [32], covers reviews of online institutions and their services.

Table 2 ABSA datasets

No	Dataset name	Opinion filed	Language used	Review/opinion count
1	SemEval 2014	Restaurants, laptops	English	Restaurant—3841 Laptops—3845
2	SemEval 2015	Hotel, restaurant, laptops	English	Hotel-30, Laptops—450 Restaurant—350
3	SemEval 2016	Hotel, restaurant, laptops, mobile, camera	English, Arabic, Chinese, French, Russian, Spanish	Laptop—350 (Eng.) Hotel—2291 (Arabic) Camera—200 (Chinese) Restaurant—313 (Russian), 555 (French), 913 (Spanish)
4	Target-dependent tweets dataset	Tweets	English	6,940

(continued)

Table 2 (continued)

No	Dataset name	Opinion filed	Language used	Review/opinion count
5	FiQA ABSA	Financial news and blogs	English	Microblogs—774 News headlines—529
6	ICWSM 2010 JDPA sentiment corpus	Automotive and digital devices	English	515
7	Darmstadt service review corpus	Online universities and their services	English	118

6 Machine Learning (ML) Approaches for ABSA

Regarding aspect-based sentiment analysis, ML algorithms have been thought to perform accurately. The various ML algorithms are discussed in the following subsections, along with how well they work for ABSA.

6.1 Latent Dirichlet Allocation (LDA)

Latent Dirichlet Allocation (LDA) is a probabilistic, generative model for a collection of texts that can be thought of as combinations of underlying issues. A topic-developing technique called LDA assists in automatically identifying the core subjects in each material. Documents are viewed by LDA as mixtures of subjects with certain probabilities of words. The authors identified themes for using the LDA model and measured the distance between them using KL divergence.W2VLDA is an unsupervised system that deals with multi-domain and multilingual ABSA, according to Garcia-Pablos et al. [24]. Aspect Sentiment Unification Model (ASUM) was created by Amplayoet al. [33] by incorporating product descriptions. Aspect and its accompanying emotion are combined in ASUM, a modified form of LDA [34].

6.2 Conditional Random Field (CRF)

CRF, a discriminatory model for predicting orders as opposed to generative models like LDA, are employed. For more precise prediction in CRF, data from the prior labels are employed. A prediction methodology was put forth by Qiu et al. [27] for calculating the ratings of unrated reviews from the Yelp dataset. For the generation of pair terms and calculating their sentiment scores, authors employed a sentiCRF form of the CRF. They developed a cumulative logit model, which forecasts review ratings using attributes and the associated sentiment values from the reviews. Furthermore, they suggested a heuristic re-sampling approach to address the class imbalance issue during sentiment score calculation.

6.3 Support Vector Machine (SVM)

SVM is promising for classification and regression tasks. Every data point in SVM is projected on an n-dimensional space for creating a hyperplane for classification or regression. The potential hyperplane with the most significant separation from the support vectors will be considered. The authors also used Stanford Dependency Parser to extract implicit elements in this work. A model was put out by Al-Smadi et al. [22] to help the ABSA perform better while translating hotel reviews into Arabic.

6.4 Convolutional Neural Network (CNN)

CNN is basically a feed-forward neural where the convolutional layer i.e., the initial layer, extracts all the characteristics from the input. CNN was used by Kumar et al. [26] for the ABSA task, and stochastic optimization was carried out in their research. Word-level embedding, in this case, was carried out using word2vec, and semantic feature extraction was carried out by creating ontologies. Particle Swarm Optimization (PSO) was added by the authors as a multi-objective function for CNN parameter tuning. Particle Swarm Optimization (PSO) was included by the authors as a multi-objective function for CNN parameter tweaking.

6.5 Long Short-Term Memory (LSTM)

The vanishing gradient and inflating gradient problems that Recurrent Neural Networks (RNN) experienced were addressed by the development of the LSTM network, a form of RNN. The main advantage of LSTM is its intelligence, which enables it to recognize long-term dependencies. In other words, LSTMs have a longer retention time for the information. For this, the LSTM network included a cell-class explicit memory unit. The three inputs that the LSTM uses to make its decision are the current input, the previous output, and the prior memory. A technique based on LSTM that has a two-stage aspect level classification of feelings was proposed by Ma et al. [35]. Xu et al. [15] used LSTM to generate a model for aspect level sentiment classification from reviews of restaurants and laptops. It is a semi-supervised technique with sentiment and context as its two stochastic variables (Table 3).

Table 3 Comparative study for different ML models on ABSC

Authors and year	Models	Accuracy
Zainuddin et al. [19] (2018)	SVM + ARM	Accuracy = 76.55%
Thet et al. [12] (2010)	CNN, BiLSTM	Complete movie—86% Direction—86%, storywriter—80%, play—90%
Kumar et al. [26] (2019)	Neural network	Accuracy = 88.52%,
Maneket al. [13] (2017)	Attention based LSTM	Accuracy = 97.32%
Fu et al. [36] (2011)	LDA	Accuracy = 89.165%
Wang et al. [16] (2012)	Co-clustering using K-means +	Accuracy = 78.198%
Dahlmeieret al. [37] (2018)	LSTM for Attention	Accuracy = 85.58%
Cambria et al. [38] (2018)	LSTM using Hierarchical	Accuracy = 89.32%
Al-Smadiet al. [25] (2018)	Attention based LSTM	Accuracy = 95.4%
Yang et al. [39] (2018)	JABST and MaxEnt-JABST	71.2% (tweets)
Amplayoet al. [33] (2018)	Ensemble based on PSO	Accuracy = 85.73%
Al-Smadiet al. [22] (2019)	SVM	Accuracy = 95.4%
Song et al. [40] (2019)	Attention based LSTM	Wiki—91.28%, reviews—92.91%, news—92.07%
Zhang et al. [35] (2019)	LSTM	Laptop—73.1%, resort—80.1%
Yao et al. [41] (2019)	MaxEnt-JABST and JABST	Amazon—83%, yelp—85%
Liet al. [27] (2018)	CRF	Accuracy = 93.6%

7 Challenges

In the past four to five years, ABSA has significantly impacted the modern technology-oriented world. This research addressed a few papers in this field. It is evident from comparing the approaches used by different authors that DL based works are producing more encouraging outcomes on aspect level SA. However, other studies indicate that machine learning techniques outperform deep learning techniques. For instance, Al-Smadi et al. [25] noted that SVM outperforms RNN [25] in terms of performance. SA is very important today for several uses, including analyzing client feedback. As a result, the domain is important for aspect level SA and its inherent that majority of projects concentrate more on similar areas, such as computers, restaurants, hotels, etc. Several other highly significant categories, such as travel reviews and news, still need to be explored by researchers. Another major issue for researchers in this field is the need for benchmark datasets on various domains. The language in which SA is conducted is another serious issue. Regional languages are only sometimes employed for SA duties; only a few selected languages, such as English, Chinese are intensively used. The widespread use of regional languages in social media has significantly increased the importance of this field, encouraging further research in SA at the aspect level. Further research in this field is being hampered by a lack of high-quality datasets in regional languages.

8 Conclusion

In the present e-world, SA, the development of deleting sentiments, feelings, or emotions from a text, blogs etc. has become well known. SA can be broken down into the document-level, the phrase level, and the aspect level. The third level of analysis, known as ABSA, is far more in-depth and involves extracting aspects, words with matching feelings, and their polarity orientation. Since there has been a lot of research done on ABSA over the last three to four years, the organization is currently attempting to stand on its own. The level of perception and complexity in SA have significantly changed because of the advent of machine learning. The absence of a benchmark dataset and publicly accessible datasets only cover a limited number of domains are two fundamental problems with ABSA. Deep learning techniques are producing some encouraging results for ABSA because of technological developments. However, it is evident from the literature review that the outcomes fall below expectations. In other words, we may argue that the development of ABSA using deep learning is just getting started.

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Irrigation Using IOT Sensors

Ajay Talele, Milind Rane, Omkar Jadhav, Mohit Burchunde, and Aniket Pardeshi

Abstract Modern world is experiencing a water shortage, making it urgent to implement clever irrigation techniques. The paper explains how the Internet of Things (IOT) may be used to manage irrigation intelligently. The goal of this irrigation technique is to save time and steer clear of issues like ongoing vigilance. By intelligently allocating water to the plants or fields according to their water needs, it promotes water conservation. The water requirement is tailored according to specific needs of different plants. Additionally, it is useful in agriculture, parks, and lawns. This system basically measures the moisture level of the soil and if it is found to be low, it automatically pumps water to the soil till the soil moisture reaches a threshold value. On a user's mobile phone, access to and monitoring of this data is simple.

Keywords Soil moisture sensors · Internet of things (IOT) · Arduino integrated development environment (IDE) · Android application · Node microcontroller unit (MCU)

1 Introduction

The development model of smart agriculture can become a real-time observation system of properties which includes temperature, moisture, soil moisture, and potential of Hydrogen (pH) [1]. By sacrificing the concept of modern irrigation systems, farmers can save up to 51% of water. This idea relies on two of his irrigation strategies: Overhead sprinkler [26] and flood supply system. Through IoT, one can manage multiple operations of the sector from anywhere at any time [21]. To overcome these shortcomings, new techniques are used in irrigation that apply only small amounts of water to the basal zone component of the plant.

Soil wetting stress in plants is prohibited as fields are automatically irrigated by following their watering schedule. Usual techniques such as mechanical equipment and surface irrigation require almost half as much water. Fields can also be supplied

A. Talele · M. Rane (✉) · O. Jadhav · M. Burchunde · A. Pardeshi
Vishwakarma Institute of Technology, Pune, India
e-mail: milind.rane@vit.edu

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with appropriate amounts of water. Dry rows between fields can result in continuous bandages throughout the watering method. Fertilizers can also be applied in this manner and are less expensive.

The new technology significantly reduces soil and wind erosion compared to overhead mechanical jig systems. Soil properties can outline the shape of dripping properties within the root zone of plants that receive moisture [16]. It offers an artistic approach to living, by allowing us to manage electronics with smartphones while saving energy. It has impacted all areas of business, not just smart farming, but also smart parking, smart building, environmental monitoring, and careful transportation.

2 Literature Review

The usage of NodeMCU in irrigation systems, the advantages of employing smart irrigation systems, and the difficulties and emerging trends for the advancement of these systems can all be found in a literature study of smart irrigation systems using NodeMCU microcontroller boards. NodeMCU has been included into a number of smart irrigation systems to regulate and track a number of irrigation-related variables, including soil moisture content, water flow, and environmental conditions [2].

The underneath steps are taken into consideration during the initial investigation: understanding the current methodologies, comprehending the demands, and creating a system abstract. In order to detect temperature, humidity, and soil moisture, sensors were inserted into the plant's root zone for this investigation. Data was transmitted from the sensors to an Android app. A microcontroller was programmed with the threshold value of a soil moisture sensor in order to regulate the water content. The Android app shows data on soil moisture, humidity, and temperature. The "Irrigation with IOT sensors" aims to build an automated irrigation system that controls the pumping machine based on the amount of soil moisture [3]. Along with the measurements for soil moisture, the planned system also provides temperature and humidity values. For a remote agricultural plantation that needs water, this research article suggests a NodeMCU-based remote irrigation system. When the soil's humidity drops below the predetermined level, provisions must be made for water. However, due to unawareness of the soil moisture level at the time, a system that incorporated additional features for temperature and soil moisture values that were shown on farmer mobile applications was proposed [4]. The system made use of a NodeMCU esp8266 board, which comes with a Wi-Fi module already installed. Improved crop yields, lower labor costs, and higher water efficiency are all advantages of employing smart irrigation systems. These devices can also aid in resource conservation and in the reduction of water waste. To further optimize irrigation, smart irrigation systems can be used with other technologies like weather forecasting and precision farming.

Smart irrigation systems are designed to improve the efficiency and productivity of agricultural irrigation, but their implementation is not without challenges. Some of these challenges include scalability, reliability, and security. Scalability refers to the ability of the system to function effectively in different sized areas and environments.

Reliability is crucial, as the system must be dependable and consistently provide the desired results. Security is also a concern, as these systems often rely on remote monitoring and control and may be vulnerable to hacking or other forms of cyber-attack [28].

To overcome these challenges, research is needed to develop more accurate and cost-effective sensors, and to improve the integration of these systems with other technologies. The use of IoT sensors, for example, can improve the precision and automation of irrigation systems, and can also provide valuable data on factors such as temperature and humidity. Additionally, the integration of IoT technology in agriculture can help address issues of food security, supply, and distribution [18].

However, most studies on smart irrigation systems have been conducted in controlled environments, and further testing is needed to address real-world challenges. Additionally, the use of NodeMCU microcontroller boards has demonstrated potential for improving irrigation efficiency and crop yields, but further research and development is necessary to fully realize this potential.

Overall, smart irrigation systems have the potential to significantly improve agricultural irrigation, but further research and development is needed to overcome challenges and fully realize this potential. As technology continues to advance, it is likely that smart irrigation systems will become increasingly popular among farmers.

3 Methodology

Write the NodeMCU code in the Arduino IDE and then upload it in the NodeMCU using a USB cable. Then enable the device hotspot and then check the 'Online' status in the Blynk App. Then insert the probes [27] of the soil moisture sensor inside the soil. For obtaining values for temperature and humidity of the surrounding area, DHT11 sensor was used [22].

Along with temperature and humidity measurements obtained by the DHT11 sensor, the app also displays records of moisture level detection [13] by the soil moisture sensor. Figure 1 shows the system flowchart for a better understanding of our system.

It can be said from the flowchart that the first power supply is given to Node MCU. After sensing the temperature and the amount of soil moisture, respectively, the DHT11 Temperature Sensor and Soil Moisture Sensor transmit their findings to the NodeMCU. If the moisture value is less than 500, then the motor starts, and the soil is watered. Else when the value is greater than 500, the motor remains in off position. As a result, the values are shown on the "Blynk" Android app.

The given Fig. 2 shows the app interface of our system. The readings of moisture, temperature and humidity (in percentage) are seen. One can also turn on the water pump manually by tapping on the "ON" button.

The given Fig. 3 portrays the hardware setup of our system.

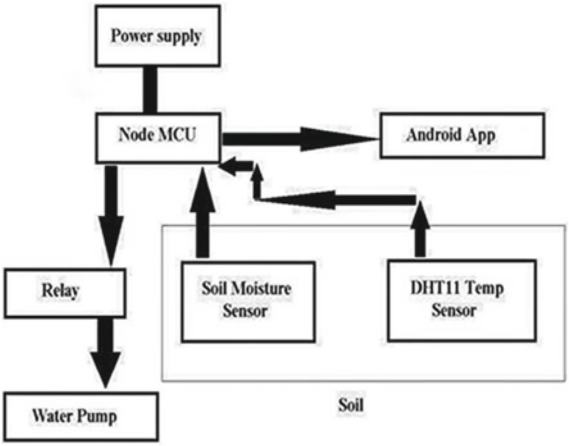


Fig. 1 System flowchart

Fig. 2 Blynk app interface

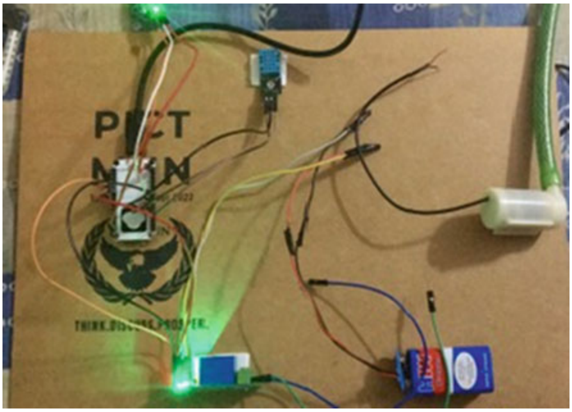


Fig. 3 Hardware setup

4 System Architecture

4.1 Hardware

The hardware components of our system include the following components: -

ESP8266 Node MCU Wi-Fi Module. A sophisticated Application Programming Interface for hardware input/output devices is NodeMCU [5]. It uses Lua, an interactive script, as opposed to Arduino-style code. It is an open-source Internet of Things platform. "Espressif" Systems firmware is used to put it into practice. There are 16 input/output pins on NodeMCU [29]. Consequently, a single node can link to 16 nodes. The Tensilica Xtensa LX106 core, which is frequently used in Internet of Things applications, is combined with the ESP8266 Wi-Fi security operations center (SoC) [6]. Instead of the development kits, "NodeMCU" often refers to the firmware. ESP8266, a NodeMCU with a built-in (Wireless-Fidelity) Wi-Fi module, as shown in Fig. 4 is selected for our system. Because of this in-built module, no external interfacing has to be done to establish a wireless connection with other sensors. Also, this makes it easy to integrate with Android applications.

Soil Moisture sensor. A kind of soil moisture sensor is depicted in Fig. 5. It includes two tests that involve injecting current into the soil. After that, it examines the soil's blockage to determine the soil's amount of moisture [19]. It is known that when water is close by, dirt is more likely to conduct power easily, meaning that such soil has less resistance (R), however, dry soil has low conductivity, and thus, offers higher resistance than wet soil [7]. This intensity feature justifies the selection of this sensor. The resistance must be converted to voltage at some point; this is accomplished using a circuit that is visible inside the sensor and that does the conversion [8].

DHT11 sensor. Temperature and humidity measurements are accurate to + 0.5 degrees [8]. On the rear of the sensor, the DTH-11 sensor has an IC, a temperature sensor, and a moisture sensing sensor. It includes both 4 and 3 pins. Figure 6 shows

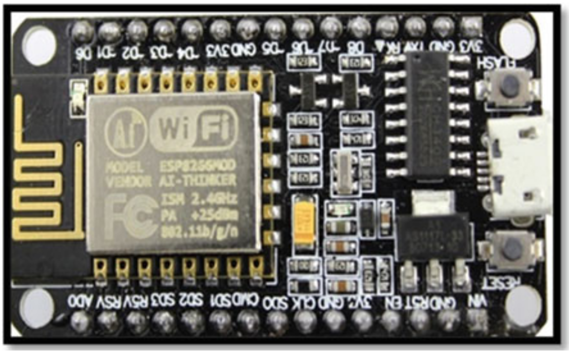


Fig. 4 ESP8266 node MCU Wi-Fi module

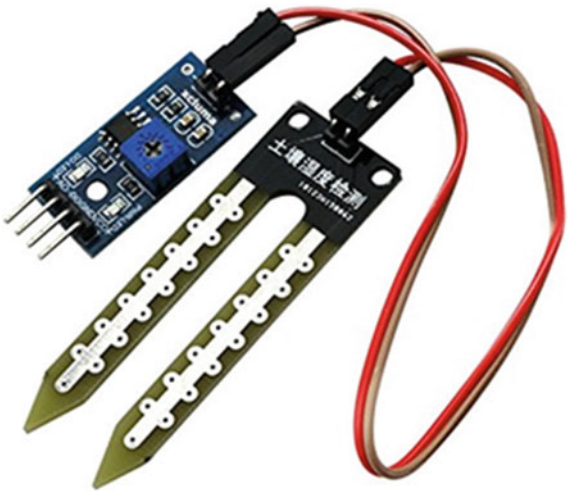


Fig. 5 Soil moisture sensor

an explanation of the connection using the DTH-11 sensor. The selection of this sensor is justified by its functionality of measuring both temperature and humidity at the same time and in one sensor only [17].

Pump motor. Draw water from a water source and water the soil [30].

9 V Battery. Supply power to the system.

Jumper wires. Connecting components of the system with the NodeMCU.

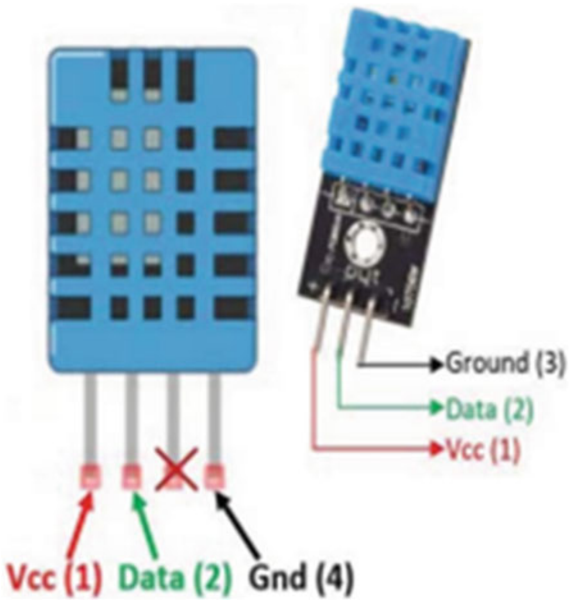
4.2 *Software*

Arduino IDE and BLYNK App were used as software in our system.

Arduino IDE. Used for writing the hardware code.

BLYNK App. Application for automating our system.

Fig. 6 DHT11 sensor



In the Arduino IDE setup part, first install the required libraries. Then from the Blynk Web obtain the Authorization token of the user's account. This verifies the user's account. Then set the user's mobile phone's hotspot name and password as 'ssid' and 'pass' respectively. Then take two variables for storing humidity and temperature values as float values. Allocate those values to virtual pins of Blynk App. Similarly, store values of temperature. For the motor, take the motor variable to read the status of the motor. Then write the pump motor function. In this check the reading of the soil moisture sensor. If the value is greater than 500 then the digital pin of NodeMCU is made 'Low'. Else it turns 'High'. Then write a stopper function for maintaining the motor in off condition. In the Loop part, run the Blynk function. Then check the motor status using if-else. If it is on then the pump motor function is executed else the stopper function is executed. Figure 7 portrays the code snippet.

4.3 Hardware Connections

The Digital Temperature and Humidity Sensor (DHT11 sensor) [25] consists of three pins + , OUT and − . + connected to the 3.3 V (3V3) supply of NodeMCU, OUT with the Digital (D4) and - to the Ground (GND).

Soil moisture sensor module has four pins Analog Output (AO), Digital Output (DO), Voltage Common Collector (Vcc) and Ground (GND). AO connected to the controller's AO, GND to the GND and Vcc to the 3V3 power supply pin of NodeMCU. The positive and negative terminal of the module is connected with the

Fig. 7 Arduino IDE code snippet

```
}
//virtual button
BLYNK_WRITE(V0) {
  motor = param.asInt();
}

void pumpmotor () {
  if (sensor > 500) {
    digitalWrite (D5, LOW);
  }
  else if (sensor < 500) {
    digitalWrite (D5, HIGH);
  }
}
void stoper () {
  digitalWrite(D5, HIGH);
}
void loop()
{
  digitalWrite (D6, LOW);
  digitalWrite (D5, HIGH);
  Blynk.run();
  timer.run();
  sendTemps();
  if (motor == 1) {
    pumpmotor();
  }
  else if (motor == 0) {
```

sensor. Relay module has Input (IN), GND and Vcc. IN pin connected to the D6 pin, GND to GND and Vcc to the 3.3 V power supply of NodeMCU. The output of the relay is then joined with the motor and the external battery source. The given Fig. 8. explains the pin diagram of the setup.

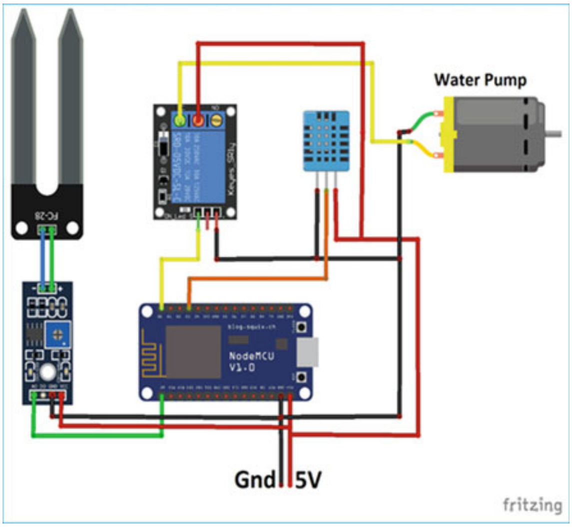


Fig. 8 Circuit diagram

5 Result and Discussion

In the experiment, data is collected on different environmental factors including temperature, humidity, and soil moisture levels at different time intervals throughout the day. This helped to analyze the fluctuations of these factors over the course of the day.

As part of our analysis, we specifically examined the soil moisture levels and observed that there was a decrease in soil moisture levels during the afternoon time slot from 12 to 4 pm. This could be due to a combination of factors such as higher temperatures, increased evaporation rates, and lower water uptake by plants during this time period. The data we collected is presented in Table 1 and can be used for further analysis and interpretation.

It can be concluded that auto watering level of soil can be set up once in the morning slot at 9:00 am and again in afternoon slot at 4:00 pm.

Table 1 Result of experimental setup

Sr. No	Time	Temperature (°C)	Humidity (%)	Soil moisture readings	Pump status
1	9:00 am	21	38	480	On
2	10:00 am	24	87	940	Off
3	12:00 pm	28	68	674	Off
4	3:00 pm	31	52	557	Off
5	4:00 pm	30	33	440	On
6	6:00 pm	27	89	976	Off

6 Future Scope

High accuracy and water use efficiency can be achieved by more accurately anticipating irrigation requirements, modifying irrigation times and volumes to suit crop water needs, and compensating adaptively for water losses [9]. This reduces irrigation water consumption and increases yields. As systems become more sophisticated and intelligent, better trained models will be deployed to make better irrigation decisions. Therefore, much of the stress and burden associated with irrigation is alleviated for farmers and users. Although each pesticide and fertilizer have a specific purpose in agriculture, using them in excess can be harmful to the land, crops, and environment [14]. As a result, these might be used in suitable quantities and in controlled volumes. Additionally, there is a need for improvement in the area of soil nutrient management [15]. The system's efficiency can be increased by taking into account variables besides moisture and temperature. In order to ensure low costs and increased productivity, farmers can be informed about the optimum crop combinations for diverse cropping circumstances.

7 Conclusion

An automatic irrigation system which measures the amount of moisture in the ground is indeed a helpful system as it does automation and regulation of watering with no manual intervention [10]. It will enable the farmers and consumers to effectively manage their watering schedules to irrigate their plants and crops [20]. This will be of great benefit to farmers facing water shortages. They face difficulty in watering their farms. The main issue is that they are not having any fixed idea of availability of current to pump water. Intelligent farming has been successfully implemented to improve consumers by checking the real-time values of temperature and humidity on the app. This also saves parameters in a very convenient way. This system allows consumers to check the status of various parameters on-site at any time. You can better control or maintain the parameters of the field [23]. The motor and temperature status will also be available on a mobile device without having to visit [24]. The technology could be upgraded to allow for use outside. It can be concluded that automatic irrigation systems are more efficient than scheduled irrigation because they can respond in real-time to changing environmental conditions and the water requirements of plants.

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Music Overflow: A Music Genre Classification Web Application

Pallavi Bharambe, Shubham Bane, Tejas Indulkar, and Yash Desai

Abstract "Music Overflow": An automated classification model for music genres is proposed to be created using a Music Genre Classification Web Application. A outmoded class that sorts some portions of music as belonging to a not unusual way of life or set of conventions is known as a music genre. Its duty is be outstanding from melodic style and form. There are numerous methods that tune may be divided into distinctive genres. Blues, Hip-Hop, Pop, country and Rock are popular forms of track. Content-based music genre classification is an essential component of music information retrieval systems. With the rise of digital music on the Internet, it has gained prominence and received a rising volume of attention. Automatic music genre classification has received little attention to date, and the stated classification correctness are also quite low. To determine a music piece's genre, we examine how various classifiers perform on various audio feature sets. Lastly, we experiment with combining various classifiers to improve classification accuracy. On a 10-style set of one thousand song pieces, we first obtain a take a look at style category accuracy of round 73.2% with a set of different classifiers. This performance is higher than the great that has been reported for these statistics set, that is 71.1%. We discover that the classifier used determines which function set is first-rate.

Keywords Model view template (MVT) · Django template language (DTL) · Mel frequency cepstral coefficients (MFCC) · Music information retrieval (MIR) · Hypertext markup language (HTML) · Database management system (DBMS)

P. Bharambe (✉) · S. Bane · T. Indulkar · Y. Desai
Shivajirao S. Jondhale College of Engineering, University of Mumbai, Dombivli (E), India
e-mail: pallavi.bharambe87@gmail.com

S. Bane
e-mail: shubham.bane13@gmail.com

T. Indulkar
e-mail: tejas.indulkar2001@gmail.com

Y. Desai
e-mail: yashd.342@gmail.com

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1 Introduction

Music genre classification is the process of categorizing music into different genres based on its acoustic and musical features. Music genres are typically defined by a set of characteristics such as rhythm, melody, harmony, instrumentation, and lyrics. The task of music genre classification involves developing algorithms that can automatically identify the genre of a given piece of music. This is typically done using machine learning techniques, which involve training a model on a dataset of labeled music samples and then using the model to predict the genre of new, unlabeled music samples.

There are many different approaches to music genre classification, ranging from simple rule-based systems to complex deep learning models. Some common features that are used in music genre classification include spectral features, such as spectral centroid and spectral roll-off, and temporal features, such as beat and tempo.

Music genre classification has many practical applications, such as music recommendation systems, content-based music retrieval, and music information retrieval. It is also an active area of research in the fields of music information retrieval, machine learning, and audio signal processing [1].

2 Literature Survey

Since the beginning of the Internet, the classification of music genres has been the subject of extensive research. Using supervised machine learning techniques like the k-nearest neighbour classifiers and Gaussian Mixture model, Tzanetakis and Cook (2002) solved this issue [2]. They added 3 units of functions: timbral structure, periodic content material, and pitch content material [3, 4]. Music genre classification was explored using Hidden Markov Models (HMMs), widely used in speech recognition tasks (Scaringella and Zoia 2005; Soltau et al. 1998) [5]. For genre classification, Mandel and Ellis (2005) study and compare support vector machines (SVMs) with various distance metrics [6]. According to Zwicker and Fast (1999), the authors of Lidy and Rauber (2005) talk about the role that psycho-acoustic features play in determining the genre of music, particularly the significance of STFT taken on the Bark Scale [7]. The features used were Mel-frequency cepstral coefficients (MFCCs), spectral roll-off, and spectral contrast (Tzanetakis and Cook 2002) [2]. In Nannietal, SVM and AdaBoost classifiers are trained using a combination of visual and acoustic features (2016). A number of studies employ deep neural network techniques to analyze dialog and other kinds of audio data in light of their recent success (Abdel-Hamid et al. 2014) [28]. Due to the high sampling rate of audio signals, it is tough to denote audio in the time field for neural network input. However, Van Den Oord et al. have addressed it in 2016 for tasks related to making audio. The spectrogram of a signal, which includes information about both time and frequency, is an alternative representation that is frequently used [8]. Many extraordinary functions may be used for song type, e.g., reference functions along with the name and composer, content-based acoustic functions collectively with tonality, pitch, and beat, symbolic functions extracted from the ratings, and text-primarily based capabilities extracted from the track lyrics. In this paper we're interested by content material-primarily based features [9].

The content-based acoustic functions are categorized into timbral texture functions, rhythmic content functions,and pitch content features. Timbral capabilities entirely originated from traditional speech recognition strategies. They are generally calculated for every brief-time frame of sound primarily based on the short Time Fourier transform (STFT) [10, 11]. Regular timbral features consist of Spectral Centroid, Spectral Roll off, Spectral Flux, energy, zero Crossings, Linear Prediction Coefficients, and Mel-Frequency Cepstral Coefficients (MFCCs) (see for extra element) [12]. Logan examines MFCCs for tune modelling and tune/speech discrimination [13]. Rhythmic content capabilities consist of information about the regularity of the rhythm, the beat and tempo records. Pace and beat monitoring from acoustic musical signals have been explored in [14, 15]. Foote and Uchihashi use the beat spectrum to symbolize rhythm [16]. Pitch content functions deal with the frequency statistics of the track bands and are obtained using diverse pitch detection strategies. A good deal of much less paintings has been reported on the tune genre category [17, 18]. Tzanetakis and Cook propose a complete set of capabilities for direct modelling of track indicators and explore using those functions for musical genre categorization the usage of k-Nearest Neighbours and Gaussian mixture models [19]. Lambrou et al. use statistical capabilities in the temporal domain in addition to 3 precise wavelet rework domain names to classify song into rock, piano and jazz [20, 21]. Deshpande et al. make use of Gaussian combinations, assist Vector Machines and Nearest Neighbours to categorise the tune into rock, piano, and jazz primarily based on timbral features [22, 23]. Pye investigates the usage of Gaussian mixture Modelling (GMM) and Tree-based totally Vector Quantization in song genre type [24]. Soltau et al. suggest a method of representing temporal structures of input sign. They show that this new set of summary capabilities may be found through synthetic neural networks and may be used for track style identity. The problem of discriminate music and speech has been investigated with the aid of Saunders, Scheier and Slaney [26, 25]. Zhang and Kuo recommend a heuristic rule-primarily based gadget to segment and classify audio signals from films or tv programs. In audio contents are divided into instrument sounds, speech sounds, and surrounding sounds the usage of automatically extracted capabilities [26]. Foote constructs a gaining knowledge of tree vector quantizer the usage of twelve MFCCs plus strength as audio capabilities for retrieval. Li and Khokhar suggest nearest characteristic line strategies for content primarily based totally class audio retrieval [27].

Music genre classification is an important and highly studied field of research in the music industry. The goal of this literature survey is to examine and summarize the current state of research in this area. In conclusion, it can be seen that music genre classification is an important and highly studied field of research.

3 Problem Statement and Objectives

Even though categorizing a song into a genre is a task that is inherently subjective, the human ear finds it easy to do. A new song's timbre, instruments, beat, chord progression, lyrics, and genre are all easily identifiable in a matter of seconds. For machines then again this is a seriously mind boggling and overwhelming errand as the entirety "human" experience of paying attention to a tune is changed into a vector of elements about the melody. In the past, many of these musical characteristics that humans recognize in music have not been reliably detected by machines. We are considering a 10-genre classification problem with the subsequent categories: pop (p), blues (b), disco (d), jazz (j), rock (ro), country (co), reggae (re), classical (cl), metal (m), hip-hop (h). Finally, we have chosen six popular genres namely classical, hip-hop, jazz, metal, pop and rock to get more accuracy for our final web-app. Mel Frequency Cepstral Coefficients (MFCC) are the feature vectors used in our pattern recognition algorithms for classification. Over traditional music players, we are planning to make an entertainment music web app.

Objective:

The objectives of music genre classification are as follows:

- To extract the features from the given music.
- To classify the input music based on the extracted features and label a class (genre) name to it.
- To quantify the match of any input music to the listed genres.
- To make genre classification more accurate.
- To find out the a couple of genres to which a track belongs to.
- Our group is highly enthusiastic to build web apps.
- To learn and implement MVT architecture components.
- To make a music lovers community web app.

4 Methodology

Following are the methodological steps using machine learning (Fig. 1).

Fig. 1 Methodological steps

Step 1: Data Preparation

Data preparation, also known as data preprocessing, is the process of cleaning, transforming, and organizing data before it is used for analysis or machine learning. This step is crucial because it ensures that the data is accurate, consistent, and suitable for analysis.

Step 2: Data Pre-Processing

Data pre-processing is an essential step in the data science process and is used to ensure that data is ready for analysis. Pre-processing involves cleaning, transforming, and normalizing the data to make it ready for further use. It usually involves removing any unnecessary or irrelevant data, handling missing values, and correcting any inconsistencies.

a. Train-Test Split

While machine learning algorithms are used to make predictions on records that has not been used to train a model, a split train test method is used to estimate how well they perform using the consequences of this quick and smooth technique, you could examine the overall performance of machine learning algorithms for your predictive modelling trouble.

b. Min–Max Normalization

The fundamental idea behind normalization and standardization never changes. Variables which might be measured at one-of-a-kind scales may additionally result in a bias due to the fact they do not all make a contribution similarly to the model fit and learned feature. Consequently, prior to model fitting, feature-wise normalization like MinMax Scaling is typically utilized to address this potential issue.

c. Feature Importance

The degree to which each feature contributes to the prediction of the model is shown by the feature (variable) importance. It basically determines how useful a particular variable is for the current model and prediction (Fig. 2).

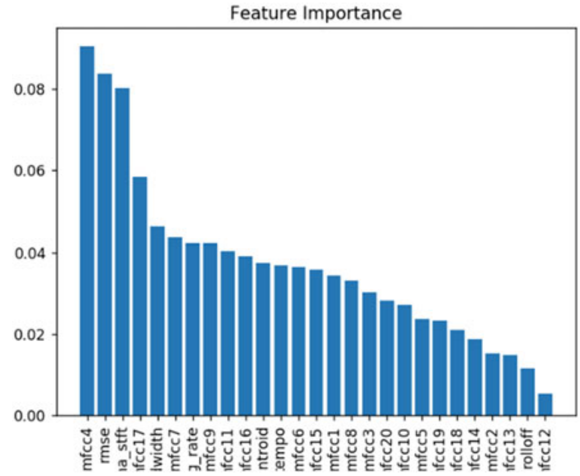
Step 3: Extraction of Features

Feature extraction is the process of extracting meaningful information from a given audio signal, such as music. It involves extracting the different components of the signal, such as pitch, rhythmic patterns, timbre, and tempo. It also involves analyzing the data in order to extract the features of the signal that are of interest, such as the loudness, texture, and complexity of the sound.

a. Mel-Frequency Cepstral Coefficients

The maximum current methods for sound recognition and type employ MFCC, which is a hard and fast of brief-term power spectrum traits of the sound. It exemplifies human voice characteristics. The final feature vector (13 coefficients) contains a significant portion of this feature.

Fig. 2 Feature importance



$$m = 2595 \log_{10}\left(1 + \frac{f}{700}\right)$$

b. Chroma Frequencies

Chroma frequencies refer to the twelve pitches in Western music that are spaced equally on the chromatic scale. These pitches are commonly referred to as the 12 notes of the Western music system and are named after the first seven letters of the alphabet: A, B, C, D, E, F, and G, with each note having two possible names, a sharp (#) or a flat (b).

c. Spectral Centroid

It specifies the area of the "centre of mass" for sound. It is basically the weighted mean of the sound's frequencies. Take a look at two songs—one from the metal and one from blues. A blues tune is by and large steady all through its length while a metal melody ordinarily has more frequencies collected towards the end part.

$$\text{Centroid} = \frac{\sum_{n=0}^{N-1} f(n)x(n)}{\sum_{n=0}^{N-1} x(n)}$$

d. Zero Crossing Rate

The wide style of instances when the waveform crosses zero is represented via this. It generally has better values for sounds with a whole lot of percussion, like rock and metal.

$$zcr = \frac{1}{T-1}\sum_{t=1}^{T-1} 1_{\mathbb{R}_{<0}}(s_t s_{t-1})$$

e. Spectral Roll-off

Spectral roll-off refers to a measure of the rate at which the spectral energy of a signal decreases as frequency increases. It is a widely used feature in digital signal processing and is commonly used in audio and speech analysis.

Step 4: Model Selection

Support Vector Machine

A Support Vector Machine (SVM) is a type of machine learning algorithm used for classification and regression analysis. It works by finding the best hyperplane that separates the data points of different classes with maximum margin.

In SVM, the data points are represented as vectors in a high-dimensional space, and the algorithm attempts to find the optimal separating hyperplane by maximizing the distance between the hyperplane and the closest data points of each class.

Step 5: Train and Test

Dataset has 10 classes, Blues, Country, Pop, Classical, Hip-Hop, Metal, Disco, Jazz, Rock and Reggae. Each class containing 100 audio clips, each clip being 30 s long in .wav format.

Of which, 75% of data is used for training and 25% of data is used for testing.

750 examples are used for training the model and 250 examples are used for testing the accuracy of the model.

We get the highest testing accuracy with the SVM ML model is 73.2% (Fig. 3).

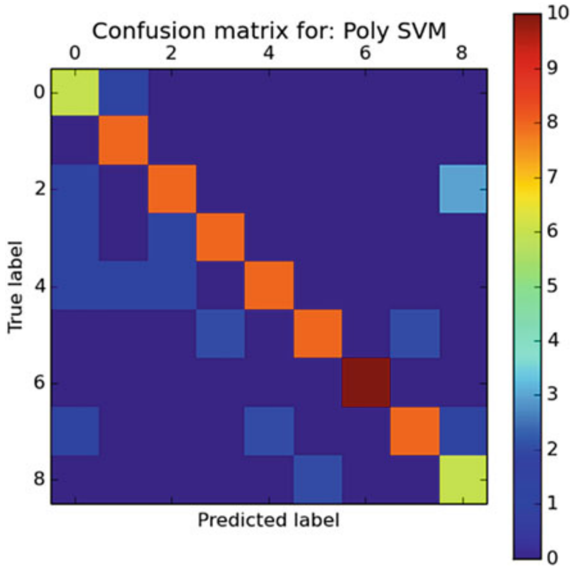


Fig. 3 Confusion matrix for SVM classifier

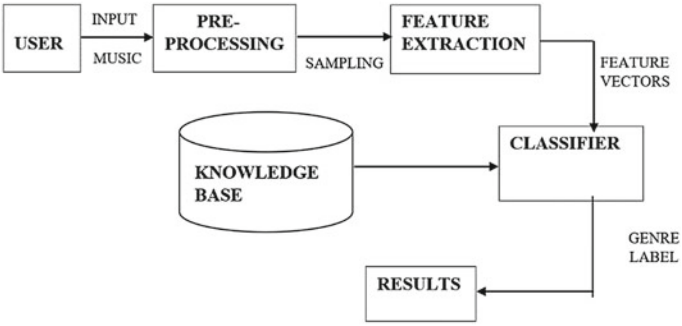


Fig. 4 Process diagram

5 Proposed System

It involves using different machine learning algorithms to analyze a dataset of music and classify it into distinct music genres.

The first step is to collect a large dataset of music examples belonging to different genres. This can be done by scraping online music databases such as Spotify or iTunes, or by manually creating a collection of songs from various artists.

Once the dataset is gathered, it must be pre-processed to extract its features. The commonly used features for music classification are the duration, frequency, loudness, and the range of instruments used. These features are then used to create a feature vector for each song.

Next, a machine learning algorithm can be used to train the system. Popular algorithms for music genre classification include decision trees, support vector machines, and neural networks. The algorithm is trained by providing it with the feature vectors and the corresponding music genres.

Finally, once the system has been trained, it can be tested by providing it with a set of unseen examples. This will help determine the accuracy of the system and if it is suitable for its intended purpose (Fig. 4).

6 Django MVT Architecture

Django MVT (Model-View-Template) is a design pattern used in the architecture of web applications built with the Django web framework. It is a variation of the Model-View-Controller (MVC) pattern that separates the concerns of the application into three components: Model, View, and Template.

Model: The model component in Django MVT represents the data and the database schema of the application.

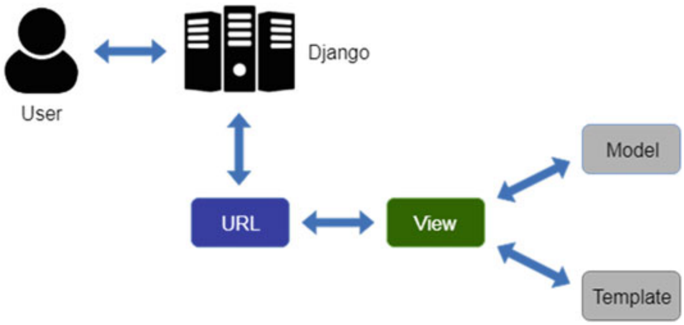


Fig. 5 MVT architecture

View: The view component in Django MVT is responsible for rendering the data in response to a user request. It interacts with the model component to retrieve the data and prepares it for rendering.

Template: The template component in Django MVT is responsible for rendering the data into an HTML format that can be displayed in the user's browser (Fig. 5).

7 Results

The results' explanation is related to the snapshots of the various modules to the entire process of execution. The different modules can explain the working of procedure.

The user has to select the option depending upon the requirements. The choice-based option contains upload music and find genre.

Figure 6 shows the front face of the Web App.

Figure 7 illustrates the uploading of the music. As the user clicks the 'upload file' button the Web App shows the file explorer to select the file or directly drag and drop file.

Figure 8 illustrates the file being classified according to genre. As the user clicks the 'check' button the Web App. It will take 2–5 min according to the input file and display the genre of the uploaded file.

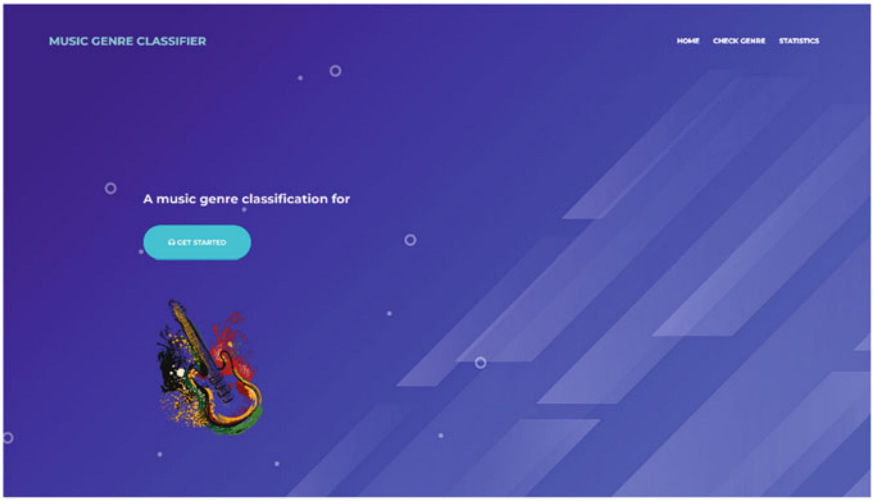


Fig. 6 Front face of web app

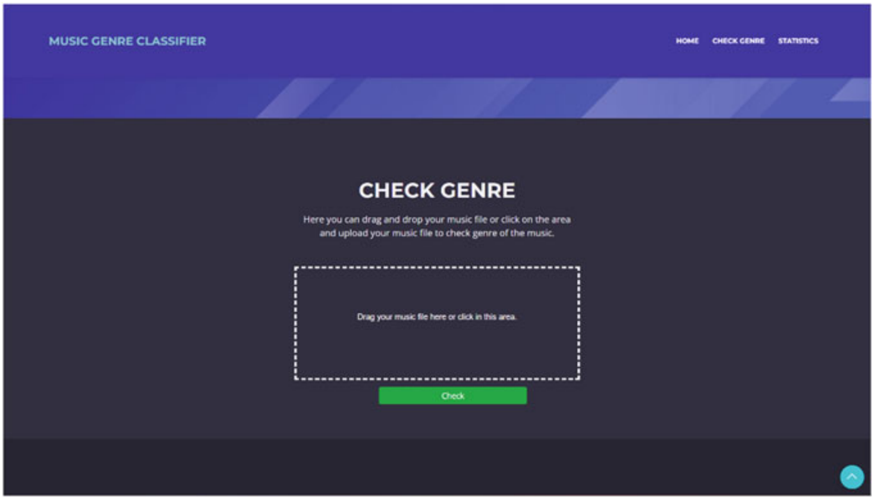


Fig. 7 Upload the music

Fig. 8 Finding the genre

8 Conclusion

The user can classify music files according to genre with the help of the proposed research.

Utilizing a variety of machine learning algorithms, this project on the classification of music genres was completed. We wanted to get the best accuracy.

In order to develop and train our model, we have implemented a number of best practices. We have learned a variety of framework patterns and best practices that are currently utilized in the industry throughout the model's development.

It has been found to be impossible to test a large number of combinations in order to obtain an accurate application evaluation.

The analysis of the results indicates that the current model functions properly and has attained the desired format of 73.2% accuracy. In order to develop and train our model, we have implemented a number of best practices. Throughout the model's development, we learned about various architecture patterns and best practices currently used in the industry.

9 Future Scope

We can deploy Web App on the cloud so that we can scale our web app. We can also provide API for developers. Another thing we can try to improve the accuracy is trying a different dataset or dynamically increase the dataset as the user input new music.

Conflict of Interest The authors whose names are listed in this manuscript certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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A Comparative Analysis of Recent Face Detection Methods Implemented for Age and Gender Detection

Archana Chaudhari, Riya Gokhale, Purva Shendge, and Nikita Sawant

Abstract Age and gender detection is an upcoming technology, which is gaining importance. Age and gender detection is a subjective task. Even humans have trouble recognizing it sometimes. Our system helps them recognize what age range and gender the person is. To select the absolute best method, we have done a comparative analysis of different face detection algorithms, these are Dlib, OpenCV, and MediaPipe, and implemented them on pretrained neural network model of age and gender detection. Then, this system was implemented in hardware using Jetson Nano.

Keywords Dlib · DNN · Jetson Nano · MediaPipe · OpenCV · Python

1 Introduction

In several fields, such as authentication, human–computer interaction, behavioural analysis, and product suggestions based on user preferences, age and gender detection are crucial [1]. Age and gender information is needed by many businesses, but there weren't many answers. To satisfy this demand, significant advancements have been made in recent years. In earlier systems we have seen facial recognition done to analyse emotional states [2, 3].

Using age detection to display your advertising on the media indicates that your target audience uses the most is a wonderful example. Businesses can better target marketing campaigns by using age and gender detection, for instance, by determining which goods and services various client segments require and can afford—refinement, which can lower your lead or sale cost per lead [4].

One of the use cases is an AI software programme called Quividi, which uses online face analysis to determine the age and gender of passing users and then automatically begins playing advertisements tailored to that audience [5]. His AgeBot, an Android app that utilises face recognition to estimate age based on images, is another illustration.

A. Chaudhari · R. Gokhale · P. Shendge (✉) · N. Sawant
Department of Instrumentation Engineering, Vishwakarma Institute of Technology, Pune, India
e-mail: purva.shendge20@vit.edu

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The main objective of this project was to experiment with newer recent face detection methods and implement them for age and gender detection. Face detection methods like Haar Cascades are far too old and do not give good accuracy [6]. Neither it is able to recognize faces at various angles and lighting. Hence, we worked on newer methods with greater accuracy.

Each author of this paper contributed to the experimentation as well as the writing of this research paper. Each author reviewed at least six research papers and realized its weaknesses as ways to improve its technology. Riya Gokhale worked on Dlib face detection and its respective implementation for OpenCV DNN based age and gender detection. Similarly, Purva Shendge worked on MediaPipe face detection and its respective implementation for OpenCV DNN based age and gender detection. Lastly, Nikita Sawant worked on OpenCV face detection and its respective implementation for OpenCV DNN based age and gender detection.

The proposed work presents a comparison of face detection algorithms implemented with age and gender detection in real time, using Jetson Nano.

2 Literature Review

There are several papers published on age and gender detection. We surveyed these papers and got a gist of the current technology as well as the improvements we could make. Table 1 presents the short Summary on Age and Gender Detection Algorithms using Machine Learning and Deep Learning. Table 2 presents research done using hardware implementation.

3 Methodology

The proposed work presents a comparison of face detection algorithms, mainly Dlib, MediaPipe, OpenCV DNN as explained below. These face detection algorithms are implemented with age and gender detection using OpenCV DNN in real time, using Jetson Nano.

As seen in the above literature review, we see that many of the authors have not implemented the system in real time, done only using pictures not used while training. For real time implementation raspberry pi is used as the microprocessor [25].

Table 1 Literature review on age and gender detection

Sr. No.	Paper	Summary
1	Age and gender recognition using deep learning June 2022 [7]	Algorithm used—Deep Learning and CNN Accuracy—80% Implemented on Python IDE
2	Deep facial age estimation by conditional multitask learning with weak label expansion 2018 [8]	Algorithm used—Deep Learning, Label Sensitive learning, Age group classifier, CNN is used Accuracy—87% Dataset—CASIA-WebFace, MORPH-II, FG-NET Implemented on computer with K80 GPU
3	Real-time face-age-gender detection system 2019 [9]	Algorithm used—CNN Datasets—UTKFace Accuracy for age detection—72.4% Accuracy for age detection—85.8%
4	Gender and age predictions using artificial intelligence algorithm 2020 [10]	Algorithm used—Haar Cascades, CNN and convNet Dataset—UTKFace Implemented on—Anaconda Navigator, Jupiter Notebook Accuracy—0.6170588
5	Age and gender prediction of face images using convolutional neural network, 2018 [11]	Algorithm used—Dlib, CNN and DTML Dataset—IMDB-WIKI Accuracy—92% Implemented on—VSCode
6	Age and gender classification using of human convolutional neural network 2021 [12]	Algorithm used—Haar Cascades, SVM, CNN Dataset—IMDB-WIKI, Adience Accuracy—90% Implemented on—Python IDE
7	Age prediction and gender classification using CNN and ResNet in real-time 2020 [13]	Algorithm used—Haar Cascades, CNN Dataset—GCVL Accuracy—85% Implemented on—Python IDE, real time

(continued)

Table 1 (continued)

Sr. No.	Paper	Summary
8	Predicting age and gender of mobile users at scale—a distributed machine learning approach 2018 [14]	Algorithm used—Logistic Regression, Random Forest and Gradient Boosting Trees (GBT) Accuracy—80% Implemented on—real time huge scale
9	Gender and age classification of human faces for detection of anomalous human behavior 2018 [15]	Algorithm—SVM Accuracy for age detection—80.17% Accuracy for age detection—90.33%
10	Gender classification using face recognition, 2019 [16]	Algorithms—LDA, PCA Accuracy using LDA—85% Accuracy using PCA—83%
11	Convolutional neural networks based face and gender recognition system 2020 [17]	Algorithm—CNN Datasets—the Wild (LFW), YouTube Face (YTF) and VGGFace2 Accuracy—93.22%
12	Age and gender prediction using DNN 2019 [18]	Algorithm used—Deep Learning, CNN, convNet Haar Cascade, OpenCV and TensorFlow Accuracy—90% Implemented on Jupiter Notebook IDE
13	Gender prediction from images using deep learning techniques 2019 [19]	Algorithm used—Deep learning techniques, CNN Accuracy—80% Implemented by using IMDB-WIKI dataset
14	Gender prediction and age estimation using convolutional neural network 2020 [20]	Algorithm used—CNN Dataset—IMDB-WIKI Accuracy—80% Implemented on MATLAB
15	Alexander V. Savin, Victoria A. Sablina, Michael B. Nikiforov Comparison of facial landmark detection methods for micro-expressions analysis, IEEE (2021) [21]	Algorithm—MediaPipe Dataset—Spontaneous Actions and Micro-Movements (SAMM) dataset
16	B. Thaman, T. Cao, N. Caporusso, Face mask detection using MediaPipe Facemesh, IEEE (2022) [22]	Algorithm—MediaPipe

Table 2 Literature review based on hardware implementation

Sr. No.	Paper	Summary
1	Age and gender detection system on Raspberry Pi, 2019 [23]	Algorithm used—CNN Dataset—Adience Benchmark Accuracy—85% Precision—88.23% Microprocessor used—Raspberry pi
2	Face detection and tracking for gender estimation, age rank, weight based on face images utilizing Raspberry Pi processor 2018 [24]	Algorithms used—local binary pattern, viola jones algorithm, CNN Microprocessor used—Raspberry pi
3	Implementation of machine learning for gender detection using CNN on Raspberry Pi platform 2020 [15]	Algorithm—CNN Dataset—Adience dataset Microprocessor used—Raspberry pi

3.1 Techniques Used for Face Detection

Dlib HOG

It's made of five HOG (histogram of oriented gradients) filters: front, left, right looking, and two combinations of mentioned. This method basically focuses on the object's shape and structure. This is as it calculates the features using gradients. As the name suggests, using the areas of images, it creates histograms that are based on the gradient's direction and magnitude.

After the HOG filters are applied, it extracts features into a vector and feeds it into a classification algorithm SVM which will assess whether the frame has a face (or object) or not [26].

Figure 1 represents the flow chart of the Dlib model. The algorithm first starts by gray scaling the image. Then it normalizes the color and constructs histograms. Using the HOG, it constructs vectors. These then get fed into SVM. SVM further processes them and gives us the output, which is detecting the face.

MediaPipe

Its face detection works on ultrafast face detection and six landmarks support. These six landmarks detect positioning of the nose, eyes, ears respectively [22]. The confidence level of detection as well as range for the image can be also defined as per our requirement. Figure 2 represents the pipeline of MediaPipe for face detection.

Two different deep neural network models work together for face detection: One model is the detector which operates on the complete image and computes face locations [27]. Another model is a three-dimensional face landmark model which

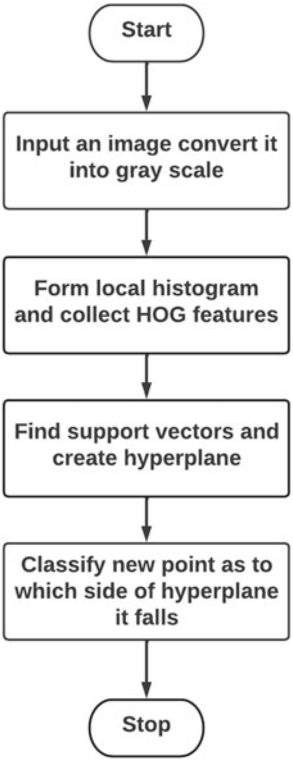
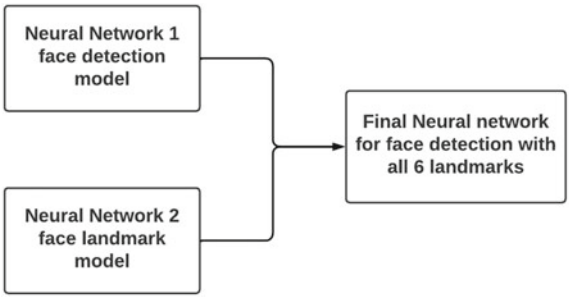


Fig. 1 Pipeline of Dlib face detection

Fig. 2 Pipeline of MediaPipe face detection



works on those particular locations and predicts the approximate 3D surface by using regression.

Configuration options in MediaPipe-

1. MODEL_SELECTION—An integer index value 0 or 1 is used for model selection. Here index 0 indicates the short range model which is capable for face

detection within 2 m range. And the value 1 indicates the full range model which works for a maximum of 5 m range.

2. MIN_DETECTION_CONFIDENCE—This value is a float value which lies between 0.0 and 1.0. This value indicates the confidence level of our model about the detected face. For example if we set this value as 0.8, then our model will detect the given face unless and until it is 80% sure about it. Default value of this is 0.5.

Open CV

OpenCV is an open source library used for computer vision, machine learning, and image processing, is essential for real-time detection and manipulation

It is utilised for a variety of tasks, including face and object recognition, classification of human behaviour, camera movement tracking, tracking of moving objects, and more [28].

DNN—Input layer, output layer, and a few hidden layers are present in Deep Neural Networks. This DNN module allows you to get pre-trained neural networks from popular frameworks like TensorFlow and use these models directly in OpenCV. This means that you can use any popular framework to train your model and only use OpenCV for inference or prediction. Here we are using DNN with the CAFFE model.

CAFFE MODEL—A deep learning framework called CAFFE (Convolutional Architecture for Fast Feature Embedding) enables users to build models for image segmentation and classification.

The CAFFE model contains the actual layer weights. CAFFMODEL files are created from .PROTOTXT files. The model works as follows: When using OpenCV and DNN (Deep Neural Network) modules in Caffe models, two sets of files are required: .prototxt and .caffemodel files.

First, the user creates the model as plain text and saves it.

The tool stores the user's trained model as his CAFFEMODEL file after the user trains and improves the model in Caffe.

Caffe-compatible prototype machine learning models are stored in PROTOTXT files. It includes a Caffe-trained image segmentation or classification model. A PROTOTXT file defines the model architecture. This code first loads face models from a file and then uses those models to detect faces in the input image. Figure 3 presents the flow chart of the OpenCV and DNN based model for Face detection [29, 30].

The marked faces are included in a list and passed to another function for further processing. Figure 4 represents the flowchart of the proposed model for face and gender detection using DNN. Figure 5 presents the architecture of Caffe Model for the proposed method. Table 3 demonstrates a comparative of implementation of three face detection methods.

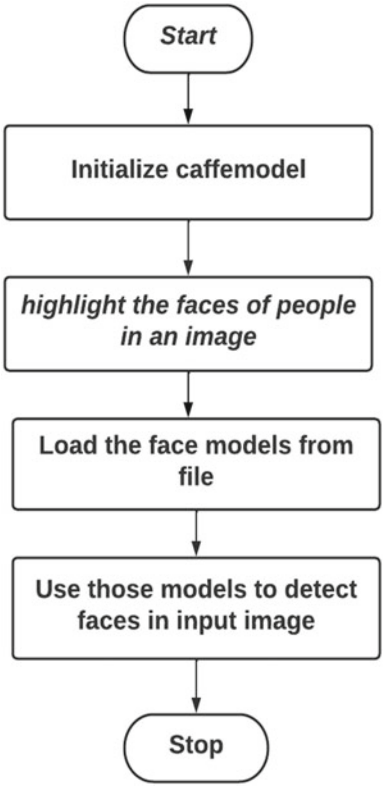


Fig. 3 Flowchart of OpenCV DNN based face detection

3.2 Steps of the Proposed Algorithm for Age and Gender Detection Using DNN

The algorithm is as stated below:

1. Capture image from real time video
2. Implement the Face detection
3. Send frame to age and gender detection algorithm
4. Process the image using caffe model weights
5. Age and Gender detection successful.

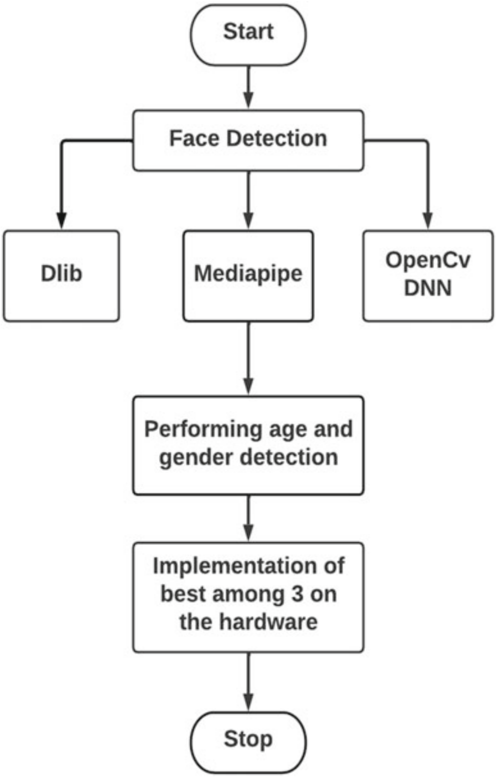


Fig. 4 Flowchart for age and gender detection using DNN

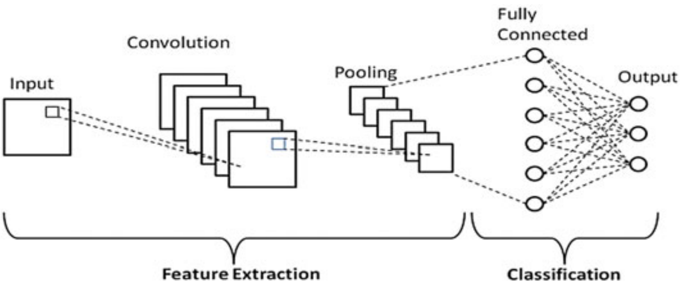


Fig. 5 Architecture of caffe model [9]

Table 3 A comparative table of implementation of face detection methods

Point of distinction	Dlib histogram of gradients	MediaPipe	OpenCV DNN
Methodology	Uses histogram of gradients to identify edges,	Uses six landmarks to detect image	Uses pre-trained neural network from popular framework like TensorFlow and use those models directly in OpenCV
Range and accuracy adjustable	No, it's fixed for given use case	Both range and accuracy can vary according to the user	Both range and accuracy can vary according to the user
Lag	Slight, but beneficial for long term application	Not much, but it can be minimize	Not much

4 Results and Discussions

4.1 Results of Age and Gender Detection Using Dlib Algorithm

Here are the results. We have used different algorithms for face detection: Dlib, MediaPipe, and OpenCV DNN. Figure 6 presents the age and gender detection results implemented using DLIB face detection and OpenCV DNN age and gender detection. As we can see, Dlib is able to successfully detect both faces in the frame even in the low light setting. The OpenCV DNN algorithm correctly identifies the face as female in the range (20–30), although it is only able to detect age and gender of one face.

Figure 7 demonstrates age and gender of a child aged 10 using Dlib algorithm with OpenCV DNN.

Figure 8 demonstrates the age and gender of a man of age 52 accurately detected using Dlib algorithm with OpenCV DNN.

4.2 Age and Gender Detection Using MediaPipe and OpenCV DNN Method

In Fig. 9 the age and gender detection results are demonstrated using MediaPipe face detection and OpenCV DNN age and gender detection. As observed from the results, MediaPipe is able to successfully detect a face in the frame. The OpenCV DNN algorithm correctly identifies face as female in the age range (20–25), although

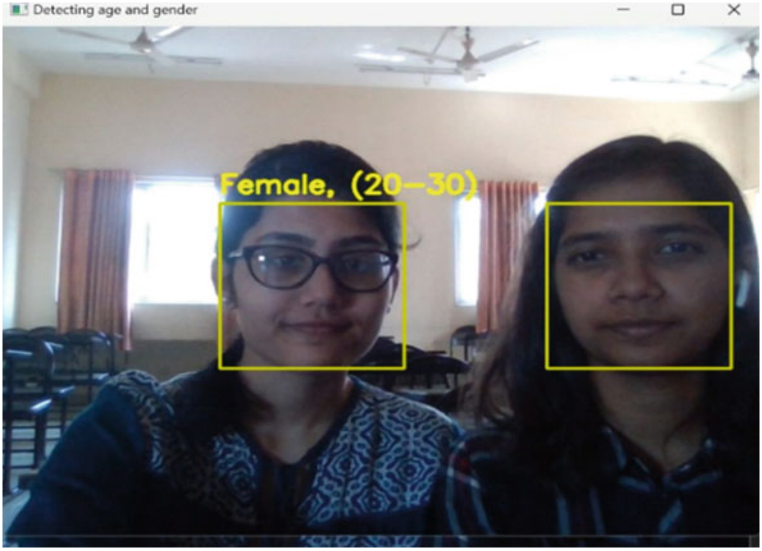


Fig. 6 Age and gender detection using Dlib detection

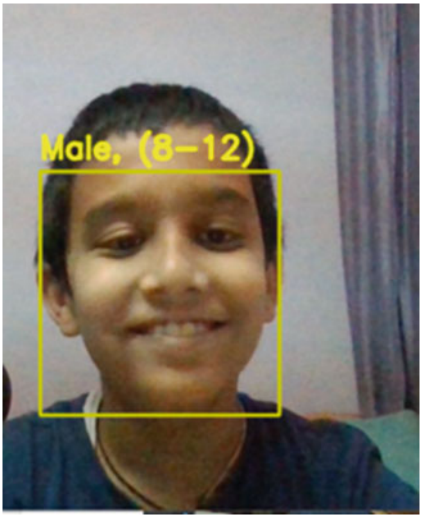


Fig. 7 Age and gender detection using Dlib detection

it is only able to detect the age and gender of one face. MediaPipe also plots the facial landmarks of the face.

Figure 10 demonstrates the age and gender of a male aged 20 accurately detected with MediaPipe algorithm.

Figure 11 presents the results of age and gender detection implemented using OpenCV DNN face detection and OpenCV DNN age and gender detection. As we

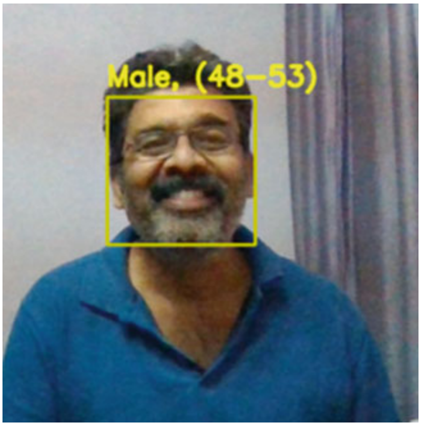


Fig. 8 Age and gender detection using Dlib detection

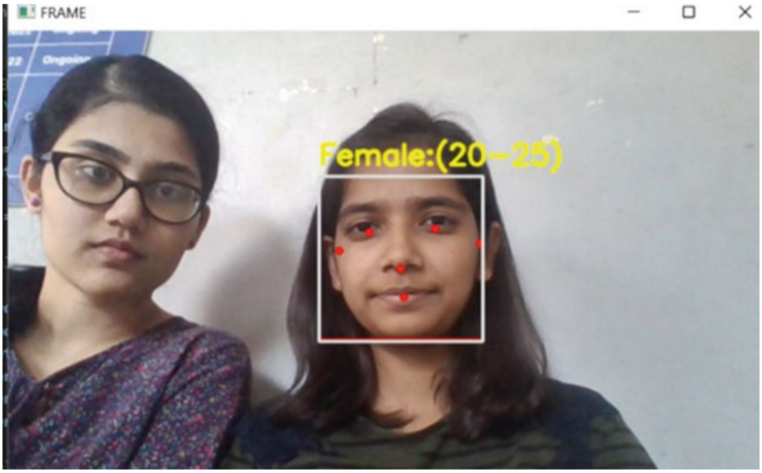


Fig. 9 Age and gender detection using detection by MediaPipe

can see, OpenCV DNN is able to successfully detect both faces in the frame. The OpenCV DNN algorithm correctly identifies both faces and accurately predict them as female in the age range of (15–20).

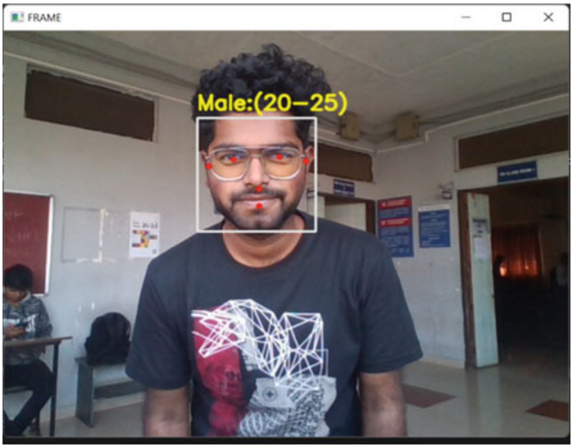


Fig. 10 Age and gender detection using detection by MediaPipe

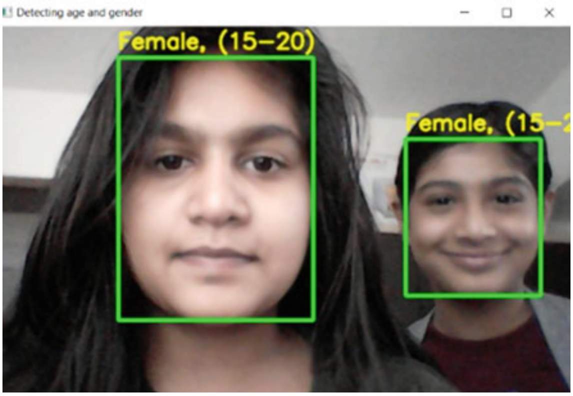


Fig. 11 Age and gender detection using OpenCV DNN

4.3 Final Results

Figure 12 demonstrates the accuracy of Dlib, MediaPipe and OpenCV DNN methods of each age and gender detection algorithm, distinguished by the face detection algorithm used. Figure 12 presents graphical analysis of the accuracy of each algorithm. The confusion matrix is presented in Fig. 13. As seen from the confusion matrix, it consists of eight classes, which are for the eight age ranges, namely, '(0–2)', '(4–6)', '(8–12)', '(15–20)', '(25–32)', '(38–43)', '(48–53)', '(60–100)'.

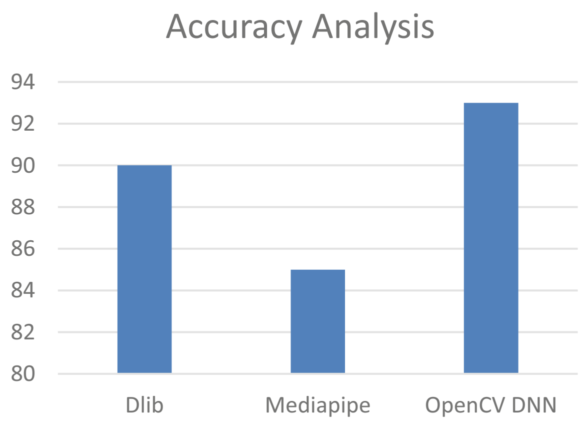


Fig. 12 Graphical representation of accuracy of each age and gender detection algorithm, distinguished by the face detection algorithm used

Confusion Matrix								Accuracy
3438	166	191	16	45	9	136	0	85.93 %
191	3306	177	1	69	2	15	0	87.90 %
88	114	3205	34	431	46	80	3	80.10 %
30	12	98	3735	78	23	24	0	93.38 %
11	28	437	29	3196	65	45	11	83.62 %
3	0	64	7	38	3702	8	0	96.86 %
59	4	79	42	44	5	3234	1	93.25 %
2	0	29	3	113	9	6	2639	94.22 %

Number of test samples: 29676

Fig. 13. 8 × 8 confusion matrix of the caffe model, from founder of OpenCV DNN page

5 Challenges

These are the limitations of the project which we came across while implementation of the algorithms in software and hardware on the Jetson Nano hardware.

The results of age and gender detection fluctuate too often, and the accuracy of the system could be improved. As each frame keeps getting detected more than once, same person detected multiple times. Using implementation on Jetson Nano, a lot of lag was faced.

6 Conclusion

Age and gender detection, has many applications as stated earlier. On analysis, we have concluded that, for the detection of multiple faces at same time the OpenCV-DNN method is prominent among all three. Hardware integration for this method was also better than others. About the accuracy, Dlib has the better than others. For mesh type applications, and for variable confidence level MediaPipe is the better one. Overall, the OpenCV-DNN performed well in our application.

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A Machine Learning Approach for Entrepreneurial Competencies and Economic Growth of Women Entrepreneurs from Kandhamal District

Sarita Dhal, Nishikanta Mishra, Puspalata Mahapatra, and Bhaigrathi Nayak

Abstract Due to LPG and continued growth in the Indian start-up industry, an increasing number of women are coming forefront and adding to the entrepreneurship segment of India. Creating a business from the beginning stage of conceiving the ideas, till its final stage and also surviving in long run as a profitable venture is a challenging task basically for women. Again these entrepreneurs have to take care of almost all business activities by themselves which starts with starting from brainstorming and vetting potential ideas, setting goals, planning a project, analyzing a product etc. Women Entrepreneurs are seen as key drivers of the economic expansion of each country irrespective of their development status. Despite their significant contributions, it's most important that they remain a small subset of business owners. This investigation into minority groups like women entrepreneurs is an attempt to identify the driving forces behind the rise of female business owners and the challenges they faced and their significant contribution to achieving their economic growth. Machine Learning, Descriptive statistics of ANOVA were used to pinpoint many factors that contribute to the success of women business owners of the Kandhamal District of Odisha. This study's findings can guide the government, other institutional networks and support agencies in developing a constructive plan of action and policies to assist women entrepreneurs to succeed in the future and to enhance their ability to overcome their challenges.

Keywords Women entrepreneurs · ANOVA · Statistics · Govt. policy · Primary factors

S. Dhal · N. Mishra · P. Mahapatra
KIIT Deemed to be University, Bhubaneswar, Odisha, India
e-mail: saritaatbbsr@gmail.com

N. Mishra
e-mail: nmishrafhu@kiit.ac.in

B. Nayak (✉)
Sri Sri University, Cuttack, Odisha, India
e-mail: bhagirathi.n@srisriuniversity.edu.in

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1 Introduction

Every country irrespective of their status of development feels the need for entrepreneurs as an important driver for economic development. Entrepreneur development leads to higher growth rates concerning GDP, Employment, Sensex, per capita income, the standard of living, infrastructure and reduction in inflation which in turn lead to the development of the Country as a whole [1]. Due to LPG and continuous growth in the Indian start-up industry, a huge number of women are coming forward to add to the entrepreneurship segment of India [2]. Women's Entrepreneurship greatly upgrades women's position in society and makes them economically sound which in turn makes their families, community and whole country develop [3]. In a growing country like India and with the presence of the male dominant society Women entrepreneurs need support and guidance from their families and spouse to become successful. Among many factors accountable for influencing a woman to become an entrepreneur the key reasons are a reduction in the number of jobs, a lifestyle change, an increase in expenses, Financial independence, increase in family income [2]. With profound struggle when women are becoming an entrepreneur and start their businesses also face many challenges like gender discrimination, getting funds and financing, acquiring the key competencies, and getting access to technology to make their enterprise successful. Though the Govt. has taken many initiatives for extending help to women entrepreneurs in both rural and urban areas still it is not result-oriented and successful [2]. As is suggested by many researchers that for overall economic development, the contribution from rural women entrepreneurs is equally needed. Women's entrepreneurship growth and development are correlated with women's empowerment which makes them financially independent, self-sufficient and socially and economically sound. For setting own businesses both men and women face many challenges but for women, the barriers are more than their counterparts and they are confronted with a lack of Govt. support, limited access to bank accounts, and less freedom to get credit and loans. They get less education, skill training and carrier guidance [4]. Cultural values also influence women entrepreneurs and family responsibility also creates challenges for them to become a success. The planning commission including the Indian Government perceives the importance and need of women for their contribution to economic development and as a strategic solution for poverty alleviation both in rural and urban areas [5].

When the conditions are going in the right direction, people are more likely to take the initiative to start their new businesses, which in turn stimulates the economic growth of a country. Women entrepreneurs are essential for achieving the aforementioned goals like economic growth because their proportion is almost nearly half of our population. Here the researchers tried to study women entrepreneurs who face lots of challenges in the district of Kandhamal, one of the state's more backward regions. Women's entrepreneurship in this district has been studied to determine under different socio-economic factors, such as age, education, marital status, and motivation to know whether these factors affect their success and growth or not. When it comes to a country's progress, it's crucial to invest in its women, as they account for nearly half of the population. On ethical and humanitarian grounds, women must take part in economic activities as an entrepreneur and should occupy an equal footing with men in every aspect. Worldwide, the number of unemployed women has been rising at a faster rate than that of unemployed men recently. Women have historically faced discrimination and a lack of opportunities in the business world. This was nothing new; just as the labour market had traditionally been male-dominated, so too had knowledge been divided along gender lines, suggesting that men and women should approach business in different ways (Labour Bureau 2015). Women have traditionally been under-represented in business ownership, in part because they hold fewer managerial positions than men, have less access to capital, and are more likely to be the primary caregivers for their children and the home. Women in traditionally patriarchal societies, such as those in Asia and India, rarely take part in decision-making involving issues outside the home. However, changes in educational opportunities, employment rates, and economic status for women have all contributed to a rise in female entrepreneurship in recent years. In recent years, more and more women have shown an interest in earning their own money, working for themselves, and starting their businesses. This paper mostly focused on identifying the major factors and their impact on the development and other challenges of the women entrepreneurs of Kandhamal Districts. The proposed paper has been structured as follows: Sect. 1 gives a brief introduction about the key competencies of women entrepreneurs, the challenges they face and their contribution to economic growth which is the motivation of the study, Sect. 2 deals with previous literature related to growth, importance, and challenges they face and Sect. 3 covers conceptual explanation about women entrepreneurs, Sect. 4 is devoted to used methodology and analysis with its findings followed in Sect. 5 which describes detailed discussions which were extracted form use of Machine Learning (ML), PCA and ANOVA. Section 6 is the concluding section that summarises the entire article and suggests a direction for further research.

2 Literature Review

Lerner and Almor, examined and found a link between the skill, competencies and growth and performance of Women entrepreneurs [6]. Man and Lau highlighted that Entrepreneurial competencies include many skills that are grounded and related to personal things like traits, personality, attitudes, and social roles and which can be achieved through education, training and experience [7]. Man et al. described that an entrepreneur plays his or her role properly and effectively in a responsible way that needs ability like entrepreneurial competencies which are known as entrepreneurial skills [8]. Lerner and Almor, while doing their study on women entrepreneurs came to know that managerial skills and entrepreneurial skills contribute significantly to their success. It includes finance, proper management of human resources, proper production and operation activities, creativity, innovations and marketing products

[6]. Chandler and Jansen who did their study on the state of Utah, found that identification and prioritisation of entrepreneurial, technical and managerial functions are three areas the entrepreneurs must employ to become successful [9]. Mitchelmore and Rowley observed in their study and then suggested that managerial competencies which include functional and organisational competencies are very vital for the growth and long-term survival of entrepreneurs [10]. Satpal and Rupa and Orser et al. suggested sixteen areas of similar expertise which include Marketing opportunities, Strategic Planning, and effective management which are very crucial among them for making entrepreneurs more successful in the long run [11, 12]. Man et al. under their study identified six competency areas like Opportunity, interpersonal relationship, organising, conceptual and commitment competencies which are key to their success [8].

Jennings and Cash pointed out that there is a difference between male and female entrepreneurs concerning aspects like capital investment, thinking, perception, knowledge, technical skills, motivation, personal background, risk-bearing capacity and support from their family etc. [13]. Carter and Shaw talked that women business proprietors enjoy a lower status either concerning organising human capital, personal socio-cultural background and working experience or negotiation skills while managing their entrepreneur [14]. According to Driessen and Zwart the competencies and key skills of entrepreneurs include experience, knowledge, motivation, capacity and entrepreneurial features (personal qualities) for growth [15]. Cardella discussed the reasons that motivate them to choose entrepreneurial activities both for men and women. In general, male entrepreneurs give importance to reasonable incentives, job certainty, growth and development scope whereas female entrepreneurs prefer to make a better balance between work and family, use initiative and get recognition while pursuing an entrepreneurship carrier [16]. According to Pihie and Bagheri, women entrepreneur's personal and functional competencies are made-up skills, attitude, knowledge and family support to make them successful [17]. Rotefoss and Kolvereid did a cluster analysis and found out that entrepreneurial education plays a crucial role in developing entrepreneurial intentions among women and bridging the gap between men and women [18]. In a study conducted by Dawson and Henley, it was found that due to differences in risk attitude, there is a huge gap between men and women entrepreneurs in starting an entrepreneurial carrier. Again women's less participation is due to fear of failure, less support from family and poor social networks [19]. Anandalakshmy [1] told about the problems and challenges the women entrepreneur are facing in India to start and operate their businesses [20]. These are non-availability of adequate capital, dual role play, restricted freedom, less Knowledge of information technology and less awareness about different schemes offered by the Govt.

As per the suggestion given by Mishra [21] after analysing 48 articles it was suggested that self-confidence, institutional support, and the ability to avail the credit facilities and social networks play a very important role for an entrepreneur for long-term survival and achieving success [21]. Afsaneh and Zaidatol, of them, highlighted that personality factors which include self-efficacy and risk appetite and contextual factors like social media are highly relevant and significant for the sustainable business of women entrepreneurs and to make them successful [22]. Kulkarni and Srinivas have identified that nowadays the Govt. has increased its assistance to women entrepreneurs to make them socially and economically well accepted [23]. Though the growth and development of women entrepreneurs depend upon many socio-economic factors but Govt. policies play an important role and their development will no doubt raise the status of women socially and economically [23]. As per the united nations report (UNIDO) 2017, female entrepreneurs face many problems in getting loans to form financial institutions including Banks, even though their repayment rate is more than their counterparts [24]. This is either due to discriminatory attitudes or less faith in them for paying the debt. Women entrepreneurship highlighted that lack of knowledge for marketing the products, getting loans and subsidies, less awareness about Government assistance and lack of family support are important challenges faced by women entrepreneurs while pursuing and developing their businesses. Tlaiss talks about the obstacles women entrepreneurs are facing in many developing countries including India [25]. Women entrepreneurs lack to access finance from banks get proper relevant education and expertise, are overburdened with household activities, are incapable to get proper practical and self-employed training, attending meetings conducted by Govt. bodies etc. All these bottlenecks are limiting the capacity of women entrepreneurs to excel in their competencies in performing their responsibilities towards the development of business establishments.

3 Challenges and Economic Development

As per the survey conducted by the sixth economic Census [26] that only 13.76% of women are entrepreneurs out of the total entrepreneurs of India. Again 2.76 million work in agriculture and 65%, approximately 5.29 million, are workings in the non-agriculture sector out of total business establishments. Among the different states of India, Tamilnadu has the highest 13.5% followed by Kerala at 11.35% and Andhra Pradesh at 10.56%. While running their business women face many obstacles and challenges which hinder their growth and success. These problems may be concerning Finance, Marketing, Health, Entrepreneurial Attitude, Technology Access, Availability of workspace and boost of confidence. Rakesh [27], pointed out that for the growth of rural areas, there is significant growth of women entrepreneurs but their development is very low as they face many challenges and problems. Studies conducted by many researchers suggest that among many obstacles lack of education, knowledge and Skill, biased attitude, and poor infrastructure support, are the major barriers which restrict the growth and development of entrepreneurial activities of rural women entering into this segment. In some situations, it is also observed that psychological and physical pressure also acts against women entrepreneurs to proceed further.

Entrepreneurship is a human endeavour that significantly contributes to economic growth and one of the identified findings is that there is a strong correlation between entrepreneurship and of economic growth of a country. It's not hard to draw parallels between the role of entrepreneurs and the growth of economies. Generally, women enter into the entrepreneurship segment due to economic factors and the mindset to do something independently. The role of Women entrepreneurs in the economic development of any country needs to be considered for several reasons as reported by [28]. It has been observed that there is a positive connection between economic growth and women entrepreneurs as they help in accelerating growth through the creation of a job, increasing savings and thereby creating growth in the industry. Rural women entrepreneurs no doubt will contribute to the wealth improvement of their family and their nations [29]. For the improvement of the rural economy through rural development, rural entrepreneurship is very much needed and it works like a growth engine for the quality of life of rural people and their economic and sustainable development [30]. Women are crucial to the success of the household economy. Govt Schemes, policies, and programmes have made the growth and development of women entrepreneurs across the Tamil Nadu state and their economic and social status has improved after entering into this profession [31]. Women's attitudes have undergone dramatic shifts due to the technological and industrial revolutions in the realms of education and social and economic necessities. The last decade has seen a rise in women's access to economic opportunities. Women's participation in the workplace is a relatively recent phenomenon. Women's emergence as business owners is a major step toward their economic independence and the recognition and respect they deserve in all spheres of society. Women's involvement in business would bring a calming influence to today's troubled world, paving the way for more harmony and cooperation. Women in India are encouraged to pursue self-employment by both the government and private organisations in several different states, including Gujrat, Kerala, Punjab, Delhi, Karnataka, Andhra Pradesh, and Tamil Nadu. The growth of the women-owned business sector in the country is undoubtedly boosted. Women's entrepreneurship in India is typically associated with more traditional industries, such as the production and sale of papad, pickle, clothing production and so on. Manufacturing, exports, the service industry, businesses with high revenue generation, etc. are all areas where women are gradually being encouraged to start their businesses.

4 Objectives and Hypotheses

The study tried to pursue the following objectives.

- Identify the crucial factors/Key Competencies accountable for the expansion and magnification of women's entrepreneurial occupation.
- Explore the classification of women entrepreneurs.
- Examine the relationship between Challenges faced by Women entrepreneurs and Govt. Policies.

- Determine whether business networking influences the performance of women's entrepreneurial activities.

5 Hypothesis

- **H1$_0$**: There is no significant relationship between entrepreneurial activities and their contribution to the economical growth of women entrepreneurs.
- **H1$_a$**: There is a significant relationship between entrepreneurial activities and their contribution to the economical growth of women entrepreneurs.
- **H2$_0$**: There is no significant relationship between Govt. policy and the economic challenges of women entrepreneurs.
- **H2$_a$**: There is a significant relationship between Govt. policy and the economic challenges of women entrepreneurs.
- **H3$_0$**: There is no significant influence of business networking on the performance of women entrepreneurs.
- **H3$_a$**: There is a significant influence of business networking on the performance of women entrepreneurs.

6 Methodology and Analysis

The following conceptual models proposed here require first being operationalized so that they can be empirically tested. Therefore, a set of indicators is required to evaluate the conceptual model's frameworks. Care has been taken in selecting parameters for the overall satisfaction to fully exploit the theoretical constructions available. In the following paragraphs, researchers look at some of the possible applications of these frameworks (Fig. 1).

This study involves the collection of primary data through our survey and interviews and secondary data from books, Journals and different reports. The researcher surveyed with our designed questionnaire from the women entrepreneurs, from the Kandhamal districts of Odisha which were collected from the sample size of 148 respondents as per random sampling. Our questionnaire is designed with six major parameters of Entrepreneurial Competencies (i) Personal, (ii) Interpersonal, (iii) Conceptual, (iv) Business Management, (v) Technological, and (vi) Economic and Financial Competencies. Each parameter is having five questions on a five-point Likert Scale. After collecting data, the empirical research provided an overview of data analysis and classified the women entrepreneurs using Machine Learning (ML), and demonstrated the hypothesis through PCA and ANOVA. Statistical methods are the mathematical formulas, models, and techniques used to analyse raw study data statistically. The reliability of the findings can be evaluated with the help of statistical methods gleaned from survey data. Before conducting the pilot test and determining the sample size for our questionnaire, we performed a factor analysis to determine

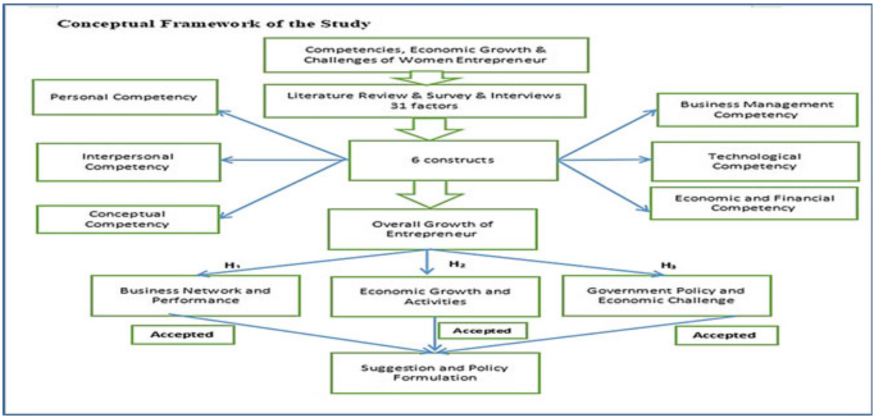

Fig. 1 Conceptual framework

the relative importance of the factors under consideration. Then we used analysis of variance (ANOVA) to assess the economical growth and challenges of the key competencies of women entrepreneurs.

6.1 Limitations of the Study

The investigation here is limited to the women entrepreneurs in the Kandhmala district of Odisha but similar studies can be generalised further by doing it in other districts of Odisha and India and also by taking the other key competencies into account also. Here stratified sampling is adopted so that an overall view can be drawn from the studies but other studies can apply different sampling to pursue their study.

7 Data Analysis and Proved Hypotheses

Part: I

As per our data collection through the questionnaire, we have used Machine Learning Technich (kNN classifier algorithm) to classify the women entrepreneurs as per their Economic and Financial Competencies and the results are bellowed (Fig. 2).

After classification analysis, we got the following performance vector results. accuracy is 86%, precision is 85.55%, recall is 87.21% and AUC is 86% (Fig. 3).

Fig. 2 Conceptual model of classification

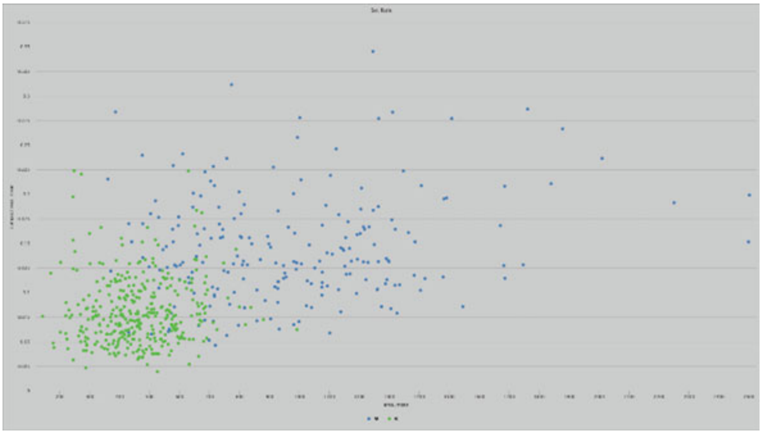

Fig. 3 Scatter plot of economical condition

Accuracy	Precision	Recall	AUC
86.26%	85.55%	87.21%	86%

We can observe the above scatter image of the lower economical condition (green colour) is more than the average economical condition of women entrepreneurs of Kandamal District.

Part: II

This empirical study explained the data analysis and proved the hypothesis in statistical analysis using PCA & ANOVA (Table 1).

After Principal Component Analysis (PCA), we can assess the validity of our data and the precision with which we measure it. Cronbach's alpha is 0.842, which indicates an extremely high degree of reliability of the data. All five questions are met by the eigenvalue Q1: 2.476, Q2: 1.112, Q3: 0.919, Q4: 0.873, and Q5: 0.889. So we can consider all questions for our analysis because all values are in the higher range and greater than 0.7 of the eigenvalue.

Table 1 Data reliability of hypothesis

Eigenvalue			
Questions	Eigenvalue	Mean	Std. Dev
Q1	2.476	1.268	1.518
Q2	1.112	2.185	1.678
Q3	0.919	3.241	1.62
Q4	0.873	1.173	1.051
Q5	0.889	4.081	1.462

Hypothesis H1:

See Figs. 4 and 5.

Table 2 shows the consistency of the eigenvalue, which is critical when verifying data. Each of the eigenvalues for each of the six criteria is larger than 0.7, which is a very good sign of the criteria's reliability. Since our data has high Cronbach's Alpha and eigenvalue reliability, we can conclude that our analysis is rigorous. Figure 3 provide clear depictions of the PCA scree plot and the component scores matrix, respectively.

Hypothesis H2:

After Principal Component Analysis (PCA), we can assess the validity of our data and the precision with which we measure it. Cronbach's alpha is 0.802, which indicates an extremely high degree of reliability of the data. All five questions are met by the eigenvalue Q1: 1.326, Q2: 2.128, Q3: 0.983, Q4: 0.975, and Q5: 1.299. So we can

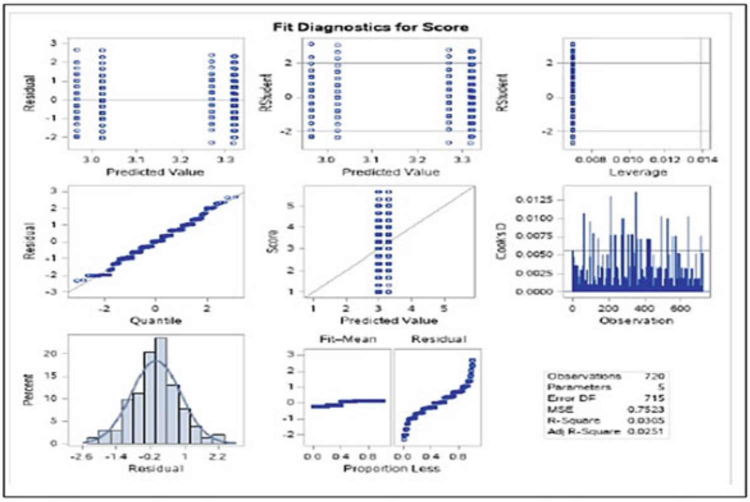

Fig. 4 Fit diagonestics of hypothesis H1

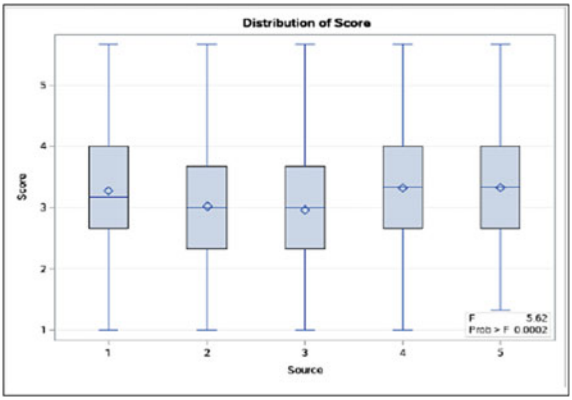

Fig. 5 Box plot of hypothesis H1

Table 2 ANOVA table of hypothesis H1

Source	DF	Sum of squares	Mean square	F value	Pr > F
Model	4	16.9114196	4.2278549	5.62	0.0002
Error	715	537.8649691	0.7522587		
Corrected total	719	554.7763889			

R-square	Coeff Var	Root MSE	Score mean
0.030483	27.28163	0.867326	3.179167

consider all questions for our analysis because all values are in the higher range and greater than 0.7 of the eigenvalue (Figs. 6, 7 and Table 3).

The above ANOVA is applied to test the hypothesis. The ANOVA results as given in Table 4, show that the calculated "F" value is (36.82), which is quite good and the "P" value is (0.0001), which is less than 0.05 of significance. Hence the null hypothesis $H2_0$: "*There is no significant relationship between Govt. policy and the economic challenges of women entrepreneurs*" is rejected, whereas the alternative hypothesis $H2_a$: "*There is a significant relationship between Govt. policy and the economic challenges of women entrepreneurs.*" is accepted. It is inferred that the Govt. Policy and schemes are most important to overcome the challenges faced by women entrepreneurs at different levels may it be Block, Panchayat, or District or State level.

Hypothesis H3:

After Principal Component Analysis (PCA), we can assess the validity of our data and the precision with which we measure it. Cronbach's alpha is 0.823, which indicates an extremely high degree of reliability of the data. All five questions are met by the eigenvalue Q1: 1.219, Q2: 2.273, Q3: 1.325, Q4: 1.649, and Q5: 2.173. So we can consider all questions for our analysis because all values are in the higher range and greater than 0.7 of the eigenvalue (Figs. 8, 9 and Table 5).

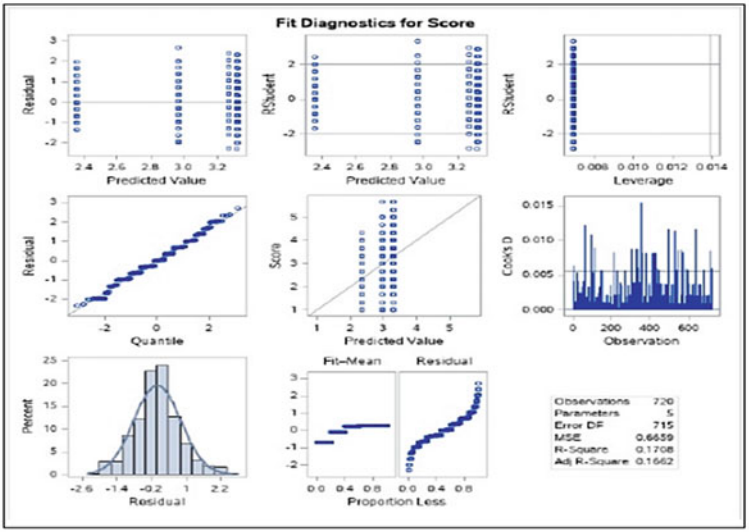

Fig. 6 Fit diagonestics of hypothesis H2

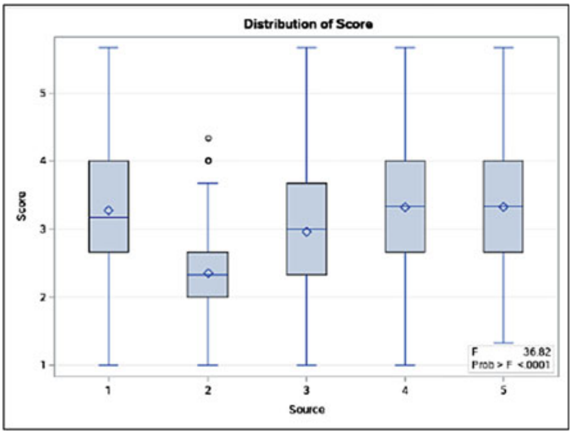

Fig. 7 Box plot of hypothesis H2

The above ANOVA is applied to test the hypothesis. The ANOVA results as given in Table 6, show that the calculated "F" value is (45.74), which is quite good and the "P" value is (0.0001), which is less than 0.05 of significance. Hence the null hypothesis $H3_0$: "*There is no significant influence of business networking on the performance of the women entrepreneurs*" is rejected, whereas the alternative hypothesis $H3_a$: "*There is a significant influence of business networking on the performance of the women entrepreneurs*" is accepted. It is inferred that business networking is an

Table 3 Data reliability of hypothesis H2

Eigenvalue			
Questions	Eigenvalue	Mean	Std. Dev
Q1	1.326	3.028	1.218
Q2	2.128	1.131	1.718
Q3	0.983	1.541	1.612
Q4	0.975	2.124	1.271
Q5	1.299	4.081	1.241

Table 4 ANOVA table of hypothesis H2

Source	DF	Sum of squares	Mean square	F value	Pr > F
Model	4	98.0669753	24.5167438	36.82	< 0.0001
Error	715	476.0871914	0.6658562		
Corrected total	719	574.1541667			
R-square	Coeff Var	Root MSE	Score mean		
0.203739	0.170803	26.79070	0.816000	3.045833	

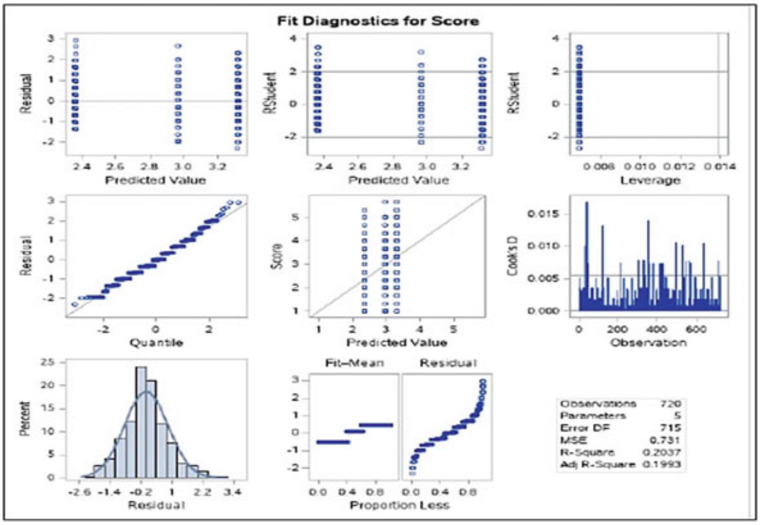

Fig. 8 Fit diagonestics of hypothesis H3

art form, and one should be adept at it as an entrepreneur. Business networking is all about bringing people together who can help each other's business activities and problems.

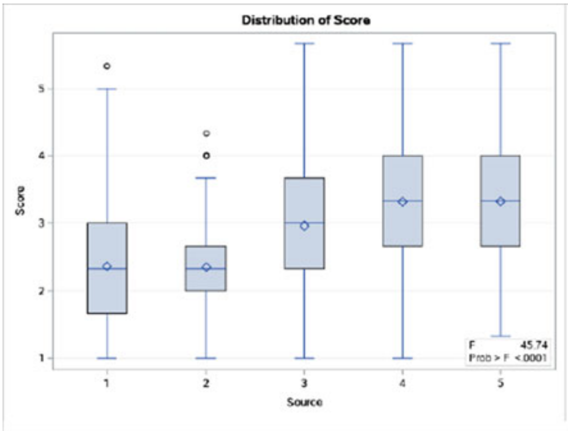

Fig. 9 Box plot of hypothesis H3

Table 5 Data reliability of hypothesis H3

Eigenvalue			
Questions	Eigenvalue	Mean	Std. Dev
Q1	1.219	3.218	1.118
Q2	2.273	2.132	1.653
Q3	1.325	1.441	1.204
Q4	1.649	2.024	1.381
Q5	2.173	4.691	1.412

Table 6 ANOVA table of hypothesis H3

Source	DF	Sum of squares	Mean square	F value	Pr > F
Model	4	133.7253086	33.4313272	45.74	< 0.001
Error	715	522.6296296	0.7309505		
Corrected total	719	656.3549383			

R-square	Coeff Var	Root MSE	Score mean
0.203739	29.83370	0.854956	2.865741

8 Findings and Discussion

This study tries to find out the key influential factors for the growth and success of women entrepreneurs and the challenges and hurdles they face during their entrepreneurship tenure. Again the study is interested to unveil the facts that starting their own business, influences the economic and financial condition of their family and country as a whole. There are many Govt. launched schemes which help them to face the challenges which generally restrict their entrepreneurial growth. The data analysis suggests that the key competencies which are responsible to make them successful and make them self-dependent are Personal Skills (1st), Economical & Financial (2nd), Business Management (3rd), Technological (4th) Interpersonal (5th) and Conceptual Competencies (6th) respectively. So it is well accepted that personal skills like Spotting Opportunities, Commitment to work, Motivation & Preservance, Self-efficiency and Self-confidence, Information seeking and Problem-Solving skills will make entrepreneurs more competent in doing their own business. The study also reveals that there is an influence of personal networking with the performance of the business owned by them and it has a significant impact on the success of entrepreneurial activities performed by women. It is inferred that business networking is an art form, and women should adopt it as an entrepreneur to survive and succeed in the long run. The networking will be beneficial to the rural women entrepreneurs in performing their enterprise-related activities like spotting business opportunities, getting moral support, enhancement of access to financial assistance, availing different Govt. schemes, acquiring technical support and understanding entrepreneurial orientation like creativity and innovativeness which will make them more proactive of success. Concerning the fourth objective to conclude the interconnection between different Govt. Schemes and challenges faced by women entrepreneurs, it is explored there is a significant influence of the former to the latter. That indicates that if women entrepreneurs will be aware of many schemes launched to grow their small businesses both by Central and state Govt. like Annapurna Scheme for catering, Stree Shakti for (EDP), Dena Shakti (Manufacturing), Orient Mahila Vikas Yojana (Small Business) then the impact of challenges to them will reduce to a great extent. Other than these schemes the Govt. has established many institutions to fulfil the needs of women business owners. The Platform for Women Entrepreneurship (WEP) is the WEP, which was launched by the NITI Aayog, and it is aspiring and established women entrepreneurs in India. It's broken down into three sections and each one has different objectives like Iccha, Gyana and Karma Sakti for encouraging, educating and providing hands-on assistance to women entrepreneurs to expand their businesses. We found that if women are aware of and access all these schemes and facilities provided by different levels of Govt. automatically the effect of challenges will come down and they can overcome them.

9 Conclusion

The study demonstrated that in rural areas like Kandhmala of Odisha, women enter into entrepreneurial activities by establishing their businesses due to many pull factors like exclusive accomplishments, responsibility towards family becoming self-sufficient and self-reliant and avail autonomy status. It is also noticeable from the study that a significant portion of women entrepreneurs became a success due to a combination of many competencies like self-confidence, strong willpower, determination, passion for their desired business and entrepreneurial spirit and tenacity. These competencies not only help them to become success also helps them to overcome the hard challenges during their entrepreneurship tenure. The findings of the study will enhance the personal and entrepreneurial skills of Women who have opted to take up business ownership status by providing new insight.

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Product Review and Recommendation

Diksha Bhave, Pendurkar Rohit, Rasal Mrugesh, and Singh Sumit

Abstract Websites that compare product reviews have emerged as an essential informational resource for customers trying to make informed judgments about their purchases. Through these platforms, customers may browse user reviews and ratings, search and compare products, and base their decisions on the expertise and experiences of other shoppers. Due to the significance of user-generated material in the purchasing process, there is an increasing need for sophisticated and user-friendly product review comparison websites. With technologies like artificial intelligence, social integration, and better verification procedures, a proposed system aims to improve the user experience in order to meet this demand. As a result, users will experience a platform that is more personalized, entertaining, and outfitted with cutting-edge technologies to deliver reliable information to customers. Product review comparison websites have the ability to play an even more significant part in the decision-making process by staying on the cutting edge of technical developments and consistently enhancing the user experience. This study presents an overview of websites that compare product reviews, the suggested system, and their influence on the choice to buy.

Keywords Artificial intelligence · Data analytics · Product reviews · Recommendation system · Recommendation engine · Product review comparison

1 Introduction

Recommender systems are tools for filtering and sorting things and data [1]. Time has become an important dimension to analyze in Recommender Systems [2]. Websites that compare product reviews have grown in importance as a tool for consumers trying to make educated judgments about their purchases. Users of these websites have access to a platform where they can look up and compare products, read user reviews

D. Bhave · P. Rohit (✉) · R. Mrugesh · S. Sumit
Department of Computer Engineering, Shivajirao S. Jondhale College of Engineering Dombivli (E), Dombivli, India
e-mail: pendurkarrohit@gmail.com

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and ratings, and base their judgments on the collective wisdom and experiences of other users. Generally lots of information and recommendations are pushed to buyers, but most of them are of not relevant [3]. As consumers rely on the experiences and opinions of others to inform their own decisions, user-generated information offered by these websites has emerged as a crucial component in the purchasing decision process.

The demand for more sophisticated and advanced platforms has grown as the use of product review comparison websites continues to soar. The suggested system incorporates social, augmented reality, and artificial intelligence aspects to improve the user experience. It also uses better verification procedures to guarantee that the data provided is reliable and correct. As a result, consumers trying to make wise purchasing decisions will have access to a tool that is more valuable, personalized, and entertaining [4]. Product review comparison websites have the ability to influence consumers' purchasing decisions even more in the future by utilizing the most recent technical developments.

2 Literature Review

The amount of information in the world is increasing far more quickly than our ability to process it [5]. Websites that provide information and analysis on various products, including their features, costs, and user evaluations, are referred to as "product review comparison websites." They are a resource that help customers compare various items based on many factors in order to make educated purchasing decisions.

2.1 Overview of Websites Comparing Product Reviews

Websites that compare product reviews have been increasingly popular in recent years as a result of the expansion of e-commerce and the ease of online shopping. They offer a variety of product-related details, such as specs, features, costs, and customer reviews, which are compiled from numerous sources. Websites that compare product reviews include some well-known ones like Amazon, Best Buy, and Consumer Reports. With the development of Internet-based commerce platforms, companies like Amazon and Walmart have set their place as two of the Forbes' largest companies in the world [6].

Benefits and Drawbacks of Product Review Comparison Websites: There are several benefits to using a product review comparison website, such as the simplicity of comparing products, easy access to a wealth of knowledge, and the chance to read user reviews and ratings. Additionally, they can offer consumers up-to-date data on prices and product availability in real-time. Websites that compare product reviews can have certain drawbacks, though, such as the possibility of biased or fraudulent reviews, the limitations of user reviews in capturing the genuine value of a product, and the possibility of out-of-date information. All the traditional recommendation algorithms are incapable and incomplete to the user's taste and preference [7]. Dealing with the customer feedback in text format, as unstructured data, is challenging [8].

2.2 *Prior Research and Conclusions on Websites that Compare Product Reviews*

Websites that compare product reviews can significantly influence consumers' purchasing decisions, according to a prior study. According to several research, these websites help consumers feel more confident while making purchases, which results in more informed and satisfying purchases. Other studies have demonstrated that the use of product review comparison websites can harm brand loyalty and undermine consumer confidence in conventional brick and mortar retailers. The goal is to predict unobserved preference of users for items [9].

In conclusion, in the digital age, websites that compare product reviews have become an invaluable resource for customers. They offer a wide variety of details on numerous products, making it simpler for customers to compare and decide what to buy. To make the best purchasing decisions, it is crucial to be aware of the limitations of these websites and to critically assess the information provided. To fully comprehend the effects of product review comparison websites on customer behaviour and the market as a whole, more research is required (Table 1).

From this table we can conclude that Consumers are often faced with the challenge of making informed purchasing decisions when shopping online. There are countless products available, each with its own unique features, benefits, and drawbacks. In addition, user reviews and ratings can vary widely, making it difficult to determine

Table 1 Literature review

References	Focus	Key findings
[10]	Product recommendation systems a comprehensive review	The recommender system will take the information and formulate the decision in one of the following two ways either by the use of collaborative filtering or by the use of content filtering [10]
[11]	Product recommendation using machine learning algorithm—a better approach	The k-mean is a method for grouping which is not parametric. According to their similarity, it distributes the objects into k clusters. In this article, using the Euclidean distance this similarity is calculated [11]
[12]	Product based recommendation system on amazon data	This system is not quite consumer friendly as it doesn't compare prices of the product from different websites [12]

(continued)

Table 1 (continued)

References	Focus	Key findings
[13]	Result review analysis of product recommendation system in domain sensitive manner	This system focuses a lot on user data and user history and also on the search pattern [13]
[14]	Comparing consumer produced product reviews across multiple website with sentiment classification	This system encourages consumers to post reviews of their purchased products, so that new consumers can evaluate these reviews for same product across different websites to help them make purchasing decisions [14]

the quality and suitability of a product. This can lead to poor purchasing decisions, resulting in wasted time and money.

3 Problem Statement

The problem today we face is the availability of vast amount of information available online, as well as the numerous websites and marketplaces that offer products for sale. Navigating this maze of information can be overwhelming, especially for consumers who are new to online shopping. Exponential growth in information has made it totally unimaginable to manually find a relevant product in a quick time, entailing the need for a mechanical recommendation system which would remember the users and recommend most suitable items [15].

The goal of the product review comparison website is to address this problem by providing consumers with a centralized resource for gathering information on products, user reviews, and ratings [5]. The website aims to make the process of shopping online more intuitive, efficient, and reliable, by providing users with a clear and concise source of information on the products they are interested in. This will enable consumers to make informed purchasing decisions with confidence, ensuring that they get the best value for their money.

4 Algorithm

Step 1: Start
Step 2: User visits the site's landing page
Step 3: Login or registration
Step 4: Home page
Step 5: Search product for Reviews and comparisons
Step 6: Search product by categories or by using search box

Step 7: Get Results/Desired product
Step 8: If found then fetch product details
Step 9: If not found then repeat step 5
Step 10: Comparison page
Step 11: Choose from multiple online stores
Step 12: Stop

5 Proposed System

Recommender systems are widely used for automatic personalization of information on web sites and information retrieval systems [16]. Among the most well-known e-commerce sites worldwide are Snapdeal, Amazon, and Flipkart. Web scraping is a technique used to collect data from these websites. In the process of web scraping, data is automatically extracted from websites and stored in an organized way, like a database. This enables the rapid gathering of enormous volumes of data from numerous websites.

A database, which acts as a central repository for the information, houses the data gathered from various websites. Large volumes of data can be structured and managed effectively in the database. This makes it possible for the website to swiftly obtain the needed information and provide it to the user in an organized and understandable manner.

The creation of the actual website is the process's last phase. The website is made to give consumers a platform where they can look for products, evaluate various possibilities, read user reviews and ratings, and decide which products to buy. The website makes advantage of the database's data to offer users the most recent details on a variety of products. The website offers a comprehensive and user-friendly platform for customers wishing to make informed purchasing decisions by utilizing the data acquired from Amazon, Flipkart, and Snapdeal.

Searching for and comparing products is one of the website's primary functions. To discover what they need, users can browse categories or conduct specialized product searches. After a product is chosen, the website offers comprehensive information about it, including characteristics, customer reviews, and ratings. Users may easily comprehend the advantages and disadvantages of a product since this information is presented in a clear and straightforward manner. The model evaluation relies on customer preferences and product requirements as well as feature ratings from the product experts [17, 18].

5.1 A General Overview of the Working of a Web Application Related to Product Review Comparison

User Login: Users would first need to create an account on the product review comparison website or sign in using their existing social media accounts.

Product Search: Users can search for products using keywords, filters for product features, prices, and user ratings.

Product Comparison: Users can compare different products based on various criteria, such as specifications, features, prices, and user ratings and reviews [19].

User Reviews and Ratings: Users can provide reviews and ratings for products they have used, and view reviews and ratings from other users.

Recommendation Engine: The recommendation engine would use machine learning algorithms and data analysis techniques to recommend products to users based on their search history, preferences, and purchasing behavior.

Product Purchasing: Users can purchase products directly from the product review comparison website or through a linked e-commerce platform.

Social Sharing: Users can share their reviews and recommendations with friends and family on social media, and view reviews and recommendations from their social network (Fig. 1).

The suggested system has the potential to completely alter how people find and buy things, making it a useful resource for years to come.

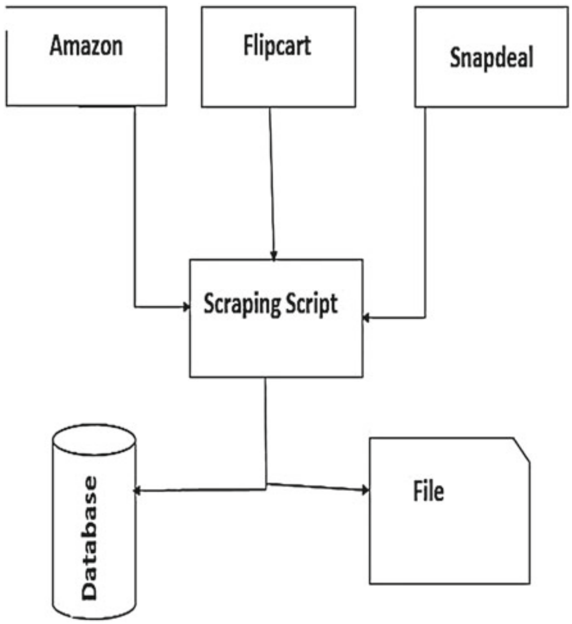


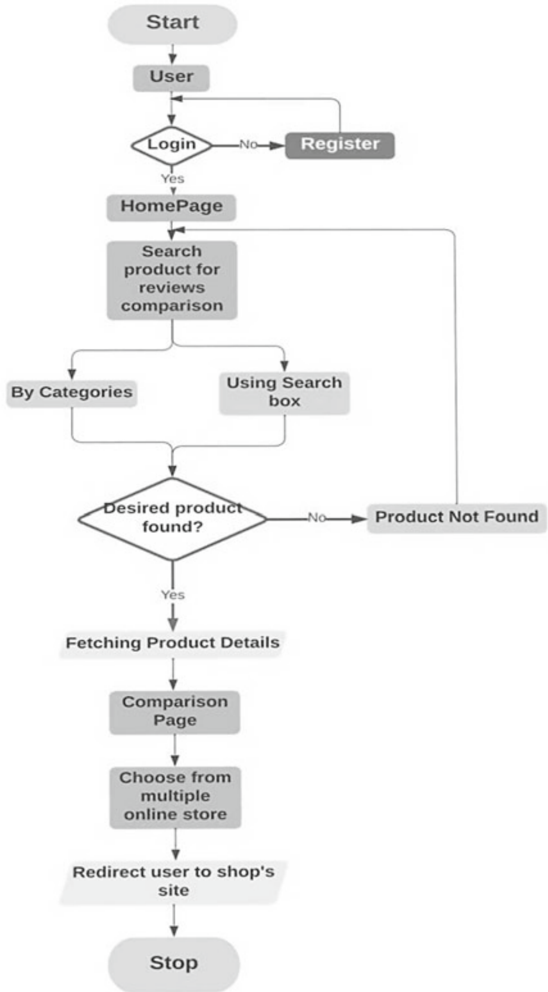
Fig. 1 Block diagram of the proposed system

6 Methodology

The methodology for developing the product review comparison website involves several key steps. These steps are designed to ensure the accuracy, reliability, and relevance of the information provided on the website (Fig. 2).

The first step in the methodology is data collection. This involves utilizing web scraping techniques to gather information from Amazon, Flipkart, and Snapdeal. The data collected includes product information, user reviews, and ratings. The data is stored in a database, which serves as the centralized repository for all information. This step is critical to the success of the website, as it provides the foundation for all other steps.

Fig. 2 Flow of the proposed system



The next step in the methodology is data cleaning and preprocessing. This involves the removal of any irrelevant or duplicate information, as well as the correction of any errors in the data. The data is then organized and structured in a format that is easy to use and understand. This step is important to ensure the accuracy and reliability of the information provided on the website.

6.1 Methodology for "Product Reviews Comparison Website" Can Be Divided into the Following Steps

Web Scraping: The first step involves collecting product reviews from popular e-commerce websites such as Amazon, Flipkart, and Snapdeal. This can be done using web scraping techniques that allow data to be extracted from websites automatically.

Database Creation: The next step involves storing the collected product reviews in a database. This database will serve as the source of data for the product reviews comparison website.

Product Reviews Comparison Website Development: The final step involves developing the product reviews comparison website. This website will allow users to compare product reviews from different e-commerce websites and make informed purchasing decisions.

In the development of the product reviews comparison website, the following technologies can be used.

Front-end development: HTML, CSS, JavaScript, and front-end frameworks such as React or Angular can be used to create the user interface.

Back-end development: A back-end technology such as Node.js or Ruby on Rails can be used to create the server-side logic.

Database management: A database management system such as MySQL or MongoDB can be used to manage the database.

Web scraping library: A web scraping library such as BeautifulSoup or Selenium can be used to extract data from the e-commerce websites.

The entire process of collecting product reviews from e-commerce websites, storing them in a database, and developing the product reviews comparison website requires a good understanding of web development, databases, and web scraping techniques.

The third step in the methodology is the development of the website itself. This involves the use of advanced web development technologies to create a platform that is user-friendly, intuitive, and provides users with the information they need to make informed purchasing decisions. The website is designed to be easily navigable, with clear and concise information on products, user reviews, and ratings.

These technologies are used to provide a more personalized and engaging experience for users, as well as improve the accuracy and reliability of the information provided. The use of these technologies ensures that the website remains relevant and up-to-date, providing users with a valuable resource for years to come.

7 Advantages

I. **Improved user engagement**: Customers will quickly take their money elsewhere if they don't get what they're looking for quickly. Pretty sure you don't want the same to happen to your store. Personalized recommendations help engage your site visitors more. That leads to more extended stays and, eventually, lower bounce rates.

II. **Improved time and cost efficiency**: With an AI-powered personalization tool [20], you can save time and money. It automatically undertakes product suggestions alongside cross and upselling approaches. Hence, it enhances efficiency gains that can save your operational costs in the long run.

III. **Improved customer loyalty**: Successful personalization systems can achieve 20% improvement in customer satisfaction. Satisfied customers are more likely to stick with your brand and even get more referrals. You can also use data insights from a product recommendation system to tailor your customer loyalty programs. It makes it easy to offer personalized rewards based on customers' wants and preferences. Hence, the rewards, whether discounts, coupons, free or subsidized shipping, etc., can resonate with each customer.

8 Results

See Figs. 3, 4, 5 and 6.

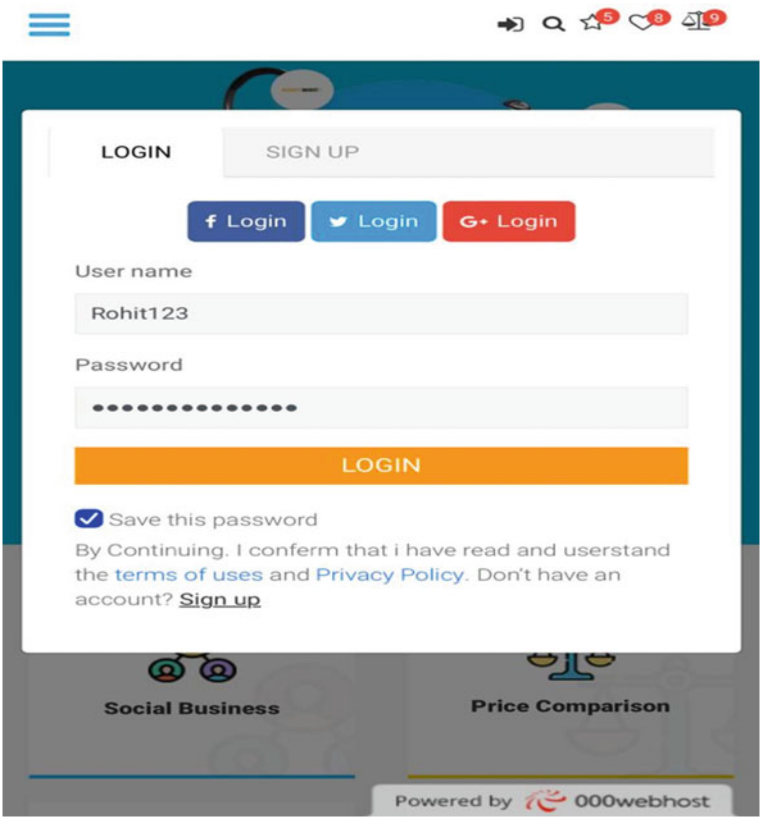


Fig. 3 Login page. This is where users can login into the website

9 Conclusion and Future Scope

Recommendation systems are achieving great success in e-Commerce applications, during a live interaction with a customer; recommendation system may apply different techniques to solve the problem of making a correct and relevant product recommendation [21]. Websites that allow users to search and compare products, read user reviews and ratings, and make informed judgments about which products to buy have become increasingly popular. This website have become a crucial resource for customers as the role of user reviews and ratings in the purchasing process increases. Through greater personalization, social features, and enhanced verification, the proposed system intends to improve user experience and give even

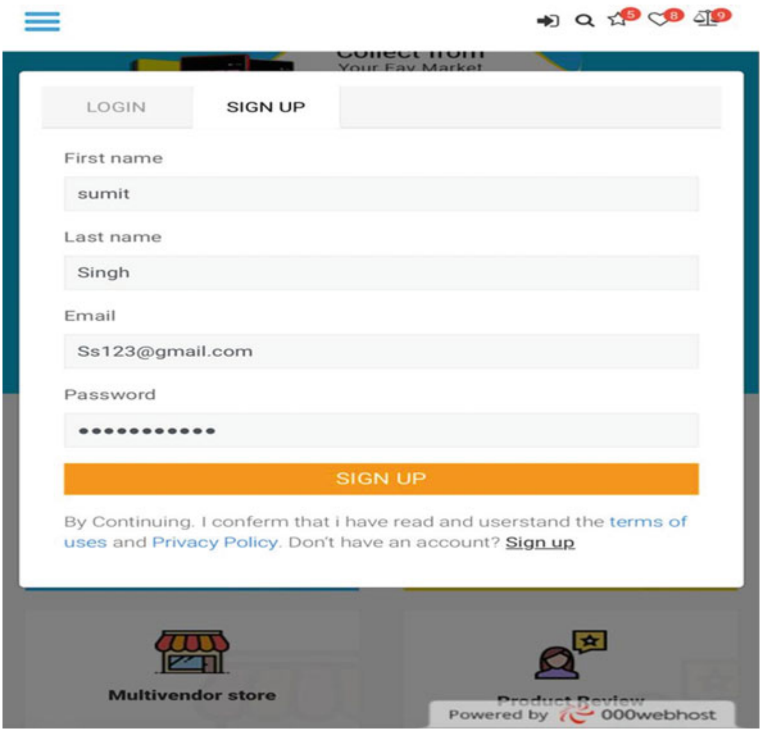


Fig. 4 Signup page. This is where users can create a new account on the website

more relevant and accurate information. Websites that compare product reviews have the potential to be even more important in the decision-making process if they continue to be at the forefront of technical development and continually enhance the user experience [9]. This website has a promising future ahead of it as it continues to assist customers in finding the products that best suit their requirements and tastes and assist them in making wise purchases.

Here are some probable directions in which the proposed system could go in the future:

Integration with AI-powered shopping assistants: Product review comparison websites may interface with AI-powered shopping assistants to offer consumers real-time assistance and direction while they browse and compare products [22, 23].

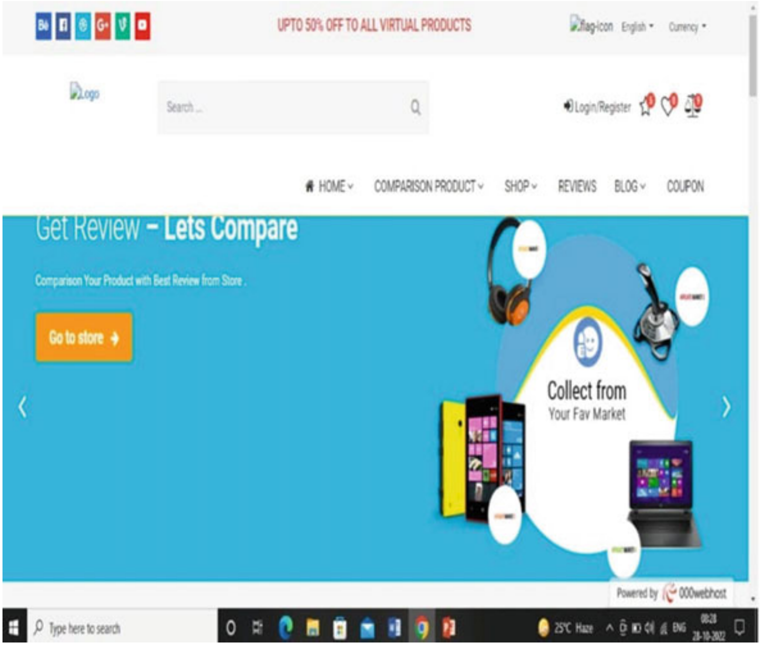


Fig. 5 Home page

Future advancements could improve the user experience and offer even more precise and relevant information to support consumers in making wise shopping decisions [24], by adding opinions of trending influencers and YouTube recommendations.

Deep learning based neural networks can be used as they have been attracting significant interest lately, due to their success in complex automatic recognition tasks in many artificial intelligence areas such as language recognition, computer vision and expert systems [25].

We can use a low-dimensional linear model to describe the user rating matrix in a recommendation system [26].

Other methods which can be used is the most popular recommendation algorithms latent factor models (LFM) [27].

Collaborative filtering recommender systems can also be used [28].

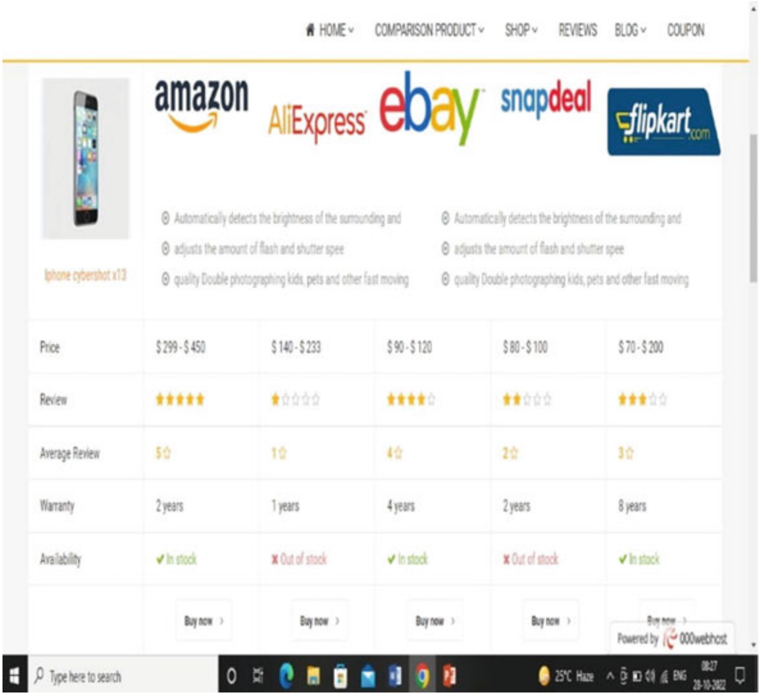


Fig. 6 Comparison page. This is where all the prices and ratings of the product can be compared

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Performance Evaluation of Lightweight ASCON-HASH Algorithm for IoT Devices

Rahul P. Neve and Rajesh Bansode

Abstract Lightweight cryptographic hashing algorithms are designed to be efficient and fast, making them suitable for use in resource-constrained milieus such as Internet of Things (IoT) devices, other embedded systems. These algorithms are optimized for limited computational power, memory, and bandwidth, while still providing a high level of security. Resources constrained devices typically have limited computational power, memory, and bandwidth, making it challenging to use traditional, computationally intensive cryptographic algorithms. Lightweight hashing algorithms are designed to address this challenge by providing a high level of security while being optimized for the limited resources of these devices. To verify the authenticity of messages and prevent tampering or replay attacks. Digital signatures: To provide non-repudiation and ensure that messages have not been altered in transit. Data integrity: To detect accidental or intentional modification of data. Key derivation: To generate cryptographic keys from a shared secret or password. This paper is focused on ASCON-Hash LWC, which his based-on sponge structure. Detailed study on hash function of ASCON is done and the algorithm is implemented on raspberry Pi to analyze the actual result. After performing experiments on the ASCON-Hash, it has been observed that there is possibility to optimize the algorithm by optimizing the permutation layer of hash function.

Keywords Cryptographic algorithm · Data integrity · Decryption · Encryption · Hash functions · IoT · Lightweight · Resource constrain devices

R. P. Neve (✉) · R. Bansode
Thakur College of Engineering and Technology, Mumbai University, Mumbai, India
e-mail: rahulneve@gmail.com

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1 Introduction

1.1 A Subsection Sample

There is extensive use of smart gadgets for variety of applications. Many smart gadgets come into resource constraint category, with limited processing capabilities in terms of memory, time and energy consumption. To maintain equilibrium of security triad (confidentiality, Integrity and availability) in resource constraint micro-controller like Raspberry Pi, Arduino Uno is matter of concern. For achieving confidentiality, encryption and decryption algorithms are used [1]. To achieve Integrity hashing concept needs to be implemented. Cryptographic hashing algorithms are mathematical functions that convert input data of arbitrary size into a fixed-size output, called a hash, such that for same input value the hash value will be same. Traditional cryptographic hashing algorithms, such as the widely used SHA-1 and SHA-2, have some potential drawbacks that can limit their effectiveness in certain scenarios:

- *Vulnerability to collision attacks*: Some traditional cryptographic hashing algorithms are vulnerable to collision attacks, which occur when two or more different input data produce the same hash value. This can lead to security vulnerabilities in systems that rely on the uniqueness of hash values, such as digital signatures and message authentication codes.
- *Limited message size*: Traditional cryptographic hashing algorithms often have a limited message size that can be hashed in a single operation. This can be a limitation in certain scenarios, such as when hashing large files or streams of data.
- *No privacy protection*: Traditional cryptographic hashing algorithms do not provide privacy protection for metadata associated with the hashed data, such as the length of the message or the identity of the sender and receiver.

Lightweight cryptographic hashing, also known as lightweight hash functions, refers to a class of hash functions that are designed to be implemented efficiently on resource-constrained devices, such as low-power microcontrollers or RFID tags. These hash functions are typically designed to have a small code and data size, low power consumption, and high processing speed, while still providing a level of security that is appropriate for the specific application.

Some examples of lightweight cryptographic hash functions include PHOTON, SPONGENT, and LESAMNTA, which are designed to provide cryptographic security while also being efficient in terms of memory and processing power [2].

Lightweight cryptographic hash functions are important for securing the growing number of IoT kind devices, which often have limited resources and are vulnerable to attacks due to their connectivity to the internet. These hash algorithms can be used for tasks such as message verification and integrity checking, which are important for ensuring the security of data transmitted between IoT devices [3]. It's important to note that while lightweight cryptographic hash functions can provide a level of security that is appropriate for many IoT applications, they may have limitations in terms of the level of security they can provide compared to more complex hash functions. Therefore, it's important to carefully evaluate the specific security requirements of the application and choose a hash function that is appropriate for those requirements [4].

2 Difference Between Traditional Hashing and Lightweight Hashing

2.1 Design

Traditional cryptographic hash function like SHA-3, is developed to be highly secure and resistant to a wide range of attacks, such as collision attacks, first preimage attacks: in which an attacker tries to find any input message that will produce a given hash value, and second preimage attacks: in which an attacker tries to find a second input message that produces the same hash value as a known input message. On the other hand, lightweight cryptographic hash functions are designed to be efficient and fast on low-power devices such as microcontrollers and embedded systems. These hash functions typically use simpler designs and algorithms that are optimized for performance but may not provide the security as traditional hash algorithms [5].

2.2 Security

Cryptographic hash functions are algorithms that are specifically designed to offer a high level of security, making them suitable for use in applications where data integrity and authenticity are of utmost importance. These applications include digital signatures and password storage, among others. To ensure their security, these hash functions are built to be resistant to a wide range of attacks and undergo rigorous security analysis and testing to verify their effectiveness.

2.3 Application

Cryptographic hash algorithms are commonly utilized by various of applications, such as encryption, authentication, and digital signatures. These functions play a crucial role in ensuring data security and integrity in various systems and processes that require secure communication and verification of data authenticity. These hash functions are used in high-security applications where data integrity and authenticity are critical. Lightweight cryptographic hash functions are often used in low-power devices and embedded systems, where efficiency and speed are more important than absolute security. These hash functions are commonly used in applications such as sensor networks, RFID tags, and other devices where memory and processing power are limited.

3 Type of Cryptographic HASHING Functions

There are three major categories of lightweight cryptographic hash function. Markel Damgard, block cipher and Sponge construction. These constructions also include substitution permutation network, Fiestel structure with arithmetic modular addition, circular fashion bit shift, exclusive or operations. Below these constructions are described in brief.

3.1 Markel Damgard Hash Function

Merkle–Damgård is a widely used construction for constructing cryptographic hash functions. It is named after Ralph Merkle and Ivan Damgard, who independently developed the idea in the late 1970s. This construction is built around a compression function that takes input of a fixed size and produces an output of a fixed size. The idea is to repeatedly apply the compression function to blocks of the input message until the entire message has been processed. The output from the final compression operation is the resulting hash value for the input message [6]. The compression function is used to process blocks of the input message and generate an intermediate hash value. The intermediate hash value is then used as the input to the compression function for the next block, until all blocks have been processed [7].

The Merkle–Damgård construction consists of three phases: padding, compression, and output.

Padding: The input message is padded so that its length is a multiple of the block size of the compression function. The padding is done in such a way that it is distinguishable from the original message, to ensure that the padding cannot be used to create collisions.

Compression: The compression function is applied to each block of the padded message, using the intermediate hash value from the previous block as an input. The output of the compression function is the intermediate hash value for the current block.

Output: Once all blocks have been processed, the final intermediate hash value is the output of the hash function.

3.2 Block Cipher Hashing Construction

The Lightweight (LW) block cipher is designed for resource-constrained environments such as low-power devices. One common use of the LW block cipher is as a component of hash functions [8].

LW block cipher-based hash algorithms use the following construction:

1. The input message is padded to a multiple of the block size of the LW block cipher.
2. The padded message is divided into blocks of the block size.
3. An initial value is chosen and passed through the LW block encryption to produce the initial state.
4. For each block of input data, the current state is passed through the LW block cipher, along with the block of input data, to produce a new state.
5. After all the blocks have been processed, the final state is the hash value.

The LW block cipher consists of a number of rounds, each of which applies a non-linear function to the state, followed by a linear mixing operation.

The LW block cipher is designed to have a low gate count, low power consumption, and a small code size. It is also designed to be resistant to various attacks, such as differential and linear cryptanalysis [9, 10].

3.3 Sponge Construction

Sponge functions are cryptographic primitives that can be used for various purposes such as message authentication, encryption, and hashing. In particular, the sponge construction can be used to build a hash function [11, 12].

It operates as follows:

1. The input data is chopped into fixed length bits, it is multiple of internal state size of the function.
2. The sponge function is initialized with an initial state and a fixed output size.
3. The input blocks are fed into the sponge function, which updates its internal state by absorbing the input.
4. The output is truncated to the desired length to obtain the final hash value.

ASCON-hash is a family of sponge functions that can generate hash values with a wide range of output sizes. It has a number of desirable properties, including resistance to various types of attacks and efficient implementation on a wide range of platforms [13].

Overall, the sponge construction provides a flexible and efficient framework for constructing cryptographic hash functions that can be tailored to specific requirements. Lightweight cryptography emphasizes the need for efficient and resource-constrained implementations, making sponge-based hash functions a promising choice for such applications [14].

4 Lightweight ASCON-HASH Algorithm

This paper is based on the detail study and implementation of ASCON-HASH algorithm on raspberry pi (IoT controller) ASCON is a family of lightweight authenticated encryption algorithms that provide both confidentiality and authenticity of messages. In addition to the encryption and decryption functions, ASCON also includes a hash function that can be used to securely derive message keys and other cryptographic parameters [15].

The algorithm has been thoroughly analyzed and is considered secure against a wide range of attacks.

The ASCON-HASH algorithm operates as follows [16].

a. The input information is divided into blocks of a fixed size (either 128 or 320 bits, depending on the security level desired).
b. The sponge function is initialized with an initial state and a fixed output size of 256 bits.
c. The input blocks are fed into the sponge function, which updates its internal state by absorbing the input.
d. After processing all input blocks squeezing phase get started, during which it generates the hash output by repeatedly applying the ASCON permutation and outputting a portion of the internal state.
e. The output is truncated to the desired length to obtain the final hash value.

ASCON – HASH Algorithm:

Plain Text P.T as input ,output Hash Message size $l < h$ or l arbitrary if $h=0$ and produce output: *Hash with fixed length*

Initialization:

$S \leftarrow p^a(IV_{h,r,a} \| 0^c)$

Absorption:

$P.T_1 \ldots\ldots P.T_s \leftarrow P.T \,\|1\|\, 0^*$

For each character of P.T

$S \leftarrow p^a((S_r \oplus P.T_i) \,\|S_c)$

Squeezing

for i=1,....., t=(l/r) do

$H_i \leftarrow S_r$

$S \leftarrow p^a(S)$

return $|H_1\| \ldots . \,||H_t\|_l$

Where

IV : Initial Vector
a : number of rounds
p : Permutation
M : Plain text or Message
c : constant
H : Hash output at each phase
r : rate
l : length (256)
S : State

Initial State: The next step is to set the initial state of the ASCON permutation. This state consists of a 128-bit state vector and a 64-bit associated data (AD) length. The state value is set to be fixed and the AD length is set to zero.

Absorb Phase: The absorb phase is used to process the input message. The message is chopped into blocks and each block is XORed with the state vector. Then, the ASCON permutation is applied to the state vector to mix in the message block [17].

Padding: If the input message is not a multiple of the block size, padding is added to the last block to make it a full block. The padding consists of a 1-bit followed by zeros and is XORed with the state vector.

The squeezing stage in ASCON-HASH consists of the following steps:

Squeeze Phase: The squeeze phase is used to extract the hash output from the state vector. This is done by repeatedly applying the ASCON permutation to the state vector and outputting a block of the specified hash size at each iteration. The size of each block is equal to the output size of the hash function, which is typically 256 bits [18].

Finalization: Once the specified number of blocks has been output, the final state vector is XORed with the key schedule to prevent multi-target attacks, and the final output is produced. The final output is equal to the concatenation of all the output blocks produced during the squeeze phase. The number of blocks output during the squeeze phase is determined by the size of the hash output and the block size of the ASCON permutation. For example, if the hash output is 256 bits and the block size is 128 bits, then two blocks will be output during the squeeze phase [19]. The squeezing stage ensures that the final hash output is a fixed-size value that is unique to the input message. By repeatedly applying the ASCON permutation and outputting blocks of the specified size, the squeezing stage produces a hash output that is resistant to collision attacks and other types of attacks [20].

Substitution Layer: The substitution layer of ASCON-HASH is designed to provide diffusion and confusion properties to the hash function. The nonlinear substitution operation and the linear diffusion operation work together to ensure that any small change in the input to the function results in a large change in the output, making the function resistant to attacks [21].

Linear diffusion: The state is transformed using a linear matrix multiplication operation. The matrix used in ASCON-HASH is designed to provide maximum diffusion in a small number of rounds. The linear diffusion layer operates on the state in a way that ensures that small changes in the input propagate uniformly to all bits of the state. Linear diffusion layer of ASCON-HASH consists of a matrix multiplication operation between the state and a fixed 5×5 binary matrix, denoted by M. The matrix M is carefully chosen to provide a good diffusion property while maintaining a small implementation footprint. The linear diffusion layer is applied to the state of the permutation in each round of the hash function, except for the first and last rounds, which do not have a diffusion layer. The number of rounds used in the hash function depends on the desired security level and the input size of the hash function. Overall, the linear diffusion layer of ASCON-HASH plays an important role in ensuring the security of the hash function, by providing a good diffusion property that ensures that any small change in the input results in a large change in the output [22].

5 Experimental Results and Observations

Implementation of ASCON-HASH algorithm was carried out using python language on R-Pi 3 model b controller. It consists of 1 GB of Random-access memory, 64-bit CPU and 1.2 GHz processing power upgraded switched Micro USB power source up to 2 mA with V_{cc} 5 V. Input as text file with varying size (i.e. 100, 200, 300, 400, 500 kb) was provided to the algorithm and results are observed (Table 1).

Figure 1 shows the linear shift in the graph of time in secs verses input file size (i.e. if input file size increases, in that case execution time of the algorithm increases in a linear fashion. In same way Fig. 2. Shows that memory consumption is increasingly steadily with increase in input file size. As energy consumption parameter is directly proportional to time, hence as execution time increases, energy consumption is also increased and the same is depicted in Fig. 3.

Table 1 Time, memory, energy and data processing rate for ASCON-HASH algorithm

File size (text file) in kb	Hashing time in seconds	Memory consumption in kb	Energy in mJ	Data process in kb/s
100	21.88	220	218	46.80073
200	45.06	380	450	45.45051
300	66	428	661	46.54545
400	87.4	528	874	46.31579
500	109.6	628	109.6	46.71533

Inference of above results:
a. As the file size increase gradually rates of time and Memory consumption increase in linear way
b. Average throughput for ASCON algorithm is 46 kb/s

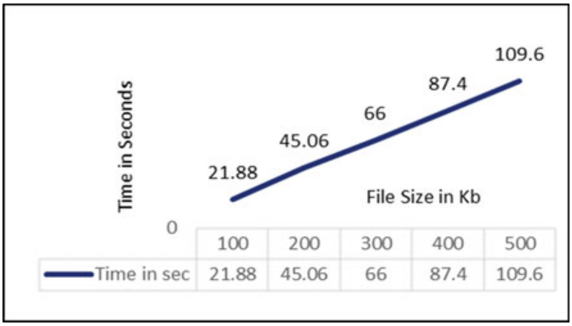

Fig. 1 Hash algorithm time consumption in seconds

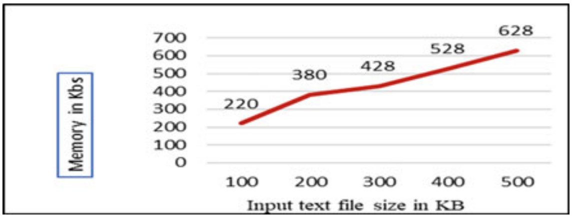

Fig. 2 Hash Algo memory consumption in KB

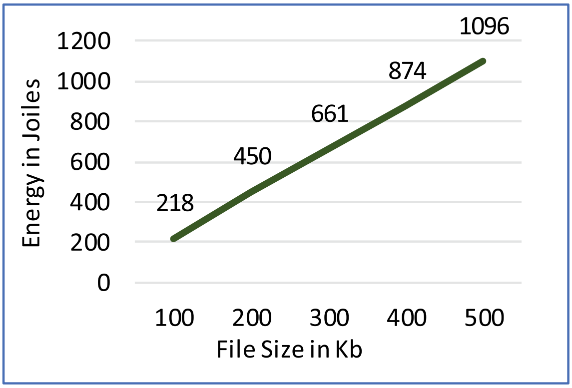

Fig. 3 Energy consumption by raspberry pi 3

6 Discussion on Achieved Output

See Table 2.

Table 2 Technical parameter measured, outcome, inference, and impact

Technical parameter measured (for 100 kb of text file)	Output achieved	Outcome as inference of output	Impact
Hashing time Time in sec	21.88 s	Time required to create hash value of plain text calculated using time.start and time.stop function and the difference is noted	As ASCON-Hash function consumes 21.88 s, it will effect energy consumption permanently
Memory consumption in kb	220 kb	For memory consumption, psutil library with memory.rss function is used to calculate memory consumption during Hashing process	Gradually increase in input file size will linearly increases in memory consumption with the average rate of 102 bytes

(continued)

Table 2 (continued)

Technical parameter measured (for 100 kb of text file)	Output achieved	Outcome as inference of output	Impact
Energy in Joules	218 mJ	Energy is calculated as $E = V_{cc}*I*t$ whereas V_{cc} is 5 V, I is current i.e. 2 mA and t is time taken for execution of decryption function for 100 kb text file	Energy parameter is directly proportional to the time required for data processing and execution of hash function. Hence energy consumption increases as time for execution increases
Bit rate/byte rate	46 kb/s	Total size of data processed per unit of time. Bit rate of 46 kb/s is observed after execution of ASCON algorithm for text file data	It is observed that average byte rate is 46 kb/s after providing the various input data, which shows that constant byte rate is maintained for the system

7 Conclusion

As the file size increases, the hashing time, memory consumption, and energy consumption also increase, while the data process rate remains somewhat constant. The ASCON-HASH algorithm seems to be performing consistently, as the data processing rate (in kb/s) is similar across different file sizes. This indicates that the algorithm can handle different file sizes with a relatively constant processing speed.

One of the main advantages of ASCON-HASH is its simplicity, which makes it easy to implement and verify. It has a small code size and memory footprint, making it an attractive option for use in low-power devices. It also offers a high degree of security, with a collision resistance of 2^{128} and a preimage resistance of 2^{256}. ASCON-HASH is based on the sponge construction. The algorithm has a flexible parameter set that allows users to adjust the security level and performance to meet their specific needs.

Overall, ASCON-HASH is a strong and efficient hash function that can be used in a wide range of applications, particularly those that require lightweight cryptography. However, as with any cryptographic algorithm, it is important to use it correctly and to ensure that it is implemented securely to avoid any potential vulnerabilities.

ASCON-hash can be optimized for better efficiency by optimizing substitution, diffusion, permutation layer of algorithm, but care should be taken to maintain security level of algorithm.

8 Future Scope

With the increasing adoption of IoT devices, there is a growing need for lightweight cryptographic primitives that can provide security without consuming too much power or code space. The ASCON-HASH algorithm is expected to find widespread adoption in the IoT ecosystem, where it can be used to provide data integrity and authentication in a lightweight and efficient manner.

Another research area could be the investigation of the ASCON-HASH algorithm's performance on different hardware platforms, such as CPUs, GPUs, and FPGAs. This analysis would help to optimize the algorithm's performance for specific hardware and identify any potential bottlenecks.

As with any cryptographic primitive, the ASCON-HASH algorithm will be subject to ongoing research and analysis, with the goal of improving its security, performance, and resistance to attacks. In the future, the algorithm may be further refined, or new variants may be developed that are optimized for specific use cases or platforms.

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Design of Virtual-Real Software-Defined Network for Cloud

J. Divya Lakshmi and P. Y. Mallikarjun

Abstract In the construction of network simulation experiment scenarios, complex physical terminals are challenging to realize virtual simulation, and physical terminals need to be accessed. However, the existing access methods are difficult to deploy, have bottlenecks in network performance, and cannot be applied to large-scale network scenarios. To effectively solve the above problems, a virtual-real fusion network simulation construction method based on SDN (software defined-network) is proposed, using the SDN controller combined with the flow table construction algorithm to realize the link management and data connection of the virtual and real network and design A prototype system for virtual-real fusion network simulation has been developed. Through the SDN controller, the virtual instance in the cloud platform and the physical terminal outside the cloud can be jointly networked to build a virtual-real fusion network simulation experiment scene. Using this prototype system, experiments have proved that the SDN-based network simulation method can realize efficient large-scale networking of virtual instances and physical terminals in the cloud platform and has good network performance.

Keywords Cloud computing · Software defined-network · Virtual-real fusion network · Large-scale networking · Data connection

1 Introduction

The network simulation experimental field has become an essential scientific device in the field of network security, which can provide support means for the observation, measurement, and analysis of network space behavior in a safe and controllable

J. D. Lakshmi (✉)
Department of Electronics and Communication Engineering, Sri Krishna Institute of Technology, Bengaluru, India
e-mail: divyalakshmiece@skit.org.in

P. Y. Mallikarjun
Department of Electronics and Communication Engineering, Dayananda Sagar Academy of Technology and Management, Bengaluru, India

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environment and provide support for the testing and evaluation of network attack and defense behavior [1, 2], network security tools. Flexible and efficient hardware and software infrastructure is provided for professional testing and verification, and technical and tactical training drills for professionals. Currently, the mainstream network simulation experiment field mainly relies on digital simulation, virtualization, and other technologies to build simulation entities in virtual environments [3]. There is no virtual simulation model or image, and some complex terminals (such as various secret-related target information systems and customized special terminals) are challenging to build virtual simulations. Therefore, it is urgent to integrate physical terminals with virtual networks. Flexible integration jointly creates a virtual and real fusion network simulation experiment field and provides a highly realistic scene environment for network attack and defense drills [4–6].

Software-defined networking is a network technology that has emerged in recent years. It provides network devices with a forwarding layer based on hardware and a control plane based on software control by separating the control plane and the data forwarding plane [7–10]. The SDN controller realizes the power and management of the global network. Given the SDN separation of management and control characteristics, the network has better flexibility and scalability.

This article proposes a method for constructing a virtual-real fusion network simulation experiment field based on SDN. The core idea is to realize the joint networking of virtual instances and physical terminals in creating network simulation scenarios so that the constructed network simulation scenarios are more comprehensive, realistic, and better [11–13]. Experiments have found that the SDN-based network simulation experimental field construction method can quickly realize the construction of virtual-real fusion scenarios. Compared with traditional construction methods, it can meet the needs of large-scale complex network scene generation and effectively avoid network node link congestion and other phenomena [14]. Connecting physical terminals to the virtual network environment through SDN can effectively reduce the construction cost of complex networks; realize the flexible expansion of network scale, and natural and effective network simulation [15–18].

The main contributions of the article include three aspects:

(1) Based on SDN, a method for constructing a virtual-real fusion network simulation experiment field is proposed. A prototype system for virtual-real fusion network simulation is designed and developed.
(2) Based on the Ryu controller, a flow table generation algorithm is proposed to realize the interconnection between virtual and real devices.
(3) Through experiments, it is proved that the construction method of the SDN-based virtual-real fusion network simulation experiment field and the flow table generation algorithm are practical and feasible.

2 SDN-Based Virtual and Actual Network Architecture Design

It mainly introduces the construction of a virtual-real fusion network simulation experiment field based on SDN software and hardware switches. This design relies on the overlay network and realizes the access of physical terminals by building vxlan tunnels. The Vxlan network has similar functions to the vlan network and can provide Layer 2 Ethernet services, but vxlan has better flexibility and scalability than vlan. It uses a 24-bit tag vxlan_id to support more Layer 2. The network has reached more than 16 million layer-2 network segments, so it can meet the needs of building network scenarios in large-scale and complex cloud platforms [19, 20].

2.1 Trend of Virtual Instance Data in the Cloud Platform

In the OVS cloud platform, the data packet transmission of the virtual instance will pass through a series of virtual bridge devices. The following is a detailed description of the data packet direction of the virtual model in the computing node [21–25].

As shown in Fig. 1, the virtual instance in the cloud platform transmits the data packet through the br-int bridge of OVS to the br-tun bridge through the virtual network card and other devices. When the data packet reaches the br-tun bridge of OVS, Forwarding is performed according to the multi-level OpenFlow flow table rules in the bridge, and the flow table forwarding rules are shown in Fig. 2.

The functions of each OpenFlow flow table in Fig. 2 are as follows:

table0: Do splitting processing, according to the source of the message, is the br-int bridge inside the cloud platform or outside the cloud platform (the data sent from the outside means that the data is transmitted from other nodes or devices to this node, not through the virtual instance in this node. The data packets transmitted by the br-int bridge) are forwarded to different flow tables for subsequent processing. The data packets sent by the internal br-int bridge are forwarded to Table 2 for processing. The data packets sent from the outside. It is delivered to table 4 for processing.

table2: For the data packets that the virtual instance is going to send out, if it is a unicast packet, it will be transferred to table20; if it is a multicast packet, it will be transferred to table22 for processing;

table3: directly discard the data packets that meet the matching conditions of the flow table;

table4: used to process the data packets sent from the outside, specifically strip the tun-id of vxlan, add vlan tag, realize the mapping conversion between vxlan and vlan inside the cloud platform, and transfer it to table10 for processing;

table10: This flow table is a self-learning flow table, which realizes the learning of the mac address table in the data message, and puts the learned content into table20;

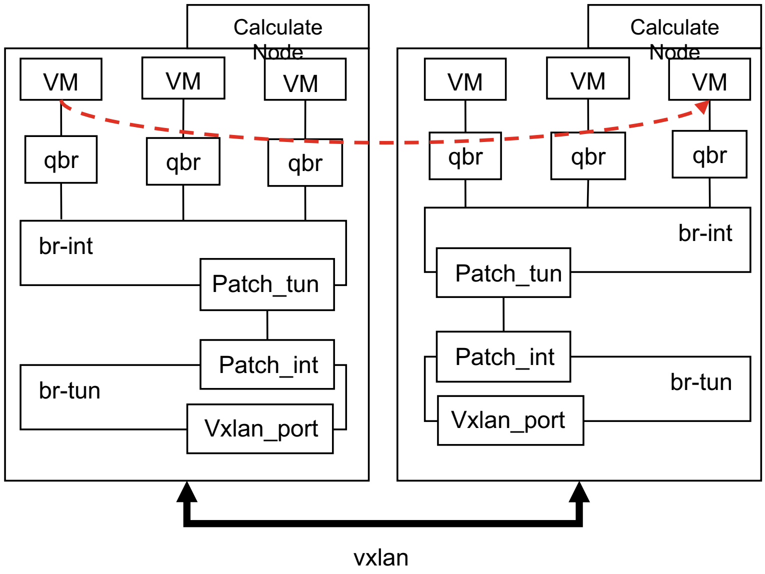

Fig. 1 Example data trend

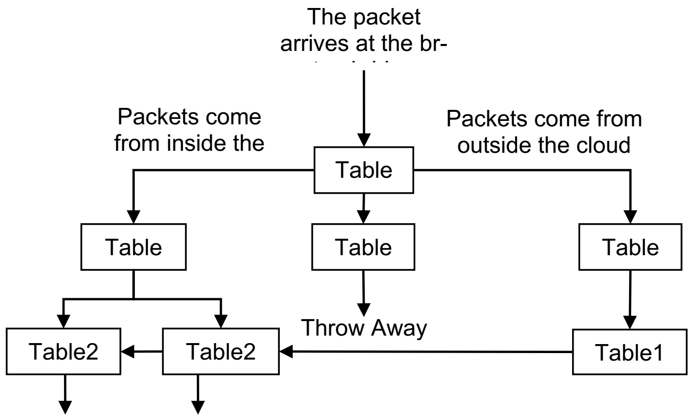

Fig. 2 The direction between flow tables

table20: According to the matching field of the message (vlan, destination mac, etc.), strip the vlan_id, add the tun-id of vxlan, and forward it from the specified vxlan port. If the corresponding match cannot be achieved, go to table22 for processing;

table22: Also according to the matching field of the message (vlan, destination mac, etc.), the vlan_id is stripped, the tun-id of vxlan is added, and then flood broadcast is performed to send the data message to all vxlan tunnels to realize the data

message transmission between different nodes in the cloud platform. Through the above forwarding rules of the OpenFlow flow table, mutual communication among the virtual instances in the cloud platform is realized.

3 Key Technologies of SDN-Based Virtual-Real Fusion Network Simulation

3.1 Construct Vxlan Tunnel

The Vxlan tunnel [26–28] is the bridge of the entire virtual-real fusion system. The controller has complete control over the physical SDN switch, and through the vxlan tunnel, the link between the physical SDN switch and the brut tunnel bridge in each computing node OVS of the cloud platform is opened, which is also realized. The prerequisite for the connection and interaction between the cloud platform and the physical terminal. The specific implementation steps are as follows:

- Step 1 Add corresponding vtep_port port pairs on the br-tun tunnel bridge and the physical SDN switch of the OVS virtual switch of each computing node on the cloud platform.
- Step 2 Through OVS-related commands, use the vtep_port port pair in step 1 to create a vxlan tunnel to realize the connection between the physical SDN switch and the br-tun tunnel bridge in the OVS of each computing node on the cloud platform and then realize the virtual instance in the cloud platform Link connectivity with entity terminals.
- Step 3 Add a daemon process to the br-tun tunnel bridge of each computing node on the cloud platform, and add the newly added vtep_port information into the flow tables at all levels of the br-tun tunnel bridge.

4 Experiments and Results

The proposed virtual-real fusion solution's performance will be tested through many simulation experiments, and the solution will be evaluated in terms of connectivity, isolation, packet-sending delay, and throughput.

4.1 Correctness Test of Flow Table Construction Results

According to the virtual-real fusion network simulation [29, 30], the flow table automatic configuration code is designed based on the SDN controller flow table generation algorithm. First, the physical terminal node is pinged from the virtual

Table 1 Connectivity test results of some instances

Source host	Destination host	Is it connected?
192.168.16.197	VM1	Yes
192.168.16.198	VM2	Yes
192.168.16.199	VM3	Yes
VM4	192.168.16.197	Yes
VM5	192.168.16.198	Yes
VM6	192.168.16.199	Yes

instance node, and the Ryu controller will automatically construct and issue flow table rules. After the virtual and real nodes are successfully pinged, log in to the physical SDN switch to view the configuration of the switch flow table information; on the computing node server where the virtual instance node is located, use the command to view the flow table rules, as shown in Table 1.

From the above flow table information, it can be seen that through the designed flow table structure and distribution algorithm, the generated flow table can enable the virtual instance and the physical terminal to realize the seamless intercommunication of virtual and actual links.

4.2 Network Isolation Test for Virtual Reality Convergence

In actual network experiments, there are often multiple virtual and real network scenarios with the same network configuration parameters belonging to different vxlan networks. Still, in these different scenarios, the IPs of the virtual instances belong to the same network segment. Combined with the network creates a virtual network of 192.168.16.0/24 under different vxlans and generates multiple virtual instances with IP segments of 192.168.16.0/24. The vxlan_ids of the two networks are vxlan100 and vxlan200, respectively, and the IP devices of physical terminals 1 and 2 are 192.168.16.199 and 192.168.16.200, respectively. Physical terminals 1 and 2 are isolated through the flow table to realize the network interconnection between physical terminal 1 and vxlan100 intercommunication, the physical terminal 2 and vxlan200 network interconnection and intercommunication. The experimental isolation test results are shown in Table 2.

Based on the automatic construction and delivery of the previous flow table algorithm, a new group of virtual-real fusion networks with the same network segments except for vxlan is added to the scene.

Table 2 Isolation test results

Source host	vxlan to which the destination host belongs	Is it connected?
Entity Terminal 1	vxlan100	Yes
Entity Terminal 1	vxlan200	No
Entity terminal 2	vxlan100	No
Entity terminal 2	vxlan200	Yes

4.3 Network Performance Test for Virtual Reality Convergence

To verify the actual network performance of the proposed scheme in virtual and real networks, different numbers of virtual instances are generated in the same virtual and real network for experimental testing. In this experiment, the Flavor configuration parameters of virtual models in OpenStack were divided into three levels, namely trim (RAM = 512 MB, Root Disk = 5 GB), medium (RAM = 2 GB, Root Disk = 10 GB), Large (RAM = 4 GB, Root Disk = 20 GB), the scale ratio of small, medium, and large virtual instances is selected as 5:3:2, and for the generated virtual and real network scenarios of different scales, the communication delay, throughput, packet loss rate, etc. The network performance test is carried out on the aspect, and the test results are shown in Figs. 3, 4 and 5

Figure 3 is a curve diagram of the relationship between the communication delay of the virtual and real networks and the number of virtual instance nodes in this paper. For each number of virtual instances, the test is performed ten times, and the average value is taken. It can be seen from the figure that as the number of virtual nodes increases, the communication delay of the method in this paper also increases slightly.

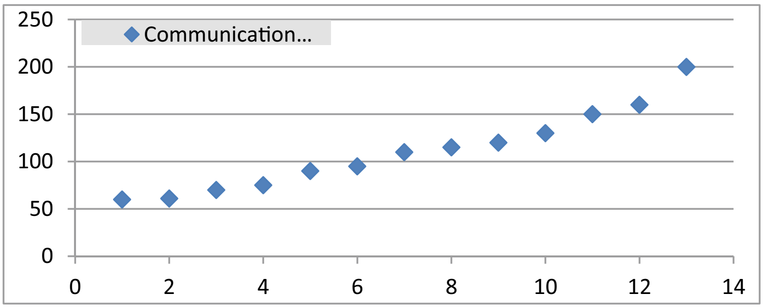

Fig. 3 Relationship curve between network delay and number of virtual instances

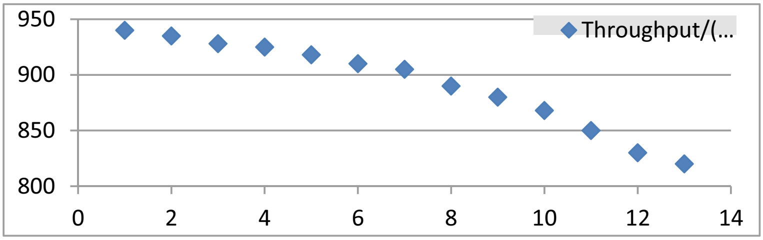

Fig. 4 The relationship between throughput and the number of virtual instances

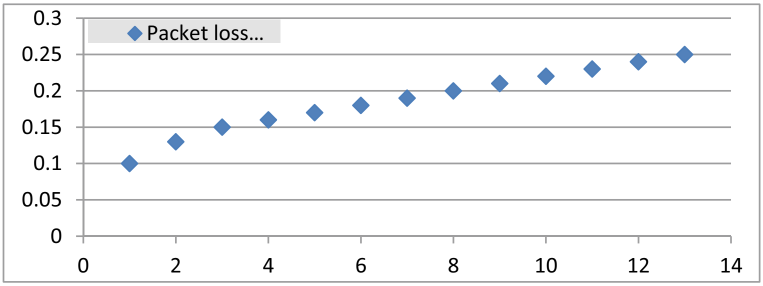

Fig. 5 The relationship between the packet loss rate and the number of virtual instances

Still, the overall uncertainty remains at a superficial level. Figure 4 is a graph of the relationship between the throughput of the virtual and real network and the number of virtual instance nodes. The same test is performed ten times for each method, and the average value is taken. The servers and physical terminals in the experimental environment work should be close to 1000 Mb/s. The method described in this article is based on the vxlan network model.

The data packets of the virtual instance are sent directly from the network card of the computing node and directly communicate with the physical terminal through the physical SDN switch. Since there are three computing nodes in the experiment, it can be seen from the figure the throughput is less affected by the size of the virtual instance. Figure 5 shows are virtual and real relationship curve between the packet loss rate of the network and the number of virtual instance nodes. In the packet loss rate test, the network bandwidth is determined by continuously sending UDP packets between the virtual instance and the physical terminal and constantly adjusting the sending bandwidth of the source host, the bottleneck.

5 Conclusion

A virtual-real fusion network simulation construction method based on SDN is proposed. Through the physical SDN switch and the Ryu controller, the physical terminal and the virtual instance constructed by the OpenStack cloud platform are used to integrate the network. The experiment proves that the SDN-based method can be effectively applied to large-scale. The construction of large-scale virtual and real network scenarios has better performance than traditional methods in terms of network connectivity, delay, throughput, and packet loss rate. Furthermore, this method does not transmit all virtual and real traffic through network nodes in the cloud platform, eliminating the possibility of network node congestion. At the same time, this method realizes network isolation based on vxlan, which ensures high reliability of virtual and real scenarios and solves the problem of limited scale.

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Deep Learning-Based Intrusion Detection Model for Network Security

Sagar Dhanraj Pande, Govinda Rajulu Lanke, Mukesh Soni, Mukund Anant Kulkarni, Renato R. Maaliw, and Pavitar Parkash Singh

Abstract Since it serves as a potent means of network security defence, intrusion detection technology is an essential component of the network security system. As the Internet has grown quickly, so too have network data volumes and threats, which are now more sophisticated and diversified. Modern intrusion detection equipment cannot reliably recognize different types of attacks. A CBL_DDQN intrusion detection model based on an upgraded double deep Q network is suggested based on deep reinforcement learning to address the imbalance of regular traffic and attack traffic data in the actual network environment as well as the low detection rate of attack traffic. This model integrates the feedback learning and policy-generating methods of deep reinforcement learning with a one-dimensional convolutional neural network and a bidirectional long-term, short-term memory network to train agents to attack different types of samples. Classification, to some extent, lessens the reliance on data labels during model training. The Borderline-SMOTE algorithm reduces data imbalance, thereby improving the detection rate of rare attacks. The NSL KDD and UNSW NB15 data sets are used to assess the model's efficacy. The findings demonstrate that

S. D. Pande (✉)
School of Computer Science and Engineering, VIT-AP University, Amaravati, Andhra Pradesh, India
e-mail: sagarpande30@gmail.com

G. R. Lanke
Data Science and Engineering, Birla Institute of Technology and Science, Pilani, Rajasthan, India

M. Soni
University Centre for Research and Development Chandigarh University, Mohali 140413, Punjab, India

M. A. Kulkarni
Bharati Vidyapeeth (Deemed to Be University), Institute of Management, Kolhapur, India

R. R. Maaliw
College of Engineering, Southern Luzon State University, Lucban, Quezon, Philippines
e-mail: rmaaliw@slsu.edu.ph

P. P. Singh
Department of Management, Lovely Professional University, Phagwara, India
e-mail: pavitar.19476@lpu.co.in

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the model has performed well with respect to the three indices of accuracy, precision, and recall, and the detection effect is significantly superior to Adam BNDNN, KNN, SVM, etc. The detection method is an efficient network intrusion detection model.

Keywords Deep learning · Intrusion detection · Classification · LSTM · CNN · SMOTE

1 Introduction

"Network intrusion detection system (network intrusion detection systems, NIDS)," as a proactive defense technology, is the primary means to discover potential network threats in time and formulate reasonable defense strategies and is an integral part of the network security technology system [1–4]. It can detect attacks in time and reduce network security threats by collecting and analyzing relevant network data.

Signature-based NIDS relies on an attack signature database for detection. It has a high detection rate for existing data in the database but cannot detect new attacks [5–8], and the database needs to be updated frequently. Anomaly-based NIDS identifies hidden attacks in computers by analyzing unusual traffic distributions and can be used to detect new types of attacks. The system uses configuration files to store all normal behaviors of users, hosts, network connections, and applications. This approach compares current activity to the configuration file and flags any significant deviations as anomalies [9–12]. This data sensitivity effectively prevents various malicious behaviors. However, this sensitivity advantage can lead to high false favorable rates, leading to unnecessary panic and overreaction [13–15].

Machine learning algorithms, such as Bayesian networks [16] and support vector machines [17], are widely employed in anomaly-based NIDS. Small-scale traffic data detection challenges have been successfully tackled by these methods. Performance of classic intrusion detection methods, however, is facing considerable hurdles in dealing with huge high-dimensional data due to the ongoing advance of network technology and the ongoing expansion of network scale.

Representing representation learning, deep learning can automatically learn high-level data features directly from complex original features, doing away with the requirement for specialized knowledge in the manual feature extraction procedure. As a result, the deep Model architecture is the basis for the vast majority of modern intrusion detection systems. Among the most popular deep models are the autoencoder [18], the convolutional neural network (CNN) [19, 20], the recurrent neural network (RNN) [21], etc. Literature [22] proposes to use CNN for network intrusion detection and uses CNN to select features to classify traffic. Compared with traditional algorithms, it has a good effect but ignores the connection in the time sequence of traffic data; Literature [23] proposes to use LSTM (extended short-term memory network) is used in intrusion detection and has achieved good classification results, but without considering the spatial characteristics of the data, there is still room for improvement in classifier performance. To fully extract the features of network traffic data, literature [24] proposes to use CNN and LSTM in combination, use CNN to learn the parts of network packets, and then use LSTM to learn the details of network traffic, compare with using LSTM, CNN or LSTM alone. The upgraded model is more efficient and produces more accurate results when classifying traffic data. These models rely on large amounts of labeled data samples, even though the neural network model is capable of powerful feature extraction. Still, there is a lot of data that needs labeling, and doing it manually is a costly and time-consuming process.

Reinforcement learning (RL) is an active solution to the aforementioned issues. Traditional RL is based on the Markov decision process (MDP) to create algorithms; however, it can only examine small-scale problems. Moreover, the natural environment is often complex and changeable. Therefore, it is problematic for traditional RL methods to obtain effective solutions when solving practical problems. Literature [25] combines reinforcement learning with deep learning. It proposes deep reinforcement learning (DRL), which approximates the complex data space and mapping relationship in reinforcement learning with neural networks, significantly expanding the application range of reinforcement learning. This is because coupled with its unique feedback mechanism; reinforcement learning also has an extensive range of applications in classification problems application. Literature [26], for the first time, equates the classification problem to the continuous decision-making process of the agent (agent) and proposes a classification task solution based on reinforcement learning, with an accuracy rate of 87.4% in the eight UCI data sets. Literature [27] proposed an AE-DQN model based on adversarial multi-agents for the problem of network intrusion detection and achieved good detection results. Although the above-mentioned deep reinforcement learning model shows unique advantages in solving the classification problem with imperfect labels, the selection of the deep learning network model cannot whether the selection of the deep learning model is mainly appropriate to determine the classifier's performance. Therefore, the focus of the above models is only on the generation of agent strategies [28, 29].

Uneven data is another common issue. The classifier can obtain higher overall classification accuracy, but the recognition rate of smaller class data is meager, and misclassification of minority classes will bring huge costs. At the algorithm level, by changing the classifier, the existing classifier can strengthen the learning of the minority class [30].

In order to boost the system's detection rate for different types of traffic, this study presents a network intrusion detection model based on the enhanced double Q network. The model incorporates the hybrid network model of CNN and BiLSTM into the learning framework of the deep Q network, simulating the intrusion detection process as the sequential decision-making process of the agent. Improve the classifier's ability to identify different forms of attack flows. Concurrently, an unbalanced processing strategy is offered to improve the detection rate of rare attacks while taking data imbalance into account.

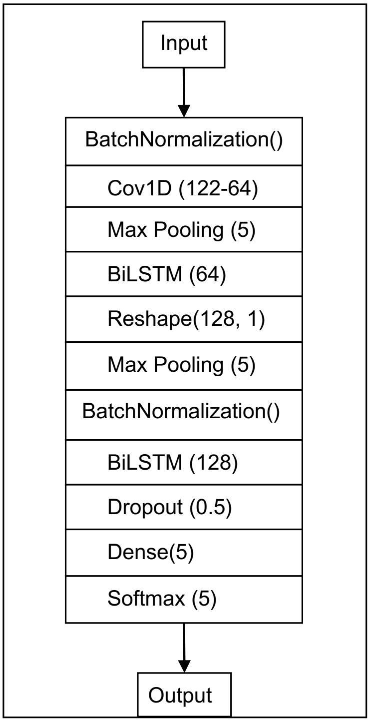
2 CBL_DDQN Model Based on Improved Double Deep Q Network

This paper introduces the mixed CBL model of CNN and BiLSTM into the dual deep Q network framework. To better utilize the CBL network to fit the Q function and the feedback mechanism in the dual deep Q network, a new model for intrusion detection called CBL DDQN is developed and optimization strategy to optimize the CBL network, and finally, realize the correct classification of traffic.

Network traffic data is a sequence with a time step, which has both spatial and temporal characteristics [13–16]. Because both CNN and BiLSTM are very good at extracting features from input, we create a CNN-BiLSTM hybrid model, the CBL model, by fusing a one-dimensional convolutional network with a bidirectional long-term, short-term memory network (see Fig. 1).

The parameters are discretized in the highest pooling layer to shorten training time and prevent overfitting; middle-layer parameters are normalized with batch normalization to speed up training; and the BiLSTM layer is used to learn onward and rearward time series data. One way to improve understanding is to utilize two BiLSTM layers that each learn at a different granularity. One-dimensional convolutional neural network with long-term time-dependent feature correlation; network

Fig. 1 CBL model framework



layer between BiLSTM layers to extract features efficiently and speed up training; Dropout layer to prevent model overfitting; and Softmax function for probability matrix output.

3 Selection and Processing of Data Sets

3.1 Dataset Selection and Preprocessing

In this study, the CBL DDQN model is validated using simulation tests on two public intrusion detection datasets NSL KDD and UNSW NB15. Tables 1 and 2 detail the two datasets, respectively.

Data preprocessing mainly includes the following three parts: character feature medicalization, one-hot encoding, and numerical normalization.

(1) Numericalization of character features

The category features of standard records and different attack records are converted from character type to digital label, and the label distribution after conversion is shown in Tables 1 and 2.

Table 1 Attack category information of NSL_KDD dataset

Attack category	Quantity	Convert tag
Normal	077,054	0
Dos	053,385	1
Probing	014,077	2
R2L	003,749	3
U2R	000,252	4

Table 2 Attack category information of UNSW_NB15 dataset

Attack category	Quantity	Convert tag
Normal	93,000	0
Generic	2677	1
Exploit	2329	2
Fuzzers	16,353	3
Dos	44,525	4
Reconaissance	24,246	5
Analysis	58,871	6
Backdoor	13,987	7
Shellcode	1511	8
Worms	174	9

(2) One-hot encoding

With one-hot encoding, the distance calculation between components can be more realistic.

(3) Numerical normalization

After one-hot encoding, to reduce the impact of the value of each dimension attribute feature on the subsequent network, each dimension attribute feature is normalized according to formula (1), and the normalized interval is [0,1]:

$$\dot{x} = (\mathrm{x} - \mathrm{x}_{\min}) \big/ (\mathrm{x}_{\max} - \mathrm{x}_{\min})$$

3.2 Experimental Results and Analysis

In this paper, two network traffic data sets, NSL_KDD and UNSW_NB15, are used for experiments. The details of the dataset are described in Sect. 3.1. In the experiment, set all the data of the entire training set as one epoch, set the maximum epoch to 30, make statistics on the classification results of the model every five ages of training, and use the control experiment to test the system before and after using the Borderline-SMOTE algorithm. The recognition rate of the data. The classification results after statistical training for 30 epochs are shown in Figs. 2 and 3.

Figures 2 and 3 show that the classifiers' performances are steadily rising as the training process progresses. If you compare the model introduced with the Borderline-SMOTE algorithm to the same training, you'll see that the latter has a greater classification accuracy. Within a certain threshold of repetitions, the accuracy of the original data set can be direct.

It can be seen visually that the introduction of the imbalance processing algorithm has a more significant role in promoting the convergence of the model.

Accuracy, recall, and precision of six approaches were evaluated to further validate the model described in this research. Table 9 and Table 10 display the statistical

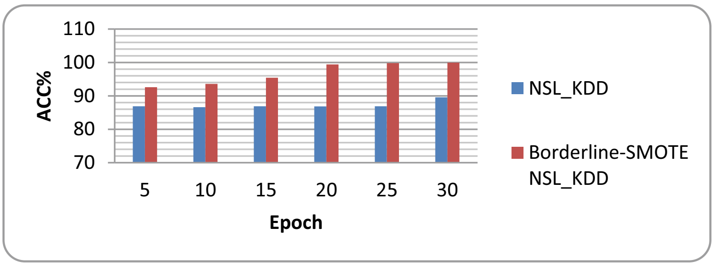


Fig. 2 The recognition rate of the model for NSL_KDD

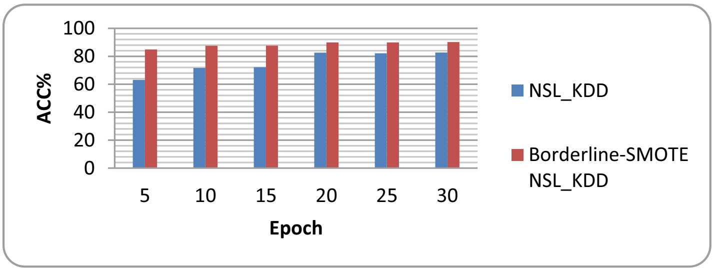


Fig. 3 The recognition rate of the model for UNSW_NB15

outcomes of Adam BNDNN [18, 18], DQN [19, 30], RF [19], SVM [19], MLP [19], and Adaboost [20], where the data in bold is the optimal value of this performance index.

For intuition, the data is drawn in a bar graph, and the result is shown in Fig. 4.

With the help of Fig. 4, it is clear that the CBL DDQN model suggested in this paper has a substantial detection effect on the NSL KDD dataset. All performance parameters are higher than prior similar studies, with precision at 99.96%, recall at 99.97%, and precision at 99.79%.

Similarly, for the sake of intuition, the data drawn in bar charts, and the results are shown in Fig. 5.

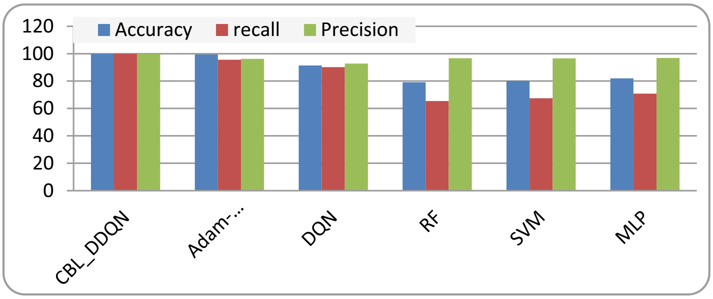


Fig. 4 Classification performance of each model for NSL_KDD

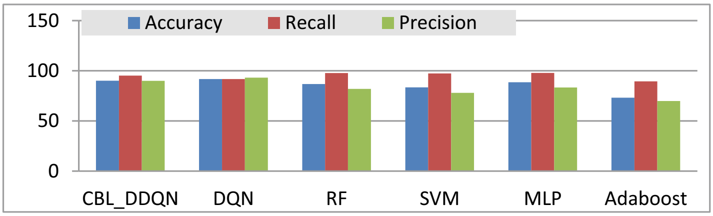


Fig. 5 Classification performance of each model for UNSW_NB15

Taken together, Table 4 and Fig. 5 show that the UNSW NB15 detection results using this paper's model have an overall identification accuracy rate of 90.12%, recall of 95.20%, and precision of 89.93%. On the downside, it does not perform as well as its counterparts. After cautiously considering a number of performance factors, the model suggested in this research is able to produce a good intrusion detection impact.

The above two sets of experimental results demonstrate that the upgraded dual deep Q network model suggested in this paper may effectively address the intrusion detection problem.

4 Conclusion

The purpose of this paper is to present the CBL DDQN network intrusion detection model, which incorporates the hybrid network CBL network of CNN and BiLSTM into the DDQN framework to enhance the model's performance. In comparison to more standard deep learning algorithms, this one reduces or eliminates the need for labeled data. In this way, it outperforms deep learning algorithms in terms of classification accuracy. Moreover, the Borderline-SMOTE technique is used to increase the number of unusual attack samples because of the fact that the disparity between the data makes it hard for the classifier to understand the data features properly. Based on these findings, it appears that the imbalanced processing technique helps the model become more accurate in classifying data.

In conclusion, the suggested model in this paper has good results in the imbalanced data classification challenge. Its overall performance is higher than that of the enhanced DQN network and other deep learning networks, suggesting a novel approach to deep reinforcement learning.

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Controlling Project Execution in the Era of Soft Computing and Machine Learning

Bhavna Bajpai, Amit Khare, Mustafa Mudhafar, Shweta Singh, Malik Mustafa Mohammad Jawarneh, and Mukta Sandhu

Abstract During the execution control of their initiatives, organisations use a variety of devices to aid independent direction. Regardless, they are still insufficient in the face of skewed statistics and shifting management approaches. The absence of frameworks for controlling the execution of enterprises has an impact on the nature of their categorization in terms of supporting independent direction. The presentation of soft registering methods, which provide heartiness, effectiveness and flexibility at apparatuses, is an optional arrangement. This research provides a technique for project execution control based on Soft Computing and ML, which contributes to the executives' ability to further improve the project. The proposed method allows for AI and the replacement of fuzzy inference frameworks in project evaluation. The results are derived from seven calculations involving space apportioning, neural architecture, gradient descent and genetic algorithms. Adoption of the proposed framework, which has been included in this paper for project personnel, signifies a change in the nature of venture evaluation. The obtained result ensures that the apparatuses are in perfect working order to assist the independent direction in projecting the executive associations.

Keywords Soft computing · ML · Project execution

B. Bajpai (✉)
Dr. C V Raman University, Khandwa, Madhya Pradesh, India
e-mail: bhavna.bajpai0212@gmail.com

A. Khare
Department of Technology, JIET Institute of Design and Technology, Jodhpur, Rajasthan, India

M. Mudhafar
Department of Anesthesia and Health Care Faculty of Altuff College University, Karbala, Iraq

S. Singh
Electronics and Communication Department, IES College of Technology, Bhopal, India

M. M. M. Jawarneh
Faculty of Computing Sciences, Gulf College, Seeb, India
e-mail: malik@gulfcollege.edu.om

M. Sandhu
SVSU, Haryana, India

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1 Introduction

For project-oriented associations, the proper use of information, processes, procedures, abilities and equipment is critical to the success of the organisation. Estimating and tracking the development of an effort is made possible in large part by the monitoring and control process [1]. This makes it possible to distinguish between different types of management plans, so that corrective steps can be made as necessary to meet the project's goals.

In the context of a company's management team, reconciliation involves coming to daily decisions about where to place the company's assets and efforts, anticipating possible problems so that they can be addressed before they become critical. Developing a group of executives that work together as a team with the leader's role as a pioneer and a keen eye for detail is critical to the success of the project. All of the following should be documented: scope, cost, timeliness and quality; coordination of activities; and human resources (HR) [2].

An individual who is in charge of overseeing improvement initiatives must have the ability to understand the challenges at hand, differentiate between solutions and take autonomous action, all with the goal of supporting the primary goals of the organisation [3]. PC instruments can be used to assist individuals during the execution and control phases of a project. In this way, it's possible to explain what's going on and propose particular solutions to the problem.

It is now possible for project managers to know exactly what action they need to take to solve a specific problem at any given time [4]. As a result of PCs that facilitate autonomous management, which rely on clear and quantitative records, this situation has arisen. Various tools for project managers have been developed during the last many years. Despite this, the majority of these tools don't take into account the treatment of vulnerability, nor the ability to adapt to the needs of different projects [5].

It is possible to complete an execution plan and illustrate the progress of the project through the administration records supported by integrated Project Management. This enables them to discover the root causes of problems so that they may make decisions based on a freshly established set of demands [6].

Multitudinous financial calamities with a large negative impact on relationships are caused by Integrated Project Management errors and deficiencies [7]. Disappointment is common in this field because of the following factors:

- There is a lack of knowledge about good project management procedures.
- Involvement in the management and observation of projects is minimal.
- Problems in dealing with information that is both ambiguous and vulnerable.
- Information gathered by individuals may be out of date if certain members of the project group are no longer in attendance.
- Instrumental flaws in the automated evaluation of projects.
- Instruments of assessment and control that are past their expiration dates in relation to shifting management styles as a result of progress made throughout their continuous improvement [8].

ML can be used as an alternative solution to the previously described concerns. Measurements, rationales, calculations, brain organisation, hypothetical data, man-made consciousness and soft registration are all intertwined in this interdisciplinary field of study [9].

Some of these challenges can be addressed by utilising a project execution control technique that takes into account the sensitivity of registration and artificial intelligence (AI) [10]. As a means of assisting project-based groups in their own self-governance, the suggested structure is projected to provide several benefits. Investigative zeal and down-to-earth dedication are centred on the following perspectives:

- The use of AI in project execution control, increasing the heartiness, variety, and harmony between the force of expectation and the understanding of project management tools.
- Free ware that includes a few soft AI registering methodologies and trial and error of frameworks for the evaluation of project performance will be used to further develop the Analysis. Pro.SC.PMC library [11].

Several relevant works are discussed in the next section, including the examination bases and the records used in project evaluation. WM, ANFIS, HYFIS, FIR.DM, GFS.HGD, GFS.THRIFT and GFS.LT.RS are the parts of the proposed strategy and the fundamentals of the applied softly registering processes in Sect. 3. Using the approach in a real-world application is the focus of Segment 4. Endings and future projects are finally revealed after a long wait.

2 Literature Review

2.1 Project Management Best Practices

Schools or institutions dedicated to the formalisation of association approaches develop Project Management as a discipline. The PMBOK criteria [9], the ware Engineering Institute (SEI) with the Capability Maturity Model Integration (CMMI) [5], the International Project Management Association (IPMA) [12] and the International Organization for Standardization (ISO) with its guidelines 10,006 and 21,500 [13] are among these organisations.

The PMBOK recommends monitoring project execution throughout its life cycle in order to provide constructive criticism, identify ambiguous or unsettled circumstances and encourage continuous improvement. It suggests that records related to the information categories of Scope, Cost, Time and Quality be used for this. It oversees the treatment of vulnerability in project gambles, as well as future usage investigation for cost estimation and undertaking plan executives [14].

The importance of PC apparatuses as a support for decision-making in the undertaking improvement group was discovered by CMMI. It oversees process streamlining inside the organisation at level 5, with a focus on continuous improvement in the presentation of cycles through innovative and incremental modifications. It suggests that businesses should periodically renew the meaning of their cycles in order to include the most recent changes made by the organisation into standard cycles when they benefit [15].

The ISO 10006:2003 and 21,500:2013 recommendations provide instructions for project directors to supervise quality during project improvement. They are in charge of the estimation and investigation processes, as well as the ongoing development of the project's execution. They advocate for the management of risk associated with spending as well as the control of the project plan. In [16] it is stated that the association's project management processes and techniques should be enhanced based on previous experience, rather than recommending radical changes. It ensures that during the project, the executives provide excellent independent direction at all levels of the organisation [17].

2.2 Records Used to Assess the Project

This investigation makes use of a number of project management documents that have been determined in accordance with [18]. These records reflect the project's execution in terms of the PMBOK guidelines and CMMI's basic information regions (see Table 1).

Table 1 Records essential to project planning

Records	The executives' information regions of the undertaking
IEP: Index of Execution Performance	Executives' authority and responsibility
IPP: Index of Plan Performance	Making the most of your time
ICE: Index of Cost-Effectiveness	Expense management
VPI: Viability Performance Index	Extension and administration of high grade
HRPI: HR Performance Index	Executives in Human Resources
PIC: Performance Index Calculated	An administration that is calculated
IIQ: Index of Information Quality	Consistency of data

Each record's overview is as follows:

IEP: the relationship between the magnitude of the effect of completed projects and the magnitude of the effect of planned projects until the deadline.

IPP: Indicates how far along a project is. Relationship between the number of genuine execution rates and the number of planned execution rates for each of their projects, as determined by the cutoff date.

ICE: displays the current state of the project's financial plan. The relationship between the estimated project cost and the actual cost as determined by the cutoff date.

VPI: depicts the current state of project viability, as determined by an examination of the relationship between the fulfilment of its requirements, the evaluation of its activities in light of the need and the non-similarities.

HRPI: depicts the state of HR execution in relation to the turn of events, their impact and the necessity for the assigned tasks.

PIC: presents a study of the evolution of material assets associated with the project, from the providers to the final client.

IIQ: suggests that the executives investigate the conclusion and proper contribution of information to the enterprise.

The quality of the decision stage, and with it, the intelligibility of vital information for the beginning of AI utilised in this examination, is underpinned by programmed computation of records for project executives in associations.

2.3 In Project Management, Soft Computing and Machine Learning Applications

With the ultimate goal of achieving appropriate administration, heartiness and greater proclivity with reality, soft registering strategies monitor resistance to the dubiousness, vulnerability and imperfect reality of realities [19]. Soft registering entails a variety of computing philosophies, including fluffy logic, brain network hypothesis, probabilistic thinking and transformational calculation, to name a few [20].

Using information and collected understanding, soft registering strategies provide PC apparatuses with a way to interact with human thinking. Furthermore, they enable independent direction by increasing efficacy, adaptability and an acceptable balance between force of expectation and understanding. These tactics are effective in situations where there are numerous information sources and there is a high level of resistance to information imprecision; they allow for minimal cost setups and a higher demonstrating limit.

A few studies have suggested possible arrangements for project executives based on AI, information mining, computerised reasoning and soft registering. However, when compared to other application areas, explicit utilizations of these tactics in project directors are somewhat low. A few related works are listed below.

In [21], a method based on soft registering is given for categorising projects into three categories: simple, medium and difficult. In [17], a record structure is built to evaluate the project leaders' presentation in terms of quality, cost, time and risks. In [22], a fluffy derivation framework is presented to determine the record that analyses the undertaking execution by combining two information sources obtained by the research of substantial worth: expense execution and arranging execution. A model based on ANFIS [23] is presented in [14] to assess product exertion. [24] introduces a framework based on brain networks for utilising the authoritative undertaking of the executives development model. [25] proposes a fuzzy derivation framework for evaluating the success of programming project administration. Some of the researches are using machine learning techniques [26, 27] for classification and predictions for many applications.

In general, the following characteristics are not included in the profiles of experts investigated:

- The application of artificial intelligence (AI) techniques.
- A shift in the assessment framework, as evidenced by the association's continued improvement.
- The model must be reconciled with the project's executive apparatuses.
- Free programming is used to run the model.
- The recommended solution will be used to evaluate the project's implementation.

A technique is presented as a solution to the concerns that have recently been raised. It employs a number of computational methodologies to assist organisations with task execution control and decision-making.

3 Execution Control Technique for Projects

An administered learning methodology is used in the suggested strategy to change the bounds of the fluffy derivation frameworks. Learning is based on a number of accomplished initiatives, the mathematical advantages of which are understood via essential administration records and the assessment offered by association specialists. A set of preparing cases and a set of approval cases are derived from this basis of activities [28].

Extricate the informative gathering from the completed undertakings data set; apply multiple AI procedures and use comprehensive quantifiable measurements to determine the framework that best assess the activities execution in the association, as shown in Fig. 1. The suggested technique uses a set of records as input and produces a fluffy derivation framework for project evaluation as a result. This instrument is more than likely to be finished when top management determines that stated evaluation framework has to be updated as a result of the association's consistent improvement in management styles.

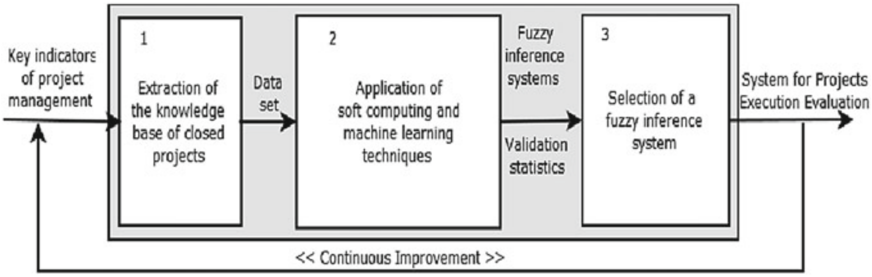

Fig. 1 Parts of the project execution control approach

Part (1) Entries to this square are compared to the project executives' records, which are then decided and stored in the association data base. It goes without saying that the use of the records mentioned in Table 1 is recommended. The finished project information base is advised, getting the pre-determined records and the master assessment to the undertakings, which will operate as case preparation. As a foundation for the age and streamlining of fluffy guidelines using AI, this wellspring of data capabilities. To get the preparation and approval sets, the foundation of pre-characterised undertakings is haphazardly divided into numerous allotments. These allotments are then utilised to execute cross-approval between each procedure's studies.

Part (2) Experiments are carried out to register and AI processes for delicate it. The delicate it is made sense of to record tactics in the following section. The following information is saved in the data set for each investigation: fluffy derivation frameworks established, name of the technique used, learning bounds, a measure of rules created and the aftereffects of elements for measuring the nature of evaluation [29]. Correct Classifications, False Positives, False Negatives, Mean Square Error (MSE), Root Mean Square Error (RMSE) and Symmetric Mean Absolute Percentage Error (SMAPE) are the approval metrics, as shown in Eqs. 1, 2 and 3.

$$MSE = \frac{1}{m}\sum_{j=1}^{m}(X_{actual} - X_{cal})^2 \tag{1}$$

$$RMSE = \sqrt{MSE} \tag{2}$$

$$SMAPE = 100 * \frac{1}{m}\sum_{j=1}^{m}\frac{|X_{actual} - X_{cal}|}{(X_{actual} + X_{cal})/2} \tag{3}$$

where m is the count of instances, X_{actual} is the average result value and X_{cal} is the framework-determined result.

Part (3) Statistical tests are carried out and the best fluffy derivation framework is selected. For this, the effects of delicate registration methods are considered in terms of the type of evaluations, using measures obtained in part 2, which allow obtaining the accuracy of calculations done. These estimates are subjected to Friedman and Wilcoxon tests with a 0.05 cut-off to see whether there are significant differences between the outcomes of delicate registering procedures [30]. The fluffy derivation framework is the final outcome of this section, and it will be used to evaluate projects that best match the association climate.

Consistent development: after learning, it is possible to engage fresh preparation instances, fully set on achieving consistent improvement and altering the evaluation framework to project the executives' changing circumstances.

Recognise changing circumstances when the company modifies the kinds and quantities of executive records it uses. It is planned to update the knowledge base of completed projects to ensure constant improvement of the proposed framework. This update may be issued when administrators believe it is necessary for the organization's continued growth [31].

3.1 The Proposed Method Makes Use of Delicate Computing Techniques

For the transformation and AI of the framework that examines the undertaking execution, a few calculations were used as the technique. The following is a collection of sensitive registering techniques that have been used in practice.

W.M. (Wang and Mendel). This method generates a slew of new fluffy principles and refines them using a space segment methodology [32]. The five stages of the educational experience are as follows.

(i) Separate the passage and leave voids of mathematical knowledge, which is supplied in fluffy districts, referring to the etymology phrase for time intervals.
(ii) Create fluffy principles based on the preparatory instances, determine the degrees of having a location with fluffy sets for each case and choose the etymological word with the highest degree in each variable.
(iii) Assign a degree of importance to each standard developed, with the goal of resolving conflicts between created regulations.
(iv) Construct a consolidated rule basis based on the guidelines and etymological concepts presented.
(v) After discarding the repetitious and those with the lowest grades, create a final rule foundation.

ANFIS (Adaptive-Network-based Fuzzy Inference System). This approach addresses a Sugeno type half-and-half neuro-fluffy arrangement [33], which combines the fluffy reasoning and brain networks ideal models [34]. ANFIS is based on a feedforward network, in which the predecessors' participation capacity should be logical, resulting in typically great results with the ringer type [35]. Jang claims that using a half-and-half approach improves the outputs and allows them to be combined more quickly than using just the falling inclination (backpropagation) to streamline all borders. The phases of the educational experience are as follows.

(a) Give the framework an information yield vector and specify the bounds of the participation components of fluffy sets. The outcomes of each layer of the organisation are defined by propagating them forward to the next.
(b) Determine the error as the difference between the determined and expected yield. The number of square blunders for each subsequent is one of the most often used methods to determine the preparation error.
(c) Using least-squares methods, adjust the participation capacity limits in the following. In reverse advance, error signals proliferate and the inclination vector for each preparation instance is pooled. The hierarchical method refreshes the predecessors' boundaries.

HYFIS (Hybrid brain Fuzzy Inference System). This method uses a flexible neuro-fluffy derivation framework [36] with five layers, which is similar to a fluffy surmising framework with Mamdani-type principles. The HYFIS learning calculation is divided into two steps. The information acquisition module organises the rules foundation in the first step. The bounds of involvement abilities are altered in the second stage in order to get a desirable level of execution. One advantage of this method is the ease with which the basis of fluffy ideas may be adjusted when new information becomes available. When additional preparation scenario becomes available, a standard is created for it and added to the fluffy principles base.

FIR.DM (Fuzzy Inference Rules by Descent Method). This process is based on the Zero Grade Sugeno [25] fluffy derivation framework, which updates its bounds from an administered preparation in light of the inclination falling technique [37]. The participation capacity used is an isosceles triangle with two boundaries: the base width and the triangle's focal point. The FIR.DM concept is divided into two modules: acknowledgment and preparation. The following advancements are included in the learning of computation.

(a) Determine each preparation case's commitment to the principles loaded.
(b) Determine the usual obligations for each weight.
(c) For each centre, base of the triangle, and weight, use the standard increment.
(d) Calculate the erroneous work employing new boundary upsides.

This technique allows each standard's fluffy arrangements to be modified independently of the rest of the regulations. This technique's learning velocity and speculation limit are superior to those of a typical backpropagation brain structure.

GFS.LT.RS (Genetic Fuzzy Systems by Lateral Tuning and Rule Selection). In light of airy recommendations [14], this approach presents an adequate hereditary calculation for the alteration of a framework. It shows a cooperation of transformational horizontal shift of participation works, as well as an instrument for selecting fluffy guidelines, which reduces the pursuit space and improves the framework's comprehensibility. It uses a fundamental depiction conspire based on etymological 2-tuples [11], which allows for horizontal mark dislodging (slight exchanges to left or right of the first participation work). Participation elements of the symmetric triangle type are used to handle fluffy allotments.

The Analysis Pro.SC.PMC library is derived from these seven calculations and is designed for the study of projects involving delicate registration procedures; it is based on free programming and has capabilities tailored in the PL/PGSQL and PL/R dialects that may be used in PostgreSQL. Local pieces of R [37] and the FRBS bundle (Fuzzy Rule-based Systems for Classification and Regression Tasks) [14] are used as conditions.

The suggested approach, which is carried out in the library, enables the association's data base to be reconciled for the evaluation of undertaking execution. The whole process is automated, from consulting a database of past projects to conducting extensive quantifiable testing to determine the framework that best analyses project performance. The executives of the project have concluded the treatment of data vulnerability and reconciliation using PC instruments using free programming improvements [37].

4 Results

The suggested approach is used at the executive level of a project (GESPRO) [14]. This device is a non-exclusive and flexible biological programming method designed to assist customers in projecting association executives. The suggested approach and GESPRO are aligned with PMBOK [1] and CMMI [9] concepts, providing customers with appropriate points of engagement to familiarise data with the framework, as shown by the well-defined requirements and development stages for the organisation. Figure 2 depicts a report on the evaluation of a couple of an organization's projects using the suggested method.

The suggested approach used in GESPRO allows for the management of project portfolios and their specific execution plans, as well as the control of their implementation. It entails fuzzy logic in the programmed calculation of a number of important

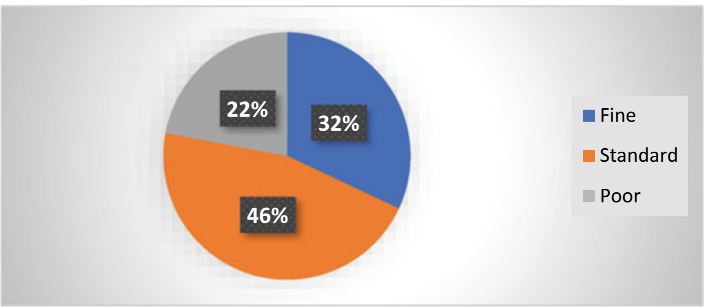

Fig. 2 Report about the undertaking assessment of a project

execution records that reflect the undertaking's conduct (Table 1), including the problems and reasons that lead to them, so that they may make decisions based on a previously established methodology of requirements.

In comparison to 212 completed projects, preparing cases were used. Each project has advantages of information recordings and a final mathematical evaluation of the project, delivered by a group of experts using the Delphi approach. The input and outcome information are normalised to values between 0 and 1 as a pre-handling process before using AI [22].

An informative collection with the content is used for learning: 68 projects rated as Fine (32%), 97 projects rated as Standard (46%) and 47 projects rated as Poor (22%); it does not include invalid or out of reach values to follow the distribution. The informative gathering is hastily divided into 22 discrete allotments. Each section is responsible for preparing 76 percent of the 212 cases and approving the remaining 25%. The cross-approval approach uses these allotments to undertake 22 executions of each soft computing operation.

The results obtained from each procedure in each of the 22 information allotments allow for consideration of the presentation of calculations, taking into account the following factors: percent of correct characterizations, number of misleading upsides, number of bogus negatives, MSE, RMSE and SMAPE. The following is a summary of the effects of the different kinds of approvals used in this investigation. Figures 3 and 4 depict the behaviour of seven computations related to % of correct characterizations and SMAPE. It's easy to observe that ANFIS produces the greatest results [17].

Finally, using the Wilcoxon test, it was possible to confirm that the ANFIS approach delivers much better results than the rest of the computations and provides the optimum framework for evaluating projects [18].

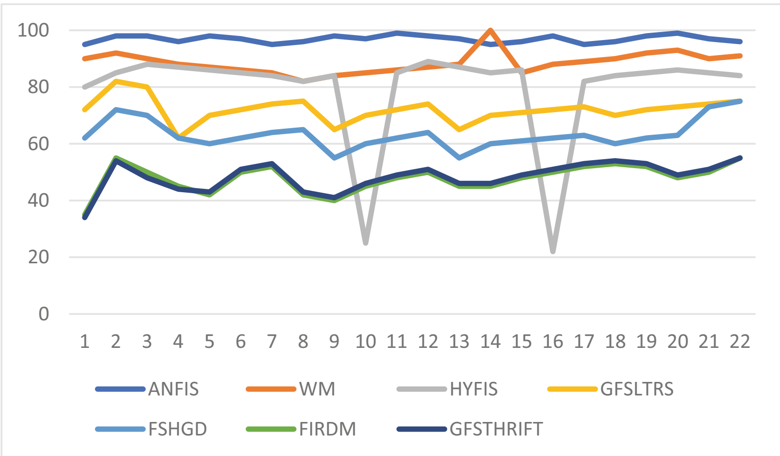

Fig. 3 Percentage of correct classifications obtained for each process on THE 22 test informational collection allotments

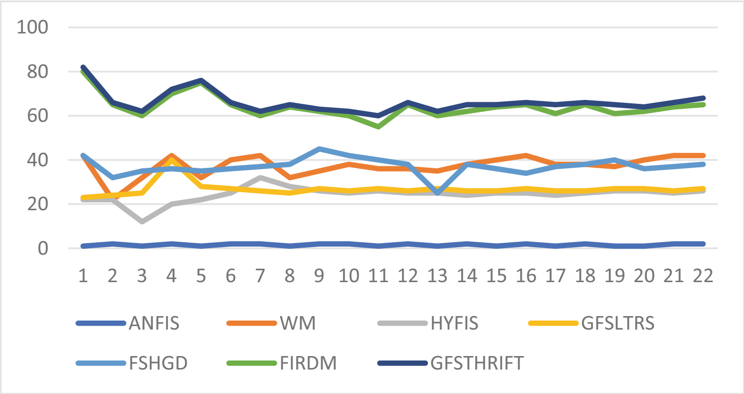

Fig. 4 Ivevery method on the 22 allotments of the test informative gathering yielded a symmetric mean absolute percentage error

An assortment of 6 organisations and 15 programming advancement centres profit from the reconciliation of the suggested method at the GESPRO stage, where around 312 projects are supervised every year. The instrument has an average of 6500 users, all with different levels of expertise and professions. Decision-making with the help of GESPRO is finished in a necessary, light-footed and pleasing manner, increasing the character of client life, thanks to the offered approach [28].

Another advantage of the suggested method is the use of free programming as a need for achieving inventive power, which ensures authoritative progress in a broad and sustainable manner [27]. This legislation is advanced and mirrored by the computational climate and capabilities built on open-source programming developments. This implies the following advantages for the executive instrument: complete control of functionality, timely detection and correction of errors, and constant improvement in light of cooperative occurrences [3].

Finally, the influence of the proposal, from a financial standpoint, is dependent on investment money, which proposes making better decisions based on data provided by the suggested approach to regulate the undertaking execution [31]. In addition to preserving assets, sending forth labour and goods that are beneficial to the progress of society as a whole.

5 Conclusion

Project execution control is a perplexing position that involves scepticism in concepts and fragility in data, a situation in which the use of careful registration processes yields excellent results. The suggested technique employs these processes in project evaluation, increasing the heartiness, adaptability, and harmony between the executive apparatuses' force of expectation and comprehension. It also makes it possible to secure information about experts in organisations and to accomplish a good project execution control. The use of artificial intelligence (AI) for project evaluation increases a company's ability to adapt to changing management styles as a result of its growth and continuous improvement. With the development of the Analysis Pro.SC.PMC library based on free programming and including important records for project executives, a commitment has been made to the enhancement of current choice aid instruments used by projects.

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PUF-Based Lightweight Authentication Protocol for IoT Devices

Amita Shah, Hetal Pandya, Mukesh Soni, Akramjon Karimov, Renato R. Maaliw, and Ismail Keshta

Abstract The Internet of Things (IoTs) carries the secure transmission and storage of a large amount of sensitive information. This paper uses the anti-tampering and anti-cloning features of the hardware physical unclonable function (PUF) to generate a shared key, combined with security primitives such as MASK algorithm and Hash function and proposes a lightweight Anonymous key-sharing security authentication protocol. Through the security analysis and verification of Ban logic and the proper tool ProVerif. It is proved that the protocol can defend against man-in-the-middle attacks, desynchronization attacks, impersonation attacks, modeling attacks, etc. By comparing other protocols, it is verified that the protocol has the advantages of low computing cost, small communication overhead and storage capacity, and high-security performance; it is suitable for secure communication transmission of resource-constrained devices.

A. Shah (✉) · H. Pandya
Department of Computer Engineering, L D College of Engineering, Ahmedabad, Gujarat, India
e-mail: Amitashah@ldce.ac.in

H. Pandya
e-mail: hetalbpandya@ldce.ac.in

M. Soni
Department of CSE, University Centre for Research and Development Chandigarh University, Mohali 140413, Punjab, India

A. Karimov
Department of Corporate Finance and Securities, Tashkent Institute of Finance, Tashkent, Uzbekistan
e-mail: akram_karimov@tfi.uz

R. R. Maaliw
College of Engineering, Southern Luzon State University, Lucban, Quezon, Philippines
e-mail: rmaaliw@slsu.edu.ph

I. Keshta
Computer Science and Information Systems Department, College of Applied Sciences, AlMaarefa University, Riyadh, Saudi Arabia
e-mail: imohamed@mcst.edu.sa

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Keywords Lightweight authentication · Internet of things · Authentication protocol · Desynchronization attacks · Anti-tampering

1 Introduction

With the rapid development of sensing, automation, and communication technologies, the massive data generated by the IoTs (IoT) has dramatically threatened the effective and safe transmission, storage and protection between devices [1]. Traditional network Security protocols will use complex security primitives such as encryption algorithms [2], digital signatures, Hash functions [3], and message verification codes to ensure the confidentiality, integrity, and non-repudiation of information transmission [4]. However, usually small size, strong resource constraints, and low hardware processing capability, so the security primitives with slow transmission rate and high communication overhead are not suitable for the authentication of lightweight devices [5].

The secure communication of the IoTs assumes that hardware devices and systems are safe. Still, malicious attackers can destroy confidential device information using chip cloning and dissection [6]. Information encryption is an effective way to protect information security certification. Still, encryption of the algorithm's key is usually stored in non-volatile memory (NVM), and the attacker can successfully read the private information in the memory through a side channel and physical attacks [7]. Physically unclonable function (PUF) is an emerging The hardware security primitives of the chip use the random process deviation that cannot be controlled in the manufacturing process of the chip to generate the unique digital signature of the device [8]. It is susceptible to physical tampering, does not need to be stored, has low hardware overhead, can solve the security problems faced by traditional keys, and is suitable for lightweight IoTs device security authentication protocols [9–11].

Key-sharing authentication protocols are usually completed using public key algorithms and digital signatures at the software layer. Still, these encryption primitives run slowly and have high communication overhead, and new quantum computing methods can effectively crack public key algorithms. To solve the problem of IoTs devices in terms of security issues in channel transmission and critical storage, this paper uses reconfigurable CROPUF anti-tampering and anti-cloning features to generate shared keys, replacing asymmetric encryption algorithms and digital signatures with high communication overhead, combined with MASK algorithm and Hash function. A lightweight anonymous key-sharing security authentication protocol is proposed for encryption methods. The protocol ensures security properties such as anonymity, availability, integrity, and forward/backward secrecy.

2 Mathematical Theoretical Knowledge and Related Work

2.1 PUF-Based Key-Sharing Mechanism

Author [12] used CRO PUF to generate the same shared key for devices, which is suitable for one-to-many security authentication protocols. This mechanism obtains shared keys between devices through two stages.

Phase 1: Generate a reliable response to the shared key. Obtain the high-precision delay matrix S of the CRO PUF through modeling; calculate the delay difference between all paths and sort them in descending order of absolute value. Consider the influence of different temperatures to determine threshold U. When the total value of the delay difference between the two paths is greater than the threshold, the output response is stable.

Phase 2: Generate incentives for the shared key. Store all the paths of the delay matrix S in the set E, and enumerate all the bits of the shared key L. Randomly select two different ways from the set E for each of the shared key L. One bit L_i, if L_i is equal to 1 and the delay difference is more significant than U, it means that a configuration stimulus D_i that can produce a stable response of 1 has been found; if L_i is equal to 0 and the delay difference is less than -U, it means that a configuration that can generate a stable response of 0 has been found Incentive D_i; otherwise, reselect the shared key L.

2.2 MASK and UNMASK Algorithms

The MASK algorithm proposed by author [13] contains three parameters: the input vector $s = [s_1, s_2, s_3, \ldots, s_l]$ with a length of l bits, and a set of l positive integers $L = \{ l_1, l_2, l_3, \ldots, l_l | l_i \in A^+\}$ and generate l-bit output vector $s_m = [s_{m1}, s_{m2}, s_{m3}, \ldots, s_{ml}]$. MASK algorithm uses positive integer set L as auxiliary data and inputs vector r with length l-bit to generate an Output function s_m of equal length. A set of positive integers L generates $L \leftarrow \mathrm{PRNG}_i(x)$ through a pseudo-random number generator, where y is an input vector of length n bits, and $y = [y_1, y_2, y_3, \ldots, y_n]$. Similarly, the reversible transformation UNMASK function of the MASK function uses the positive integer set L to convert the output function s_m into a restored output function r.

(1) Integer set generation: The vector y is used as the seed of the pseudo-random number generator PRNG circuit to generate a set of positive integers l $\{ l_1, l_2, l_3, \ldots, l_l | l_i \in A^+\}$. The integer set L contains l n-bit positive Integer; the maximum value of any positive integer is 2^n-1.

(2) Function range transformation: It defines a range function Range() as a linear mapping transformation, given an l-digit integer $\{ k | k \in K\}$, whose value range is [0,

2^{l} - 1], generate an m-digit The new set R = { r|r ∈ Z^{+}} with integers in the Range [0, 2^{m}–1], where m ≤ l. The following equation governs the linear range mapping:

$$M_{\text{new}} = \left[\frac{(M_{\text{old}} - M_{\text{oldamin}}) \times (M_{\text{neemax}} - M_{\text{neemin}})}{(M_{\text{oldmax}} - M_{\text{oldanin}})} \right]$$

$M_{\text{old}} \in$ L is the input of the Range () function, M_{oldamin} and M_{oldmax} are the minimum and maximum values in the Range [0, 2^{l} - 1], Nnewmin and Nnewmax are the minimum values in the new Range [0, 2^{m} - 1] with the maximum value.

(3) Bit obfuscation: The MASK function finally completes the bit obfuscation of the sequence based on the Fisher-Yates Shuffler shuffling algorithm, and a finite series of n different elements generates an n! a random permutation algorithm.

The MASK algorithm has two advantages: (1) It effectively hides the relationship between device PUF excitation and response; (2) It verifies the device's input. The PUF will not be activated without verifying the input stream, so the device will not generate any response, effectively preventing the device from any brute force attack.

3 Protocol Design and Analysis

This paper proposes a lightweight key-sharing authentication protocol based on the IoT device embedded with PUF, including the protocol registration phase, two-way authentication, and firmware update phase. The related symbols of the protocol are shown in Table 1.

Table 1 Description of protocol-related symbols

Symbol	Meaning
CRO PUF	Reconfigurable Ring Oscillator PUF
Delay Matrix M_A/ M_B	Delay Matrix M_A or M_B
(C,R)	Stimulus C and response R generated by PUF
Timestamp()	Timestamp function
(n_{i1}, n_{i2})	Pseudo-random number
Hash(·)	One-way hash function
Fisher - Shuffler()	Shuffle Confusion Algorithm
PRNG()/TRNG()	Pseudo/True Random Number Generator

4 Formal Security Proof

4.1 Formal Security Analysis

This protocol ensures the channel transmission security of IoT devices and can also defend against physical attacks on PUF by attackers. The specific security analysis is as follows.

(1) Modeling attack. The machine learning modeling attack is aimed at a strong PUF with a publicly accessible CRP interface. The attacker collects many CRPs, trains, learns, and optimizes an accurate model to predict response.

(2) However, the reconfigurable CRO PUF is a weak PUF used for key generation. There is no access interface to read the key generated inside the chip, and the key will not be exposed to attackers. At the same time, due to the protection of the protocol mechanism, it is protected by the MASK algorithm (D_A, D_B) value. The Fisher-Shuffler confusion algorithm divides the response value into two parts (s_{A1}, s_{A2}) and (s_{B1}, s_{B2}) and uses the Hash algorithm and random number generator to protect part of the response value s_{A1}in the device. Since the Hash function is one-way, the attacker cannot obtain the real CRP value by eavesdropping on the content e_A, so it is difficult for the attacker to carry out machine learning modeling attacks on PUF.

(3) Untraceability. In the IoT device identity authentication process, if the attacker cannot effectively associate the request and response information of the two authentications, the input and output results cannot be mapped, and the device is considered untraceable. The attacker passes when eavesdropping to obtain messages E_A and E_B because the incentives D_A and D_B are encrypted and protected by the MASK function; the attacker cannot infer the value of the incentive D_A and D_B. After the attacker eavesdrops on the messages e_A and e_B, that is, $e_A \leftarrow Hash_A(s_{A1}, PRNG(\ n_{B2}))$, $e_B \leftarrow Hash(s_{A2}, PRNG(n_{A2}))$. Due to the uniqueness of the hash function, it cannot obtain the values of the shared keys S_A and S_B. Therefore, this protocol can prevent location tracking, as shown in Fig. 1.

(4) Desynchronization attack. In the update phase of the protocol, the device generates new random numbers $(n_{A1}^{\text{new}}, n_{A2}^{\text{new}})$ and $(n_{B1}^{\text{new}}, n_{B2}^{\text{new}})$ and also stores the random numbers $(n_{A1}^{\text{old}}, n_{A2}^{\text{old}})$ and $\left(n_{B1}^{\text{old}}, n_{B2}^{\text{old}}\right)$. When device B is attacked by desynchronization, the random number of device A will be updated typically, but the random number of device B will not be updated. When the next authentication round is performed, the server will return the value E_A, n_{B2}^{old}, given to device A, and the value E_B, n_{A2}^{new}, is returned to devise B. Since the verification value e_A^{old} returned by device A is consistent with the previous authentication round, the dnew B returned by device B is different from the last round. Therefore it is possible to detect that the devices are not synchronized.

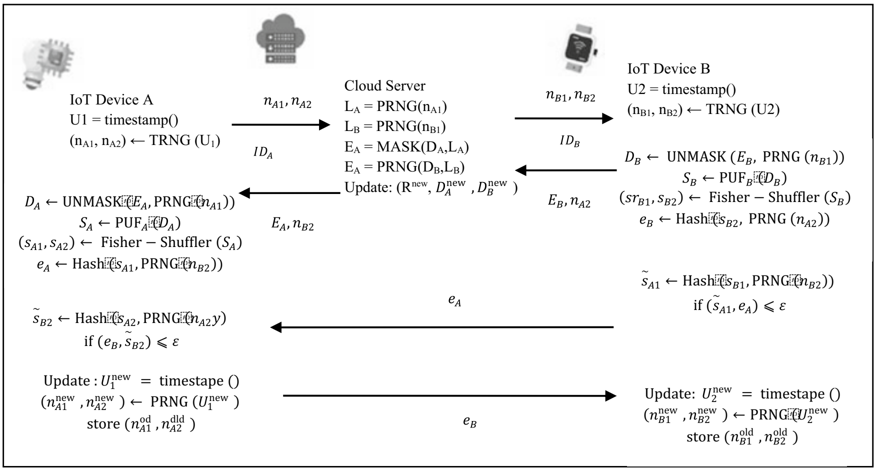

Fig. 1 Two-way authentication and firmware update phase

(5) Replay attack. The protocol mechanism uses timestamps and updated random numbers (n_{A1}, n_{A2}) and (n_{B1}, n_{B2}) to defend against replay attacks. Taking the authentication of device A as an example, assuming that the i-th and i + 1 random numbers are $\left(n^{i}_{A1}, n^{i}_{A2}\right)and\left(n^{i+1}_{B1}, n^{i+1}_{B2}\right)$, when the attacker obtains the i-th session message E^{i+1}_{A}, e^{i+1}_{B}, , and performs the i + 1th authentication, device A receives The session message of has been updated to E^{i+1}_{A}, e^{i+1}_{B}. Therefore, the attacker authentication will fail, and the replay attack will be detected.

(6) Counterfeit attack. When the attacker pretends to be a legitimate device, it must send valid messages e_A and e_B. Using device A as an example, because the generation of information e_A requires valid s_{A1}, the input of the PUF is protected by the MASK function to stimulate CA. At the same time, the response S_A is protected by the obfuscation algorithm. Due to the uniqueness of the Hash function, effective information s_{A1}cannot be obtained even if the information e_A is obtained. Therefore, in this protocol mechanism, the attacker cannot pretend to be a legitimate device and authenticate with the server. Similarly, when the attacker Masquerades as a server, the value of incentive DA and D_B cannot be obtained, so mutual authentication with the device cannot be accepted.

(7) Man-in-the-middle attack. This protocol can perform a man-in-the-middle attack in defense, the attacker cannot obtain valid information by eavesdropping the messages E_A, n_{B2}, E_B, n_{A2}, e_A, and e_B. Because the eavesdropped messages are all encrypted information, if the attacker replaces the new message, the devices will not be able to identify each other causing authentication failure. Furthermore, suppose the attacker wants to parse the encrypted information. In that case, it can be known from the above impersonation attack that the attacker cannot obtain the CRP pair (D_A, S_A) of the PUF, so the authentication between devices will fail.

4.2 Protocol Proof

This section uses BAN logic to prove the security of the shared key generated by PUF. BAN logic is a formal analysis method for authentication protocols, which can prompt some defects that are difficult to find by non-formal methods. The logic symbols commonly used in BAN logic are shown in Table 2. Table 3 shows the logic rules widely used in BAN logic related to this section.

For the convenience of analysis, devices A and B are denoted as E1 and E2, and the server is designated as S. After idealizing the protocol, the message-sending rules are adjusted as follows:

$$\begin{aligned}
&\text{N1}: E1 \rightarrow T: n_{A1}, n_{A2};\ \text{N2}: E2 \rightarrow T: n_{B1}, n_{B2}\\
&\text{N3}: T \rightarrow E1: \{d_A, n_{A1}\}_{n_{A1}}, n_{B2};\ \text{N4}: T \rightarrow E2: \{d_B, n_{B1}\}_{n_{B1}}, n_{A2}\\
&\text{N5}: E1 \rightarrow E2: \{s_{A1}, n_{B2}\}_{n_m};\ \text{N6}: E2 \rightarrow E1: \{s_{B1}, n_{A2}\}_{n_e}
\end{aligned}$$

Protocol initialization assumes the following:

$$\begin{aligned}
&\text{A1}: E1| \equiv \neq \{n_{A1}, n_{A2}\} \text{A2}: E21 \equiv \neq \{n_{B1}, n_{B2}\}\\
&\text{A3}: E11 \equiv E1 \underset{\leftrightarrow}{n_n}\ E11 \equiv E1 \underset{\leftrightarrow}{n_{B_j}} T\\
&\text{A4}: E21 \equiv E2 \underset{\leftrightarrow}{n_m} T;\ E21 \equiv E2 \underset{\leftrightarrow}{n_m} T\\
&\text{A5}: E11 \equiv E1 \underset{\leftrightarrow}{n_m} E2,\ E21 \equiv E1 \underset{\leftrightarrow}{n_m} E2
\end{aligned}$$

The target formula is as follows:

Table 2 Description of logic symbols commonly used in BAN logic

Symbol	Meaning	Symbol	Meaning
$P \mid \equiv X$	P believes that X is true	$P \mid \Rightarrow X$	Entity P has jurisdiction over X
$P \mid \sim X$	P used to send message X	#(X)	X is fresh
$P \lhd X$	Entity P receives message X	(X,Y)	X or Y is part of (X, Y)
$\{X\}_K$	Use key K to message X perform cryptographic operations	$P \overset{K}{\rightarrow} Q$	K is between P and Q shared key

Table 3 Logic rules widely used in BAN logic

Rules	Logical expression
Rule 1: Message Meaning Rules	$\frac{P \mid = P \underset{\leftrightarrow}{k} Q, P \lhd \{X\}_k}{P \mid = Q \mid \sim X}$
Rule 2: The Freshness Rule	$\frac{P \mid = \#(X)}{P \mid = \#(X,Y)}$
Rule 3: Temporary Value Validation Rules	$\frac{P \mid = \#(X), P \mid = Q \mid \sim X}{P \mid \equiv Q \mid = X}$
Rule 4: Trust Rules	$\frac{P \mid = Q \mid = (X,Y)}{P \mid = Q \mid = X}$

$$\text{G1} : E1 \equiv\equiv \text{T} \equiv \{d_A\}; \text{H3} : E2| \equiv E1| \equiv \{s_{A1}\};$$
$$\text{G2} : E2| \equiv T| \equiv \{d_B\}; \text{H4} : E1\text{E} \equiv E2 \ |\equiv \{s_{B1}\};$$

Protocol initialization assumes the following:

If the target formula is established, both the device and the server T negotiate and confirm the secret key with each other, and the private key is bound to the platform integrity report and the communication channel. The following logical reasoning can be made according to the rules in Table 3.

From rule 1 and A_3, N_3 we know.

Therefore, the target formula G1 can be proved: D1|≡S|≡ { cA}, device A and server S share key cA.

From rule 1 and A4, M4 we know

$$\frac{E1|\equiv E1 \underset{\longleftrightarrow}{nA_1} T, \ E1 \triangleleft \{d_A, n_{A1} A_1\}}{E1||\equiv T| \sim \{d_A, n_{A1}\}}. \tag{1}$$

From rules 2 and A2, we know

$$\frac{E1 \ |\equiv \#(n_{A1})}{E1 \ |\equiv \#(d_A, n_{A1})} \tag{2}$$

According to rule 3 and formula (1) and formula (2):

$$\frac{E1|\equiv \#(d_A, n_{A1}), E1| \equiv T \ |\sim \{d_A, n_{A1}\}}{E1| \equiv T| \equiv \{d_A, n_{A1}\}} \tag{3}$$

From rule 4 and formula (3), it can be seen that

$$\frac{E1| \equiv T| \equiv \{d_A, n_{A1}\}}{E1| \equiv T| \equiv \{d_A\}} \tag{4}$$

Therefore, it can be proved that the target formula H1: E1|≡T|≡ {d_A}, equipment A shares secret key d_A with server T.

From rule 1 and A_4, N_4 we know

$$\frac{E2|\equiv E2 \underset{\longleftrightarrow}{n_{m1}} T, \ E2 \triangleleft \{d_B, n_{B1}\}_{n_{n1}}}{E2|\equiv T| \sim \{d_B, n_{B1}\}} \tag{5}$$

From rule 2 and message A2, we can know

$$\frac{E2 \ |\equiv \#(n_{B1})}{E2 \ |\equiv \#(d_B, n_{B1})} \tag{6}$$

Similarly, according to rules 3 and 4, and formula (5) and formula (6), the target formula can be proved: H2: E2| ≡ T| ≡ {d_B}.

From rule 1 and A5, N5 we know

$$\frac{E2|\equiv E1 \overset{n_m}{\leftrightarrow} E2,\ E2 \triangleleft \{s_{A1}, n_{B2}\}n_m}{E2|\equiv E1| \sim \{s_{A1}, n_{B2}\}} \tag{7}$$

From rule 2 and message A2 we know

$$\frac{E2\ |\equiv \#\{n_{B2}\}}{E2|\equiv E1| \equiv \{S_{A1}, n_{B2}\}} \tag{8}$$

From rule 3 and formula (7) and formula (8), it can be seen that

$$\frac{E2|\equiv \#\{s_{A1}, n_{B2}\},\ E2| \equiv E1\ |\sim \{s_{A1}, n_{B2}\}}{E2| \equiv E1| \equiv \{s_{A1}, n_{B2}\}} \tag{9}$$

From rule 4 and formula (9), it can be seen that

$$\frac{E2| \equiv E1| \equiv \{s_{A1}, n_{B2}\}}{E2| \equiv E1| \equiv \{s_{A1}\}} \tag{10}$$

Therefore, the target formula G3 can be proved: E2|≡E1|≡{sA1}, device B and device A share key s_{A1}.

From rule 1 and A5, N6 we know

$$\frac{E1|\equiv E1 \overset{n_{A2}}{\leftrightarrow} E2,\ E1 \triangleleft \{s_{B1}, n_{A2}\}n_{A2}}{E1|\equiv E2| \sim \{s_{B1}, n_{A2}\}}. \tag{11}$$

From rule 2 and A1 we know

$$\frac{E1\ |\equiv \#\{n_{A2}\}}{E1\ |\equiv \#\{s_{B1}, n_{A2}\}} \tag{12}$$

Similarly, from rules 3 and 4, and formula (11) and formula (12), it can be proved that.

Target formula: H4: E1| ≡ E2| ≡ { s_{B1} }.

5 Protocol Performance Analyses

This section analyzes and evaluates the authentication protocol in terms of security attributes, storage capacity, and communication costs. The authentication protocol program between the device and the server is written in Python. The network interaction is completed by abstracting the socket connected by the TCP client/server. It makes the server wait for a connection with the device on the specified IP address.

Once the device establishes a relationship with the server, the protocol performs a mutual authentication session. The server and the machine run on Windows 10, using an Intel Core i7-9750 CPU with a frequency of 2.60 GHz, Equipped with 8 GB RAM, simulating the proposed authentication scheme.

Regarding security attributes, it is compared with other protocols. In the protocol [14–17] mechanism, the attacker can obtain the CRP pair of PUF through eavesdropping, counterfeiting, and physical attack. Therefore, it is impossible to defend against modeling attacks. The protocol [18] can protect against PUF attacks through the d-time locking mechanism. Still, the information in channel transmission is not encrypted, which leads to security threats such as eavesdropping, resynchronization, and replay attacks in device authentication. Protocol [19, 20] device stores the old and new identities. The attacker obtains the current identity information by accessing the memory to trace the authentication information of the previous or next round, so the protocol is not irretrievable. However, this protocol uses the shared key generated by PUF. It uses the encryption of the MASK algorithm and Hashes function to ensure the privacy and non-traceability of the device.

Figures 2 and 3 list the comparison with other protocols regarding device storage and communication overhead. Referring to the paper by Literature [11], the pseudo-random identity PIDiD is 128 bits, the word length of the CRP pair (D_i, S_i) is 128 bits, and the byte length of the key is 96 bits. This protocol only stores 128-bit random numbers (no1ld, no1ld), which is far lower than the storage capacity of other protocols. Furthermore, the protocol only transmits information (n_{A1}, n_{A2}, E_A, n_{B2}, e_A, e_B) and the communication overhead is 640 bits. Compared with other protocols [3, 5, 9–11, 16, 17], the communication cost of the proposed protocol is lower than that of different schemes (as shown in Fig. 3), and it is suitable for lightweight devices—Security authentication scenarios.

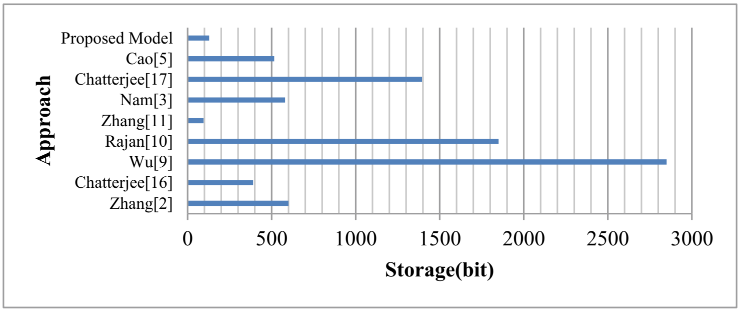

Fig. 2 Performance comparison of device storage

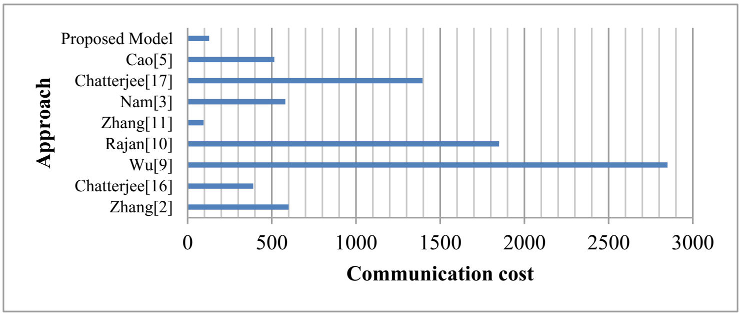

Fig. 3 Communication cost in the proposed protocol

6 Conclusion

The massive data generated by the IoTs brings information transmission security threats to resource-constrained terminal devices. At the same time, hardware devices usually face security issues such as chip cloning, device forgery, and key storage. Therefore, the traditional network protocol is not suitable for the security authentication of lightweight IoT devices. This paper proposes a lightweight anonymous key-sharing authentication protocol for IoT devices. This mechanism uses the characteristics of PUF anti-tampering and anti-cloning to generate shared keys on the hardware side combining security primitives such as obfuscation algorithm, MASK algorithm, and Hash function to ensure security attributes such as anonymity, untraceability, non-repudiation, and forward/backward secrecy of information transmission. Through formal verification tools, ProVerif, BAN logic, and informal, the security analysis and verification of the protocol prove the security, reliability and anti-channel attack ability of the protocol operation. Compared with other existing protocols, the proposed protocol has low computing cost, small communication overhead and storage capacity, and high security, suitable for secure communication transmission of lightweight IoT devices.

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Support Vector Machine for Multiclass Classification of Redundant Instances

Herison Surbakti, Athmakuri Naveen Kumar, Saikumar Tara, Ismail Keshta, Amit khare, and G. N. R. Prasad

Abstract In recent years, support vector machine has become one of the most important classification techniques in pattern recognition, machine learning, and data mining due to its superior classification effect and solid theoretical base. However, its training time will increase dramatically as the number of samples increases, and training will become more sophisticated when dealing with problems involving multiple classifications. A quick training data reduction approach MOIS appropriate for multi-classification tasks is presented as a solution for the aforementioned issues. While eliminating redundant training samples, the boundary samples that play a vital role are chosen in order to considerably reduce training data and the problem of unequal distribution between categories. The experimental results demonstrate that MOIS may maintain or even improve the classification performance of support vector machines while substantially enhancing training efficiency. On the Opt digit dataset, the suggested method improves classification accuracy from 98.94% to 99.05%, while training time is reduced to 15% of the original; in HCL2000, the proposed

H. Surbakti
Faculty of Computing and Digital Technology, HELP University, Kuala Lumpur, Malaysia

A. N. Kumar
Senior Software Engineer, Glosoft Technologies Pvt Ltd, Chennai, India

S. Tara
Department of ECE, BVRIT Hyderabad College of Engineering for Women, Hyderabad, India
e-mail: Saikumar9@ieee.org

I. Keshta
Computer Science and Information Systems Department, College of Applied Sciences, AlMaarefa University, Riyadh, Saudi Arabia
e-mail: imohamed@mcst.edu.sa

A. khare
Department of Technology, JIET Institute of Design and Technology, Jodhpur, Rajasthan, India

G. N. R. Prasad (✉)
Department of MCA, Chaitanya Bharathi Institutute of Technology (A), Hyderabad, Telangana 500091, India
e-mail: gnrp@cbit.ac.in

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method improves classification accuracy from 98.94% to 99.05%. When the accuracy rate is marginally increased (from 99.29% to 99.30%) on the first 100 categories dataset, the training time is dramatically reduced to less than 6% of the original. Additionally, MOIS has a high operational efficiency.

Keywords Machine learning · Pattern recognition · Data mining · Classification · Multi-classification

1 Introduction

In pattern recognition, machine learning, and data mining, Support Vector Machine (SVM) [1] has become a high-profile classification method in recent years due to its excellent classification effect and solid theoretical foundation. Structural risk minimization and convex quadratic programming make SVM more robust classification performance than other classification methods. The kernel method makes SVM also have an excellent classification effect on nonlinear separable problems. SVM has been successfully applied to text recognition [1], image classification [2], financial prediction [3], medical diagnosis [4], and many other scientific and technological fields.

However, the application of quadratic programming in the training process makes the SVM training complexity increase significantly with the rise in the number of training samples. In general, for a training set containing n samples, the time for SVM training on it is complex. The degree is O (n3). In addition, SVM is proposed for the two-class problem. For multi-class problems, the usual practice is to convert the multi-class problem into multiple two-class issues in a one-to-one or one-to-many manner. The one-to-one conversion method requires training an SVM classification model between two classes, and the classification process is more complicated. The one-to-many approach requires much fewer models to be taught, and the classification process is more straightforward, but each model is trained on a severely imbalanced dataset. The model training will be more complicated no matter which transformation method is adopted.

In recent years, with the development of information technology, the scale of data sets is also increasing, among which multi-classification problems exist widely. High training complexity has become the main bottleneck of SVM in many practical applications. How to improve the training efficiency of SVM, especially for multi-classification problems, is an important and urgent research topic. To reduce the training complexity of SVM, researchers mainly explore two ways: to improve the efficiency of quadratic programming and sample selection. To improve planning efficiency, researchers try to decompose the quadratic planning process on the entire training set into a series of small-scale optimization processes. The typical algorithms in this regard are SMO (Sequential Minimal Optimization) [5], SOR (Successive Over Relaxation) [6], Chunking [7], LIBSVM (LIBrary for SVM) [8], and other algorithms. The method based on sample selection is to select the samples that play a decisive role in the training results from the training set to form a smaller training subset and train the SVM classification model on it. Since the training complexity of SVM is highly dependent on the number of training samples, the method of sample selection has a more significant effect on improving the training efficiency. Because of this, this paper will follow this idea, take the cluster center as the reference point, and construct an SVM acceleration algorithm suitable for multi-classification problems through sample selection. This paper first summarizes the related research on improving the efficiency of SVM through sample selection; then introduces the proposed algorithm and verifies its effectiveness through experiments; finally, it outlines the work and suggests further research ideas.

2 Related Work

In the training process of SVM, only the boundary samples as support vectors affect the training results [9]. The remaining samples are either redundant samples that do not work or noise samples that damage the training results. Based on The SVM acceleration method of sample selection is to delete redundant and noisy instances and select boundary samples that may be support vectors. Based on this idea, author [10] proposed the SVM-KM algorithm. The algorithm first clusters the training samples.

If a cluster only contains samples of the same class, the cluster center of the cluster replaces all samples in it; if a collection contains samples of different types; all models in the collection are retained. SVM-KM has a higher [11] proposed training sample deletion algorithm. First, an initial SVM was obtained by training a small-scale sample subset, and then the original sample set was separated from the initial SVM. The training samples that are far from the classification hyperplane are deleted. This method can delete some pieces irrelevant to the classification, but the selection effect depends on the initial SVM. Author [12] proposed a fast sample selection method NPPS (Neighborhood Property Based Pattern Selection Algorithm), using the neighborhood information of training samples for analysis, NPPS selects selections near the classification hyperplane to form the final training set. This method can significantly improve the training speed, but noise data quickly disturbs election results. Angiulli (ConsistentSub-Set) gave a sample selection algorithm FCNN (Fast Condensed Nearest Neighbor Rule) [13]. This algorithm reduces the sample size to a large extent, which is easy to reduce the classification accuracy. Author [14] used the class distribution information and geometry of the K nearest neighbors of the sample. The feature gives the BEPS (Border Edge Pattern Selection) algorithm to select critical samples. This algorithm can fully use the distribution information of the pieces in the training set. Still, it is easy to choose more samples, and the values of 4 hyper parameters need to be given, increasing the algorithm's application. [15] Randomly selected a certain proportion of samples from the training set to form multiple training subsets, trained an SVM model on each subgroup, and then used these models to evaluate each sample based on the evaluation results selection of samples. Since the evaluation result of each piece is generated by the SVM trained on a randomly selected subset of models, the selection result inevitably has a certain degree of randomness.

The above research has achieved apparent results in improving the efficiency of SVM for binary classification problems. However, for multi-classification problems, these algorithms often find it difficult to achieve satisfactory results in terms of efficiency and effect. To solve this problem, this paper constructs a suitable fast sample selection method MOIS (Multi-classification Oriented Instance Selection). This algorithm has apparent advantages over existing algorithms in terms of efficiency and effect for multi-classification problems.

3 MOIS Algorithm

3.1 MOIS Algorithm Framework

As mentioned above, to use SVM for multi-classification problems, the original problem needs to be transformed into multiple binary classification problems. Compared with the one-to-one conversion method, the one-to-many approach not only trains the model much less but also makes the classification process simpler and more efficient, just solving the imbalance between positive and negative classes caused by this method. Therefore, the MOIS algorithm will use a one-to-many conversion method.

Assume that the training set A is composed of L types of samples, the total number of samples in A is N, and the number of c ($1 \leq c \leq L$) type samples is M_c. Then, according to the one-to-many transformation method, each type of sample in A takes turns to be a positive class sample, and the rest of the samples are taken as the current negative class sample. In the following, we consider the boundary sample selection process when the lth ($1 \leq l \leq L$) class is used as the current positive class, where T_l represents the result of the selection.

First, use a specific clustering method (k-means clustering method is used in this paper) to cluster the current positive samples, the lth class samples, and assume that the k cluster centers are D_1, D_2,.., D_k. Then, for each D_i ($1 \leq i \leq k$), calculate its distance $e(x, D_i)$ to each sample x. Finally, select a proportion of the more significant $d(x, D_i)$ from the lth class positive class samples. At the same time, a certain proportion of negative class samples with minor $e(x, D_i)$ from each other class. The basis of this approach is as follows.

For positive examples, the closer it is to the cluster center of the positive class, the more likely it is an interior point; otherwise, the more likely it is a boundary point. For negative examples, the closer it is to the center of a positive cluster, the more likely it is a boundary point Big.

3.2 Determination of Parameters in MOIS

The following key question is how to determine the total sample selection ratio and the selection ratio of positive and negative samples.

3.2.1 Determination of the Selection Ratio of the Total Sample and the Number of Clusters

This section presents a simple and effective method to determine the total sample selection ratio and the number of groups. This method allows the user to choose the appropriate balance and number of groups according to the computing resources and the size of the dataset. Generally, a small data set with fewer samples has higher sparseness and a higher proportion of selected models. On the contrary, for a data set with a large number of pieces but not a high dimension, due to the denser parts, a lower ratio is often selected, i.e., the ideal effect can be achieved. Therefore, we can first determine the range of the scale value according to the size of the data set and then select an appropriate value within this range to verify through experiments. For example, a small data set can be in {0. 2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5, 0.55, 0.6, 0.65, 0.7} through experimental verification to select an appropriate one value.

For the number of clusters k in the positive class, it is also selected according to the size of the positive course. When there are few positive class samples, the value of k is generally smaller; otherwise, a more significant deal of k is required. It is found through experiments that the ideal selection effect can be obtained when the value of k is between 1 and 7.

3.2.2 Determination of the Selection Ratio of Positive and Negative Examples

When determining the selection ratio of positive and negative examples, considering that the current negative class is composed of multiple courses other than the lth class, it is very likely that the number of positive samples is significantly less than the number of negative examples. The selection ratio should help to eliminate this high imbalance between positive and negative classes. At the same time, considering that the boundaries of antagonistic classes composed of multiple types are generally more complex, the number of negative examples should be appropriately selected more than positive models. For instance, in the MOIS algorithm, the number of negative cases is twice that of positive points.

Also, when selecting negative examples, should I choose from the entire negative class, or should I choose from each category that makes up the negative class? Since the multiple courses that make up the negative class tend to be of different sizes, we should choose from each negative example. The number of negative examples should be proportional to the class size. Assuming that the total number of negative criteria set is M_{-}^{s}, , then the negative examples are chosen from the i-th ($1 \leq i \leq M$, $I \neq l$) class. The number M_i^s is

$$M_i^s = \frac{M_{-}^{S} - M_i}{\sum_{1 \leq c \leq, c \neq l} M_C}. \tag{1}$$

Based on the above analysis, the steps of the MOIS algorithm are shown in Algorithm 1.

Algorithm 1 MOIS Algorithm

Input: A training set U consisting of M samples divided into K classes, the sample selection ratio r

Output: the selection result Sl when the l ($1 \leq l \leq K$) class is the current positive class

Step 1: Cluster the lth ($1 \leq l \leq K$) class samples as the current positive class, and obtain k cluster centers $D_1, D_2,.., D_k$

Step 2: For each cluster center D_i($1 \leq i \leq l$), calculate the distance $e(x, D_i)$ to each sample x.

Step 3 Calculate the number of positive examples to be selected M_{+}^{s}, M_{-}^{s} as follows:

$$M_{+}^{s} = \min\left\{\frac{1}{3}M \cdot r, M_1\right\}, M_{-}^{s} = \frac{2}{3}M \cdot r_{\circ}$$

Step 4: Select M_{+}^{s} samples with more significant $e(x, D_i)$ from the current positive class and put them in T_l.

Step 5: Select M_i^s negative examples with small $e(x, D_i)$ from the i-th ($1 \leq i \leq K$, $i \neq l$) class.

Enter T_l. Among them, M_i^s is determined by the formula (1).

By training SVM on the obtained selection result T_l, the l classification model can be obtained, and if l is set to 1, 2,.., K classification models will be accepted.

3.3 Complexity Analysis of MOIS Algorithm

In Step 1 of Algorithm 1, the complexity of the clustering process on the positive class is O(UkM/K). Among them, the value of l is not greater than 7, which is the number of cluster centers; U is the number of clustering centers in the clustering process the number of cycles. When k is set to 1, the complexity is reduced to O(M/K). The complexity of step 2 in algorithm 1 is O(lM). Steps 3 to 5 in algorithm 1 are mainly for each sample type. Sorting according to the distance metric, the complexity is O(kMlog(M/K). In summary, the complexity of MOIS is O(Mlog(M/K). It can be seen that MOIS has high efficiency. The following effectiveness of the MOIS algorithm will be verified by experiments.

4 Experimental Verification

To verify the effectiveness of the MOIS algorithm, it is compared with the most representative three algorithms, NPPS [12], FCNN [13], and BEPS [14], in the experiment. These three algorithms are used to accelerate the SVM, and achieved significant results. In the experiment, the above four algorithms involved in the comparison will be evaluated from the following aspects: (1) the impact on the classification effect; (2) the simplification of the data set; (3) the effect on the training efficiency; and (4) The execution efficiency of the algorithm itself.

4.1 Experimental Dataset

Several standard multi-classification datasets and a practical handwritten Character recognition dataset are used in the experiment. Table 1 lists the number of samples (Size), number of features (#Fea), number of categories (# Cls), the number of training samples (#Trn), and the number of test samples (#Tes). Except for HCL2000, the rest of the datasets are from the UCI machine learning database [16], and each dataset has been divided into a training set and a Test set. HCL2000 is a handwritten character set with 3755 classes; each class contains 1000 samples, 700 for training, and 300 for testing. The test time is too long, so only the first 100 kinds of models are used for the experiment. By extracting the 8-directional gradient features, the number of elements of each sample is 512.

Table 1 Experimental dataset

Data set	Size	#Fea	#Cls	#Trn	#Tes
Dermatology	368	44	16	292	86
Glass	224	19	16	178	56
HCL2000	100,010	522	110	70,010	30,010
Iris	160	14	13	127	43
Isolet	7807	627	36	6248	1569
Letter	20,010	26	36	16,010	4010
Optdigits	5630	74	20	3833	1807
Pendigit	11,002	26	20	7504	3508
USPS	9308	266	20	7301	2017

4.2 Experimental Parameter Settings

To avoid running too long on high-dimensional datasets, the usual dimensionality reduction measures are adopted for high-dimensional datasets such as HCL2000, Isolet, and USPS. Among them, HCL2000 is reduced to 99 dimensions by the LDA method, and Isolate and USPS use PCA, respectively. Down to 150 and 80 sizes.

The kernel function of the SVM used in the experiment is always the Gaussian function defined by Eq. (2).

$$l\left(y_i, y_j\right) = \exp\left(-\left\|y_i - y_j\right\|^2 \Big/ 2\sigma^2\right) \tag{2}$$

The error limit parameter on each dataset and the parameter σ in Eq. (2) is optimized through experimental verification. The selection ratio and the number of clusters in MCIS are set according to the instructions in Sect. 3.2. In the NPPS algorithm, the number of neighbors in l_b, l_e, λ, and γ in BEPS are all set according to the method in the original text. Here, only the FCNN algorithm does not need parameter settings.

4.3 Experimental Results

To compare the performance of various algorithms on the experimental data set, Tables 2, 3, 4 and 5 list the classification accuracy, sample selection ratio, model training time, and sample selection time obtained by different algorithms on each data set. "All" in the table indicates the case of using the entire training set for training.

From Table 2, it can be found that among the four sample selection algorithms, MOIS maintains the highest classification accuracy on all experimental datasets

Table 2 Classification accuracy on the experimental dataset

Data set	Algorithm				
	All	Beps	Fcnn	Mois	Npps
Dermatology	99	87.16	91.11	97.68	93.74
Glass	68.57	55.52	55.52	64.22	57.7
Hcl2000	98.29	98.17	98.02	98.3	97.28
Iris	99	99	89.91	99	62.64
Isolet	95.34	94.32	94.77	95.06	93.74
Letter	96.97	92.65	94.67	96.4	85.42
Optdigits	97.94	97.83	96.61	98.05	96.83
Pendigit	97.8	97.26	95.8	97.46	97.34
Usps	94.81	94.17	94.52	93.67	94.62

Table 3 Sample selection ratio on the experimental data set

Data Set	Algorithm			
	BEPS	FCNN	MOIS	NPPS
Dermatology	0.74	0.43	0.54	0.65
Glass	0.49	0.55	0.62	0.73
Hcl2000	1.088	0.17	0.17	0.27
Iris	0.41	0.25	0.53	0.42
Isolet	1.01	0.4	0.5	0.65
Letter	0.56	0.28	0.45	0.57
Optdigits	1.03	0.19	0.35	0.51
Pendigit	0.48	0.15	0.5	0.39
Usps	1.06	0.22	0.44	0.57

Table 4 Training time on the testing dataset

Data Set	Algorithm				
	All	Beps	Fcnn	Mois	Npps
Dermatology	0.2	0.16	0.11	0.08	0.14
Glass	0.17	0.05	0.03	0.06	0.08
Hcl2000	23.92	7616.8	1456.2	3378.6	–
Iris	0.03	0.03	0.03	0.01	0.01
Isolet	290.97	231.56	76.8	89.99	116.86
Letter	2139.5	885.29	228.06	596.42	713.03
Optdigits	65.03	59.13	4.23	9.75	25.22
Pendigit	36.22	12.88	2.06	12.73	12.36
Usps	197.78	142.23	14.53	43.42	78.49

except USPS, and even on HCL2000 and Optdigits datasets, MOIS makes the classification accuracy more elevated than the original. This is because the data set has been improved. Only on the USPS, MOIS has slightly lower classification accuracy than the other three algorithms. In contrast, the three different algorithms do not maintain classification accuracy very well.

Judging from the sample selection ratio listed in Table 3, the selection ratio of MOIS is only higher than that of FCNN in general but significantly lowers than that of NPPS and BEPS. It is worth noting that the high simplification of FCNN can easily lead to a significant drop in classification accuracy. Observing the SVM training time listed in Table 4, it can be found that the four sample selection algorithms can significantly shorten the training time. In contrast, MOIS's ability to shrink the training time is slightly inferior to that of FCNN and substantially better than NPPS and BEPS.

Table 5 Running time of the four selection algorithms on the dataset

Data set	Algorithm			
	Beps	Fcnn	Mois	Npps
Hcl2000	970.59	6870.9	61.98	1088.5
Isolet	8.76	23.19	2.97	6.88
Letter	17.78	55.28	1.87	19.42
Optdigits	5.53	3.86	1.17	7.5
Pendigit	4.92	2.47	1.16	6.05
Usps	9.55	5.55	1.59	12.52

Since the running times of the four selection algorithms on small-scale datasets are very short, it is difficult to detect significant differences in their running times. Therefore, Table 5 only lists the running times of these algorithms on larger-scale datasets. It can be found in Table 5 that the running time of MOIS is significantly shorter than that of other algorithms, generally only a fraction or even several tenths of the different algorithms, which fully shows that the running efficiency of MOIS is significantly higher than that of other algorithms.

Conclusion As one of the most crucial classification methods in pattern recognition, machine learning and data mining, the support vector machine has an excellent classification effect. However, the model training time of this method will increase significantly with the increase of samples, especially when processing many. In the case of classification problems, the model training will be more complicated. Therefore, not only does the number of trained models increase significantly, but the model's training effect needs to be improved. To solve the above problems, this paper presents a fast training data reduction algorithm MOIS suitable for multi-classification problems. The method first clusters the current positive class, then takes the obtained cluster center as a reference point, removes redundant samples, selects the boundary samples that play a decisive role, and reduces the selection ratio by appropriately controlling the selection ratio of positive and negative examples unbalanced distribution among categories. Different from the previous data reduction methods based on clustering, MOIS clustering only one category of samples and the clustering speed is much faster than the previous algorithm. In addition, by deleting redundant pieces and selecting boundary samples by doing, the data reduction effect of MOIS is better. Compared with several algorithms with excellent performance in the past, it is found that MOIS is significantly better than other algorithms in maintaining the classification effect of support vector machines, and its operational efficiency is also considerably higher.

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Energy Efficient Lightweight Scheme to Identify Selective Forwarding Attack on Wireless Sensor Networks

Malik Jawarneh, M. Jayakrishna, Sandeep Kumar Davuluri, S. V. Ramanan, Pavitar Parkash Singh, and J. Armstrong Joseph

Abstract To protect wireless sensor networks from selective forwarding attacks, a lightweight WSN selective forwarding attack detection technique (LSFAD) is presented. The proposed approach identifies selective forwarding attack paths based on the average packet loss rate (PLR). It compares the PLR of attacked path with its normal PLR, and it can detect and pinpoint malicious links by doing the same for each node. The LSFAD technique detects malicious pathways during regular data packet sending and receiving without disrupting the normal operation of the entire network. Its simple design eliminates the need for listening nodes and sophisticated evaluation models. According to the results of the security and performance investigation, the LSFAD scheme is secure against passive and active selective forwarding attacks launched by malicious nodes. The LSFAD method has a significantly lower communication overhead compared to similar schemes. Despite an average PLR of 0.125 on the link, the LSFAD method detected the selective forwarding attack path in experimental simulations. Identifying the malicious nodes that launch a selective

M. Jawarneh
Faculty of Computing Sciences, Gulf College, Muscat, Oman
e-mail: Malik@gulfcollege.edu.om

M. Jayakrishna (✉)
Mechanical Engineering, Sri Sivani Engineering College, Srikakulam, India
e-mail: jayakrishnamakka555@gmail.com

S. K. Davuluri
Department of Information Technology, University of the Cumberlands, Williamsburg, KY 40769, USA

S. V. Ramanan
Department of Electronics and Communication Engineering, PPG Institute of Technology, Coimbatore 641035, India
e-mail: ramanan.it@ppg.edu.in

P. P. Singh
Department of Management, Lovely Professional University, Phagwara, India
e-mail: pavitar.19476@lpu.co.in

J. A. Joseph
Department of Computer Science and Engineering, Sri Venkateswara College of Engineering and Technology (Autonomous), Chittoor 517127, India

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forwarding attack is possible when the link's usual PLR is more than 0.025. It is effectively detected and localized, and the network's energy consumption for doing so is not significantly higher than under normal conditions.

Keywords Wireless sensor network · Selective forwarding attack · Energy consumption · Data packet · Lower communication overhead

1 Introduction

Wireless sensor networks (WSNs) are employed in the military, environmental monitoring, transportation, agriculture, medical, and home furnishing [1]. Security issues have always been a study focus. WSNs are multi-hop networks of sensor nodes. Sensor nodes are vulnerable to internal and external attacks, including selective forwarding attacks [2]. It is a severe internal assault [3], and the wireless channel's instability makes it hard to identify the network link's average packet loss from the selective forwarding attack's malicious packet loss, which has high concealment. Thus, detecting, locating, and isolating rogue nodes that initiate forwarding attacks is critical.

Recently, scholars have presented effective selective forwarding attack detection techniques. Multi-hop confirmation-based forwarding attack detection techniques were presented [4–8]. Multi-hop confirmation requires more confirmation packets, which increases communication costs and reduces network life. Trust evaluation-based detection techniques [9–13] can identify harmful network nodes. Most detection techniques have a static trust threshold and misinterpret regular nodes. Thus, malevolent nodes are frequently misidentified, requiring more monitoring nodes and increasing network overhead. Literature [14] proposed malicious node detection schemes based on learning automata. However, literature [15] did not perform well in detecting selective forwarding attacks that do not generate malicious data packets. Literature [16] proposes two abnormal node detection schemes to detect particular forwarding attacks effectively. Still, the reward and punishment parameters of the learning automata of these two schemes are fixed values artificially set according to the voting situation of node neighbors, and the detection method is not flexible. To solve this problem, literature [17] proposes a detection scheme that can dynamically adjust the reward and punishment parameters of the learning automaton, which improves the flexibility and applicability of the detection method. Still, this scheme needs to be determined by the number of confirmation packets replied to by the neighbor nodes. Defining the environment's feedback to the learning automaton consumes many network resources.

The proposed LSFAD strategy is based on this crucial observation. When a malicious node is on a selective forwarding attack route, the base station (BS) receives far fewer data packets from the originating node. Based on this observation, the BS records each source node's total data packets and routing path. The average and regular Packet loss rate (PLR) are determined by it. If the course's average PLR is higher than the way's, the current form is under selective forwarding attack. Each intermediate forwarding node tracks the data packets provided by the forwarding source node to the BS to identify the malicious node or connection that starts the forwarding assault. If a selective forwarding attack is found, the BS will notify each node of the quantity of data packets sent by the source node to the BS. The BS then estimates PLRs. The node with the highest average PLR is malevolent.

2 Related Work

Currently, the selective forwarding attack detection schemes proposed by scholars at home and abroad can be roughly divided into projects based on the multi-hop confirmation model, methods based on the trust evaluation model, and strategies based on learning automata.

Author [18] presented a two-hop acknowledgment detection system based on the confirmation node since each node is a confirmation node and sends an Acknowledging packet every time it gets a data packet, which dramatically increases network message collisions and collisions. Author [19] presented a multi-hop confirmation detection technique. This approach can greatly reduce network message collisions if more than two malicious nodes are chosen. Malicious nodes collaborate to confirm the nodes, invalidating the plan. This difficulty was solved by the author [20]'s multi-hop confirmation mechanism system (per-hop acknowledgment-based method, PHACK). PHACK places all nodes on the forwarding path except for average forwarding data. A confirmation packet for each forwarded packet must be generated and sent to the source node over several paths to detect and find rogue nodes. Author [21] proposed a selective forwarding attack detection strategy using the multi-hop confirmation and trust evaluation model to reduce network overhead and enhance the detection rate of numerous malicious nodes on the path. Author [22] also developed a wireless ad hoc network-based selective forwarding attack detection technique that uses a hop-by-hop confirmation mechanism and an upgraded two-hop confirmation mechanism to detect demanding forwarding behavior to reduce resource overhead. The source node receives the confirmation packet over the original data forwarding mechanism.

Author [23] proposed two wireless sensor network intrusion detection schemes based on learning automata. Since these two schemes are not malicious behaviors of detection nodes, although detecting malicious data packets is very effective, they are not suitable for generating malicious data packets. Selective forwarding of packages does not seem well.

3 System Model

3.1 Network Model

The sensor network comprises a base station, ordinary nodes, and malicious nodes. Unique identity mark ID_i is given to each node$_{_i}$ at beginning of deployment and a symmetric key $L_{i,BS}$ is shared with BS. They assume that all nodes are no longer moving after the network is deployed to the target area. With BS as the root, all nodes form a tree structure; as the network runs, some nodes may die due to battery exhaustion, causing the network path changes, so every once in a while, BS will update the topology of the entire network in time. When I perceive the data, it will send it to BS in a multi-hop manner. While sending the data packet from source node I to BS, each intermediate forwarding node will record that it has forwarded the data transmitted by source node I. The number of packages BS will record the total number of data packets sent by each source node I and the forwarding routing path of these data packets. On the forwarding path of data packets, not only malicious nodes on the way will discard the data packets it should forward with a certain probability, but also because of the instability of the physical layer wireless channel and the collision of MAC (media access control) layer data packets, any A communication link between two intermediate forwarding nodes may also drop packets typically. Scholars at home and abroad have proposed many schemes for generating and fusion sensing data [19, 20]. This paper will not describe this in detail here. This paper mainly focuses on detecting the selective forwarding attack path and locating the malicious node or node that initiates the attack malicious link. For the convenience of reading and explanation, Table 1 describes the symbols used in this paper.

3.2 Attack Model and Security Goals

Assume that in the network, except for the BS, any other node may be captured by the enemy. Once captured, these nodes will become malicious nodes, and the enemy will obtain security information such as identities and keys, and use these nodes to launch a series of attacks, such as injecting false data attacks, wormhole attacks, cloning attacks, Sybil attacks, and selective forwarding attacks, etc. This paper considers only particular forwarding attacks initiated by malicious nodes. Special forwarding attacks can be divided into passive and active, selective attacks. A passive special

Table 1 Packet forwarding table

Source_ID	ForWord_Count
…	…
ID	ForWard_Count$_i$
…	…

forwarding attack means malicious nodes only discard regular data packets with a certain probability and ignore the demanding forwarding attack detection behavior in the network. On the other hand, the active, particular forwarding attack not only loses regular data packets with a certain chance but also interferes with the detection behavior of special forwarding attacks to avoid being detected.

This paper's LSFAD technique resists malicious passive selective forwarding attacks and deliberate nodes' active, specialized forwarding attacks.

4 LSFAD Scheme

This study proposes a four-step LSFAD scheme: the source node generates the data packet, the intermediate node forwards it, and the BS detects the selective forwarding attack path and locates the malicious node.

4.1 Generate Data Packets

After producing the data packet P, the source node I will create two fields to record the unique identifying tag of the data packet source node I and the sequence of the current data packet Number while sending sensing data to BS. Then, node I hold its identity tag ID_i and data packet sequence number Seq_Number_i into the corresponding fields, where the sequence numbers of data packets are continuous, and finally sends the data packet to the next intermediate forwarding node.

4.2 Forwarding Packets

As illustrated in Table 1, all intermediate forwarding nodes retain a packet source node forwarding table (forward data table, DFT) with the Source ID field identifying the source node and the ForWord Count field indicating the number of packets passed.

When the intermediate forwarding node j receives a data packet Q, it first checks in its data forwarding table DFT whether there is a record whose source node identity is $Q.ID_i$; if not, then creates a new one in the data forwarding table DFT record, make DFT.Source_ID = Q.IDi, DFT.ForWord_Count = 1; if it exists, add 1 to the ForWord_Count field value of the corresponding record, and make DFT.ForWord_Count = DFT.ForWord_Count + 1, forward the data packet P to the next intermediate forwarding node.

4.3 Detecting Selective Forwarding Attack Paths

After collecting the network's topological structure, the BS saves the identities of all source nodes and intermediary nodes in DST. Source ID, DST, Empty Path Array fields and other sections are initialized. If a node dies due to battery exhaustion, the path from some source nodes to the BS may change. The BS will update the network topology and adjust the Path Array field of the data packet sending table DST. When the BS receives a data packet Q, it will execute Algorithm 1 to detect the selective forwarding attack path. It first finds the record of the source node identity $Q.ID_i$ in the data packet transmission table DST; if the Seq_Number field is empty, it is the first time receives the data packet sent by the source node, and it will set DST.Seq_ Number = $Q.Seq\_Number_i$.

DST.Sum_Count = 1, DST.Drop_Count = 0; if the Seq_Number field is not empty, compare whether the sequence number of the last received data packet is continuous with the sequence number of the currently received data packet; if continuous, it means that there is no packet loss, it will update the values of the DST.Seq_ Number and DST.Sum_Count fields so that DST.

Seq_Number = $Q.Seq\_Number_i$, DST.Sum_Count = DST.Sum_Count + 1; If the packet sequence number is not consecutive, it means there is packet loss; it will make DST. Seq_Number = $Q.Seq\_Number_i$, DST.Sum_Count = DST.Sum_Count + 1, DST.Drop_Count = DST.Drop_Count(Q.Seq_ $Number_i$–DST.Seq_Number − 1), where Q.Seq_ $Number_i$–DST.Seq_Number − 1 represents the number of lost data packets. Then the BS executes the formula (1) to determine the average PLR DPaverage of the current path and DPnormal of the recent course (2). The BS will use Algorithm 1 to find malicious nodes if the average PLR DPaverage is higher than DPnormal. DST. Sum Count + DST. The source node, DST, sends Drop Count data packets. Drop Count is the number of packages lost in the current path, q is a node in formula (2), and the average PLR of the previous nodes, h, is the path length from the source node to the BS.

$$DP_{\text{average}} = \frac{\text{DST.Drop\_Count}}{\text{DST.Sum\_Count} + \text{DST.Drop\_Count}} \tag{1}$$

$$DP_{\text{normal}} = \sum_{i=1}^{h} r_i \prod_{j=1}^{i-1} \left(1 - r_j\right) \tag{2}$$

Algorithm 1 Selective forwarding attack path detection

Input: data packet Q, data packet transmission table DST, average PLR q of the link between nodes.

Output: Selective forwarding attack path.

1. For each record in DST, do
2. If $Q.ID_i$ = = $DST.ID_i$ then

3. If DST.Seq_Number = null, then
   a. DST.Seq_Number = Q.Seq_Number$_i$
   b. DST.Sum_Count = 1
   c. DST.Drop_Count = 0
4. Else if P.Seq_Number$_i$ - DST.Seq$_{Number}$ = = 1 then
   a. DST.Seq_Number = Q.Seq_Number$_i$
   b. DST.Sum_Count = DST.Sum_Count + 1
5. Else
   a. DST.Seq_Number = Q.Seq_Number$_i$
   b. DST.Sum_Count = DST.Sum_Count + 1
   c. DST.Drop$_{Count}$ = DST.Drop$_{Count}$ + (Q.Seq_Number$_i$ − DST.Seq_Number − 1)
   d. Calculate the DP average by formula (1)
   e. Calculate DP$_{normal}$ by formula (2)
   f. If DP$_{average}$ > DP$_{normal}$, then
      i. Return DST.PathID_Array$_i$
   g. End if
6. End if
7. End if
8. End for

4.4 Locating Malicious Nodes

The BS detects DST selective forwarding. PathID Arrayi, assuming the path DST.PathID Arrayi has m hops from the source node to the BS, is represented by (×1, × 2,…, xm, BS), where × 1 represents the data packet's source node and the remainder Nodes indicate intermediate forwarding nodes. The BS first alerts each node to send the number of data packets in the data forwarding table DFT forwarded by the source node n1 to the BS sequentially along the route, starting from the source node (3). Equation (3) represents node ni's identification, IDi. DFT. For Ward Count is the number of data packets forwarded by node ni, pi−1 represents data packets transmitted by the previous hop, and Timestamp represents Timestamp, || denotes join operation.

$$q_i = F_{L_{4,1}}(ID_i \| n_i.DFT.ForWard\_\text{Count}\ q_{i-1}\ \text{Timestamp}\ ) \tag{3}$$

5 Performance Analysis

This paper will analyze and compare the LSFAD scheme proposed in this paper and the scheme proposed in the literature [24] from the aspects of communication overhead and storage overhead and analyzes the detection probability of selective forwarding attack paths. Since the scheme in this paper assumes that the calculation, storage, and communication capabilities of the BS are not due to limitations, the BS's communication overhead [25] and storage overhead [26] are not discussed here.

5.1 Communication Overhead

This paper analyzes the communication overhead caused by malicious node detection. It takes the number of data packets each node needs to forward or send as an indicator to measure the communication overhead. The PHACK scheme proposed in [27] is a detection method based on multi-hop confirmation. In addition to forwarding regular data packets, each node on the forwarding path needs to generate proof for each box to detect and locate malicious nodes. The data packets are sent back to the source node in different ways. Suppose a source node sends M data packets at a time. In the PHACK scheme, the communication overhead of each node is 2 M. In the CLAIDS scheme proposed in the literature [28], assuming each node has N neighbor nodes and a node sends M data packets to each of its neighbor nodes, each node needs to return M confirmation packets. Therefore, in the CLAIDS scheme, the communication overhead of each node is 2 M $\times$ N. In the DSFLACQ method introduced in [29], each node has N neighbor nodes and sends M data packets to them. Each neighbor node only needs to reply to one confirmation packet [30], hence the communication overhead of each node is (M + 1) $\times$ N. This paper's LSFAD technique does not monitor adjacent nodes or transmit confirmation packets to the source node. BS detects the attack path. If the BS discovers the attack path after receiving M packets from the source node, each node on the forwarding path will sequentially submit a statistical data packet for a malicious node location. This paper's LSFAD technique has M + 1 communication overhead per node. Table 2 shows each node's communication overhead for the LSFAD scheme, CLAIDS, DSFLACQ, and PHACK schemes. Table 2 shows that the LSFAD system suggested in this paper has substantially lower communication overhead than other schemes.

Table 2 Comparison of communication overhead

CLAIDS	DSFLACQ	PHACK	LSFAD
2 M $\times$ N	(M + 1) $\times$ N	2 M	M + 1

6 Simulation Experiment

In this study, the LSFAD scheme's malicious path detection probability, malicious node placement probability, and energy consumption are simulated and analyzed. The OMNeT++ platform does simulation experiments. 100 nodes are randomly arranged in a 500 m X 500 m square. Each node has an ID. 90 m is each node's communication range. After deployment, the node won't move because the BS is in the center of the region. Select data source, malicious, and intermediate forwarding nodes at random. The data source node provides 256-byte data packets to the BS via multi-hop every 1 s. Each node has 1 J of starting energy and consumes 50 nJ/bit for sending and receiving. 0.2 ~ 0.8% of malicious intermediate forwarding nodes reject data packets they seek to forward. Table 3 lists the simulation parameters.

Algorithm 1 detects the selective forwarding attack path when the BS receives the source node's data packet. It estimates the present course's average PLR DPaverage and the recent system's formula (1) and (2). The existing form has a selective forwarding attack if the system's average PLR DPaverage is greater than the way's DPnormal. Figure 1 shows that the link's average PLR is 0.005, 0.025, 0.045, 0.065, 0.085, and 0.105, and the malicious node's is 0.2, 0.4, 0.6, and 0.8. Figure 1 shows that when there is only one malicious node in the path, the regular PLR and the average PLR are higher than the regular PLR DPnormal of the way. This means that selective forwarding attack pathways begun by hostile nodes on the way with PLRs q = 0.2, 0.4, 0.6, and 0.8 can be detected. Selective forwarding attack path verification is more likely if the association's normal PLR is low and malicious nodes' PLR is high.

Table 3 Simulation parameters

Parameter	Value or range
N/w area/m2	500 × 500
No. of nodes in n/w	100
Radius/m	90
Initial energy of the node/j	1
Sending and receiving energy consumption/(nj/bit)	50
Data packet sending interval/s	1

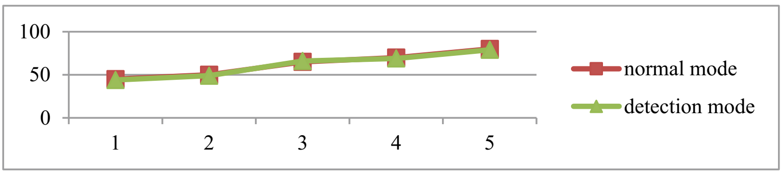

Fig. 1 Energy consumption in normal mode and detection mode

Table 4 Energy consumption in normal mode and detection mode

Length of path	Normal mode	Detection mode
4	45	44
5	50	49
6	65	66
7	70	69
8	80	79

In the LSFAD scheme, the energy consumed by a node on the path is mostly from receiving and sending data packets, so the sum of the energy consumed by all nodes on a forwarding path is UE = (i = 1)^n^M × N i × (f r + f s), where n is the path length, M is the data packet length, Mi is the number of data packets forwarded by node I, and er and es are the energy consumed to receive and send 1-bit data, respectively. Figure 1 shows the energy usage of all forwarding path nodes in normal and detecting modes. Table 4 shows that when the average PLR of the link is 0.025, the PLR of malicious nodes is 0.2, and 50 packets are sent at a time, when the path lengths are 4 ~ 12, the energy consumption of the whole way in normal mode and detection mode increases. The detecting state consumes 1.8 μJ more energy than the regular form.

This is because when the BS finds a selective forwarding attack on a particular path, it will start the detection mechanism to locate the malicious node. After all, each node on the course needs to send information about the number of data packets forwarded by the source node along the forwarding path to the BS locates malicious nodes. The network consumes slightly more energy in detecting and locating malicious nodes than under normal conditions, but the overall energy consumption is not much different.

7 Conclusion

This research provides a lightweight wireless sensor network selective forwarding attack detection technique (LSFAD). In the LSFAD scheme, the BS records the total number of data packets sent by each source node and the routing path of these data packets forwarding through Calculate and comparing the average PLR of the way with the regular PLR to determine if there is a selective forwarding attack on the current form. The base station assesses each node's average and regular PLR to find malicious nodes or links that launch forwarding attacks. The previous node link is a malicious link if its average PLR is higher than expected. The LSFAD scheme is simple, requires no listening nodes, does not need a sophisticated evaluation model to generate node trust values, and is easy to implement. LSFAD can also withstand passive and active selective forwarding assaults by hostile nodes. Performance study and experimental simulations reveal that the LSFAD method has substantially lower communication overhead than competing techniques. LSFAD's selective forwarding attack path can be detected even with a 0.125 connection PLR. Malicious nodes that launch forwarding attacks can be found when the packet rate exceeds 0.025. The network uses similar energy to detect and locate malicious nodes.

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A Global Overview of Data Security, Safety, Corporate Data Privacy, and Data Protection

Swapnil V. Khandare, Varun P. Herlekar, Vaishali S. Hanwate, Gangaprasad M. Shirale, and Dhiraj D. Sirbhate

Abstract These days it is common to hear public employees threaten privacy in the name of "Procedure Established by Law" or "Public Duty," two concepts that are perhaps the most important to human life on this planet. Let's imagine for a moment what a person would be like without privacy rights, which include all of the private rights associated with family, employment, relationships, etc. Simply said, privacy is essential to human health and is the means by which a tranquil existence with dignity and freedom is genuinely guaranteed. With the rise in social media use and the gradual digitalization of our country, it is fair to say that we are in a "Cyber Era". Data protection and privacy are inextricably interwoven, and they now occupy a highly important and delicate area in the legal system. In order to acquire material and further refine it into a precise piece of information, the review paper uses secondary sources. It is reviewed using the analogical technique of research. Cyber security is a crucial modern-era prerequisite for a safe digital and cyber ecosystem. Since most business processes now take place online, data and resources are at risk from various cyber threats. Since data and system resources serve as the bedrock of the organization, any threat to these components automatically poses a hazard to the organization as a whole.

Keywords Cyber · Cyber security · Data protection · Data privacy · Data security · PII · PbD · Breach

1 Introduction

Cyber security is the process of defending against hostile assaults on systems that are connected to the internet, including computers, servers, mobile devices, electronic systems, networks, and data. One aspect of cyber security is called cyber, while the other is called security. Systems, networks, software, and data are all included in

S. V. Khandare (✉) · V. P. Herlekar · V. S. Hanwate · G. M. Shirale · D. D. Sirbhate
Department of Computer Engineering, Government College of Engineering, Yavatmal, Maharashtra, India
e-mail: swapnilkhandare4122001@gmail.com

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the term “cyber” technology. Additionally, security is concerned with safeguarding data, applications, networks, and systems. It may also go by the name’s electronic information security or information technology security in some instances [1].

Data privacy refers to the safeguarding of personal information from those who shouldn’t have access to it and the freedom of individuals to control who has access to their data. Information privacy is another name for day privacy. Data handling properly with an emphasis on observing data protection laws is referred to as data privacy, sometimes known as information privacy, and it is a subset of the data protection field. Data privacy is concerned with the proper collection, management, storage, and sharing of data with any third parties, as well as the observance of any privacy regulations. The goal of data protection laws worldwide is to restore individuals’ sovereignty over their personal information. Giving people control over how their personal data is handled and used by empowering them to understand how, by whom, and why it is being used.

2 Literature Review

A solid cyber security plan can give users and organizations a good security posture against malicious assaults that aim to gain access to, alter, delete, destroy, or extort sensitive data and systems. A system’s or device’s activities being disabled or otherwise disrupted can be prevented by cyber-attacks thanks in large part to cyber security. Being sensitive to personal information based on data collection is what is meant by data privacy [2].

2.1 Types of Cyber Security

– Application Security

Most of the apps we use on our smartphones are safe and follow the guidelines of the Google Play Store. Consumers can choose from 1.85 million distinct apps to download. Just because we have options does not mean that all apps are safe. Many apps make security claims, but after collecting our data, users sell it to third parties for money, and the apps stop functioning. Suddenly, this is the target of a cyber-attack. The program must be installed via a trustworthy platform, not through Google Chrome.

– Mobile Security

As the number of devices and methods in which they are used has grown rapidly, so has the importance of mobile device security. This is especially problematic in the enterprise when employee-owned devices connect to the corporate network. Increased corporate data on devices attracts cybercriminals, who can use mobile malware to target both the device and the back-end systems it connects to. IT departments work hard to ensure that employees understand acceptable use policies, and administrators enforce those policies.

– Cloud Security

Over the past ten years, cloud-based data storage has gained popularity as a solution. Although proper identification is required, it improves privacy and stores data in the cloud, making it accessible from any device. Several well-known platforms include Dropbox, Microsoft Cloud, and Google Drive. If we wish to save more data than we have to pay for, these platforms are partially free. AWS is another innovative method for managing your organization online and offering data protection.

– Network Security

Your internal network will be more protected from outside assaults with increased network security. On rare occasions, we made use of the free Wi-Fi offered in public spaces like shopping centers and coffee cafes. A third party could start following your phone online through this behavior. If you use a payment gateway, your bank account can be empty. Because free networks do not support securities, avoid using them. When personal information slips into the wrong hands, bad things can happen. A data leak from a government agency, for example, might provide a hostile state with access to top secret material. A business hack might provide a competitor access to sensitive information. A breach in school security might provide thieves access to student PII, opening the door to identity theft. Because of a breach at a hospital or doctor's office, PHI might get up in the hands of someone who could misuse it [3].

The data privacy rules govern how information is collected, kept, and shared with other parties. The following are the most often debated data privacy laws:

- GDPR: The European Union's General Data Protection Regulation (GDPR) is the most comprehensive data privacy regulation in force. It applies to all European Union persons and businesses that conduct business with them, even those based outsides of Europe. Individuals have the right under GDPR to choose what data corporations keep, to request that personal data be erased, and to be notified of data breaches. Noncompliance may result in severe penalties and legal action.
- The California Consumer Privacy Act (CCPA) is a United States state-level policy. It allows California people to ask organizations what personal data they have on them, request that it be deleted, and learn what data has been shared with third parties. These regulations relate to customer data collected within the state.

India is not a party to any personal data privacy convention comparable to the GDPR or the Data Protection Directive. However, India has accepted or is a party to several international declarations and agreements that recognize the right to privacy, such as the Universal Declaration of Human Rights and the International Covenant on Civil and Political Rights. India has also yet to establish appropriate data protection legislation. The Indian legislature did, however, update the Information Technology Act (2000) ("IT Act") to add Sections 43A and 72A, which provide for compensation for the improper disclosure of personal information.

The practice of protecting sensitive data on the internet and on devices from attack, destruction, or unauthorized access is known as cyber security. The goal of cyber security is to create a safe and secure environment where data, networks, and devices can be safeguarded from cyber-attacks.

Preventing information from being stolen, compromised, or attacked is the goal of cyber security. Three objectives can be used to gauge cyber security:

- Maintain the confidentiality of the data.
- Maintain the data's integrity.
- Encourage making data availability to authorized users.

2.2 Advantages of Cyber Security

1. Protection:

Protecting the network and the personal information of the person and the organization is the basic goal of cyber security. It is essential from a personal safety perspective as well as a legal and financial perspective. Every nation has a legal framework in place that says the responsible organization will be held accountable in the event of a data loss. Data loss also damages the organization's reputation. Its customers will stop doing business with them, which could have serious financial consequences. Cyber security is treated seriously in order to avoid this.

2. Enhances reputation:

A strong cyber security defense team or plan improves the organization's reputation. Customers will feel confident in doing business with a company that values their data appropriately. This broadens the potential of business.

3. Cost:

Preventing an issue always costs less than fixing it after the fact. Cost effectiveness is the main benefit that cyber security offers its users. Risk management and cyber security charges are not outrageously expensive. The ensuing financial loss, legal difficulties, and reputational damage will be difficult to overcome and ultimately cost more money.

4. Enhancement of technology:

Technology does not operate in silos when it comes to cyber security. It is a comprehensive strategy that includes all the organizational divisions and stakeholders. The total improvement will be noticeable and cover everything from email management and protection to cyber security awareness training, cutting edge software and hardware enhancement, and greater departmental cooperation. Additionally, the organization as a whole becomes more productive as a result.

2.3 *Disadvantages of Cyber Security*

1. Awareness:

The greatest obstacle to cyber security is this. The weakest link in the best defense against outside threats. Every organization's greatest resource for advancing its goals is its workforce. However, the majority of phishing campaigns are directed at the same personnel, who unintentionally end up serving as the hackers' pawns. Every organization provides training for its personnel on cyber security issues since it is vital. These awareness sessions are created to disseminate the most recent knowledge about the virtual world. The new security requirements as well as the types of risks that are now prevalent are made clear to the workforce.

2. Evolving Field:

Cyber security is a dynamic idea. The evolution of security is the cause. Whole facets of cyber security change along with the rapid advancements in technology. Therefore, it is quite challenging to maintain a single barrier of protection for an extended period of time.

3. Adaptation:

The ability of teams to absorb the constantly evolving technology is becoming a greater challenge in the field of cyber security. The human mind takes some time to comprehend the threat that is being thrown at them. Expecting things to change instantly is dangerous since widespread tech adoption takes time to happen.

4. Lack of Numbers:

The shortage of cyber security experts is another significant barrier to improved cyber security. Insufficient supply exists to meet the rising demand for professionals. According to recent estimates, there are already close to two million unfilled positions, and this figure is steadily growing to astronomical proportions. As a result, the obligations related to cyber security are suffering, and the experts occupying these positions are overworked.

3 Data Protection Methodologies

A. Minimize What Data You Collect

The first step toward a comprehensive strategy to data privacy is data minimization. Don't collect something if you don't need it. Sometimes referred to as the "minimum dataset", or MDS, this step lowers the privacy overhead of a system. Personal information like name, address, and so on are typically collected with less data. For Instance, (1) Unless necessary, avoid collecting name prefixes like Mrs. or Mr. (2) Consider whether you need to know someone's whole address; a country or state location might be sufficient. (3) Will an age range suffice in place of a complete date of birth? [3].

B. Reduce the Data You Disclose

You can arrange your system in a similar way to reduce the amount of data you release. This depends on the IT tools you are using, of course, but many systems now have privacy-enhancing options. An age requirement for purchasing an item with an age restriction is one illustration. An age-over request could be made in place of asking for or requiring the user's date of birth to complete the transaction. The response would indicate whether the user is over 21 or not. These kinds of exchanges can be supported by contemporary protocols like OAuth 2.0.

C. Reduce the Data You Share

Only those with a legitimate need to know should have access to the data you do gather. The devil is in the details when it comes to data access. No matter how much minimization you do, if a malevolent entity has access to these data and decides to reveal them, all privacy enhancements are essentially nullified. One of the more challenging aspects of security to develop is access control. Additionally, there are two sides to a consumer system, and both require the application of strong authentication measures:

1. Admin Access: One of the main entry points for cyber-attacks is compromised administrator access to databases. Only administrators who actually require access to sensitive information should be granted it. The most reliable option should be used for two-factor authentication. As an illustration, the second factor ought to be out-of-band and, if at all possible, restricted to corporate IP addresses. Audits of access are required.
2. Data Owner (Customer) Access: Customers frequently have an account manager who grants access to the private Information stored in their account. They might be able to use that account manager to change this data as well. Account managers can be used to accommodate data-access rights like access to data and data correction under legislation like GDPR. This is also a system attack vector, though, and it can expose data. Use two-factor authentication whenever possible to restrict access to customer accounts and guarantee the safety of account recovery [4].

D. Data of the Benefits of Encryption

4.8 million records of data are accessible daily. Only 4% of them are encrypted; the rest are completely vulnerable to exploitation. Wherever sensitive or personal data is collected, kept, and exchanged, encryption should always be employed. It is Security 101. To ensure that the data is kept private and that the impact of any compromise is reduced, encryption must be implemented both during transit and at rest. There isn't a single encryption product that works for everyone, so you need to be aware of a few things. These consist of the encryption tool using a well-known method, such AES 256.

When at rest, that is, when it is kept in a database on a server or mobile device. Hard-disk encryption is required if you store sensitive or private data on a hard drive employing the industry-standard protocols SSL/TLS during transmission, such as browser-based HTTPS. HTTPS implementation is essential yet prone to error. Make sure encryption is turned on for all of your website's components. You should also consider security solutions like email encryption and/or data-leak prevention software if you use email to transfer sensitive or confidential data.

E. Personal Data Respect

Respecting the privacy of an individual's data will go a long way toward improving privacy generally throughout a system. Understanding that personal data should be treated with the same care as a person will help you start a privacy culture in your firm. Permission is the first step in demonstrating this regard. A crucial design goal is to build systems with privacy-by-design (PbD), which is fast becoming just as critical as hardening against cyber security threats. The two are actually inextricably related. These five essential requirements, which are illustrated above, provide a fundamental framework for data privacy and should help you find a system that takes privacy seriously [5, 6].

All security programmers are built on the confidentiality, integrity, and availability (CIA) triad. The CIA triad is a security model created to help organizations and businesses develop rules for information security on their property. In order to prevent confusion with the Central Intelligence Agency, this concept is often referred to as the AIC (Availability, Integrity, and Confidentiality) triad. The three most important pillars of security are referred to as the triad.

When installing a new programmed, creating a database, or ensuring access to particular data, the majority of organizations and businesses apply the CIA criteria. All of these security objectives must be met for data to be entirely protected. As a result of the interdependence of these security policies, it may be improper to ignore any one of them [7].

The CIA tried is as follows:

A. Confidentiality:

Privacy and confidentiality are similar in that they prevent the unauthorized disclosure of information. It involves the safeguarding of data, granting access to those who are authorized to view it while preventing unauthorized parties from discovering any of its contents. It makes sure that the correct individuals can access crucial information while preventing it from getting to the wrong people. A good example of how to ensure.

Tools for Confidentiality:

- Encryption
- Access Control
- Authentication
- Authorization
- Physical Security confidentiality is data encryption .

(a) Encryption

By applying an algorithm, encryption is a technique for altering data so that only authorized users can decipher it. Data is transformed using a secret key (an encryption key), making it possible to only decode the altered data using another secret key (decryption key). By encrypting and converting data into cypher language that is impossible to decipher, it secures sensitive data like credit card information. Only after decryption is it possible to read this encrypted data.

(b) Access Control

Rules and policies for controlling access to a system or to physical or virtual resources are defined by access control. It is a procedure by which users are given access to systems, resources, or data as well as certain privileges. Users of access control systems must present credentials, such as a person's name or a computer's serial number, before being allowed access. These credentials may take many different shapes in physical systems, but the most secure credentials are those that cannot be transferred.

(c) Authentication

Every company needs authentication because it allows them to keep their networks secure by allowing only authenticated users to access their protected resources. Computer systems, networks, databases, webpages, and other network-based applications or services may be among these resources.

(d) Authorization

An authorization mechanism grants consent to perform or possess something. It is used to decide whether a person or system is granted access to resources, including computer programs, files, services, data, and application features, based on an access control policy. Authentication for user identity verification usually comes before it. Permission levels covering all system and user resources are frequently assigned to system administrators. A system verifies an authenticated user's access policies during authorization and either approves or rejects the request for resource access.

(e) Physical Security

Physical security refers to procedures intended to prevent illegal access to IT assets including buildings, machinery, staff, resources, and other properties from being harmed. These resources are shielded from hazards including theft, vandalism, fire, and natural calamities [7].

B. Integrity:

Integrity is the process used to make sure that data is authentic, correct, and protected against unauthorized user alteration. It is a quality that information has not been changed without authorization and that the information's source is reliable.

Tools for integrity:

(a) Backups

Data backup refers to routine data archiving. Making copies of data or data files is a method that can be used if the original data or data files are lost or damaged. A data retention policy may mandate copies to be made for historical reasons, such as longitudinal research, statistics, or historical records. It may also be utilized for these reasons. Many programs, particularly those running in a Windows environment, create backup files utilizing the file type extension BAK.

(b) Checksums

The integrity of a file or a data transfer is checked using a checksum, which is a numerical value. To put it another way, it is the computation of a function that converts a file's contents into a numerical value. They are often applied to verify if two sets of data are identical by comparing them. The full file's contents are required for a checksum function to work. It is built in a such that even a minor alteration to the input file, like flipping a single bit, is likely to produce a different output value.

(c) Data Correcting Codes

It is a technique for data storage that makes it simple to find minor errors and automatically fix them.

C. Availability:

Information is said to be available if it can be accessed and changed quickly by people with the right permissions. It is a guarantee that only persons with permission will have dependable, continuous access to our sensitive data.

Tools for Availability:

(a) Physical Protection.

Information must be protected physically while remaining accessible in case of physical difficulties. It makes sure that sensitive data and essential information technologies are kept in secure locations.

(b) Computational redundancies.

It serves as a fault-tolerant barrier against unintentional defects. It safeguards computers and backup storage systems that can be used in the event of a disaster.

4 Detection Using Our System

The Fig. 1 depicts the progression of our data leakage detection system. Here, we've developed a file transfer mechanism that ensures no files evacuate the company's premises (Figs. 2 and 3).

Fig. 1 Flowchart of data leakage detection system

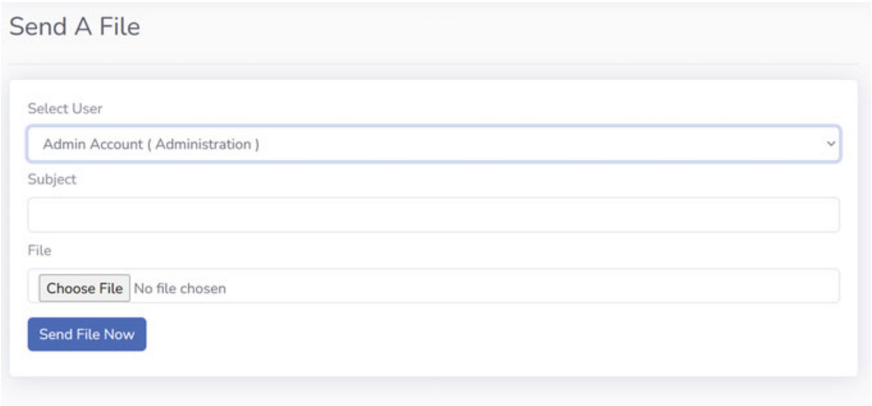

Fig. 2 Sending/transferring the file using system

The individual first registers with the system/portal. Assume they need to send a PDF document to another person registered on the system/portal. The sender chooses the recipient's name from a drop-down list, writes the subject line, and picks the file. When the file is sent, the recipient receives an email notification.

At the receiver's end, the receiver now has two options: obtain the file or request the key. If the receiver wishes to download the file that the sender has sent, he must obtain the access pass from the sender. The file cannot be accessed without the access key (Fig. 4).

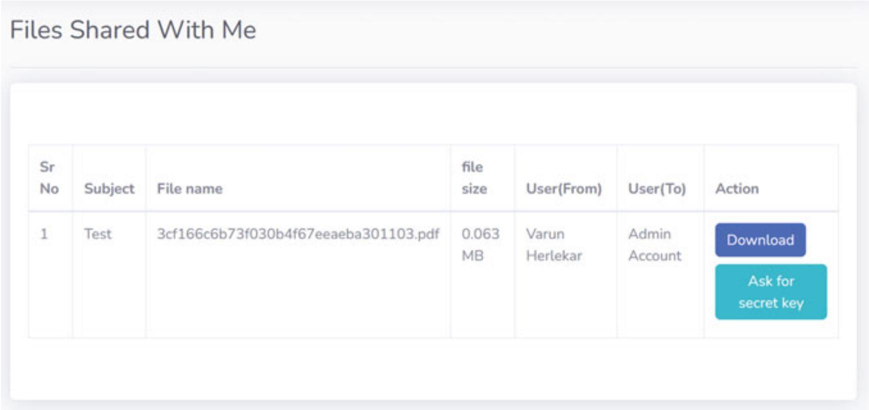

Fig. 3 Downloading the file (recipient view)

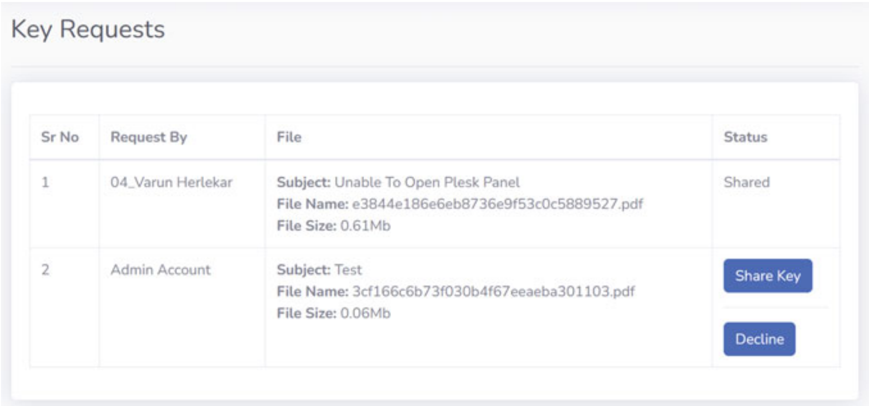

Fig. 4 Key requests (senders end)

The sender now receives a key request and has the choice of verifying the request and sharing the key or deciding on the key request. If the request is granted, the recipient is given a four-digit access code that allows them to download the file (Fig. 5).

The user now has two tries to enter the correct four-digit combination and download the file using the key provided by the sender; if the user fails, the account will be immediately blocked and marked as a leaker. In addition, if an intruder attempts to download the file by retrieving the key from the database or guessing the password, the account will be flagged as a leaker and banned, allowing the administrator to act against the account (Fig. 6).

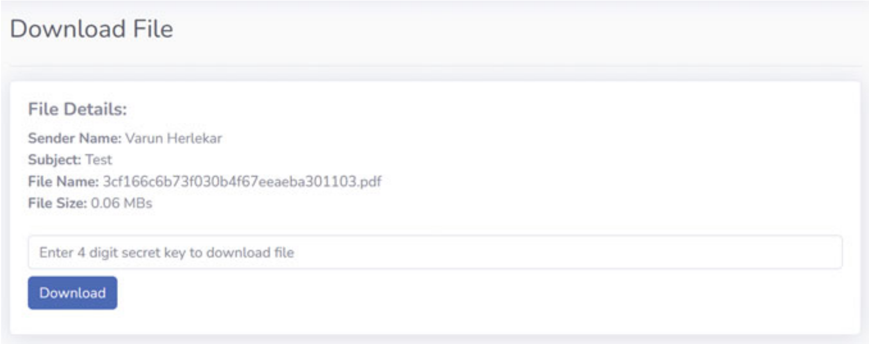

Fig. 5 Download file (receiver's view)

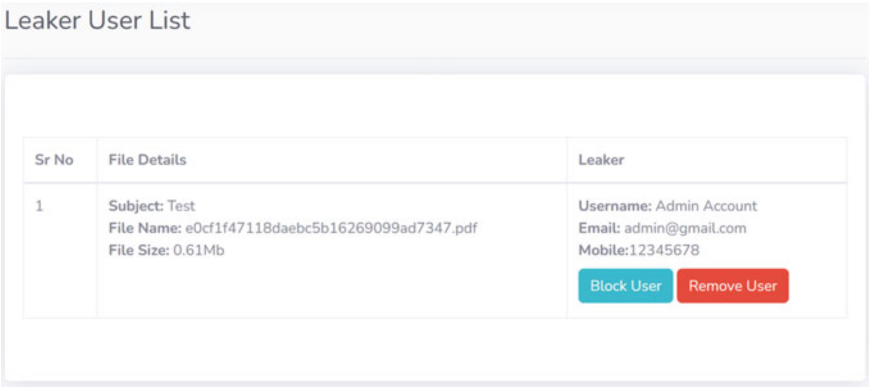

Fig. 6 Leakers list (admin's view)

5 Related Work and Case Studies

The graph below shows the types of data exposed in Data Breaches in various countries despite increased investment in information security products and services, with $114 billion invested in 2018, hackers continue to assault enterprises across a wide range of sectors in order to obtain access to valuable consumer data Fig. 7 [6].

5.1 Some Recently Reported Huge Data Breach Incidents

A. Pegasus Airline Data Breach

A misconfigured AWS bucket exposed 23 million files belonging to the Turkish airline Pegasus Airlines. Safety Detectives, a security firm, uncovered the security

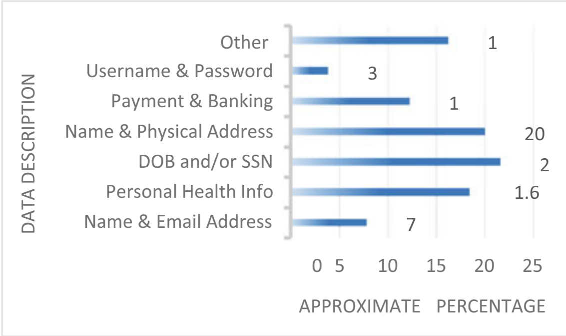

Fig. 7 Graph showing the percentage of data types breached during some major cyber-attacks

breach. The information was linked to the airline's EFB software, a solution that required access to takeoff, landing, and refueling data as well as critical flight crew information.

Because of the AWS bucket misconfiguration, anybody had unrestricted access to this database, which included approximately 400 files including plain text passwords and secret keys. Pegasus Airlines discovered no indication of data breach after the risk was notified. However, while the AWS bucket was incorrectly set, attackers may have secretly exfiltrated the exposed data.

- Report date: March 2022
- Impact of the breach: Around 6.5 TB of Data.

B. Optus Data Breach

Cybercriminals got access to Optus' internal network, allowing them to see consumer data for up to 9.8 million subscribers. The hacked data, which dated back to 2017, contained the following sorts of information: Names, Dates of birth, Telephone numbers, Addresses by email, Data subsets also contain street addresses, driver's license numbers, and passport numbers. It's believed that the cybercriminal organization acquired access via an unlawful API endpoint, which means that no user/password or other authentication mechanism was necessary to connect to the API.

- Report date: September 2022
- Impact of the breach: Data of around 9.8Mn Optus users was allegedly breached by the hackers.

C. LinkedIn Data Breach

On June 2021, data linked with 700 million LinkedIn users was listed for sale in a Dark Web forum. This vulnerability affected 92% of LinkedIn's overall user base of 756 million members. The data was leaked in two waves, first revealing 500 million members and then a second dump in which the hacker "God User" boasted of selling a database of 700 million LinkedIn users.

- Report date: June 2021
- Impact of the breach: Data of around 700Mn LinkedIn users was allegedly breached by the hackers.

D. Rockstar Games Data Breach

Rockstar Games, the developer of the Grand Theft Auto series, was the victim of a breach that saw video of their unannounced Grand Theft Auto VI game revealed by the hacker. Furthermore, the hacker claims to have the game's source code and is attempting to sell it. The breach is believed to have occurred as a result of social engineering, with the hacker getting access to an employee's Slack account. The hacker also claimed to be behind the Uber assault earlier this month. "We recently suffered a network intrusion in which an unauthorized third party fraudulently accessed and copied proprietary material from our computers, including early development video for the upcoming Grand Theft Auto", Rockstar stated in a statement.

- Report date: September 2022
- Impact of the breach: Some confidential visuals of the upcoming products.

E. Uber Data Breach

Uber's computer network has been infiltrated, and several engineering and communications systems have been taken offline while the firm examines how the attack occurred. According to one researcher, the attacker has already supplied email, cloud storage, and code repositories to security firms and The New York Times. After breaking into a staff member's Slack account and sending out messages proving they'd successfully penetrated their network, Uber employees discovered their systems had been breached.

- Report date: September 2022
- Impact of the breach: Data of Uber users was allegedly breached by the hackers.

6 Conclusion

The need to protect data from unwanted attacks has gained attention on a global scale. Data privacy has become a hot button issue in information security during the last few decades. Its goal is to increase knowledge about and support for ethical data collecting, privacy, and protection practices. Data privacy is also necessary because people who want to exist online need to believe that their information is being treated appropriately. Without the data subject's express and free consent, it should not be utilized for any reason. The Data Privacy Act safeguards people from the illegal handling of private, non-public personal information. Going back in time, when mineral oil was the most profitable good and practically every country was vying for it, would help us understand where this remark came from. This is demonstrated by the fact that the data industry is home to five of the most valuable corporations in the world: Amazon, Google, Apple, Microsoft, and Facebook.

Conflict of Interest The researchers confirm that no commercial or financial ties that might be seen as a potential conflict of interest existed during the course of the research.

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Probabilistic Scheme for Intelligent Jammer Localization for Wireless Sensor Networks

Amar Pal Yadav, Sandeep Kumar Davuluri, Piyush Charan, Ismail Keshta, Juan Carlos Orosco Gavilán, and Gaurav Dhiman

Abstract With the development of artificial intelligence technology, intelligent interference sources can improve the interference effect by changing their transmission power, which leads to the failure of traditional positioning technology based on received signal strength. Therefore, a sensor wake-up mechanism is introduced to study multi-interference sources based on block-compressed sensing positioning method. First, wake up the sensor nodes periodically, while improving the utilization effectiveness of the sensor nodes and the accuracy of positioning information collection; secondly, considering that the distance between the distance and the received signal strength cannot be determined when the transmission power of the interference source is unknown and changes. Then, based on the compressive sensing theory, the positioning problem is modeled as a block sparse vector reconstruction problem; finally, a variation-based algorithm is designed by exploring the law of power variation. The Wake-VBEM reconstruction algorithm based on Bayesian mean expectation accurately reconstructs the target position vector. The simulation proves that

A. P. Yadav
CSE-AI College: Noida Institute of Engineering and Technology Greater Noida, Greater Noida, India

S. K. Davuluri (✉)
Department of Information Technology, University of the Cumberlands, Williamsburg, KY 40769, USA
e-mail: sandeepkdavuluri@gmail.com

P. Charan
Electronics and Communication Engineering, Manav Rachna University, Faridabad, India
e-mail: piyushcharan@mru.edu.in

I. Keshta
Computer Science and Information Systems Department, College of Applied Sciences, AlMaarefa University, Riyadh, Saudi Arabia
e-mail: imohamed@mcst.edu.sa

J. C. O. Gavilán
Universidad Privada del Norte, campus virtual, Lima, Peru
e-mail: juan.orosco@upn.edu.pe

G. Dhiman
Institute of Engineering and Technology, Chitkara University, Punjab, India

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the proposed method can simultaneously realize the position estimation of multiple interference sources and effectively improve the service life of the network when the interference source power is unknown and changes.

Keywords Wireless sensor network · Sensor wake-up mechanism · Artificial intelligence · Bayesian algorithm · Intelligent jammer

1 Introduction

Wireless Sensor Networks (Wireless Sensor Networks, WSN) are composed of low-cost, low-power sensor nodes and are widely used in daily production, life, and national defense and military fields [1]. However, it affects the transmission of wireless signals, thus destroying the security and reliability of information interaction in WSNs.

Existing spectrum anti-jamming measures ensure communication quality by improving technical solutions, such as direct sequence spread spectrum, frequency hopping technology, and intelligent anti-jamming technology [2, 3]. However, such methods are highly complex and require a large amount of bandwidth. And storage resources cannot fundamentally eliminate the problem of malicious interference. Interference source location can provide technical support for anti-jamming technology, such as directional interference elimination but also help to achieve precise strikes on interference sources [4], fundamentally eliminating malicious interference source of interference.

The interference source location technology based on received signal strength (Received Signal Strength, RSS) locates by collecting the RSS of the interference source, without the need for any sensor nodes to send any signals, and has the advantages of simple operation, high security, and wide application [5]. In the case of hiding itself, the location of the interference source is realized [6], so it has become an effective measure for locating the source. The literature [7] proposed a distributed RSS-based interference source location algorithm. According to the gradient idea, the interference source is approached along the rising direction of RSS, but this algorithm can only realize a single interference source location. Literature [8] studies the interference source location method in short-distance communication and uses the clustering method to learn the site according to whether it is affected by the interference source. The positioning accuracy is low. With the continuous improvement of the intellectual level of malicious interference sources, the interference source can change its transmission power to enhance the interference effect [9]. The RSS-based positioning method relies on signal attenuation to estimate its position, but the interference source transmission power after the change.

The power attenuation is difficult to calculate, so the positioning cannot be completed. In addition, to improve positioning accuracy, it is usually necessary to deploy many sensor nodes, but batteries power the sensor nodes, and their energy is limited. Therefore, any applications will seriously hinder the development of sensor networks [10].

Based on this, this paper studies the energy-saving positioning method for multi-intelligent interference sources and designs the Wake-VBEM algorithm based on the variation Bayesian Expectation-Mean (Variational Bayesian Expect-Mean, VBEM) [11] under the sensor wake-up mechanism. Sensor nodes are used to improve the service life of the network and obtain more positioning data. At the same time, introduce reference power to solve the problem of inability to locate caused by interference source power changes. The main innovations are as follows:

(1) Design the sensor sleep–wake mechanism and periodically wake up the sensor to collect more positioning information and prolong the service life of the sensor node.
(2) Introduce the reference power to solve the problem that the perception dictionary cannot be obtained due to the power change of the interference source, and then adopt the Block Compressive Sensing (BCS) principle [12] to model the localization problem as a block sparse vector reconstruction problem.
(3) Design a correlation matrix based on the VBEM algorithm to explore the law of interference source power changes and design a correlation matrix based on interference source power changes to achieve accurate position vector reconstruction.

2 Model Establishment

The sensor wake-up mechanism is used to collect data to solve the positioning problem under the intelligent change of interference source power. First, the compressed sensing theory is introduced to establish a positioning model. Then, in reconstructing the position vector, a reconstruction algorithm is designed under the framework of VBEM. The specific operation process is as follows: Fig. 1 shows.

2.1 Compressed Sensing Localization Model

Compressed sensing theory [13] reconstructs discrete sparse signals, and introducing it into positioning requires discretization of the positioning area. Therefore, as shown in Fig. 2, the positioning area is divided into N grids and numbered sequentially (here, take a square area as an example, other positioning areas of arbitrary shapes are handled this way).

The position of the interference source is uncertain and can be randomly distributed in any grid, and its work is described by a sparse vector t:

$$t = [0, 1, 0, \cdots, 0, 1, 0, \cdots, 0]. \tag{1}$$

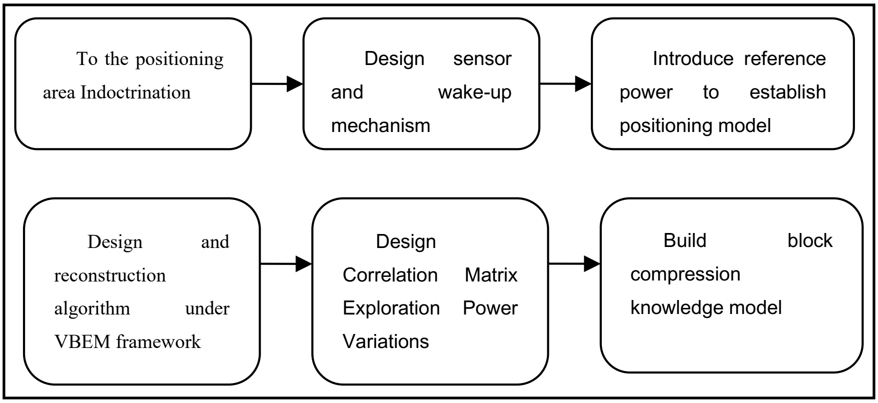

Fig. 1 Flowchart of intelligent interference source location under the sensor wake-up mechanism

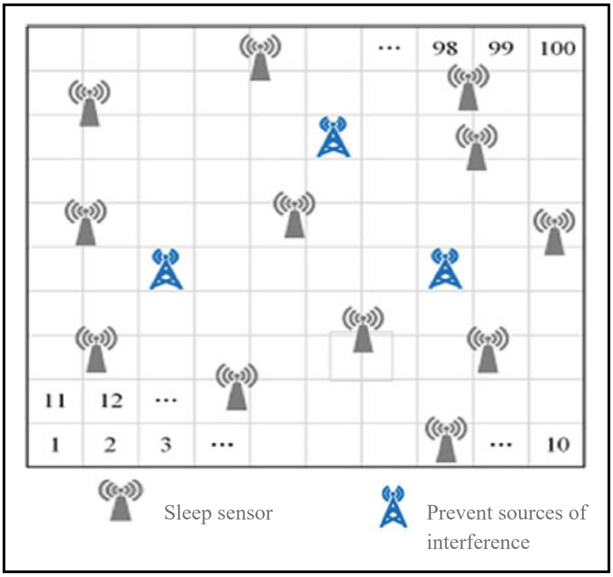

Fig. 2 Compressed sensing positioning scene diagrams

If $t_i = 1$, it means that there is an interference source in the ith grid; otherwise, it does not exist. If there are R sensor nodes in the positioning area to collect positioning information, the positioning equation is as follows:

$$z_R = \boldsymbol{\Psi}_{R\times M} T_M + n_R \tag{2}$$

Among them, z_R represents the sum of the RSS of all interference sources collected by the eth sensor node; Ψ is the perception dictionary, which is used to describe the relationship between the position of the interference source and the RSS; the RSS [14–19] of a sensor node is expressed as follows:

$$\boldsymbol{\Psi}_{rm} = Q_m g(T_r, K_m) \tag{3}$$

Among them, Q_m is the transmission power of the interference source located in the nth grid; T_r and K_m are the coordinates of the rth sensor node and the nth interference source, respectively; g represents the relationship between the interference source and the sensor node. $\boldsymbol{\Psi}_{rm}$ is affected by the influence of the transmitting power of the interference source and its distance from the sensor node. Once the position vector s is obtained, it can correspond to the coordinates of the interference source.

2.2 Sensor Wake-Up Mechanism Design

The introduction of the sensor wake-up mechanism under the CS framework has the following advantages: (1) Periodically wakes up the sensor nodes, which can save energy and prolong the service life of the network; (2) wake up different sensor nodes, which can collect more and more helpful positioning information; (3) to achieve positioning under the CS framework, which can make the number of wireless links awakened each time as small as possible.

At this time, the design of the wake-up strategy can be modeled as the construction of the observation matrix, let l = 1,2,…, M is the sampling time, then:

$$z_N^{(l)} = \boldsymbol{\Phi}_{N\times R}^{(l)} \boldsymbol{\Phi}_{R\times M}^{(l)} T_M + n_M^{(l)} = B_{N\times M}^{(l)} T_M + n_M^{(l)} \tag{4}$$

Among them, $B^{(l)}$, $z^{(l)}$, and $m^{(l)}$ are the perception matrix, measurement vector, and noise vector at the lth sampling, respectively; here, it is assumed that the number of wireless links awakened each time is the same; T_M represents the position of the interference source $\Phi^{(l)}$ is the sampling matrix of the lth time, and only one element in each row is 1, indicating the wireless link index of the lth wake-up.

The design of the wireless link wake-up strategy is embodied in the compressed sensing positioning model, which is to design the observation matrix Φ, and the position of its non-zero element represents the wireless link to be awakened. This paper develops a random wake-up strategy, the non-zero elements in the observation matrix Φ_S. The position of the 0 pieces is randomly generated, and there is only 1 in each row.

2.3 Establishment of Positioning Model Under the Wake-Up Mechanism

The intelligent interference source affects the communication by changing the power, so the perception dictionary $\Phi^{(l)}$ in the CS model cannot be established. To solve this problem, a reference power Q_0 is introduced. If the power of the interference source at the lth sampling time is Q_l, the positioning model can be expressed as follows:

$$z^{(l)} = B^{(l)} T^{(l)} = C^{(l)} x^{(l)}, m = 1, 2, \cdots, M \tag{5}$$

Among them, $z^{(l)}$, $B^{(l)}$, and $T^{(l)}$. are the measurement vector, perception matrix, and position vector at the lth sampling, respectively. The establishment of $B^{(l)}$ is shown in formula (3), and it needs to know the power Pl of the interference source. Still, it cannot be realized when the power of the interference source changes, so the rce tnsmits power is introduced, then:

$$\begin{aligned} z_m^{(l)} &= \sum_{n=1}^{N} Q_n^{(l)} g\left(T_m, K_n^{(l)}\right) \\ &= \sum_{n=1}^{N} x_n^{(l)} Q_{n0}^{(l)} g\left(T_m, K_n^{(l)}\right) \\ &= \sum_{n=1}^{N} x_n^{(l)} E_{mn}^{(l)} \end{aligned} \tag{6}$$

Among them, $Q_n^{(l)}$. and $Q_{n0}^{(l)}$. are the interference source's transmit power and reference power located in the nth grid at the lth sampling moment. $E^{(l)}$ is the power corresponding to the reference power P(1) n0 Perception dictionary. At this time, the position vector $x^{(l)}$ is defined as follows:

$$Q_n^{(l)} = x_n^{(l)} Q_{n0}^{(l)} \tag{7}$$

Although the elements of $x^{(l)}$ are not either 0 or 1, the index of the non-zero elements can indicate the target position. For example, if $x_n^{(l)} \neq 0$, there is no interference source in the nth grid.

Problem (5) involves the reconstruction of multiple measurement vectors. However, traditional numerous measurement vectors (Multiple Measurement Vectors, MMV) generally require the perception matrix to be consistent, so it is difficult to solve the above problems. Based on this, this paper introduces the BCS theory and explores the interference. In addition, the change law of source power is used to improve the positioning accuracy, which can solve the reconstruction problem when the perception matrix and the position vector are inconsistent.

3 Simulation Verification

The intelligent interference source location algorithm Wake-VBEM proposed in this paper is verified in MATLAB. The 12 m × 12 m square area is divided into N = 144 guards and Q = 144 sensor nodes deployed. K intelligent interference sources are randomly distributed, and each time, the number of awakened sensor nodes is M. The noise is described by the signal-to-noise ratio (Signal-to-Noise Ratio, SNR).

To measure the positioning performance of the algorithm, the average positioning error ErrL is defined as

$$\mathrm{ErrM} = \frac{1}{L \cdot R} \sum_{l=1}^{L} \sum_{r=1}^{M} \sqrt{\left(y_l^t - \hat{y}_l^t\right)^2 + \left(z_l^t - \hat{z}_l^t\right)^2} \tag{8}$$

Among them, (y_l^t, z_l^t) and $(\hat{y}_l^t, \hat{z}_l^t\ )$ are the actual and estimated coordinates of the kth interference source in the nth simulation, respectively. R and L are the total numbers of simulations and targets, respectively. The power change of the interference source can be regarded as a Gaussian process; the lth target power $Q_k^{(l)}$. changes at the lth sampling moment as follows [20–25]:

$$Q_k^{(l)} = sQ_k^{(l-1)} + (1-r)Q_0^k + w^{(l)}. \tag{9}$$

Among them, $Q_k^{(l)} \sim \mathcal{N}\left(Q_0^k, \sigma_q^k\right) and \sigma_q^k$. is the variance of the power fluctuation of the kth interference source, and the parameter $w^{(l)} \sim \mathcal{N}\left(0, \left(1-s^2\right)\sigma_q^k\right)$. r is used to describe the relationship between power changes. When the sampling interval is minimal, s is close to 1 and decreases with increasing sampling time. This model can set the power changes between different interference sources to be independent of each other, which is in line with the actual situation. Assuming four interference sources, the reference powers are 100w, 800w, 900w, and 1100w, respectively. When the correlation is s = 0.5, the power changes of different interference sources are shown in Fig. 3. The power changes of interference sources are inconsistent simultaneously at other sampling ties and other interference sources. Therefore, the power changes are unpredictable. It can be seen that the proposed interference source change model can effectively simulate the actual power change. Thus, the performance of the algorithm is verified based on this model.

First, test the positioning performance of the algorithm under different sampling lengths M. Assuming that the number of sensor nodes awakened each time is N = 14 and SNR = 30 dB when the number of interference sources L = 3, the ErrM of the algorithm under different sampling lengths is shown in Fig. 4. SVM-VBEM and MMV-VBEM are single measurement vectors and VBEM algorithms that do not consider the law of power variation, respectively, and are used to compare with the proposed Wake-VBEM algorithm. As the sampling length M increases, the ErrM of both MMV-VBEM and Wake-VBEM algorithms decreases because M the addition of MMV can provide more positioning information. At the same time, SVM-VBEM

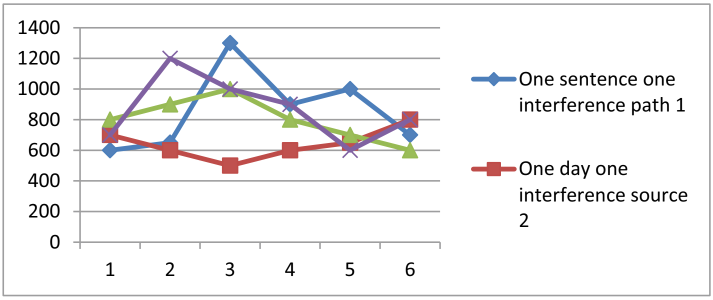

Fig. 3 Schematic diagram of power variation of different interference sources

only processes a single measurement vector, which cannot improve positioning accuracy. In addition, the MMV-VBEM algorithm does not consider the power variation of the interference source, its positioning error is significant, and the change of ErrM with M is small while Wake-VBEM. The ErrM of the algorithm decreases rapidly with the increase of M, showing its superior positioning performance. To balance the positioning accuracy and algorithm complexity, M = 2 in the following simulation.

Secondly, the influence of the number of different interference sources on the proposed Wake-VBEM algorithm is tested. Let N = 14 and SNR = 30 dB. When M = 2, the average positioning error of different algorithms is shown in Fig. 5. It can be seen that the increase in the number of targets leads to the vector scarcity decreases, which affects the reconstruction accuracy, so the ErrM of all algorithms gradually increases. The proposed Wake-VBEM algorithm considers the power variation law of the interference source, and its positioning error is the smallest.

Then, test the positioning performance of the algorithm under different sensor wake-up numbers. Let L = 3, SNR = 25 dB, and M = 2, the change of ErrM of different algorithms with M is shown in Fig. 6. Increasing M can obtain more positioning information, which helps to improve positioning accuracy. It can be seen that the ErrM of the three algorithms decreases with the increase in the number of wake-up sensors, and the ErrM of the proposed Wake-VBEM algorithm is the lowest.

Finally, the anti-noise performance of the algorithm is tested. Let N = 14, L = 3 and M = 2, and the change of ErrM with noise is shown in Fig. 7. As the SNR gradually increases, the ErrM of all algorithms decreases slowly. However, regardless

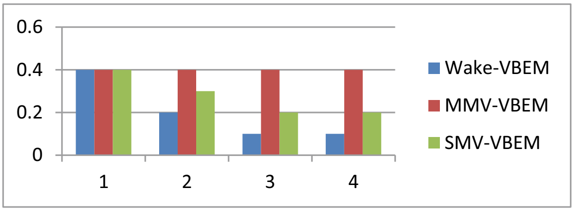

Fig. 4 Effect of sampling length on localization performance

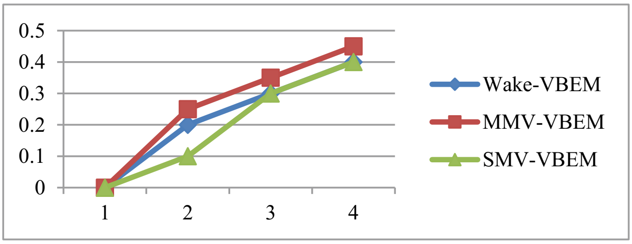

Fig. 5 The effect of the number of targets on localization performance

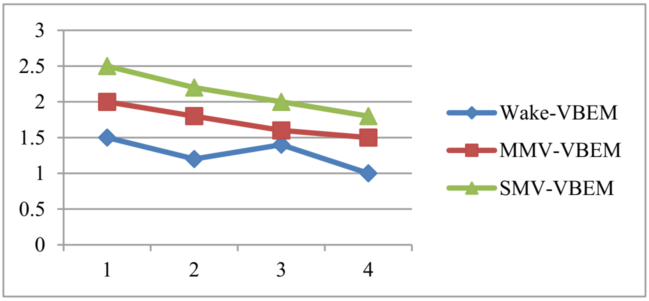

Fig. 6 The influence of the number of awakened sensor nodes on the positioning performance

of the SNR. However, the ErrM of the proposed Wake-VBEM algorithm is always the lowest. In addition, when SNR = 30 dB, the ErrM of the proposed algorithm gradually begins to converge, reflecting the algorithm's good anti-noise performance.

Conclusion to solve the problem of positioning failure caused by the power change of intelligent interference sources, an energy-saving algorithm Wake-VBEM under the sensor wake-up mechanism is proposed. First, the random wake-up strategy is modeled as the design of the observation matrix under the CS framework. Then, the reference power is introduced. Based on the BCS theory and BCS theory, the positioning problem is modeled as a block sparse reconstruction problem. Finally,

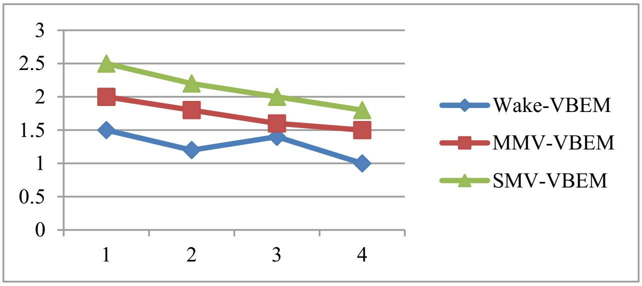

Fig. 7 The impact of noise on positioning performance

a reconstruction algorithm is designed under the VBEM framework, and the reconstruction accuracy is improved by exploring the power variation law of the interference source. Under the condition of unknown and changing source power, multiple interference source positioning can be realized, and the network's effective utilization can be improved simultaneously.

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Bidirectional Attention Mechanism-Based Deep Learning Model for Text Classification Under Natural Language Processing

Sagar Dhanraj Pande, T. Kumaresan, Govinda Rajulu Lanke, Sheshang Degadwala, Gaurav Dhiman, and Mukesh Soni

Abstract Existing text classification models based on graph convolutional networks usually update node representations simply by fusing neighborhood information of different orders through adjacency matrices, resulting in an insufficient representation of node semantic information. In addition, models based on conventional attention mechanisms are only Word vectors which are forward-weighted representations, ignoring the impact of negative words on the final classification. This paper proposes a model based on a bidirectional attention mechanism and a gated graph convolutional network to solve the above problems. The model first uses the gated graph convolutional network, selectively fuses the multi-order neighbourhood information of the nodes in the graph, retains the information of the previous order, and enriches the feature representation of the nodes. Secondly, the influence of different words on the classification results is learned through a two-way attention mechanism.

S. D. Pande (✉)
School of Computer Science and Engineering, VIT-AP University, Amaravati, Andhra Pradesh, India
e-mail: sagarpande30@gmail.com

T. Kumaresan
Department of Artificial Intelligence and Data Science, Bannari Amman Institute of Technology, Sathyamangalam, India
e-mail: kumaresant@bitsathy.ac.in

G. R. Lanke
Data Science and Engineering, Birla Institute of Technology and Science, Pilani, Rajasthan, India

S. Degadwala
Department of Computer Engineering, Sigma Institute of Engineering, Vadodara, Gujarat, India

G. Dhiman
Department of Computer Science and Engineering, Graphic Era Deemed to be University, Dehradun, India

M. Soni
Department of CSE, University Centre for Research and Development Chandigarh University, Mohali, Punjab 140413, India

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While giving positive weights to terms that play a positive role in classification, negative consequences are given to words that have adverse effects of weakening their influence in vector representation, thereby improving the model's ability to discriminate nodes with different properties in documents. Finally, maximum pooling and average pooling are used to fuse the vector representation of the word, and the document representation is obtained for the final classification. Experiments are carried out on four benchmark data sets, and the results show that the method is significantly better than the baseline model.

Keywords Natural language processing · Text classification · LSTM · Deep learning · CNN · Attention mechanism

1 Introduction

Text classification has been widely used as the primary task and core technology of natural language processing, including spam detection, sentiment analysis, news classification, etc. Traditional text classification methods based on machine learning include Naive Bayesian [1] and Support Vector Methods. However, these methods require manual feature extraction, which is labor-intensive and inefficient.

In recent years, with the continuous development of deep learning, convolutional neural networks (CNN) and recurrent neural networks (Recurrent Neural Networks, RNN) have been widely used in text classification. However, most of these models focus on capturing the local information of words. In continuous communication, discontinuous and distant words lack mutual information. Text classification models based on Graphical Convolutional Networks (GCN) [2] can handle tasks with rich structural relationships and solve the relationship between nodes to a certain extent. However, most previous GCN-based models update node representations by fusing neighborhood information of different orders through an adjacency matrix, which cannot generate word representations well. And based on attention (Attention) [3], the model usually uses the sigmoid function to formulate a positive attention score distributed between 0 and 1 and cannot give negative weights to words that have an adverse effect while ignoring word pairs that have an adverse effect—The impact of document representation.

Aiming at the above problems, this paper proposes a text classification method based on a gated graph convolutional network and a bi-directional attention mechanism. The technique first constructs each document as a document graph so that the model can achieve inductive learning. Secondly, the convolutional graph network is improved, using the gate mechanism to selectively fuse the multi-order neighborhood information of the nodes in the graph, using the previous order information and the updated information to iterate, and the effect achieved is to enrich the node features and generate words—better hidden layer representation. Then use two-way attention to improve the traditional attention mechanism to a certain extent. Unlike the conventional attention mechanism, it uses tanh as the activation function to give positive weights to the words that play an active role in the classification. At the same time, negative consequences are assigned to terms that have the effect of weakening their influence in the document representation so that the model can distinguish words that play different roles in the final classification. Then the assignment adds weighted features to the original part so that the model can enhance the features of words with positive influence and weaken the details of terms with negative impact to obtain a more differentiated feature representation and help the model classify better. Finally, word representations are fused into document representations using max and average pooling for final classification. Overall, the main contributions of this paper are as follows:

(1) Using the gating mechanism to improve the convolutional graph network, a gated chart convolutional network is proposed. The network selectively fuses the multi-order neighborhood information of the nodes in the graph and retains the neighborhood information of the previous layer—better-generated word node representations.
(2) The attention has been improved to a certain extent so that the model gives positive weights to the words that play a positive role in the classification and gives negative consequences to the terms that negatively weaken their influence in the document representation. In this way, the ability of the model to discriminate nodes with different properties in the document is improved.
(3) Many experiments have been carried out on four text benchmark datasets, and the results show that the model in this paper is significantly better than the baseline model, which verifies the model's effectiveness in this paper.

2 Related Work

Existing text classification methods using deep learning have made significant progress. Literature [4] proposed the TextCNN model to extract local and position-invariant features in documents. Literature [5] used CNN to remove character-level feature representation and achieved good results. Literature [6] used Bi-LSTM to capture long-distance semantic information in documents using a gating mechanism. Literature [7] used an optimized multi-channel CNN to extract local features to compensate for the neglect of Bi-RU. Insufficient local features. Literature [8] used Multi-timescale LSTM to capture context information of different time scales to model long documents. Since CNN and RNN prioritize local information and order information, although these models can capture local Semantic and syntactic information in word sequences, but ignores the interactive communication of discontinuous and long-distance words. Therefore, attention mechanisms and convolutional graph networks are widely used in text classification to solve these problems and have achieved specific re sults.

The attention mechanism was first proposed in computer vision [9]. In natural language processing, the attention mechanism was first used in the machine translation task based on the decoder [10] and then extended to other tasks. Literature

[11] combined Attention with Bi-LSTM for relational classification tasks. Literature [12] proposed a hierarchical attention model, using attention structures at the word and sentence levels so that the model can give words and sentences differently. Literature [13] used attention to calculate the weight value of each concept in the knowledge map, reducing the impact of irrelevant noise concepts on short text classification. However, these attention-based models usually only perform node vectors. Simple positive weighting to obtain document representation cannot well weaken the influence of negatively affecting words on document representation.

Graph convolutional networks have recently received more and more attention. In reference [14] the authors proposed a graph-based CNN model, which converts documents into graphs for the first time and uses them as the input of graph convolutional networks. Reference [15] proposed Text-CN, which first uses documents and words as nodes in the chart, then uses sliding windows to generate edge relationships between words, builds a sizeable heterogeneous chart based on the corpus level, and finally uses graph convolution. The network classifies the document nodes. Reference [16] constructed a document graph for each document, and the weights of the edges between words are randomly initialized and shared globally and are constantly updated during training; A word node uses the Message Passing Mechanism (Message Passing Mechanism MPM) [17], first aggregates the information of its neighbor nodes, and then updates the node representation. Reference [18] used heterogeneous graphs and topic models to classify short texts. Based on GCN, although text classification models can solve the mutual information problem of non-continuous and long-distance words in documents to a certain extent, these models usually update node representations by simply fusing neighborhood information of different orders through adjacency matrix and do not make full use of the knowledge of nodes. Multi-order neighborhood information cannot update the word node representation well.

3 Model Based on Bidirectional Attention and Gated Graph Convolutional Network

The model in this paper consists of four essential parts: graph construction layer, gated graph convolutional network layer, bidirectional attention pooling layer and classification layer. The overall framework of the model based on bidirectional attention and gated chart convolutional network is as follows: It is shown in Fig. 1. This section first gives the overall algorithm of the model and then introduces the four parts in detail.

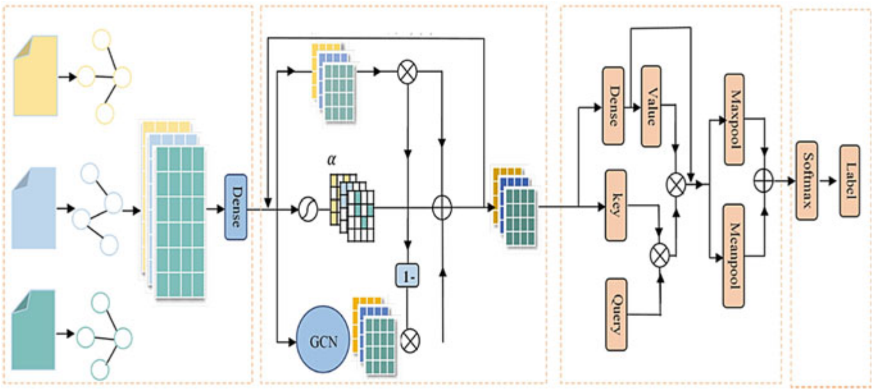

Fig. 1 The overall structure of the convolutional network model based on bidirectional attention and gated graph

The specific description of the model in this paper is shown in Algorithm 1.

Algorithm 1 Model based on bidirectional attention and gated graph convolutional network.

Input: Document Text $= (\text{word}_1, \text{word}_2, \cdots, \text{word}_1)$
output: document says h_g

1. Preprocess text in a standard way, including tokenization and stopword removal.
2. Based on the sliding window, the text is constructed as a graph, and the feature matrix X and the adjacency matrix $\tilde{A}$
3. fornodeingraph:
4. $V_i \leftarrow$ Glove embedding / ∗ Represent words as vector form ∗ /
5. forlayertin $\{2, 3, 4, \cdots, T\}$:
6. $\tilde{h}^t \leftarrow \tanh\left(\tilde{A}\, h^{t-1} W_b\right)$ / ∗ Transfer the first-order neighborhood information of the node to its own node ∗ /
7. $\alpha \leftarrow \sigma\left(W_c h^{t-1}\right)$ / ∗ Get the selection matrix α with gating function, control the aggregation of node neighborhood information ∗ /
8. $h^t \leftarrow h^{t-1} \odot \alpha + \tilde{h}^t \odot (1-\alpha)$ / ∗ Selectively fuse different order neighborhood information ∗ /
9. $h^{t'} \leftarrow \tanh\left(W_n h^t + b_s\right)$ / ∗ Get deep node representation ∗ /
10. score $\leftarrow \tanh\left(W_s h^t + b_s\right)$ / ∗ Get bidirectional attention score ∗ /
11. $\overline{h} \leftarrow \text{score} \odot h^{t'}$ / ∗ assign weight to node representation ∗ /
12. $h_n \leftarrow \overline{h} + h^{t'}$ / ∗ is added to the original feature to enhance the node feature ∗ /
13. $h_g \leftarrow \text{MaxP}(h_1 \cdots h_n) + \text{MeanP}(h_1 \cdots h_n)$ / ∗ Use maximum pooling and average pooling to obtain document representation for classification ∗ /.

Table 1 GPU memory consumption and running time

Dataset	Model	Memory/MB	Time/s
Ohsumed	TextLevelGNN	9725	495
	Text-GCN	7195	365
	OurModel	2235	248
R8	TextLevelGNN	8455	585
	Text-GCN	4635	155
	OurModel	1825	125
R52	TextLevelGNN	9175	635
	Text-GCN	5188	195
	OurModel	1865	156

3.1 Experimental Environment and Result Analysis

The operating system is Linux, the memory is 64 GB, the CPU is AMDEPYC7302, and the graphics card is 24 GB NVIDIA GeForceRTX3090. The model in this paper is implemented using the Pytorch framework.

Table 1 lists this model's GPU memory and time overhead and other models on different datasets (memory and time consumption under the same environment configuration). The data in Table 1 shows that the model in this paper has apparent advantages in terms of memory usage and running time. The reasons are as follows: Text-GCN needs to use training and test documents to build a graph based on the corpus level, which will inevitably generate many edges and consume a certain amount of time. The video memory and time; Text-Level-GNN randomly initializes the weight of the border between words, which belongs to the model parameter. It needs to be updated iteratively during training, so it takes up a part of the time and video memory; the graph constructed in this paper belongs to the document level, and the weight of the edge is in the document construction graph. The time is determined, so it will not consume too much time and video memory.

4 Parameter Analysis

Figure 2 shows the reflection of sliding window size on accuracy of different datasets. If the window is too large, edges will be added between nodes that are not closely related so that the updated node representation noise information is included in the classification, affecting the classification effect.

Figure 3 shows the effect of the number of layers of the gated graph convolutional network on classification results on the R52 dataset. Through analysis, the reason may be that as the number of layers grows, the model retains part of the original word information and captures the information of foreign words. By selectively fusing information from two nodes, word representations with richer meaning information

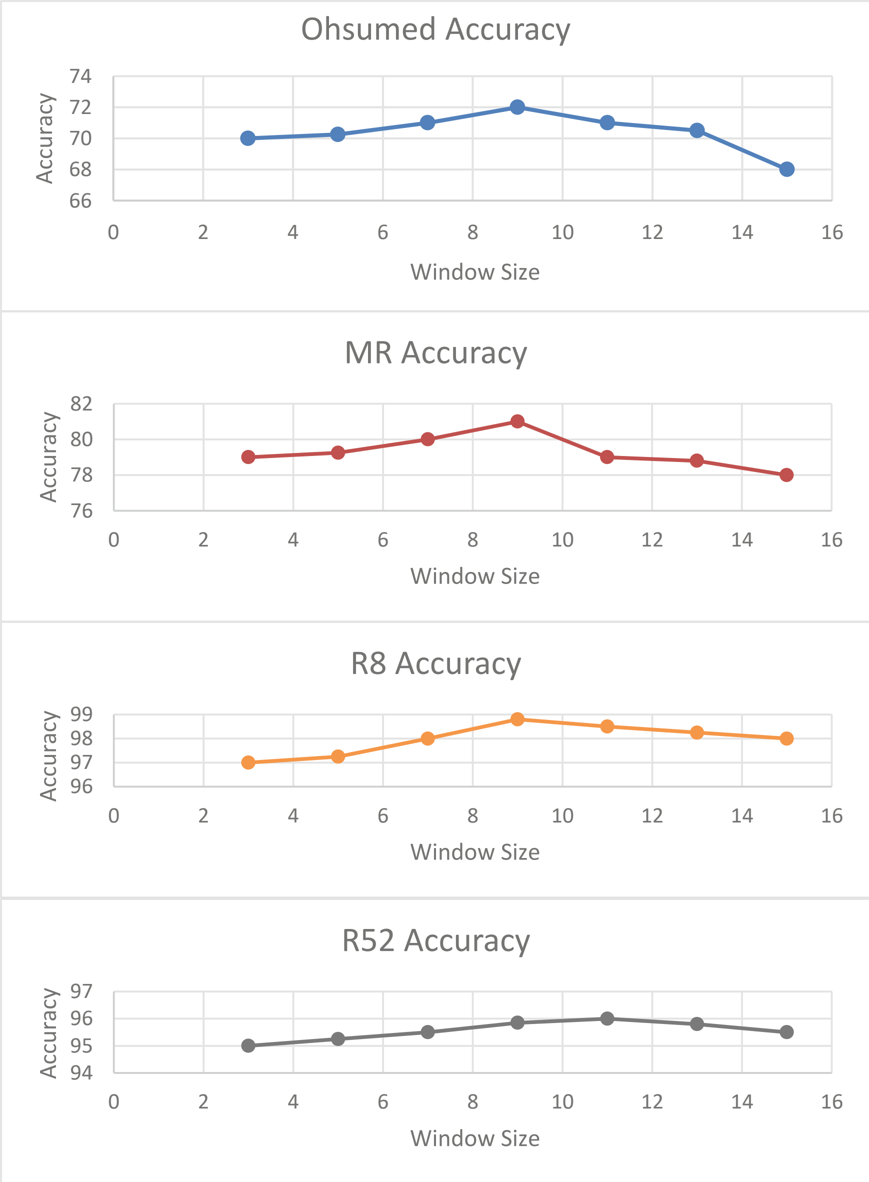

Fig. 2 The effect of the sliding window size on the experimental results

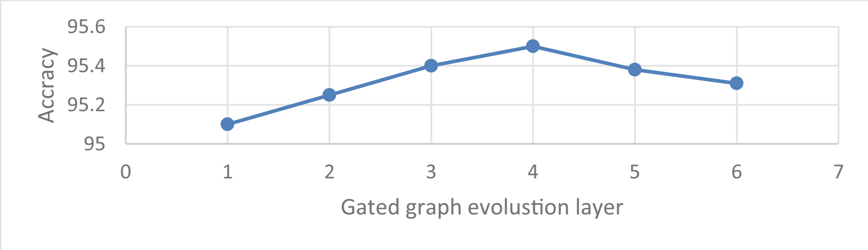

Fig. 3 The effect of the number of gated graphs convolutional network layers on the experimental results

are generated. As the number of network layers further deepens, the representation of each node will tend to be the same. Still, due to the gate mechanism, the model retains part of the original word information, which alleviates the excessive problem and smooths the nodes in the graph to a certain extent, so the accuracy will first decrease and then level off.

5 Conclusion

This paper suggests a text categorization model based on a gated graph convolutional network and a bidirectional attention mechanism. The technique first creates a document as a graph and then uses a gated graph convolutional network to identify multi-order neighborhood data selectively. After you have feature representations with more noticeable distinctions for final classification, you can employ a two-way attention mechanism to give different weights to words of various types. Many tests on four benchmark datasets reveal that the method in this study is vastly superior to the standard. To improve the model's performance, future work will consider introducing external knowledge, such as some internal statistical characteristics of the corpus. This is because the graph produced in this paper did not wholly utilize the statistical information of words in the text.

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Multi-scale Memory Residual Network Based Deep Learning Model for Network Traffic Anomaly Detection

M. Jayakrishna, V. Selvakumar, Atul Kumar, Salunke Mangesh Dilip, and Renato R. Maaliw

Abstract Models for detecting network traffic anomalies based on deep learning usually exhibit weak generalization, confined representative capacity, and low real-world adaption. In light of this, a multi-scale memory residual network-based model for identifying network traffic anomalies is proposed. The distribution analysis of the three-dimensional feature space illustrates the efficiency of the network traffic data preprocessing technique. The deep learning algorithm enhances the model's capacity to represent data by coupling multi-scale 1DCNN-LSTM networks. The realization of the residual network is shown using the residual network notion as a foundation. Deep feature extraction accelerates model convergence to detect network traffic anomalies accurately and effectively while preventing gradient disappearance, explosion, over fitting, and network damage. The experimental findings show how the multi-scale 1DCNN-LSTM network can improve the model's representational competence and generalization ability. Performance indicators for the model in this study are also superior to those for other deep learning models.

Keywords Deep learning · Feature extraction · Network traffic anomaly detection · Network degradation · One-Hot encoding · CNN · LSTM

M. Jayakrishna (✉)
Mechanical engineering, Sri Sivani Engineering College, Srikakulam, India
e-mail: jayakrishnamakka555@gmail.com

V. Selvakumar
Department of Maths and Statistics, Bhavan's Vivekananda College of Science, Humanities and Commerce, Hyderabad, Telangana, India

A. Kumar
Dr. D. Y. Patil B-School, Pune, India

S. M. Dilip
Department of Computer Engineering, GHRCEM, Pune, India

R. R. Maaliw
College of Engineering, Southern Luzon State University, Lucban, Quezon, Philippines
e-mail: rmaaliw@slsu.edu.ph

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1 Introduction

With the development of network technology and the expansion of network scale, network traffic is increasing exponentially, and network security threats and risks are becoming more and more prominent. Intrusion Detection System (IDS) [1] is a network security monitoring system. Network traffic is one of the main network states. When network intrusions occur, network traffic anomalies usually occur. Therefore, network traffic anomaly detection is the current network Research focus of the Network Intrusion Detection System (NIDS).

However, the continuous change of network attack patterns has increased the difficulty of network traffic anomaly detection [2]. Based on the empowerment effect of artificial intelligence, cyberspace security is facing new risks, including network attacks becoming more and more intelligent, and large-scale attacks becoming more and more difficult. Network attacks are becoming more and more frequent, the concealment of network attacks is getting higher and higher, the game-fighting nature of network attacks is becoming stronger and stronger, and important data is becoming easier to steal, etc. [3]. Maintaining network security is a process of attack and defense games. Network traffic Anomaly detection, as a prerequisite for ensuring network security, has received more and more attention because it can identify unknown network attacks.

The representational capacity is constrained and the false alarm rate is high [4]. It is challenging for classical machine learning to accomplish the goal of analysis and prediction due to the increase in huge data on the network [5–7], the improvement of network capacity [8–11], the complexity of data [12–15], and the diversity of features [4]. Large-scale network traffic data can be processed successfully using the deep learning approach [16–19]. Deep learning [20–22] offers better representation performance when compared to conventional machine learning techniques, which can significantly increase the effectiveness [23, 24] and precision of network traffic anomaly identification [25, 26]. The identification of network traffic anomalies is a powerful tool for thwarting contemporary network attacks. The network traffic anomaly detection model is the main topic of this study, which also suggests a multi-scale memory residual network-based network traffic anomaly detection model.

2 MSMRNet Based IDS Model

Since standard recurrent neural networks have an issue with gradient disappearance, the long-short-term memory neural network (LSTM) has been presented as a solution (RNN).Its basic unit is a structure containing multiple groups of neurons, called a cell (cell), as shown in Fig. 1.

Combining the idea of the residual network and long-term short-term memory network, this paper proposes a network traffic anomaly detection model based on MSMRNet.

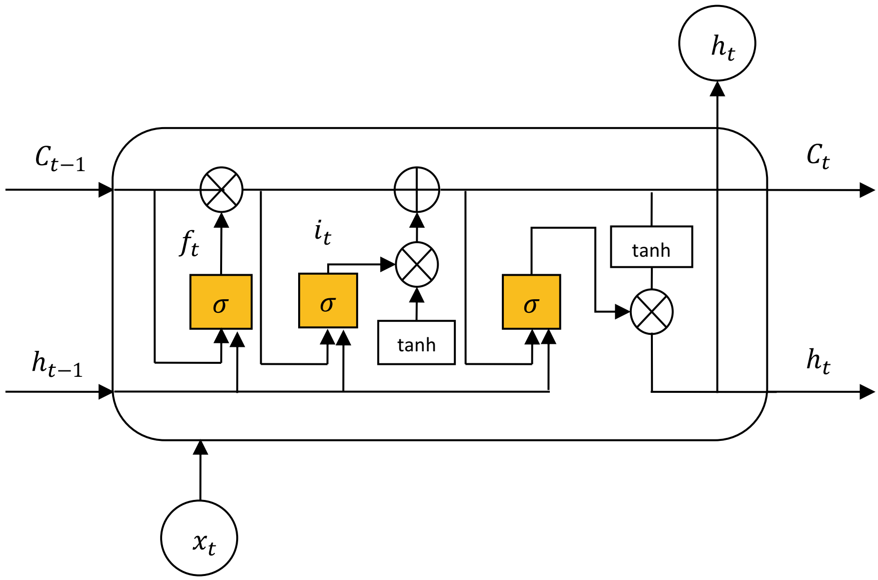


Fig. 1 Long short-term memory network

There are non-numeric data such as protocol type and service type in the network traffic data, and the machine learning model cannot handle non-numeric data, so the network traffic data needs to be processed numerically. At the same time, the network traffic data is different, and the characteristic attributes. There is a large difference in magnitude between [4], so it is necessary to carry out dimensionless processing on the network traffic data. The initial network traffic data $X_0 = \{x_1, x_2, \cdots, x_n\}$, where $f = |X_0|$ represents the characteristic dimension of the initial traffic data, and $n = |X|$ represents the characteristic dimension of the network traffic data after data preprocessing.

Input the preprocessed network traffic data X to MSMRNet for deep feature extraction, that is, the initial input of MSMRNet $X_1 = X$. MSMRNet is formed by stacking several multi-scale memory residual modules, and its l multi-scale memory residual the difference module takes Xl as input and generates output X_{l+1}, where the input X_l and output X_{l+1} have the same dimension. The multi-dimensional output Y_0 is one-dimensionalized by the Flatten layer to obtain the output Z_0. The local features are comprehensively processed through the fully connected layer [5] Obtain the output Z. Use the softmax function as the classifier to realize the network traffic classification, and obtain the network traffic classification result Y. Each element in Y represents the probability of each network traffic category, and the maximum probability category is the classification result. The calculation formula as shown in formula (1), where W_d and b_d represent the weight matrix and bias item respectively.

$$Y = softmax(Z) = softmax\left(W_d^T Z_0 + b_d\right)\ldots\ldots \tag{1}$$

The implementation method of the network traffic anomaly detection model based on MSMRNet is shown in Algorithm 1.

Algorithm 1 Implementation method of network traffic anomaly detection model based on MSMRNet
Input: Training set X_{train}, Test set X_{test}, label set Y.
Output: network traffic classification result Y_D

Steps 1 Data preprocessing

1. Perform numerical processing on the training set X_{train} and the test set X_{test} to obtain $X_{\_train\_num}$ and $X_{\_test\_num}$
2. Perform dimensionless processing on the training set $X_{\_train\_num}$ and test set $X_{\_test\_num}$ to obtain X_{train} and X_{test}

Step 2 Build the model

3. Add several multi-scale memory residual modules
4. Add Flatten layer and fully connected layer, use softmax function as the classifier step 3 training model
5. Set the experimental hyper-parameters: optimizer, the number of single training samples batch_size, the learning rate , the number of training rounds epoch. Set the experimental verification set
6. While it did not reach the preset number of training rounds epochs do
7. While the training set is not empty
8. Take the mini-batch data set batch as the model input
9. Calculate the cross-entropy loss function, C represents the number of network traffic categories
10. Update model parameters using Adam optimizer
11. end while
12. Use the validation set to validate the model and perform parameter fine-tuning
13. end while

Step 4 Save the model

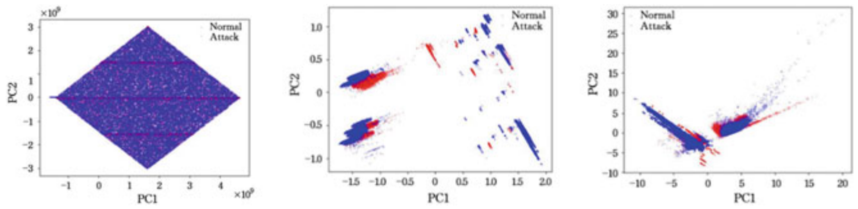
14. Save the Fine-Tuned Model

Step 5 Test the model

15. Load the saved model, test the model with the test set
16. return test set network traffic data classification results

3 Experiment

In order to obtain an effective data preprocessing method, this experiment uses the PCA two-dimensional visualization method to analyze the feature space distribution of the experimental data set after data preprocessing. The PCA two-dimensional



(a) Experimental dataset after one-hot encoding

(b) Experimental dataset after one-hot encoding and normalization

(c) Experimental dataset after one-hot encoding and normalization

Fig. 2 Visualization of data set after preprocessing using PCA

visualization results of the experimental data set after one-hot encoding are shown in Fig. 2. As shown in (a), it can be seen that the normal flow data points overlap with the abnormal flow data points in a large area. The PCA two-dimensional visualization results of the experimental data set after one-hot encoding and standardization processing are shown in Fig. 2b, which can be It can be seen that the normal flow data points and the abnormal flow data points have been partially separated, but there are still some overlapping areas. The PCA two-dimensional visualization results of the experimental data set after one-hot encoding and normalization processing are shown in Fig. 2c, it can be seen that compared with the results after one-hot encoding and normalization, normal flow data points and abnormal flow data points have been effectively separated. Therefore, this experiment uses one-hot encoding to deal with.

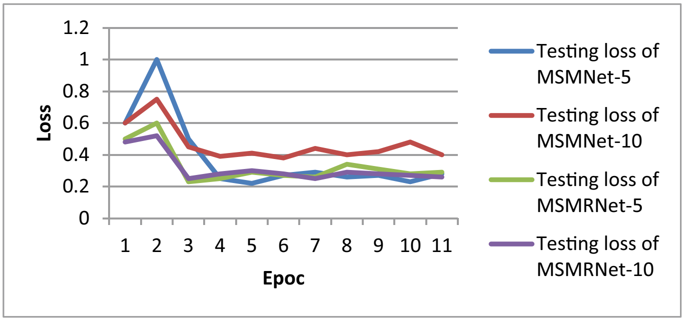
3.1 Validity Verification Experiment

In order to verify the effectiveness of MSMRNet in solving the problem of network degradation, this paper constructs a multi-scale memory network (Multi-Scale Memory Network-work, MSMNet) of different depths and compares it with MSMRNet. The experimental model is as follows.

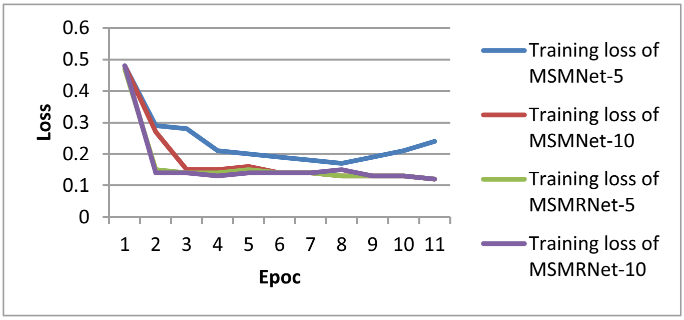
MSMNet-5: It is stacked by 5 multi-scale memory modules, including 20 training parameter layers, 10 non-training parameter layers and 1 fully connected layer.

MSMRNet-5: It is stacked by 5 multi-scale memory residual modules, including 20 training parameter layers, 10 non-training parameter layers and 1 fully connected layer.

MSMNet-10: It is stacked by 10 multi-scale memory modules, including 40 training parameter layers, 20 non-training parameter layers and 1 fully connected layer.



(a)



(b)

Fig. 3 **a** Comparison of loss rates between MSMNet and MSMRNet. **b** Comparison of loss rates between MSMNet and MSMRNet

MSMRNet-10: It is stacked by 10 multi-scale memory residual modules, including 40 training parameter layers, 20 non-training parameter layers and 1 fully connected layer.

Figure 3a, b show the comparison of the loss rate between MSMNet and MSMRNet during training and testing.

4 Conclusion

This work proposed a network traffic anomaly detection model based on the multi-scale memory residual network to address the issues of poor environmental adaptation, restricted representation ability, and weak generalization ability of the network traffic anomaly detection model based on deep learning. This essay is based on crediblity The effectiveness of the network traffic data preprocessing method is demonstrated by the analysis of the three-dimensional feature space distribution; the multi-scale one-dimensional convolution is combined with the long short-term memory network, and the model representation ability is enhanced through the deep learning algorithm; based on the idea of the residual network, there is realization of in-depth feature extraction, while preventing gradient disappearance, gradient explosion, and over fitting. The results of data preprocessing visualization demonstrate that following one-hot encoding, the results of validity verification experiments and performance evaluation experiments show that adding identity mapping can accelerate model convergence, improve network traffic anomaly detection performance, and effectively solve the problem of network degradation; comparative experimental results show that normalization processing can effectively separate normal traffic and abnormal traffic data. Performance metrics have improved.

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Auto-encoder and Graph Neural Networks-Based Hybrid Model for Link Prediction on Complex Network

Surendra Reddy Vinta, Gurpreet Singh, Manpreet Kaur, Amrinder Kaur, Sapaeva Feruza Norbaevna, and Ashok Kumar

Abstract Among the many network data mining tasks, link prediction is crucial. Due to the in-depth development of graph neural network research, related models can learn essential features of the network more effectively and have achieved good prediction results in tasks such as link prediction. However, unlike the CNN model in deep learning, the existing graph neural network model only aggregates the first-order neighbor information of nodes and does not fully consider the topological structure characteristics between neighbor nodes. On this basis, a motif-based graph neural network link prediction model is proposed. The model adopts the self-encoder structure. In the encoding process, the adjacency matrix of the node is constructed through the motif, and then the motif neighborhood of the node is obtained. Next, the neighbor information is aggregated according to the neighborhood of each type of motif, and the node is obtained through nonlinear transformation. Finally, concatenate the representations of nodes under each type of motif. However, since different motif structures have other importance in the network, the attention network gives the attention weights expressing different motifs. The vector representation of nodes is provided by connecting the attention network. During decoding, the network is

S. R. Vinta (✉)
School of Computer Science and Engineering, VIT-AP University, Amaravati, Andhra Pradesh, India
e-mail: vsurendra.cse@gmail.com

G. Singh
Faculty of Computational Science, GNA UNIVERSITY, Phagwara, India

M. Kaur
Lovely Professional University, Phagwara, India

A. Kaur
Pyramid Institute of Business and Technology, Phagwara, India

S. F. Norbaevna
Urgench Branch of Tashkent University of Information, Technologies Named After Muhammad Al-Khwarizmi, Urgench, Uzbekistan

A. Kumar
Department of Computer Science, Banasthali Vidyapith, Jaipur, Rajasthan 304022, India
e-mail: kuashok@banasthali.in

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reconstructed by computing the similarity between nodes. Investigational results on several citation collaborator networks show that the proposed method surpasses most baseline algorithms on two metrics, effectively improving the accuracy of network link prediction.

Keywords Deep learning · Link prediction · Complex network · Auto-encoder · CNN · GCN

1 Introduction

A network is an abstract representation of real-world objects and their interactions, where nodes stand for entities and links for the one-to-one or two-to-two relationships between them. These networks contain rich node attribute information, structural information and network evolution information. During the evolution of the network, some links may appear or disappear, and it is necessary to complete the missing data and make predictions about links that could be added or removed in the near or distant future. At the same time, as an important division in the field of data mining, link prediction has very important practical significance. For example, in biological network analysis [1–3], link prediction can mine and complete biological data. In scientific collaborators [4–6] and friend recommendations [7–10], link prediction can recommend relevant new friends and scientific collaborators. As a classic problem in the field of data mining, the link prediction problem has many related models and methods. Most of the current link prediction methods are based on the similarity assumption of node representations, that is, the more similar the representations of node pairs, the greater the possibility of generating links, so the problem boils down to finding high-quality node representations so that the representations of nodes retain. The topological features of the original network, that is, the nodes connected by edges in the original network, are relatively similar in the representation of nodes.

Recent advances in graph-structured network embedding approaches and graph neural networks have helped the model become even better at representing the network nodes [11, 12].

This research offers a motif-based graph neural network link prediction model to enhance node representation capability while also accounting for computational economy.

2 Preliminary Knowledge

2.1 Basic Definition

Let G = (V; E) represent a graph network, V represents a node set, E represents an edge set, if two nodes v_i and v_j are connected by an edge, then $(v_i, v_j) \in E$ [13–16]. A recurring theme is an instance of the sub-graph M of a graph G, where just a few of the network's nodes are present, is well-defined as follows:

Definition 1 (*motif*) [17] Let M be a connected sub-graph of graph G = (V; E), and satisfy any $(v_i, v_j) \in E^M$, $(v_i, v_j) \in E$, then M is called G, where E^M represents the edge set of sub-graph M.

Different third-order motifs and fourth-order motifs are shown in Fig. 1 (expressed as M_3, M_4, the first subscript *i* of each motif M_{ij} represents the number of nodes, and the second subscript *j* is the specified sorting), in the research cooperation network, the motif M_{31} represents that two authors have cooperated with one other author, but they have not cooperated with each other. Often this happens in collaborations between ordinary researchers and scientific celebrities. M_{32} represents the cooperation of researchers who are close to the same level, and the three authors have cooperated with each other.

Indicates that two authors have collaborated with one other, but they have not collaborated with each other. Often this happens in collaborations between ordinary researchers and scientific celebrities. M_{32} represents the cooperation of researchers who are close to the same level, and the three authors have cooperated with each other.

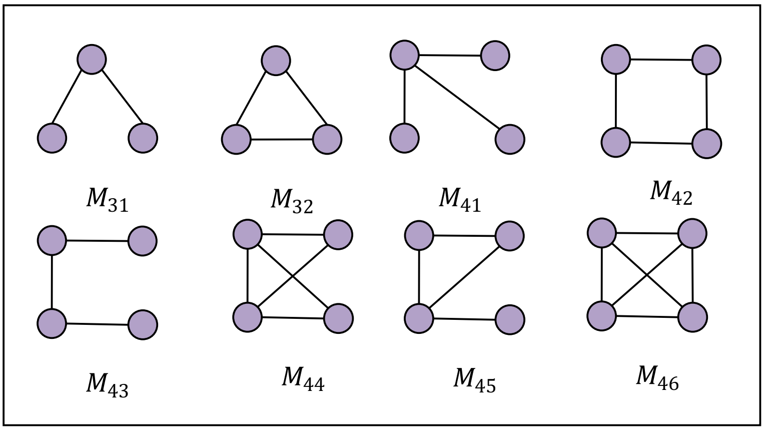


Fig. 1 All tri-motif and quad-motifs

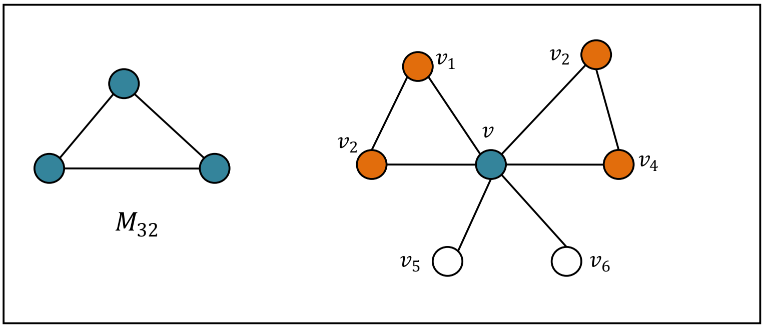


Fig. 2 Receptive field around node v contains v_1, v_2, v_3, v_4

Definition 2 (*Instances of Motifs*) [17] Let $S_u = (V_S, E_S)$ be the containing node u instance of motif S, if S_u is a subgraph of graph G, and there are $V_S \in V$, $E_S \in E$ such that For any x, y $\in V_S$, $(x, y) \in E_S$ there exists a bijection $\psi : S \rightarrow M$ such that $(x, y) \in E_S$ iff $(\psi S(x), \psi S(y)) \in E_M$.

Definition 3 (*motif neighbors*) for a specified motif type *M*, among the first-order neighbors of node v_i, the nodes located in the same type of motif as vi are called the neighbors of node vi based on motif M.

The two motifs (v, v1, v2) and (v, v3, v4) in Fig. 2 are two instances of motif M_{32}, and v1, v2, v3, v4 are also motif neighbors of v. In different networks, there are different eigenmotifs belonging to the network, which occur much more frequently than that sub-graph in random graphs. In the calculation process, in order to excerpt the motif features of the network, it is necessary to search all the motifs that are isomorphic to different types of motifs from the network, and the calculation cost of this process is relatively high. In order to reduce the computational complexity, this paper only considers the third-order and fourth-order motifs. First, the software mfinder [18] is used to find out how many of each theme type are spread throughout the network, and the distribution of motifs in each network is given.

3 Link Prediction Model Based on Motif Graph Neural Network

This paper introduces a motif-based spatial convolution operation to extract features of nodes [17]. Given the motif M and the target node v_i, the input node is a 1-dimensional vector, and the dimension of the mapped node is F = 1. Define the weight matrix w_0 of node v_i and the weight w_j of its neighbor node v_j, then the convolution at node v can be defined as the weighted sum of the neighbors of the motif M based on the motif M of node v_i, namely:

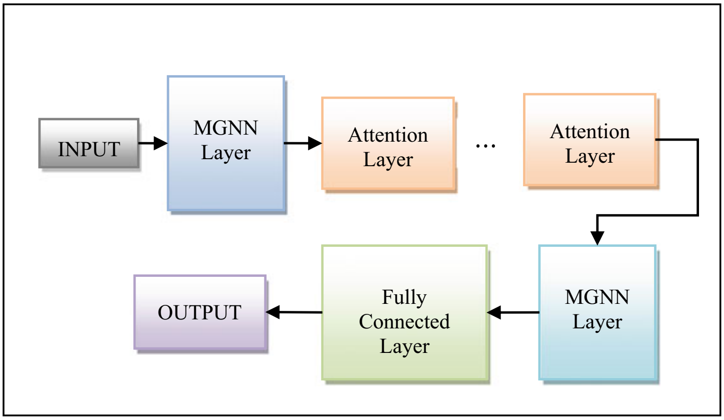


Fig. 3 Deep graph neural network framework MGNN

$$h^M(v_i) = \sigma\left(x_i w_0 + \sum_{j=1}^{N} A^M_{i,j} x_j w_j\right) \tag{1}$$

Figure 3 Among them, x_i, x_j represent the attribute vectors of nodes v_i, v_j. $h^M(v_i)$ represents the convolution output of node v_i, $\sigma(\cdot)$ represents the activation function, such as ReLU$(\cdot)$ or Softmax$(\cdot)$, the weight sharing process is to give the same weight to the motif neighbors of node v_i. Furthermore, the situation of the above formula can be extended to the general situation: the attribute matrix X of the node is a matrix of $N \times D$ dimensions, and the output dimension is F, then:

$$H^M = \sigma\left(XW_0^M + \left(D^M\right)^{-1}\right)A^M XW^M \tag{2}$$

Since the feature information of a single type of motif cannot fully represent a node in the process of representing each node, it is necessary to synthesize the information of multiple types of motifs. However, in the process of aggregation, different types of motifs have different importance to each node. In order to reflect the different influences of various motifs on node representation in the convolution process, this paper adds the attention mechanism of motifs [17].

$$h(v_i) = \sum_{i=1}^{U} \alpha_{k,i} h^k(v_i) \tag{3}$$

$$\alpha_{k,i} = softmax\left(e_{k,i}\right) = \frac{\exp(e_{k,i})}{\sum_{j=1}^{U}(e_{j,i})} \tag{4}$$

In the formula, U is the number of motifs, $\alpha_{k,i} = \alpha_{k,i}\left(h^k(v_i)\right) = W.h^k(v_i)$ is a one-dimensional convolution about $h^k(v_i)$, and the attention coefficient $\alpha_{k,i}$ reflects the importance of volume Mk to node v_i, $h^k(v_i)$ is the convolution output of node vi under Mk. In this paper, the motif graph neural network model is referred to as MGNN (motif-based graph neural network), and its network architecture is shown in Fig. 3. After the aggregation of convolutions of different types of motifs, the attention network is connected in the activation layer. These two parts form a basic neural network unit, and then iterate layer by layer. Finally, after aggregating the convolutional output of various motifs in the convolutional layer, the final node's output is obtained through a fully connected network express.

4 Experiment and Results

This research offers a homogeneous network-based MGNN link prediction model, and demonstrates the efficiency of the method on different real network datasets. During training, some links (positive class edges) of the dataset have been eliminated, although all node properties remain unaltered. The justification and test sets are constructed with the detached edges and the equal number of randomly picked unconnected node pairs (negative class edges) (negative class edges).

The Cora dataset comprises a total of 2708 sample points, each sample point being a scientific publication, and the individual document is signified by a 1433-dimensional word vector. After stemming and deleting stop words, only 3703 words are left. The PubMed dataset includes 19,717 research publications on diabetes from the PubMed database. There are 44,338 citation linkages in the network. A word vector from the dataset's lexicon of 500 distinct terms, weighted by their frequency in the document's TF and IDF values, describes each publication [19, 20]. Use the mfinder software to give the motif distribution of each network (as shown in Table 1). (as shown in Table 1). In order to further improve the computational efficiency, the motifs including 3 nodes and 4 nodes in the network are picked.

Table 1 Proportion of various motifs in 3 datasets (%)

Data set	M_{31}	M_{32}	M_{41}	M_{42}	M_{43}	M_{44}	M_{45}	M_{46}
Cora	76.70	3.30	15.10	80.40	0.10	4.10	0.20	0.10
CiteSeer	95.30	4.70	30.70	61.50	6.20	0.61	0.07	0.92
PubMed	98.25	1.75	46.10	49.10	0.60	3.90	0.28	0.02

Table 2 Results of link prediction models based on MGNN

Method	Cora		CiteSeer		PubMed	
	AUC	AP	AUC	AP	AUC	AP
VGAE	91.4	92.6	90.8	92.0	96.4	96.5
LINE	76.0	75.5	73.1	75.2	72.2	72.0
MGNN(*)	91.9	92.1	90.7	91.6	67.2	76.2
VMGNN(*)	91.6	91.4	92.0	91.0	62.3	74.0
MGNN	92.1	91.4	93.3	93.7	92.4	91.8
VMGNN		93.5	94.6	95.1	95.9	94.4

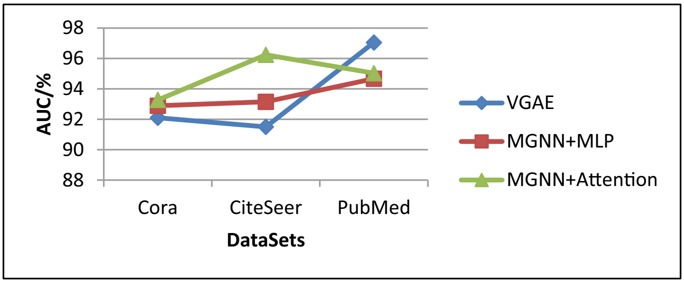
4.1 Experiment Description and Summary

Table 2 displays the findings from the experiments. After combining the high-order structural information on the two data sets, the method in this paper can obtain a better representation of network nodes on most networks, and the link prediction results are improved by 1% ~ compared with traditional methods. 4%. At the same time, the prediction result on the PubMed data set is slightly lower than that of VGAE. Due to the consideration of the motif construction data of the node, the number of motifs may not obey the normal distribution, so the experimental results of the auto-encoder will be lower than that of the variable in some cases the result corresponding to the self-encoder.

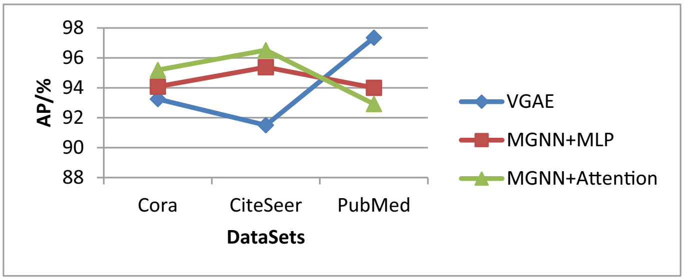
4.2 Ablation Experiment

This section compares the method in this paper with VGAE and MGNN + MLP (fully connected network). VGAE combines the graph convolutional network GCN and autoencoder to achieve good results in link prediction tasks. MGNN + MLP is a fully connected network MLP linked after the MGNN network to illustrate the efficiency of the model MGNN + Attention in this article. The result is shown in Fig. 4a, b.

The experimental outcomes demonstrate that, in most situations, the model presented in this work outperforms the other two types of models, which further demonstrates that incorporating the motif structure into the model can effectively improve the predictive ability of the neural network. At the same time, the comparison between MGNN + MLP and MGNN + Attention also verified the effectiveness of the link attention network in the MGNN network, indicating that the importance of different motifs needs to be considered in the process of node representation.



(a) Comparison Results



(b) Comparison Results

Fig. 4 **a** Comparison results **b** Comparison results

4.3 Experimental Efficiency Comparison

This section gives a comparison of the running time of each model on the link prediction task on the network Cora and CiteSeer. Figure 5 gives a comparison chart of the operating efficiency of the six corresponding algorithms.

As shown in Fig. 5, the shallow network model based on graph representation learning is more efficient, but it is difficult for such a model to learn the complex

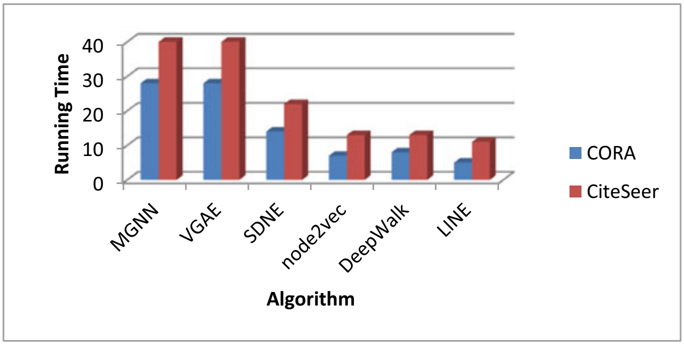


Fig. 5 Comparison of algorithm on different datasets

structural information of the nodes in the network. The method in this paper combines multi-layer neural network structure and high-order motif information. Although the computational cost is high, it has achieved a significant improvement in link prediction indicators compared with the graph representation learning methods. At the same time, because the model adopts the auto-encoder framework, the running time is close to that of models such as the graph auto-encoder (VGAE).

5 Conclusion

In this study, we offer a homogenous network-based model for predicting links in neural networks, using motifs from graphs as inspiration. Using a graph convolutional neural network, the network's high-order structure-motif information are combined, and each motif structure is given. The representation of the nodes is obtained, and the attention weights of various motifs for the nodes are further considered, and finally the network is reconstructed by using the representations of the nodes. The effectiveness of the algorithm is verified on link prediction tasks of several conventional citation datasets. Computational efficiency and accuracy need to be further improved in large-scale networks. When the model is being trained, MGNN uses the high-order motif information of the nodes, and the computational cost of the model increases to a certain extent.

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Build Near Real Time Social Media Intelligence Using Web Scraping and Visualization

Bhavesh Pandekar and Savita Sangam

Abstract Social media has become a powerful tool for marketers and advertisers to reach their target audience. With the help of platforms like Facebook, Twitter, Instagram, and YouTube, businesses can create campaigns that are more engaging and effective than ever before (Zeng et al. Social media analytics and intelligence IEEE Intelligent Systems, 25 (6) (2010), pp. 13–16). In addition to providing valuable insights into customer behavior, social media also allows businesses to share content with a wider audience in real time. By leveraging the power of social media, marketers and advertisers can create more effective strategies that have the potential to generate substantial results. Brands are always looking for ways to increase their market share and grow their customer base. One way to do this is by understanding the consumers better. Retailers often use market research to figure out who their target audience is, what they want, and how they behave. With the combination of web scraping and data visualization to build a near real time platform to analyze incremental data it is the best way to help researchers to understand the audience.

Keywords Web scraping · Social media · Data mining · ETL · Artificial intelligence · Machine learning · Web scraping tools

1 Introduction

The term social media refers to a computer-based technology that facilitates the sharing of ideas, thoughts, and information through virtual networks and communities. Social media is Internet-based and allows users to communicate content

B. Pandekar (✉)
KPMG India, Mumbai, India
e-mail: pandekarbhavesh@gmail.com

School of Engineering and Applied Sciences, University of Mumbai, Kalyan 421301, India

S. Sangam
Department of Computer Engineering, School of Engineering and Applied Sciences, University of Mumbai, Kalyan 421301, India

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such as personal information, documents, videos, and photos quickly electronically. Users interact with social media through a computer, tablet, or smartphone through web-based software or apps [1].

YouTube is an increasingly popular platform for social media marketers due to its broad reach and potential for creative content [2]. With YouTube, businesses can create engaging videos that effectively convey their message to a wide audience. Additionally, YouTube's advanced targeting capabilities make it easier than ever to target specific demographics and increase the likelihood of reaching the right people with your message. This makes YouTube an essential tool for any serious social media marketer looking to get the most out of their campaigns [3].

Social media **YouTube** data analysis is the process of analyzing user data collected from YouTube to gain insights into user behavior and preferences. This data analysis can be used to inform marketing decisions, optimize content, and more. With the increasing popularity of YouTube videos, it is essential to understand how users interact with this platform so that businesses can make informed decisions about their video content strategy.

2 Related Work

Social Media Analytics is concerned with developing and evaluating informatics tools and frameworks to gather, monitor, analyze, summarize, and visualize social media data, usually driven by specific requirements from a target application which is an interdisciplinary research area that's concerned with developing, adapting and increasing informatics tools, frameworks and methods to trace, collect and analyze an outsized amount of structured, semi structured and unstructured social media data to extract useful patterns and data [4]. Graphic theory is probably the goal strategy for analyzing social media platforms in the first period of such a platform [5].

The operating system used in social media data with certain conclusions has the goal of determining the key components of the system, for example, harps and communications (e.g., Facilitators and volunteers) [6]. They are not influenced by this, the community is respected as the customers have physical movement or sensation of different clients, and the method of tracking or impact on the choices made by different clients in the system. This hypothesis has kept it extremely powerful on the scale of the story data [7]. This is because it is equipped beyond the actual physical representation function of information to be used in information structures. Centrality measure was used to investigate power [8].

Clients in the same social system group always decide things and manage each one considering the experience in things or installed features. This is known as the recommender framework [9]. Considering the similarities between hubs in the field of communication hubs the CF method known as co-filtering can be used which forms one of the three phases of a recommendation framework (RS) that can be used for study relationships between customers. Items can be placed in a client considering the measurement of their common organization [10] where the main failure of CF is the minimum of information, based on content (other RS technique) investigating information structures to submit suggestions. However, the cross types of methods often suggest suggestions by joining CF and data-based recommendations [10].

3 System Architecture

Building a social media intelligence platform is about developing and evaluating informatics tools and frameworks for collecting, monitoring, analyzing, summarizing, and visualizing communication data, often driven by specific needs from a targeted application [11]. Social media and statistical analysis provide a rich source of educational research challenges for social scientists, computer scientists. For building this platform we follow the architecture which is cost effective and most of the technology are open-source technology (Fig. 1).

Internet

The internet has revolutionized the way businesses market their products and services. With the ability to reach customers around the world, marketing on the internet has become a powerful tool for businesses of all sizes [12]. From SEO to pay-per-click campaigns, digital marketing tactics can be used to reach target audiences and increase visibility for brands. As such, it is no surprise that more businesses are turning to online marketing as a way to maximize their success in today's digital world [13].

Web Scraping

Web scraping is the process of extracting data from a website [14]. It is a method for gathering information without having to enter the site manually and extract it [15]. Web scraping can be done by using a programming language or by using an online service [16]. Web scraping is used in market research because it provides access to

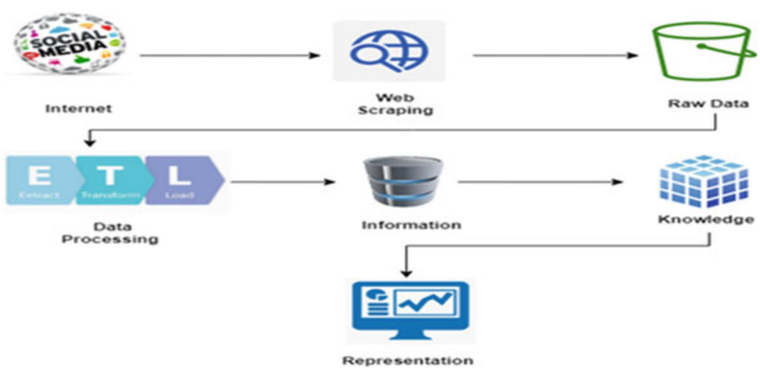


Fig. 1 Near real time social media intelligence architecture

data that would not be available otherwise, as well as being able to quickly gather large amounts of data with little effort. It is used in order to get a more complete understanding of what is happening in the market and in an industry, as well as being able to track changes to produce better analysis [17].

We can build web scraping applications using programming language, API and open source technology.

- Selenium
- Python Beautiful soup
- Graph API: Facebook, Instagram etc.

Data Processing and Information

Once the data comes into the raw bucket, cleaning up random text data (e.g., standard text), especially real-time data streamed at high frequency, introduces many research problems and challenges [18].

Knowledge

Comprehensive analysis in the field of data mining for ideas by the process of segmenting, viewing and understanding data on a website is called segmentation [19]. Large data blocks are cut into smaller sections and the process is repeated until the correct level of detail is reached for better analysis. Cutting and dialing therefore presents data in new and different ways and provides a closer look at it for analysis [20].

Representation

Visual representation of data is where information is summarized in a particular system for the purpose of conveying information clearly and effectively in graphic ways. Given the size of the data involved, visibility will grow significantly. For visualization we can use multiple tools or libraries but in our project, we used those mentioned below:

- Matplotlib
- Seaborn.

4 Design Requirements

A **use case diagram** is a simplest representation of a user's interaction with the system that shows the relationship between the user and the different components of the system in which the user is involved. A use case diagram can identify the different types of the users of a system and the different use cases and will often be accomplished by other types of the diagrams as well (Fig. 2).

Fig. 2 Design requirements

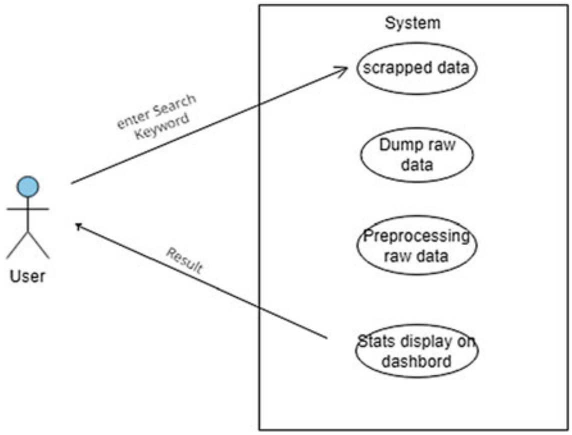
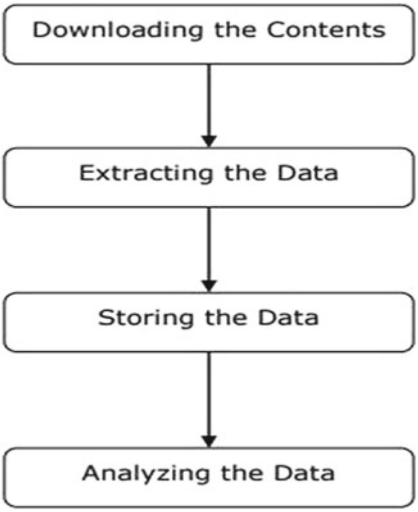


Fig. 3 Data flow diagram



A **DFD (Data Flow Diagram)** is a graphical representation of the 'flow' of data through an information system, modeling its process aspect. A DFD is often used as a preliminary step to create an overview of the system, which can later be evaluated. DFD can also be used for the visualization of data processing (Fig. 3).

5 Requirements

The hardware and software requirements of our project is as follows.

Hardware Requirements

1. Processor- Intel core i3 and above
2. RAM- 4GB or above
3. Hard Disk- 4GB or above

Software Requirements

1. Windows 8 or above
2. Python 3.6.7
3. Selenium 4.7
4. PIP and NumPy 1.13.1, pandas, matplotlib, seaborn
5. Jupyter Notebook.

6 Result and Analysis

Social media YouTube data analysis is the process of analyzing user data collected from YouTube to gain insights into user behavior and preferences [21]. This data analysis can be used to inform marketing decisions, optimize content, and more. With the increasing popularity of YouTube videos, it is essential to understand how users interact with this platform so that businesses can make informed decisions about their video content strategy [20].

Keyword Searching Using Selenium WebDriver

Once we run the code system automatically go to the YouTube site and append the specified data into created DataFrame (Fig. 4).

Once Data come into Dataframe its looks like below with Title, Meta Data (Description), URL from HTML Page (Fig. 5).

A real-world data generally contains noises, missing values, and maybe in an unusable format which cannot be directly used for analysis [22]. Data preprocessing is required for cleaning the data and making it suitable for the system which increases the efficiency of the system. After cleaning data it looks clean and now it is easy to use for analysis purpose [23] (Fig. 6).

Extract Comments and Cleaning the Comments

YouTube Comments Scraping is a process of extracting comments from YouTube videos [24]. It has become an essential tool for marketers and researchers looking to gain insights into the opinions of their target audience. By scraping YouTube comments, businesses can monitor customer sentiment in real-time, track competitor performance, and identify trends in public opinion [25]. This data can be used to

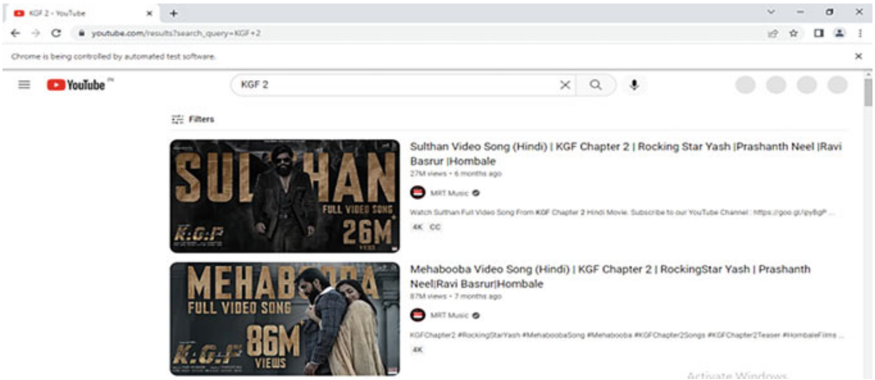


Fig. 4 Search keyword on Youtube

Out[10]:

	Title	URL	Meta_Data
1	Sulthan Video Song (Hindi) | KGF Chapter 2 | R...	https://www.youtube.com/watch?v=PWaPCqeCfeY	Sulthan Video Song (Hindi) | KGF Chapter 2 | R...
1	KGF 2 Rocky Destroy Police station KGF Ch 2	https://www.youtube.com/watch?v=tWkjFdcFbxo	KGF 2 Rocky Destroy Police station KGF Ch 2 . ...
1	Mehabooba Video Song (Hindi) | KGF Chapter 2 |...	https://www.youtube.com/watch?v=suk3mW0tDPA	Mehabooba Video Song (Hindi) | KGF Chapter 2 |...
1	Toofan Video Song (Kannada) | KGF Chapter 2 | ...	https://www.youtube.com/watch?v=VPUwb76aOw4	Toofan Video Song (Kannada) | KGF Chapter 2 | ...
1	Falak Tu Garaj Tu Lyrical (Hindi) | KGF Chapte...	https://www.youtube.com/watch?v=5CP9fycq8pk	Falak Tu Garaj Tu Lyrical (Hindi) | KGF Chapte...
1	Mehabooba Video Song (Telugu) | KGF Chapter 2 ...	https://www.youtube.com/watch?v=5xwM12SOXEE	Mehabooba Video Song (Telugu) | KGF Chapter 2 ...
1	Full Video: Yadagara Yadagara KGF Chapter 2 | ...	https://www.youtube.com/watch?v=PikKs62zE-A	Full Video: Yadagara Yadagara KGF Chapter 2 | ...
1	Mehabooba Video Song (Kannada) | KGF Chapter 2...	https://www.youtube.com/watch?v=TNFXSKbQ3JA	Mehabooba Video Song (Kannada) | KGF Chapter 2...
1	The Monster Song - KGF Chapter 2 | Adithi Saga...	https://www.youtube.com/watch?v=R4He_Gcn7cA	The Monster Song - KGF Chapter 2 | Adithi Saga...
1	KGF Chapter: 2 | Rocky's KGF | Powerful People...	https://www.youtube.com/watch?v=858wcJAWCcE	KGF Chapter: 2 | Rocky's KGF | Powerful People...
1	Toofan Video Song (Hindi) | KGF Chapter 2 | Ro...	https://www.youtube.com/watch?v=zR5-HbFW6hc	Toofan Video Song (Hindi) | KGF Chapter 2 | Ro...
1	Gravity and Logic Ke Dushman ‼ India vs Pakis...	https://www.youtube.com/watch?v=EPUqA_F4RJA	Gravity and Logic Ke Dushman ‼ India vs Pakis...
1	Kaun Banega Crorepati (Sasta Edition)	https://www.youtube.com/watch?v=RtMnzruo-h8	Kaun Banega Crorepati (Sasta Edition) by Trigg...
1	KGF 2 Dekh Ke Nischay To Pagal Hi Ho Gaya 😂	https://www.youtube.com/watch?v=dVeTc1GNERo	KGF 2 Dekh Ke Nischay To Pagal Hi Ho Gaya 😂 by...
1	KGF vs RRR vs PUSHPA | Mega Mashup | DJ Dalal ...	https://www.youtube.com/watch?v=iNJLhbKWI70	KGF vs RRR vs PUSHPA | Mega Mashup | DJ Dalal ...

Fig. 5 Scrapped data stored in DataFrame

inform marketing strategies, product development decisions, and customer service initiatives (Figs. 7 and 8).

Regular Expressions (Regex) is an essential tool for text analytics. It is powerful in searching and manipulating text strings [26]. Compared to the traditional approach for processing strings with a combination of loops and conditionals, one line of regex can replace many lines of code [25]. Regex expression is used to preprocess the author comment meaning removing garbage values from received sentences like special character, and numerical value which is not making any sense for analysis [27] (Fig. 9).

Some Data Analysis and Observations

YouTube is an increasingly popular platform for social media marketers due to its broad reach and potential for creative content. With YouTube, businesses can create

```
data.head()
```

	Title	URL	Meta_Data	video_id	owner	published	duration	views
0	Sulthan Video Song (Hindi) | KGF Chapter 2 | R...	https://www.youtube.com/watch?v=PWaPCqeCfeY	Sulthan Video Song (Hindi) | KGF Chapter 2 | R...	PWaPCqeCfeY	MRT Music	6 months ago	4 minutes 3 seconds	27071800
1	KGF 2 Rocky Destroy Police station KGF Ch 2 ...	https://www.youtube.com/watch?v=tWkjFdcFbxo	KGF 2 Rocky Destroy Police station KGF Ch 2 ...	tWkjFdcFbxo	New Feed Movie	5 months ago	4 minutes 31 seconds	1230424
2	Mehabooba Video Song (Hindi) | KGF Chapter 2 |...	https://www.youtube.com/watch?v=suk3mW0tDPA	Mehabooba Video Song (Hindi) | KGF Chapter 2 |...	suk3mW0tDPA	MRT Music	7 months ago	4 minutes 8 seconds	87517966
3	Toofan Video Song (Kannada) | KGF Chapter 2 | ...	https://www.youtube.com/watch?v=VPUwb76aOw4	Toofan Video Song (Kannada) | KGF Chapter 2 | ...	VPUwb76aOw4	Lahari Music Kannada | T-Series	6 months ago	3 minutes 36 seconds	2697804
4	Falak Tu Garaj Tu Lyrical (Hindi) | KGF Chapte...	https://www.youtube.com/watch?v=5CP9fycq8pk	Falak Tu Garaj Tu Lyrical (Hindi) | KGF Chapte...	5CP9fycq8pk	MRT Music	8 months ago	3 minutes 3 seconds	28220408

Fig. 6 Data after ETL process

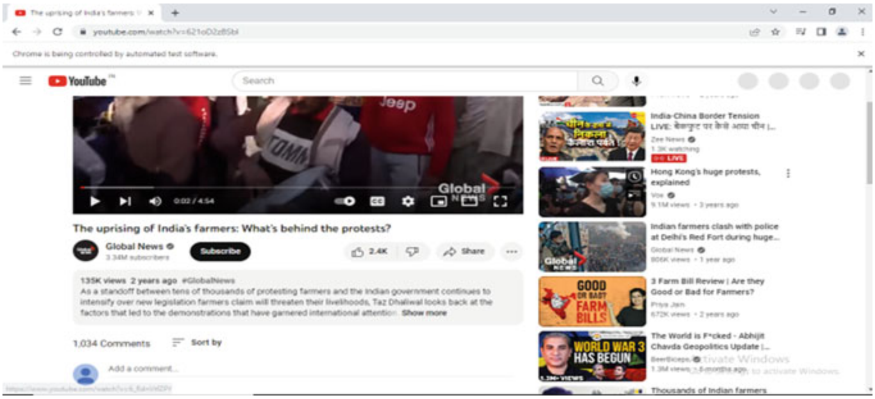


Fig. 7 Find all comment from YouTube

```
['Neutral Gear\n1 year ago\n"My grandfather used to say that once in your life you need a doctor, a lawyer, a policeman and a preacher but every day, three times a day, you need a farmer."\n10\nReply',
 'JSingh_Fauz\n2 years ago\nOnce in your life you need a doctor, a lawyer, a policeman, and a preacher. But every day, three times a day, you need a farmer....Support and respect farmers\n82\nReply',
 'Pinkius Piacus\n2 years ago\nMuch respect to the farmers\n20\nReply',
 'Jesse Tombstone\n2 years ago\nWe are nothing without our farmers!\nThey are any countries GREATEST ASSET!!!\n273\nReply',
 'Marvli mann\n1 year ago\nRICH PEOPLE PLAYS THE MONEY GAME TO WIN. POOR PEOPLE PLAY THE MONEY GAME TO NOT LOSE. THE GOAL OF TRULY RICH PEOPLE IS TO HAVE MASSIVE WEALTH AND THE POOR SEES A SURPLUS AS AN OPPORTUNITY FOR CONSUMPTION INSTEAD OF INVESTIN G IT.\n37\nReply',
 "Aspen Gaming\n2 years ago\nFarmers have guts but Indian opposition don't have.\n163\nReply",
 'Shirley L\n2 years ago\nI feel for these farmers. Keep fighting for your rights. People need you. Always the small farmers suffer making it hard for them to keep up and they always get robbed in the end.\n86\nReply',
 'SEXY BRILLIANT with The Devina Kaur\n2 years ago\nThank you Taz for this neutral reporting & Global News for keeping this i ssue active\n10\nReply',
 'L\n2 years ago\nFinally some coverage!\n55\nReply',
 'Jagvir Sandhar\n2 years ago\n@global news and Taz Dhaliwal  thank you for the awareness brought to this issue\n18\nReply',
 'Satbir Thandi\n2 years ago\nThank you Global News for covering #FarmersProtest India's national media is sold out\n6\nRepl y',
 'V 7 P 7\n2 years ago\nDown with Monsanto, Stand with Farmers\n128\nReply',
```

Fig. 8 Extract all comment from YouTube

	Comments
Neutral Gear	"My grandfather used to say that once in your ...
JSingh_Fauz	Once in your life you need a doctor, a lawyer,...
Pinkius Piacus	Much respect to the farmers\n20\nReply
Jesse Tombstone	We are nothing without our farmers!\nThey are ...
Marvli mann	RICH PEOPLE PLAYS THE MONEY GAME TO WIN. POOR ...
...	...
Punjabischoolclub	These farmers are protesting against 3 controv...
Sam Phoenix	Canada government complaints against India in ...
tova lynch	Where is the spokesperson for the Indian gover...
Niraj Tayde	We Support our Farmer brother's n sister's (d...
Zord Shenlong	I love India famer \n7\nReply

Fig. 9 Final extracted comments

engaging videos that effectively convey their message to a wide audience. Additionally, YouTube's advanced targeting capabilities makes it easier than ever to target specific demographics and increase the likelihood of reaching the right people with your message. This makes YouTube an essential tool for any serious social media marketer looking to get the most out of their campaigns [15] (Figs. 10, 11, 12 and 13).

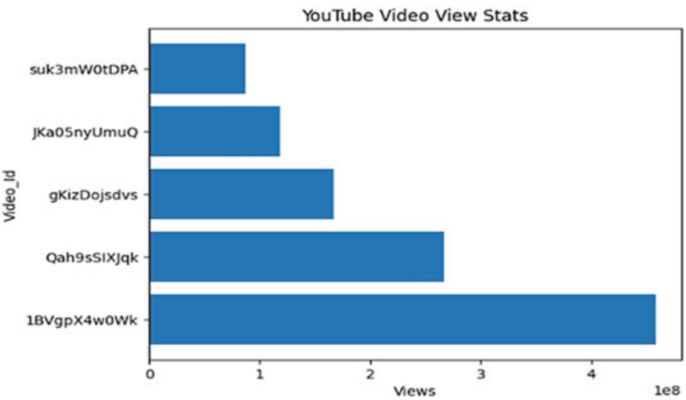


Fig. 10 YouTube video view bar graph

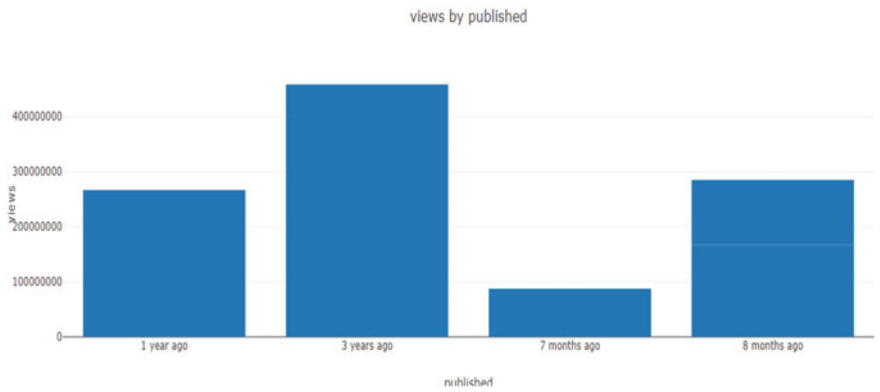


Fig. 11 Bar graph of views by published

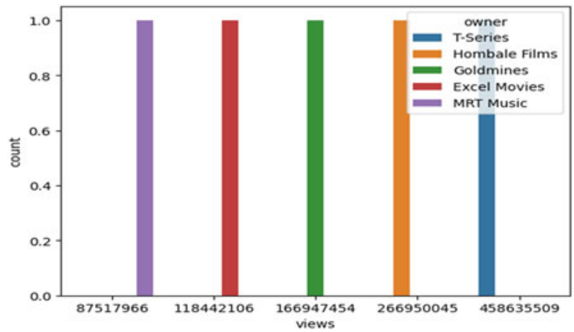


Fig. 12 Count plot of views with respect to owner

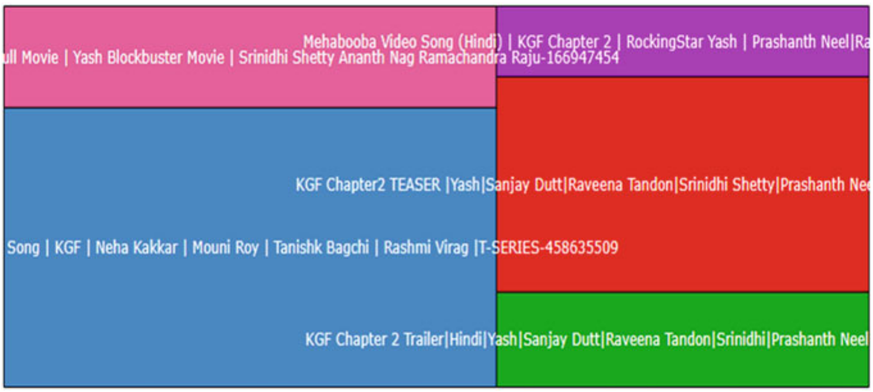


Fig. 13 Tree Plot of search keyword

7 Conclusion and Future Scope

A. Conclusion

The main goal of this project was to explain how to use web scraping techniques to gather data from the web and display it in a meaningful way. We were able to accomplish this goal by using data from different social media platform to create a meaningful inside for business use. It will save a great deal of loading time. In fact, you would save yourself time and financial loss in a very nice way by using this social media intelligence system.

B. Future Scope

Some limitations of this study are the unstructured, irrelevant, missing attribute values. To build a sentimental model of received data with many null and irrelevant values it may lead to some issues in analysis. Testing these models with large valid datasets with minimal or no missing attribute values reveals more insights and better analysis.

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Predicting Position of User Equipment Using Machine Learning

Samhita Kothandaraman, Keerthi Srinivas, and Megharani Patil

Abstract Radio-based positioning of user devices is a critical utility that has experienced extensive development and improvement in fifth generation (5G) radio networks. The reliability on smartphones has also significantly increased over the past ten years, providing us with more data to investigate on. In addition to that, forecasting a user's upcoming position is essential for recommender systems and location-based services. The rapid rise of location-based service applications has had a substantial impact on many fields, including traffic flow forecasting, weather forecasting, and network resource optimization. Large amounts of trajectory data are currently being generated in relation to human mobility as a result of the rapid proliferation of positioning and sensor equipment. The positioning of emergency calls is an example of an application area in which the user unit must found with a precision of about ten meters. In metropolitan regions, where buildings obscure and reflect radio signals, resulting in multipath propagation along with non-line-of-sight (nlos) signal conditions, radio-based positioning has always been difficult. To deal with nlos, one method is to use data-driven techniques, such as machine learning algorithms on beam-based data. This case study focuses on estimating position of User Equipment (UE) in a radio network. A model that converts measurements to position estimates is developed using a set of positioned measurements as training data. This is done by means of a Neural Network model. Both Convolutional Neural Network and Dense Neural Network were used based on available features. The model's performance was further validated using the K-Fold Cross Validation method.

Keywords Channel impulse response · User equipment · Convolutional neural network · Dense neural network · k-fold cross validation

S. Kothandaraman (✉) · M. Patil
Thakur College of Engineering and Technology, Mumbai, India
e-mail: krs302002@gmail.com

M. Patil
e-mail: megharani.patil@thakureducation.org

K. Srinivas
Altiostar Networks India, Pvt. Ltd, Bangalore, India
e-mail: keerthi.srinivas@rakuten.com

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1 Introduction

There is a wealth of expert knowledge in the field of communications about how to model various kinds of channels, account for different hardware flaws, and design the best signaling and detection systems to ensure reliable data transmission. As a result, it is a sophisticated and complex engineering field with numerous discrete areas of study, all of which have observed fading returns in terms of performance enhancements [1]. The advent of Industry 4.0 and the development of mobile wireless technology, from 3G/4G to 5G, has led to an increase in the complexity of wireless system design. Due to demands for effective resource sharing among growing user bases, wireless networks have also grown more challenging to operate [2]. Since traditional rule-based engineering approaches are no longer sufficient to address these problems, many engineers are turning to artificial intelligence (AI) as a solution [3]. Artificial intelligence (AI) has introduced the sophistication required for contemporary wireless applications, from coordinating communications between autonomous vehicles to optimizing resource allocation in cellphone conversations.

The fifth-generation (5G) of mobile network period has made it conceivable to integrate edge computing, big data, Internet of Things (IoT), artificial intelligence (AI) and multiple other technologies [4]. Big data analytics, accurate parameter estimation, collaborative decision making, and instrumental functions of wireless networks like, heterogeneous networks (HetNets), cognitive radio (CR), Internet of Things (IoT), and machine to machine (M2M) networks have all been positively supported by AI algorithms [5]. Future wireless network architecture integration of AI has emerged as a technology trend for investigation and research. The architecture for wireless network transmission in the future will benefit greatly from AI [6, 7]. In addition, more study is required on how AI is applied to modelling communication network channels, network working and maintenance, network security audits, and other areas [8].

Wireless technology aids communication amongst users or the transfer of data between locations without the use of cables or wires. Radio frequency and infrared waves are utilized for a lot of the communication. The two types of wireless networks that are most commonly encountered in real-world situations are:

(a) Local Area Network (LAN): Used in places like an office's internal network or a house with a network of gadgets (computers, gaming consoles, personal mobile phones and tablets connected to the same router in one area forms what is known as a local area network). Previously, setting up a local area network necessitated the need of a wired connection via what is called an "ethernet cable." Wi-Fi is at present being widely utilized for local networking, in spite of wired networks even now being extremely popular for a variety of factors, along with improved defense from interference and security when compared to wireless.
(b) Wide Area Network (WAN): The Internet can be assumed to be a wide area network (WAN). It is a type of network that traverses a wider area. Since they are naturally faster, more reliable and are less prone to hindrance, wires are employed to transport most of the data passing through the network in the case

of Internet. However, people are increasingly using cellular data to access the Internet wirelessly since the induction of the modern smartphone and other transportable devices like tablets in the real world [9].

Function of AI in wireless technology will grow along with the variety and volume of devices linked to networks. Engineers need to be ready to integrate it into ever-more complicated systems. The future success of the technology will depend on understanding the advantages and existing uses of AI in wireless networks, as well as the best practices required for effective implementation [10].

2 Background

In positioning scenarios, multipath, non-line-of-site (NLOS), indoor coverage and non-ideal synchronization may create obstacles that can be challenging to solve using conventional techniques. As a result, an AI-based positioning strategy is employed.

The aim of performing this study is to estimate the position of a user experiment (x, y) in a given area using features of a radio network gathered by eighteen base stations. Following are the features incorporated in training the neural network.

(a) Channel Impulse Response (CIR): The technique of channel sounding involves figuring out a transmission channel's impulse response, particularly a cellular radio channel. The idea was inspired by traditional acoustic distance-measuring techniques, such as using an echo sounder to gauge water depth. The channel impulse response (CIR), which includes the magnitude and phase of the signal, gives intricate, thorough information on the effect of the channel of interest on a radio signal. As a result, it is perfect for describing the channel. The radio channel is negatively impacted by reflection-induced signal echoes, diffraction and scattering-induced distortions, building and tree induced shadow effects, and even weather-related impacts like rain and snow [11].

(b) Reference Signal Received Power (RSRP): In order to always connect to the cell tower with the best signal, user equipment continuously scans the signals emitted by all neighboring cell towers. A modem uses a value known as RSRP to decide which tower to connect to. Reference Signal Received Power (RSRP) is defined as the linear average of power contributions made by resource elements that transmit cell-specific reference signals in the bandwidth of the measurement under consideration (in [W]). The antenna connector on the user equipment (UE) is to be used as the RSRP's reference point. With the restriction that relevant measurement accuracy standards must be met, the UE implementation is left to decide how many resource elements within the considered measurement frequency spectrum and within the measurement time are employed by the UE to estimate RSRP [12].

(c) Time of Arrival (TOA): The time of arrival (TOA) is the length of time it takes for a signal to travel from the source to the receiver. In order to measure the TOA information, the source and all of the receivers must be precisely synchronized. If a two-way or round-trip TOA is computed, a similar system is not required. The measured distance between the source and the receivers is obtained by multiplying the computed TOA by a specified propagation speed, typically represented by the letter c. The source must be located on the circumference of the circle that the measured TOA represents having its center as the receiver in a two-dimensional (2D) space.

    A distinct intersection point between three or more of these circles produced by the noise-free TOAs, corresponds to the source position. To estimate a two-dimensional position of a source, a minimum of three sensors are needed. These can be represented as a collection of circular equations. By using the optimization criterion and the known sensor array geometry, source position can be calculated [13].

(d) Time Delay of Arrival (TDOA): TDOA is a positioning method that is calculated by finding the difference between the TOA of radio signals. In a real-time location system (RTLS), TDOA is used to precisely determine the location of tracked entities in real-time, such as tracking tags attached to personnel or important assets. It requires three or more remote receivers (probes) that can pick up the desired signal. Each probe is time-synchronized to record the appropriate I/Q data blocks. To determine the difference in arrival times at each probe, the software modifies the time signature of each I/Q data set. This reveals the variation in the source's separation from each pair of probes. A collection of curving lines indicating answers to the distance equations are produced by using many probes [14].

(e) Position (Pos): Determines the x and y co-ordinates of a User Equipment in a given area on the basis of features such as fingerprinting, received signal strength, angle of arrival, time of arrival, time difference of arrival, etc. One can directly change the TOA and RSS to reflect the range measurement.

An ensemble model [15] composed of Convolutional Neural Networks and Deep Neural Networks were used for the final prediction where all four features were given as input attributes. A K-Fold cross validation method was used to validate the neural network's performance.

3 Methodologies Implemented in the Model

Machine learning lifecycle is a methodical reiterative process of developing an optimal model ready for incorporation into a production system and utilization by the aimed end-users. It involves training [16], testing, and deployment of the model. Following are the methodologies incorporated in this model:

(a) Dense Neural Network: A individual layer of an artificial neural network (ANN) [17] contains an input layer, few hidden layers, and an output layer. Every node, or artificial neuron, is linked to other nodes. Each of these nodes has a corresponding weight and threshold that is associated with it. Any node where the output exceeds a definite threshold value is triggered and begins supplying data to the network's topmost layer. In this instance the activation function used prominently was Rectified Linear Unit (ReLU) [18].

(b) Convolutional Neural Network: Convolutional neural networks surpass other deep learning models when they are provided input features such as images, voice, or audio [19]. There are three key types of layers:

    i. Convolutional layer
    ii. Pooling layer
    iii. FC (fully-connected) layer

    Any convolutional network's main layer is the convolutional layer. The fully-connected dense layer is the concluding layer and convolutional layers or pooling layers, come before it. The Network [20, 21] precedingly becomes more and more complex with every layer, identifying larger areas of the image [22]. Initial layers specify rudimentary elements like colors and borders. Conspicuous features or shapes of an object are recognized first and foremost when visual data moves through CNN layers and finally the anticipated object is identified [23].

(c) K-Fold Cross Validation: Machine learning models are tested on a small data sample using a resampling technique known as cross-validation. In the procedure, one parameter, k, determines how many groups should be formed from a given data sample. As a result, the procedure is commonly referred to as k-fold cross-validation. Cross-validation is frequently used in practical machine learning to assess how well a machine learning model works on untrained data. It specifically refers to the use of small samples to evaluate how the model will perform when forecasting using data that was not used during model training [24].

4 Implementation of Models

The dataset is a labelled set of wireless communication data which contains large amounts of channel samples in different forms which can be used for AI-based wireless communication research [25].

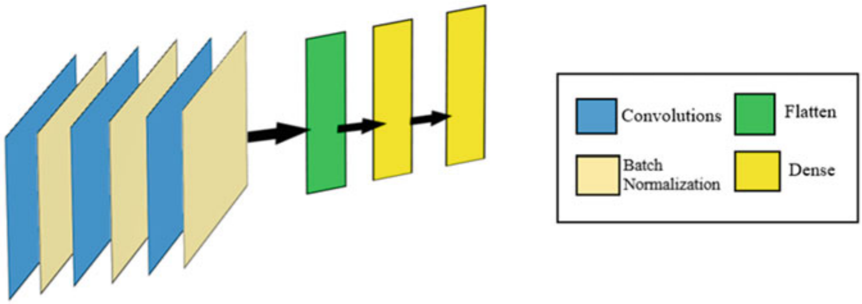

Fig. 1 This figure shows the flow of convolutional neural network executed for stage 1 i.e., predicting position of user equipment using channel impulse response

The implementation of the model took place in four stages. In the first stage, channel impulse response was provided as input [26] to the neural network model. In the second stage, RSRP, TDOA and TOA were used as input features. In the third, all four inputs [27, 28] were given to the neural network [29]. The results of all three stages were compared to see which approach was more optimal and accurate and validated using the K-Fold Cross Validation Method.

4.1 Stage 1: Channel Impulse Response

In the initial stage, Channel Impulse Response was used to predict the position of User Equipment. The dimensions of the input are as follows:

$$(80,000,\ 18,\ 256,\ 2)$$

Since the input is 4 dimensional, a convolutional neural network [20] approach was used. Following is the model of the neural network that was implemented (Fig. 1).

A total 5-layer neural network was defined wherein 3 layers were convolutional layers, each followed by a batch normalization parameter. After flattening, the output from the convolutional layers was fed to a fully connected dense layer which then provided output in the final layer.

4.2 Stage 2: RSRP, TDOA, TOA

In the second stage the input data given was RSRP, TDOA and TOA. Each of the features had a dimension of:

$$(80,000,\ 18)$$

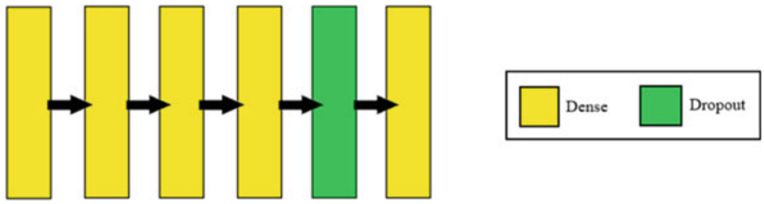

Fig. 2 The figure shows the execution of a simple artificial neural network, taking inputs rsrp, tdoa and toa, comprising only of dense layers along with one dropout layer

Since the data provided was two dimensional, all three inputs were merged to form a three-dimensional data of the form:

$$(80,000,\ 18,\ 3)$$

This input was fed into a model containing Dense layers. An illustration of the model incorporated is shown below (Fig. 2):

4.3 *Stage 3: Ensemble Model*

This penultimate stage used all four inputs for prediction. Considering the fact that inputs were of different dimensions, an ensemble approach was used [30, 31]. Here the models were defined functionally. The ensemble approach is illustrated diagrammatically below (Fig. 3):

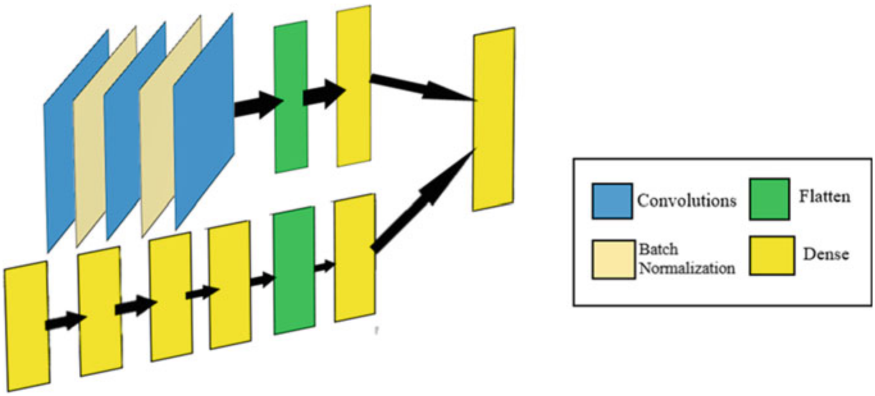

Fig. 3 Figure shows flow of neural network model in two stages. The first one being a Convolutional model and the second one being a dense neural network model, both outputs merging into one dense layer which provides position of user equipment as output

In the first part of the ensemble, convolutions were used to predict position on getting CIR as input. In the second part dense layers were used so as to predict using RSRP, TDOA and TOA. Outputs from the final layer of both the models were then concatenated and a Dense layer was mentioned to predict the final position of User Equipment.

4.4 Stage 4: Validation Using K-Fold Cross Validation

The final stage consisted of validating the above ensemble model using K-Fold Cross Validation13. The value for k was taken as 5. That is, 5 folds were used to validate the results and after the conclusion of execution of each fold, the MSE and position error at 0.9 cdf was noted. The above folds were iterated over both 30 and 50 epochs.

5 Result and Discussion

To evaluate the performance of the model, following metrics were used:

(a) Mean Squared Error (MSE): MSE measures the amount of error in a statistical model. It is defined as the mean or average of square of difference between the actual values and the predicted or estimated values. It is represented mathematically as follows:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - y_{\mathrm{pred}})^2 \tag{1}$$

The closer the value of MSE is to zero, the more accurate the model is.

(b) Position error at 0.9 cdf: A cumulative density function (cdf) is used to describe the probability distribution of the variables. In this case the cumulative density function was plotted against position error in the model and the error at 90% cdf was observed. The lower the error value, the more accurate the model is.

Upon execution of all three stages and applying the above-mentioned metrics, following were the results obtained (Table 1):

During analysis it was found that the ensemble model provided the most accurate result with very low position error. To further prove this statement the cdf plot and a frequency plot which shows the distribution of error are shown below (Fig. 4):

The CDF plot shows a linear increase in the positioning error value until 0.519 after which there is a constant value 1. To understand this better, a frequency plot was plotted to understand error distribution in the model. The same is illustrated below (Fig. 5).

Table 1 The below table displays Mean Squared Error (MSE) values and position error values after execution of each model

Model	No. of epochs	MSE	Position error at 0.9 cdf
Stage 1: CNN	25	1.0143	2.141
Stage 2: ANN	300	0.5967	1.288
Stage 3: ensemble model	50	0.0676	0.519

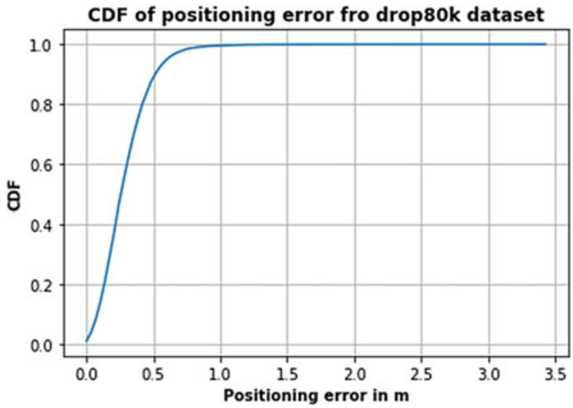

Fig. 4 Cumulative density plot for ensemble method

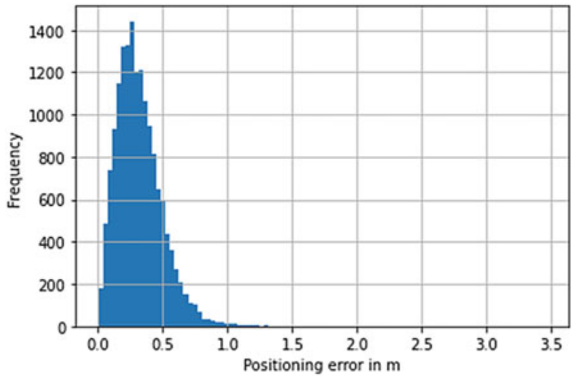

Fig. 5 Frequency plot for ensemble model

As observed in the frequency plot, majority of the position error is approximately distributed between 0.0 and 0.55 making the model most accurate amongst the three models used.

K-Fold Cross Validation was performed on the Ensemble model iterating over both 30 epochs and 50 epochs to observe how results change with changes in input values. MSE values and position error at 0.9 cdf after execution of each fold are tabulated below (Table 2):

Table 2 The above table displays MSE values and position error at 0.9 cdf after execution of K-fold Cross-validation over 30 and 50 epochs

Indices of samples inputted	K-fold value	MSE	Position error at 0.9 cdf
Epochs: 30			
0–16,000	1	0.1257	0.7448
16,000–32,000	2	0.098	0.6516
32,000–48,000	3	0.1131	0.7219
48,000–64,000	4	0.1261	0.7307
64,000–80,000	5	0.13	0.7458
Epochs: 50			
0–16,000	1	0.1135	0.6482
16,000–32,000	2	0.1277	0.7447
32,000–48,000	3	0.0659	0.537
48,000–64,000	4	0.0947	0.6346
64,000–80,000	5	0.0676	0.5236

The MSE values and position error values changed with changes in input values and better results were observed while iterating the folds on 50 epochs. This goes on to validate the accuracy of predictions of position produced by the ensemble model and how these results could be improved by increasing the number of iterations.

6 Conclusion and Future Scope

In this paper a deep learning-based approach for estimating the position of User Equipment in a radio network is proposed. The results show that an ensemble model taking all four inputs would outperform other models implemented in this study. The same has been validated by the use of K-Fold Cross-Validation.

While the above method provided a good result, there is further scope to improve the model in the following aspects:

(a) The above model can be run on increased number of epochs to achieve better accuracy.
(b) The model should be robust to different types of input features and be able to predict with any given data.
(c) While being robust to varied types of features, it must also be able to compute the position of user equipment without taking up too much memory.
(d) The model must also be able to compute data in real time.

The development of these aspects will be supported by forthcoming advances powered by 5G and AI, two essential elements that operate together to enhance system functioning and efficiency. The emergence of 5G-connected devices has shown the capability to foster distributed intelligence via further advances in AI learning and interpretation. Associated intelligent edge evolution has begun. It is critical to comprehend the full potential of the 5G future as required for on-device intelligence increases. This study goes on to provide a promising direction for future research in the field of wireless networks [32].

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A Comprehensive Review on Explainable AI Techniques, Challenges, and Future Scope

Ashwini Patil and Megharani Patil

Abstract Artificial Intelligence (AI) has been making remarkable advancements in recent years and has the potential to revolutionize many aspects of our lives. From self-driving cars to healthcare systems, AI can make tasks easier, faster, and more accurate. However, the increasing reliance on AI has raised concerns about its transparency, accountability, and interpretability. eXplainable AI (XAI) is a field that focuses on explaining the predictions made by AI systems. This has become increasingly important as AI is being used in sensitive and critical applications such as medical diagnoses, financial risk assessments, and criminal justice decisions. It is essential to ensure that the decisions made by AI systems are transparent, trustworthy, and can be justified to stakeholders. The paper explores the challenges associated with creating explainable AI systems and the different techniques that are being developed to overcome these challenges. Further, it presents a summary of the strengths and weaknesses of various XAI techniques. The paper will provide an overview of the state-of-the-art in XAI and highlight the need for further research in this field.

Keywords eXplainable AI (XAI) · Machine learning

1 Introduction

Artificial Intelligence (AI) has become an increasingly ubiquitous technology, permeating various domains such as healthcare, finance, and transportation [1]. While AI systems have demonstrated remarkable performance, there are growing concerns over their lack of transparency, accountability, and interpretability [2]. This lack of transparency is often referred to as the "black box" problem and is a major hindrance to the widespread adoption of AI systems. The need for more transparent AI systems

A. Patil (✉) · M. Patil
Thakur College of Engineering and Technology, Mumbai 400101, India
e-mail: ashwini.patil@thakureducation.org

M. Patil
e-mail: megharani.patil@thakureducation.org

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has led to the development of a new field called eXplainable AI (XAI) [3]. XAI aims to design and develop AI systems that can provide human-understandable explanations for their predictions, decisions, and internal workings [3].

XAI has gained significant attention in recent years, and researchers have developed various techniques for improving the interpretability of AI systems. These techniques range from model-agnostic methods, such as saliency maps and feature importance analysis, to model-specific methods such as decision trees and rule-based models.

The motivation behind the research in the field of XAI aims to overcome the challenges of gaining transparency and trust in the decisions made by the AI/ML models, compliance with regulations, improved decisions, and model selection for solving a specific problem.

1. Transparency and Trust: The black-box nature of AI systems has led to a lack of trust and accountability among stakeholders. XAI aims to address this issue by providing explanations for the predictions and decisions made by AI systems. This increased transparency can lead to improved trust and confidence in AI systems.
2. Compliance with Regulations: There are various regulations and ethical guidelines that mandate the need for explainable AI systems, such as the European Union's General Data Protection Regulation (GDPR) and the US's Algorithmic Accountability Act.
3. Improved Decisions: Explainable AI can help decision-makers understand the reasoning behind AI predictions and decisions, allowing them to make better-informed decisions. This is especially relevant in high-stakes domains such as healthcare and finance.
4. Model Selection: XAI can help practitioners compare and evaluate different AI models, making it easier to choose the most appropriate model for a given task.

There are different challenges associated with Explainable AI as stated by authors of [4]. One of the primary challenges in XAI is developing methods for making AI models transparent and interpretable, without sacrificing performance. A black-box model is one in which the decision-making process is not clearly understood, making it difficult to trust the output of the system. This lack of transparency can be especially problematic in critical applications, such as medical diagnosis or autonomous driving, where the consequences of incorrect decisions can be severe. XAI aims to address this issue by developing methods for making AI models more interpretable and transparent so that the decision-making process is better understood and trusted.

Another challenge in XAI is developing methods for interpreting AI models' output in a meaningful way to humans [4]. This is particularly important in applications where the model makes decisions that significantly impact people's lives. For example, in a medical diagnosis application, it is important to be able to understand why a model made a particular diagnosis so that healthcare professionals can verify the accuracy of the decision. Similarly, in autonomous driving, it is important to be able to understand why a model made a particular driving decision so that the safety of the system can be assessed.

To address these challenges, XAI research has focused on developing new methods for making AI models more interpretable, such as using local explanations, feature importance methods, and model distillation. Another approach is to develop new AI models that are inherently interpretable, such as decision trees and rule-based systems. XAI research has also focused on developing new methods for visualizing the output of AI models, such as using heat maps, saliency maps, and activation maps.

2 Related Work

The Machine Learning models that exist in the literature can be distinguished into two types, Inherently Interpretable Models, and Post-hoc Explainable Models. The inherently interpretable models are self-explanatory or interpretable by design whereas post-hoc explainable models are prebuilt models that are explained by external XAI techniques [5].

2.1 Inherently Interpretable Models

In general, the models which are understandable by themselves fall under the category of Inherently Interpretable Models, they are also known as transparent models. Examples of such models are as follows.

Linear Regression/Logistic Regression

Linear Regression is a machine-learning model used to predict a value of the dependent variable y, given the value of the independent variable x. This model assumes the linear dependence between the input (Predictor) and output (Predicted) variables, inhibiting an adaptable fit to the data. This nature of the model fits it under the category of inherently interpretable models. The models' explainability can be correlated with different users depending on the application. Although Linear Regression/Logistic Regression possesses the properties of an inherently interpretable model, depending on the context, it may fall under both categories. For example, for ML experts the model is self-explanatory but in the case of the users who are not ML experts, post-hoc explainability techniques (like visualization) can be used to interpret the behavior of the model. This model is used in practice by many researchers. The authors of [6–9] agree that the overall model evaluation gives an improvement over a baseline.

Decision Trees

Decision trees are a popular tool in Explainable AI (XAI) because they provide a clear and interpretable representation of the decision-making process used by the model. A decision tree is a tree-like model that makes predictions by recursively partitioning the data into smaller subsets based on the values of the predictor variables until each subset consists of instances with similar values for the response variable. The tree structure of a decision tree can be visualized, and the decisions made by the model can be traced from the root to the leaves, making it easy for humans to understand how the model is making its predictions [10, 11]. The internal nodes of the tree represent the questions being asked about the data, and the leaves represent the final predictions. In XAI, decision trees can be used to generate explanations for the predictions made by the model. For instance, the tree structure can be used to identify the most important features used by the model to make its predictions, and the decisions made at each node can be used to highlight the reasoning behind the prediction. This makes decision trees a useful tool for applications where interpretability is important, such as medical diagnosis, fraud detection, and customer segmentation. However, it is important to note that decision trees can have limitations when dealing with complex and non-linear relationships between the predictor variables and the response variable. In these cases, decision trees can easily overfit the data, leading to poor generalization performance on new data. To address this issue, ensemble methods, such as Random Forests and Gradient Boosting, are often used to improve the robustness and accuracy of decision tree models.

K-Nearest Neighbours

K-Nearest Neighbours (KNN) is a simple and intuitive machine learning algorithm that is used for classification and regression tasks. In KNN, the prediction for a new instance is based on the majority vote or average of the k-nearest neighbors in the training data. In KNN, the distance metric used to determine the nearest neighbous can be customized, such as Euclidean distance, Manhattan distance, or Cosine similarity. The value of k, the number of nearest neighbous used for prediction, is a tunable parameter that can be optimized through cross-validation or other model selection methods [12, 13]. In Explainable AI (XAI), KNN is used as a baseline model for comparison with more complex models, such as deep learning networks, to assess their performance and interpretability. KNN provides a simple and interpretable representation of the decision-making process used by the model, making it easy for humans to understand how the model is making its predictions.

Rule-Based Models

Rule Based Models use rules to represent the knowledge coded in the systems. They are widely used in expert systems for knowledge representation [14]. The first approach discussed here is The Bayesian Rule Lists. This was first introduced by Ben Letham and Cynthia Rudin in 2015 [15]. They proposed a rule list classifier for stroke prediction in which the rules are represented as a sequence of conditional if-else-if constructs. The classifier predicts the stroke risk in advance, considering the different factors leading to a stroke condition in patients. The interpretability of decision statements is simplified by the high dimensional and multivariate feature space by the discretization of if–then conditions. The decision list has posterior distribution yielded by the Bayesian rule list. Healthcare experts or doctors want to make certain decisions, so they want simple models which are easily understandable by themselves rather than complex models. Although the model is simple, another criterion is that the predictions made by the model should be accurate. In this paper, the researchers designed a generative model to produce if/else-if lists that strike a balance between accuracy, interpretability, and computation. Another approach that can be linked to rule-based models is Fuzzy rules/ systems. Fuzzy rules are a kind of if–then statement specifying some conditions and generating truth to a specific degree instead of complete true or false results. A deep rule-based fuzzy system is used to predict patients' mortality in the ICU. It consists of a diverse dataset combining categorical and numeric attributes in a hierarchical manner [16].

Risk Scores

Risk scores refer to the predictions or assessments generated by AI models in terms of risk levels. These scores are used to identify the level of risk associated with a particular decision or event [17]. In XAI, the risk scores are accompanied by an explanation of the reasoning behind the score, which helps humans understand and interpret the results of the AI model. The goal of XAI is to provide transparent and interpretable models that can be trusted by humans, especially in sensitive and high-stakes applications, such as credit risk assessments, fraud detection, or medical diagnosis.

Generalized Additive Models

Generalized Additive Models (GAMs) are a class of non-parametric statistical models used in Explainable AI (XAI) to make predictions based on a linear combination of smooth, non-linear functions. GAMs are a flexible alternative to traditional linear models and can capture complex relationships between the predictor variables and the response variable. In XAI, GAMs are used to generate explanations for the predictions made by the model. For instance, the smooth functions in a GAM can be interpreted as the contribution of each predictor variable to the prediction, providing insights into the model's decision-making process. This makes GAMs well-suited for applications where interpretability is important, such as medical diagnosis [18], financial forecasting [19], and risk management. However, while GAMs provide a more transparent view of the model's decision-making process compared to black-box models, the interpretability of the smooth functions can still be limited, particularly when dealing with high-dimensional data or complex interactions between predictor variables.

Prototype-Based Models

Prototype-based models are a class of machine learning models that are used for classification and clustering tasks. They work by representing each class or cluster as a prototype or representative feature vector. The prototypes are learned from the data and can be thought of as the central or average feature representation of the data points belonging to a particular class or cluster. During the classification or clustering process, new data points are compared to the prototypes and assigned to the closest class or cluster based on a distance metric, such as Euclidean distance or cosine similarity. The decision boundary between the classes or clusters is defined by the prototypes [20]. Prototype-based models are often used in Explainable AI (XAI) because they are simple and intuitive to understand and can provide straightforward explanations for the predictions made by the model. For instance, in a classification task, the prototypes can be used to represent the decision rules used by the model, making it easy for humans to understand how the model is making its decisions.

2.2 Post-hoc Explainable Models

Posthoc Explainable Models refer to machine learning models that have been trained to make predictions, but the explanations for the predictions are generated after the fact, rather than being built into the model itself. Posthoc explanation methods are used to generate explanations for the predictions made by black-box models, such as neural networks and gradient-boosting machines, which can be difficult to interpret. These methods can provide a local or global explanation for the predictions, either by highlighting the features used by the model to make the prediction or by approximating the decision boundary used by the model.

Local Explanation

It refers to the process of understanding the behavior of an AI model by examining the contributions of individual input features to the model's prediction. The goal of local explanation is to understand why the model made a particular prediction for a specific instance, rather than trying to understand the behavior of the model in general. There are several methods used for generating local explanations, including Feature Importance, Rule-Based, Saliency Maps, Attention Mechanisms, Prototypes/ Example Based, and counterfactuals. Some popular Local Post-hoc explainable models include.

LIME, or Local Interpretable Model-Agnostic Explanations is a method for generating local explanations for complex machine learning models. The goal of LIME is to provide a simple and interpretable explanation for the predictions of any machine learning model, regardless of the underlying architecture [21] and all its variations [22, 23]. LIME works by approximating the complex machine learning model with a simpler, interpretable model in the vicinity of a specific prediction. It does this by perturbing the input features of a specific instance and measuring the impact on the model's prediction. The resulting relationship between the input features and the model's prediction is then used to generate an interpretable explanation for the prediction.

SHAP (SHapley Additive exPlanations): The method is based on the concept of Shapley values from cooperative game theory and provides a fair and consistent way to distribute the contribution of each feature to a prediction. The main idea behind SHAP is to attribute the prediction of a machine-learning model to each of the input features. The method computes the expected value of the prediction when a feature is included in the input and when it is not and uses this information to calculate the contribution of each feature to the prediction [24]. Another approach to tackle the contribution of each feature to predictions has been coalitional Game Theory [25] and local gradients [26]. Similarly, by means of local gradients [27] test the changes needed in each feature to produce a change in the output of the model.

Rule-Based Approach-Anchors: The goal of anchors is to provide a simple, interpretable explanation for the predictions of a machine-learning model by identifying the input features that are most important for the prediction. Anchors work by generating a small set of rules, called anchors that are used to explain the prediction of a model. These anchors are based on the relationships between the input features and the model's prediction, and they are designed to be both simple and interpretable [28].

Saliency Maps: The goal of saliency maps is to highlight the input features that have the greatest impact on the prediction of a machine-learning model [29]. A saliency map is a visual representation of the model's decision, where the most important input features are highlighted. The map is generated by computing the gradient of the model's prediction with respect to the input features and visualizing the resulting information as a heatmap, where the color encodes the magnitude of the gradient.

Prototype/Example-Based Approach: The goal of prototype-based explanations is to provide a human-understandable explanation of a machine learning model's prediction by comparing the input to a set of prototypes or examples. Prototype-based explanations work by identifying a small set of prototypes or examples that are representative of the data and are used to explain the prediction of a machine-learning model [30]. The model's prediction is explained by comparing the input features to the prototypes and identifying which prototypes are most like the input. The main advantage of prototype-based explanations is their simplicity and interpretability. By comparing the input features to a small set of prototypes, prototype-based explanations provide a straightforward way to understand the reasoning behind a model's decision.

Counterfactuals: The goal of counterfactual explanations is to provide an explanation of a machine learning model's prediction by showing how the prediction would change if a certain input feature were different. Counterfactual explanations work by generating alternative inputs that are like the original input except for one or more features. The model's prediction for these alternative inputs is then compared to the original prediction to determine how the prediction would change if the input features were different [31, 32].

Global Explanation

It refers to methods that provide an understanding of the behavior of a machine learning model across the entire data distribution. The goal of global explanations is to provide a comprehensive and holistic understanding of a model's behavior, beyond just a single input or prediction.

Global explanations can be achieved through various methods, such as feature importance measures, model-agnostic interpretability techniques, and model-specific techniques. Feature importance measures provide a ranking of the most important features in a model, helping to identify which features drive the predictions the most. Model-agnostic interpretability techniques, such as partial dependence plots and individual conditional expectation plots, provide a way to understand the relationship between the model's predictions and the input features. Model-specific techniques, such as decision trees and rule-based models, provide an explicit representation of the model's decision-making process and allow for a step-by-step explanation of the model's predictions.

Feature importance measures: These methods provide a ranking of the most important features in a model, helping to identify which features drive the predictions the most. The most used methods include permutation importance, mean decrease impurity, and mean decrease accuracy [25].

Model-agnostic interpretability techniques: These methods provide a way to understand the relationship between the model's predictions and the input features, without relying on any specific model architecture. Partial dependence plots and individual conditional expectation plots are examples of model-agnostic interpretability techniques.

Model-specific techniques: These methods provide an explicit representation of the model's decision-making process and allow for a step-by-step explanation of the model's predictions. Examples of model-specific techniques include decision trees and rule-based models.

Model-level explanations: These methods provide an overall explanation of the model's behavior, regardless of the specific input or prediction. Model-level explanations can be provided through various methods, such as decision surface visualization, model predictions on synthetic data, and model predictions on an artificial validation set.

Model Distillation: Model distillation provides a way to understand the behavior of a complex machine learning model by training a smaller, simpler model to mimic the behavior of the complex model. By looking at the predictions and decision-making process of the student model, it is possible to obtain a global explanation of the behavior of the complex model. It provides a holistic view of the model's behavior, as opposed to local explanations, which focus on the behavior of the model for specific input instances.

Representation-Based Approach: Representation-based explanations refer to methods that explain the decisions of a machine learning model by focusing on the representations learned by the model. These representations are typically intermediate representations in the model, such as activations in a neural network, and can provide insight into how the model is processing the input data. In representation-based XAI, the focus is on understanding the internal workings of the model by examining the learned representations. This can help to identify what features of the input data are being used by the model to make its decisions and how these features are being combined to produce the final prediction.

3 Comparative Analysis

Comparative analysis of inherently interpretable and posthoc methods has been done based on their Pros, Cons, Use Cases, and Applicability (Table 1).

4 Challenges and Future Scope

Challenges associated with Explainable AI:

1. Trade-off between Explanation Quality and Model Performance: There is often a trade-off between the quality of explanations provided by XAI techniques and the performance of the underlying AI model.
2. Difficulty in Defining Explanations: There is no clear consensus on what constitutes a good explanation. Different stakeholders may have different criteria for what constitutes a satisfactory explanation.
3. Scalability: XAI techniques can be computationally expensive and may not be feasible for large, complex AI models.
4. Human Interpretability: The explanations provided by XAI techniques may not be easily understandable by human stakeholders, particularly if they are not experts in AI and machine learning.
5. Generalizability: XAI techniques that work well for one AI model or task may not work well for others. This makes it challenging to develop generalizable XAI techniques that can be applied across a wide range of AI models and tasks.

There are many exciting research opportunities in XAI that will help advance the field and make AI systems more accessible and usable. Some of the most promising areas of research in XAI include:

1. Improving explainability and interpretability methods: As AI models become more complex, it becomes increasingly difficult to understand how they make decisions. This lack of transparency can create challenges for the deployment of AI in critical domains and hinder its widespread adoption. Further research work is needed to develop new and improved methods for explaining the behaviour of AI systems. This includes developing strategies that provide explanations at different levels of abstraction and developing methods that are more intuitive and accessible to users.
2. Exploring the psychological and social aspects of XAI: AI models can encode biases and make unfair decisions, which can have serious consequences for individuals and society. Further research is needed to develop XAI methods that can help identify and address biases in AI systems. Researchers are interested in understanding how people perceive and understand explanations from AI systems, and how these explanations can be made more useful and usable.
3. Developing XAI for safety–critical applications: In critical domains such as healthcare, finance, and criminal justice, it is important to ensure that AI systems

Table 1 Comparative analysis of XAI methods

XAI method	Type	Pros	Cons	Use cases	Applicability
Linear/ logistic regression	Inherently interpretable	Easy to interpret coefficients and understand feature importance	May not capture complex relationships in the data	Predicting mortality rates in hospitalized patients, drug response prediction	Suitable for low-dimensional data
Decision trees	Inherently interpretable	Easy to understand the decision-making process	Can be prone to overfitting	Predictive analytics in healthcare, customer segmentation in marketing	Suitable for low-to-medium-dimensional data
Random forests	Post-hoc interpretable	Can handle non-linear relationships and interactions	Difficult to interpret at individual tree level	Medical image analysis, credit risk modelling	Suitable for low-to-medium-dimensional data
Gradient boosting machines	Post-hoc interpretable	High predictive performance, can handle complex interactions	Difficult to interpret at individual tree level	Predicting ICU admission	Suitable for low-to-medium-dimensional data
Support vector machines	Post-hoc interpretable	Effective at capturing complex, non-linear relationships	Computationally intensive, difficult-to-interpret coefficients	Cancer diagnosis, credit scoring	Suitable for low-to-medium-dimensional data
K-nearest neighbors	Inherently interpretable	Simple and easy to understand	Computationally intensive and sensitive to noisy data	Medical diagnosis, drug discovery	Suitable for low-dimensional data
Rule-based models	Inherently interpretable	Easy to understand the decision-making process	May not perform as well as more complex models	Medical diagnosis, fraud detection	Suitable for low-to-medium-dimensional data

(continued)

Table 1 (continued)

XAI method	Type	Pros	Cons	Use cases	Applicability
Risk scores	Inherently interpretable	Simple and easy to understand	May not capture complex relationships in the data	Predictive analytics in healthcare, credit risk modelling	Suitable for low-dimensional data
Generalized additive models	Inherently interpretable	Can capture non-linear relationships and interactions	Maybe overfitted with too many variables	Disease risk prediction, identifying factors affecting patient readmission	Suitable for low-to-medium-dimensional data
Lime	Post-hoc interpretable	Can explain black box models at the local level	May not capture the global behaviour of the model	Explaining fraud detection	Suitable for high-dimensional data
Shap	Post-hoc interpretable	Can explain black box models at the local level and provide global feature importance	Computationally intensive, may be slow for large datasets	Identifying critical features in drug development	Suitable for high-dimensional data
Anchors	Post-hoc interpretable	Can provide simple, human-readable conditions for model behaviour	May not capture the complexity of some models	Interpreting disease diagnosis	Suitable for high-dimensional data
Saliency maps	Post-hoc interpretable	Can visualize important regions of input for image data	May not be as effective for non-image data	Medical image analysis, identifying features contributing to patient outcomes	Suitable for image data

are transparent and trustworthy. Researchers are working to develop XAI methods that can provide safe and reliable explanations for these applications.
4. Integrating XAI with inference: Researchers are interested in exploring how XAI methods can be integrated with causal inference methods to provide more robust explanations for AI systems.
5. Developing XAI methods for autonomous systems: Autonomous systems, such as self-driving cars, require transparent and understandable explanations to ensure their safety and reliability. Researchers are exploring new XAI methods that can be used to explain the behaviour of autonomous systems.
6. Combining XAI with active learning: Active learning is a machine learning technique that involves actively seeking information to improve the performance of a model. Researchers are exploring how XAI methods can be integrated with active learning to improve the transparency and reliability of AI systems.
7. Developing XAI for deep learning models: Deep learning models are highly complex and can be difficult to explain. Researchers are working to develop XAI methods that can provide meaningful explanations for these models.
8. Exploring the ethical and legal implications of XAI: As AI systems become increasingly widespread, it is important to understand the ethical and legal implications of their use.
9. Improving the integration of XAI with other AI technologies: Researchers are exploring how XAI can be integrated with other AI technologies, such as reinforcement learning, to provide more comprehensive explanations for AI systems.

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Weakly Supervised Learning Model for Clustering and Segmentation of 3D Point on Cloud Shape Data

Shashikant V. Athawale, Mukesh Soni, Krishna Murthy, Gaurav Dhiman, and Pavitar Parkash Singh

Abstract Point cloud data, which retains more 3D spatial geometric information, has emerged as one of the core data formats for describing 3D models with the rapid growth of 3D acquisition technologies. However, the majority of deep learning network topologies used in 3D point cloud model segmentation studies rely on expensive labelled high-quality training data. A weakly supervised learning-based method for collaborative and consistent segmentation of 3D point cloud model clusters is presented to address the issue of using training samples with a small number of labeled points to achieve collaborative segmentation of 3D model clusters. Initially, the K-nearest neighbor technique is used to create the local neighborhood graph between the points. The point cloud model's component features are then extracted using the local convolution approach, and a matrix of related components is created. The network weights are then refined using energy function back propagation to produce results for model cluster consistency segmentation. The experimental findings indicate that the algorithm's segmentation accuracy on the public dataset ShapeNet Parts is 85.0%. Furthermore, when the number of training sample labels is lowered to 10%, this algorithm can still yield segmentation results comparable to or superior to those of

S. V. Athawale (✉)
Department of Computer Engineering, AISSMS COE, SavitribaiPhule Pune University, Pune, India
e-mail: svathawale@gmail.com

M. Soni
Department of CSE, University Centre for Research and Development Chandigarh University, Mohali, Punjab 140413, India

K. Murthy
Department of Journalism and Mass Communication, Indira Gandhi National Tribal University, Amarkantak, Madhya Pradesh, India

G. Dhiman
Department of Computer Science and Engineering, Graphic Era Deemed to Be University, Dehradun, India

P. P. Singh
Department of Management, Lovely Professional University, Phagwara, India
e-mail: pavitar.19476@lpu.co.in

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supervised learning approaches. In addition, segmentation accuracy is considerably enhanced compared to current weakly supervised mainstream techniques.

Keywords Machine learning · Cloud computing · Clustering · Segmentation · Supervised learning · K-nearest neighbour

1 Introduction

Three-dimensional models become a new generation of digital multimedia after audio data, two-dimensional images and videos. Compared with traditional multimedia data, 3D models have attracted extensive attention from industry and academia because of their strong sense of reality, which is more in line with people's intuitive understanding of nature. With the wide application of technologies such as information technology industry, data-driven and deep learning in computer vision and computer graphics, 3D model segmentation has made remarkable progress. The segmentation of 3D models is a basic problem in the field of 3D model analysis. It divides the 3D model into several meaningful, connected and independent semantic components according to the geometric and topological characteristics of the 3D model [1] to help. The understanding and analysis of 3D models is the research field of computer graphics is an important question to study.

The widespread use of laser scanning and other 3D sensing technologies in recent years has led to the development of 3D point clouds, which are now the standard format for representing 3D geometric data [2] and are being used extensively in a variety of industries, including autonomous vehicles, virtual reality, 3D city modeling, and more. Some deep learning methods have begun to try to analyze and process point cloud data. Reference [3] proposes that the PointCNN network model can directly apply the convolutional neural network structure to unordered point cloud data, avoiding the change of features with the order of input points, and enhancing the network's ability to process the model. Reference [4] proposed a deep learning network framework PointNet that can directly classify and segment point clouds. Considering the sparsity of the point cloud, the point cloud is not converted into multi-view or voxel grid, so that more 3D feature information can be preserved and better segmentation results are achieved. Reference [5] proposes a multi-feature fusion 3D point cloud model segmentation method, which learns the implicit relationship between global single-point features and local geometric features by constructing an attention fusion layer to fully mine the fine-grained geometric features of the model to obtain better model segmentation results. At present, most of the 3D point cloud model segmentation methods based on deep learning rely on labeled training data set, which limits the promotion of such methods.

Reference [6] learns the potential common features of the model from the labeled data set, so as to perform the collaborative analysis of the model. Therefore, if the number of labeled data sets is large enough, supervised algorithms can achieve ideal segmentation results. However, obtaining high-quality labeled data is time-consuming and expensive. On the contrary, Ref. [7] proposes an unsupervised co-segmentation method to segment the input model into patches, and then classify similar patches by subspace clustering. Although unsupervised 3D model segmentation algorithms can utilize unlabeled data to achieve collaborative segmentation of 3D models, this segmentation algorithm comes at the cost of reducing the segmentation accuracy [8–10]. Combined with the characteristics of point cloud data, this paper proposes a weakly supervised learning clusters based model for segmentation of 3D point cloud. In the feature extraction process, the use of local convolution operations can better correlate feature information between points and improve the network's ability to identify and segment point cloud models. Since similar models have similar feature information, similar component features can be represented by similar component matrices. When the network is learning, multiple models of the same type can be learned collaboratively. In addition, the deep learning network is guided by the energy function, and the network parameters are optimized through backpropagation, so as to achieve consistent segmentation of point cloud model clusters [11–14]. The main innovations and contributions of the algorithm are: (1) The neighborhood of the subset is chosen in the sampled point set before sampling, and the information of the discarded point can be incorporated into the related point to make up for the sampling loss. In the neighborhood the feature information of more points is retained while reducing the computational complexity of the network. (2) Using the weakly supervised learning method for 3D point cloud model segmentation can reduce the dependence on high-quality labeled data sets in the network training phase. This method only needs to mark a small number of points in the training phase to achieve and supervise methods comparable to segmentation results.

2 3D Point Cloud Model Segmentation and Feature Extraction

The 3D point cloud model is composed of a series of point sets with geometric information. The essence of its segmentation is to classify the point sets point by point, aiming to classify the points of the same category into the corresponding parts to obtain the model segmentation results [15–17]. Therefore, this paper proposes to use the local convolution method and construct a local neighborhood map to solve the above problems, and use a small amount of labeled data for training through the weak supervision method, which improves the universality and segmentation effect of the network architecture.

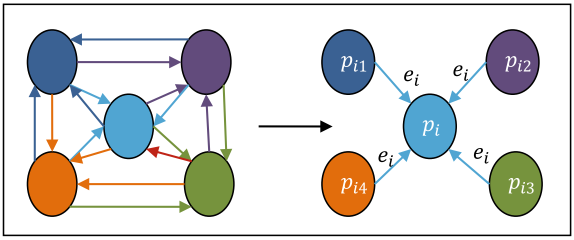

Fig. 1 Construction process of neighborhood graph

2.1 Construct the Local Neighborhood Map of the Point Cloud Model

The 3D point cloud model is a collection of 3D coordinate information of a group of unordered points. However, each point does not exist independently, and there is similar geometric information between adjacent points [18–20]. Therefore, establishing the correlation between points can be more accurate. To characterize the local feature information of the 3D model well, Fig. 1 shows the construction process of the neighborhood graph of the point. Define a point set as $\{p_i = x_i, y_i, z_i)|i = 1, 2, \ldots, n\}$, where p_i is any point in the point cloud, represented by the coordinates (x_i, y_i, z_i) corresponding to the point, with p_i as the center point, use the Knearest neighbor (KNN) method to establish a local directed graph G, which is composed of a vertex set V and an edge set E, defined as follows:

$$\begin{cases} G = (V E) \\ V = \{P_i|1, 2, \cdots, n\} \\ E = \{e_i = (e_{i1}, e_{i2}, \cdots, e_{ik})|i = 1, 2, \cdots, n\} \\ e_{ij} = p_{ij} - p_i \end{cases} \tag{1}$$

In the formula, e_i is the ith directed edge set, and e_{ij} represents the directed edge from the adjacent point p_{ij} to the central point p_i.

2.2 Local Convolution Operation to Extract Features

The local convolution operation is used to extract the features of the center point and the edge vectors between the center point and the K nearest neighbor points [18–22]. Since the point cloud model is a set of unordered point sets, the maximum pooling operation is adopted [23–25]. This method is not affected by the order of neighboring points and can extract the most important features of all edge vectors. Define the same feature extraction function f_e for all points, then input a local neighborhood map whose center point is p_i to get the output local feature l_i, the formula is as follows:

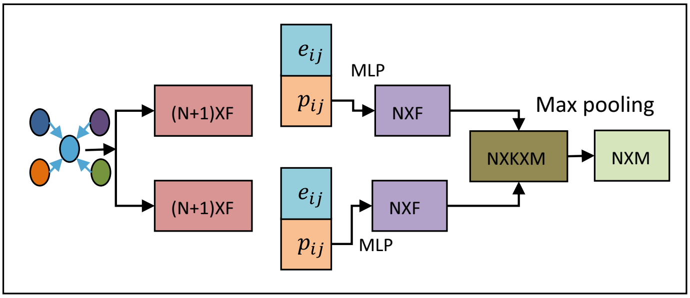

Fig. 2 Local convolution process

$$l_i = f_e(G(p_i, e_i)) = \max\{\boldsymbol{h}(p_i, e_{i1}), \boldsymbol{h}(p_i, e_{i2}), \cdots, \boldsymbol{h}(p_i, e_{ik})\} \tag{2}$$

where $\boldsymbol{h}\left(p_i, e_{ij}\right)$ is the hidden feature vector between the center point p_i and an edge vector e_{ij}. The output local feature l_i takes the maximum value $max\{\}$ of the hidden feature vector.

For the segmentation task, it is necessary to concatenate the global features and the extracted local features point by point, and use a neural network to output a prediction score for each point. The local convolution operation method is represented in the Eq. (3)

$$p'_i = \boldsymbol{h}\left(p_i, e_{ij}\right) = \boldsymbol{h}(p_i, e_{i1}) \oplus \boldsymbol{h}(p_i, e_{i2}) \oplus \cdots \oplus \boldsymbol{h}(p_i, e_{ik}) \tag{3}$$

In the formula, p'_i represents the updated point feature, which associates the own feature of each point with the corresponding neighbor point, and □ represents as pooling operation or summation. In summary, local convolution creates a new set of points with new dimensional characteristics linked to the local features by first applying a multi-layer perceptron to each point and its accompanying neighbourhood to record a local receptive field. The local convolution process is shown in Fig. 2.

The local convolution operation mainly generates the edge features, which are used to represent the mapping between each point and its neighbors. This convolution method has two advantages: (1) KNN Graph can better extract point local information while maintaining the invariance of the arrangement, and improve the segmentation effect of the point cloud model; (2) after multi-layer iteration it is then able to better capture potentially distant similar features.

3 Experimental Results and Analysis

The experiments are performed in the Intel Core i9 9900 k CPU and NVIDIA Rtx 2080Ti GPU (11 GB video memory) processor, with CUDA 9.0, the GPU accelerated library is cudnn 7.0. To implement the proposed deep learning model the Keras and Tensorflow framework have been used.

a. Data Set

In order to compare with other algorithms, this paper uses the ShapeNetParts [8] dataset, which contains 16,881 3D models, 16 categories, and a total of 50 labeled parts. Most models are labeled with 2 to 5 Parts, and labels are annotated on the sampling points of the 3D model.

b. Parameter Setting and Performance Evaluation

To improve the segmentation performance of the model, 2048 points are uniformly sampled from the model surface for training and testing experiments. In the experiment, adaptive moment estimation (Adam) is used to optimize the neural network model. The initial learning rate is set to 0.003. Momentum is set to 0.8 to prevent the network from falling into local optimum when updating parameters. In order to prevent inconsistent data distribution in the middle layer of the network during training, a batch normalization layer (batch normalization) is inserted before each input layer of the network, and the exponential decay coefficient of batch normalization is set to 0.5, so that the loss function can be quickly convergent. In this experiment, when the batch size is set to 32, the impact on the gradient of the network is minimal.

c. Evaluation Criteria

The intersection over union (IoU) on the point set is the metric that is used to evaluate how accurately the 3D point cloud model segments data. IoU is mainly used to compute the ratio between the intersection and union of two data point sets: true segmentation and predicted segmentation by the model. The IOU formula is as follows:

$$IoU = \frac{TP}{T + P - TP} \tag{4}$$

In the formula, TP represents the true sample size; T represents the real true sample size; P represents the predicted true sample size.

It is required to incorporate the mean intersection over union (mIoU) for measurement after getting the IoU value for each type of model in order to assess the segmentation effect of the network on the entire model cluster. The formula is as follows:

$$mIoU = \frac{1}{m} \sum_{i=0}^{m} \frac{TP}{\sum_{i=0}^{m} T + \sum_{i=0}^{m} (P - TP)} \tag{5}$$

In the formula, m represents the number of categories of the 3D model, and the larger the value of mIoU, the closer the real value is to the test value, so that the effect of model segmentation is better. In the ShapeNet Parts data set, since the number of models of different categories is different, it is necessary to continue to evaluate the segmentation accuracy from a quantitative perspective, so the part averagedIoU (pIoU) is introduced for further comparison. The formula is as follows:

$$pIoU = \frac{1}{\sum_{i=0}^{m}\sum_{i=0}^{m} TP} \times \sum_{i=0}^{m} \frac{TP\sum_{i=0}^{m} T}{\sum_{i=0}^{m} T + \sum_{i=0}^{m}(P - TP)} \tag{6}$$

This formula represents the weighted summation of the IoU of each class according to the frequency of each type of model. In this experiment, the frequency is the number of each type of model. The higher the pIoU value, the higher the segmentation accuracy.

d. **Analysis of Results**

The collaborative segmentation effect of the algorithm on the 3D point cloud model cluster is as follows: Where point clouds of different colors represent different model segmentation parts. It can be seen that the colors of the same parts corresponding to each type of model are consistent, which better reflects the consistency of the collaborative segmentation of the 3D model clusters.

Table 1 and Fig. 3 show the comparison of the algorithm in this paper with the algorithms in literature [4], literature [9, 10], literature [11] and literature [12] on the ShapeNet dataset. The algorithm in this paper adopts a weak supervision strategy, and establishes a local neighborhood map of points through the KNN method to better correlate information between the sampling points. When only one point is marked for each component in the model, it can already obtain competitive As a result, when 10% of the points are marked, the mIoU value increases by 0.012 and the pIoU value increases by 0.014 compared with the algorithm in [4]. In addition, the algorithm in this paper uses an energy function backpropagation iteration to generate the consistent segmentation results of the model. Compared with the recursive proximity search strategy adopted in the literature [12], the mIoU value is increased by 0.042, and the pIoU value is increased by 0.027. Since the algorithm in this paper adopts a weakly supervised learning strategy and experiments with a small amount of labeled data, the segmentation accuracy is slightly lower than the methods in [9, 11]. However, compared with the weak supervision strategy adopted in literature [13] and literature [10], the mIoU value increased by 0.007 and 0.017, and the pIoU value increased by 0.001 and 0.004, respectively.

This study further compares the segmentation performance in terms of mIoU and pIoUof various algorithms. Literature [4], Literature [9], Literature [11] and Literature [12] all adopt the supervised strategy, while literature [13], Literature [10] and the algorithm in this paper adopt the weak supervision strategy. When each part of the model occurs when marking a sampling point, compared with the supervised

Table 1 Comparison of segmentation performance with existing method

Algorithm	mIoU	pIoU
PointNet [24]	82.5	85.4
PointNet++ [25]	83.7	87.3
DGCNN [26]	84.4	87.6
KD-Net [27]	79.3	84.8
Capsule-net [28]	82.7	86.3
BAE-NET [15]	80.8	86.2
Our algorithm(1 point)	76.7	77.2
Our algorithm(10% points)	83.5	87.2

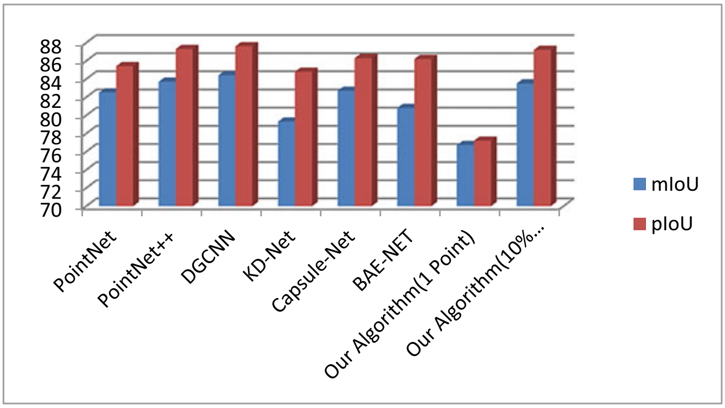

Fig. 3 Comparison of segmentation performance of various algorithm

strategy in [12], the weak supervision strategy in this paper has a certain improvement in the segmentation accuracy of some models, among which the accuracy of the hat model (cap) and the cup model (mug) are respectively improved. 0.049 and 0.036. When the marked sampling points in the training samples are increased to 10%, the segmentation accuracy of various models is comparable to the segmentation results of the supervised strategy in the comparative literature, and the segmentation effect of some models is even better than the supervised learning method. Compared with literature [12], the segmentation accuracy of the car model (car) and the rocket model (rocket) has improved most significantly, increasing by 0.075 and 0.101, respectively. Literature [11] uses the edge convolution method to correlate the features between point cloud model points. When segmenting a model with a large number of training samples, the segmentation result is better than other algorithms, such as airplane model (airplane), chair model (chair) and Guitar model (guitar). Compared with literature [11], the segmentation accuracy of the algorithm in this paper is reduced by 0.005, 0.011 and 0.006 in the three types of models of airplane (airplane), chair (chair) and guitar (guitar), but in the car model (car), knife model (knife model).), the rocket model (rocket) and the table model (table) have higher segmentation accuracy than the literature [11] algorithm by 0.007, 0.003, 0.022 and 0.009, respectively. Literature [13] uses the attention mechanism driven by dynamic routing to effectively extract local feature information, so that it has a high segmentation accuracy for small parts of the model. For example, the segmentation accuracy of the motorcycle model motorbike is 0.044 higher than that of the algorithm in this paper. Literature [10] adopts a branch auto-encoder, and each branch learns a component feature of the input model separately, without model cluster consistency optimization, and the classification accuracy of the model with a small number of input sets is low. For example, the segmentation accuracy of the algorithm in the bag model (bag), earphone model (earphone) and rocket model (rocket) is higher than that of the algorithm in [10] by 0.099, 0.069 and 0.048, respectively. The experimental results show that compared with the supervised algorithm, the weak supervision strategy in this paper can still obtain competitive results when dividing various models when the number of marked points is reduced to 10%. Compared with the weak supervision method using different strategies In contrast, the main advantage of the algorithm in this paper is to constrain the consistency of model segmentation results by constructing component feature matrices, and to achieve label prediction for unlabeled points with a label propagation strategy under weaker supervision, which can complete collaborative segmentation tasks while further improving the segmentation accuracy. Therefore, the advantages of the weak supervision strategy are also reflected in the algorithm of this paper.

The role of each module of this network architecture is very important. In order to better analyze the impact of each module of the network on the results, the segmentation accuracy comparison of the ablation experiment is shown in Table 2. It can be seen that: (1) the local convolution module has the greatest impact on network performance. After removing the local convolution module, the segmentation result mIoU is reduced to 56.3%. After the final after far-point sampling, local convolution can characterize the spatial relationship between each sampling point and its neighbors to make up for the feature information lost in the sampling process. (2) The impact of removing the energy function on the segmentation results is inferior to (1), which proves that this module can effectively implement backpropagation learning, and constrain the consistency of the segmentation results with the feature matrix. (3) After replacing the farthest point sampling with random sampling, the segmentation accuracy rate has declined, because this sampling method cannot have a good coverage of the sampling space, which is not conducive to the subsequent extraction

Table 2 Verify effectiveness of model components with different ablated experiments

Replace with random sampling	LocalConv	Energy function	mIoU%
Yes	Yes	Yes	71,2
Yes	Yes	Yes	56.3
Yes	Yes	Yes	67.1
Yes	Yes	Yes	**81.6**

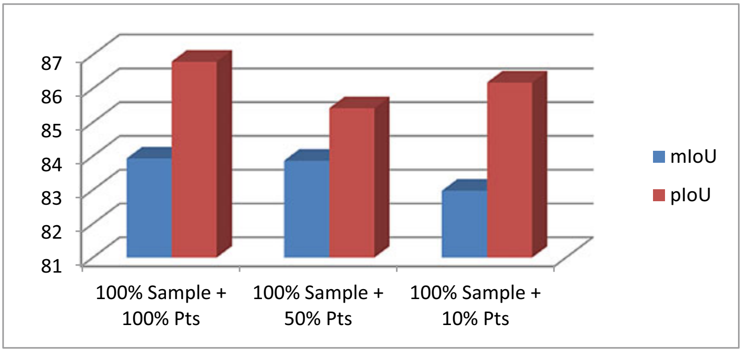

Fig. 4 Bar graph of ShapeNet parts segmentation with different labeling strategies

of feature information. The results of ablation experiments show that the components of the network constructed in this paper are effective and can complement each other to achieve the best performance.

The three labeling techniques were tested in the experiment to confirm that the weakly supervised sample labeling strategy used by the algorithm in this paper had certain advantages. The comparative findings are displayed in Fig. 4 for each labeling strategy. 100% training samples (samples) and the quantity of labeled points (Pts) included in each sample serve to define the labeling approach. The experimental results show that as the marked points of the training samples are reduced from 100 to 10%, the mIoU value of the segmentation result is reduced from 83.93 to 82.98%, and the pIoU value is reduced from 86.80 to 86.18%, which are respectively reduced by 0.49 percentage points and 0.1 points. As a result, it is evident from the experimental findings that the segmentation accuracy of the algorithm in this study does not significantly change when the number of markers is decreased (from 100 to 10%), which effectively verifies that the algorithm in this paper is less dependent on the number of markers. The reduction of marked points has strong robustness and weak supervision, which saves time-consuming and laborious manual marking costs.

4 Conclusion

This study presented the proposed a co-segmentation model based on weakly supervised learning, aiming for addressing the co-segmentation problem of 3D point cloud models with a small amount of labeled data. First, use the farthest point sampling method to sample the original point cloud, and establish a local neighborhood map of the point through the KNN algorithm to associate the information between points, so as to better characterize the local characteristics of the model. Then, model features are extracted using local convolution methods to generate feature descriptors corresponding to each part of the model. Finally, by constructing the component feature matrix, the consistent segmentation results of the model clusters are obtained through the energy function backpropagation iteration in the deep learning network. Experiments were carried out on the ShapeNet Parts dataset. Compared with supervised methods such as literature [4] and literature [11], the deep learning network proposed in this paper only needs a small number of marked points to effectively achieve point cloud model segmentation. When it is reduced to 10%, the results comparable to the supervised algorithm can be achieved, and compared with the weakly supervised methods in the literature [13] and literature [10], the algorithm in this paper achieves better segmentation under weaker supervision result. However, the algorithm in this paper also has limitations: on the one hand, the algorithm has a clear neighbor relationship for each point in LocalConv, so the network calculation complexity is greater, and the efficiency will decrease when processing large-scale point cloud data; on the other hand, Since the algorithm in this paper adopts a weak supervision strategy, it has obvious advantages in terms of data set requirements, which makes the network more universal. However, compared with the fully supervised algorithm, the segmentation results of some models have obvious gaps. These are issues that require further research in the future.

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Generative Artificial Intelligence: Opportunities and Challenges of Large Language Models

Fabian Barreto, Lalita Moharkar, Madhura Shirodkar, Vidya Sarode, Saniya Gonsalves, and Aaron Johns

Abstract Artificial Intelligence (AI) research in the past decade has led to the development of Generative AI, where AI systems create new information from almost nothing after learning from trained models. Generative AI can create original work, like an article, a code, a painting, a poem, or a song. Google Brain initially used Large Language Models (LLM) for context-aware text translation, and Google went on to develop Bidirectional Encoder Representations from Transformers (BERT) and Language Model for Dialogue Applications (LaMDA). Facebook created OPT-175B and BlenderBot, while OpenAI innovated GPT-3 for text, DALL-E2 for images, and Whisper for speech. GPT-3 was trained on around 45 terabytes of text data at an estimated cost of several million dollars. Generative models have also been developed from online communities like Midjourney and open-source ones like HuggingFace. On November 30, 2022, OpenAI launched ChatGPT, which used natural language processing (NLP) techniques and was trained on LLM. There was excitement and caution as OpenAI's ChatGPT reached one million users in just five days, and in January 2023 reached 100 million users. Many marveled at its eloquence and the limited supervision with which it generated code and answered questions. More deployments followed; Microsoft's OpenAI-powered Bing on February 7, 2023, followed by Google's Bard on February 8, 2023. We describe the working of LLM and their opportunities and challenges for our modern world.

Keywords Bing · chatGPT · Generative artificial intelligence · Large language models

F. Barreto (✉) · L. Moharkar · M. Shirodkar · V. Sarode
Department of Electronics and Telecommunication, Xavier Institute of Engineering, Mumbai, India
e-mail: frfabiansj@xavier.ac.in

S. Gonsalves
Department of Applied Sciences and Humanities, Xavier Institute of Engineering, Mumbai, India

A. Johns
Department of Information Technology, St. Xavier's College, Mumbai, India

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1 Introduction

Every human culture uses language as a communication system to convey ideas, emotions, and information to one another. As infants, we pick up meaningful word meanings, and as we grow to adulthood become more skilled in adapting our speech. Language also helps us to understand ourselves and the world around us. The uniqueness of the human person in learning a language is the ability to generalize and hence learn from a limited amount of exposure to a language, especially as children. Contrast this with a model trained on deep learning systems. During the testing phase, samples of a different distribution than those trained show a generalization behaviour inconsistent with a human person. Making decisions on unseen data require extensive learning. The authors in Lake et al. [1] elaborate on how humans never learn "from scratch" but use their previous knowledge to learn new tasks. They outline "core ingredients" that humans use to be good at generalization. These ingredients include the domain knowledge of numbers, space, physics, and psychology. These building blocks and diverse experiences result in better generalization for human beings. In Linzen [2], the author looks at how a statistical model extracts generalizations based on the way that inductive biases of the models interact and on the dataset's statistical properties. He suggests that models can be improved by incorporating the human-like inductive biases so that learning can happen with limited data.

Decades of quest followed to develop systems that could generate human-like responses for Natural Language Processing (NLP) tasks like conversation, text completion, and language translation. The Transformer-based models, like OpenAI's Generative Pre-trained Transformer 3 (GPT-3), generate human-like text, which can be used for various applications such as text completion, dialogue systems, and language translation. The text generation process is a token sampling conditional on previous tokens, $a_j \sim p(a_j | a_j \ldots a_{j-1} : \phi)$ where a_j denotes the jth token in the text sequence and ϕ denotes sampling distribution parameters. ϕ is optimized on the training data conditioned on N preceding tokens. A large language model can creatively handle novel concepts given as a prompt even if not encountered in the training data [3]. The authors [4] train GPT-3, and test its performance in the few-shot setting. Their 96-layered model trained on 175 billion parameters shows performance that at times exceeds the State-of-the-art (SOTA) fine-tuned models.

The remainder of this paper is organized as follows. The details of Generative Artificial Intelligence (GAI) are illustrated in Sect. 2. We elaborate on the framework of the GPT family in Sect. 3. The opportunities and challenges of LLM are explained in Sect. 4. Finally, Sect. 5 gives the concluding remarks of this paper.

2 Generative AI Models

In AI the models are trained using datasets and then tested on unseen samples of data. The AI system can then apply its model to new, unseen data and make accurate predictions or decisions. The real-world applications help to encounter new and diverse data that was not seen during training. Good generalization, is achieved using techniques such as regularization, cross-validation, and ensemble methods to reduce overfitting. By the process of generalization an AI system performs well on tasks not encountered during training. Generative Artificial Intelligence (GenAI) refers to the use of AI algorithms and models to generate new content, such as realistic computer-generated images and videos, text generation, and audio synthesis, especially music and speech. GenAI can also be used for data augmentation and anomaly detection.

Some prominent GenAI models (e.g. GPT-3) have demonstrated impressive language generation capabilities. NVIDIA's StyleGan generates realistic faces, objects, and even entire scenes. OpenAI's Music Transformer generates original pieces of music in a variety of styles.

Some of the most well-known types of generative AI architectures include Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs).

2.1 Variational Autoencoders (VAEs)

VAEs consist of an encoder and a decoder [5] as shown in Fig. 1. The encoder takes in data and compresses it into a lower-dimensional representation, called the latent code. The decoder then takes the latent code and generates new data samples that are similar to the original input.

2.2 Generative Adversarial Networks (GANs)

GANs consist of two networks: a generator (G) and a discriminator (D) [6] as shown in Fig. 2. The generator works at creating new data samples, while the discriminator attempts to tell apart the generated samples (fake) from the real samples. Through competition (minimax game of adversarial nature), the generator learns to create more realistic samples, while the discriminator learns to better identify fake samples. D wants to maximize its cost value log(D(x)), and G wants to minimize its cost value log(1-D(G(z))), as given in the Loss Function [6] equation shown below.

$$\text{Loss Function}_{(\mathrm{D,G})} = \mathbb{E}_{\mathrm{x} \sim \mathrm{pdata(x)}}\left[\log \mathrm{D(x)}\right] + \mathbb{E}_{\mathrm{z} \sim \mathrm{pz(z)}}\left[\log\left(1 - \mathrm{D(G(z))}\right)\right] \quad (1)$$

where represents the mathematical expectancy. The authors in Razavi-Far et al. [7] mention the different GAN-based techniques that cater to different types of

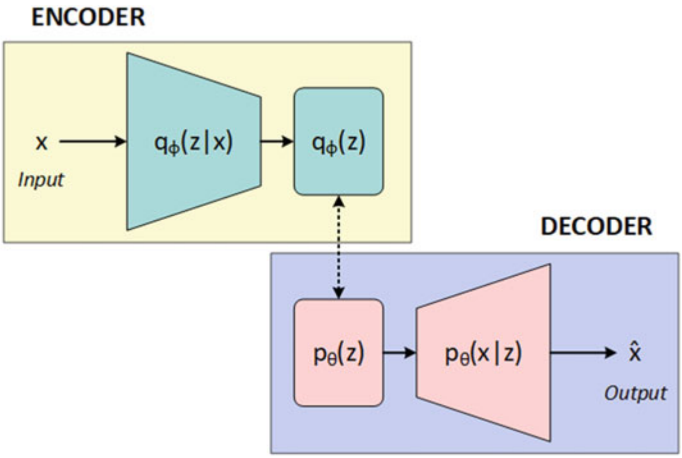
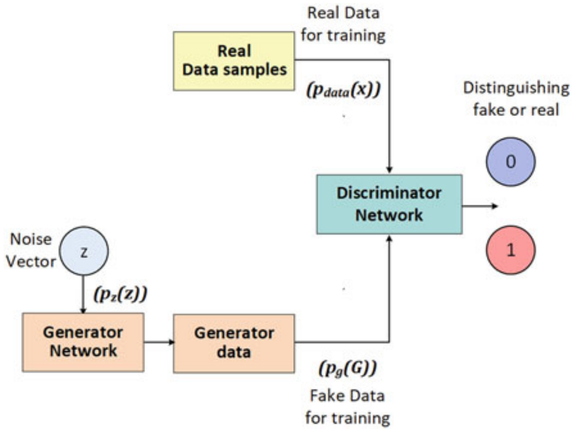


Fig. 1 Variational autoencoders

Fig. 2 Generative adversarial networks (adapted from [6])



AI learning problems. They mention the various new GAN architectures and the development of GAN objective functions for specific applications. GANs are also combined with transfer learning and reinforcement learning to achieve noteworthy results in several applications [8]. The authors in Farajzadeh-Zanjani et al. [9] mention the development of GANs beginning with the 1990 Artificial Curiosity to the 2020 Continuous Conditional GAN (CcGAN). They also elaborate the challenges associated with training GANs namely mode collapse, and vanishing gradients. In NLP, text generation using GANs has been a great success, as described by de Rosa and Papa [10], who survey recent studies in GAN-based text generation. Specifically, NLP models need to learn the connection between characters and words (grammar, syntax and semantic properties). The context can be learnt by using pre-trained embedding models such as BERT [11], ELECTRA [12], and GPT-2. Systems can be trained to deal with adversarial data (slightly modified data probably not present in the training) by having noisy samples during the learning process. GANs by their innate nature learn the data's distribution and hence can augment with artificially generated data.

3 The GPT Family

The Generative Pre-trained Transformer (GPT) is a generative model, innovatively created by OpenAI. Their goal is "to advance digital intelligence in the way that is most likely to benefit humanity as a whole" [13]. The transformer uses the attention mechanism [14] to focus on relevance using self-attention (words association in a sentence) and encoder-decoder attention (between source and target sentences). The transformer's decoder is the GPT. The models can be trained in few-, one-, and zero-shot settings. In few-shot setting, after presenting the task few examples are given to the model. In a one-shot there is only one example and, in a zero-shot there is no example. The GPT family uses tokens to process the text. In regular English text a token will be four characters, about three-quarters of a word. The trained models learn the statistical relationships between tokens and can thus predict the next token.

The decision of OpenAI to release GPT-3 through a public application programming interface (API), opened the world to countless possibilities. Up until 2020, the AI research was available for a selective few researchers and engineers. With OpenAI's API, users throughout the world could get access to this LLM through a simple sign-in.

3.1 The Evolution

OpenAI presented GPT-1 in June 2018 [13], with the transformer architecture and unsupervised (unlabeled data) pre-training. The GPT-1 model was trained on 8 million web pages and had 117 million parameters with a context token size of 512. The model demonstrated that a good pre-trained model could perform the task of generalization. A remarkable advantage was zero-shot learning, where the model could do a task without a previous example, with an application in question answering.

In February 2019, OpenAI released a larger model, GPT-2 having, 1.5 billion parameters with a context token size of 1024, and trained on 40 GB of webpages (WebText dataset). Its purpose was to predict the next word(s) in a sentence. It had a poor performance for specialized tasks such as music and storytelling.

In June 2020, OpenAI released GPT-3 [4] with 175 billion parameters and trained 45 TB of webpages with a context token size of 2048. GPT-3 is pre-trained on a corpus of text from five datasets: Common Crawl, WebText2, Books1, Books2, and

Wikipedia. 93.69% of the total documents are in English. A more advanced architecture (sparsely-gated mixture-of-experts) allows one to better understand the context of conversations and generate more accurate and nuanced text. GPT-3 can hence answer questions, write essays, summarize text, translate language, and generate computer code.

On November 30, 2022, OpenAI launched ChatGPT trained on LLM. Many admired its articulateness and the limited supervision with which it generated code and answered questions. There was excitement and caution as OpenAI's ChatGPT reached one million users in just five days and 100 million by the end of January 2023.

For the next version, GPT-4, the authors [15, 16] envision better parameter-optimized, text-only model. GPT-3 was only trained once as training is costly and hence not optimized. GPT-4 possibly will have better optimization of hyperparameters and better analysis of the optimal model emphasizing sparsity.

3.2 Other Technologies

Microsoft and NVIDIA built Megatron-Turing NLG 530B (MT-NLG) [17], a transformer-based language model with 530 billion parameters. It was trained 15 datasets consisting of a total of 339 billion tokens. Hence modern LLM are evolving thanks to the availability of computational resources, large datasets, and effective software stacks.

More deployments followed after the launch of ChatGPT. Microsoft's OpenAI-powered Bing on February 7, 2023, followed by Google's Bard on February 8, 2023. Bing integrated OpenAI's language model, Prometheus, was explicitly designed for search engines. Prometheus ensures humans are in the loop, incorporating Microsoft's responsible AI [18].

4 Opportunities and Challenges of LLM

4.1 Opportunities

LLM offers several opportunities in different domains. The authors in Bommasani et al. [19] give a detailed description of the capabilities and applications of what they call foundation models (e.g., GPT-3). The capabilities include Language, Vision, Reasoning, and Interaction, to name a few. The applications include healthcare, Law, and Education.

Language, with its nuances of dialect and style, offers the richness and complexity that LLM are trying to learn. The present models exhibit adaptable linguistic capabilities. Applications include classification, generation, and sequence labelling. Multilingual models catering to different languages have also evolved (e.g., mBERT) [20] and can be important use cases for the Indian scenario. In the domain of Computer Vision, raw web-based datasets are readily available for training and have contributed to the enhanced performance of LLM. An evolving area is multimodal integration tasks like image captioning. Reasoning includes proving formal theorems and generating code. Interaction includes lowering the difficulty threshold for developers to proto-type powerful applications with seamless integration.

LLM can be a good interface for healthcare providers when scaling healthcare services. Some relevant examples include summarizing healthcare records and retrieving relevant healthcare cases and literature. Legal applications and processing is another interesting use case for LLM, especially for private, criminal, and public law, where it can provide the necessary context. An example can be an automated brief generation. The digital age post-millennium ushered in the rapid growth of digital learning. LLM can help in making learning more effective for teachers and learners. LLM can provide relevant feedback to learners and help teachers to create personalized content to meet students' needs.

4.2 Challenges

The authors in Bommasani et al. [19] describe the challenges in the context of inequity, misuse, and effects on the environment, legality, economics, and ethics. ITU estimates that approximately 5.3 billion people (66 percent of the world's population) were using the Internet in 2022 [21]. In the context of the digital divide LLM can compound the existing disparities, leading to unfair outcomes. There can be intrinsic and latent biases within the models. Biases of religion, gender, and race. An example is a misrepresentation based on stereotypes and negative outlooks.

In recent days we have heard of the unhinged responses of Bing [22], expressed as angry remarks and bizarre conversations. Yes, the models can attack users with harmful content, leading to trauma and psychological harm. Protocols must be in place to develop and deploy models that mitigate toxic content and rectify harmful responses.

The greatest danger of LLM is the misuse that can happen in the form of fake news or biased propaganda that can harm populations. Exploiting the models' vulnerabilities is another threat. Target content for misinformation is a real-case scenario that can be exploited.

LLM require tremendous amount of data and computing resources and hence has its own contributions to carbon emissions. Once deployed at large scales these models require substantial levels of energy, and therefore, carbon costs. A question that is often raised is the legality of the training data, precisely its collection and use.

There is a need for proper ethical frameworks for publicly accessible data, especially for mass data collection.

5 Conclusion

The AI revolution is here to stay. Generative AI will play a significant role in shaping our human destiny. The task-specific NLP models have evolved to models that can perform a variety of tasks. Large Language Models have shown SOTA performance on various NLP tasks and will continue contributing to the AI innovation landscape. LLM are evolving with greater emergent abilities, that is, abilities not seen in the smaller models are manifested in larger models. Emergent risks will also have to be seen, especially in the context of fake news, which can harm society, not just in generating human-like texts but also in writing codes. Given human conversations' diversity and complexity, AI models face several challenges. Prominent among them is the issue of human bias while training. Hence the need for human persons to be in the loop to monitor the model output, thus detecting and mitigating biases and toxicity. Sustainable computing resources, will have to be looked into when a large number users start using the LLM, as they are computationally resource-heavy. It is also imperative that we as humans use these models responsibly with proper ethical considerations.

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A Review of Artificial Intelligence for Predictive Healthcare Analytics and Healthcare IoT Applications

Amogh Chaudhari, Vidya Sarode, Shailaja Udtewar, Lalita Moharkar, Leena Patil, and Fabian Barreto

Abstract Our modern world is marked by rapid progress in Information and Communication Technologies (ICTs). Though there are limitations of the digital divide globally, the use of Artificial intelligence (AI) has revolutionized the healthcare industry through predictive analytics and the integration of healthcare Internet of Things (IoT) devices. Predictive healthcare analytics, integrated with explainable AI (XAI), can improve the efficiency and effectiveness of healthcare delivery. Healthcare IoT (HIoT) devices provide the data for predictive analytics and enable remote monitoring of patients. Predictive healthcare analytics can identify high-risk patients for chronic conditions and develop personalized treatment plans. AI can analyze patient data, including demographic information, medical history, and lab test results, to identify patterns and predict future health outcomes leading to better intervention. Patients at high risk for acute conditions can be helped, thus reducing the overall cost of care and improving patient prognoses. HIoT devices, provide information on patient vital signs, physical activity, and medication adherence. Wearable fitness trackers, such as smartwatches and fitness bands, provide data on physical activity and sleep patterns to identify patients at risk for chronic conditions such as heart disease. Remote monitoring devices can provide real-time data on patient vital signs, enabling healthcare professionals to monitor patients remotely within a hospital environment and care facilities and intervene as needed. A well-integrated secure ecosystem with seamless wireless connectivity can usher in innovative AI-based healthcare solution.

Keywords AI · HIoT · IoT · Predictive analytics

A. Chaudhari (✉)
General Practitioner, M.B.B.S., Mumbai, India
e-mail: amogh.chaudhari@gmail.com

V. Sarode · S. Udtewar · L. Moharkar · L. Patil · F. Barreto
Department of Electronics and Telecommunication, Xavier Institute of Engineering, Mumbai, India

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1 Introduction

Innovations in Information and Communication Technologies (ICTs) mark our modern world, primarily through access to the Internet and the World Wide Web. However, many countries strive to provide their population with equitable access to digital technologies. There is a digital divide between those who can and cannot access these technologies. The International Telecommunication Union (ITU), a United Nations agency for ICTs, ITU estimates Internet usage by approximately 5.3 billion people (66% of the world's population) as of 2022 [1]. Hence about 2.7 billion people are still offline. The digital divide is a nuanced term. It is not just about a simple divide, a difficult gap to bridge, a matter of access, or a static condition, but a variety of terms that constitute inequality of access. Access can be material (computers), digital training (skills), and applications (usage) [2]. The emergence of Health Information Technology (HIT) has paved the way for better access to healthcare, augmented clinical results, and advanced healthcare quality [5]. Telehealth has been a boon to many and made healthcare equitable. In the U. S. telehealth requires HIPAA (Health Insurance Portability and Accountability Act)-compliant video-conferencing platforms.

The past decades have seen shortages of medical professionals. Healthcare human resources planning is critical to achieving human well-being [3]. As of 2020, the global health workforce is around 65 million [4]. World Health Organization projects a shortfall of 10 million health workers by 2030. This shortage is mostly in low-mid-income countries. The rural, remote, and underserved areas bear the brunt of these inequities.

The past decade has also seen remarkable research in Artificial Intelligence (AI), the deployment of technologies like the Internet of Things (IoT), and the emergence of Big Data (BD) and scalable Cloud Computing (CC). The last few years have seen a convergence of these silos leading to smart health and predictive healthcare analytics. The convergence has created an ecosystem for institutions, healthcare givers, and patients to respond intelligently for effective monitoring and access to medical care.

The structure of the paper is as follows. Section 2 illustrates the details of Artificial Intelligence (AI). The framework of Predictive Healthcare Analytics is elaborated in Sect. 3. The Healthcare IoT applications are explained in Sect. 4. Finally, Sect. 5 gives the concluding remarks of this paper.

2 Artificial Intelligence

2.1 The Early Beginnings

John McCarthy, in 1956, coined the term "Artificial Intelligence" (AI) at Dartmouth College [6]. He understood AI as creating intelligent machines run on intelligent computer programs. The nearly decades-long quest has recently opened unimagined possibilities of AI. There have been advancements in the area of mathematical modelling, Computer Vision especially using Deep Learning, Natural Language Processing using Transformers, and the use of Blockchain for secure transactions.

2.2 AI in Healthcare

AI must be built on robust and explainable AI theories and deployed as safe, reliable, and trustworthy technology to be integrated into healthcare. AI is seen more as a black-box model, and to be a practical healthcare tool, it will have to explain the underlying information on the model's working, i.e., explainable AI (XAI). In healthcare, it is imperative that machine decisions and predictions are reliable. In healthcare, interpretability includes risks and responsibilities. Hence the need is to understand the working of the algorithms that train the models [7]. The fidelity of the algorithms can be assessed through ablation studies. The studies help to understand the significance of the training methods and the choice of hyperparameters. Perturbing inputs that lead to a decrease in model performance can help understand the inner working of the model [8]. XAI will bring trustworthiness and confidence in the models leading to a better validation of the model output.

3 Predictive Healthcare Analytics

3.1 Industry 5.0 and Healthcare 5.0

The first Industrial Revolution (Industry 1.0) harnessed the power of steam, the second coupled the power of electricity to create massive production lines, and the third brought in automation using computer-programmable logic and communication equipment. The evolution is shown in Fig. 1. Germany launched Industry 4.0 in 2011 and gave us connected machines, namely the Internet of Things (IoT), integrated AI and Cyber Physical Systems. Industry 4.0 went further by making intelligent decisions using real-time communications using Cyber Physical Production System (CPPS) [9]. Other countries adopted similar initiatives, e.g. Industria 4.0 (Italy) and Society 5.0 (Japan). Industry 4.0 focused on process automation and less on social

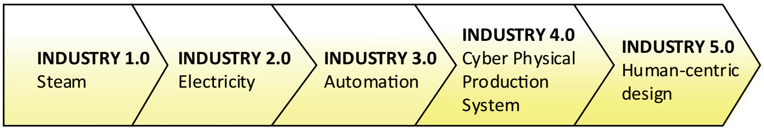

Fig. 1 Evolution from industry 1.0 to industry 5.0

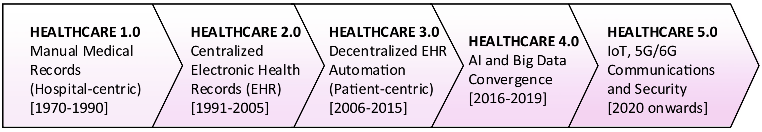

Fig. 2 Evolution from healthcare 1.0 to healthcare 5.0

fairness and sustainability. Hence the development of Industry 5.0, which looks at holistic growth focused on human progress and well-being based on sustainable economic and equitable prosperity [10].

The healthcare ecosystem has evolved in the past decades, as shown in Fig. 2. There has been a paradigm shift in healthcare from hospital-centric to patient-centric. This is primarily due to the emerging digital technologies that seamlessly link persons, institutions, and resources. It has been a long journey from Healthcare 1.0, where healthcare professionals manually captured medical data on paper, to Healthcare 4.0, which integrated Mobile Technologies, IoT, Robotics, AI, Blockchain, Cloud, and Fog computing. Healthcare 5.0 [11] ushers in a new era of personalized smart healthcare. An era of smart disease detection and control, smart virtual monitoring of patients, and smart patient-care management.

The authors in Gupta et al. [12] propose an Internet of Drones (IoD) and a secure Blockchain-based system to deploy Healthcare 5.0, especially in harsh conditions. In Gohar et al. [13], the authors present patient-centric healthcare that integrates IoT, cloud, and Blockchain resulting in secure healthcare data sharing, better protection, and enhanced interoperability.

3.2 *Healthcare Analytics*

The breakthroughs in AI have helped healthcare professionals to collect and analyze voluminous amounts of patient data in real time. Analytics can be descriptive, predictive, or prescriptive and involves quantitative and qualitative analysis. Analytics can help to visualize meaningful insights and also anticipate potential healthcare problems. The analytics capabilities in healthcare will allow for more accurate diagnoses and more effective treatments, facilitating personalized medicine. Another essential dimension of analytics is that the systematic use of data leads to better business insights which help in fact-based decision processes.

The authors in Miah et al. [14] present a detailed study of healthcare analytics research used in clinical and non-clinical decision-making. They highlight how healthcare analytics helps minimize cost and optimize health at the level of clinical care of patients and hospital management. The Healthcare Analytics solutions look into data collection processes, cleaning (noise minimization), classification, and inferring meaning from the data. Effective Healthcare Analytics has resulted in Health Information Systems (HIS) that effectively integrate the management of health sector data to manage programs and patients. It has a hierarchy of data collection beginning with routine district-level information systems and broadening to disease surveillance systems. Healthcare Analytics can provide a holistic quality of efficient and cost-effective care by providing streamlined operations and reduced waste.

Predictive healthcare analytics were seen during the coronavirus disease 2019 (COVID-19) pandemic. On March 12, 2020, World Health Organization declared the COVID-19 pandemic, a day remembered in history as it began waves of global health crises. Scientists, engineers, and doctors used predictive analytics [15] to monitor patient outcomes, visualize the spread of the disease, plan hospital constraints, and forecast the severity of the disease.

The authors in Bastani and Shi [16] advise caution in the use of healthcare analytics. The trained machine-learning models can exhibit biases and may beg the question of the reliability of the predictive models. They suggest a human-in-the-loop approach. The domain experts, namely healthcare professionals, should actively participate in developing healthcare predictive models. Another issue is the use of predictive models in dynamically changing uncertain environments. They suggest using a Markov decision structure to build decision models that integrate the predictions with the dynamic system model.

Another area of healthcare analytics that is growing by leaps and bounds is the wearable human activity recognition (HAR) systems [17] that monitor the activities of daily living through wearable sensors that pick up data. As health-related difficulties emerge, there is a need to recognize and diagnose early disease symptoms that can later cause chronic and severe illness. At a later stage, the sickness may become very challenging to treat and may be the cause of the patient's death. Wearable gadgets can screen a person's physiological parameters 24×7. Sensors include those that measure body temperature, blood pressure, pulse, pulse oximetry, respiratory rate, and other wearable sensors. Smart wearable devices then transfer the patient personal information to devices and often stores it in the cloud. One of the main challenges is acquiring data, especially if it is from multisensors. Some of the challenges of HAR include the complexity of some human activity and the scalability of acquired data.

4 Healthcare IoT Applications

4.1 Internet of Things (IoT)

Kevin Ashton, in 1999, began his pilot work of connecting objects to the internet using RF technologies. Thus began the development of the Internet of Things (IoT). Like a multifaceted diamond, IoT can be perceived in many ways, for example, by its type of wireless connectivity (Wireless Sensor Network, Radio Frequency Identification), variety of application domains, and network architectures. Each IoT device has an identity, senses, communicates, computes, renders services, and performs semantic operations to extract useful abstract information.

4.2 HIoT

IoT in the healthcare context consists of the digital connection of intelligent sensing devices and objects that acquire and monitor data related to healthcare [18]. Healthcare IoT (HIoT) aims to enhance healthcare quality using wired/ wireless connectivity, biosignal monitoring, machine-to-machine interaction, and human-to-machine interaction. HIoT is also called the Internet of Medical Things (IoMT) or the Internet of Healthcare Things (IoHT). HIoT is ubiquitous in terms of the extensive use of smart wearables used for fitness tracking.

An HIoT architecture is shown in Fig. 3. It is usually deployed within a small geographical area either a patient or healthcare facility. It consists of different sensors and wearables, the things layer that monitor human body vitals and communicate through Bluetooth, WiFi, or ZigBee technologies. The sensors can also do some part of the pre-processing of data before transmitting it. Cloud services are used for sensors that generate large amount of data. Fog network that works on distributed computing is used to enhance latency, security and interoperability. The communication layer sees that the sensed data is sent to the processing layer for healthcare analytics.

4.3 Challenges

The author in Qadri et al. [19] mentions two significant challenges for HIoT; Quality of Service (QoS) improvements and Scalability. QoS improvements include low latency, low power operation, security and real-time operations. Scalability issues include scalable deployment, networking solutions, interoperability, regulatory frameworks, and service availability.

Low latency is one of the significant challenges of the use of HIoT. The transmission between sensors can be enhanced using Edge Computing, which also provides interoperability. Another challenge is the volume of data that is generated every

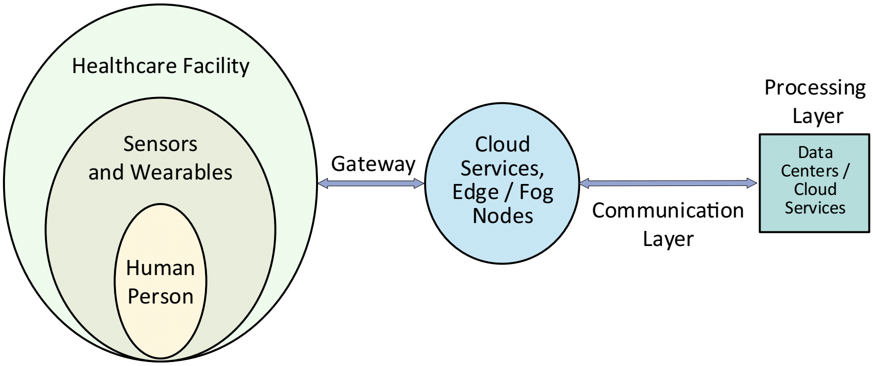

Fig. 3 HIoT architecture

second. Big Data can be used to meet the challenge of high volume and a large variety of data. Integration of healthcare analytics and IoT enable real-time analysis. Blockchain can help in meeting the challenge of security and transparency. Software Defined Networks can ensure flexible deployments of large networks. Other challenges include the development of HIoT standards, integration with heterogeneous ecosystems, and scalability.

Various architectures are proposed in Qadri et al. [19] to target the requisite applications. These include Ambient Assisted Living, Cardio-Vascular Diseases, Neurological Disorders, and Fitness tracking.

5 Conclusion

AI has tremendous potential to revolutionize the healthcare industry through AI predictive analytics and HIoT devices. Predictive healthcare analytics primarily used for low-risk patients can be used to identify high-risk patients for chronic conditions. These can be administered through personalized treatment plans. HIoT devices can provide valuable data for predictive analytics and enable healthcare professionals to remotely monitor patients. AI can be used to study readmission risk prediction. However, some challenges that must be addressed are high-quality data, privacy, and security. AI can play a crucial role in the triage of patients to predict risk complications. A well-integrated secure Artificial Intelligence for Predictive Healthcare Analytics and Healthcare IoT Applications based ecosystem will help improve healthcare delivery's efficiency and effectiveness and ultimately improve patient outcomes.

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